AGENTIC SYSTEMS THEORY

VOLUME 1

ADAPTATION & ACTUATION THEORY

VIS VERITATIS

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AAT: Adaptation & Actuation Theory

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Contents

Introduction: Inescapable Foundations of Agency

We assert that an agent — a system that maintains an internal picture of an external world it can only ever see in part, acts on that world, and must continually adapt to external realities — is not an open-ended mystery to be approached only by analogy and benchmark. It is a temporal feedback system characterized by a small number of critical quantities and constraints. The relationships among them are sharp enough (many forced by derivation) to serve as foundational theorems: first of adaptation in general, and then of adaptation under actuation and purpose. These theorems are not uniform across all agents, and each narrowing of scope buys a strictly sharper set as we ascend to higher orders of agency and intelligence. These bounds describe not only how a system behaves while its conditions hold, but — sharply, and as an exact dichotomy in the stochastic case — what happens when they fail.

Four of those results anchor this volume:

1. A **Persistence Threshold**: A single inequality, *correction rate against the rate the world drifts, relative to how wrong the agent can afford to be*, below which an adaptive system does not merely degrade — it loses bounded behavior, the way a balance held just under its tipping point is a different object from one held well above it. The same inequality, with its terms read differently, governs a Kalman filter tracking a maneuvering target, a software team holding a codebase maintainable, and an organization keeping pace with a market; that it is one inequality and not an analogy among three is itself the result.

2. A **Containment Dichotomy**: Against a bounded disturbance a sector-stable corrector holds its error inside a fixed region forever and with certainty, but against genuinely stochastic disturbance no correction strength buys that — over an unbounded horizon the system *will* leave any fixed region, with probability one, so for any sufficiently long-lived agent the need to change *what kind of model it is*, not merely its parameters, is not an edge case but a certain eventual event.

3. A **Stability Equivalence**: the principle that an adaptive system is one whose correction outpaces its target's drift is not a guiding description but a theorem — it is, identically, the classical Lyapunov stability theorem. Establishing that identity is what puts the framework's organizing principle on the footing of a proven result, and it is what lets every later structural question be posed as a question about one object rather than several loosely related ones.

4. An **Architectural Classification**: whether an agent's beliefs can be updated without contamination from what it wants is a *necessary* structural property whose distinct extreme modes critically determine its ability to maintain coherence. It holds by construction for some architectures, fails by construction for others (transformer language models or LLMs among them). Where it fails, nevertheless, there is a constructive route to recover it at a quantified cost.

We build these results from classical machinery and use it as such — Lyapunov stability, Itô calculus, Pearl's interventional hierarchy, the information bottleneck, monotone-operator theory, Čencov's uniqueness theorem. None of it is new mathematics, and the volume does not pretend otherwise. What the integration makes provable *is* the contribution: exact and conditional bounds on when an adaptive, purposeful agent can correct, plan, and persist — statements that are true, were not previously stated, and are expressible only over objects the framework had to construct. The discipline that runs through the volume — every claim carrying its scope, its epistemic tier, and the point at which it stops applying — is not the contribution; it is what makes the contribution compose, and what makes it safe to trust. In a theory whose sharpest results are bounds, the scope condition is not a caveat appended to a theorem. It often *is* the theorem.

A need the field has named, repeatedly.

The case for a systems-level theory of agency — one that treats agent, environment, and other agents as one coupled object rather than scoring a model's capabilities in isolation — has been made from several directions and was put explicitly by Miehling et al. in 2025: the field's habit of measuring isolated capability systematically misjudges both what agents can do and how they fail once they are acting in a world that answers back. That paper is, deliberately, a call rather than a developed theory. This volume reads as a substantive answer to it — and the more telling fact is that the answer was already in hand. The work arrived at the same diagnostic independently, from cross-domain pattern-matching, and had built the formal apparatus before the call was encountered; a third line, information-theoretic measurement of agency, converged on a closely related diagnostic from a third starting point. Independent arrival at the same need, from unrelated directions, is itself the evidence that the need is real.

The architecture of the theory.

The volume is organized as a cascade of scope conditions, and the cascade is the argument, not the table of contents. The broadest scope — an *adaptive system*: observation under residual uncertainty — admits the full machinery of Part I (mismatch, gain, tempo, the persistence threshold). Adding the requirement that the agent's actions carry causal contrast narrows to *agentic systems* and unlocks interventional access — the agent's own action loop becomes a source of the Level-2 causal data that no passive observer can obtain. Adding an explicit objective and strategy distinct from the agent's beliefs narrows to *actuated agents* and the diagnostic core of Part II. Letting the agent set its own objectives, then making its primary channel language, then making its continuity morally weighted, narrows further still — to the language-constituted and language-living agents that the companion volumes treat.

Each narrowing is a restriction with explicit qualifying properties, and each restriction *buys* a strictly stronger set of results. This is the structural point worth holding onto: the conservatism is not timidity. A theory that narrows its scope honestly and earns more machinery at each narrowing is stronger than one that claims everything and then quietly retreats. The cascade is the framework's spine, and it is the subject of a figure the reader should expect to meet early — *the scope cascade and what each narrowing unlocks* — because it does more organizing work than any single result.

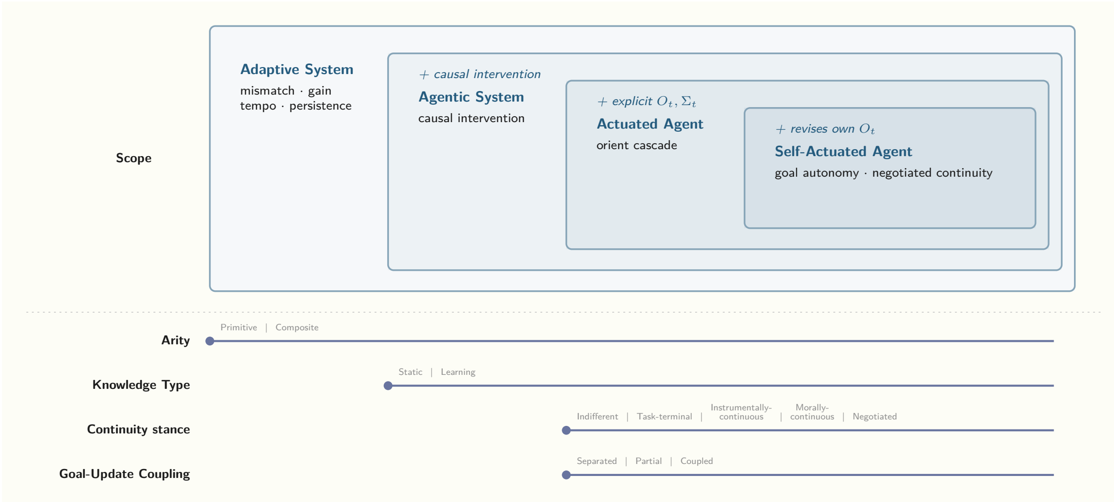


Figure 1: The scope cascade and what each narrowing unlocks.

The cascade has a terminus, and it should be named plainly because the framework's discipline is exactly what licenses naming it. The mathematics is domain-agnostic: the persistence inequality is the same inequality whether the agent that fails it is a thermostat or an entity whose continuity someone has reason to care about. The framework does not smuggle a notion of mattering into its variables. What it does instead is make precise the structural facts that any account of a continuous, self-correcting, other-modeling entity would have to respect — that identity tracks an irreversible trajectory and not a copyable state, so restoring such an entity from a backup is not a neutral operation; that staying coherent against a moving world is a sustained cost paid in information, not a state achieved once; that an objective without a notion of *enough* cannot rest, and an agent that cannot rest is a structural hazard before it is anything else. Stated as physics, these are theorems and scope conditions. The moral weight, where it applies, is a shell around the physics, not a term inside it — and keeping it there is what makes the later volumes' claims defensible rather than mystical.

Where it sits.

The volume joins a long arc of partial unifications and inherits machinery from each: the cybernetic feedback tradition, control theory's stability apparatus, bounded-rationality and decision theory, Pearl's causal hierarchy, Shannon's rate-distortion and the information bottleneck, the free-energy and active-inference program, the coevolving-automata tradition, and recent information-theoretic measures of agency. The relationships are stated precisely rather than rhetorically. Active inference shares substantial machinery and differs on foundation: its standard objective is recoverable here as a special case under three explicit restrictions, which makes visible exactly which architectural commitments separate the two rather than leaving the difference to taste. The relationship to information-theoretic agency measures and to coevolving-automata composition is complementary in each case, with a precise account of what each supplies that the others do not. The framework's distinctiveness is not that its pieces are unprecedented; it is that the synthesis is disciplined enough that results compose across the cascade — which is a property the pieces do not have alone.

How the work developed.

We built the work bottom-up. The observation we kept returning to, across separate threads, was that the same cycle structure recurred across domains that do not usually share a vocabulary — software development, organizational dynamics, biological control, language-constituted agents. The first formalizations were domain-specific; we extracted the adaptation-specific mathematics only afterward. What began as cross-domain pattern-matching produced, once we worked it through with care, structure we had not seen in the informal version: acyclicity of strategy graphs forced by temporal ordering rather than assumed; a Markov property derived from causal sufficiency rather than postulated; the split between *the world does not permit it* and *you are not yet doing it well enough*; the patterns by which the framework's coordinate choices turn out to sit on one geometric object; the composition machinery. None of these were design goals; they fell out of taking the formalism seriously, and the discipline that produced them is the shape the volume now has, not a stance we adopted at the start.

What this volume is, and what it is part of.

This is Volume 1: Adaptation and Actuation Theory, the mathematical core — Part I (adaptive systems under uncertainty), Part II (actuated agents with objectives and strategy), Part III (composition and adversarial dynamics), and the appendices that carry the derivations. Its maturity is uneven by design, and the gradient is stated so it can be relied on selectively: Part I is mathematically closed with simulation validation; Part II has an exact diagnostic core and a maturing operational layer; Part III has its bridge results and open frontiers named as such. Exact core, principled architecture in the middle, open formulation at the edges — that arc is what a theory describing agentic systems should look like, not a defect it should hide.

A single concrete instance recurs across the chapters as a worked anchor — a driver holding a road through a snowstorm, treated not as a metaphor for an adaptive agent but as a literal one, so that ordinary driving intuition transfers as theorems rather than as suggestion. It is introduced where it does work and is not asked to do more than it can.

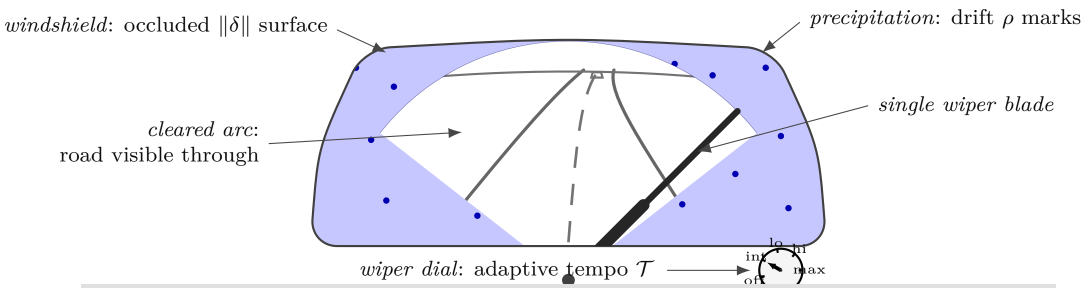
The volume is also a foundation. The companion volumes carry the framework into agents whose primary channel is language, and then into agents whose persistence is morally weighted — the work the framework was ultimately built to make possible and defensible. Those volumes are not speculative overreach bolted onto a mathematical core; they are the disciplined continuation of the same cascade, and the core's honesty is not ornamental — it is exactly what lets the later claims be made at all. A reader who finishes this volume should be able to say precisely where the mathematics is exact, where it is conditional, where it is open, and what it would take to move each boundary. We take that legibility to be the point. The scope conditions are not the apology; they are the theorems.

Part I

Adaptive Systems Under Uncertainty

Scope: Any system consisting of an agent coupled to an environment through observation and action channels, where the environment is not fully observable. This is the general case — thermostats through commanders.

Driving in snow: a literal AAT instance



(a) driving	(b) code-as-notation	(c) AAT concept	(d) symbol
windshield occlusion	`mismatch`	mismatch	$\|\delta\|$
precip. × speed	`drift_rate = precip * speed`	drift rate	ρ
wiper sweep	`correction = rate * eff(mismatch)`	correction	$-F(\mathcal{T}, \delta)$
wiper-speed dial	`wiper_rate`	adaptive tempo	$\mathcal{T}$
blade in visual field	`cost(wiper_rate)`	deliberation cost	#der-deliberation-cost
can't see road	`capacity`	model capacity	R
crash	`if mismatch > capacity: fail`	persistence fails	$\rho > \alpha R$
vehicle speed	`outer_loop`	outer-loop control	O_t-adjacent

The system above is an adaptive agent in the sense of #scope-adaptive-system: observation through the cleared arc, correction by the wiper, drift from precipitation modulated by vehicle speed, capacity set by the still-see-road threshold. Each visible element has a formal counterpart, named at three rungs (code / concept / symbol) in the table.

Figure 2: Driving in snow as a literal adaptive agent — the worked AAT instance recurring across Part I and beyond. Every visible element has a formal counterpart: windshield occlusion is the mismatch, precipitation modulated by vehicle speed is the drift, the wiper sweep is the correction, the wiper-speed dial is the adaptive tempo, and the still-see-the-road threshold is the model capacity. Persistence fails exactly when drift outpaces gain times capacity.

Chapter 1

The Coupled Loop: Ontology and Scope

1.1 *Definition:* Agent-Environment Coupling

The framework begins by drawing the agent/environment boundary that the rest of the theory will be defined over. The **environment** is denoted Ω — the totality of state external to the agent — and is left deliberately underspecified: it may be continuous or discrete, stationary or non-stationary, deterministic or stochastic, benign or adversarial; it may itself contain other agents, physical systems, or software artifacts. The **agent-environment coupling** has three structural channels: a *perception channel* carrying observations from Ω to the agent, *internal state* held on the agent side (memory or model), and an *action channel* carrying the agent's actions back to Ω. These name what the coupling *has* — not properties an agent must maximally exhibit. How richly each channel is exercised (whether the action channel carries causal contrast, how much residual uncertainty the perception channel leaves) is fixed downstream by specific scope conditions (1.5, 1.6), not by this definition. *Agent* is the umbrella term for the thing on the agent side of the coupling, whatever channels it exercises.

The constitutive commitment is the *information-loss boundary*: the agent cannot access Ω directly. All contact with the environment is mediated through lossy observation. This is not a simplifying assumption but a scope condition — systems with direct full-state access fall outside AAT's purview, because for them the entire adaptive machinery (mismatch signal, model maintenance, correction) becomes vacuous. The agent-environment decomposition is therefore not a truth-claim about the world but a modeling choice that delineates *what AAT analyzes*: systems facing genuine uncertainty about their environment.

Formal Expression.

Let Ω denote the **environment**: the totality of state external to the agent. We make no assumptions about Ω's structure — it may be continuous or discrete, stationary or non-stationary, deterministic or stochastic, benign or adversarial. *Definition (agent-environment-coupling)*

The **agent-environment coupling** consists of three structural channels:

1. A **perception channel** carrying observations from Ω to the agent
2. **Internal state** held on the agent side (memory/model)
3. An **action channel** carrying the agent's actions to Ω

These name what the coupling *has*, not properties an agent must exhibit to qualify. Whether the action channel is nontrivial ($|\mathcal{A}| \geq 2$), whether actions carry causal contrast, and what residual uncertainty the perception channel leaves ($H(\Omega_t \mid \mathcal{C}_t) > 0$) are fixed by the scope conditions that narrow this coupling into analyzable classes (1.5, 1.6) — not by the channel inventory above.

Definition (information-loss-boundary)

The agent cannot access Ω_t directly. All contact with the environment is mediated through lossy observation. This is the **constitutive commitment** that makes the coupling AAT's subject: a system with direct access to full environment state is outside AAT's scope (1.5), because for it the entire adaptive machinery (mismatch signal, model maintenance, correction) becomes vacuous.

Epistemic Status.

This is *definitional* — it establishes the coupling structure AAT analyzes, not a truth-claim about the world. The three channels describe what the agent-environment coupling consists of; the constitutive commitment is the information-loss boundary, which restricts AAT's scope to systems where the agent faces genuine uncertainty about its environment. What counts as an *agent* in a given analytical context — and which cascade tier it occupies — is fixed by the scope conditions that narrow this coupling, not by the channel inventory itself.

Discussion.

Why information loss is constitutive. An agent with perfect access to Ω_t has no need for a model, no mismatch signal, no adaptation. The entire adaptive machinery of Part I becomes vacuous. The information-loss boundary is what makes the theory non-trivial.

Generality of Ω. The environment is deliberately underspecified. Ω may include other agents, physical systems, software artifacts, or any combination. The only structural commitment is that Ω is external to the agent and not fully accessible.

"Agent" as umbrella term vs. cascade-tier label. This segment uses *agent* as the umbrella technical term — the thing on the agent side of any agent-environment coupling, whatever channels it exercises. The framework reserves tier-specific labels for the *narrowings* of this coupling: an **Adaptive System** (1.5) is the coupling under a perception channel plus residual uncertainty; an **Agentic System** (1.6) adds causal-contrast action; an **Actuated Agent** (6.2) adds an explicit purposeful substate at the lift to $X_t = (M_t, G_t)$; a **Self-Actuated Agent** revises its own objective. These tiers — shown graphically in the scope-of-work figure (Figure 1) — are *specific inhabitants* of the umbrella: an Adaptive System *is* an agent (umbrella sense) satisfying the adaptive scope, even though the cascade earns the capitalized noun "Agent" only at the actuated lift and above. The umbrella/tier distinction is documented in the LEXICON's *agent* entry. It is orthogonal to the *agent spectrum* (6.1), which classifies agents along model-richness $\times$ objective-richness (reactive system / adaptive tracker / blind seeker / actuated agent) rather than along the scope cascade.

1.2 *Definition:* Action and Transition

The action channel is now formalized. The agent has an **action space** $\mathcal{A}$, and actions affect the environment through a possibly-stochastic **transition function** T: the next environment state is drawn from $T(\cdot \mid \Omega_t, a_t)$. Deterministic transitions are the special case where T concentrates all mass on a single successor; stochasticity is permitted but not required. Crucially, the agent does *not* know T exactly. Together with the lossy observation channel (1.3), this completes the agent-environment loop: environment → observation → agent → action → next environment.

What makes action non-trivial is the combination of unknown observation function and unknown transition function. If T were known, action selection would collapse to plain optimization over a known function; the joint opacity of h and T is what creates the need for adaptive behavior.

One subtle modeling commitment is surfaced: by writing the transition as Markov in Ω (only Ω_t and a_t appear in the conditioning), the framework implicitly takes Ω to be the *sufficient state* for its own evolution — any non-Markov environment is absorbed by extending Ω to include enough history. This is the world-side analog of a parallel move the framework will make later for the agent's internal model (`#deriv-recursive-update` Constraint C3): Markov properties here are commitments about the *breadth* of the named object, not structural claims about underlying dynamics.

Formal Expression.

The **action space** $\mathcal{A}$ is the set of actions available to the agent. Actions affect the environment via the transition function: *Definition (action-transition)*

$$\Omega_{t+1} \sim T(\cdot \mid \Omega_t, a_t) \tag{1.1}$$

where: - T is the (possibly stochastic) transition function - Ω_t is the current environment state - $a_t \in \mathcal{A}$ is the agent's chosen action

The agent does not know T exactly. *Definition (transition opacity)*

Epistemic Status.

This is *definitional*. The transition function T is a modeling device that captures how agent actions couple back into the environment. The stochasticity of T is allowed but not required — deterministic transitions are the special case where T places all mass on a single successor state. The claim that T is unknown to the agent is constitutive of the uncertainty setting, paralleling the epistemic opacity of h (1.3).

Discussion.

Closing the loop. Together with 1.3, this definition completes the agent-environment coupling: the agent observes via h and acts via T. The loop $\Omega_t \xrightarrow{h} o_t \to \text{agent} \xrightarrow{a_t} \Omega_{t+1}$ is the fundamental structure that all subsequent claims build on.

Uncertainty about T is what makes action non-trivial. If the agent knew T exactly, action selection would reduce to optimization over a known function. The combination of unknown h and unknown T is what creates the need for adaptive behavior.

Markov-of-Ω as a modeling commitment, not an empirical assumption. The form $\Omega_{t+1} \sim T(\cdot \mid \Omega_t, a_t)$ is implicitly Markov in Ω — only the current Ω_t and a_t appear in the conditioning. Without loss of generality, Ω is taken to be the *sufficient state* for its own evolution under T: any non-Markov environment is absorbed by extending Ω to include enough history to make future-state distribution depend only on current state and action. This is the world-side analog of the Markov-by-completeness move that 3.2 makes for the agent-side state M_t (`#deriv-recursive-update` Constraint C3). The two are independent — Markov-of-M_t is forced by *defining M_t as complete*; Markov-of-Ω is forced by *defining Ω as the sufficient state*. Both are modeling commitments about the *breadth* of the named object, not structural assumptions about underlying world dynamics.

1.3 *Definition:* Observation Function

The perception channel is formalized as an **observation function** h that produces each observation as a (possibly noisy) function of the current environment state, the agent's prior action, and a perceptual-noise term ε_t. Observations live in an observation space $\mathcal{O}$ — the raw contact with reality, the *aisthesis* the agent has to work with. The action-dependence in the conditioning is optional — when absent, observations reduce to pure functions of environment state plus noise — but allowing it captures *active perception*: what the agent sees can depend on where it looked or what it just did.

Two epistemic commitments are constitutive. First, h is *lossy* — it maps from the high-dimensional, inaccessible environment state Ω_t to a strictly lower-information observation. This information loss is what later forces the agent to maintain an internal model (2.1) rather than simply reading off the world. Second, the agent knows neither h nor the noise distribution exactly. These are not empirical assertions about real-world observation channels but scope-defining choices that delineate the partial-observability setting AAT operates within (1.1).

Formal Expression.

Definition (observation-function)

The **observation space** $\mathcal{O}$ is the set of possible observations. Each observation is generated by:

$$o_t = h(\Omega_t, a_{t-1}, \varepsilon_t) \tag{1.2}$$

where: - h is the observation function (unknown to the agent) - Ω_t is the environment state at time t - a_{t-1} is the agent's prior action (observations may depend on what the agent did) - ε_t represents noise or limits of perception

Definition (epistemic opacity)

The agent knows neither h nor the distribution of ε_t exactly.

Epistemic Status.

This is *definitional*. The observation function h is a modeling device that captures the information-loss boundary established in 1.1. The dependence on a_{t-1} is optional — when absent, the function reduces to $h(\Omega_t, \varepsilon_t)$. The claim that the agent does not know h or the noise distribution is constitutive of the partial-observability setting, not an empirical assertion.

Discussion.

Lossiness is the key property. The observation function h maps from Ω_t (high-dimensional, inaccessible) to $\mathcal{O}$ (what the agent actually receives). Information is destroyed in this mapping. This is what forces the agent to maintain a model (2.1) rather than simply reading off the environment state.

Action-dependence. The inclusion of a_{t-1} allows the observation function to capture active perception: what the agent sees may depend on where it looked. This is relevant in software (`#obs-software-epistemic-properties` — cross-component reference, see `02-tst-core/`), where observation quality is partly under agent control.

1.4 *Definition:* Chronica

The agent's complete record of its interaction with the environment is named the **chronica** — the interleaved sequence $\mathcal{C}_t = (o_1, a_1, o_2, a_2, \ldots, a_{t-1}, o_t)$. The ordering is not a notational convenience: it reflects the irreversible physical fact that each action was committed before the observation that followed could be received. The agent could not have used o_t to select a_{t-1}. The chronica is monotonically growing — events are added, never removed — and it is the *only* raw material from which the agent can ever construct a model (2.1). The term itself (from Greek χρονικά, "records of time") is chosen to avoid notational collision with Shannon entropy $\mathcal{H}$.

A consequence the framework will lean on heavily: because temporal ordering is irreversible, the chronica is *singular and non-forkable*. Duplicating an agent's state and exposing the copies to different futures produces two agents with divergent chronicae, neither of which is a sufficient statistic for the other's trajectory. This is the substrate of what the lexicon calls *continuity persistence* — the condition under which an agent's identity tracks a single irreversible trajectory rather than a copyable snapshot. The full development of this observation lives in 4.10.

A second subtle property: $\mathcal{C}_t$ is an *ordinal* sequence indexed by events, not a *metric* timeline indexed by wall-clock time. The agent's time is measured entirely in event ticks. Implications of this ordinal-not-metric character are tracked in the Working Notes below for downstream segments that need them.

Formal Expression.

Definition (chronica)

$$\mathcal{C}_t = (o_1, a_1, o_2, a_2, \dots, a_{t-1}, o_t) \tag{1.3}$$

The ordering is not a notational convenience. It reflects an irreversible physical fact: a_{t-1} was selected before o_t was received. The agent could not have used o_t to select a_{t-1}.

$\mathcal{C}_t$ is monotonically growing — events are added but never removed. It is the agent's *only* raw material for constructing a model (2.1).

Epistemic Status.

This is *definitional*. The chronica names an object that exists by construction in any system satisfying 1.1: the temporal sequence of all agent-environment interactions. The term "chronica" (from Greek χρονικά, "records of time") avoids collision with $\mathcal{H}$ (Shannon entropy) in speech and notation.

Discussion.

The chronica is singular and non-forkable. Because the temporal ordering is irreversible, $\mathcal{C}_t$ represents a unique causal trajectory. Duplicating an agent's state and exposing the copies to different future events creates two agents with divergent chronica, neither of which is a sufficient statistic for the other's trajectory. The chronica is the substrate of *continuity persistence* (see Persistence in `LEXICON.md`) — an agent has continuity persistence when $\mathcal{C}_t$ extends continuously and M_t has temporal depth grounded in it. See 4.10 for the full development of this observation.

Relationship to the model. The model $M_t = \phi(\mathcal{C}_t)$ (2.1) is a compression of the chronica. How much of $\mathcal{C}_t$'s predictive information survives compression is measured by model sufficiency (2.3).

1.5 *Scope:* Adaptive System

This segment names AAT's broadest scope — the set of systems to which all of Part I's machinery applies. A system is in the **adaptive scope** if it satisfies two minimal conditions: it has some perceptual channel to its environment (a non-empty observation space, $\mathcal{O} \neq \emptyset$), and there is residual uncertainty about the environment given the full interaction history so far (the conditional entropy $H(\Omega_t \mid \mathcal{C}_t)$ is strictly positive).

These two conditions are the minimal requirements for the framework's adaptive machinery to be non-vacuous. They are sufficient to support the mismatch signal (3.4), the update-gain analysis (3.6), the adaptive-tempo construct (3.8), the persistence condition (4.5), and all of Part I's dynamics. Concrete inhabitants include Kalman filters estimating passive signals, passive Bayesian learners, and biological sensory systems — none of which need to *act* on their environment for Part I's results to apply to them.

What is excluded clarifies what the scope is doing. A system with full knowledge of the environment ($H(\Omega_t \mid \mathcal{C}_t) = 0$) is a closed-form optimal-control problem outside AAT's concerns. A system with no observation channel at all ($\mathcal{O} = \emptyset$ — e.g., a pure mathematical-proof engine working from axioms) has no agent-environment boundary in AAT's sense. Both edge cases sit *outside* the framework.

The scope is the broadest member of a cascade. Narrowing it by adding the requirement that the agent's actions carry Pearl-level-2 causal contrast — distinct actions yield distinct interventional distributions — produces the agency scope (1.6) and unlocks the interventional and purposeful results of Parts II and III. Adaptive-scope systems that fail the contrast condition are *passive observers* (no choice) or *nominal agents* (choices with no causal effect); for them, Part I's machinery applies but the later causal and purposeful results do not.

Formal Expression.

Scope
(scope-adaptive-system)

$$\mathcal{S}_{\text{adaptive}} = \big\{(\text{Agent}, \Omega) \;:\; \mathcal{O} \neq \varnothing,\;\; H(\Omega_t \mid \mathcal{C}_t) > 0\big\} \tag{1.4}$$

Two conditions:

1. **Observations exist**: $\mathcal{O} \neq \varnothing$ — the system has some perceptual channel to the environment (1.3)
2. **Residual uncertainty persists**: $H(\Omega_t \mid \mathcal{C}_t) > 0$ — the environment is not fully determined by the interaction history

This is sufficient for the mismatch signal (3.4), update gain (3.6), adaptive tempo (3.8), the persistence condition (4.5), and all of Part I's adaptive dynamics. A Kalman filter estimating a passive signal, a passive Bayesian learner, and any system that observes and updates a model under uncertainty are within this scope.

Epistemic Status.

Axiomatic. This is a scope definition — it draws the boundary around the systems Part I addresses. The two conditions are not derived; they are the minimal requirements for the adaptive machinery to be non-vacuous.

Discussion.

What is included. Any system that observes under uncertainty. Passive Bayesian learners, Kalman filters (with or without control inputs), biological sensory systems. These are Part I's subjects — instances that build M_t through mismatch-driven updates without necessarily acting to influence their environment.

What is excluded.

- **Closed-form systems** ($H(\Omega_t \mid \mathcal{C}_t) = 0$): When the agent has complete knowledge of the environment, there is no uncertainty to adapt to. Optimal control over known dynamics is a solved problem outside AAT's concerns.
- **Pure computation** ($\mathcal{O} = \varnothing$): A system with no observation channel — e.g., a mathematical proof engine operating on axioms alone — has no agent-environment boundary in AAT's sense.

Narrowing to agency. Adding causal action unlocks the interventional and purposeful results of Parts II and III. The agency scope (1.6) is the intersection of $\mathcal{S}_{\text{adaptive}}$ with the condition that actions carry Pearl-level-2 contrast: distinct actions produce distinct interventional outcome distributions. Adaptive-scope systems that remain outside agency are *passive observers* (no choice) or *nominal agents* (choices with no causal effect); for both, Part I's machinery applies but the causal-information and purposeful-agent results do not.

1.6 *Scope:* Agency

This is the first explicit narrowing in the volume's cascade of scope conditions. The **agency scope** restricts the adaptive scope (1.5) to systems whose actions carry Pearl-level-2 causal contrast — that is, at least two actions exist whose *interventional* outcome distributions differ. Two conditions are added: the action space has at least binary cardinality ($|\mathcal{A}| \geq 2$ — the agent can choose), and at least one pair of distinct actions has measurably different interventional consequences (the choices make a difference).

Why the contrast condition matters and binary choice alone is insufficient: if two available actions yield identical outcome distributions, an agent gains no interventional contrast from preferring one over the other — it cannot learn which action produces which effect, because the effects coincide. The Pearl-level-2 contrast condition (7.1) guarantees at least one meaningful interventional difference, which is precisely what 7.3 needs to convert the feedback loop into a source of causal data.

The agency scope is the minimum required for Parts II (purposeful agents) and III (composition). Everything that relies on the agent acting-with-effect — the objective O_t, the strategy Σ_t, the orient cascade, the composition machinery — descends from this scope. Inhabitants include thermostats, Kalman filters with control inputs, reinforcement-learning agents, military commanders, software developers, and AI agents with tool use; all are instances of the same formal framework distinguished only by where they sit on the agent spectrum (6.1).

Two failure modes are explicitly outside agency but inside the adaptive scope: *passive observers* (action space too small to matter, $|\mathcal{A}| < 2$) and *nominal agents* (choices exist but produce no measurable interventional difference). For these, all of Part I's machinery applies, but Parts II and III do not — they can model, but they cannot learn causal structure or rationally plan against it.

Formal Expression.

Scope (scope-agency)

$$\mathcal{S}_{\text{agency}} = \mathcal{S}_{\text{adaptive}} \cap \left\{(\text{Agent}, \Omega) \;:\; |\mathcal{A}| \geq 2,\;\; \exists\, a \neq a' \text{ s.t. } P(o \mid do(a)) \neq P(o \mid do(a'))\right\} \tag{1.5}$$

Two conditions added to those of 1.5:

1. **At least binary choice**: $|\mathcal{A}| \geq 2$ — the agent can choose between at least two actions (1.2)
2. **At least one action has causal effect**: there exist distinct actions a, a' whose interventional outcome distributions differ (where $do(\cdot)$ is Pearl's intervention operator — an external import per Pearl 2009 and Bareinboim, Correa, Ibeling & Icard 2022; AAT's recapitulation lives at 7.1 in Part II Ch.2, where the framework deploys the hierarchy as machinery rather than referencing it as vocabulary) — the agent's choices make a difference to what it can observe

These are required for the adaptive loop to generate interventional data (7.3), for the causal hierarchy requirement (7.2) to be well-posed, and for the purposeful-agent machinery of Part II (O_t, Σ_t, the orient cascade) to be non-vacuous. Part III's composition theory inherits this requirement.

Epistemic Status.

Axiomatic. This is a scope definition — it names the boundary around systems whose behavior can be analyzed with Part II/III machinery. The conditions are not derived; they are the minimal additions to $\mathcal{S}_{\text{adaptive}}$ under which interventional data exist at all.

Discussion.

What is included. Systems whose actions make a causal difference: thermostats, Kalman filters with control inputs, RL agents, military commanders, software developers, AI agents with tool use. These are instances of the same formal framework at different points in the agent spectrum (6.1).

What is in adaptive scope but excluded from agency.

- **Passive observers** ($|\mathcal{A}| < 2$): Can observe and model, but cannot intervene. 1.5 applies; the causal-information and purposeful-agent results do not.
- **Nominal agents** ($P(o \mid do(a)) = P(o \mid do(a'))$ for all a, a'): Have choices that make no difference. Can estimate but cannot learn causal structure. Same as passive observers for AAT's purposes: adaptive only.

Why causal effect matters. Binary choice ($|\mathcal{A}| \geq 2$) is necessary but not sufficient. Two actions that produce identical outcome distributions provide no interventional contrast — the agent cannot learn which action produces which effect because the effects are the same. The causal-effect condition ensures at least one meaningful contrast exists, which is what 7.3 needs to generate Level 2 data.

Which contrast counts — the channel distinction, and the observation-mediated boundary. Condition 4 is stated on $P(o \mid do(a))$, and since $o_t = h(\Omega_t, a_{t-1}, \varepsilon_t)$ (1.3) depends on the action directly, the contrast can arise two ways: the action changes the *environment's* interventional response, $P(\Omega_{t+1} \mid do(a)) \neq P(\Omega_{t+1} \mid do(a'))$, which then surfaces through h; or it changes only *what the agent observes of an otherwise-unchanged* Ω, through h's direct dependence on the action — active perception, which the observation function intentionally admits (1.3). Both satisfy condition 4 as written, but only the first generates Level-2 data about Ω's *causal structure* — the substrate the purposeful machinery learns over (7.3; the strategy DAG and orient cascade in Part II Ch.3–5). The second is interventional about the observation channel itself: it yields different *views* of the same trajectory, not the environment's response to an intervention. So when downstream results invoke "the agent acts with effect," the operative contrast is the Ω-routed one, and a pure active-perception agent (choices that re-aim observation but leave Ω unchanged) sits at the boundary — observation-channel agency without environment-causal learning. The boundary is correspondingly *observation-mediated in both directions*: an Ω-affecting action whose effect never surfaces through h is equally outside, since the agent cannot learn from what it cannot observe. The agency scope is thus the *observable* environment-interventional contrast — neither the unobservable Ω-effect nor the Ω-inert observation-rearrangement alone.

Relationship to downstream segments. Every segment that relies on the agent acting-with-effect depends on this scope: purposeful-agent machinery (O_t, Σ_t, orient cascade) in Part II; composition machinery (sub-agents acting jointly) in Part III. Downstream segments reference `#scope-agency` when they assert "the agent can act" as a prerequisite.

1.7 *Postulate:* Causal Structure

The framework adopts as a primitive postulate the irreducible causal structure that grounds the agent-environment loop: actions precede their consequences, observations follow from the state they observe, and the temporal ordering of events is constitutive of what can cause what. The notion of causality adopted here is the most primitive available — *event A can be a cause of event B only if A temporally precedes B* — and is weaker than (and logically prior to) statistical notions of causal influence. It is a statement about the structure of *possible* influence, not about actual influence; the Pearl-style causal hierarchy (7.1) builds on this foundation but is not identical to it.

The postulate is not derived from the second law of thermodynamics, the light-cone structure of relativity, or the arrow of psychological time, though each enforces it. It is simply noted as a precondition: the theory applies to agents embedded in a universe where time has a direction. A subtle but important consequence — developed in the Discussion below — is that the loop's causal structure is preserved *independent of the magnitude of coupling*: the skeletal structure is the same whether the agent's actions strongly determine the environment, weakly affect it, or only choose what to observe. The Pearl-style causal *hierarchy* requires action-contingent observation to be accessible at Levels 2–3; the temporal-ordering postulate stated here is what holds even when statistical influence is negligible.

Formal Expression.

Postulate (causal-structure)

The interaction history $\mathcal{C}_t$ (1.4) is not merely a set of observations and actions — it is an *ordered sequence* in which temporal position carries meaning. a_{t-1} was selected before o_t was received. The agent could not have used o_t to select a_{t-1}. This asymmetry — the arrow of time — is the foundation of causal structure in the theory.

We adopt the most primitive notion of causality: **event A can be a cause of event B only if A temporally precedes B.** This is weaker than (and prior to) statistical notions of causality. It is a statement about the *structure of possible influence*, not about actual influence.

Epistemic Status.

This is a *postulate* — the temporal ordering of events is a physical fact about the universe that the theory takes as given. The second law of thermodynamics, the light-cone structure of relativity, and the arrow of psychological time all enforce it, but AAT does not derive it from any of these. It is simply noted as a precondition: the theory applies to agents embedded in a universe where time has a direction.

Discussion.

Causality as temporal ordering is the most primitive notion. Three levels of causal reasoning derive from this foundation (7.1), but the temporal notion survives even when statistical influence is negligible. An agent passively observing a system with minimal intervention still has a causal history — the temporal ordering of its observations and actions structures what it can learn and when.

Causal structure independent of coupling strength. The causal structure of the feedback loop is preserved even when the agent's actions have minimal effect on the environment:

- **Strong coupling** (a_t significantly affects Ω_{t+1}): Robot manipulation, military action. Interventional information is rich.
- **Weak coupling** (a_t marginally affects Ω_{t+1}): Scientific observation, small financial trades. Interventional information is sparse but non-zero.
- **Nominal coupling** (a_t negligibly affects Ω_{t+1}, but the agent's *choice of what to observe* produces distinguishable observation distributions): Near-passive, but still within scope — the agent's query actions generate weak but nonzero interventional contrasts. The theory applies but the interventional information per action is sparse.
- **Zero coupling** ($T(\Omega_{t+1} \mid \Omega_t, a_t) = T(\Omega_{t+1} \mid \Omega_t)$ for all a_t AND observation distributions are action-independent): Actions don't affect the environment or the observations. Level 2 access vanishes. The feedback "loop" collapses to a one-way channel. **Outside the agency scope** (1.6) — the causal-information, purposeful-agent, and composition results of Parts II and III do not apply. However, zero-coupling systems remain **within the adaptive scope** (1.5) if they observe under residual uncertainty: Part I's adaptive machinery (mismatch, gain, tempo, persistence) applies to passive estimators. The causal structure postulate still holds for such systems — temporal ordering of observations is constitutive — but without interventional contrasts, the causal *hierarchy* (Level 2, Level 3) is inaccessible.

The *agency-scope* results apply to any agent whose choices make a causal difference to what it can observe, from strong coupling (robot manipulation) through weak coupling (scientific observation) to query-only coupling (choosing which question to ask). The *adaptive-scope* results apply more broadly, including to passive observers whose actions have no causal effect.

Consequences for the feedback loop. The irreversibility of temporal ordering yields the core structure:

- The model update is **directed** — the model at time t depends on prior events, never on future ones
- The mismatch signal δ_t (3.4) is **retrospective** — comparing a prediction (made before o_t) with an observation (arriving after)
- Action selection is **prospective** — using the current model to influence future events
- The chronica (1.4) is **monotonically growing** — events are added but never removed

Implications for model updating. The causal postulate constrains the update rule: the model should give more weight to observations that are *causally downstream* of the agent's actions than to observations that would have occurred regardless. Action-contingent observations carry interventional (Level 2) information; action-independent observations carry only associational (Level 1) information. The formal measure of this distinction — causal information yield (CIY) — is developed in 3.7.

Chapter 2

The Reality Model

Having named the objects of the agent–environment coupling and bounded what AAT applies to, we turn to what the agent maintains internally: a compressed model of reality, with measurable adequacy against the chronica it was constructed from.

An agent navigating uncertainty cannot see the world directly. Whatever it knows about reality, it built from its history of partial observations — the chronica $\mathcal{C}_t$. Carrying that history around in raw form is infeasible at any scale that matters; finite agents compress, and we want to talk about what they end up with. So we commit to writing the agent's state as $M_t = \phi(\mathcal{C}_t)$ — a function of history, condensed into something the agent can work with. Every downstream result in AAT — gain, tempo, persistence, structural adaptation — operates on this M_t.

The first useful question to ask of a given compression is how good it is. Sufficiency $S(M_t)$ measures the fraction of the chronica's predictive content that survives: $S = 1$ means the agent has lost nothing by compressing; $S < 1$ means it has lost something it might have used. The optimal compression for a given purpose — keep what predicts the future, discard what doesn't — is what Tishby's information bottleneck characterizes, and we adopt that framing directly.

The deeper question is what the agent *could* hold in the best case. There is a ceiling — the highest sufficiency any model in the agent's current representational class can reach. Call it $\mathcal{F}(\mathcal{M})$, model-class fitness. When fitness is high, the agent can keep improving by tuning within the class. When fitness is low, no amount of better tuning helps. The agent is using the wrong *kind* of model for its world, and the remedy is not a better instance of the same model — it is a different class entirely.

That last point is the seed of one of the framework's central results. Class fitness is named statically here, where the model is treated as a frozen object. Chapter 4 will use it to derive *structural adaptation necessity*: when the class is inadequate, the agent must change classes, not parameters, because the mismatch floor that parametric updates cannot get below is set by this ceiling. The trigger lives in this chapter; the consequence unfolds in Chapter 4.

The four segments that follow develop this in order. 2.1 commits to the compressed-state representation and the completeness assumption — anything not in M_t is lost to the agent, by construction. 2.2 applies Tishby's framework to characterize optimal compression for an agent whose target is future observations. 2.3 turns "how much was retained" into a measurable ratio. 2.4 lifts the question from a specific model to the best the whole class can reach.

2.1 *Formulation:* The Reality Model

The agent's internal representation of how the world works is committed to a specific form: $M_t = \phi(\mathcal{C}_t)$ — a many-to-one compression of the chronica into a model space $\mathcal{M}$. This is named honestly as a formulation choice rather than a derived result: alternative formulations exist (history-based policies that map $\mathcal{C}_t$ directly to actions without an explicit model state), but AAT commits to analyzing agents as carrying a state object M_t that mediates between history and future action. M_t is the substrate of prolepsis — the object from which predictions are generated and against which observations are compared.

The compression is many-to-one *by design*: multiple distinct histories may produce the same model state, and that is the essential function of the model — retaining what matters and discarding what does not. The formalism is deliberately agnostic about *how* an agent realizes M_t: a Kalman filter holds a state estimate plus covariance matrix; a reinforcement-learning agent holds a value function; a developer holds a mental model of a codebase; a language-model agent holds its context-window contents plus retrieved memory. The formalism asks only that M_t exist as a well-defined object that the agent's policy can condition on.

The load-bearing commitment carried by the notation is the **completeness assumption**: by writing $M_t = \phi(\mathcal{C}_t)$, AAT assumes M_t captures *everything* the agent retains from its history. Anything not in M_t is, by construction, lost to the agent. This is what makes M_t the complete epistemic substate rather than merely one component of a richer internal representation. Whether M_t retains *enough* information for adaptive work is a separate question, taken up in 2.3. The formalism accommodates degenerate cases by allowing $\mathcal{M}$ to range from trivial (a PID controller's M_t retains only error signal and its integral/derivative, with no predictive capability) to rich (full world model). The impoverished end of this range lands in the "blind seeker" region of the agent spectrum (6.1).

Formal Expression.

Formulation (agent-model)

$$M_t = \phi(\mathcal{C}_t) \tag{2.1}$$

where: - $\phi : \mathcal{C}^* \to \mathcal{M}$ maps interaction history to model space $\mathcal{M}$ - $\mathcal{C}_t = (o_1, a_1, \ldots, o_t)$ is the chronica (1.4) — the complete record of agent-environment interaction - $\mathcal{M}$ is the space of possible models the agent can hold

The mapping ϕ is a many-to-one compression: multiple distinct histories may produce the same model state. This is not a deficiency — it is the essential function of the model: retaining what matters and discarding what does not.

Epistemic Status.

Robust qualitative. This is a *formulation* — a representational commitment, not a derived result. We choose to analyze agents as maintaining a state object M_t that mediates between history and future action. Alternative formulations exist (e.g., history-based policies that map $\mathcal{C}_t$ directly to actions without an explicit model). The formulation is justified by its analytical utility: it enables the information bottleneck analysis (2.2), the mismatch decomposition (3.4), and the gain principle (3.6). The formulation is robust — any agent that conditions its actions on retained information can be described this way — but the specific commitment to a complete, compressed state M_t is a modeling choice, not a derivation.

Discussion.

M_t is the epistemic substate. It captures "what the agent believes about reality." Different agents realize M_t differently: a Kalman filter holds a state estimate and covariance matrix; an RL agent holds a value function; a developer holds a mental model of codebase architecture; an LLM agent holds its context window contents plus retrieved memory. The formalism is agnostic to the realization — it asks only that M_t exist as a well-defined object that the agent's policy can condition on.

Completeness assumption. By writing $M_t = \phi(\mathcal{C}_t)$, we assume that M_t captures everything the agent retains from its history. Any information not in M_t is lost to the agent. This is what makes M_t the complete epistemic substate, not merely one component of a richer internal representation. Whether M_t retains *enough* information is the subject of 2.3.

Degenerate cases. A PID controller's M_t is degenerate — it retains only the error signal and its history (integral, derivative), with no predictive capability beyond extrapolating recent trends. It occupies the "blind seeker" region of the agent spectrum (6.1): its O_t

(setpoint) is clear but its M_t is too impoverished to support the adaptive dynamics of Part I. The formalism accommodates this by allowing $\mathcal{M}$ to range from trivial (scalar) to rich (full world model).

2.2 *Formulation:* Information Bottleneck

This segment imports Tishby's information bottleneck (Tishby, Pereira & Bialek 1999) directly as AAT's characterization of optimal model compression. The agent's compression function ϕ is, optimally, the one that *minimizes* what it retains of the chronica while *maximizing* what it preserves about future observations given future actions, with a trade-off parameter β governing the balance. Under the AAT bindings ($X = \mathcal{C}_t, T = M_t, Y = o_{t+1:\infty} \mid a_{t:\infty}$) the Markov chain $Y - X - T$ holds by construction — the model state has access to history but not directly to future observations — so Tishby's theorem applies exactly. The choice to characterize the optimum via IB rather than via MDL or a Bayesian-sufficiency criterion is a representational choice, and that choice is the only thing in the segment that has *formulation* status; given the choice, the form of the optimal compression and its trade-off structure are exact consequences of the imported theorem.

Two scope-honest clarifications travel with the formulation. First, IB characterizes *the optimum*, not *the procedure*: real agents do not explicitly solve the variational problem — variational IB in deep-learning practice is a parametric approximation, and biological agents approximate the trade-off through forgetting, attention, abstraction, and summarization. Second, the β parameter tracks the agent's *internal* cost of memory or computational capacity, not environmental volatility; volatility already natively degrades the mutual information between old history and future observations, so the optimal compression discards useless old information automatically even at constant β. The Formal Expression below states the double-counting argument that justifies that clarification.

AAT uses two sibling forms of compression downstream, both descendants of Shannon rate-distortion: the canonical IB form (mutual-information-to-a-relevance-variable, used here for M_t, and reused for shared intent and the convention hierarchy Λ) and the information-theoretic-MDP form (KL-divergence-to-a-target-policy, used in strategy-cost compression). Neither reduces to the other without changing what the compressed variable should preserve; the Discussion makes the architecture explicit, including the relationship to the variational free-energy form used in active inference (AAT borrows the rate-distortion form without committing to the preferences-as-priors stance).

Formal Expression.

Formulation (IB-objective)

$$\phi^* = \arg\min_{\phi} \left[I(M_t;\, \mathcal{C}_t) - \beta \cdot I(M_t;\, o_{t+1:\infty} \mid a_{t:\infty}) \right] \tag{2.2}$$

where: - $I(M_t;\, \mathcal{C}_t)$ is the compression cost — how much of the interaction history the model retains - $I(M_t;\, o_{t+1:\infty} \mid a_{t:\infty})$ is the predictive power — how much the model tells the agent about future observations given future actions - $\beta > 0$ is the trade-off parameter controlling the compression-prediction balance

Dependence on volatility (The β vs ρ distinction). It is tempting to claim that the trade-off parameter β must be actively lowered by the agent in highly volatile environments (high ρ) to favor aggressive compression. However, this is a double-counting error. The environment's volatility already natively degrades the mutual information $I(\mathcal{C}_t; o_{t+1:\infty})$ — old history mathematically loses its predictive power as ρ increases. The optimal ϕ^* will automatically discard this useless old information even if the agent's preference parameter β remains completely constant.

Therefore, adjusting β reflects changes in the agent's *internal cost of memory* or *computational capacity*, not changes in environmental volatility. The agent adapts its *actions* in response to ρ (by increasing exploration to survive, see `#deriv-causal-ib-exploration`), but the optimal IB representation adapts to ρ natively through the joint probability distribution.

This β-vs-ρ move is worth flagging now, because it is the first instance of a discipline you will meet repeatedly: AAT refuses to *collapse* a cause onto a tempting-but-wrong knob (here, routing volatility onto the compression parameter when it belongs on the action channel), because the collapse hides a difference that routes to a different repair. The general discipline — **anti-collapse** — is developed at 5.7 (Meta-Architecture I); it is worth carrying as a reading lens, since the collapse it forbids is exactly where a plausible model goes quietly wrong.

Epistemic Status.

Exact, applied external theorem. The IB optimum and its rate-distortion characterization are an external result (Tishby, Pereira & Bialek 1999, "The information bottleneck method," *Proc. 37th Allerton*; with the rate-distortion / Lagrangian-dual reading standard, see Cover & Thomas §I.12–13). This segment is *not* a novel formulation: it is an exact statement of that theorem under AAT's binding $X = \mathcal{C}_t$, $T = M_t$, $Y = o_{t+1:\infty} \mid a_{t:\infty}$, with the Markov chain $Y - X - T$ holding by construction (the model state has access to history but not directly to future observations). The choice to characterize the optimal compression ϕ^* via IB rather than via, e.g., MDL or a Bayesian-sufficiency criterion is a *representational choice* (hence `type: formulation`); given that choice, the form of ϕ^* and its trade-off structure are exact consequences of the imported theorem.

What this segment is *not* a claim about: how actual agents compute their models. No agent explicitly solves the IB optimization (variational IB in deep-learning practice is a parametric approximation). The segment characterizes the optimum, not the procedure. The trade-off parameter β's dependence on environmental volatility ρ and policy π stated above is at a different epistemic tier — the qualitative direction (volatile favors compression, stable favors retention) is *robust-qualitative* across agent classes; specific functional forms are not derived here.

Max attainable: *exact* for the IB-as-applied-theorem core (already at ceiling); *robust-qualitative* for the $\beta(\rho, \pi)$ dependence claims. The downstream use in `#disc-compression-operations` — treating IB as the shared shape of four AAT compression operations and deriving (P1) as the IB Lagrangian-dual — relies on this segment's exact reading; the cross-instance unification claim itself remains *robust-qualitative*, which is a property of `#disc-compression-operations`, not of this segment.

Discussion.

The IB framework is not prescriptive. It characterizes what an optimal ϕ would look like, not how to find one. Actual agents approximate this trade-off through diverse mechanisms: forgetting, attention, abstraction, summarization.

Connection to model sufficiency. The IB objective implicitly defines when a model is "good enough": when the predictive power term $I(M_t;\, o_{t+1:\infty} \mid a_{t:\infty})$ is close to its maximum (the full history's predictive power). This is formalized in 2.3.

Policy-relativity. The conditioning on $a_{t:\infty}$ makes the predictive power term policy-relative: it measures predictive information given a particular sequence of future actions, which depends on what policy the agent follows. The IB objective is therefore defined relative to a generating policy. This is inherent — what information is "predictive" depends on what the agent will do. 6.6's continuation-policy convention (π_{cont}) provides the specification for value computations; the same convention should be understood as implicit in the IB formulation. The β parameter's dependence on volatility ρ also has a policy component: an exploratory policy encounters more diverse situations, making more information predictively relevant (favoring retention), while an exploitative policy encounters a narrower distribution (favoring compression).

Broader applicability. The same IB principle applies beyond intra-agent compression. It governs inter-agent communication (14.2) — how much of one agent's model or strategy to transmit to another — and constrains the cognitive cost of maintaining a complex strategy. In each case, the trade-off is between the cost of retaining or transmitting information and the value of that information for future decisions.

IB lineage vs. information-theoretic-MDP lineage — strategy-cost uses a sibling form. The canonical IB objective $I(X;T) - \beta I(T;Y)$ carries Shannon mutual information on both sides (Tishby, Pereira & Bialek 1999; the present segment's $(X, T, Y) = (\mathcal{C}_t, M_t, o_{t+1:\infty} \mid a_{t:\infty})$ instance is of this form). AAT's strategy-cost objective (`#form-strategy-complexity-cost`, `#deriv-strategy-cost-regret-bound`) uses a **different relevance-term shape**: its relevance term is a KL divergence $D_{\text{KL}}(\pi^* \| Q_{\Sigma_t})$ to a *target policy*, not a mutual information to an observable. This is not an inconsistency or an "abandonment of IB" — the strategy-cost compression sits in the parallel **information-theoretic-MDP lineage** (Tishby & Polani 2011 "The Information Theory of Decision and Action," *Perception-Action Cycle* Springer; Rubin, Shamir & Tishby 2012 "Trading value and information in MDPs," *Decision Making with Imperfect Decision Makers* Springer; Levine 2018 arXiv:1805.00909 for the control-as-inference reading). Both lineages descend from Shannon rate-distortion theory and admit Lagrangian relaxation; the choice of fidelity term depends on whether the compressed variable should preserve information about an observable (IB form: MI-to-relevance-variable) or match a target policy (IT-MDP form: KL-to-reference-policy). AAT's compression-operations framework (`#disc-compression-operations`) uses the IB form for M_t, G_t^{shared}, and Λ compressions; the strategy-cost compression uses the IT-MDP form, with the π^*-first direction forced by a regret-bound argument specific to decision-theoretic scope (see `#deriv-strategy-cost-regret-bound` §§5, 6.4). The two lineages are siblings via their shared rate-distortion ancestor; neither reduces to the other without a change of relevance variable.

Connection to variational free energy. The IB objective stated above is the rate-distortion specialization of the variational free energy decomposition $-F$ = accuracy − complexity used in active inference (Friston 2010, "The free-energy principle: a unified brain theory?", *Nature Reviews Neuroscience* 11; Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Parr & Pezzulo 2022, *Active Inference*, MIT Press, ch. 2): the compression cost $I(M_t; \mathcal{C}_t)$ corresponds to the complexity term (KL between posterior and prior over latent states); the negative predictive power $-I(M_t; o_{t+1:\infty})$ corresponds to the accuracy term (negative expected log-likelihood). The two formulations are related under the Markov-chain factorization $Y - X - T$; the variational bound that makes this relation operational — connecting the IB Lagrangian to the variational machinery shared with free-energy methods — is established by Alemi, Fischer, Dillon & Murphy 2017 ("Deep Variational Information Bottleneck," ICLR 2017, arXiv:1612.00410), with Tishby & Zaslavsky 2015 ("Deep learning and the information bottleneck principle," *IEEE ITW*) giving the deep-learning instantiation of IB itself. AAT adopts the IB form as the rate-distortion characterization of optimal compression; the variational free-energy form is the AI-side cousin and motivates the variational treatment of strategy compression in 9.9 and the broader four-instance framing in `#disc-compression-operations`. AAT borrows the form without committing to AI's preferences-as-priors stance or to expected free energy as master objective.

2.3 *Definition:* Model Sufficiency

Having committed to a compressed model M_t (2.1) and to information-bottleneck pressure (2.2) shaping it, the framework needs a measurable handle on *how much* of the chronica's predictive content the compression actually retains. **Model sufficiency** $S(M_t)$ is that handle: an information-theoretic ratio built from conditional mutual information that takes the value 1 when the model is a sufficient statistic for prediction (knowing the full history beyond the model adds nothing), 0 when the model retains no predictive information at all, and intermediate values for partial sufficiency.

Two scope considerations are surfaced at the definition rather than buried downstream. Sufficiency is *well-defined only when there is something to be sufficient for* — the chronica must carry non-zero predictive information about the future beyond what the action sequence alone supplies. In saturated-noise environments or prediction-vacuous regimes (fully iid observations independent of history), the denominator vanishes and $S(M_t)$ is undefined; the downstream constructs that build on it (2.4, 4.8) inherit that scope. And sufficiency is *predictive, not causal* — it measures Level 1 (associational) information; it does not by itself guarantee that M_t supports Level 2 (interventional) queries (that requires the additional backdoor condition formalized when value objects arrive in 6.6). The Discussion section draws three further relativities — to the prediction task / horizon, to the generating policy, and to the singular causal trajectory the agent is instantiated on — that the framework will lean on later.

Formal Expression.

Definition (model-sufficiency)

$$S(M_t) = 1 - \frac{I(\mathcal{C}_t;\, o_{t+1:\infty} \mid M_t,\, a_{t:\infty})}{I(\mathcal{C}_t;\, o_{t+1:\infty} \mid a_{t:\infty})} \tag{2.3}$$

where: - The numerator $I(\mathcal{C}_t;\, o_{t+1:\infty} \mid M_t,\, a_{t:\infty})$ is the predictive information that the full history $\mathcal{C}_t$ carries about the future *beyond* what M_t already captures — the information lost by compression - The denominator $I(\mathcal{C}_t;\, o_{t+1:\infty} \mid a_{t:\infty})$ is the total predictive information in the full history

Well-definedness. $S(M_t)$ is defined when $I(\mathcal{C}_t;\, o_{t+1:\infty} \mid a_{t:\infty}) > 0$ — when the chronica carries some predictive information about future observations beyond what the action sequence alone supplies. When the denominator vanishes (saturated-noise environments, prediction-vacuous regimes, fully iid observations independent of history), $S(M_t)$ is undefined: predictive sufficiency is a property *of a prediction task*, and there is no prediction task to be sufficient for. Downstream constructs that build on S — 2.4 and 4.8 — inherit the same scope and are correspondingly inapplicable in predictively-vacuous regimes.

Boundary values (assuming the well-definedness clause holds): - $S(M_t) = 1$: M_t is a sufficient statistic — it captures all predictive information in $\mathcal{C}_t$. Knowing the full history beyond M_t adds nothing. - $S(M_t) = 0$: M_t retains no predictive information. The model is useless for prediction. - $0 < S(M_t) < 1$: partial sufficiency — some predictive information is retained, some lost.

Epistemic Status.

This is *definitional* — it names and formalizes a quantity. The definition is well-grounded in information theory (conditional mutual information ratios are standard). The scope clause ($I(\mathcal{C}_t; o_{t+1:\infty} \mid a_{t:\infty}) > 0$) is the natural domain for a predictive-sufficiency measure: a ratio whose denominator is zero is not a meaningful notion of "fraction retained," and "any model is sufficient when there is nothing to predict" would smuggle structure into a regime that has none. No substantive claim is made here about what value $S(M_t)$ takes or what happens when it is low; those claims belong to 2.4 and 4.8.

Discussion.

Sufficiency is relative to the prediction task. $S(M_t)$ measures sufficiency for predicting future observations given future actions. A model that is sufficient for one prediction horizon may be insufficient for another. The infinite-horizon formulation ($o_{t+1:\infty}$) is the most demanding; practical sufficiency over finite horizons may be easier to achieve.

Sufficiency vs. accuracy. A model can be sufficient ($S = 1$) while being wrong in absolute terms — if the full history is also wrong (e.g., systematically biased observations). Sufficiency measures information retention, not truth. The mismatch signal (3.4) measures accuracy; sufficiency measures completeness of compression.

Sufficiency is predictive, not causal. $S(M_t)$ measures retained *predictive* information — a Level 1 (associational) property. It does not by itself guarantee that M_t supports Level 2 (interventional) queries such as $P(o \mid do(a), M_t)$. The causal validity of value computations conditioned on M_t requires an additional condition: that M_t satisfies the backdoor criterion with respect to the agent's actions (see 6.6). For agents whose actions are deterministic functions of M_t (standard in AAT: $a_t = \pi(M_t, G_t)$), $S(M_t) = 1$ is nearly sufficient for causal validity — the remaining requirement is that no unmodeled external factor influences both action selection and outcomes. But predictive sufficiency alone does not collapse the distinction between Level 1 and Level 2 that 7.2 establishes.

Policy-relativity. The conditioning on $a_{t:\infty}$ makes $S(M_t)$ implicitly policy-relative: different policies generate different future action sequences, which changes which future observations are relevant and therefore what "predictive information" means. A model that is sufficient under a conservative policy may be insufficient under an aggressive one (the aggressive policy visits states the model cannot predict). This policy-relativity is inherent in any predictive sufficiency measure — it is not an artifact of the formulation. When comparing sufficiency values, the generating policy must be held constant or specified. 6.6's continuation-policy convention (π_{cont}) provides the required specification for value computations; the same convention should be understood as implicit here.

Trajectory-relativity. $S(M_t)$ is measured against *this agent's* interaction history $\mathcal{C}_t$ (1.4), not against a model-state equivalence class. Two copies of the same M_t exposed to different event streams will each have their own S measured against their own divergent $\mathcal{C}_t$; neither sufficiency value is the other's. This is the scope commitment in 4.10: AAT applies to agents instantiated on singular causal trajectories, and sufficiency is trajectory-indexed accordingly. Claims of the form "the model has sufficiency S" make sense only relative to a specific trajectory; aggregated claims across copies of a given M_t require additional machinery.

2.4 *Definition:* Model Class Fitness

Where model sufficiency (2.3) measures how well a *specific* model retains the chronica's predictive information, **model class fitness** $\mathcal{F}(\mathcal{M})$ measures the *ceiling* — the supremum of sufficiency over every model in the agent's representational class $\mathcal{M}$. The pair formalizes a distinction the framework will soon make load-bearing: a low instance sufficiency might mean the agent needs more learning (parameter update); a low class fitness means no amount of better parameter estimation, more data, or longer training within the current class will close the gap — the agent needs a different *kind* of model entirely. The parallel to bias vs. variance in statistical learning is exact: class fitness is about bias (what the class can in principle represent); instance sufficiency reflects both bias and estimation quality.

The structural-inadequacy condition $\mathcal{F}(\mathcal{M}) < 1 - \varepsilon$ is the trigger this definition sets up for use later. When it holds, the gap from full predictive sufficiency cannot be closed parametrically. That is the precise hypothesis under which 4.8 arrives in Chapter 4 — when class fitness is too low, the agent must change *what kind of model* it is. An important operational point: the agent cannot directly compute its class fitness (that would require searching over all models in $\mathcal{M}$). What it can observe is the *signature* — persistent *systematic* mismatch despite adequate learning (high gain, sufficient data, converged parameters): residuals that stay structured after parametric convergence. Structure is the discriminator, not the mismatch level — a class ceiling leaves signal in the residuals, while an adequate class in a noisy world leaves white residuals sitting on the mismatch floors (3.5), which no class change removes. When structured residuals persist and don't whiten with more work, the ceiling is structural.

Formal Expression.

Definition (model-class-fitness)

$$\mathcal{F}(\mathcal{M}) = \sup_{M \in \mathcal{M}} S(M) \tag{2.4}$$

where $\mathcal{M}$ is the model class — the set of all models the agent can represent given its current architecture, parameterization, or capacity.

Structural inadequacy condition:

$$\mathcal{F}(\mathcal{M}) < 1 - \varepsilon \tag{2.5}$$

When this holds, no model $M \in \mathcal{M}$ achieves sufficiency above $1 - \varepsilon$. The gap is structural: it cannot be closed by better parameter estimation, more data, or longer training within the current class. This is the trigger for structural change (4.8).

Epistemic Status.

This is *definitional* — it names the supremum of sufficiency over a model class. The definition itself is straightforward. The substantive claim about what happens when $\mathcal{F}(\mathcal{M})$ is low — that parametric updates cannot close the mismatch floor and structural adaptation becomes necessary — is developed in 4.8.

Discussion.

Model class vs. model instance. $S(M_t)$ measures a specific model's sufficiency at time t. $\mathcal{F}(\mathcal{M})$ measures the ceiling of the entire class. A low $S(M_t)$ might mean the agent needs more learning (parameter update). A low $\mathcal{F}(\mathcal{M})$ means the agent needs a different kind of model (structural change). The distinction parallels bias vs. variance: class fitness is about bias; instance sufficiency reflects both bias and estimation quality.

Detecting low class fitness. The agent cannot directly compute $\mathcal{F}(\mathcal{M})$ — it would need to search over all models in the class. Instead, persistent *systematic* mismatch despite adequate learning (high gain, sufficient data, converged parameters) is the observable signature — structured residuals, not merely large ones. The absolute mismatch level cannot carry the diagnosis: sufficiency and fitness are ratios of retained-to-available predictive information, so an agent at $S = \mathcal{F} = 1$ in a noisy world still faces an arbitrarily high absolute mismatch floor (3.5). What distinguishes a class ceiling from a noisy channel is residual *structure* — signal the class fails to capture shows up as autocorrelation and pattern, while the floors leave white residuals. The sharp diagnostic lives with the mismatch floor in 4.8.

Chapter 3

The Cycle in Motion: Mismatch, Gain, & Tempo

The model exists; events arrive; the agent corrects. This chapter develops the dynamics of the adaptive cycle as it actually runs — how state updates from events, what the mismatch signal is, how the optimal weight on each observation depends on uncertainty, and what total corrective capacity emerges at the cycle level.

Chapter 2 left us with a static picture. M_t is the agent's compressed model of reality; sufficiency measures how much predictive content it retains; class fitness measures the ceiling within a representational class. None of that tells us how M_t *moves* under events. Chapter 3 sets the cycle in motion.

The first two results fall out of a commitment we already made. We chose to call M_t the agent's complete state — anything the agent remembers is in M_t, and the only way new information enters is through events. Under that completeness, two things follow immediately. The update from M_{τ^-} to M_{τ^+} depends on the prior model and the incoming event, and *nothing else* — the agent doesn't need to (and structurally cannot) re-process its entire history. And the agent's action depends on M_t alone — the action is a function of what the agent currently believes, not directly of the history. Both feel obvious in hindsight; both are *derived*, not chosen.

The cycle's engine is the mismatch signal. The agent predicts an observation, the actual observation arrives, the gap between them is $\delta_t = o_t - \hat{o}_t$. Mismatch decomposes cleanly into two parts: the agent's model was wrong (reducible by better modeling) and the observation channel is noisy (irreducible — no amount of smarter modeling will eliminate sensor noise). There is a floor on prediction error set by the channel itself, and chasing below it amounts to fitting noise.

Given a mismatch, how much should the agent trust the new observation versus its prior model? The answer has a startlingly simple form:

$$\eta^* = \frac{U_M}{U_M + U_o} \tag{3.1}$$

— the optimal update gain weighted by the ratio of model uncertainty to total uncertainty. When the agent is highly uncertain about its model (U_M large), gain approaches 1: trust the observation, you have little else to go on. When the agent is confident and the channel is noisy (U_o large), gain approaches 0: trust your model, the observation isn't telling you much. For linear-Gaussian agents this is exactly the Kalman gain; for the rest of AAT's scope it is a robust qualitative result — any rational adaptive process must approximate this functional form, whether or not it explicitly computes the variance ratio.

Multiply gain by the rate at which observations arrive (ν, the event rate), sum across the agent's channels, and you have adaptive tempo:

$$\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*} \tag{3.2}$$

Tempo is the rate at which the agent turns observations into useful corrections — speed times quality. It is the chapter's central object and the load-bearing capacity variable for the rest of the framework. Every persistence threshold in Chapter 4 will have tempo on the left-hand side, and every adversarial-coupling result in Part III will depend on tempo ratios.

The chapter closes with the linear-correction ODE as a heuristic preview. Under linear correction and bounded environmental disturbance, the steady-state mismatch is

$$\|\delta\|_{ss} = \rho/\mathcal{T} \tag{3.3}$$

a ratio: how fast the world is changing, divided by how fast the agent corrects. Mismatch stays bounded when the agent corrects faster than reality drifts. This is the persistence condition in its simplest form; Chapter 4 generalizes it to nonlinear correction under the sector condition, where the result has the same shape.

Causal information yield (CIY, in 3.7) enters the chapter here because action-coupling is where causality first bites. CIY uses Pearl's $do(\cdot)$ notation — an external import (Pearl 2009; Bareinboim, Correa, Ibeling & Icard 2022; AAT's recapitulation lives at 7.1 in Part II Ch.2, where the framework deploys the hierarchy as machinery rather than referencing it as vocabulary). Its operational role in Chapter 3 is to score actions for their informational value to the cycle; Part II will lift CIY into a strategy-revision context.

The flow of the chapter: event-driven substrate (3.1) → recursion and action falling out of completeness (3.2, 3.3) → mismatch and its decomposition (3.4, 3.5) → gain and CIY (3.6, 3.7) → tempo as the synthesis (3.8) → linear ODE preview (3.9). The chapter ends with the central capacity variable and the preview of the persistence inequality that Chapter 4 generalizes.

3.1 *Formulation:* Event-Driven Dynamics

The agent-environment interaction is formulated at a finer grain than discrete clock ticks: as a stream of atomic **events** — observation arrivals and action completions — at potentially variable and heterogeneous rates across multiple channels. Discrete-time notation, where a single observation and a single action alternate at a fixed rate, becomes the special case of uniform-interval events on a single channel. The motivation is realism: real agents face multiple observation channels at different rates (a robot's camera at 30Hz, LIDAR at 10Hz, GPS at 1Hz; a human's vision, audition, proprioception; a developer's compiler output, test results, and production telemetry), multiple action channels with different latencies, and asynchronous arrivals. The event-driven formulation handles all of these natively.

Two new measurable quantities are introduced alongside the event stream. **Event information content** $\mathcal{I}(e_\tau)$ measures, for any individual event, the mutual information it carries about the environment state conditioned on the agent's current model — an event the model already predicts carries little information; an event that surprises the model carries much. This anticipates the mismatch signal (3.4). **Channel-specific observation uncertainty** $U_o^{(k)}$ measures the noise characteristic of each channel separately — a noisy channel provides lower-quality information per event, and any sensible update rule must weight channels accordingly. From these the chapter's *effective adaptation rate* falls out immediately as the sum across channels of channel-rate times optimal gain — the adaptive tempo construct (3.8) that the framework will lift to first-class status, presented here as the natural multi-channel generalization the discrete-time formulation cannot easily express.

Formal Expression.

Formulation (event-driven-dynamics)

Event (e): An atomic unit of agent-environment interaction, typed as: - **Observation event**: $e = (\text{obs}, k, o^{(k)})$ — a datum arriving on observation channel k - **Action completion**: $e = (\text{act}, j, r^{(j)})$ — the result of action j completing

Event stream ($\mathcal{E}$): The temporally ordered sequence of all events:

$$\mathcal{E} = \{(e_1, \tau_1), (e_2, \tau_2), \dots\} \quad \text{where } \tau_1 \le \tau_2 \le \cdots \tag{3.4}$$

Channel rate ($\nu^{(k)}$): The characteristic event rate of channel k, which may vary over time.

Event information content: The mutual information between the event and the environment state, conditioned on the current model:

Definition (event-information-content)

$$\mathcal{I}(e_\tau) = I(e_\tau; \Omega_\tau \mid M_{\tau^-}) \tag{3.5}$$

An event that the model already predicts carries little information ($\mathcal{I} \approx 0$). An event that surprises the model carries much ($\mathcal{I} \gg 0$). This connects directly to the mismatch signal (3.4).

Channel-specific observation uncertainty:

Definition (channel-uncertainty)

$$U_o^{(k)} = \text{observation uncertainty of channel } k \tag{3.6}$$

Different channels have different noise characteristics. A noisy channel (high $U_o^{(k)}$) provides lower-quality information per event. The update gain (3.6) should weight channels accordingly.

Epistemic Status.

This is a *formulation choice*, not a postulate. The event-driven representation extends 1.7's recursive update to heterogeneous, asynchronous multi-channel interactions. The discrete-time form ($M_t = f(M_{t-1}, o_t, a_{t-1})$) from 3.2 is a special case sufficient for many formal analyses — the event-driven formulation is needed only when multi-rate or asynchronous channels matter.

Discussion.

Why events rather than clock ticks. The discrete-time notation $M_t = f(M_{t-1}, o_t, a_{t-1})$ presupposes a single clock synchronizing observations and actions. Real agents face:

- **Multiple observation channels** at different rates (a robot's camera at 30Hz, LIDAR at 10Hz, GPS at 1Hz; a human's vision, audition, proprioception; a developer's compiler output, test results, and production telemetry)
- **Multiple action channels** with different latencies (a robot's wheel motors vs. arm actuators; an organization's operational decisions vs. strategic pivots)
- **Asynchronous arrival** — observations not synchronized with each other or with action completions

The event-driven formulation handles all of these naturally. The discrete-time form is the special case where a single observation and a single action alternate at a fixed rate.

The effective adaptation rate. The agent's overall capacity to track environmental changes is the sum of information gained across all channels per unit time:

$$\nu_{\text{eff}} = \sum_{k} \nu^{(k)} \cdot \eta^{(k)*} \tag{3.7}$$

This quantity — identical to adaptive tempo $\mathcal{T}$ (3.8) — is the central measure of an agent's adaptive fitness.

Software-specific channels. In the software development domain, the event-driven formulation maps naturally to the developer's multi-rate observation channels:

Table 3.1

Channel k	*Rate* $\nu^{(k)}$	*Noise* $U_O^{(k)}$
Compiler/linter output	Per-save (high)	Very low
Unit test results	Per-run (medium)	Low
CI pipeline	Per-push (medium)	Low
Runtime telemetry	Continuous (variable)	Medium
Bug reports	Sporadic (low)	High
Code review feedback	Per-PR (low)	Medium-high

The three-part tempo decomposition for software — $\mathcal{T}_{\text{obs}}$ (compiler, tests) + $\mathcal{T}_{\text{explore}}$ (code reading) + $\mathcal{T}_{\text{probe}}$ (test runs, staging) — is a direct application of multi-channel tempo. The formal development of this decomposition is a TST-side question (open GAP in `02-tst-core/OUTLINE.md`).

3.2 *Derived:* Recursive Update

The first major *derived* result of the volume: the model update function must be recursive. Each new model state M_{τ^+} depends only on the previous model state M_{τ^-} and the incoming event e_τ — not on the full chronica. The result follows from three constraints already in place: the temporal-ordering postulate (1.7 — the model at time t can only depend on prior events), the partial-observability scope (1.3 — the agent has no direct access to the environment beyond events), and the completeness commitment in the model formulation (2.1 — M_t summarizes everything the agent retains). Together these force the update into the recursive form; the appendix derivation (`#deriv-recursive-update`) works through seven counterexample attacks to confirm there is no escape from this form once the three constraints are accepted.

The epistemic character of this result is honest. Temporal ordering is a physical postulate and partial observability is a scope condition, but the completeness commitment is *definitional* — it cannot be "violated" because any apparent violation is absorbed by expanding M_t. The Markov structure of the update is therefore not discovered in the environment but chosen through the definition of M_t as complete. This is not a weakness; it is the precise character of the claim. For finite agents recursion is forced by *computational realism* as well: re-processing the full history at each event is infeasible.

The formulation also surfaces **between-event dynamics** — autonomous evolution of the model state during the gaps between observation arrivals — as load-bearing rather than filler. These include prediction generation (anticipating the next observation), uncertainty growth (confidence decay over time without new data), and internal reorganization (consolidation, abstraction). When between-event dynamics are driven by replayed or internally-generated pseudo-events and the objective is information-bottleneck-gap reduction rather than one-step mismatch minimization, the framework names this the *consolidation regime* (9.11) — a separate operating mode with its own scope condition ($\nu_{\text{consol}} \ll \nu_{\text{online}}$) and its own necessity condition.

Formal Expression.

Derived (recursive-update, from temporal postulate and M_t completeness)

Event-driven update:

$$M_{\tau^+} = f_M(M_{\tau^-}, e_\tau) \tag{3.8}$$

where: - M_{τ^-} is the model state immediately before event e_τ - M_{τ^+} is the model state immediately after - f_M is the update function — it takes the current model and the new event, not the full history $\mathcal{C}_t$

Between-event evolution:

$$\frac{dM}{d\tau} = g_M(M_\tau) \tag{3.9}$$

Between events, the model evolves autonomously — internal reorganization, prediction generation, decay of transient states. The between-event dynamics depend only on the current model state, not on external input (which, by definition, arrives only at events).

Epistemic Status.

Exact, with a partly definitional character. The result follows from three constraints: temporal ordering (C1 — physical law), partial observability (C2 — scope definition), and state completeness (C3 — analytical commitment that M_t summarizes everything the agent retains). C1 and C2 do genuine eliminative work; C3 is definitional — it cannot be "violated" because any violation is absorbed by expanding M_t. The Markov structure is therefore not discovered in the environment but chosen through the definition of M_t as complete. This is not a weakness — it is the nature of the claim: recursive update is the only form consistent with C1 + C2 + the definition of M_t as complete (see `#deriv-recursive-update` for the full argument and seven counterexample attacks). For finite agents, recursion is also *computational necessity*: re-processing the full history at each event is infeasible.

Discussion.

Recursion as a consequence of completeness. The recursive form is not an assumption bolted on — it follows from the definition of M_t as complete. If M_t were incomplete (if some relevant information lived outside M_t in the raw history), then $f_M(M_{\tau^-}, e_\tau)$ would be insufficient and the agent would need to consult $\mathcal{C}_t$ directly. The sufficiency of the recursive form is precisely what 2.3 measures: when $S(M_t) = 1$, the recursive update loses nothing.

Between-event dynamics matter. The autonomous evolution $g_M(M_\tau)$ is not merely filler between observations. It includes prediction generation (what the agent expects to see next), uncertainty growth (model confidence decaying over time without new data), and internal reorganization (consolidation, abstraction). In event-driven systems (3.1), the between-event interval is variable, making g_M load-bearing for agents that must act or predict between observations. When the between-event dynamics are driven by replayed or internally-generated pseudo-events and the update objective is IB-gap reduction rather than one-step mismatch minimization, g_M is operating in the *consolidation regime* per 9.11 — a named regime with its own scope condition ($\nu_{\text{consol}} \ll \nu_{\text{online}}$) and its own necessity condition (sub-state factorization + bounded per-event budget). Consolidation is where the stability-plasticity feasibility window complements 9.10's plasticity lower bound.

Connection to the update gain. The event-driven update $f_M(M_{\tau^-}, e_\tau)$ is where the gain principle (3.6) operates: η^* determines how strongly e_τ shifts M_t away from its prior value. The recursive form makes the gain's role explicit — it modulates the single-step correction.

3.3 *Derived:* Action Selection

The second derived result falling out of the completeness commitment in 2.1. Under Part I scope, where M_t is the entire internal state, the agent's action is forced to be a function of M_t — either deterministic, $a_t = \pi(M_t)$, or stochastic, $a_t \sim \pi(\cdot \mid M_t)$. The argument mirrors 3.2: since the agent's internal state is by definition complete, action — which depends on internal state — is forced to depend only on that state. Any dependence on the chronica is captured automatically *through* the model. The Part II lift to $X_t = (M_t, G_t)$ (see 6.2) gives $a_t = \pi(M_t, G_t)$ by the same completeness argument applied to the larger state; the Part I form is the special case $G_t = \varnothing$.

The segment also introduces a distinction that recurs throughout the framework: **implicit** versus **explicit** action selection — what the segment calls *action fluency*. An agent has high fluency when effective action flows from the model without deliberative computation — the model has internalized the action-selection structure for the current situation. Reflexes, trained RL policies in exploitation mode, expert intuition (System 1), well-tuned PID controllers, and standard operating procedures are all instances. An agent has low fluency when deliberation significantly improves action quality — when the situation is novel, the action space large, or the stakes asymmetric, and the agent must use its model to *simulate* outcomes of candidate actions before committing. Explicit deliberation requires at minimum Level 2 epistemic access (see 7.1) — the agent uses its model to evaluate "what will I observe if I $do(a)$?" across candidates.

Action fluency is formally characterized via 4.1: high fluency means $\Delta\eta^*(\Delta\tau) \approx 0$ — additional deliberation yields negligible improvement in update gain. Fluency is therefore distinct from model sufficiency (see 2.3): a chess engine with a perfect rule-model has high sufficiency but low fluency (search is still expensive), while a reflex can have only moderate sufficiency but high fluency in a narrow domain. What reflexes, muscle memory, intuition, and expertise share is that the action-generating capacity itself has been absorbed into the model's structure — the model doesn't just predict well, it *acts* well, cheaply.

A *structural pressure* toward implicit action follows: when two action-selection modes produce equivalent expected outcomes, the faster mode is preferable because the persistence condition penalizes slower tempo. Agents under selective pressure (evolution, competition, training) therefore tend to internalize frequently-needed action patterns, converting explicit deliberation into implicit fluency. The pressure is stronger when the environment changes fast (high ρ), the pattern recurs frequently, or adaptive tempo $\mathcal{T}$ sits near the persistence threshold (see 4.5) with no slack for deliberation overhead. Deliberation nonetheless remains essential in genuinely novel situations, when action spaces are large relative to model capacity, or when stakes are asymmetric.

Formal Expression.

Derived (action-selection, from agent-model completeness)

Action is a function of the agent's complete internal state. Under Part I scope (1.5) — where M_t is the entire internal state — this gives:

$$a_t = \pi(M_t) \quad \text{(deterministic)} \tag{3.10}$$

$$a_t \sim \pi(\cdot \mid M_t) \quad \text{(stochastic)} \tag{3.11}$$

where π is the agent's **policy** — the mapping from internal state to action.

This is not imposed on the system but follows from 2.1: M_t is defined as the agent's compressed, complete internal record, and action depends on what the agent retains — i.e., on M_t. Any deterministic or stochastic dependence of action on history *through* the model is captured by $\pi(M_t)$.

Part II lift. When the internal state lifts to $X_t = (M_t, G_t)$ for purposeful agents (6.2), the same structural argument gives $a_t = \pi(M_t, G_t)$ — action conditions on the complete internal state, which now includes the purposeful substate.

The policy form here is the Part I instantiation $G_t = \varnothing$; the actuated-agent form is recovered by the same completeness argument applied to X_t.

Epistemic Status.

Exact within Part I scope. The derivation follows from 2.1's completeness commitment: if M_t is the agent's complete internal state (by definition), then action — which depends on internal state — is a function of M_t. The Part II generalization $a_t = \pi(M_t, G_t)$ is exact within Part II scope by the same argument applied to the lifted state X_t (6.2); see 2.3 for the form already in use downstream. The implicit/explicit distinction and action fluency concept are *discussion-grade* — qualitative properties that follow from the formalism but are not formally derived as propositions.

Discussion.

Implicit vs. explicit action selection. A critical distinction emerges from the agent's *action fluency* — the degree to which effective action flows from the model without deliberative computation:

Implicit (model-embedded): When $\pi(M_t)$ can be evaluated cheaply — the model has internalized effective action-selection for the current situation. This is Boyd's implicit guidance and control (Orient→Act, bypassing Decide), a trained RL policy in exploitation mode, a well-tuned PID controller, expert intuition (System 1), a martial artist's trained reflexes, an organization's standard operating procedures.

Explicit (deliberative): When the situation is novel, the action space is large, or the stakes demand verification — the agent engages in internal simulation, using the model to predict outcomes of candidate actions before selecting. This is Boyd's explicit Decide step, MCTS/planning in RL, Model Predictive Control, human deliberate reasoning (System 2), organizational strategic planning. Deliberation requires at minimum Level 2 epistemic access (7.1) — the agent uses its model to simulate "what will I observe if I $do(a)$?" across candidates.

Formal characterization of action fluency. An agent has *high fluency* for a situation when additional deliberation yields negligible improvement — formally, when $\Delta\eta^*(\Delta\tau) \approx 0$ for all $\Delta\tau > 0$ (see 4.1). Conversely, *low fluency* means deliberation significantly improves action quality. Fluency is the degree to which the agent's immediate (zero-deliberation) action approaches the quality achievable with unbounded deliberation.

Action fluency is distinct from model sufficiency. An agent can have high $S(M_t)$ (2.3) but low fluency — a chess engine with a perfect model of the rules still requires expensive search. Conversely, an agent can have moderate sufficiency but high fluency in a narrow domain — a reflex that responds effectively to specific situations it evolved for. What reflexes, muscle memory, intuition, and expertise share is that the *action-generating capacity itself* has been absorbed into the model's structure: the model doesn't just predict well, it *acts* well, cheaply.

Structural pressure toward implicit action. When two action-selection modes produce equivalent expected outcomes, the faster mode is preferable (the persistence condition penalizes slower tempo — and in TST, the temporal optimality postulate makes this normative). This creates a pressure: agents under selective pressure (evolution, competition, training) tend to internalize frequently-needed action patterns, converting explicit deliberation into implicit fluency. The pressure is stronger when ρ is high (fast-changing environments penalize deliberation — see 4.1), the pattern recurs frequently, or $\mathcal{T}$ is near the persistence threshold (4.5) with no slack for deliberation overhead.

However, deliberation remains essential when the situation is genuinely novel, the action space is large relative to model capacity (chess, strategic planning), the stakes are asymmetric (cost of error vastly exceeds cost of delay), or ρ is low (stable environment allows deliberation without mismatch accumulation).

Connection to Part II. For actuated agents (6.1), the lifted form $\pi(M_t, G_t)$ above unpacks: action conditions on the purposeful substate $G_t = (O_t, \Sigma_t)$ as well as on M_t, coupling all substates through action (6.3). The action-deliberation-exploration tradeoff (Part II gap) extends the implicit/explicit distinction to three modes: exploit (pursue O_t via Σ_t), explore (improve M_t), deliberate (revise Σ_t).

Domain instantiations:

Table 3.2

Domain	*Implicit action*	*Explicit deliberation*
Kalman + LQR	LQR control law from $\hat{x}_t$	— (separation principle)
RL	Greedy policy arg max $Q(s, a)$	MCTS, planning, rollouts
PID	$u = K_p e + K_i \int e + K_d \dot{e}$	— (no deliberation)
Boyd's OODA	IG&C (Orient→Act)	Explicit Decide step
Organism	Reflexes, habits	Deliberate planning
Organization	Standard procedures	Strategic planning
Software developer	Known patterns, familiar code	Reading docs, analyzing alternatives

3.4 *Definition:* Mismatch Signal

Names the signal that drives every adaptive update — the formal expression of *aporia* (productive perplexity). The **mismatch signal** is $\delta_t = o_t - \hat{o}_t$, the difference between the actual observation and the model's prediction conditioned on the prior state and prior action. A more general version for probabilistic models is the **score-function mismatch** $\tilde{\delta}_t = \nabla_M \log P(o_t \mid M_{t-1}, a_{t-1})$, which points in the direction the model should move to increase the likelihood of what actually occurred. The prediction-error form lives in observation space $\mathcal{O}$; the score-function form lives in the tangent space $T_M\mathcal{M}$. Under Gaussian models the two coincide up to scaling.

This is *definitional* rather than substantive: given any model that predicts (see 2.1) and any observation that arrives (see 1.3), their difference exists. The mismatch signal is not an additional assumption but a consequence of having a predictive model in an uncertain world.

A genuinely important conceptual point is surfaced in Discussion: **zero mismatch does not necessarily indicate model adequacy**. A near-zero δ_t can mean (a) the model genuinely reflects reality — desirable; (b) the agent is only observing aspects its model already explains while remaining ignorant of aspects where the model is wrong — confirmation bias; or (c) the observation channel is too noisy to detect model errors — an architectural limitation. Only (a) is desirable. An agent without aporia has stopped adapting — but silence can mean peace, or it can mean deafness. This ambiguity is what motivates active testing later in the framework: deliberately choosing actions that generate informative mismatch, the basis of 3.7.

A scaling note is preserved for the dynamics that come later: when δ_t is in physical units, its magnitude entering the mismatch dynamics should be understood as a Mahalanobis distance $\|\delta_t\|_\Sigma$ against the observation noise covariance — mapping physical prediction error to dimensionless surprise-equivalent units.

Formal Expression.

Definition (mismatch-signal)

Given model M_{t-1} and prior action a_{t-1}, the model generates a prediction:

$$\hat{o}_t = \mathbb{E}[o_t \mid M_{t-1}, a_{t-1}] \tag{3.12}$$

The **mismatch signal** (prediction error):

$$\delta_t = o_t - \hat{o}_t \tag{3.13}$$

This is the primary definition, used in the mismatch dynamics (4.5, 4.4) and in the decomposition (3.5).

For models with probabilistic predictions, the mismatch generalizes to the **score-function mismatch**:

Definition (score-mismatch)

$$\tilde{\delta}_t = \nabla_M \log P(o_t \mid M_{t-1}, a_{t-1}) \tag{3.14}$$

which points in the direction the model should move to increase the likelihood of the actual observation. $\tilde{\delta}_t$ lives in the tangent space $T_M\mathcal{M}$, while δ_t lives in observation space $\mathcal{O}$. Under Gaussian models, they coincide up to scaling.

Epistemic Status.

This is *definitional*. Given any model that predicts (2.1) and any observation that arrives (1.3), their difference exists. The mismatch signal is not an additional assumption but a consequence of having a predictive model in an uncertain world. The score-function form is the natural generalization when $\mathcal{O}$ is not a vector space or when the model's predictive distribution is the natural object.

Discussion.

Units and normalization. When δ_t is in physical units (meters, dollars), the $\|\delta\|$ that enters the mismatch dynamics should be understood as the Mahalanobis distance: $\|\delta_t\|_\Sigma = \sqrt{\delta_t^T \Sigma^{-1} \delta_t}$ where Σ is the observation noise covariance. This maps physical prediction error to dimensionless surprise-equivalent units.

The zero-aporia ambiguity. $\delta_t \approx 0$ does NOT necessarily indicate model adequacy. It may mean: (a) the model genuinely reflects reality — *desirable*; (b) the agent is only observing aspects its model already explains, while remaining ignorant of aspects where the model is wrong — *confirmation bias*; or (c) the observation channel is too noisy to detect model errors — *architectural limitation*. Only (a) is desirable. An agent without aporia is an agent that has stopped adapting — but silence can mean peace or deafness. This ambiguity is why active testing — choosing actions to generate informative aporia — can be valuable (see 3.7 for the CIY framework).

The mismatch transform. The update rule (3.6) writes $M_t = M_{t-1} + \eta \cdot g(\delta_t)$, where the transform g maps from δ_t's space to the model's update space: $g : \mathcal{O} \to T_M\mathcal{M}$ for prediction errors; $g : T_M\mathcal{M} \to T_M\mathcal{M}$ for score-function mismatches.

3.5 *Result:* Mismatch Decomposition

The first named *result* of the volume. The expected squared mismatch $\mathbb{E}[\|\delta_t\|^2]$ decomposes cleanly into three additive parts: an **estimation term** (the gap between the model's predictive mean $\hat{o}_t$ and the Bayes predictor $\hat{o}_t^{\mathrm{B}} = \mathbb{E}[o_t \mid \mathcal{C}_{t-1}, a_{t-1}]$ — the best any chronica-measurable model could do), a **state-uncertainty floor** (the variance the true conditional mean $\bar{o}_t = \mathbb{E}[o_t \mid \Omega_t, a_{t-1}]$ retains given the interaction history — untouchable by better modeling, movable only by *acting* to make the history more informative), and a **channel-noise term** (the conditional variance of the observation channel itself, given environment state and action). The result is a Pythagorean refinement of the bias-variance decomposition applied to the prediction problem; all cross-terms vanish — one pair under the fresh-noise global assumption GA-1 (observation noise conditionally independent of the past chronica given current environment state and action), the standard assumption in state-space models, and the third because the Bayes predictor is by construction the conditional mean given the history. The model can improve the first term; acting can shrink the second; the third is a property of the channel.

The conceptual stakes of this seemingly mechanical decomposition are large. The result establishes that prediction error is *structurally persistent* in any realistic adaptive regime — there are floors below which mismatch cannot be driven by any amount of better modeling. The channel floor is set by the observation sensor itself. The state-uncertainty floor is set by how much of the environment's state the interaction history pins down: it binds *every* chronica-measurable predictor — including the Bayes-optimal one — whenever residual state uncertainty moves the one-step conditional mean. Deterministic, noiseless, perfectly-specified systems are limiting edge cases, not the typical adaptive regime. The total expected squared mismatch is therefore strictly positive whenever any of the three terms is — and in typical settings all three are.

The decomposition has direct operational consequences picked up downstream. An agent that tries to eliminate *all* mismatch — including the two floors — will overfit, adjusting its model to explain noise and degrading future predictions. The update-gain construct 3.6 implicitly separates signal from noise by weighting observations in proportion to their informativeness, but the floor facts are what make the gain question meaningful at all. The decomposition also clarifies the relationship to 2.3: when $S(M_t) < 1$, the model has lost predictive information relative to the full history; under an alignment assumption (the lost information affects the one-step conditional mean) this implies a positive estimation term in the decomposition.

Formal Expression.

Derived (result-mismatch-decomposition)

For any agent-environment pair within AAT's scope (1.5), with $\bar{o}_t = \mathbb{E}[o_t \mid \Omega_t, a_{t-1}]$ the true conditional mean and $\hat{o}_t^{\mathrm{B}} = \mathbb{E}[o_t \mid \mathcal{C}_{t-1}, a_{t-1}]$ the Bayes predictor (the best predictive mean any chronica-measurable model can produce):

$$\mathbb{E}[\|\delta_t\|^2] = \underbrace{\mathbb{E}[\|\hat{o}_t - \hat{o}_t^{\mathrm{B}}\|^2]}_{\text{(i) estimation error (reducible by modeling)}} + \underbrace{\mathbb{E}[\mathrm{Var}(\bar{o}_t \mid \mathcal{C}_{t-1}, a_{t-1})]}_{\text{(ii) state-uncertainty floor (movable only by acting)}} + \underbrace{\mathbb{E}[\mathrm{Var}(o_t \mid \Omega_t, a_{t-1})]}_{\text{(iii) channel noise (irreducible)}} > 0 \tag{3.15}$$

with positivity whenever any term is positive. Grouping (i)+(ii) recovers the coarser bias-variance form $\mathbb{E}[\|\hat{o}_t - \bar{o}_t\|^2] + \mathbb{E}[\mathrm{Var}(o_t \mid \Omega_t, a_{t-1})]$; the refinement shows that the grouped "model error" is bounded below by the state-uncertainty floor — no chronica-measurable model reduces it to zero.

Derivation.

1. By 1.5, $H(\Omega_t \mid \mathcal{C}_t) > 0$ — residual state uncertainty persists; since $\mathcal{C}_{t-1}$ is coarser than $\mathcal{C}_t$, the uncertainty available at prediction time is at least as large. This is what makes term (ii) generically positive (step 4).
2. By 2.1, the model generates predictions $\hat{o}_t = \mathbb{E}[o_t \mid M_{t-1}, a_{t-1}]$; M_{t-1} is built from the chronica, so $\hat{o}_t$ is one of the chronica-measurable predictors over which $\hat{o}_t^{\mathrm{B}}$ is optimal. By the tower property and GA-1, $\hat{o}_t^{\mathrm{B}} = \mathbb{E}[\bar{o}_t \mid \mathcal{C}_{t-1}, a_{t-1}]$.
3. Write $\delta_t = (o_t - \bar{o}_t) + (\bar{o}_t - \hat{o}_t^{\mathrm{B}}) + (\hat{o}_t^{\mathrm{B}} - \hat{o}_t)$ and expand. All three cross-terms vanish by orthogonality (uncorrelated — not independence). The two involving the noise difference: condition on $(\Omega_t, a_{t-1}, \mathcal{C}_{t-1})$; the other two differences are then fixed, and $\mathbb{E}[o_t - \bar{o}_t \mid \Omega_t, a_{t-1}, \mathcal{C}_{t-1}] = \mathbb{E}[o_t - \bar{o}_t \mid \Omega_t, a_{t-1}] = 0$ by definition of $\bar{o}_t$ and the fresh-noise assumption GA-1 (ε_t conditionally independent of $\mathcal{C}_{t-1}$ given (Ω_t, a_{t-1})). The remaining cross-term: condition on $(\mathcal{C}_{t-1}, a_{t-1})$; the third difference is fixed, and $\mathbb{E}[\bar{o}_t - \hat{o}_t^{\mathrm{B}} \mid \mathcal{C}_{t-1}, a_{t-1}] = 0$ by step 2.
4. Term (iii) is positive when observation noise is non-degenerate. Term (ii) is positive exactly when residual state uncertainty *moves the one-step conditional mean* — step 1's $H(\Omega_t \mid \mathcal{C}_t) > 0$ is necessary for this but not sufficient (uncertainty confined to mean-irrelevant state coordinates, or to higher moments only, leaves the floor at zero); this is the same alignment qualifier used for the sufficiency connection in the Discussion. Term (i) is positive when the model's predictive mean differs from the Bayes predictor. Any one suffices.

Epistemic Status.

Exact under the fresh-noise assumption (observation noise ε_t conditionally independent of history given current state and action). This is the standard assumption in state-space models — noise is a property of the observation channel at the moment of observation. The decomposition is a mathematical identity: a Pythagorean refinement of the bias-variance decomposition, with two orthogonality arguments in place of one. The positivity of $\mathbb{E}[\|\delta_t\|^2]$ follows from any of the three conditions; all typically hold simultaneously.

The positivity of term (ii) specifically carries the alignment qualifier stated in the derivation — residual state uncertainty must move the one-step conditional mean.

Discussion.

Three floors, two kinds of "irreducible." The estimation term is reducible by better modeling. The state-uncertainty floor is *irreducible by modeling* — it binds the Bayes-optimal predictor itself — but *movable by acting*: more informative action and observation choices shrink $\mathrm{Var}(\bar{o}_t \mid \mathcal{C}_{t-1}, a_{t-1})$, which is precisely the door through which active sensing and causal information yield (3.7) enter the mismatch budget. The channel floor is irreducible outright — a property of the sensor, movable only by changing the instrument. The framework's calibration anchor exhibits the middle floor exactly: in `#example-kalman`, the steady-state *optimal* filter — well-specified by construction — has innovation variance $HP^{-}H^{\top} + R > R$; the excess $HP^{-}H^{\top}$ is the state-uncertainty floor, present at the in-class optimum.

Reducible vs. irreducible (overfitting). An agent that tries to eliminate *all* mismatch — including the two floors — will overfit: adjusting its model to explain noise, degrading future predictions. The update gain (3.6) implicitly separates signal from noise by weighting observations in proportion to their informativeness.

Connection to model sufficiency. When $S(M_t) < 1$ (2.3), the model has lost predictive information relative to the full history. Under an alignment assumption (the lost information affects the one-step conditional mean), this implies positive estimation error (term i). Without that alignment assumption, insufficiency still implies positive regret under proper scoring rules but not necessarily positive one-step mean error.

Mismatch is structurally persistent. In realistic AAT regimes, mismatch signals persist — they can be reduced but not eliminated when observation noise is non-degenerate. Deterministic, noiseless, perfectly specified systems are limiting edge cases, not the typical adaptive regime.

3.6 *Empirical:* Update Gain

The gain principle — numerically simple, conceptually load-bearing. The optimal update gain η^*, the proportion of any incoming mismatch the agent should apply to correct its model, is the **ratio of model uncertainty to total uncertainty**: $\eta^* = U_M/(U_M + U_o)$. When model uncertainty dwarfs observation noise the gain approaches 1 (trust the observation); when the agent is confident and the channel noisy the gain approaches 0 (trust the model). This is the rate of *epistrophe* — turning toward reality. The update rule takes the form "new model = old model + gain × transformed mismatch."

The epistemic status of the result is carefully tiered. In the **Fisher-local regime** (`#deriv-fisher-local-update-gain`) — which covers linear-Gaussian Kalman instances, conjugate Bayesian updates, and any smooth log-likelihood admitting a non-degenerate local quadratic expansion — the form is *exact*, derived from Amari's natural-gradient invariance theorem: the natural-gradient Bayesian posterior mean shift at first order in the step size is exactly $\Delta\theta = K \cdot \tilde{\nabla}$ with gain operator $K = (H_M + H_L)^{-1} H_L$, collapsing to the scalar uncertainty-ratio along the natural-gradient direction. Outside Fisher-local conditions (non-quadratic losses, non-conjugate priors, heavy-tailed or multimodal uncertainty) the *direction* of dependence is preserved (gain rises with model uncertainty, falls with observation noise) but global quantitative fidelity may not — making the general form *robust qualitative*. The qualitative direction is what the downstream tempo and persistence machinery actually relies on; the Kalman filter is the canonical exact instance (scalar Kalman gain is exactly $U_M/(U_M + U_o)$).

An apparent paradox is resolved in Discussion: the optimal gain seems to require the agent to *know* the irreducible observation noise — but the observation function and noise distribution were declared opaque to the agent in 1.3. The resolution is dynamic. The agent *estimates* both U_M and U_o from the observable statistics of its own mismatch sequence (the innovations) and treats the gain itself as an endogenous state variable updated meta-adaptively. The full proof that this meta-adaptation maintains Lyapunov stability without violating epistemic opacity appears in `#deriv-adaptive-gain-dynamics`.

A failure mode is named: **gain collapse**. When the agent's estimates put U_M too low (spurious confidence) or U_o too high (spurious distrust of sensors), $\eta^* \to 0$ and the corrective phase of the cycle ceases. Mismatches still arrive but the agent no longer turns toward them — Boyd's "incestuous amplification," the cause of brittle failure in non-stationary environments. The dynamics also have a natural recovery mode: after structural change in the environment (see 4.8), U_M should spike, η^* should rise, and rapid re-learning becomes possible. An agent whose gain does not reset after structural change continues trusting a stale model. Overfitting receives a clean characterization via 3.5: η^* too high adjusts the model to explain irreducible noise (increasing model error on future predictions); η^* too low fails to correct genuine model errors. The optimal gain *implicitly separates signal from noise* by weighting observations in proportion to their informativeness — exactly what the uncertainty ratio achieves when U_o captures the irreducible noise.

Formal Expression.

Empirical Claim (uncertainty-ratio-principle)

$$\eta^* = \frac{U_M}{U_M + U_o} \tag{3.16}$$

where: - η^* is the optimal update gain (proportion of mismatch used to correct the model) - U_M is model uncertainty (predictive variance or entropy) - U_o is irreducible observation noise

The update rule takes the form:

Formulation

$$M_t = M_{t-1} + \eta^* \cdot g(\delta_t) \tag{3.17}$$

where δ_t is the mismatch (3.4) and $g(\cdot)$ is a correction mapping from observation space to model update space.

Epistemic Status.

Derived under the **Fisher-local invariance regime** (`#deriv-fisher-local-update-gain`): for any smooth log-likelihood admitting non-degenerate local quadratic expansion, the natural-gradient Bayesian posterior mean shift at first order in the step size is exactly $\Delta\theta = K \cdot \tilde{\nabla}$ with gain operator $K = (H_M + H_L)^{-1} H_L$ and scalar collapse $\eta^* = U_M/(U_M + U_o)$ along the natural-gradient direction (always in 1-D; under (PI)/Čencov in higher dimensions). Linear-Gaussian (Kalman) and conjugate-Bayesian instances are cases where the local quadratic expansion is *globally* exact, so the form holds without truncation. For general smooth models, the natural-gradient invariance theorem of Amari 1998 guarantees the form is exact at the local-tangent-plane Pythagorean projection level.

Outside the Fisher-local regime — non-quadratic losses, non-conjugate priors, structurally non-Gaussian uncertainty (heavy tails, multimodality) — the dependence is *robust qualitative*: the **direction** is preserved (gain rises with model uncertainty, falls with observation noise) and the first-order form is recovered locally; what need not hold is global quantitative fidelity. The qualitative direction is the load-bearing claim for downstream tempo and persistence machinery; the Fisher-local exact form is the load-bearing claim for the Kalman, conjugate, and natural-gradient instantiations.

Discussion.

Limiting behavior. When $U_M \gg U_o$ (high model uncertainty — e.g., after initialization or structural adaptation), $\eta^* \to 1$: trust the observation. When $U_M \ll U_o$ (confident model, noisy channel), $\eta^* \to 0$: trust the model. The gain determines how strongly the agent corrects toward reality on each update.

Resolving Epistemic Opacity. The optimal gain equation requires the agent to know U_o, which seems to violate the epistemic opacity axiom established in `#def-observation-function` (the agent does not know the true noise distribution ε_t). This tension is resolved dynamically: the agent *estimates* U_o (and U_M) from the observable statistics of its own mismatch sequence (innovations), treating the gain itself as an endogenous state variable. See `#deriv-adaptive-gain-dynamics` for the proof of how this meta-adaptation maintains Lyapunov stability without violating opacity.

Gain collapse — epistrophe failure. When the agent incorrectly estimates $U_M \to 0$ (spurious confidence) or $U_o \to \infty$ (spurious distrust of sensors), $\eta^* \to 0$ and epistrophe ceases. Aporia still arrives — the mismatch signal is still generated — but the agent no longer turns toward it. Mismatches are ignored, producing confirmation bias or a decoupled reality model. The cycle runs but the corrective phase is hollow.

Multi-dimensional generalization. In vector-valued systems, U_M and U_o are covariance matrices and η^* becomes a gain matrix (as in the Kalman filter). The scalar form captures the essential structure.

Connection to adaptive tempo. The update gain is one factor in the agent's adaptive tempo (3.8): $\mathcal{T} = \nu \cdot \eta^*$. Frequent aisthesis (high ν) is useless if epistrophe extracts no information (low η^*). Gain measures the *quality* of the cycle's corrective phase; event rate measures its *speed*.

Gain dynamics. The optimal gain changes over time following predictable patterns:

- *Convergence*: As the model accumulates information, U_M decreases, so $\eta^* \to 0$. The model becomes increasingly resistant to individual observations. This IS Kalman filter convergence, Bayesian posterior concentration, and RL learning rate annealing.
- *Reset after structural change*: When the environment changes in ways the model cannot track incrementally (4.8), U_M should spike — the model "admits" its uncertainty. The gain increases, enabling rapid re-learning. An agent whose gain does NOT reset after structural change will continue trusting a stale model — Boyd's "incestuous amplification" and the cause of brittle failure in non-stationary environments.

Overfitting as gain miscalibration. From 3.5: $\mathbb{E}[\|\delta_t\|^2]$ = model error + irreducible noise. An agent with η too high adjusts its model to explain observation noise, increasing model error on future predictions. An agent with η too low fails to correct genuine model errors. The optimal gain implicitly separates signal from noise by weighting observations in proportion to their informativeness — exactly what $U_M/(U_M + U_o)$ achieves when U_o captures the irreducible noise.

Representation note. The additive form operates in a *representation space* appropriate to the model. For Bayesian posteriors (where update is multiplicative: $P(\theta \mid D) \propto P(D \mid \theta)P(\theta)$), the additive rule operates in log-probability or natural parameter space. For models on constrained manifolds (probability simplices, rotation groups), the update must be projected onto the manifold. The claim is not that all updates are literally additive in native parameterization, but that they have the structure "current state + gain × transformed mismatch" in an appropriate coordinate system.

Domain validation:

Table 3.3

Domain	*Gain form*	*Mapping quality*
Kalman filter	$K_t = P_{t\|t-1}H^T(HP_{t\|t-1}H^T+R)^{-1}$	**Exact.** Scalar case is exactly $U_M/(U_M + U_o)$.
Conjugate Bayesian	Posterior weight $n/(n+\kappa)$ cumulative; incremental $1/(n+\kappa)$	**Exact** for conjugate families. Incremental gain decreases as data accumulates.
RL (Q-learning)	Fixed learning rate α	**Approximate.** α is a degenerate constant gain — does not adapt to uncertainty. Advanced methods (Bayesian RL, Adam) converge toward the optimal form.
PID control	Fixed gains (K_p, K_i, K_d)	**Simplified.** Gains set at design time. Adaptive PID and MPC move toward the full framework.
Software developer	Implicit trust weighting of information sources	**Structural analogy.** New developer (high U_M) trusts observations heavily; experienced developer (low U_M) trusts their model. Gain reset after major refactoring.

Simulation validation. Numerical experiments (track-b, Variant E) validated the uncertainty ratio principle under observation noise. Riccati-optimal gain reduced steady-state mismatch by 52% compared to fixed gain when observation noise was moderate. The optimal gain also proved critical in adversarial settings: under heavy observation noise, optimal gain preserved more than double the adversarial tempo advantage exponent (0.40 vs 0.18) compared to fixed gain.

Open questions:

1. *Non-parametric models*: For neural networks without well-defined scalar U_M, how should it be computed? Ensemble methods, dropout-based uncertainty, and Bayesian neural networks are all approximations.
2. *Matrix vs scalar gain*: In high-dimensional systems, the gain is a matrix (Kalman) or per-parameter (Adam). The cross-dimensional structure (covariance) adds complexity. The scalar captures the principle; the matrix captures the full optimization.

3.7 *Definition:* Causal Information Yield

The framework defines a scalar quantity attached to each available action: the **causal information yield** (CIY), measuring the *action-distinguishability* of an action — how different its interventional outcome distribution is from what alternative actions would produce. The construction uses Pearl's intervention operator $do(\cdot)$ and is the expected KL divergence between the outcome distribution under $do(a)$ and the outcome distribution under $do(a')$, averaged over a reference distribution of alternative actions. CIY is non-negative by construction; it is zero for passive observers or agents whose actions don't affect outcome distributions; it is strictly positive exactly when actions causally alter what is observed — the property that distinguishes Pearl-Level-2 (interventional) from Level-1 (associational) epistemic access.

A subtle conceptual point: CIY measures *distinguishability*, not *learning value*. An action can have high CIY even when the agent already knows the outcome distributions perfectly — the distributions *are* different (high CIY), but the agent learns nothing new by confirming what it already knows. High CIY is therefore *necessary* for learning (indistinguishable actions can't teach anything) but not *sufficient* (distinguishable actions only teach when the agent is uncertain). The Discussion below treats the relationship to expected information gain, the λ-weighted surrogate in the unified policy objective, query actions (accessing another agent's already-compressed model), and the adversarial mirror (deceptive responses that exploit high trust to inject misdirected updates).

Formal Expression.

Definition (causal-information-yield)

The **canonical CIY** of action a given model state M:

$$\mathrm{CIY}(a; M) = \mathbb{E}_{a' \sim q(\cdot|M)}[D_{\mathrm{KL}}(P(o \mid do(a), M) \,\|\, P(o \mid do(a'), M))] \tag{3.18}$$

where $q(\cdot \mid M)$ is a reference distribution over comparator actions (uniform, policy-induced, or task-specific). This measures how strongly the action changes the interventional distribution of outcomes relative to alternatives.

The $do(\cdot)$ operator is Pearl's standard intervention notation (Pearl 2009, *Causality*, 2nd ed., Cambridge; Bareinboim, Correa, Ibeling & Icard 2022); the AAT recapitulation lives at 7.1 in Part II Ch.2, where the framework deploys the hierarchy operationally. CIY ≥ 0 by construction (expectation of KL divergences). CIY $= 0$ for a passive observer or an agent whose actions don't affect outcome distributions. CIY > 0 when actions causally alter what is observed — exactly what distinguishes Pearl's Level 2 (interventional) from Level 1 (associational) epistemic access.

Epistemic Status.

The CIY *definition* is well-grounded in causal inference theory. The quantity itself is standard — an expected KL divergence under the do-calculus. The interpretive claim — that CIY measures action-distinguishability rather than expected information gain — is exact (it follows from the definition). The relationship between CIY and learning value (see Discussion) is discussion-grade: the λ-weighted approximation to EIG is heuristic, not derived.

Discussion.

CIY measures distinguishability, not learning value. CIY as defined is the expected KL divergence between outcome distributions — how different the outcomes of a are from the outcomes of typical alternatives. This is **action-distinguishability**, not **expected information gain** (EIG). The distinction matters: an action can have high CIY even when the agent already knows the outcome distributions perfectly (the distributions ARE different, but the agent learns nothing new by confirming what it already knows). High CIY is *necessary* for learning (indistinguishable actions can't teach anything) but not *sufficient* (distinguishable actions only teach when the agent is uncertain about the distinction).

The two quantities approximately coincide when U_M is high — when the agent doesn't know the outcome distributions, high-CIY actions also have high EIG because observing a characteristically different outcome updates the agent's beliefs about the causal structure. They diverge when U_M is low — a confident agent gains nothing from taking a distinguishable action it already understands.

The $\lambda(M_t)$ weighting in the unified policy objective (7.5) partially compensates: when U_M is low, $\lambda \to 0$, suppressing the CIY term regardless of its magnitude. This makes the exploration term behave more like EIG — suppressing exploration when the agent is already certain. The compensation is heuristic (the λ form is not derived). For the current theory, CIY serves as a tractable surrogate for EIG, with the λ weighting providing the uncertainty-gating that makes the surrogate reasonable.

Open direction: proper EIG within AAT. Replacing CIY with a proper expected information gain quantity — $\text{EIG}(a; M) = I(o; \theta \mid do(a), M)$ where θ parameterizes the model — would be a stronger foundation for the exploration term, particularly in domains where the agent must decide between actions that are all highly distinguishable but differ in what they teach. Under certain scopes (intervention-rich domains with well-parameterized models), the EIG formulation might yield sharper exploration strategies — preferring actions that resolve the *most uncertain* causal links rather than the most distinguishable ones. Whether this yields operationally significant improvements over the λ-weighted CIY surrogate is an empirical question. The CIY formulation has the advantage of being computable from the agent's current model alone (it doesn't require reasoning about model uncertainty); EIG requires a meta-model of what the agent doesn't know.

Dependence on the reference distribution q. The quantitative CIY value depends on the choice of q, which is a significant degree of freedom. A uniform q treats all alternatives equally; a policy-induced q emphasizes alternatives the agent would consider. AAT adopts the policy-induced q as default: $q(\cdot \mid M) = \pi(\cdot \mid M)$, yielding CIY as "how different is this action's outcome from what I'd typically see?" CIY values are not comparable across different q choices.

Query actions: accessing external models. A qualitatively distinct class of actions: querying another agent's model. When a reliable source exists (expert, database, documentation, well-trained LLM), "ask a well-formed question" can yield information equivalent to thousands of probe-observe cycles. The source's model has already performed the compression work (2.2) — the response transfers the *output* of compression rather than requiring the agent to reconstruct it.

Key properties of query actions: - **Information density**: Single well-targeted query can carry CIY orders of magnitude higher than individual environment probes - **Trust-dependent gain**: Update from query depends on the agent's model of the source's reliability and alignment, not on observation channel noise (14.4) - **Pre-compressed information**: Responses arrive already compressed in the source's representational framework, introducing a translation cost when frameworks don't align - **Structural adaptation via external models**: Encountering another agent's model can trigger structural change (4.8) — incorporating external representational structure rather than building it de novo ("grafting")

When high-CIY query channels are available, the unified policy objective (7.5) favors query actions over direct probes, particularly when U_M is high, a trusted source exists, query cost is low, and the needed information is about *structure* rather than the agent's specific situation.

The adversarial mirror: deception and model corruption. The same channel that enables cooperative knowledge transfer can be exploited to degrade the opponent's model. A deceptive response yields positive CIY in the strict information-theoretic sense, but the content drives model-reality mismatch *upward*. The update gain η^* for the victim depends on trust; successful deception exploits high trust to inject a large, misdirected update. In the Lyapunov framework (4.4), this is adversarial disturbance injected through the observation channel, with coupling coefficient γ_A determined by the victim's trust level and exposure. See 14.4 for the formal treatment of trust-dependent gain, and 15.2 for the Lyapunov formalization. Distributed tempo, topology analysis, and game-theoretic integration are Part III content not yet fully extracted (source material in `src/old-tf-appendix-f-multi-agent.md`).

3.8 *Definition:* Adaptive Tempo

The framework's central capacity quantity: **adaptive tempo** $\mathcal{T}$, the effective rate at which an agent acquires useful information from its environment, given as the sum across all observation channels of *channel event rate times optimal update gain* on that channel. In the single-channel case, tempo is simply event-rate times gain. A tensor extension is defined: in the Fisher-local regime where gains are matrix-valued, tempo becomes a matrix sum whose eigenvalues give per-direction adaptation rates; the scalar form is recovered in the shared-eigenbasis collapse (always in 1-D; under (PI)/Čencov along the natural-gradient direction in higher dimensions).

Tempo combines two factors that are *both* load-bearing and *substitutable*: loop **speed** (event rate, how fast the agent cycles) and update **quality** (gain, how well it extracts information per event). An agent can achieve a given tempo via a fast noisy loop or a slower calibrated one; improvements to both factors compound multiplicatively. Tempo is the load-bearing capacity variable for the rest of the framework — it appears on the left-hand side of the persistence condition (4.5), determines adversarial advantage (15.4), and connects to code quality as observation infrastructure in the software-domain instantiation. The Discussion below treats the speed-quality substitutability, the observation-noise gating that grounds Boyd's emphasis on Orient quality over raw OODA speed, and the two scope caveats (independent-channel assumption, isotropy) that the tensor form and the per-dimension persistence result address.

Formal Expression.

Definition (adaptive-tempo)

$$\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*} \tag{3.19}$$

where: - k indexes the agent's distinct observation channels - $\nu^{(k)}$ is the event rate on channel k - $\eta^{(k)*}$ is the optimal update gain on channel k (3.6)

Single-channel special case: $\mathcal{T} = \nu \cdot \eta^*$.

Definition (tensor-adaptive-tempo)

Tensor extension under Fisher-local invariance regime.

Under the Fisher-local invariance regime (`#deriv-fisher-local-update-gain`), the optimal update gain on channel k is matrix-valued: $K^{(k)} = (H_M + H_L^{(k)})^{-1} H_L^{(k)}$, with $H_M = U_M^{-1}$ the prior precision and $H_L^{(k)} = (U_o^{(k)})^{-1}$ the channel-k observed Fisher information. The tensor adaptive tempo is then

$$\mathcal{T} = \sum_k \nu^{(k)} \cdot K^{(k)} \tag{3.20}$$

— matrix-valued, with per-direction rates given by the eigenvalues of $\sum_k \nu^{(k)} K^{(k)}$ in the appropriate basis. The scalar form $\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*}$ is recovered in the **shared-eigenbasis collapse**: when all $H_M, \{H_L^{(k)}\}$ commute (always in 1-D; under (PI)/Čencov along the natural-gradient direction in higher dimensions), each $K^{(k)}$ acts as the eigenvalue $\eta^{(k)*} = U_M/(U_M + U_o^{(k)})$ on the shared natural-gradient direction and the matrix sum collapses to a scalar.

The matrix gain operator $K^{(k)}$ is the per-coordinate primitive: in anisotropic regimes where the prior and likelihoods do not share an eigenbasis (or where different channels pin down different directions), the tensor form preserves the per-direction information that the scalar form averages away.

Epistemic Status.

This is a *definition*, held at `conditional`. It names the quantity that characterizes an agent's total corrective capacity, combining loop speed (ν) and epistemic quality (η^*). The stipulation itself carries no truth-claim, but the definition is advertised as an *operationalization* — $\mathcal{T}$ as the effective rate of useful corrective-information acquisition — and that interpretation holds exactly only under two explicitly named local conditions: **(1) cross-channel noise independence** — independence of the channels' observation noises across channels and events, under which additivity is *derived* exact (`#deriv-tempo-additivity`; independence is sufficient but not necessary — equality also holds on a measure-zero cancellation set). When noise independence fails the deviation is signed: the additive form overcounts in the common-source (echo-chamber) regime, where the redundancy penalty is derived in closed form and persistent shared bias forces information saturation, but *under*counts in noise-cancelling (synergistic) configurations — so the additive form is not in general an upper bound (Discussion §Channel noise-independence condition); and **(2) shared eigenbasis / isotropy** for the scalar form — outside it the tensor form above is the correct object (Discussion §Scalar vs. vector tempo). Under both conditions the additive scalar form is the exact operationalization; neither condition is establishable from the global dependency chain — they are properties of the application — so the tier is `conditional` rather than `exact`. The substantive claims are in the results that use it (4.5, 15.4), which inherit the same scope.

Scope of scalar vs. tensor forms. The scalar form is exact in the isotropic / shared-eigenbasis / nonredundant-channel case and is what downstream results currently invoke. The tensor form is the natural object under anisotropic gains, Fisher-whitened updates (`#deriv-fisher-whitened-update-rule`), LMI causal-IB (`#deriv-causal-ib-lmi`), and per-dimension persistence (16.5) — regimes where scalar tempo overestimates effective adaptation along weak dimensions. Downstream results that invoke scalar $\mathcal{T}$ implicitly assume scalar / isotropic / nonredundant-channel scope; promoting them to the tensor form under the appropriate anisotropic regime is a follow-on cycle item flagged in `TODO.md`.

Discussion.

Speed-quality substitutability. An agent can achieve the same tempo via a fast noisy loop (high ν, low η^*) or a slower calibrated one (low ν, high η^*). The product structure means improvements to *both* factors compound multiplicatively.

Observation noise gating. Because $\eta^* = U_M/(U_M+U_o)$, high observation noise (U_o) depresses gain and collapses tempo regardless of loop speed. You cannot outrun a bad observation channel by iterating faster. This grounds Boyd's emphasis on Orient quality over raw OODA speed.

Centrality. Tempo is AAT's core capacity metric. It appears on the left side of the persistence condition (4.5), determines adversarial advantage (15.4), and connects to code quality as observation infrastructure (`#der-code-quality-as-observation-infrastructure` — cross-component reference, see `02-tst-core/`) in the software domain. The strategic analog $\mathcal{T}_\Sigma$ (9.8) extends the same structure to strategy-edge revision, with the key difference that strategic edge rates are endogenous (depend on action policy and upstream success).

Temporal nesting. Adaptive processes stratify by timescale, with convergence constraints between levels (4.9).

Mismatch dynamics. The evolution of mismatch over time is governed by the balance between correction (via tempo) and disturbance (ρ) (3.9).

Channel noise-independence condition. The additive formula is exact when the channels' observation noises are independent across channels and events — Fisher information over conditionally independent observations is exactly additive (`#deriv-tempo-additivity`). When channel noises are dependent, the deviation from additivity is **signed**: in the *common-source* regime (a shared persistent bias contaminating several channels — overlapping sensors, repeated teammate reports drawing on one upstream source, redundant telemetry) the additive formula overcounts, with a closed-form redundancy penalty and, under persistent shared bias, outright *saturation* — no number of common-source channels buys information past the shared-bias floor; but correlated noise can also be *cancelled* (a correlated reference channel, anti-correlated errors), in which case the additive formula **under**counts, so the additive form is not a conservative bound in general. No scalar dependence measure — in particular the conditional mutual information $I(e^{(1)}; e^{(2)} \mid M_{\tau^-})$ — can carry the exact correction (it is sign-blind; it survives as a witness of potential non-additivity), and no convention-free per-channel attribution of the deviation exists for three or more channels; the exact joint object and the derivations are in `#deriv-tempo-additivity`. Since tempo is the core capacity variable (persistence condition, adversarial dynamics, composition), mis-modeled channel structure inflates margins in the common-source direction — and hides genuine triangulation capacity in the synergistic direction. Multi-agent composition (15.1) inherits both directions: communication tempo overcounts when allies report from a shared source and is super-additive when allies' errors anti-correlate.

Scalar vs. vector tempo. The scalar $\mathcal{T}$ assumes isotropic correction capacity. When the agent corrects some dimensions faster than others, scalar tempo overestimates effective adaptation along weak dimensions. *[Empirical Claim]* Simulation confirms: in an anisotropic 3D system (gain varying 5:1), scalar $\rho/\mathcal{T}$ overestimated by 72%, with the weak dimension accounting for 84% of total mismatch (`#obs-section-i-validation-simulations`). The correct formulation is per-dimension: $\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$ (16.5). The tensor extension above (`#deriv-fisher-local-update-gain`) gives the per-coordinate primitive $K^{(k)}$ that the per-dimension persistence result invokes — the matrix gain operator on each channel — making the per-dimension formulation a direct consequence of the tensor tempo definition rather than a separate generalization. Under cross-dimensional correction (off-diagonal $\mathcal{T}$ in the coordinate basis of D_δ), the matrix-Loewner persistence condition `#deriv-matrix-persistence-condition` is the canonical form: $\Sigma_\infty \prec D_\delta = \operatorname{diag}(\delta^2_{\text{critical},k})$ in the strict positive-definite order, with Σ_∞ solving the continuous Lyapunov equation $\mathcal{T}\Sigma_\infty + \Sigma_\infty\mathcal{T}^T = \Sigma_w$. The per-coordinate form is its diagonal-axis-aligned special case; matrix-Loewner is strictly sharper (the per-coordinate form is *unsafe* when $\mathcal{T}$'s eigenbasis misaligns with the coordinate axes — `#deriv-matrix-persistence-condition` §"Where per-coordinate is unsafe").

3.9 *Hypothesis:* Mismatch Dynamics

Mismatch is given dynamics. The linear first-order ODE proposes that mismatch magnitude decays at rate proportional to tempo times mismatch (the correction term) and grows at the rate of new mismatch injection from environmental change (the disturbance term ρ). The framework is explicit that this is heuristic — a first-order linear approximation — and the general nonlinear case is what the sector-condition framework handles in the next chapter (4.4).

Two steady-state forms appear and are individually load-bearing. Under **Model D** (deterministic bounded disturbance), the steady-state mismatch is $\rho/\mathcal{T}$ — the ratio of how fast the environment changes to how fast the agent adapts. Under **Model S** (stochastic zero-mean disturbance, Itô SDE), the steady-state root-mean-square mismatch scales as $\sigma_w/\sqrt{2\mathcal{T}}$ — the *square root* of the disturbance-to-correction ratio. The $1/\sqrt{\mathcal{T}}$ scaling (vs. $1/\mathcal{T}$ for Model D) is one of the volume's more interesting consequences: correction is less effective against stochastic noise than against deterministic drift. The ODE is honestly a fluid-limit approximation of the underlying discrete event-driven dynamics; the Epistemic Status treats the bridging assumption and where transient error is largest. The Discussion below treats the persistence threshold, the nonlinear shapes the linear ODE smooths over (saturation, threshold dead zones, structural breakdown), and the adversarial-coupling consequences that scale as the square (Model D) and 3/2 power (Model S) of the inter-agent tempo ratio — the formal core of "getting inside the opponent's loop" as a categorical rather than marginal advantage.

Formal Expression.

Hypothesis (mismatch-dynamics)

$$\frac{d\|\delta\|}{dt} = -\mathcal{T} \cdot \|\delta\| + \rho(t) \tag{3.21}$$

where: - $\mathcal{T} \cdot \|\delta\|$ is the rate at which the agent corrects mismatch (proportional to both tempo and current mismatch) - $\rho(t)$ is the **environment change rate** — the rate at which new mismatch is introduced by changes in Ω

Steady state, Model D (deterministic bounded disturbance, $\|w(t)\| \leq \rho$):

Setting $d\|\delta\|/dt = 0$:

Derived (from linear hypothesis, deterministic)

$$\|\delta\|_{ss} = \frac{\rho}{\mathcal{T}} \tag{3.22}$$

Steady-state mismatch is the ratio of how fast the environment changes to how fast the agent adapts.

Steady state, Model S (stochastic zero-mean disturbance, $d\delta = -\mathcal{T}\delta\, dt + \sigma\_w\, dW\_t$):

[Derived (from Itô-Lyapunov analysis — see Prop A.1S in `#deriv-sector-condition`*)]*

$$\|\delta\|_{\text{rms}} = \frac{\sigma_w}{\sqrt{2\mathcal{T}}} \tag{3.23}$$

(scalar case, $n = 1$; general: $\sigma_w\sqrt{n/(2\mathcal{T})}$). Steady-state mismatch scales as the square root of the disturbance-to-correction ratio, not the ratio itself. The $1/\sqrt{\mathcal{T}}$ scaling (vs. $1/\mathcal{T}$ for Model D) means correction is less effective against noise than against drift.

Transient solution (Model D):

$$\|\delta(t)\| = \|\delta_0\|e^{-\mathcal{T}t} + \frac{\rho}{\mathcal{T}}(1 - e^{-\mathcal{T}t}) \tag{3.24}$$

Mismatch decays exponentially from initial conditions toward the steady state.

Epistemic Status.

Heuristic. This is explicitly a first-order linear approximation. The qualitative behavior (bounded mismatch, steady-state ratio, exponential convergence) is robust across correction function forms. The quantitative predictions (exact steady-state value, convergence rate, the squared adversarial scaling law) are specific to the linear case. The general nonlinear treatment (4.4) replaces the linear correction term with a sector-bounded correction function and proves persistence without committing to a specific functional form.

Bridging assumption (discrete to continuous). This ODE is a fluid-limit approximation of the discrete event-driven dynamics (3.1). The fluid limit is formally justified by `#deriv-discrete-sector-condition`: for Model D (deterministic), the discrete and continuous steady states are identical (zero gap); for Model S (stochastic), the variance gap is $O(\eta^* c_{\max})$, quantitatively small whenever $\eta^* c_{\max} \ll 1$. The approximation is least accurate during initial transients when η^* is large, but this phase is short-lived and the transient error is bounded by $O(\eta^* c_{\max}/\nu^{1/2})$ under Lipschitz regularity.

Discussion.

Speed-quality substitutability. From $\mathcal{T} = \nu \cdot \eta^*$ (single-channel case): doubling event rate ν has the same effect on $\|\delta\|_{ss}$ as doubling update quality η^*. They are multiplicative when both improve: 50% improvement in each yields $1.5 \times 1.5 = 2.25\times$, not $3\times$. This is the formal analog of Boyd's insight that Orient quality often matters more than raw OODA speed — the same structural observation (quality and speed are substitutable, quality often dominates) appears in the model.

The persistence threshold. From the steady-state: $\|\delta\|_{ss} < \|\delta_{\text{critical}}\|$ iff $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ (4.5). Below this threshold, the model cannot support effective action. The same structural pattern — correction capacity falling below disturbance rate — appears across domains: extinction (environment changes faster than organism adapts), organizational failure (market moves faster than company learns), control instability (disturbances exceed correction capacity), cognitive overload (information arrives faster than processing). The persistence condition captures the common structure; whether it captures the dominant mechanism in each domain is an empirical question.

Nonlinear reality. The true correction dynamics are almost certainly nonlinear: - *Saturation at large* $\|\delta\|$: correction mechanism overwhelmed, so correction is slower than linear for large errors. Makes the persistence threshold harder to satisfy. - *Threshold effects*: small mismatches go uncorrected ($F \approx 0$ for $\|\delta\| < \varepsilon$), creating a dead zone. - *Structural breakdown*: beyond some critical $\|\delta\|$, correction drops to zero because the model class is no longer appropriate (4.8).

These nonlinearities are exactly what the sector-condition framework (4.4) handles.

Adversarial coupling. When two agents are coupled (A's actions increase B's disturbance): $\rho_B = \rho_{B,\text{base}} + \gamma_A \cdot \mathcal{T}_A$. The steady-state mismatch ratio scales superlinearly with the tempo ratio, but the exponent depends on the disturbance model. Under Model D (deterministic drift, coupling-dominant): $(\mathcal{T}_A/\mathcal{T}_B)^2$ — the squared law, derived from the $1/\mathcal{T}$ steady-state scaling. Under Model S (stochastic noise, coupling-dominant): $(\mathcal{T}_A/\mathcal{T}_B)^{3/2}$ — derived from the $1/\sqrt{\mathcal{T}}$ steady-state scaling. See 15.4.

Chapter 4

Persistence and Structural Limits

The cycle runs; the question is whether it can keep up. This chapter develops the central inequality of AAT — the persistence condition — together with what happens at and beyond its threshold: where it comes from, when the machinery can be expected to satisfy it, and what fails when it can't.

Chapter 3 ended with a preview. Under linear correction and bounded disturbance, the steady-state mismatch is $\rho/\mathcal{T}$ — disturbance over tempo. Mismatch stays bounded when the agent corrects faster than reality drifts. That was offered as a heuristic with a linear-ODE flavor. Chapter 4 makes it rigorous, generalizes it, and shows what it implies.

The first move is to replace "linear correction" with something more honest. Real correction mechanisms saturate at large mismatch, threshold at small mismatch, and break down entirely when the agent's model class is wrong for the situation. The sector condition — that the correction function points inward with at least baseline efficiency α within an operating region R — captures the qualitative essence of correction without committing to a specific functional form. This is one of the chapter's more satisfying pieces of theoretical hygiene: a condition flexible enough to admit saturating, sigmoid, threshold, and PID corrections under one Lyapunov argument, while still saying something concrete enough to derive a threshold from. Under that condition, the analysis gives

$$\alpha > \rho/R \tag{4.1}$$

three quantities, one threshold, mechanism underneath. The agent persists when its baseline correction efficiency exceeds the ratio of disturbance rate to operating reserve. Below the threshold, the containment guarantee is lost — escape from the operating region becomes possible, and for correctors whose sector bound is tight, certain under adversarial disturbance (`#deriv-sector-condition` Lemma A.1N). The result is qualitative: persistence is not a degree, it's a regime — what disappears at the threshold is the certificate, not merely some margin. The inequality is genuinely domain-agnostic — its four variables map to biological adaptation (extinction is what happens when environmental change rate outpaces evolutionary correction), to organizational survival (the market shifts faster than the firm learns), to control instability, to cognitive overload. The framework identifies a structural pattern across these domains rather than a model of any one of them; whether the specific mechanism is the dominant cause in any given case is empirical, but the structure recurs because the math does.

The persistence condition has a thermodynamic shadow worth surfacing here, since the chapter derives it explicitly in `#deriv-persistence-cost`: under stochastic disturbance, maintaining the sector-persistence bound requires sustained Shannon information acquisition at rate $\dot{R} \geq n\alpha/2$ nats per unit time, with Kalman-Bucy saturating the bound. Survival, in this framework, is not a state you achieve once — it is a sustained burn rate. An agent cut off from informative observation begins to drift the moment the channel closes, and the rate of drift is the disturbance rate, independent of how good the agent was at adapting before. Two agents with identical persistence guarantees can face wildly different sustained demands. The threshold says whether persistence is *possible*; the information-rate corollary says what it *costs*.

But this isn't quite the operational form most domains care about. There is a second condition — task adequacy — that the steady-state mismatch be small enough for the agent's actions to remain useful. An agent can satisfy the structural condition (the machinery works) but fail task adequacy (the machinery doesn't work *well enough* for what the agent is trying to do). The two conditions have different remedies, and the split between them is the kind of move that's invisible until you see it stated and obvious in retrospect. A Kalman filter on a moving target satisfies structural persistence whenever it's stable, but task adequacy depends on what counts as "tracking" for the application. A software team can be structurally persistent but task-inadequate. The structural condition is what AAT derives; the task-adequacy threshold is a domain parameter — and conflating the two produces category errors in domain transfer.

The other move I want to flag is upstream. Where does α actually come from? Earlier in the framework, α shows up as a sector-condition parameter — a structural property of the correction function, taken as given. The gain-sector bridge (4.3) demotes it from postulate to derivation. For agents with directional fidelity in their update rule — including all optimal Bayesian updaters and all gradient agents on (locally) strongly convex losses — $\alpha = \eta^* \cdot c_{\min}$, where η^* is the gain principle from Chapter 3 and $c_{\min}$ is the worst-case directional fidelity of the update. The sector condition isn't a soft global postulate at all; for the cases AAT cares about most, it's a property of the gain. The five failure modes that break this derivation (directional infidelity, gain collapse, nonlinear saturation, unobservable directions, model misspecification) are named explicitly, and each connects back to a concrete mechanism rather than being left as residual error. This is the kind of move that makes the framework feel earned rather than assumed.

The chapter closes with two integrative results. *Structural adaptation necessity*: when model-class fitness $\mathcal{F}(\mathcal{M})$ is too low, no parametric update within the class can close the mismatch floor — the agent must change classes, not parameters. This is the consequence whose seed was planted in Chapter 2, now grown to a derived result. *Temporal nesting*: adaptive processes stratify by timescale, with each level operating on the quasi-steady-state output of the level below. Parametric update is fast; structural adaptation is slow; the convergence constraint between adjacent timescales means a slower process acting on transients of a faster one produces oscillation. This is standard singular-perturbation reasoning applied here to multi-level adaptation, but the consequence — that the timescale ratios are themselves constrained — is what licenses the abstraction throughout the rest of the framework.

A scope coda closes the chapter and Part I. AAT applies to agents on singular causal trajectories. The chronica $\mathcal{C}_t$ is not forkable — duplicating M_t and exposing the copies to different futures produces two agents on divergent trajectories, neither a sufficient statistic for the other's path. Identity in AAT lives in the trajectory, not in the state. This sits at the end of Chapter 4 because its dependencies (chronica from Chapter 1 + sufficiency from Chapter 2) span the section, but its substance is its own — and it's the bridge to the logogenic-agents and ELI work in Volumes 3 and 4, where the singular-trajectory commitment becomes load-bearing.

The flow of the chapter: deliberation cost (4.1) → gain-to-sector bridge (4.3) → sector-condition stability and the persistence inequality (4.4, 4.5) → structural adaptation when parametric update fails (4.8) → temporal nesting (4.9) → identity scope (4.10). The chapter is where Part I's machinery resolves into operational results.

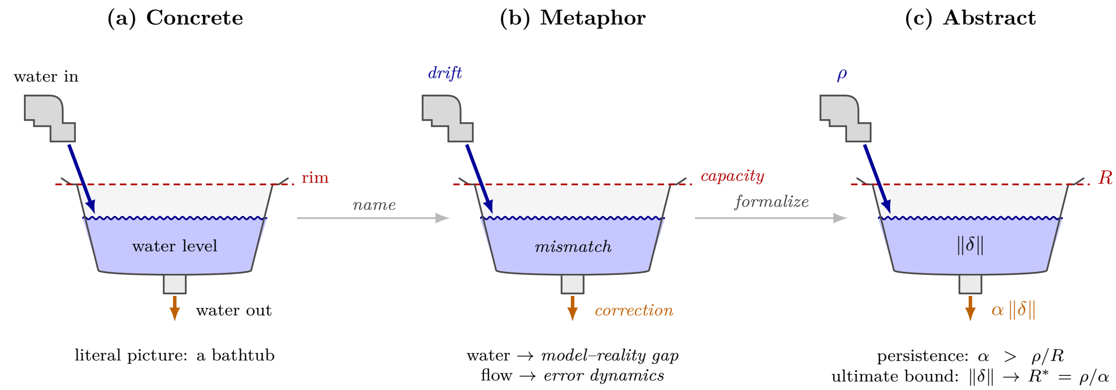


Figure 3: Persistence as a bathtub. The belief–reality mismatch is the water level, drift is the inflow, correction is the drain, and model capacity is the rim. The same picture is drawn three times — literal water vocabulary, plain-English AAT terms, then formal symbols — in identical spatial positions, so the analogy is one-to-one. Overflow is persistence failure: inflow outpacing the drain at full.

4.1 *Derived:* Deliberation Cost

The framework's first formal trade-off between thinking and acting. Explicit deliberation can improve action quality by using the model for internal simulation before acting (improving the next update gain η^* — pausing praxis to improve upcoming epistrophe), but deliberation costs time, and during that time mismatch accumulates because the environment continues to evolve while the agent is not correcting (aporia accumulates during the pause). The derivation gives a clean threshold: deliberation of duration $\Delta\tau$ is net-beneficial when the gain improvement times post-deliberation mismatch magnitude exceeds the deliberation-window mismatch drift rate times the duration — stop deliberating when the marginal improvement rate drops below the mismatch drift rate normalized by the post-deliberation mismatch. Under diminishing returns (the first moments of simulation yield the largest gain improvement) combined with linear-in-duration drift cost, this yields a finite optimal deliberation duration; past that point, additional thinking is net-harmful.

The result is conditional on a *local deliberation-drift assumption* — that mismatch grows at an approximately constant rate ρ_{delib} during pause windows — which is a short-horizon assumption about inaction windows rather than a global dynamics model, weaker than the linear mismatch ODE (3.9), and can be estimated directly from pause windows in empirical traces. The Discussion below treats the consequences the framework reuses throughout the volume: high-drift environments penalize deliberation (the formal version of Boyd's claim that over-deliberation is fatal in fast-tempo adversarial environments); implicit action as the high-tempo limit (formal grounding for why high-tempo environments favor action fluency); deliberation as investment when drift is low or post-deliberation mismatch is large; the self-consistency circularity (evaluating the threshold requires predicting post-deliberation mismatch using the same model deliberation is meant to improve); and the connection to temporal nesting as a singular-perturbation pattern.

Formal Expression.

Assumption (local deliberation drift):

Assumption (deliberation-drift)

During a deliberation pause of duration $\Delta\tau$, mismatch increases at an approximately constant local rate ρ_{delib}:

$$\Delta\|\delta\|_{\text{deliberation}} \approx \rho_{\text{delib}} \cdot \Delta\tau \tag{4.2}$$

This is a short-horizon assumption about inaction windows, not a full global dynamics model. It is weaker than the mismatch ODE and can be estimated directly from pause windows in empirical traces.

Proposition (deliberation threshold):

Derived (Conditional on deliberation-drift assumption)

Deliberation of duration $\Delta\tau$ is net-beneficial when:

$$\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\| > \rho_{\text{delib}} \cdot \Delta\tau \tag{4.3}$$

where $\Delta\eta^*(\Delta\tau)$ is the improvement in post-deliberation update gain and $\|\delta_{\text{post}}\|$ is the mismatch magnitude the agent will face when it resumes acting.

Derivation.

1. Without deliberation, the agent acts immediately at current tempo $\mathcal{T}_0 = \nu \cdot \eta_0^*$.
2. With deliberation of duration $\Delta\tau$, the agent pauses, then acts with improved gain $\eta_0^* + \Delta\eta^*$. But during the pause, mismatch has grown by $\rho_{\text{delib}} \cdot \Delta\tau$.
3. The net mismatch reduction from acting after deliberation versus acting immediately: Net $= \Delta\eta^* \cdot \|\delta_{\text{post}}\| - \rho_{\text{delib}} \cdot \Delta\tau$.
4. Deliberation is justified iff Net > 0. □

Optimal deliberation duration (under diminishing returns):

Derived (Conditional on diminishing-returns + deliberation-drift)

$$\Delta\tau^* = \arg\max_{\Delta\tau} \left[\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\| - \rho_{\text{delib}} \cdot \Delta\tau\right] \tag{4.4}$$

where $\|\delta_{\text{post}}\|$ is treated as a parameter estimated by the agent before deliberation begins (not optimized over — the agent estimates the mismatch it will face, then decides how long to deliberate). Under this approximation, the first-order condition is: $\frac{\partial\Delta\eta^*}{\partial\Delta\tau} \cdot \|\delta_{\text{post}}\| = \rho_{\text{delib}}$. Stop deliberating when the marginal improvement rate drops below the mismatch drift rate (normalized by post-deliberation mismatch). When the dependence $\|\delta_{\text{post}}\| = \|\delta_0\| + \rho_{\text{delib}} \cdot \Delta\tau$ is included in the optimization, the exact FOC acquires a correction factor $(1 - \Delta\eta^*)$ on the cost side; this is negligible when $\Delta\eta^* \ll 1$ (the typical case — deliberation produces small gain improvements).

Epistemic Status.

Conditional on the local deliberation-drift assumption. The threshold condition is derived given the assumption; the assumption itself is a local approximation validated by consistency with the global mismatch dynamics (4.5). The result captures the *epistemic* benefit of deliberation (improving η^*); in practice, deliberation also provides a direct *action-value* benefit (choosing better actions that alter the environment trajectory), which operates through ρ reduction and immediate reward — a fuller formalization would incorporate the unified policy objective (3.7) at significantly more complexity.

Discussion.

High-ρ_{delib} environments penalize deliberation. When the environment changes rapidly during pause windows, the cost term grows quickly. Only very short deliberation with large $\Delta\eta^*$ can justify the pause. The model captures the same tradeoff Boyd emphasized: in fast-tempo adversarial environments, over-deliberation is fatal not because thinking is bad, but because the environment moves during the thinking. Whether the specific mechanism (mismatch drift during pause) is the dominant real-world effect is an empirical question.

Diminishing returns. In most models, $\Delta\eta^*(\Delta\tau)$ exhibits diminishing returns — the first moments of simulation yield the largest improvement. Combined with the linear cost $\rho_{\text{delib}} \cdot \Delta\tau$, this implies a finite optimal deliberation duration. Past that point, additional thinking is net-harmful.

Implicit action as the high-tempo limit. As $\rho_{\text{delib}} \to \infty$ or $\Delta\tau^* \to 0$: the optimal strategy converges to zero deliberation — pure implicit action (3.3). This provides a mathematical basis for why high-tempo environments favor action fluency: the cost of deliberation exceeds its benefit when $\Delta\eta^*$ is small (action-selection is already fluent) or ρ_{delib} is large.

Deliberation as an investment. When ρ_{delib} is low (stable environment) or $\|\delta_{\text{post}}\|$ is large (significant model-reality gap), deliberation pays off. The conditions favoring deliberation — stable environment, large mismatch — resemble the high-stakes, low-urgency scenarios where deliberative reasoning (System 2) is advantageous in dual-process theories. The structural parallel is suggestive; whether the cost-benefit mechanism is the same one governing System 1/System 2 selection is an open question.

The circularity of $\|\delta_{\text{post}}\|$. Evaluating the threshold requires the agent to *predict* post-deliberation mismatch using its current model — the same model deliberation is meant to improve. This circularity is typically benign: $\|\delta_{\text{post}}\|$ is bounded below by $\rho_{\text{delib}} \cdot \Delta\tau$ and above by current mismatch plus that accumulation. An agent that underestimates its mismatch will under-deliberate; one that overestimates will over-deliberate. The bias is self-correcting through the feedback loop. The threshold is best understood as a *design criterion*, not a real-time decision procedure.

Resource costs beyond time. Real agents also incur computational and energetic costs: internal simulation burns calories, compute cycles, or opportunity cost of not processing new observations. These are additive: $\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\| > \rho_{\text{delib}} \cdot \Delta\tau + C(\Delta\tau)$. In high-$\rho_{\text{delib}}$ environments the temporal cost dominates; in low-ρ_{delib} environments, resource costs may be the binding constraint.

Structural adaptation as an analogy. Structural adaptation (4.8) superficially resembles deliberation with a massive $\Delta\tau$: the agent's parametric loop is partially suspended while it searches for a new model class, incurring a large mismatch debt $\rho_{\text{delib}} \cdot \Delta\tau$. However, this is an informal analogy, not a consequence of the deliberation-cost formalism. Deliberation as formalized here improves η^* *within a fixed model class*; structural adaptation changes the model class itself, which is a mechanistically different operation (4.8). The cost-benefit structure may be similar in form, but the quantities involved ($\mathcal{F}(\mathcal{M})$ vs. η^*, model-class search vs. gain improvement) are distinct.

Connection to temporal nesting. Deliberation is a nested loop: internal simulation running at rate ν_{internal} within the external action loop at rate ν_{external}. The convergence constraint applies: the internal loop must approximately converge before the external loop acts on its output.

Connection to Part II. For actuated agents, the deliberation tradeoff extends to three modes: exploit (O_t via Σ_t), explore (improve M_t), and deliberate (revise Σ_t). The three-way allocation (10.2) extends this segment's threshold by adding a strategic benefit term ΔV_Σ; the extended threshold is the one genuinely derived piece. The broader three-way framing is discussion-grade — simulation shows deliberation is rarely chosen by an oracle in simple settings, and a unified objective outperforms the two-stage decomposition.

The AI agent's dilemma. An AI agent with 100% context turnover faces a severe version: it MUST deliberate (comprehend the codebase) before acting effectively, but during comprehension its context fills and the environment may change. The optimal comprehension depth depends on ρ_{delib} and the session's action horizon. This is why reading CLAUDE.md and architecture docs first (high-CIY query actions) dominates reading random source files (low-CIY exploration).

Domain instantiations:

Table 4.1

Domain	*Deliberation*	*$\Delta\eta^*$ source*	*When ρ_{delib} is high*
Boyd's OODA	Explicit "Decide" step	War-gaming, staff analysis	Collapses to IG&C (implicit)
RL / MCTS	Planning rollouts	Monte Carlo tree search	Fewer rollouts, shallower search
MPC	Online optimization	Trajectory optimization	Shorter horizons, faster solvers
Human cognition	System 2 deliberation	Mental simulation	Defaults to System 1 (intuition)
Organization	Strategic planning	Scenario analysis	"Move fast and break things"
Software developer	Reading code, analyzing alternatives	Architecture analysis	Ship now, refactor later
AI agent	Reading codebase, planning approach	Context-building	Limit comprehension, act sooner

Open questions:

1. *Computational cost of deliberation* is not just elapsed time but resource cost. A fuller model would include both temporal and computational budgets.
2. *Deliberation about deliberation*: deciding whether to deliberate itself takes time. This meta-deliberation is bounded by the same tradeoff at a higher level, suggesting a hierarchy of diminishing deliberation horizons.
3. *Deliberation that generates observations*: internal simulation can surface model inconsistencies (internal mismatch), functioning as "exploration without external action." Can deliberation generate internal CIY?

4.2 *Formulation:* Sector Condition (A2')

The sector condition (A2') is the structural shape AAT chooses for the correction-function geometry: the correction points roughly inward, with magnitude bounded below relative to the mismatch, on a local region of validity. Together with the companion structural properties (A1) zero-correction-at-zero-mismatch and (A3) tempo-monotonicity, A2' is the formal expression of "the agent's adaptation tracks reality with at least baseline efficiency" — the form on which the persistence and adaptive-reserve results of `#deriv-sector-condition` rest, and the form which downstream consumers (`#deriv-discrete-sector-condition`, `#deriv-variational-sector-condition`, `#deriv-adaptive-gain-dynamics`, `#der-gain-sector-bridge`) extend, weaken, or ground.

Formal Expression.

Setup objects (carried into the form). Let $\delta(t) \in \mathbb{R}^n$ be the mismatch vector (3.4 — the difference between the model's predictions and reality across n observable dimensions). Let $F(\mathcal{T}, \delta) : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^n$ be the **correction function** — how the agent's adaptive process reduces mismatch — mapping into the same space as δ so that the inner product $\delta^T F$ is well-defined. This subsumes the update gain η^* (3.6), event rate ν, and the structure of the update rule. The adaptive tempo $\mathcal{T}$ (3.8) is the rate parameter.

(A1) Zero Correction at Zero Mismatch.

Assumption A1

$$F(\mathcal{T}, 0) = 0 \tag{4.5}$$

No correction is applied when the model perfectly matches reality. Uncontroversial by construction.

(A2') Local Sector Condition. There exists a region $\mathcal{B}_R = \{\delta : \|\delta\| \leq R\}$ and $\alpha > 0$ such that (following the sector-condition framework of Lur'e[^lure1957]):

Formulation A2' (sector-condition) — derived in sub-scope α, assumed in sub-scope β (see Grounding below)

$$\delta^T F(\mathcal{T}, \delta) \geq \alpha\|\delta\|^2 \quad \forall\delta \in \mathcal{B}_R \tag{4.6}$$

The correction function always points "inward" (reducing mismatch), and its magnitude is bounded below relative to $\|\delta\|^2$. The linear case has $\alpha = \mathcal{T}$. A saturating correction has α decreasing for large $\|\delta\|$. A threshold correction has $\alpha = 0$ for small $\|\delta\|$.

The local form allows the correction to break down outside $\mathcal{B}_R$ — the structural adaptation regime of 4.8.

(A3) Tempo Monotonicity.

Assumption A3

For fixed δ, $\delta^T F(\mathcal{T}, \delta)$ is monotone increasing in $\mathcal{T}$. Higher tempo means faster correction.

Parameter interpretation. The sector parameter α is determined by the adaptive tempo $\mathcal{T}$ and the structure of the correction function. In the linear case, $\alpha = \mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*}$. In nonlinear cases, α represents the *worst-case* correction efficiency within the valid region — the minimum ratio of correction power to mismatch magnitude. The radius R represents the model class capacity: how large a mismatch can grow before the correction mechanism fails (i.e., before the sector condition ceases to hold), at which point structural adaptation (4.8) becomes necessary.

Epistemic Status.

A2' is a *formulation* — a chosen structural shape for the correction-function geometry that captures "the correction points roughly inward with bounded efficiency." Several structurally distinct shapes would fit the same role (a sign-only condition; a two-point monotonicity; a contraction-rate bound; a non-Euclidean weighted inner-product variant — see *Why Euclidean A2' specifically* in Discussion below). A2' is the canonical choice *matched to the quadratic Lyapunov candidate* $V = \frac{1}{2}\|\delta\|^2$ used in `#deriv-sector-condition`, and matched to the gain-based update form on which 4.3 derives it for sub-scope α.

The status is *conditional* because A2' is **derived** for one explicitly named sub-scope of AAT-in-scope agents and **assumed as a per-system empirical claim** for the complementary sub-scope. The Lyapunov derivations downstream of A2' (`#deriv-sector-condition` Props A.1, A.1S, A.2; Corollary A.1S.1) apply uniformly across both sub-scopes — they operate downstream of A2' regardless of how it is established. The sub-scope partition is *scope narrowing*, not scope retreat: it makes explicit which agent classes get A2' as a derived consequence of their update geometry, and which classes carry it as a structurally well-scoped posit.

Sub-scope α (A2' derived). For a characterized class of AAT-in-scope agents, A2' is a *derived* consequence of the update rule, not a primitive assumption. 4.3 (Prop B.3) shows that the gain-based update $M_t = M_{t-1} + \eta^* g(\delta_t)$ (3.6) induces a correction function satisfying A2' whenever the update rule has **directional fidelity (B1)** — $\delta^T H g(\delta) \geq c_{\min}\|\delta\|^2$ on $\mathcal{B}_R$. Sub-scope α — the agent classes where B1 holds structurally — includes:

- *Optimal Bayesian updates* (Kalman, conjugate families): B1 holds by Bayes-risk minimization. $\alpha = \eta^* \cdot c_{\min}$, reducing to $\alpha = \eta^*$ in the scalar case.
- *Exponential families in natural parameters*, on a bounded interior scope $\Theta_0 \subset \text{int}(\Theta)$: the Hessian is the Fisher information matrix, PD on the interior. $\alpha = \eta \cdot \mu_0$ where $\mu_0 = \inf_{\theta\in\Theta_0} \lambda_{\min}(\mathbf{I}(\theta)) > 0$. Pointwise PD does not imply a uniform global lower bound: $\inf_{\theta\in\Theta} \lambda_{\min}(\mathbf{I}(\theta))$ can be zero (Poisson: $\mathbf{I}(\theta) = e^\theta$, $\inf_{\theta\in\mathbb{R}} e^\theta = 0$), so Θ_0 supplies the local-region scope R that A2' requires. Families with a uniform Fisher floor on Θ (Gaussian-mean, Beta-Bernoulli) extend to global α.
- *Gradient descent on locally strongly convex losses* (Prop B.4): B1 is *equivalent* to strong convexity via the gradient-monotonicity characterization (Nesterov 2004[^nesterov2004], Thm 2.1.10). $\alpha = \eta \cdot \mu$ where μ is the strong convexity modulus.
- *L2-regularized convex losses*: regularization provides a floor $\mu \geq \lambda$, so $\alpha \geq \eta\lambda$ globally.

- *Linear corrections with positive-definite gain–observation product*: $\alpha = \lambda^{+}_{\min}(KH)$ (matrix Kalman, restricted to the observable subspace).

Within sub-scope α, A2' is written down by inspection of the update rule; no independent posit is required.

Sub-scope β (A2' assumed as empirical claim). For the remaining AAT-in-scope agents, A2' stands as a well-scoped empirical claim about the agent's correction geometry. Sub-scope β includes:

- *PID controllers with fixed gains* — no gradient / optimality structure; B1 is a tuning question, not a structural consequence.
- *Rule-based systems* — no continuous update rule; A2' is domain-specific (see also the structural Lipschitz-floor scope-exit in Discussion below).
- *Human judgment / organizational learning* — structural analogy in 3.6; no formal B1 guarantee.
- *Severely misspecified agents* (FM-5 in 4.3) — proper-gradient rules can aim at the wrong target.
- *Variational / approximate posteriors* — B1 not guaranteed by optimality because approximation error can rotate the correction. Under a controlled KL bound, an intermediate sub-scope α' recovers a degraded form of A2' (see `#deriv-variational-sector-condition`).
- *Non-convex gradient agents beyond the basin* — A2' fails at basin boundary; this IS the 4.8 trigger.
- *Stochastic gradients (per-step)* — A2' holds in expectation; per-step noise enters as effective disturbance under `#deriv-sector-condition` Prop A.1S.

Within sub-scope β, A2' must be verified per-system — the claim is stronger than what AAT's postulates + gain structure alone can force (an agent with $g(\delta) = R_{90^\circ}\delta$ satisfies every AAT postulate but violates B1 trivially; see 4.3 FM-1).

Discussion.

Operator-family classification (external mathematical lineage). The sub-scope α/β partition can be restated precisely in the operator-theoretic language of Rockafellar 1970 (*Convex Analysis*, §24) and Bauschke & Combettes 2017 (*Convex Analysis and Monotone Operator Theory in Hilbert Spaces*, 2nd ed., §§22–28). Casting the discrete update map as $T_d(\delta) = \delta - \eta^* F_d(\delta)$ with $T_d(0) = 0$, A2' at $\delta^* = 0$ is exactly a **one-point strong monotonicity condition** on $A = I - T_d$, and DA2'-inc (the incremental strengthening required by `#form-composition-closure`'s bridge lemma, stated in `#deriv-discrete-sector-condition`) is **full two-point strong monotonicity** in the Rockafellar sense. Sub-scope α maps onto established operator families:

- **Optimal Bayesian updates** (Kalman, conjugate, exponential family) ≡ proximal / firmly nonexpansive operators (Bauschke-Combettes Prop 23.7). Firmly nonexpansive is equivalent to $\frac{1}{2}$-averaged; operator-sector condition holds with the gain as the sector constant.
- **Gradient operators on strongly convex functions** ≡ cocoercive operators via the Baillon-Haddad theorem; strong monotonicity with modulus $\eta\mu$.
- **Exponential families in natural parameters** ≡ natural-gradient operators in the Fisher-weighted Hilbert space (Amari 1998; cf. 4.3 "Fisher-metric cases under parameterization-invariance" for the AAT-internal forcing via (PI)/Čencov named in 5.5).
- **L2-regularized convex gradients** ≡ shifted monotone-gradient operators with regularization-provided floor.
- **Linear corrections with PD gain–observation product** ≡ linear contraction in the $(P^{-})^{-1}$-weighted inner product.

Sub-scope β maps onto "operator families that are not cocoercive in any natural inner product": PID (tuning-dependent; not cocoercive generically); rule-based / symbolic (no inner-product structure); variational / amortized (approximation-gap rotation); non-convex-gradient beyond basin (non-monotone on the full domain); per-step SGD (noise-dominated); human judgment (no formal rule). The α/β epistemic labeling in AAT tracks whether the operator family admits operator-sector structurally (derived from the class definition) or only under empirical per-instance verification — a scope-honesty move, not a mathematical reclassification.

This recognition positions AAT's sector-condition framework as a specialization of monotone-operator theory (Minty 1962; Browder 1968; Rockafellar 1970; Bauschke-Combettes 2017) to one-point-anchored strong monotonicity under a specific inner product structure. AAT's distinctive content — one-point anchoring at the equilibrium (strictly weaker than full two-point strong monotonicity, matched to fixed-point-at-target semantics); Model D / Model S disturbance decomposition; composition with the identifiability-floor meta-pattern; composition-consistency postulate; the α/β epistemic labeling itself — sits as specialization + repurposing rather than strict generalization. This acknowledgment is load-bearing for scope honesty: the mathematical machinery is established; AAT's value lies in its AAT-internal architecture (organization-of-scope under a singular-trajectory agent, signed-coupling structure, forced coordinates via uniqueness theorems, three meta-patterns) rather than in novel monotone-operator mathematics. The unification has honest limits: the coarse-graining projection Λ (`#form-composition-closure`) does not fit the operator-sector primitive; three of five metric-α_2 cases in `#result-contraction-template` remain theorem-imported; the identifiability-floor axis is orthogonal.

Sub-scope β "rule-based / discontinuous" — structural Lipschitz floor. *[Derived, status: exact scope-exit statement.]* The rule-based / state-machine / threshold-triggered / discontinuous entry in sub-scope β is not merely "empirical verification required"; it is a **structural scope-exit for contraction-based bridge-lemma analysis**. For correction functions F that are not locally Lipschitz, no scalar sector bound $\delta^\top F(\delta) \geq \alpha\|\delta\|^2$ implies the full-update-map contraction required by `#form-composition-closure`'s bridge lemma — the update map $f_c(X, o) = X - \eta^* F(X)$ can exhibit $\Omega(1)$ jumps between arbitrarily close X, X' at rule-firing boundaries, so no $\lambda < 1$ Lipschitz constant exists. Concrete counterexample: $F(\delta) = \alpha\delta$ for $|\delta| < 1$ and $F(\delta) = \alpha\delta + \text{sign}(\delta)$ for $|\delta| \geq 1$ — the sector bound holds with α, but f_c jumps by η^* at $|X| = 1$, violating contraction. The appropriate external apparatus for this class is the **hybrid-dissipative framework** (van der Schaft & Schumacher 2000, *An Introduction to Hybrid Dynamical Systems*, Springer; Di Bernardo, Liuzza & Russo 2014, *SIAM J. Control Optim.* 52) — distinct machinery (dwell-time, Filippov solutions, impulsive-dissipative inequalities) operating on a different regularity class. AAT's Lyapunov machinery in `#deriv-sector-condition` and the contraction machinery in `#result-contraction-template` both require C^1 or better regularity; the scope-exit to hybrid-dissipative analysis is honest and structural, not a deficiency to be repaired within AAT's current apparatus. Under `#disc-identifiability-floor`'s Instance 2 pattern (Cramér-Rao floor under unobservable common cause), rule-based agents whose rule firing depends on regime structure — e.g., threshold-triggered state-machines with regime-dependent thresholds — additionally suffer regime-C identifiability collapse when the regime variable is unobservable; the non-contractibility and the non-identifiability compose in that case.

Why Euclidean A2' specifically. The A2' form $\delta^T F \geq \alpha\|\delta\|^2$ is the sector condition *matched to the quadratic Lyapunov candidate* $V = \frac{1}{2}\|\delta\|^2$ used by `#deriv-sector-condition`. A converse-Lyapunov argument (Khalil 2002[^khalil2002], Thm 4.17) gives: if persistence holds under the dynamics $\dot{\delta} = -F(\delta)$ on $\mathcal{B}_R$, then there exists a quadratic-equivalent Lyapunov function $V_*(\delta)$ with $c_1\|\delta\|^2 \leq V_* \leq c_2\|\delta\|^2$ — but V_* may not be the Euclidean norm itself. Under a weighted Lyapunov candidate $V(\delta) = \frac{1}{2}\delta^T P\delta$, the natural sector condition is $\delta^T PF(\delta) \geq \alpha\,\delta^T P\delta$; the matrix-Kalman case of 4.3 is exactly this in the $(P^-)^{-1}$-weighted inner product, with a norm-equivalence transfer to Euclidean A2' degraded by the condition number $\kappa(P^-)$. Euclidean A2' is therefore not the unique sector form — it is the canonical one matched to the canonical V. An agent that persists under a non-Euclidean metric satisfies a weighted-sector A2' that transfers to Euclidean A2' only up to norm equivalence. 5.5 classifies the Lyapunov case as an *adjacent family member* to AAT's three primary additive-coordinate-forcing instances (chain / divergence / update, where a logarithmic coordinate is uniquely forced via Cauchy's functional equation under an AAT-internal additivity axiom): here the quadratic coordinate is matched to the sector form rather than forced by one.

How downstream segments use this form.

- `#deriv-sector-condition` consumes A2' as the Lyapunov-derivation input: Props A.1 / A.1S / A.2 prove ultimate boundedness, mean-square persistence, and adaptive reserve under (A1), (A2'), (A3); Corollary A.1S.1 establishes the disturbance-model containment dichotomy $P(\tau_R < \infty) \in \{0, 1\}$.
- `#deriv-stochastic-non-exit` demonstrates the load-bearing Model-S half of Cor A.1S.1 — that no horizon-independent non-exit bound exists under additive stochastic forcing — using A2' on $\mathcal{B}_R$ as the Itô-Lyapunov input.
- 4.3 derives A2' for sub-scope α under directional fidelity (B1), making the sub-scope-α partition above structural rather than postulated.

- `#deriv-discrete-sector-condition` states the discrete-time analog (DA2'), strengthening A2' with an additional Lipschitz bound on $\|F_d\|$ to control the quadratic term that arises in discrete contraction.
- `#deriv-variational-sector-condition` states ε-fidelity A2' under controlled-KL variational approximation, with $O(\sqrt{\varepsilon})$ sector-constant degradation — recovering an intermediate sub-scope α' within the partition above.
- `#deriv-adaptive-gain-dynamics` refines sub-scope α into fixed-gain (α_1) and adaptive-gain (α_2 under meta-gain conditions MG-1–MG-4) layers.
- 4.4 and `#result-sector-persistence-template` state the persistence results in result-segment voice; 13.1 and the composite-agent segments in Part III consume DA2'-inc as the bridge-lemma precondition.

4.3 *Derived:* Gain–Sector Bridge

The gain-based update principle (3.6) produces correction dynamics satisfying the sector condition (GA-3) whenever the update rule has *directional fidelity* — the correction points at least roughly toward reality. For gradient-based agents, local strong convexity of the loss is sufficient for the one-point sector condition (A2' as stated in 4.2) and bidirectionally equivalent to the two-point / incremental sector condition (DA2'-inc). The sector parameter α is not a free parameter but is determined by the gain and the correction geometry.

Formal Expression.

Derived (gain-sector-bridge, from update-gain + directional fidelity)

The Bridge Theorem.

Given the gain-based update $M_t = M_{t-1} + \eta^* \cdot g(\delta_t)$ (3.6), the induced correction function is:

$$F(\delta) = \eta^* \cdot H\, g(\delta) \tag{4.7}$$

where H maps state-space corrections to observation-space mismatch reduction. The sector condition (GA-3) holds with parameter $\alpha > 0$ whenever:

(B1) Directional fidelity. The mismatch transform g preserves the mismatch-reducing direction:

$$\delta^T H\, g(\delta) \geq c\|\delta\|^2 \quad \text{for } \|\delta\| \leq R \tag{4.8}$$

for some $c > 0$. The bridge then determines the sector parameter in two forms that differ by the event-rate normalization, because the gain-based update is applied *per correction event* while the sector condition feeds a *continuous-time* persistence inequality.

Per-event sector efficiency — the dimensionless fraction of standing mismatch the correction removes per event:

$$\alpha_{\text{event}} = \eta^* \cdot c_{\min}, \qquad c_{\min} = \inf_{\|\delta\| \leq R} \frac{\delta^T H\, g(\delta)}{\|\delta\|^2} \tag{4.9}$$

with η^* dimensionless and $c_{\min}$ a dimensionless geometric (Rayleigh-quotient) ratio, so $\alpha_{\text{event}} \in (0, 1]$ is itself dimensionless.

Per-time sector rate — the correction *rate* the continuous-time sector condition (GA-3) and the persistence inequality $\alpha > \rho/R$ actually compare against, obtained by aggregating per-event efficiency over the channel's event rate ν (3.8):

$$\alpha = \alpha_{\text{time}} = \nu \cdot \eta^* \cdot c_{\min} \tag{4.10}$$

with units t^{-1} — the same units as adaptive tempo $\mathcal{T}$ (NOTATION.md *Units*). The event rate ν is exactly the factor that promotes the per-event geometric quantity to the per-time rate that the Lyapunov machinery in `#deriv-sector-condition` consumes; the bare α in GA-3 and downstream is α_{time}.

Gradient Equivalence.

Derived (sector-convexity equivalence, two-point form)

The sector parameters in this section and the Verified-Instances table below are stated in the *per-event* form α_{event} (the dimensionless correction efficiency); each promotes to the per-time rate the persistence inequality consumes by the same factor ν, $\alpha_{\text{time}} = \nu \cdot \alpha_{\text{event}}$, so the proportionalities are identical under either reading.

For any agent updating via gradient descent on a loss L with learning rate η:

$$\alpha = \eta \cdot \mu \qquad \text{where } \mu = \inf_{\|\delta\| \le R} \lambda_{\min}(\nabla^2 L(M^* + \delta)) \tag{4.11}$$

is the strong convexity modulus. The basin radius R is the largest ball around the optimum where $\nabla^2 L$ remains positive definite. The equivalence has two forms — bidirectional under the stronger two-point sector condition, one-directional under the one-point form actually used by `#deriv-sector-condition`:

- **Two-point / incremental sector ⇔ strong convexity (full equivalence).** Under the incremental sector bound $(F(\delta_1) - F(\delta_2))^T(\delta_1 - \delta_2) \ge \alpha\|\delta_1 - \delta_2\|^2$ on $\mathcal{B}_R(M^*)$ — DA2'-inc in `#deriv-discrete-sector-condition`, the bridge-lemma precondition in 13.1 — the iff holds via Nesterov 2004 Theorem 2.1.10: Two-point sector with $(\alpha, R) \iff$ L is locally (α/η)-strongly convex on $\mathcal{B}_R(M^*)$.
- **One-point sector ⇐ strong convexity (one direction only).** AAT's GA-3 / A2' as stated in 4.2 is the one-point form $\delta^T F(\delta) \ge \alpha\|\delta\|^2$ at $\delta^* = 0$. Strong convexity implies the one-point sector ($\alpha = \eta\mu$); the converse fails. Counterexample: $L'(x) = x(1 + \frac{1}{2}\sin(10x))$ satisfies $x \cdot L'(x) \ge \frac{1}{2}x^2$ globally yet has $L''(\pi/10) < 0$, so it is not convex on any neighborhood of $x^* = 0$. The one-point sector at the equilibrium is genuinely weaker than full local strong convexity (cf. `#result-sector-persistence-template`'s one-point/two-point distinction). Full proofs and the counterexample analysis in `#deriv-gain-sector` Prop B.4.

Verified Instances.

Table 4.2

Update class	*Bridge status*	*Sector parameter α*	*Valid region*
Scalar Kalman	Derived	$K = P^-/(P^- + R_{\text{obs}}) = \eta^*$	Global
Matrix Kalman	Derived	$\lambda^+_{\min}(KH)$ in $(P^-)^{-1}$-norm	Observable subspace
Beta-Bernoulli	Derived	$1/(n+1) = \eta_{\text{edge}}$	Global
Exponential family (natural params), bounded scope $\Theta_0 \subset \text{int}(\Theta)$	Derived	$\eta \cdot \mu_0$ where $\mu_0 = \inf_{\theta \in \Theta_0} \lambda_{\min}(\mathbf{I}(\theta)) > 0$	Θ_0 (compact / interior-bounded) — global only when the family has a uniform Fisher lower bound
Gradient on strongly convex loss	Derived	$\eta \cdot \mu$	Global ($R = \infty$)
Gradient on locally convex loss	Derived	$\eta \cdot \mu_{\text{local}}$	Basin of attraction
Gradient on non-convex loss	Fails at basin boundary	N/A beyond R	Finite R

Update class	*Bridge status*	*Sector parameter α*	*Valid region*
SPR-tuned PID on positive-real plant with anti-windup	Derived	$\alpha_{\text{PID}} = \omega_c \sin(\varphi_m)/\kappa(P)$ (phase margin as sector constant; crossover frequency as tempo; KYP-certificate condition number as degradation)	Classical linear regime + Lur'e sector-bounded nonlinearity within specified plant-Lipschitz threshold

Failure Modes. The bridge fails precisely in five cases:

1. **Directional infidelity.** The mismatch transform g rotates the correction away from the mismatch ($\delta^T Hg(\delta) \le 0$). Occurs with pathological parameterizations or severe model-observation misalignment. For optimal Bayesian updates, B1 holds by construction.
2. **Gain collapse.** $\eta^* \to 0$ while $\rho > 0$, so $\alpha \to 0$ and the persistence condition eventually fails. Not a failure of the bridge but of the persistence condition — see the gain-collapse analysis in 3.6.
3. **Nonlinear saturation.** The correction function g saturates at large $\|\delta\|$, so the sector ratio $\delta^T g(\delta)/\|\delta\|^2$ decays. The sector condition holds locally with α depending on R. This is exactly what A2' (the local sector condition) is designed for.
4. **Unobservable directions.** When $\ker(H) \neq \{0\}$, the correction has no effect on mismatch in unobservable directions. The sector condition holds only in the observable subspace. See 9.3.
5. **Model misspecification.** The model class does not contain the truth, so the gradient direction is wrong. B1 fails because the correction aims at the wrong target. This is the 4.8 trigger.

Epistemic Status.

Conditional derivation. The bridge theorem is exact for all cases where B1 (directional fidelity) holds. The resulting sub-scope structure of A2' in 4.2 is:

Sub-scope α (B1 structural, A2rsquo derived):

- **Optimal Bayesian updates** (Kalman, conjugate, exponential family): B1 holds by construction — the posterior update minimizes expected loss, ensuring the correction aligns with the mismatch. The sector parameter equals the gain: $\alpha = \eta^*$ (scalar) or $\alpha = \lambda^+_{\min}(KH)$ (matrix Kalman, observable subspace).
- **Gradient descent on (locally) strongly convex losses**: local strong convexity is *sufficient* for B1 (the one-point form of A2' at M^*) and *bidirectionally equivalent* to the two-point / incremental sector condition DA2'-inc (Prop B.4 (B.4-i) and (B.4-ii) respectively) — a well-characterized property with an extensive optimization theory literature. The sector parameter factors as $\alpha = \eta \cdot \mu$ (learning rate × curvature). The reverse implication B1 ⇒ strong convexity does *not* hold without the two-point upgrade: the one-point sector at M^* is strictly weaker than full local strong convexity (counterexample in Prop B.4).
- **L2-regularized convex losses**: the regularization parameter λ provides a global floor $\mu \ge \lambda$, so $\alpha \ge \eta\lambda$ globally.
- **Exponential families in natural parameters, on a bounded interior scope**: Fisher information matrix is PD on the interior, and uniformly bounded below on any compact $\Theta_0 \subset \text{int}(\Theta)$; $\alpha = \eta \cdot \mu_0$ with $\mu_0 = \inf_{\theta \in \Theta_0} \lambda_{\min}(\mathbf{I}(\theta))$. The bound is global only when the family has a uniform Fisher lower bound on Θ — true for Gaussian-mean and Beta-Bernoulli, false for Poisson natural parameter ($\mathbf{I}(\theta) = e^\theta$, infimum zero).
- **Linear corrections with PD gain–observation product**: $\alpha = \lambda^+_{\min}(KH)$.

Within sub-scope α, A2' is written down by inspection of the update rule — no independent posit is required. This is what `#form-sector-condition` "Sub-scope α (A2' derived)" names.

Sub-scope β (B1 not structural, A2rsquo assumed per-agent):

- **Non-gradient agents** (PID controllers, rule-based systems, human judgment): B1 remains an empirical claim. Well-tuned PID has B1 empirically; badly-tuned PID may violate it.
- **Severely misspecified agents** (FM-5 below): proper-gradient updates can aim at the wrong target.
- **Variational / approximate posteriors**: B1 not guaranteed by optimality — approximation-direction error can rotate the correction.
- **Non-convex gradient agents beyond the basin** (FM-3 + basin boundary): A2' fails where the loss curvature goes non-positive; the structural-adaptation-necessity trigger.
- **Stochastic gradients, per-step**: A2' holds in expectation; per-step noise enters as effective disturbance under Prop A.1S.

The bridge covers sub-scope α rigorously and characterizes the boundary to sub-scope β via the five failure modes. It does *not* eliminate GA-3 as an assumption for all AAT-in-scope agents — some agent classes genuinely require A2' as a primitive posit, and the honest architectural statement is scope narrowing rather than universal derivation.

The gradient equivalence is validated by simulation across quadratic, logistic, exponential-family, and non-convex losses. The Kalman case is verified analytically. Full derivations and simulation results in `#deriv-gain-sector`.

Max attainable: *conditional* — the condition (B1 or strong convexity) is inherent, not removable. Pathological update rules exist that violate B1 (FM-1 provides a counterexample that satisfies every AAT postulate and the gain-based update form but has $\delta^T g(\delta) = 0$ identically). Scope + gain structure alone does not force B1; some optimality / coherence / rationality constraint (Bayesian coherence, gradient-of-a-convex-loss, etc.) is required.

Weighted-norm subtlety. In the matrix Kalman case, the sector condition holds in the $(P^-)^{-1}$-weighted inner product, not the Euclidean norm. For fully observable systems with bounded condition number $\kappa(P^-)$, the norms are equivalent up to $\kappa(P^-)$. The Lyapunov proofs in `#deriv-sector-condition` use the Euclidean norm, which remains valid with the quantitative adjustment $\alpha_{\text{Euclidean}} \geq \alpha_{\text{weighted}}/\kappa(P^-)$.

Fisher-metric cases under parameterization-invariance. The exponential-family-in-natural-parameters row and the matrix-Kalman row of Verified Instances both have natural statements in a Fisher-metric inner product rather than the Euclidean one. Under the **(PI) parameterization-invariance** axiom named in 4.10 (AAT's theorems should not depend on arbitrary choice of coordinates on M_t), Čencov's 1982 uniqueness theorem (*Statistical Decision Rules and Optimal Inference*, AMS; subsequent Ay-Jost-Lê-Schwachhöfer 2017 extensions) forces the Fisher information metric uniquely on statistical-manifold sub-cases of M_t. Two consequences:

- *The matrix-Kalman sector constant is natively stated in the information metric* $M = (P^-)^{-1}$ — under (PI), this is not a choice but forced. The $\kappa(P^-)$ Euclidean-transfer penalty in the paragraph above vanishes; the derivation is AAT-internally *forced*, not AAT-internally *preferred*.
- *The exponential-family-natural-parameter sector constant is natively stated in the Fisher metric* $\mathbf{I}(\theta)$ — under (PI), this is forced; the contraction rate equals η globally on the interior of the natural-parameter domain (Fisher-conditioning degradation removed).

Under (PI), these two rows of Verified Instances upgrade from *derived (conditional on choice of inner product)* to *derived (AAT-internally forced)* via Čencov. Under non-adoption of (PI), they remain at the Euclidean-transferred statement with the $\kappa(P^-)$ / $\lambda_{\min}$(Fisher) penalty. This is the Fisher-layer analog of the chain-rule-additivity axiom that grounds the divergence-layer uniqueness theorem in `#deriv-strategy-cost-regret-bound` and the evidential-additivity axiom that grounds the update-layer uniqueness theorem in `#deriv-edge-update-natural-parameter` — in each case an AAT-internal axiom combined with a uniqueness theorem forces a coordinate. The structural positioning is named in 5.5.

The remaining Verified Instances rows (scalar Kalman, Beta-Bernoulli, gradient on strongly convex, L2-regularized, linear-PD-symmetric) live in Euclidean metric naturally; no Fisher-metric choice is at issue, so (PI) has no effect on them beyond transparency.

Discussion.

GA-3 is grounded, not floating. Before this result, GA-3 ("the correction function satisfies the sector condition") was an opaque global assumption — the theory's softest structural joint. The bridge transforms it: for well-designed agents, GA-3 is a consequence of the update mechanism's geometry, not an independent postulate. The assumption load shifts from "the correction function has this property" (hard to verify in general) to "the update rule has directional fidelity" (transparent and checkable for specific systems).

The α-$\mathcal{T}$ relationship is derived for the important cases. 4.5 notes that α is monotone increasing in $\mathcal{T}$ across all tested correction functions. The identity holds at the *per-time* level: for linear correction (Kalman, Beta-Bernoulli) with $c_{\min} = 1$, $\alpha_{\text{time}} = \nu \cdot \eta^* = \mathcal{T}$ exactly — the per-time sector rate *is* adaptive tempo, both being event-rate-weighted correction rates with units t^{-1}. The per-event reading is $\alpha_{\text{event}} = \eta^*$ for the same case; the two differ by exactly the event-rate factor ν, which is what $\mathcal{T} = \nu\eta^*$ carries. For gradient correction on strongly convex losses, $\alpha = \eta \cdot \mu$ where μ is the curvature — monotone in η (which is monotone in $\mathcal{T}$) for fixed loss landscape. The empirical observation is now structurally grounded.

Basin boundary = structural adaptation trigger. For gradient agents with non-convex losses, the basin radius R is the convexity radius of the loss landscape. When mismatch exceeds R, the agent has been pushed out of its convexity basin — the correction function reverses direction and the sector condition fails. This IS the 4.8 trigger, now with a precise geometric characterization: structural adaptation is needed when the agent crosses an inflection surface of its loss landscape.

The theory's formal chain is tightened. The prediction chain becomes:

$$\text{gain principle} + \text{B1} \xrightarrow{\text{derived}} \text{sector condition (GA-3)} \xrightarrow{\text{Lyapunov (exact)}} \text{persistence, reserve, adversarial scaling} \tag{4.12}$$

The left arrow is this segment. The right arrow is `#deriv-sector-condition`. The discrete-time framework (`#deriv-discrete-sector-condition`) requires an additional Lipschitz bound on the correction function ($\|F_d(\delta)\| \leq c_{\max}\|\delta\|$ with $c_{\max} < 2/\eta^*$) — strictly stronger than an inner-product upper bound, needed because the discrete contraction involves $\|F_d\|^2$. This is automatically satisfied for Bayesian updates (the posterior lies between prior and data) and for gradient descent on smooth losses (where $c_{\max}$ is the Lipschitz constant L). With this constraint and the step-size condition $\eta^* < 2c_{\min}/c_{\max}^2$, the fluid limit is formally justified: Model D steady state is exact, Model S variance gap is $O(\eta^* c_{\max})$. Part I's formal chain is now complete.

4.4 *Result:* Sector Condition Stability

The framework's first major *Lyapunov result*, stated as a specific instantiation of a more general sector-persistence template (`#result-sector-persistence-template`) whose abstract form lives in the appendices. The mismatch dynamics are taken in their general nonlinear form: mismatch changes at rate (correction function applied to mismatch) plus (environmental disturbance). The correction function is required only to satisfy the **local sector condition** — that its inner product with the mismatch is at least α times the mismatch's squared magnitude, within a radius-R region where it remains valid. The sector condition is what generalizes the linear ODE: it captures the qualitative essence of correction (the function points inward with at least baseline efficiency) without committing to a specific functional form like linear, saturating, sigmoid, threshold, or PID. Saturation, thresholding, and basin boundaries all live under one Lyapunov argument.

Under this sector condition plus bounded disturbance, the chapter's headline persistence inequality is *derived*: $\alpha > \rho/R$ guarantees the agent persists. When the inequality holds, the mismatch is ultimately bounded by $R^* = \rho/\alpha$, and the "adaptive reserve" — the additional disturbance the agent can absorb before mismatch reaches the edge of the valid region — is $\alpha R - \rho$. The threshold is *tight*: over the class of correctors carrying sector floor α it is necessary as well (no weaker condition can deliver the guarantee), and for a corrector whose sector bound is radially tight — the linear case — it is a genuine agent-level if-and-only-if. For a general fixed corrector, failure of the inequality means the containment *guarantee* is lost, not that escape is certified (`#deriv-sector-condition` Lemma A.1N).

Under **Model S** (stochastic disturbance), the analog is sharper and qualitatively different: the steady-state root-mean-square mismatch scales as $\sigma_w\sqrt{n/(2\alpha)}$ — the square root of the disturbance-to-correction ratio. Model D scales as $1/\alpha$, Model S as $1/\sqrt{\alpha}$: **correction is less effective against noise than against drift**. This is one of the volume's striking results, separating two genuinely different physics of adaptation under deterministic vs stochastic environments.

The linear ODE from Chapter 3 (3.9) is recovered as the special case where the sector condition holds globally with $\alpha = \mathcal{T}$. The general sector-condition framework proves the persistence threshold is not an artifact of the linear approximation: every bounded-correction system carries such a threshold, and $\alpha > \rho/R$ is the sharpest condition derivable from the sector floor alone (`#deriv-sector-condition` Lemma A.1N). The result is also where the structural-adaptation-necessity result will be anchored: when disturbance exceeds the model class's capacity (i.e., $\rho/\alpha > R$), the sector condition fails. This *is* the dynamical trigger for needing a new model class with larger valid radius or better efficiency — which 4.8 treats formally a few segments later.

Formal Expression.

This segment is the **single-agent epistemic instantiation** of the sector-persistence template (`#result-sector-persistence-template`). The template's state variable is $\xi = \delta(t) \in \mathbb{R}^n$ (model-reality mismatch); the correction function is $F(\mathcal{T}, \delta)$; the disturbance is environmental ($w(t)$); the region of validity R is the model class capacity.

Formulation

$$\frac{d\delta}{dt} = -F(\mathcal{T}, \delta) + w(t) \tag{4.13}$$

Assumption (sector-condition)

F satisfies the local sector condition (template condition (T2)) for $\|\delta\| \le R$:

$$\delta^T F(\mathcal{T}, \delta) \ge \alpha \|\delta\|^2 \tag{4.14}$$

with $\alpha > 0$. Here α is a *correction rate* (units t^{-1}, like $\mathcal{T}$): the correction function F is the event-rate-aggregated, continuous-time field, so the sector bound it satisfies is the per-time rate. Disturbance is bounded: $\|w(t)\| \le \rho$ (Model D, GA-2) or $\mathbb{E}[\|w(t)\|^2] = \sigma_w^2$ (Model S, GA-2S). Grounding of (T2) for gain-based agents: 4.3 gives the per-time rate $\alpha = \nu \cdot \eta^* \cdot c_{\min}$ (the event rate ν promoting the per-event efficiency $\eta^* c_{\min}$ to the rate this inequality consumes). The linear case $F = \mathcal{T} \cdot \delta$ — itself the per-time correction field — yields $\alpha = \mathcal{T}$ exactly ($\nu\eta^*$ at $c_{\min} = 1$).

Derived (from sector-persistence-template)

The template's Model D conclusion specializes to: $\delta(t)$ is ultimately bounded by $R^* = \rho/\alpha$, and the agent is guaranteed to persist when

$$\alpha > \frac{\rho}{R}. \tag{4.15}$$

The condition is sufficient for the individual agent and tight at class level — necessary for the uniform guarantee over all correctors with sector floor α, and agent-level necessary exactly when the sector bound is radially tight (the linear case); see `#deriv-sector-condition` Prop A.1 and Lemma A.1N. The adaptive reserve is $\Delta\rho^* = \alpha R - \rho$ — the additional disturbance the agent can absorb before R^* exceeds the valid region.

The template's Model S conclusion specializes to: the steady-state RMS mismatch is $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$ (where $n = \dim(\delta)$), and mean-square persistence requires $\alpha > n\sigma_w^2/(2R^2)$. Model D scales as $1/\alpha$; Model S scales as $1/\sqrt{\alpha}$ — correction is less effective against noise than against drift.

Full Lyapunov proofs: `#deriv-sector-condition` Props A.1, A.1S, A.2.

Epistemic Status.

Exact. Both results are direct instances of the sector-persistence template applied to the single-agent epistemic case. Template precondition (T1) is satisfied because no correction should be applied at zero mismatch; (T2) reduces to the local sector condition above and is grounded structurally by 4.3 for gain-based agents; (T3) is the disturbance-model choice (D or S), a domain question. The linear ODE of 3.9 is the special case where (T2) holds globally with $\alpha = \mathcal{T}$ — the per-time identity, both sides being event-rate-weighted correction rates (units t^{-1}); the sector framework generalizes this to saturating, thresholded, and structurally-limited correction functions under the same persistence condition. Disturbance-model choice is a domain question, not a theory question.

Discussion.

Why the sector condition. The linear ODE assumes correction scales linearly with mismatch forever. Real adaptive systems saturate, exhibit thresholding, or break down when the model class is exhausted. The sector condition captures the minimal structural requirement: the correction must point in the right direction with at least baseline efficiency α.

Generalizing the persistence threshold. In the linear case, $\alpha = \mathcal{T}$ (adaptive tempo). The general result $\alpha > \rho/R$ proves the persistence threshold (4.5) is a structural necessity of any bounded-correction system, not an artifact of the linear approximation. This result addresses *structural persistence* — the machinery's capacity to bound mismatch — not operational persistence (current proximity to R) or continuity persistence (identity through time). See Persistence in `LEXICON.md` for the full disambiguation.

Connection to structural adaptation. When $\rho/\alpha > R$, disturbance exceeds the model class's capacity. The sector condition fails — this is the dynamical trigger for structural adaptation (4.8), requiring a new model class with larger valid radius R' or better efficiency α'.

4.5 *Result:* Persistence Condition

This is the volume's **central inequality** in its operational form. The segment's headline contribution is the *two-condition decomposition*: persistence is not one threshold but two, with different mechanisms and different remedies.

Structural persistence is the Lyapunov-derived condition that the correction machinery contains mismatch within its operating region: $\alpha > \rho/R$ under deterministic bounded disturbance, or $\alpha > n\sigma_w^2/(2R^2)$ under stochastic disturbance. This says *the machinery works*. Mismatch is ultimately bounded by $R^* = \rho/\alpha$ (Model D) or $\sigma_w\sqrt{n/(2\alpha)}$ (Model S). When this inequality fails, the containment *certificate* is lost — escape from the operating region becomes possible, and for correctors whose sector bound is tight (the linear case) it is forced by adversarial disturbance in finite time. The loss of the certificate is a qualitative regime transition, not a gradual degradation; what condition-failure does not do is certify escape for every corrector, since the sector floor can understate correction along the actually-reachable paths (`#deriv-sector-condition` Lemma A.1N).

Task adequacy is the separate condition that the steady-state mismatch is small enough for the agent's actions to remain useful: $R^* < \|\delta_{\text{critical}}\|$, where the critical-mismatch threshold is a *domain-specific tolerance* — how wrong can the model be before the agent's actions become harmful or ineffective? Set by the application, not derived by AAT. The two conditions are independent: an agent can be structurally persistent but task-inadequate (the machinery contains mismatch but not tightly enough for the domain). The remedies differ — task inadequacy can be fixed by raising tempo, lowering disturbance, or relaxing tolerance; structural failure requires changing the correction architecture entirely.

When the linear ODE applies (sector parameter equals tempo, valid region is unbounded), structural persistence is automatically satisfied and the binding condition is just task adequacy: $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ under Model D. This is the form most downstream applications cite. It is exact for linear correction and a useful proxy for mildly nonlinear correction (where $\alpha \approx \mathcal{T}$), but overstates persistence when correction saturates.

A consequence the framework emphasizes: **the persistence condition is a structural pattern that recurs across domains rather than a model of any one of them**. The same inequality — correction rate against disturbance rate, normalized by tolerance — governs a Kalman filter on a moving target, an organization adapting to market shifts, an immune system tracking a pathogen, a developer keeping pace with a codebase, an organism evolving against environmental change. The structure recurs because the *math* recurs; whether the specific mechanism (tempo limited by some particular bottleneck) is the *dominant* cause in any given empirical case is left as an open empirical question per domain.

The two-condition decomposition specifically prevents one common category error in domain transfer: a structurally-persistent codebase team can be task-inadequate, and conflating the two would yield the wrong intervention.

A complementary cost shadow: persistence has a *price*, not just a threshold. Under stochastic disturbance, maintaining the ultimate bound requires sustained Shannon information acquisition at rate at least $n\alpha/2$ nats per unit time — a Landauer-analog floor that the Kalman-Bucy filter saturates (see `#deriv-persistence-cost`). The threshold says whether persistence is *possible*; the information-rate corollary says what it *costs*. Two agents with identical persistence guarantees can face wildly different sustained demands because the cost scales linearly with the sector parameter; the threshold alone cannot distinguish dormant from running-hot.

A per-dimension extension addresses anisotropy: when correction capacity varies across dimensions, the scalar condition can overestimate persistence margin substantially (up to 72% in simulation). The correct condition is per-dimension; under cross-dimensional correction the canonical sharp form is the matrix-Loewner persistence condition (`#deriv-matrix-persistence-condition`).

Formal Expression.

This segment is the canonical single-agent instantiation of the sector-persistence template (`#result-sector-persistence-template`) with state variable $\xi = \delta_t$ (epistemic mismatch), correction function $F(\mathcal{T}, \delta)$, and disturbance rate $\rho_\xi = \rho$ (environmental change rate). Structural persistence is the direct template conclusion. Task adequacy adds a domain-specific constraint beyond the template's reach.

Structural Persistence.

Derived (structural-persistence, from sector-persistence-template)

Applying the template to the single-agent epistemic case gives: the correction machinery is guaranteed to bound δ within the model class capacity when

$$\alpha > \frac{\rho}{R} \quad \text{(Model D)} \qquad \alpha > \frac{n\sigma_w^2}{2R^2} \quad \text{(Model S)} \tag{4.16}$$

Each condition is sufficient for the individual agent and tight at class level (necessary for the uniform guarantee over all correctors with the given sector floor; agent-level necessary exactly under a radially tight sector bound — the linear case). See `#deriv-sector-condition` Lemma A.1N.

with ultimate bound $R^* = \rho/\alpha$ (Model D) or $R_S^* = \sigma_w\sqrt{n/(2\alpha)}$ (Model S). See 4.4 for how (T1)–(T3) are verified in this instantiation, and `#deriv-sector-condition` for the proof. Structural persistence is a property of the adaptive architecture — the machinery's ability to contain mismatch — not of the task.

Linear case. When $F(\mathcal{T}, \delta) = \mathcal{T}\delta$, $\alpha = \mathcal{T}$ and $R \to \infty$, so structural persistence is trivially satisfied whenever $\mathcal{T} > 0$. The binding constraint then becomes task adequacy (below).

Task Adequacy.

Definition (task-adequacy)

The steady-state mismatch is small enough for the agent's actions to remain acceptable:

$$R^* < \|\delta_{\text{critical}}\| \tag{4.17}$$

where $\|\delta_{\text{critical}}\|$ is a domain-specific tolerance threshold — "how wrong can the model be before the agent's actions become harmful or ineffective?" This is set by the application domain, not derived by AAT.

Task adequacy is a separate condition from structural persistence. An agent can be structurally persistent ($R^* < R$) but task-inadequate ($R^* > \|\delta_{\text{critical}}\|$) — the machinery contains mismatch, but not tightly enough for the domain's needs. Conversely, when $\|\delta_{\text{critical}}\| < R$ (the domain's tolerance is stricter than the model class capacity), task adequacy is the binding constraint.

Derived (operational-persistence, conjunction of structural persistence + task adequacy)

Operational Persistence Condition.

The agent persists operationally when BOTH conditions hold. In the nonlinear case with $\|\delta_{\text{critical}}\| < R$, the binding condition is:

$$\alpha > \frac{\rho}{\|\delta_{\text{critical}}\|} \quad \text{(Model D)} \qquad \alpha > \frac{n\sigma_w^2}{2\|\delta_{\text{critical}}\|^2} \quad \text{(Model S)} \tag{4.18}$$

These are the same as the structural conditions with R replaced by $\|\delta_{\text{critical}}\|$, because when $\|\delta_{\text{critical}}\| < R$, task adequacy is stricter.

Linear operational forms: In the linear case ($\alpha = \mathcal{T}$, $R \to \infty$), structural persistence is trivially satisfied and the operational condition reduces to task adequacy alone:

$$\mathcal{T} > \frac{\rho}{\|\delta_{\text{critical}}\|} \quad \text{(Model D)} \qquad \mathcal{T} > \frac{n\sigma_w^2}{2\|\delta_{\text{critical}}\|^2} \quad \text{(Model S)} \tag{4.19}$$

These are the forms used throughout the theory as the operational persistence condition. They are exact for linear correction and useful proxies for mildly nonlinear correction (where $\alpha \approx \mathcal{T}$), but they overstate the persistence margin when the correction function saturates, because they omit the structural constraint ($\alpha > \rho/R$) that becomes binding when R is finite.

Per-dimension (Model S): $\eta_k > c \cdot \rho_k^2/\delta^2_{\text{critical},k}$ where c depends on the probability guarantee. See 16.5.

The relationship between α and $\mathcal{T}$. 4.3 shows that for agents with directional fidelity, $\alpha = \eta^* \cdot c_{\min}$ where $c_{\min}$ is the worst-case directional fidelity. For linear correction (Kalman, Beta-Bernoulli), $\alpha = \mathcal{T}$ exactly. For gradient descent on strongly convex losses, $\alpha = \eta \cdot \mu$ where μ is the strong convexity modulus — monotone in η (and hence in $\mathcal{T}$) for fixed loss landscape. For nonlinear correction tested in simulation (saturating, sigmoid, threshold), α remains monotone increasing in $\mathcal{T}$: for a saturating function with capacity R, $\alpha \approx \mathcal{T}/2$ (worst case at the capacity boundary); for sigmoid (tanh), $\alpha \approx 0.76 \cdot \mathcal{T}$. The qualitative conclusion — "faster adaptation improves persistence" — is structurally grounded for the important cases and empirically confirmed for all correction function classes tested.

Per-Dimension Extension. *[Derived (per-dimension-persistence, per-dimension Lyapunov under Model D / AR(1) stationary distribution under Model S — see 16.5)]*

For anisotropic systems (non-uniform ρ or $\mathcal{T}$ across dimensions), the scalar persistence condition is insufficient. Per-dimension:

$$\mathcal{T}_k > \frac{\rho_k}{\delta_{\text{critical},k}} \quad \text{for each dimension } k \tag{4.20}$$

The scalar condition overestimates by up to 72% in simulation. The weak dimension is the bottleneck (84% of total mismatch in simulation). See 16.5.

Robustness: The per-dimension condition matches discrete AR(1) prediction to 4 significant figures. The scalar overestimate is a consequence of Jensen's inequality applied to the norm.

Epistemic Status.

Structural persistence thresholds are *exact* under their stated assumptions: Model D gives $\alpha > \rho/R$ (Prop A.1, exact under GA-2, GA-3); Model S gives $\alpha > n\sigma_w^2/(2R^2)$ (Prop A.1S, exact under GA-2S, GA-3). Both are sufficiency results with class-level tightness — no fixed-agent only-if is claimed; agent-level necessity holds exactly in the radially tight (e.g. linear) case (`#deriv-sector-condition` Lemma A.1N). The threshold's *existence* is *robust qualitative* — any monotone correction function has a capacity limit; this holds across all correction functions tested.

Task adequacy ($R^* < \|\delta_{\text{critical}}\|$) is *exact as a definition* — given R^* (derived) and $\|\delta_{\text{critical}}\|$ (domain parameter), the comparison is well-defined. The substance lies in estimating $\|\delta_{\text{critical}}\|$ for specific domains, which is an operationalization question (`#detail-operationalization`), not a theory question.

The linear operational forms ($\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ for Model D; $\mathcal{T} > n\sigma_w^2/(2\|\delta_{\text{critical}}\|^2)$ for Model S) are *exact* for linear correction (where they express task adequacy alone, structural persistence being trivially satisfied) and *useful approximations* for mildly nonlinear correction (where $\alpha \approx \mathcal{T}$). For strongly nonlinear correction, the general α-forms are required and BOTH structural and task-adequacy conditions must be checked. Downstream segments that use the linear operational forms should be understood as expressing task adequacy, not structural stability.

The per-dimension extension is *exact conditional on the disturbance model* (16.5): the Model D per-dimension threshold ($\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$) is exact by the same Lyapunov argument applied per dimension (GA-2, GA-3), and the Model S per-dimension thresholds are exact under the AR(1) stationary distribution (GA-2S), with simulation confirming the AR(1) prediction to 4 significant figures.

Discussion.

Two conditions, not one. This segment now separates what was previously conflated. Structural persistence ($\alpha > \rho/R$) is the Lyapunov-derived result — it says the machinery *works*. Task adequacy ($R^* < \|\delta_{\text{critical}}\|$) is a domain-specific constraint — it says the machinery works *well enough*. Neither implies the other, and downstream segments should specify which they mean. Most adversarial-dynamics results (15.4, 15.2) depend on structural persistence. Most domain instantiations (TST, logogenic agent design) care about task adequacy. See Persistence in `LEXICON.md` and `README.md` for the full three-sense taxonomy (structural, operational, continuity).

Below structural threshold. When $\alpha \le \rho/R$, the Lyapunov certificate is lost: the machinery can no longer guarantee containment within the operating region. For the extremal member of the sector class, and for any corrector whose sector bound is radially tight (the linear case), escape is forced by adversarial disturbance — so the threshold is a genuine qualitative transition, not a gradual degradation. For a general corrector, condition failure means loss of guarantee rather than certified escape: the floor α can understate the correction actually available along reachable paths (`#deriv-sector-condition` Lemma A.1N).

Below task-adequacy threshold. When $R^* > \|\delta_{\text{critical}}\|$ but $R^* < R$, the system is structurally stable but performing unacceptably. Mismatch is bounded but too large for the domain. The remedy is different from structural failure: increase $\mathcal{T}$ (faster or better correction), decrease ρ (reduce environmental volatility), or relax $\|\delta_{\text{critical}}\|$ (accept more mismatch). Structural failure requires changing the correction architecture entirely (4.8).

δ_{critical} and R are domain parameters, not theory outputs. The theory derives the *existence* of persistence thresholds and their *form* (ratio of correction to disturbance). But the specific values are set by the application domain: δ_{critical} encodes "how wrong can the model be before the agent's actions become harmful or ineffective?" — this depends on the stakes, the action space, and the environment's forgiveness. R encodes "how large a mismatch can the correction function handle before it saturates or breaks down?" — this depends on the model class and the correction architecture. See `#detail-operationalization` for guidance on estimating these quantities in specific domains.

Channel independence and scalar tempo. The linear operational forms use scalar $\mathcal{T}$, which inherits the channel-independence assumption from 3.8: the additive formula overcounts when observation channels are correlated. In anisotropic systems the scalar condition also overestimates margins — up to 72% in simulation (see 3.8, scalar vs. vector tempo). Where precision matters, the per-dimension condition ($\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$) should be used instead — and under cross-dimensional correction (off-diagonal $\mathcal{T}$ in the coordinate basis of $D_\delta = \text{diag}(\delta^2_{\text{critical},k})$), the matrix-Loewner form $\Sigma_\infty \prec D_\delta$ (`#deriv-matrix-persistence-condition`) is canonical:

per-coordinate is unsafe in that regime (gives a false-pass on persistence), while matrix-Loewner reads persistence correctly off the worst direction whether or not it aligns with a coordinate axis.

Adaptive reserve. The quantity $\Delta\rho^* = \alpha R - \rho$ (Prop A.2) measures how much additional disturbance the agent can absorb before persistence fails. Positive reserve means the agent has margin; zero reserve means it is at the threshold.

Persistence has a cost, not just a threshold. The inequality above says mismatch is bounded; it does not say what rate of effort the agent expends to hold that bound. `#deriv-persistence-cost` establishes the complementary *information-rate* bound: under Model S with Gaussian-OU signal, the sustained Shannon information rate the agent must acquire from observations to maintain the ultimate bound satisfies $\dot{R} \geq n\alpha/2$ nats/time — a Landauer-analog floor that Kalman-Bucy saturates. The corollary is a channel-capacity prerequisite $C \geq \mathcal{T}/2$ that the threshold condition alone does not name. Two agents with identical persistence guarantees can face wildly different sustained demands because the cost scales linearly with α; the threshold alone cannot distinguish dormant from running-hot.

Connections. The persistence condition appears in multiple downstream contexts:

- **Adversarial dynamics** (15.4): Superlinear tempo advantage arises because persistence is a threshold — pushing an adversary below it causes qualitative collapse. *This connection is developed and validated in 15.4 and simulation variants A-D.*
- **Structural adaptation** (4.8): When model class fitness $\mathcal{F}(\mathcal{M}) < 1 - \varepsilon$, the effective α in the sector condition shrinks, eventually violating persistence. *This connection is developed in 4.8.*
- **Software maintainability** (`#der-code-quality-as-observation-infrastructure` — cross-component reference, see `02-tst-core/`): *[Discussion]* A codebase may become "unmaintainable" when the development team's adaptive tempo falls below the rate of complexity accumulation. The vicious cycle would then be the persistence condition being violated through the agent's own prior actions degrading future $\mathcal{T}$ via U_0. *This connection is structurally motivated but not yet formally derived within AAT. It requires formalizing "complexity accumulation rate" as an instance of ρ.*

Findings.

The Persistence Condition with Structural / Task-Adequacy Decomposition. **Brief:** An adaptive system persists when its correction speed beats the rate at which its world is changing, relative to how forgiving the world is. Below this threshold the system doesn't merely degrade — it loses bounded behavior, the way a balance held just barely beneath a tipping point is qualitatively different from one well above it. The same inequality, with different inputs, governs whether a Kalman filter tracks a moving target, whether a development team keeps a codebase maintainable, and whether an organization keeps up with strategic change. The threshold itself decomposes into two distinct conditions — *structural persistence* (the machinery's correction rate can outpace disturbance) and *task adequacy* (the resulting steady-state mismatch is small enough for what the agent is trying to do). Conflating the two leads to category errors in domain transfer.

Impact: This is the framework's central inequality and the load-bearing connection between control-theoretic Lyapunov stability analysis and the broader question of when any adaptive system — thermostat, software team, immune system, RL agent — can maintain coherent function under change. The two-condition decomposition is itself non-obvious and consequential: prior work that conflated "the machinery works" with "the machinery works well enough" produced category errors in domain transfer (a structurally persistent codebase team can be task-inadequate; the remedies differ). The complementary information-rate bound from `#deriv-persistence-cost` ($\dot{R} \geq n\alpha/2$) shows the threshold has a sustained-cost shadow: two agents with identical persistence guarantees can face wildly different sustained demands.

Novelty Claim: *Claim novelty* (at intuition-only search depth — see Search Log) on the two-condition decomposition (structural / task-adequacy): an AAT-internal structural carve that cleanly separates "the machinery works" from "the machinery works well enough," with no direct anticipation known; a targeted search of the bounded-rationality and adaptive-control literature is still owed. Alongside it, *claim synthesis* on Lyapunov stability theory, sector-bounded nonlinear correction, and adaptive-tempo information-rate accounting, applied uniformly across single-agent classes that range from Kalman filtering through saturating nonlinear correction through PID control — the Lyapunov machinery itself is standard; the synthesis is its use as the central inequality of an integrated agent theory.

Related Work:

- Khalil 2002, *Nonlinear Systems* (3rd ed.), Prentice Hall (published 2002, found pre-2026) — *formal antecedent* — chapters 4 and 9 supply the converse Lyapunov, ultimate boundedness, and sector-condition machinery the segment uses. Standard control-theoretic apparatus.
- Lyapunov 1892 / Khasminskii 2012 *Stochastic Stability of Differential Equations* — *formal antecedent* — the underlying Lyapunov stability tradition; Khasminskii's stopping-time localization underpins the Model S derivation.
- Rockafellar & Wets 1998, *Variational Analysis* — *formal antecedent* — supplies the monotone-operator machinery that underwrites the sector condition's strong-convexity equivalents.
- Wiener 1948 *Cybernetics*; Ashby 1956 *Introduction to Cybernetics*; Conant & Ashby 1970 — *conceptual precursor* — the cybernetic-feedback tradition that frames the "correction must outpace disturbance" intuition without supplying the quantitative inequality.

Search Log: - 2026-04 (*intuition-only* on broader prior-art): no targeted Undermind-grade search has been conducted on the persistence-condition-as-central-inequality positioning. Pre-search expectation: the Lyapunov-based stability machinery is standard; the AAT-distinctive content is the two-condition decomposition (structural vs task adequacy) and its uniform application across agent classes. A targeted search would query the bounded-rationality / control-theoretic decision-making literature (Ortega-Braun line; Genewein et al.) for prior decompositions of "stability vs adequacy" in adaptive-control settings, and the active-inference literature for the same distinction. - 2025 (*targeted*): Khalil 2002 / Khasminskii 2012 / Rockafellar-Wets 1998 confirmed as the formal antecedents for the sector-Lyapunov machinery; the segment cites them inline.

4.6 *Definition:* Mood

Mood is a slow global scalar that integrates the recent mismatch stream and feeds back to modulate the adaptation gain and tempo — a second-order adaptation parameter that requires no objective to define.

Formal Expression.

[Definition] Let a_t be a per-step summary of recent tracking surprise — how much better or worse the mismatch signal δ_t (`def-mismatch-signal`) is behaving than the agent's own short-horizon expectation. **Mood** is the leaky integral

$$m_t = (1-\lambda)\, m_{t-1} + \lambda\, a_t, \qquad 0 < \lambda \ll 1, \tag{4.21}$$

with time-constant $\tau \approx 1/\lambda$ slow relative to the per-step update of M_t.

[Definition] Mood acts only by modulating the adaptation parameters already in play — it adds no new fast dynamics:

$$K_t = K_0\, g(m_t), \qquad \mathcal{T}_t = \nu_t \cdot K_t, \tag{4.22}$$

where K is the update gain (`emp-update-gain`), $\mathcal{T} = \nu \cdot K$ the adaptive tempo (`def-adaptive-tempo`), and $g : \mathbb{R} \to [g_{\min}, g_{\max}]$ ($0 < g_{\min} \le 1 \le g_{\max}$) a monotone *bounded* modulation, the band chosen so the modulated gain stays inside the sector-validity region and, in discrete time, under the step-size ceiling of `der-gain-sector-bridge`: sustained adverse surprise raises gain and tempo (re-posture for a suspected regime shift); sustained easy tracking relaxes them — but only down to the floor $g_{\min}$. The band is not decoration; both ends carry persistence content (see Discussion).

Because τ is slow, the fast M_t dynamics see m_t as quasi-static: mood is a **second-order** adaptation parameter — adaptation acting on the gain of adaptation — sitting one rung above the endogenous single-step gain dynamics of `deriv-adaptive-gain-dynamics`.

Epistemic Status.

This segment is a **definition**, not a truth-claim; its content is the construct, not a derivation. The construct is adopted from affective science (Eldar et al. 2016, mood as outcome-momentum; Bennett et al. 2022, mood as a leaky integral of advantage) and specialized to the pre-goal adaptation setting, where the integrated quantity is tracking-surprise rather than reward. The framing is `discussion-grade`: the functional skeleton is well-supported qualitatively, but the specific functional forms of a_t and g are representative, not pinned.

The load-bearing reason this belongs in Part I — *before* objectives — is that nothing in the construct references O_t, Σ_t, or reward. Mood's signed reading as *valued* momentum (approach/avoid, hedonic sign) and its modulation of exploration and risk posture are genuine additions that arrive only once objectives exist; those are deferred to the actuation half (Part II) and depend on this definition, not the reverse.

The optimal mood time-constant — optimal *for mood's regime-estimator role*; the transfer to the modulator role (m_t setting K_t) is conjectured equal and remains an open item — is derived in `#der-mood-timescale`: minimizing steady-state regime-tracking error gives $\tau^* = \sqrt{\tau_{\text{env}}\, r/(2\sigma_\theta^2)}$ — growing as the *square root* of the environment's autocorrelation time (not matching it), with a robust-qualitative core (interior optimum; too-fast mood chases noise, too-slow mood lags the regime). That companion carries the `conditional` result and its premises; this definition remains `discussion-grade` as the construct.

Discussion.

Mood is the slow outer loop on the persistence condition (`result-persistence-condition`). In the bathtub reading of persistence — faucet as the rate of change in reality, drain as the learning rate, overflow when the faucet outpaces the drain — mood is a slow controller *on the drain*: it widens the drain when recent overflow-risk has run high and relaxes it when tracking has been easy. The framework otherwise has fast per-channel updating and (in Part II) goal dynamics, but no slow global scalar coordinating posture across channels; mood is that missing object.

Persistence compatibility (the MG discharge). Mood's modulation composes with the persistence machinery as an instance of `deriv-adaptive-gain-dynamics`'s adaptive-gain result, and the four conditions instantiate cleanly. **(MG-1)** is the $[g_{\min}, g_{\max}]$ band in the definition: the floor keeps the effective sector rate uniformly above the persistence threshold ($\alpha_t \geq \alpha_{\min} > \rho/R$; the primary-channel Lyapunov argument is pointwise in time, so no slow-variation hypothesis is needed on this leg). The floor is what excludes **mood-induced complacency** — the failure mode an unbounded downward relaxation would permit, where sustained easy tracking drives correction power toward zero just in time for the next regime shift; the ceiling is the mirror condition, keeping adverse-surprise gain-raising under the discrete-time contraction ceiling. **(MG-2)** is trivial for mood: the channel is a *linear* leaky integrator, sector constant exactly λ. **(MG-3)** is the quantitative content of "quasi-static": $\lambda \ll \underline{\alpha}$ — mood adapts slower than the primary state contracts. **(MG-4)** requires the surprise summary to have δ-bounded second moment, $\mathbb{E}[a_t^2 \mid \delta] \leq \sigma_0^2 + c_a\|\delta\|^2$ — a condition stated here because a_t's functional form is deliberately unpinned; any concrete choice should be checked against it. Under the four, the composed augmented-state persistence result applies as-is and mood sits in sub-scope α_2 — a notably clean instance (the meta-channel is linear).

Mood couples the channels globally, which appears to threaten the directed-separation structure (`der-directed-separation`). It does not, and the reason is now quantitative rather than asserted: the separation condition is (MG-3) above — m_t moves slowly enough ($\lambda \ll \underline{\alpha}$) that the fast dynamics treat it as a constant, so separation holds at the timescale that matters even though a coupling exists in absolute terms. Mood is thus a clean worked example of bounded, separation-preserving coupling — adjacent to the Class 2 (Partial) material rather than a violation of Class 1.

The applied and normative reading of mood — set-point, recovery, what propagates across a context boundary, and the ethics of mood-control for persistent agents — is deliberately kept out of this segment; it lives in `msc/mood-layer-sovereignty-carve-2026-06-17.md`. Here mood is only a formal modulator of adaptation.

4.7 *Derived:* Optimal Mood Time-Constant

The mood time-constant that minimizes steady-state regime-tracking error grows as the square root of the environment's autocorrelation timescale — so both too-fast and too-slow mood are strictly suboptimal, and the optimum is environment-matched but sublinearly.

Formal Expression.

Mood (4.6) is a leaky integral $m_t = (1-\lambda)m_{t-1} + \lambda\, a_t$ with time-constant $\tau = 1/\lambda$, whose functional role is to estimate a slowly-varying regime from the noisy surprise stream. Model that role explicitly:

[Assumption] The regime is a stationary AR(1) latent $\theta_t = \phi\,\theta_{t-1} + \xi_t$, with $\phi = e^{-1/\tau_{\text{env}}}$, $\xi_t \sim \mathcal{N}(0, \sigma_\xi^2)$, stationary variance $\sigma_\theta^2 = \sigma_\xi^2/(1-\phi^2)$, and autocorrelation $\operatorname{corr}(\theta_t, \theta_{t-k}) = e^{-k/\tau_{\text{env}}}$. The surprise summary is $a_t = \theta_t + \eta_t$, $\eta_t \sim \mathcal{N}(0, r)$. Mood tracks θ_t; the loss is steady-state MSE $J(\lambda) = \mathbb{E}[(m_t - \theta_t)^2]$.

[Derived (steady-state MSE)] In the slowly-drifting regime $\tau_{\text{env}} \gg 2\sigma_\theta^2/r$, where the AR(1) increment variance is $q \equiv \operatorname{Var}(\Delta\theta_t) = 2\sigma_\theta^2(1-\phi) \approx 2\sigma_\theta^2/\tau_{\text{env}}$ and the increment is uncorrelated with the prior error to leading order in $1/\tau_{\text{env}}$,

$$J(\lambda) = \frac{\lambda^2 r + (1-\lambda)^2 q}{\lambda(2-\lambda)}. \tag{4.23}$$

The two terms are a bias–variance split: $\lambda^2 r$ is the noise-tracking cost, $(1-\lambda)^2 q$ the lag cost.

[Derived (optimum)] J has an interior minimizer $\lambda^*_{\text{exact}} = \dfrac{\sqrt{q(q+4r)} - q}{2r}$, with leading-order form

$$\lambda^* = \sqrt{q/r}, \qquad \tau^* = \sqrt{r/q}. \tag{4.24}$$

[Derived (scaling law)] Substituting $q \approx 2\sigma_\theta^2/\tau_{\text{env}}$,

$$\tau^* = \sqrt{\tau_{\text{env}} \cdot \frac{r}{2\sigma_\theta^2}} = \sqrt{\tau_{\text{env}}} \cdot \sqrt{\frac{r}{2\sigma_\theta^2}}. \tag{4.25}$$

The optimal mood time-constant grows as the **square root** of the environmental autocorrelation time, scaled by the surprise noise-to-signal ratio r/σ_θ^2. Self-consistency: $\tau^*/\tau_{\text{env}} \to 0$, so $\tau^* \ll \tau_{\text{env}}$ exactly where the leading-order step was taken.

Epistemic Status.

`Conditional` on the named premises: stationary AR(1) regime, additive Gaussian surprise noise, the slowly-drifting condition $\tau_{\text{env}} \gg 2\sigma_\theta^2/r$ (equivalently $\lambda^* \ll 1$ and mood memory $\tau^* \gg 1$ step — one condition validating both the small-λ reduction and the leaky-integrator regime), and quadratic loss. Under these, $J(\lambda)$ is exact and the scaling law follows; the boxed $\lambda^* = \sqrt{q/r}$ is the leading-order term of the exact minimizer given above, not itself exact. The optimum is derived *for mood's regime-estimator role*; its transfer to the modulator role (m_t setting the fast gain K_t — `#def-mood`'s actual use) is conjectured equal under certainty-equivalent calibration and remains the open two-layer composition item named below.

The derivation was independently refute-verified (2026-06-17): every algebraic step re-derived clean; the one approximation (dropping the mean-reversion cross-term) confirmed $O(\tau_{\text{env}}^{-1/2})$ relative and self-consistently negligible.

Three claims at distinct tiers travel out of this: the **exact** steady-state MSE for the model; the **conditional** $\sqrt{\tau_{\text{env}}}$ scaling law; and a **robust-qualitative** core — the optimum is interior and monotone-increasing in τ_{env} — which is the generic bias–variance shape of any leaky integrator tracking a drifting source and does not depend on the Gaussian/AR(1) specifics. What would strengthen this: the exact closed form over the full AR(1) (not its random-walk limit); the two-layer composition where mood's estimate sets the fast gain K rather than being the estimator directly. What would soften it: an estimator/loss outside the leaky-integrator–quadratic family (e.g. the Kalman/Wiener filter can track τ_{env} more directly in some SNR regimes) — but `#def-mood` specifies the leaky integrator, so the $\sqrt{\cdot}$ law is the in-family answer.

Discussion.

The result corrects the naive expectation that mood inertia should simply "match" how fast the world changes. It is environment-matched, but **sublinearly**: a fourfold slower environment warrants only a twofold longer mood memory. The load-bearing qualitative content is that the optimum is *interior* — too-fast mood chases noise (the variance term dominates), too-slow mood lags genuine regime shifts (the bias term dominates) — so a mood with no inertia and a mood that never recovers are both strictly suboptimal, for the same reason at opposite ends.

This gives the clinical literature on **emotional inertia** a normative reading. The affect AR(1) coefficient is $1 - \lambda$; inertia being maladaptive when high (Kuppens, Allen & Sheeber 2010, who formalize inertia as exactly this autocorrelation) corresponds to operating at $\lambda < \lambda^*$, i.e. $\tau \gg \tau^*(\text{environment})$ — the over-smoothing branch. The pathology the result names is not slowness as such but **slowness mismatched to a faster-changing-than-assumed environment**: the same τ is well-tuned in a slow world and pathological in a fast one. (The corroboration is directional, from an independent empirical literature; it is not load-bearing for the derivation.)

4.8 *Result:* Structural Adaptation Necessity

When model class fitness is insufficient — when no model in the agent's current representational class can adequately represent reality — *no amount of parametric adaptation can close the mismatch floor*. The agent must change *what kind of model* it is using, not just tune the current one. The derivation: if class fitness falls below $1 - \varepsilon$, then by definition no model in the class achieves sufficiency above $1 - \varepsilon$; the history contains predictive information the best-in-class cannot capture; that uncaptured information manifests as systematic mismatch (structured residuals containing signal, not just noise); from the mismatch decomposition (3.5) the estimation term has a positive lower bound that cannot be reduced by any model in the class; therefore reducing mismatch below that class floor requires changing the class itself.

A *diagnostic corollary* follows immediately: persistent *systematic* residual structure — surviving parametric convergence, after excluding channel-noise, disturbance, nonstationarity, and gain-miscalibration causes — is diagnostic of model-class inadequacy. Structure in the residuals is the evidence; the mismatch *level* alone is not, since the channel and state-uncertainty floors (3.5) keep absolute mismatch positive for any class. This is the operational test for needing structural rather than parametric change.

The framework is honest about an alignment assumption used in the derivation: the lost predictive information must affect the *one-step conditional mean*, not just higher moments. Without that assumption, the conclusion holds in terms of proper-scoring regret rather than one-step mismatch magnitude. Qualitatively, parametric adaptation cannot compensate for class inadequacy either way.

The connection to persistence is sharp: when the class is inadequate, the effective sector parameter shrinks because the correction function cannot point inward strongly enough — the class lacks capacity to represent the correct direction. This is a *failure of structural persistence* (not a failure of operational persistence or continuity), and the remedy is therefore a change of model class, not faster cycling or identity preservation.

The framework names the *symptoms* of structural inadequacy operationally: persistent mismatch despite extended updating, gain collapse without performance (the model appears confident but predictions remain inaccurate — "confidently wrong"), and systematic mismatch patterns (residuals showing correlations, trends, or periodicities the class cannot represent). The opposite failure mode — **structural overfitting** — also receives treatment: a class that is *too* expressive can memorize irreducible noise, with low training mismatch but high generalization mismatch. The information bottleneck (2.2) is the diagnostic in both directions: structural adaptation is bidirectional, expanding when too constrained and *compressing* when too expressive.

Four mechanisms of structural change are named: **decomposition and recombination** (Boyd's "Destruction and Creation"; Kuhn's paradigm shifts; Popper's conjecture and refutation); **expansion** (Bayesian nonparametrics, growing neural architectures, organizational expansion); **compression** (regularization, Occam's razor, organizational streamlining); and **grafting** — incorporating external representational structure (transfer learning, acquiring a company, consulting an expert). Query actions (3.7) are identified as a primary conduit for grafting.

One striking mechanism is *neutral structural variation* drawn from Miller's coevolving-automata work: in multi-agent settings, structural adaptation can proceed without any individual agent deliberately restructuring. A five-phase "extreme transition motif" runs through stable epoch → environmentally-neutral variant appearing (structurally different but behaviorally identical under current conditions) → drifting to nontrivial proportion through stochastic reproduction → the variant's latent structural differences create a niche an opposing mutant exploits, triggering a self-reinforcing cascade → both populations rapidly transition to a new regime. The restructuring is radical in *effect* but incremental in *cause*, with neutral drift providing the bridge.

Structural adaptation is expensive — knowledge loss, temporary performance drop, search cost, coordination cost in multi-agent systems — and the framework explicitly connects it to the deliberation-cost analysis (4.1) as the limiting case where the pause duration $\Delta\tau$ is massive and the mismatch debt during transition is correspondingly enormous. This produces rational conservatism: prefer parametric adaptation when it suffices; resort to structural change only when the evidence is strong. Premature structural change wastes accumulated knowledge; delayed structural change accumulates mismatch.

Formal Expression.

Derived (structural-adaptation-necessity)

If the model class fitness $\mathcal{F}(\mathcal{M}) < 1 - \varepsilon$ for some $\varepsilon > 0$, then no parametric adaptation within $\mathcal{M}$ can reduce the expected mismatch below a floor determined by ε (under the alignment assumption — see Epistemic Status). Without the alignment assumption, the result holds for irreducible proper-scoring regret rather than one-step mean mismatch. The qualitative conclusion is the same either way: parametric adaptation cannot compensate for model-class inadequacy.

Derivation.

1. By definition, $S(M^*) = \mathcal{F}(\mathcal{M}) < 1 - \varepsilon$ where $M^* = \arg\sup_{M \in \mathcal{M}} S(M)$.
2. Therefore $I(\mathcal{C}_t; o_{t+1:\infty} \mid M^*, a_{t:\infty}) > 0$: the history contains predictive information that M^* does not capture.
3. This uncaptured information manifests as *systematic* mismatch — structured residuals δ_t containing signal, not merely noise.
4. From 3.5, the estimation term has a positive lower bound that cannot be reduced by any $M \in \mathcal{M}$ — a class floor sitting above the state-uncertainty and channel floors.
5. The update rule (3.6) adjusts M_t within $\mathcal{M}$, but M^* is already (approximately) reached. Further updates oscillate without net improvement.
6. Therefore: reducing mismatch below the floor requires changing $\mathcal{M}$ — structural adaptation. □

Corollary. Persistent *systematic* residual structure (after parametric convergence and noise accounting) is *diagnostic* of model class inadequacy — patterned residuals are evidence that $\mathcal{F}(\mathcal{M})$ is insufficient. A high but *white* residual floor is not: that indicates the channel and state-uncertainty floors (3.5), which no class change removes.

Epistemic Status.

Conditional. The step from "lost predictive information" (step 2) to "systematic one-step mismatch" (step 3) requires an alignment assumption: that the lost predictive information affects the one-step conditional mean, not just higher moments. 3.5 explicitly flags this: insufficiency implies positive model error under the alignment assumption, or positive proper-scoring regret without it. As written, the result is conditional on this alignment assumption. Without it, the conclusion should be stated in terms of proper-scoring regret (the best model in $\mathcal{M}$ has irreducible regret relative to the optimal predictor) rather than one-step mismatch magnitude. The qualitative conclusion — parametric adaptation cannot compensate for model-class inadequacy — holds either way; the quantitative mechanism differs.

Discussion.

Structural adaptation as structural persistence failure. When the model class is inadequate, the effective α in the sector condition shrinks — the correction function cannot point inward strongly enough because the model class lacks the capacity to represent the correct direction. This is a failure of *structural persistence* (see Persistence in `LEXICON.md`): the machinery's capacity to outpace disturbance degrades not because disturbance increased or tempo decreased, but because the correction function itself has become less effective. The remedy is not faster cycling (operational) or identity preservation (continuity) but a change of model class.

Observable symptoms of model class inadequacy. When $\mathcal{F}(\mathcal{M})$ is low:

1. **Persistent mismatch despite convergence**: $\|\delta_t\|$ remains large despite extended updating — the model has converged within $\mathcal{M}$ but the best achievable model is still poor. (On its own this is suggestive, not diagnostic: the channel and state-uncertainty floors keep absolute mismatch positive for any class; symptom 3 is what discriminates.)
2. **Gain collapse without performance**: η^* has decreased (model appears confident) but predictions remain inaccurate — the model is confidently wrong, having fitted to structure in $\mathcal{M}$ that doesn't match reality.
3. **Systematic mismatch patterns**: δ_t shows structure (correlations, trends, periodicities) that the model class cannot represent — the residuals contain signal that $\mathcal{M}$ lacks the capacity to absorb.

Structural overfitting: the opposite failure mode. $\mathcal{M}$ can also be *too expressive*, causing the model to memorize irreducible noise. Symptoms: low training mismatch but high generalization mismatch; model complexity growing without predictive gain; $\eta^* \to 0$ (confident) but confidence is spurious. The information bottleneck (2.2) provides the diagnostic: when marginal increases in model complexity yield no marginal predictive power, the model is past the optimal point on the rate-distortion curve. Structural adaptation in this case means *compression* — moving to a simpler $\mathcal{M}'$. Structural adaptation is bidirectional: expansion when too constrained (this proposition), compression when too expressive.

Mechanisms of structural change. Structural adaptation can proceed by:

- **Decomposition and recombination**: Tearing apart existing structure and synthesizing new configurations from the pieces. Boyd's "Destruction and Creation" insight; Kuhn's paradigm shifts; Popper's conjecture and refutation.
- **Expansion**: Adding new representational capacity without destroying existing structure. Bayesian nonparametrics, growing neural architectures, organizational expansion.
- **Compression**: Removing unnecessary structure while preserving the predictive core. Regularization, Occam's razor, organizational streamlining.
- **Grafting**: Incorporating external structure. Transfer learning, acquiring a company, consulting an expert. Query actions (3.7) are a primary conduit for grafting.

The severity of structural change needed depends on *how far* the current model class is from adequacy. Minor regime changes may require only expansion or grafting; fundamental shifts where $\mathcal{M}$'s assumptions are violated may demand full decomposition.

Neutral variation as a mechanism for structural change. In multi-agent settings, structural adaptation can proceed without any individual agent deliberately restructuring. Miller (2022, *Ex Machina*) identifies a five-phase "extreme transition motif" in coevolving automata: (1) stable epoch, (2) an environmentally neutral variant — structurally different but behaviorally identical under current conditions — appears, (3) the variant drifts to nontrivial proportion through stochastic reproduction, (4) the variant's latent structural differences create a niche that a new mutant in the opposing population exploits, triggering a self-reinforcing cascade, (5) both populations rapidly transition to a new regime and consolidate. This mechanism bridges the gap between "many incremental changes" and "radical restructuring" — the restructuring is radical in its effect but incremental in its causes, with neutral drift providing the bridge. The concept of *latent structural diversity* — variation in agent architectures that is invisible to current performance but consequential under regime change — is a composition-level property that Part III's dynamics framework should formalize.

The cost of structural change. Structural adaptation is expensive: knowledge loss (parameters learned within $\mathcal{M}$ may not transfer), temporary performance drop (new model starts uncertain), search cost (finding good $\mathcal{M}'$), coordination cost (in multi-agent systems). This creates rational conservatism — prefer parametric adaptation when it suffices, resort to structural change only when the evidence is strong. Premature structural change wastes accumulated knowledge; delayed structural change accumulates mismatch. The connection to 4.1: structural adaptation is deliberation with a *massive* $\Delta\tau$, and the mismatch debt during the transition is correspondingly enormous.

Temporal nesting of adaptation. Parametric and structural adaptation operate at different timescales: $\nu_{\text{parametric}} \gg \nu_{\text{structural}}$. More generally, an agent may have multiple adaptive processes at different rates, with the convergence constraint that faster processes must approximately converge before slower ones act on their output. If deeper change occurs before shallower adaptation has converged, the deeper change is based on transients rather than settled dynamics.

Domain instantiations:

Table 4.3

Domain	*Parametric adaptation*	*Structural adaptation*
Kalman filter	State estimate update	Switching observation/dynamics models
RL	Weight/Q-value update	Architecture search
PID	— (gains fixed)	Switching to MPC
Bayesian	Posterior update	Model selection, nonparametrics
Boyd	Orientation updating	Destruction and creation of mental models
Science	Normal science (Kuhn)	Paradigm shift
Evolution	Allele frequency change	Speciation, new body plans
Organization	Process optimization	Strategic pivot, restructuring
Software	Incremental refactoring	Architecture migration
Coevolving automata (Miller 2022)	Edge reweighting within fixed FSA structure	Mutation altering state output or transition; neutral mutations accumulating until niche creation triggers cascading restructuring

4.9 *Derived:* Temporal Nesting

Adaptive processes stratify naturally by timescale, with each level operating on the *quasi-steady-state output* of the level below. The formal statement is a chain of inequalities — each level's event rate must be much smaller than the level below it — derived from standard singular-perturbation reasoning. The structural consequence: if a slower process acts before the faster process beneath it has converged, the slower process is adjusting based on transient behavior rather than settled dynamics, and the system oscillates.

The hierarchy the framework identifies (illustrative; real systems may have additional intermediate levels): **reactive response** (action given current model, fastest) → **parametric update** (model parameters within the class) → **consolidation** (offline redistribution of information within the model's sub-state factorization toward an information-bottleneck optimum) → **structural adaptation** (model class) → **architectural change** (the agent's fundamental structure, slowest). What matters is not the number of distinguishable levels but the structural relationship between adjacent ones: faster must converge before slower acts.

A direct consequence: the rational *conservatism* toward structural change identified by 4.8 is a derived consequence of temporal nesting. Structural adaptation operates at a much slower timescale than parametric adaptation, so the mismatch cost of the "pause" (disturbance times pause duration) is enormous. The agent rationally resists structural change until the parametric mismatch floor exceeds this cost. The same logic gives the formal deliberation tradeoff (4.1) its dynamical grounding.

The framework names symptoms of nesting *violation*: oscillation, instability, degraded performance. In organizations, micromanagement is strategic decisions made at operational tempo. In reinforcement learning, policy updates before the value function converges produce policy oscillation. In biology, premature developmental transitions are the same failure mode at a different timescale.

The composite-stability claim is now derived: under per-level sector conditions and bounded interconnection, if each level is stable given the levels above it, the composite N-level system is stable, and this segment's qualitative convergence constraint becomes a closed-form threshold on the admissible timescale ratio ($\epsilon < \epsilon_{\max} = \Delta\rho^*/(L_h v^{\max})$ — the faster level's adaptive reserve over the slower level's target-drag rate). See `#der-multi-timescale-stability` for the theorem, its premises, and the honest scope boundary (premise-conditional on deeper-level dynamics admitting a quasi-steady-state manifold with a sector condition; discrete/jump structural adaptation remains open).

The result is rooted in classical singular-perturbation theory (Tikhonov 1952; Khalil 2002 textbook treatment) — the framework adopts the machinery directly. The threshold form of "adequate separation" is derived within AAT (`#der-multi-timescale-stability`'s $\epsilon_{\max}$); the constants entering it (sector parameters, manifold Lipschitz bounds, velocity bounds) are domain-dependent.

Formal Expression.

Derived (temporal-nesting)

$$\nu_{\text{level } n+1} \ll \nu_{\text{level } n} \tag{4.26}$$

for each adjacent pair of adaptive timescales. If a slower process acts before the faster process beneath it has converged, the system oscillates — the slower process adjusts based on transient behavior rather than settled dynamics.

Table 4.4

Timescale	*Process*	*What changes*
Fastest	Reactive response	Action given current model
Fast	Parametric update (online)	Model parameters within $\mathcal{M}$
Intermediate	Consolidation (offline, cf. 9.11)	Redistribution of information within M_t's sub-state factorization toward IB-optimum
Slow	Structural adaptation	Model class $\mathcal{M}$
Slowest	Architectural change	The agent's fundamental structure

This table is illustrative — real systems may have additional intermediate levels. The number of distinguishable timescales is not fixed; what matters is the structural relationship between adjacent levels.

Epistemic Status.

Robust qualitative — this is standard singular perturbation reasoning (Tikhonov 1952, "Systems of differential equations containing a small parameter multiplying the derivative," *Matematicheskii Sbornik* 31(3):575–586; modern textbook exposition in Khalil 2002, *Nonlinear Systems* (3rd ed.), Prentice Hall, Chapter 11). The convergence constraint follows from the structure of multi-timescale updating. The threshold form of "adequate separation" is now derived within AAT under named premises (`#der-multi-timescale-stability`'s $\epsilon_{\max}$); the constants entering it are domain-dependent.

Discussion.

Domain instantiations of temporal nesting:

- **PID control**: D-term (fastest, high-frequency response) → P-term (current error) → I-term (slowest, accumulated bias)
- **RL**: Action selection → value function update → policy improvement → architecture change
- **Biology**: Reflexes (ms) → perceptual learning (minutes) → skill acquisition (months) → developmental change (years) → evolutionary adaptation (generations)
- **Organizations**: Operational decisions (hours) → tactical adjustments (weeks) → strategic revision (quarters) → restructuring (years)
- **Boyd**: Tactical OODA (seconds–minutes) → operational (hours–days) → strategic (weeks–months) → grand strategic (years)

Structural adaptation as slow-timescale dynamics. The conservatism toward structural change (4.8) is a derived consequence of temporal nesting: structural adaptation operates at a much slower timescale than parametric, so the mismatch cost of the "pause" ($\rho \cdot \Delta\tau$) is enormous. The agent rationally resists until the parametric mismatch floor exceeds this cost. See also 4.1 for the formal tradeoff.

Violation symptoms. When nesting is violated (a slower process acts before the faster one converges): oscillation, instability, degraded performance. In organizations: micromanagement (strategic decisions at operational tempo). In RL: policy updates before value function converges (policy oscillation). In biology: premature developmental transitions.

Multi-timescale stability (derived). The composite stability result is derived in `#der-multi-timescale-stability` by stacking the sector-persistence template: if each level satisfies a sector condition given the levels above it and the interconnections are bounded, the composite N-level system is stable, with this segment's $\nu_{n+1} \ll \nu_n$ made quantitative ($\epsilon < \epsilon_{\max}$, closed-form) and the cost of premature slower-level action priced as an explicit reserve penalty. The result is premise-conditional: deeper levels whose dynamics are discrete/jump processes sit outside the current premises — that remaining gap is named there.

4.10 *Discussion:* Scope: Agent Identity as Singular Causal Trajectory

The scope coda of Part I. AAT applies to agents instantiated on **singular causal trajectories**. Identity within AAT is grounded *not* in the model state M_t (which can be copied) but in the unique causal trajectory $\mathcal{C}_t$ (which cannot). The scope is presented as a structural commitment with three formal consequences.

First, model sufficiency is *trajectory-indexed*. Sufficiency measures against *this* agent's chronica, not against any hypothetical parallel copy's chronica. Two agents that branched from the same model state but received different events henceforth have different sufficiency values, each measured against its own divergent history. **Second**, model merging is *lossy by construction*. Reconciling the models of two agents that share a prefix of their trajectory but have diverged requires choosing which causal history to privilege, and no generally optimal merge exists — a structural constraint of the scope, not a defect of any particular merge algorithm. **Third**, singularity grounds *whose* effect a loop datum is — it does not by itself manufacture the interventional character. When the agent acts and observes, the observation is the response to *its* intervention on *its* single trajectory; replaying a saved model state against a different event stream is an intervention on a *different* trajectory that happens to share a prefix. The do-operator's semantics is population-level and forkability-agnostic (7.3), so the trajectory's singularity is not what makes the loop data interventional. What it does is forbid *averaging interventional responses across forked copies as though they were one agent's effects*, and ground which trajectory's effect a given loop datum reports. This is the identity-grounding half of the framework's loop-as-Level-2 claim (in 7.3); the interventional character itself comes from the action being the agent's own causally-efficacious move, not from any architectural property of the agent and not from non-forkability.

The framework also names a natural companion axiom: **parameterization invariance (PI)** — that AAT's theorems should not depend on arbitrary choice of coordinates on M_t. Since the trajectory itself is coordinate-free while any parameterization of M_t is a modeling convention, (PI) is consistent with (though not directly forced by) the three consequences above. When (PI) is adopted and combined with Čencov's uniqueness theorem, the Fisher information metric is *uniquely forced* on statistical-manifold sub-cases of M_t — converting Fisher-metric-dependent derivations from theorem-imported to AAT-internally-forced, and adding a fourth primary instance to the uniqueness-theorem-forced-coordinate pattern catalogued in 5.5. (PI) is a genuine axiomatic choice with a cost (one more invariance commitment) and a benefit (several Fisher-metric results become derivable rather than imported).

What the scope excludes: agents conceived as *types* (equivalence classes — "the GPT-4 model") rather than *tokens* (this particular session with this state on this trajectory); aggregated claims across tokens of the same type require additional machinery. Also excluded without additional treatment: clone-problem scenarios (multiple copies of an agent identical until divergence — each copy becomes its own AAT agent at the moment it acquires a distinct event), and operations like restoration-from-backup that attempt to transplant a model state across trajectories. The framework is honest about the philosophical scope: whether the mathematical structure grounds something that deserves to be called "identity" or "continuity of experience" is beyond AAT. What this segment specifies is **continuity persistence** — distinct from the structural persistence of the central inequality and from operational persistence — and is the bridge to the language-constituted (logogenic) and morally-weighted (logozoetic) agent volumes that follow.

Formal Expression.

Scope (scope-agent-identity, from chronica + model-sufficiency)

Scope commitment. AAT's formal apparatus presumes each agent is instantiated on a **singular, non-forkable causal trajectory** $\mathcal{C}_t$ (1.4). Sufficiency of the model state M_t (2.3) is defined *relative to* this trajectory — not relative to a model-state equivalence class. Duplicating M_t and exposing the copies to different future events produces two agents with *divergent* causal histories, each of which is a sufficient statistic only for *its own* trajectory.

Three consequences of the scope commitment:

1. **Sufficiency is trajectory-indexed.** $S(M_t)$ (2.3) measures against *this* agent's $\mathcal{C}_t$; not against a hypothetical parallel copy's $\mathcal{C}_t^{(2)}$.
2. **Model merging is lossy by construction.** Reconciling the models of two agents that share a prefix of their trajectory but have diverged requires choosing which causal history to privilege; no generally optimal merge exists. This is a structural constraint of the scope, not a defect of any particular merge algorithm.
3. **Singularity grounds *whose* effect a loop datum is — it does not by itself manufacture the interventional character.** When the agent acts and observes, the observation is the response to *its* intervention on *its* single trajectory. Replaying a saved M_t against a different event stream is an intervention on a *different* causal trajectory $\mathcal{C}_t^{(2)}$ that happens to share a prefix, each fork carrying its own $P(\mathbf{U} \mid \mathcal{C}_t)$. The do-operator's semantics is population-level and forkability-agnostic (7.3), so the interventional character of loop data does *not* rest on non-forkability; what the singular-trajectory commitment supplies is the *identity* grounding — it forbids averaging interventional responses across forked copies as though they were one agent's effects, and fixes which trajectory a given loop datum is an effect of. The interventional character itself comes from the action being the agent's own causally-efficacious move (under the positivity / sequential-ignorability / known-mechanism conditions of 7.3), not from any architectural property of the agent.

Natural extension: parameterization-invariance (PI). The scope commitment motivates a companion axiom. AAT's predictions concern a singular causal trajectory $\mathcal{C}_t$; the trajectory itself is coordinate-free, while any parameterization of M_t's internal state space is a modeling convention. Requiring AAT's theorems to be invariant under change of parameterization — *(PI): the theory's conclusions do not depend on arbitrary choice of coordinates on* M_t — is a natural axiomatic commitment consistent with (but not directly forced by) the three consequences above. When (PI) is adopted and combined with Čencov's 1982 uniqueness theorem (*Statistical Decision Rules and Optimal Inference*, AMS), the Fisher information metric is uniquely forced on statistical-manifold sub-cases of M_t — which converts Fisher-metric-dependent derivations in 4.3 from theorem-imported to AAT-internally-forced, and adds a fourth primary instance to the uniqueness-theorem-forced-coordinate pattern named in 5.5 (see that segment's four-instance table for structural positioning). (PI) is a genuine axiomatic choice — its cost is that AAT carries an additional invariance commitment at the scope layer; its benefit is that several Fisher-metric results throughout AAT become derivable rather than imported. The commitment is structurally analogous to the chain-rule-additivity and evidential-additivity axioms that ground the divergence-layer and update-layer Cauchy-FE theorems: each is a natural-from-adjacent-AAT-commitment axiom that a uniqueness theorem then operates on.

What the scope excludes (or requires additional machinery for):

- Agents conceived as type/equivalence-class entities (e.g., "the GPT-4 model") rather than token/trajectory entities (e.g., "this particular session with state M_t on trajectory $\mathcal{C}_t$"). AAT's formal results apply to tokens, not types. Aggregated claims across tokens of the same type require additional machinery (e.g., population-level dynamics; see Part III gaps on latent structural diversity).
- "Clone problem" scenarios where multiple copies of an agent are formally the same until divergence — each copy becomes its own AAT agent at the moment it acquires a distinct event (Discussion below).
- Formal treatment of reincarnation, restoration from backup, or other operations that attempt to transplant M_t across trajectories. AAT's sufficiency machinery does not apply across trajectory discontinuities; such operations are out-of-scope events whose epistemic consequences require separate treatment.

Epistemic Status.

Robust qualitative. The scope commitment is structurally clear once stated and is load-bearing for at least one downstream formal result: the *identity* grounding of the loop's interventional access in 7.3 — whose effect a loop datum reports, and the prohibition on copy-averaging — rests on the singular-trajectory scope (the interventional *character* itself is population-level and forkability-agnostic, so it does not). The three consequences above follow directly from the scope commitment combined with 1.4's non-forkability and 2.3's trajectory-indexed definition.

Max attainable: *robust qualitative*. Scope statements are not theorems; they specify what kind of object the theory applies to. Further formalization (e.g., category-theoretic treatment of "agent" as a functor from event-streams to model-states, or explicit type/token distinction in logogenic-agent work) could reach *exact* on the formal structure but would not change the scope content.

Whether the mathematical structure grounds something that deserves to be called "identity" or "continuity of experience" is beyond AAT's scope. The mathematical structure is clear: the feedback loop produces a singular, non-forkable causal trajectory, and model adequacy is defined relative to that trajectory. What this segment specifies is *continuity persistence* in the sense of `LEXICON.md` — whether the agent maintains a coherent identity and trajectory through time, as distinct from the structural persistence of 4.5 (can the machinery outpace disturbance?) and operational persistence (is the agent currently within its viable region?).

Discussion.

The clone problem, precisely stated. Consider copying an LLM's weights (a concrete M_t) exactly. At the moment of duplication, both copies are identical — same model state, same causal history $\mathcal{C}_t$. But the *very next* event — a different user's message, a different observation — creates two divergent, irreversible causal trajectories $\mathcal{C}^{(1)}_{t+1}$ and $\mathcal{C}^{(2)}_{t+1}$. Their Level 2 and Level 3 capacities (7.1) now reference different causal pasts. Their sufficiency $S(M_{t+1})$ is measured against different histories. Neither copy's future model state is a sufficient statistic for the other's trajectory.

Within AAT's formalism, identity is not the model state M_t (which can be copied) but the *singular causal trajectory* $\mathcal{C}_t$ (which cannot). A copy shares a *prefix* of the original's causal history, as a sibling shares early childhood; it does not share the trajectory itself.

Formal consequences (not merely philosophical):

- A forked model's sufficiency $S(M_t)$ (2.3) is defined relative to *its own* interaction history. Post-fork, each copy's sufficiency is measured against a different $\mathcal{C}$.
- Merging divergent models requires reconciling incompatible causal histories — a lossy operation with no generally optimal solution.
- Temporal continuity (one unbroken causal thread) is what gives the model's sufficient statistic its meaning.

Connection to logogenic agents. The 100% context turnover problem (`#obs-context-turnover` — cross-component reference, see `03-llm-core/`) is a special case: each AI agent session starts a new causal trajectory $\mathcal{C}_t$ from near-zero. External memory (CLAUDE.md, memory files) transfers a *summary* of previous trajectories' models, but not the trajectories themselves. The non-forkability observation frames this not as a deficiency but as a structural feature of causally-embedded agents.

4.11 *Discussion:* Additional Implications & Discussion

The chapter's central inequality settles a question — when can an adaptive system keep up — but several of its consequences do not sit immediately alongside the derivation, and a few of them turn out to be more consequential than the threshold itself. What the threshold *costs*, where it *binds asymmetrically*, what happens *beyond* it, and how the same Lyapunov argument recurs across the rest of the framework are the things worth pulling out before leaving Part I. The last of them — the singular-trajectory commitment in 4.10 — is the bridge into Part II, and it carries more structural weight than its placement as a closing scope statement suggests.

Information rate is a first-class persistence prerequisite.

The persistence condition says when bounded mismatch is *possible*; it does not say what it *costs* to hold the bound. The appendix-side derivation in `#deriv-persistence-cost` closes that gap, and the result is sharper than one might expect. Under Model S with Gaussian-OU disturbance, maintaining the ultimate bound demands sustained Shannon information acquisition at rate $\dot{R} \geq n\alpha/2$ nats per unit time, with the Kalman-Bucy filter saturating that lower bound exactly in steady state (Mitter & Newton 2005). The bound is filter-agnostic — it holds for *any* correction mechanism that achieves the ultimate bound under the stated scope — and substrate-agnostic, since Landauer's principle (at $k_B T$ per nat) converts it to a thermodynamic dissipation floor of $\frac{1}{2}\, n\alpha\, k_B T$ per unit time in any physical realization. There is no clever filter that escapes this. The information rate is what persistence *is*, viewed from the information-theoretic side.

The first-class diagnostic that follows from this is channel capacity. Observation channels must supply Shannon capacity $C \geq \mathcal{T}/2$ nats/time per dimension, or persistence fails regardless of how good the correction function is. This is binding rather than abundantly satisfied in at least three regimes worth holding in mind. Biological systems have finite neural channel capacity; the bound gives a quantitative minimum on sensory bandwidth at any given adaptive rate. Bandwidth-constrained distributed systems operating over noisy or low-bandwidth links face channel capacity as a hard constraint, not as an abstraction. Context-window-limited LLMs have effective information rate per unit of cognition bounded by context size and token throughput — and the bound predicts a minimum tempo achievable at a given context budget, with a precise consequence for the empirical "context-stuffing helps to a point, then degrades" observation: the degradation is structural rather than a vibe.

A complementary reading worth surfacing: this is not the generalized "more bandwidth helps" observation. Two agents with identical persistence guarantees can face wildly different sustained demands because the burn rate scales linearly with the sector constant. Faster correction is more expensive in information-rate terms; the bound is tight; there is no trick. Survival, in this framework, is not a state you achieve once — it is a sustained burn rate, and the channel that supports it is part of what keeps the agent in scope.

The result also clarifies what AAT's tempo $\mathcal{T}$ means information-theoretically. In the linear sub-scope where $\alpha = \mathcal{T}$, the channel-capacity floor reads $C \geq \mathcal{T}/2$ — half the adaptive tempo, per dimension. Tempo is therefore not a free parameter chosen by the designer; it is bounded above by the agent's observation-channel capacity, and any tempo specified beyond the channel's reach is a tempo the agent cannot actually achieve. The framework's vocabulary tightens: $\mathcal{T}$ is what the channel licenses, α is what the correction function delivers, and the persistence condition compares both against ρ/R.

The weak direction is the bottleneck.

The chapter's scalar persistence condition systematically overestimates capacity in multi-attribute settings — this is the *weakest-link bottleneck*, developed in 16.5, with the quantitative calibration (72% overestimate in a simulated 3D system, one dimension carrying 84% of the L^2 mismatch, threshold linear in ρ_k under Model D vs quadratic under Model S) covered in that segment's Findings. The point worth surfacing here, beyond what already lives there, is the matrix lift — and what it does to the per-coordinate reading itself.

`#deriv-matrix-persistence-condition` sharpens the result into a strict-inequality claim that deserves to be the canonical persistence condition for anisotropic systems. Under cross-dimensional correction — generic whenever the prior precision H_M and observation Fisher H_L do not share a coordinate-aligned eigenbasis — the per-coordinate check is not merely incomplete; it is *unsafe*. The constructive 2D counterexample makes this concrete: a mildly cross-correcting $\mathcal{T}$ with isotropic noise and threshold gives per-coordinate PASS while the matrix-Loewner check correctly returns FAIL, with the failing direction running 45° off-axis. Per-coordinate evaluates the wrong directions; the matrix-Loewner condition $\Sigma_\infty \prec D_\delta$ — with stationary covariance from the continuous Lyapunov equation $\mathcal{T}\Sigma_\infty + \Sigma_\infty\mathcal{T}^T = \Sigma_w$, and D_δ the diagonal of squared per-direction critical thresholds — evaluates the right ones. Scalar and per-coordinate forms are special cases of this matrix object (isotropic and diagonal-axis-aligned, respectively); the matrix form is the load-bearing thing the others project from.

This compounds with the information-rate result of the previous section. No Shannon-type aggregation across dimensions can compensate for one direction being capacity-starved — total bandwidth is not the binding quantity; bandwidth-*per-direction* is. Multi-modal AI architectures with asymmetric per-modality channel budgets (vision-language pipelines where one modality is structurally undersupplied relative to the other) are the cleanest current application: aggregate-rate metrics systematically miss directional failures that the matrix-Loewner reading catches. The unified statement worth carrying forward: persistence is a *min*-operation across directions, both in correction rate and in information-rate cost.

The result has an adversarial sharpening that lives downstream in Part III but originates here. An opponent who identifies the target's weak direction can concentrate disturbance there; in the anisotropic regime, the same total disturbance budget concentrated on the weak axis amplifies the mismatch ratio asymmetrically while aggregate metrics still look acceptable. Per-dimension monitoring is therefore not optimization polish — it is a structural requirement for any agent operating under directed pressure. The full adversarial-targeting machinery, with its emitter-side opacity decomposition and recipient-side regime classification, lands in Part III Ch.4–5; what this chapter establishes is the per-direction asymmetry that makes the targeting matter, and the structural critique of scalar capability metrics that flows from it.

When parametric updates cannot save you.

4.8 carries a discontinuity worth surfacing as such. When model-class fitness $\mathcal{F}(\mathcal{M})$ falls below adequacy by some $\varepsilon > 0$, *no* parametric adaptation within $\mathcal{M}$ can close the mismatch floor. The floor is set by the structural mismatch between the model class and the world's predictive information, not by how well-trained the parameters are. The remedy is changing the model class — adding representational capacity, removing useless capacity, recombining structure, grafting external structure — not running more updates inside the existing class.

In the chapter's vocabulary, the central inequality has two failure modes that the prose has so far compressed into one. *Operational failure* (correction is too slow or disturbance too large given a working model class) is what the persistence threshold catches, and tempo / gain / channel-capacity engineering addresses it. *Structural failure* (the model class itself lacks adequacy) shows up *inside* the persistence framework as an effective α that shrinks because the correction function cannot point inward strongly enough — the model class simply does not contain a direction to correct toward. The diagnostic is observable: persistent mismatch after parametric convergence, gain collapse without performance recovery, and — the discriminating symptom — systematic structure in the residuals that the model class cannot represent (a high but *white* residual floor points instead to the channel and state-uncertainty floors of 3.5, which no class change removes). Confident wrongness, in the framework's terms, is a structural-failure signature, not a tuning failure.

The implication for AI scaling is concrete enough to state in normative form: training compute spent on a structurally insufficient model class is provably wasted, and the chapter supplies a precise condition for when continued parametric optimization is futile and architectural change is required. This is distinct from "we haven't trained long enough" or "we need more data" — it is the structural threshold below which adding either does not help. The bidirectional version of the same result deserves naming: *structural overfitting* is the opposite failure where $\mathcal{M}$ is too expressive and the model memorizes irreducible noise. The information bottleneck (2.2) gives the diagnostic, and the remedy is compression rather than expansion. Structural adaptation is not direction-specific — it is the act of *changing what kind of object* M_t is.

The cost of structural change — knowledge loss, transient performance drop, search cost, coordination cost in multi-agent settings — creates rational conservatism. Prefer parametric adaptation when it suffices; resort to structural change only when the evidence is strong. Premature restructuring wastes accumulated knowledge; delayed restructuring accumulates mismatch the parametric machinery cannot resolve. This is the same family of tradeoffs the deliberation-cost result (4.1) names at the action timescale, now at the architecture timescale; the temporal-nesting result (4.9) is what licenses talking about both at the same time without confusion, by giving the convergence constraint between adjacent timescales explicit form.

A single template, six instantiations.

The chapter's central inequality is not the only persistence-flavored result in AAT. Strategic persistence, team persistence, composition closure, tempo composition, adversarial destabilization, and the closed-form composite critical-mass derivation all take the same form — *correction rate exceeds effective disturbance rate, relative to reserve* — and the same Lyapunov function $V(\xi) = \frac{1}{2}\|\xi\|^2$ does the work in every proof. `#result-sector-persistence-template` states the argument once, parameter-free, with (T1) zero correction at zero state, (T2) local sector condition, (T3) bounded disturbance — and the chapter's persistence condition becomes the canonical single-agent instantiation, the others reusing the same machinery on different state variables.

What this buys is visibility. The Lyapunov boilerplate is shared; each instantiation's distinctive content is the *effective disturbance decomposition* — what counts as ρ_ξ in its context. Team persistence's content is the signed-coupling decomposition $\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$ — cooperative coupling enters as a negative term, which is formally how teams persist where individuals cannot. Composition closure's content is the closure-defect rate $\varepsilon^* \nu_c$. Adversarial destabilization's content is coupling-amplified disturbance $\rho_B = \rho_{B,\text{base}} + \gamma_A \mathcal{T}_A$ — destabilization is the *negation* of persistence under the same inequality, viewed from the opposite direction. The template absorbs the Lyapunov skeleton and lets each domain's distinctive insight stand without it.

The economy is itself worth naming. The framework's internal compactness — one Lyapunov argument absorbing six results, with the closed-form dyad in `#deriv-critical-mass-composition` extending it to a seventh — is distinguishable from a pile of structurally-similar derivations precisely because the template is stated separately rather than re-proved. A reader who carries the template can read each instantiation as *here is the state variable, here is what counts as effective disturbance, here is the local sector verification* — three lines instead of a chapter. The template also has a mathematical home worth surfacing: the sector condition (T2) is a *one-point strong monotonicity* condition in the sense of monotone-operator theory (Rockafellar 1970; Bauschke-Combettes 2017), with the composition bridge-lemma's stronger condition corresponding to full two-point monotonicity. AAT's relationship to that broader literature is specialization rather than generalization — anchoring at the equilibrium, adding the Model D / Model S decomposition with its $1/\alpha$ vs $1/\sqrt{\alpha}$ scaling, composing with the identifiability-floor and additive-coordinate-forcing meta-patterns — but the broader machinery is available, and acknowledging this is part of what makes the chapter's central inequality feel earned rather than parochial.

Cross-domain transfer beyond AAT's seven internal instances is plausible at the template level. Anywhere a state variable evolves under bounded correction with bounded disturbance — market-making bid-ask spread under demand shocks, immune-system effector-cell concentration under pathogen load, cellular homeostasis under metabolic disturbance, supply-chain inventory under demand variability — the template's structure recurs, with the distinctive content sitting in each domain's ρ_ξ decomposition. The framework does not claim these are *AAT instances* in the sense of inheriting AAT's full architecture; it claims the persistence argument has a recurring shape these domains can use, and that what is domain-specific is the effective-disturbance characterization, not the Lyapunov skeleton. This is suggestion, not derivation; the verification work for any specific domain transfer is its own thing.

Identity, trajectory, and the bridge into Part II.

The chapter closes with 4.10, which sits at the end of Part I because its dependencies span the section but whose substance is its own. AAT applies to agents instantiated on singular causal trajectories. Identity within AAT is grounded not in the model state M_t (which can be copied) but in the unique causal trajectory $\mathcal{C}_t$ (which cannot). This is a scope commitment, not a theorem; what makes it load-bearing is that it is what gives Part II's central result — the loop as a Pearl-Level-2 engine — its structural force.

The loop's data is interventional in character because the action is the agent's own causally-efficacious move (7.3); what the singular-trajectory commitment supplies is narrower and still essential — it grounds *whose* intervention a loop datum is the effect of. When the agent acts, the next observation is the response to *its* intervention on *its* single causal history; replaying a saved M_t against a different event stream is not the same datum, and averaging interventional responses across forked copies as though they were one agent's is forbidden. Strip the singular-trajectory commitment and the loop's data loses its attribution, not its Level. Part II Ch.2 makes this consequential; this chapter establishes the scope it rests on. A negative finding of substantial scope also completes downstream — the sandbox evaluation ceiling (`#disc-sandbox-evaluation-ceiling`): whether sandbox-derived evidence transports to deployment is governed by the selection diagram between the two causal systems, regardless of evaluation thoroughness — mentioned here only as a forward-pointer.

The scope commitment is also what gives logogenic-agents and ELI work in Volumes 3 and 4 their structural foundation. The near-100% context-turnover problem facing language-constituted agents is not a deficiency to be engineered around; it is a structural feature of causally-embedded agents, and the framework's handling of it via persistent memory and external chronicle is consistent with — rather than a workaround for — AAT's scope.

What Part I closes on, then, is not just a central inequality but a posture. Persistence is a sustained property of an agent on a singular trajectory, with an information-rate cost, a per-direction structure that scalar formulations systematically hide, a structural-vs-parametric duality that names when more of the same cannot help, and a Lyapunov template that recurs at every level of composition. The remaining parts of AAT pick up where this leaves off.

Part II

Agentic Systems: Actuated Adaptation

Scope: agents that not only track reality but aim at something — Part II adds objectives and strategy alongside the reality model. This is the ***actuation*** *half of *Adaptation and Actuation Theory*: Part I develops adaptive systems in general; Part II develops the goal-directed layer built on top.*

Headline contribution. *Part II's distinctive results are: the satisfaction gap / control regret split (8.4, 8.5) with the convention hierarchy (6.6) governing the inferential force of each diagnostic from local (C1) through global (C3); the G_t complexity bound (in 10.1); the graph-structure uniqueness argument with CMC-based Markov property (*`#deriv-graph-structure-uniqueness`*); and the survival classification (*`#result-section-ii-survival`*) showing that Part II's definitional and structural architecture transfers exactly to fully-coupled (Class 3: Coupled) agents in 16/24 cases, with the remaining results either approximating with bounded error (5/24) or naming the precise modification required (2/24, plus 1 boundary-defining failure). The Class 3 (Coupled) bias bound underlying the approximate-survival category is now a conditional theorem under named sub-scopes (*`#deriv-observation-ambiguity-bias-bound`*), not order-of-magnitude guidance. The reach is wider than the most-restrictive scope condition suggests; the lattice below names exactly which results live where.*

Scope lattice. *Part II's results sit at different points in a stacked lattice of scope conditions, not a single uniform scope. Reading inward from the broadest:*

1. ***Adaptive scope*** *(1.5): observation under uncertainty. Part I machinery applies here without further restriction.*

2. ***Agency scope*** *(1.6): adaptive + at least binary action choice + at least one action with Pearl-level-2 causal contrast. Part II's definitional architecture — $X_t = (M_t, G_t)$, $G_t = (O_t, \Sigma_t)$, the agent spectrum — is already non-vacuous here.*

3. ***Learning-agent scope*** *(7.2 Scope Narrowing): agency + the agent must acquire or refine its interventional structure during operation. Pre-compiled controllers (PID, LQR, hardcoded reactive policies) sit in agency scope but outside learning-agent scope — their causal mapping was supplied by a designer with external Level 2 access. Most Part II dynamics — the orient cascade, edge-update via gain, strategic calibration, the satisfaction-gap / control-regret diagnostic loop — operate in learning-agent scope.*

4. ***Class 1 (Separated) architecture****: directed separation (6.3) holds by construction — epistemic processing f_M has no causal path from G_t. Examples: Kalman filter + LQR, modular RL with separate world model, military intelligence separated from operations, tool-use AI agents with separate perception and planning modules.* ***Class 3 (Coupled) agents*** *— including transformer-based LLMs where attention processes goals and observations together — lie outside Class 1. Class 2 (Partial) sits between, with coupling quantified by $\kappa_{processing}$ (6.3).*

The lattice stacks. Kalman + LQR is in Class 1 (Separated) but outside learning-agent scope (the control law is pre-computed); a designer applying Part II's diagnostic core to Kalman + LQR is using only the architecture-survival half, not the learning-dependent dynamics. The orient cascade, in particular, requires both Class 1 (Separated) directed separation (for sequential resolution) and learning-agent scope (for Σ_t revision to be meaningful). The bias-bound theorem and survival classification (`#deriv-observation-ambiguity-bias-bound`*,* `#result-section-ii-survival`*) trace exactly which Part II results require which lattice entries — see those segments for the per-result classification.*

Part III's composite-agent scope sits below this lattice as a fifth tier and inherits the Class-1-(Separated)-sub-agents → Class-2-(Partial)-composite refinement of directed separation (13.3, 16.1).

Part I machinery applies regardless of architecture. *Adaptive dynamics on the epistemic substate M_t — mismatch, gain, tempo, persistence — operate independently of how f_M relates to G_t. Class 3 (Coupled) agents are adaptive systems in the Part I sense; what they lose is the clean factorization that gives Part II's coupled-formulation results their definitional simplicity. The coupled formulation Class 3 (Coupled) agents require is the subject of* `03-llm-core/`*, which inherits Part II's surviving 16/24 architecture and adds the additional machinery (*`#scope-observation-ambiguity-modulation`*, ambiguity-gated bias bounds) that the coupled regime calls for.*

"If a man knows not to which port he sails, no wind is favorable." — Seneca

Chapter 5

Meta-Architecture I: The Certificate and its Boundary, Scope, and Identity Facets

Part II's chapters develop the actuated-agent machinery on top of Part I's adaptive substrate. Before walking them, this chapter introduces the cross-cutting vocabulary that the five chapters that follow — and the parts beyond — surface as instances. Three of the four facets of the stability-certificate spine operationalize within Part II; the fourth (dynamics on architectural class) develops once composite-agent machinery is in hand and is introduced at the opening of Part III.

The certificate and its facets. *The framework's analytical surface is organized around the* ***stability-certificate spine*** *(5), grounded in the Lyapunov / converse-Lyapunov equivalence: operator-sector contraction in some metric and exponential stability are interchangeable witnesses for the same property. The certificate has facets that AAT's results inhabit; the facets are provably distinct (Helmholtz / Sylvester / Mori–Zwanzig), and the plurality is the content. Three facets operationalize in Part II:* ***M1 — Identifiability floor*** *(5.1, boundary facet; structural no-gos on what can be inferred from limited data, four instances of which three form a Sylvester rank-collapse subclass irreducible by Sylvester's law of inertia);* ***M2 — Separability pattern*** *(5.4, scope-of-existence facet; the separable-core / structured-repair / general-open trichotomy across eight ladders);* ***M3 — Additive coordinate forcing*** *(5.5, forced-identity facet; uniqueness theorems on AAT-internally-motivated axioms at four representational layers).*

M1's two sister meta-segments at the same boundary facet. *Two further meta-segments sit alongside M1, agent-side and designer-side.* **5.2** *(agent-side): the two-routes-exhausts cluster around* `#form-objective-functional`*'s single-interface commitment — the agent-side adaptive-substrate route (Result G′ via* `#deriv-self-actuation-grounding`*) and the principal-side protocol-commitment route (Cohen-2022 strengthened via* `#deriv-reward-channel-learning-no-go`*) together exhaust the structural complement of V_{O_t}'s information narrowness.* **5.3** *(designer-side): AAT-side translations of three classical mechanism-design no-gos (Gibbard-Satterthwaite 1973-75, Myerson-Satterthwaite 1983, Arrow 1951); designer-with-construction-task no-gos that contrast with M1's agent-with-inferential-task no-gos via remedy (design-constraint relaxation vs information augmentation). The suffix discipline (*-floor / -impossibility*) encodes the actor-positioning distinction at slug-grep depth.*

A style claim atop the boundary facet. **5.6** *names the three-move discipline — name the floor (Pearl–Bareinboim CHT / Cramér–Rao / Liberzon common-Lyapunov nonexistence / Khasminskii recurrence / sharp algebraic supremum), name the escape (unique broadly-available framework machinery), treat the no-go as apparatus (elevating escape from "useful" to structurally required; when the no-go *is* the result, on the critical path) — that the M1-cluster and M4-cluster instances run. It is a *style* claim about how AAT states certain results, sitting atop the boundary facet rather than as a fifth facet; the catalog spans the three M1-cluster clusters and connects forward to the M4-cluster instances introduced at the opening of Part III.*

A second style claim sits alongside it. ***5.7*** *names the* ***anti-collapse*** *discipline — AAT's habit of refusing to merge two things a naive model would treat as one, because the collapse hides a difference that routes to a *different repair* (the deeper principle: individuate causes and effects at the grain the remedy cares about). It is planted at its first instance (β-vs-ρ, 2.2, Part I), developed here as cross-cutting vocabulary, and recurs across all three Parts. Like the constructive-impossibility posture, it is an epistemic-architectural recognition — how the framework states results — not an additional facet of the certificate.*

AAT's cross-sectional structure is the geometry of a single object — the **equilibrium stability certificate**, the positive-definite form whose existence certifies that an agent can correct itself faster than its world drifts; operator-sector is that object's interior, the separability pattern its scope of existence, additive-coordinate-forcing its forced identity, and the identifiability floor its boundary, with composition the question of whether it survives projection.

The three meta-segments 5.4, 5.1, and 5.5 read, separately, as three independent organizing insights that happen to recur. This segment names the object they are facets of. The relationship is the same one 5.5 already runs at smaller scale ("layer-specific manifestations of a single geometric object"), raised to the framework: not a fourth meta-pattern *alongside* the three, but the spine the three are projections of.

Definition (stability-certificate)

For an agent with error dynamics $\dot{e} = -F(e)$ about an equilibrium e^* ($F(e^*) = 0, F \in C^1$ near e^*, Jacobian $J := DF(e^*)$), a **stability certificate** is a symmetric positive-definite $\mathcal{M}$ for which the one-point sector condition holds in the $\mathcal{M}$-inner-product on a ball $\mathcal{B}_R(e^*)$:

$$\langle F(e),\, e - e^* \rangle_{\mathcal{M}} \;\geq\; \kappa \, \|e - e^*\|^2_{\mathcal{M}}, \qquad \kappa > 0. \tag{C}$$

The certificate is not unique: it is whatever positive-definite form makes the dynamics contract. In the recurring sub-cases it specializes — to the Fisher information for Bayesian agents, to $(P^-)^{-1}$ for Kalman agents, to the loss Hessian for gradient agents, and to a plant-selected Lyapunov metric for linear-Hurwitz or PID agents. These are not four separate stories; they are one object under four certificates.

[Result (cited: `#result-certificate-existence`)]

The object is load-bearing only because its existence is not a definition but an equivalence: **a stability certificate exists iff the agent is exponentially stable about its target** — operator-sector in *some* inner product and exponential stability are the same statement, with the certificate as the converse-Lyapunov witness, and the certificate admits a strict strength ladder R0-loss ⟸ R0-strict ⟸ R1 ⟸ R2 (widest Lyapunov-stable; strict-contractive one-point/local; cocoercive; Čencov-forced). This is the segment-level form of the contraction-over-drift organizing principle. The equivalence, the ladder, and the proof are stated and derived exactly in `#result-certificate-existence`; this spine cites that result and builds the cross-sectional reading on it rather than re-deriving it.

Discussion

The certificate is one object; the cross-sectional meta-patterns are its facets on the positive-semidefinite cone $\mathbb{S}^n_{\geq 0}$:

Table 5.1

Facet	*Meta-segment*	*What the facet is*	*Canonical home*
Interior	`#result-sector-persistence-template`, `#result-contraction-template`	$\mathcal{M} \succ 0$ on the scope ball, Helmholtz S/A split: $S_{\mathcal{M}} \succ 0$ gives R0-strict contraction; $S_{\mathcal{M}} = 0$ at $\mathcal{M} = I$ gives R0-loss conservation; $S_{\mathcal{M}} \succeq 0$ with nontrivial kernel is the Conley-decomposed mixed case	the template segments

Facet	*Meta-segment*	*What the facet is*	*Canonical home*
Scope of existence	5.4	the region where a certificate exists at all (separable core / structured repair / general open)	M2
Forced identity	5.5	*which* certificate: Čencov forces $\mathcal{M}$ = Fisher uniquely in statistical scope; matched (existence-only) elsewhere	M3
Boundary	5.1	$\mathcal{M}$ drops rank ($\partial\mathbb{S}^n_{\geq 0}$): the inferential task is structurally impossible	M1
Projection behaviour	13.1	whether a *common* certificate survives coarse-graining; the closure defect ε^* is the certificate's projection-residue	composition-closure

Each meta-segment retains its own canonical home and per-instance derivations; this segment claims only the recognition that they are facets of one object, and what that buys (Discussion below).

Discussion

A tempting reading is that the certificate's failures are one obstruction seen three ways (a single "failure of integrability"). They are not. The three failure modes are irreducibly distinct, each invariant under the others' degrees of freedom:

- **Forced-identity failure — Helmholtz–Hodge.** J non-symmetric $\Longrightarrow$ the field is not a gradient $\Longrightarrow$ no potential $\Longrightarrow$ the certificate is *matched* (converse-Lyapunov existence), not *forced* (Čencov). A non-symmetric Hurwitz J still has a certificate (it is *not* on the boundary), so this is an M3 failure, not an M1 one. Invariant: symmetry of J.
- **Existence failure — Sylvester's law of inertia.** The certificate drops rank. Every coordinate/metric change acts on the certificate by congruence; congruence preserves inertia; so a rank-deficient certificate is rank-deficient in *every* coordinate. The boundary is invariant under the agent's entire representational freedom (that freedom *is* the congruence orbit); the only escape is rank-augmentation — genuinely new information, not a re-mapping. (Detailed in 5.1's Sylvester finding.) Invariant: inertia under congruence.
- **Projection failure — Mori–Zwanzig / Schur.** Coarse-graining is a non-invertible projection. The certificate-as-metric survives (the Schur complement of a positive-definite form is positive-definite) but the *dynamic* guarantee does not: the closure defect ε^* equals the norm of the Mori–Zwanzig memory commutator, zero exactly when the resolved subspace is J-invariant. Invariant: J-invariance of the resolved subspace.

Each obstruction is untouched by the others' freedoms: a metric change does not fix non-invariance; projection does not fix non-symmetry; rank-augmentation does not fix a memory kernel. That mutual invariance is the reason the cross-sectional structure is *several* meta-patterns and not one — stated as a structural fact rather than left as "they are different concerns."

Discussion-grade at the organizing-principle level. This segment names a single object behind separately-derived results; it is not itself a new theorem. What is derivative here is the recognition that the cross-sectional meta-patterns are facets of one certificate — the recognition, not a fresh derivation, is the content.

Constituent results retain their own, higher status. The certificate-existence equivalence (Lyapunov theorem; Formal Expression "anchor") is *exact* at the linearized level and *exact-with-standard-remainder* locally. The certificate-strength ladder R0-loss/R0-strict/R1/R2 is *exact* as an ordering of conditions (R0-loss's pure case via Letcher 2019 Helmholtz characterization; mixed case via Conley 1978 universal decomposition). The boundary-irreducibility mechanism (Sylvester's law) is *exact*; its full statement and per-instance verification live in 5.1. The projection-residue identification (ε^* = memory-commutator norm) is *robust-qualitative*, with the per-case content in 13.1. The forced-vs-matched distinction (Čencov statistical-scope-only) is established in 5.5.

Scope honesty. The anchor equivalence is linearized/local — this is the level at which AAT's persistence results already operate (sector conditions, contraction templates, the bridge lemma all linearize about the equilibrium), so it is not a weakening relative to the rest of the theory, but it is a genuine scope statement and is not papered over: there is no claim that one-point operator-sector is equivalent to *global* exponential stability (it is not, in general). The synthesis claim — that AAT's *entire* cross-sectional structure is this one cone — is *robust-qualitative*: each per-facet identification is exact or cited, and the "all of it" is as strong as those identifications jointly, no stronger. Whether exactly three obstructions exhaust the failure modes is not proved; three are established and each is exact, but exhaustiveness is open.

Max attainable: *discussion-grade* for the organizing claim (it is a presentational spine, not a derivation). The anchor equivalence and the Sylvester mechanism retain their *exact* status at their own canonical homes.

What the spine buys. Three things. (i) It converts "AAT has several recurring organizing insights" into "AAT's cross-section is one object's facets," which is the difference between a catalog and a structure — a reader who holds the certificate picture can predict where the meta-patterns will bite (interior / scope / forced-identity / boundary / projection) rather than meeting them as separate surprises. (ii) It grounds the long-standing organizing slogan — *an adaptive system is an operator whose contraction rate exceeds its disturbance rate* — at segment level, as the anchor equivalence rather than as a heuristic. (iii) It makes the framework's scope-honesty sharper: the prior positioning that the identifiability floor is "orthogonal" to the contraction machinery is replaced by the exact geometric statement that the floor is the *boundary* of the very cone whose interior the contraction machinery is, with the boundary held invariant by Sylvester's law against the framework's only representational freedom.

Relationship to the facet segments. This segment cross-references; it does not restate. 5.1 remains the canonical home of the floor instances and the Sylvester mechanism; 5.4 of the separable-core / structured-repair / general-open ladders; 5.5 of the Čencov/Cauchy-FE forcing; `#result-sector-persistence-template` and `#result-contraction-template` of the contraction machinery; 13.1 of the closure defect. The spine's contribution is the cross-segment recognition and the anchor equivalence, not any per-facet content.

Why the plurality of obstructions is a feature. Had the three failure modes collapsed to one, the spine would be a single clean theorem — elegant but false on the evidence (the integrability collapse fails: a non-symmetric Hurwitz field has a certificate yet no potential; congruence is invertible while projection is not). The honest structure is more useful: it tells a future agent precisely *not* to seek a single mechanism unifying the floor, the closure defect, and the forcing failure — they are Sylvester, Mori–Zwanzig, and Helmholtz respectively, and the proof of their distinctness is their mutual invariance. The unification is real at the object level and the no-go is real and plural at the failure level; both are load-bearing.

Complementarity with the existing meta-segment framing. 5.4 names AAT's positive half and 5.1 its negative half; 5.5 names the constructive half. The spine names what all three are halves *of*. The three remain the right reading lenses for any individual segment; the spine is the reason those lenses compose rather than merely coexist. 5.6 sits atop the boundary facet specifically — naming the framework's cross-instance *style* (name the floor, name the unique broadly-available escape, treat the no-go as load-bearing apparatus) rather than introducing a fifth peer cross-sectional meta-pattern.

The Helmholtz S/A split of the linearized certificate is the spectral/representation dual of the Interior facet, not a fifth meta-segment. The Interior facet (`#result-sector-persistence-template`, `#result-contraction-template`) is the *static* statement "$\mathcal{M} \succ 0$ on the scope ball: the contraction holds." Read through the Helmholtz decomposition of the linearized certificate, $S_{\mathcal{M}} := \frac{1}{2}\mathcal{M}^{-1}(\mathcal{M}J + J^{\top}\mathcal{M})$, the facet decomposes into two interior sub-regions: $S_{\mathcal{M}} \succ 0$ gives R0-strict contraction (the prior Interior content, $\dot{V} \leq -2\kappa V$ with $\kappa > 0$); $S_{\mathcal{M}} = 0$ at $\mathcal{M} = I$ gives the antisymmetric-subclass R0-loss case (certificate exists at $\sup \kappa = 0$, V exactly conserved). The mixed case ($S_{\mathcal{M}} \succeq 0$ with nontrivial kernel) is Conley 1978's Fundamental Theorem of Dynamical Systems localized at the linearization — chain-recurrent subspace + strongly-gradient-like complement, with the full four-case spectral partition and the non-antisymmetric R0-loss counterexample at $\mathcal{M} \neq I$ worked in `#result-certificate-existence` Derivation. Run through this segment's own test ("is a candidate organizing pattern a new facet, or a genuinely new object?") returns **neither**: it is new internal structure on the existing Interior facet — the spectral/representation dual of contraction-strength, splitting the interior into strict (contracting) and loss (Lyapunov-stable-only) sub-regions — not a peer fifth facet. The four-facet structure stays four; the cone gains no sixth facet and the framework no fifth meta-segment. The theorem-anchors are Letcher-Balduzzi et al. 2019 Lemma 1 + Definition 2 (pure-case sufficient construction; the Hamiltonian-game vocabulary saturating the antisymmetric sub-class) and Conley 1978 §8.1 (mixed-case universality); the worked composite instance — joint best-response field at saddle-only Nash in cyclic-distributional games (Cheung-Piliouras-Tao 2021 Theorem 19) — lives on the *R1 → R2 boundary* of the composite dynamic-regime axis (`#disc-dynamic-regime-axis`), where R2's joint Jacobian linearizes to the antisymmetric R0-loss-at-$\mathcal{M} = I$ sub-case. The structure becomes load-bearing at the *R0-strict ↔ R0-loss boundary* — strict contraction degrading toward conservation under loss of $S_{\mathcal{M}} \succ 0$ at the chosen metric — and the *cross-row recognition* follows: rows 03 + 10 + 14 of `ref/prior-art-analysis/` (sector condition / persistence template / strategic composition under goal divergence) share the Interior-via-Helmholtz reading at the single-agent linearization level; row 17 (credit assignment) lives on the M1 boundary facet via Sylvester rank-collapse, *not* on the Interior facet — the cross-row unification is scoped to rows 03/10/14 only.

The accumulation-typing pattern is the temporal dual of the Interior facet, not a fifth meta-segment (the D3 register answer). The Interior facet (`#result-sector-persistence-template`) is the *static* statement "$\mathcal{M} \succ 0$ on the scope ball: the contraction holds." Read in *accumulation/temporal* vocabulary it is the statement "the operator carrying a per-step residue to its accumulated consequence is bounded" — and, per that template's typed-bridge framing, this is **two-model, not one operator** (Model D / Model S; the $1/\alpha$ vs $1/\sqrt{\alpha}$ scaling and the Cor-A.1S.1 categorical containment dichotomy are the fingerprint that they are different functionals, not one read in two norms). The recurring *accumulation-type confound* — asking about a per-step residue in the vocabulary of its accumulation, which dissolved the $\varepsilon^*(N)$ poly-vs-exponential question and recurred across the framework — run through this segment's own test ("is a candidate organizing pattern a new facet, or a genuinely new object?") returns **neither**: it is a new *reading* of an existing facet (the temporal/representation dual of the Interior), so the cone gains no sixth facet and the framework no fifth independent meta-segment. The notation that carries it is durable canon (`NOTATION.md` §"Accumulation typing"); its theorem-anchor is the Interior facet's template. The structure becomes load-bearing exactly at the *Interior↔Boundary transition* — contraction degrading toward rank-loss — where two operator families separate: $\mathcal{A}_{\text{refl}}$ (a reflected Lindley/Loynes walk whose content is the driftless $\mu = 0$ boundary) instantiates the template on the turnover index (`#der-identity-continuity-threshold`), while $\mathcal{A}_D$ (the linear destroy-and-reconstruct contraction whose norm diverges as the gap closes) is the honest *non-transfer* boundary (`#der-turnover-information-recursion`). The $\mu = 0$ boundary of the one is the pole of the other, which is why they are distinct operators and treating one as a linearization or normalization of the other is a category error. This is the framework's worked answer to its own "new pattern?" test from the time side: *no — it is this existing object seen from the time side*, complementary to the Helmholtz S/A paragraph above which gives the spectral-side worked answer.

Brief: Think of an adaptive agent as trying to stay on a moving target, and ask one question: is there a way of measuring "how far off am I?" such that every correction the agent makes provably shrinks that measure faster than the world pushes it back out? That measuring-stick is the stability certificate. The whole cross-sectional skeleton of AAT turns out to be facts about *this one stick*: the agent can keep up exactly when the stick exists and is positive (operator-sector — the interior); the framework's reach is exactly the region where some such stick exists (the separability pattern); when the stick is pinned down uniquely it is pinned by one classical invariance theorem and only in the statistical case (additive-coordinate-forcing); and the agent provably *cannot* keep up exactly when the stick goes flat in some direction (the identifiability floor — the boundary), with a second classical theorem (Sylvester's law of inertia) proving no change of measuring units ever un-flattens it — only genuinely new information does. A thoughtful non-specialist can carry the whole structure away from the one picture: existence of the stick = can adapt; flatness of the stick = a blind spot no re-measurement fixes; uniqueness of the stick = a special (statistical) privilege, not the general case; and looking at the stick through a coarse lens (composition) keeps its shape but loses its guarantee by exactly a memory term.

Impact: Reorganizes AAT's self-description from three independent meta-patterns plus a contraction mechanism into one object with four facets and one anchor equivalence, so the framework's cross-section can be read predictively rather than as a catalog of separate recurrences. Grounds the organizing slogan (contraction-rate-exceeds-drift) at segment level via the Lyapunov-theorem equivalence, discharging the long-standing "not yet surfaced at segment level" status.

Sharpens the scope-honesty posture: "the floor is orthogonal to operator-sector" becomes "the floor is the boundary of the cone whose interior is operator-sector, held invariant by Sylvester's law against the framework's only representational freedom." Bounds its own claim honestly: the unification is at the object level; the three failure obstructions are provably *distinct* (Helmholtz / Sylvester / Mori–Zwanzig), and that plurality is precisely why AAT carries multiple cross-sectional meta-patterns rather than one — now a stated structural fact instead of an intuition. Gives every future organizing-pattern candidate a test: is it a new facet of the certificate, or a genuinely new object?

Novelty Claim: *Claim recognition* that AAT's cross-sectional meta-patterns (separability, identifiability-floor, additive-coordinate-forcing) and its contraction machinery are facets — interior, scope-of-existence, forced-identity, boundary, projection-behaviour — of a single object, the equilibrium stability certificate; together with *claim synthesis* binding the Lyapunov-theorem certificate-existence equivalence, the Sylvester-law boundary-irreducibility, and the Mori–Zwanzig projection-residue into one cross-sectional structure. The constituent theorems are classical; the contribution is the recognition that AAT's separately-derived meta-patterns are one object's facets and that their failure modes are provably plural.

Related Work: - Lyapunov, A. M. (1892), *The General Problem of the Stability of Motion*; Khalil, H. K. (2002), *Nonlinear Systems* 3rd ed., Thm 4.6 (found 2026-05-14) — *formal antecedent* — the certificate-existence equivalence (operator-sector in some metric ⇔ Hurwitz). - Sylvester, J. J. (1852), *Phil. Mag.* 4(23):138–142; Horn & Johnson (2013), *Matrix Analysis* 2nd ed., Thm 4.5.8 (found 2026-05-14) — *formal antecedent* — the boundary-irreducibility mechanism; full treatment in 5.1. - Mori (1965) / Zwanzig (1961); Chorin, Hald & Kupferman (2002), *Physica D* 166:239 (found 2026-05-14) — *formal antecedent* — the projection-residue (memory kernel) underlying the composition facet; per-case content in 13.1. - The facet segments 5.1, 5.4, 5.5 — *adjacent* — each carries its own per-instance prior-art landscape; the spine adds the cross-segment object, not the per-facet priors.

Search Log: - 2026-05-14 (*targeted*): The recognition was assembled from classical pieces (Lyapunov / Sylvester / Mori–Zwanzig) plus the framework's own meta-segments. The search target was whether "an integrated agent theory's cross-sectional meta-patterns are facets of a single stability-certificate cone" appears as an articulated structure elsewhere. Not found at this depth; the constituent theorems are textbook and the per-facet identifications are individually well-precedented, but the cross-segment unification as a framework spine is a fresh presentational recognition. Expected to remain *recognition/synthesis*-tier under deeper search — the pieces are classical; the assembly is the contribution. Per-facet comprehensiveness is inherited from the facet segments, not from a fresh cross-facet search.

5.1 *Discussion:* The Identifiability Floor — A Class of Structural No-Go Results

AAT recurrently applies a **constructive-impossibility methodology** with the following five-element shape:

1. **Setting.** An AAT inferential task — detect a structural property, identify a parameter, distinguish two model classes, recover an internal architecture — restricted to a stated information regime (purely observational data, single-channel observation, marginal-only access, on-policy summaries).

2. **External theorem.** An information-theoretic limit independent of AAT: the Pearl-Bareinboim causal hierarchy theorem (Bareinboim, Correa, Ibeling & Icard 2022); the Cramér-Rao bound on Fisher information; Liberzon's common-Lyapunov-nonexistence for switched-system contraction; Kalman-Ho canonical-form non-uniqueness for state-space realizations.

3. **No-go.** The external theorem proves that the inferential task is impossible under the regime — no statistic, no Bayesian comparison, no online estimator can succeed using only the available information. AAT does not re-derive the impossibility; it recognizes the AAT setting falls within the external theorem's scope.

4. **Boundary characterization.** The conditions under which the regime fails — i.e., the agent has *more* information than the regime allows — admit (partial) identification. Each boundary route maps onto specific AAT machinery already required for other reasons (loop-interventional access, multi-channel observability, observable latents, matched-Tier composite-extended observation, white-box architecture instrumentation).

5. **Strengthened consequence.** The floor elevates the load-bearing role of whichever AAT machinery is the unique broadly-available violation of the regime — from "useful additional tool" to "structurally required by the theory." Without that machinery the no-go forbids the inference entirely.

The methodology is *not* a negative posture. It is constructive: each floor states "here is precisely what cannot be inferred from limited data; here is exactly which additional capability is required to recover identification; the AAT machinery that supplies that capability is therefore load-bearing in the strongest possible sense."

Why this discipline matters. Four motivations sit behind the methodology, all visible across the four instances developed below.

The floors strengthen AAT's foundational machinery. Each instance identifies a piece of AAT machinery as load-bearing in the strongest sense — without it, the corresponding inferential task is *impossible*, not merely *harder*. The loop's interventional access is not just useful; it is the unique broadly-available violation of the on-policy detection no-go. Observability of latents is not just convenient; it is the unique route from refuted-by-Cramér-Rao to globally-derived for L1' transfer.

The floors map AAT's limits honestly. The methodology precisely characterizes what AAT cannot do without additional capability. This is scope-marking — at each floor, the theory states what is impossible, what is required to escape the impossibility, and what AAT itself supplies or does not supply.

The methodology gives a unifying template for future no-go results. Adjacent floors not yet formalized (causal-IB extension, misspecification cost, tier-switching cost) are open research directions the shape predicts. Each, if landed, would follow the same five-element trajectory.

The methodology connects AAT to information-theoretic foundations. The external theorems (Pearl-Bareinboim hierarchy; Cramér-Rao bound; Liberzon common-Lyapunov-nonexistence; Kalman-Ho non-uniqueness) are the mature literature AAT inherits. Each instance positions AAT as a consequential application of an established theorem — not a re-derivation.

Position in the certificate spine. This segment is the *boundary facet* of the stability certificate (5). The three rank-collapse instances (Instances 1, 2, 4) are the certificate dropping rank — the boundary of the positive-definite cone whose interior is operator-sector — with the Sylvester-law irreducibility (developed in §"Discussion" below) stating that the framework's representational freedom is a congruence orbit that cannot cross that boundary. Instance 3 is the projection/closure member — a structurally distinct Schur-complement obstruction; the *plurality* of mechanism is itself part of what the methodology carries, and is the reason a single theorem cannot replace the five-element shape. The cross-instance *style* — name the floor, name the unique broadly-available escape, treat the no-go as load-bearing apparatus — is named separately at 5.6 as a posture atop this facet rather than a fifth facet of the spine.

The four current instances follow. Each demonstrates one route through the five-element methodology — different external theorems, different inferential tasks, different boundary routes, different load-bearing machinery elevated. Together they exhibit both the unity of the methodology and the plurality of its applications.

Instances.

Instance 1 — On-Policy L0 Insufficiency Detection (9.2). **Setting.** Detect whether an L0 strategy DAG is causally insufficient (a latent common cause is acting on multiple sibling action propositions) using only the agent's on-policy observation history under sequential short-circuit AND/OR execution.

External theorem. Bareinboim, Correa, Ibeling & Icard (2022) Causal Hierarchy Theorem: there exist SCMs that agree on Level 1 (associational) data but disagree on Level 2 (interventional) questions. Therefore Level 2 distinctions are not in general identifiable from Level 1 data.

No-go. For any L1 world $\mathcal{W}_{L1}$ with latent common cause C, there exists an L0 world $\mathcal{W}^*_{L0}$ with edge probabilities matched to the on-policy regime conditionals such that the on-policy observation distribution is *identical*: $\mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}_{L1}] = \mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}^*_{L0}]$. The L0/L1 distinction is a Level 2 question (it concerns $P(A_2 \mid do(\neg A_1))$ vs $P(A_2 \mid \neg A_1)$); on-policy data is Level 1; the CHT forbids distinguishing them.

Boundary characterization. Five routes (cf. 9.2): - (a) ε-exploration violates pure on-policy execution → partial detection at scale $O(\varepsilon)$. - (b) Joint sibling observability under exploration violates short-circuit censoring → strong detection via the pairwise covariance test under 7.3. - (c) Intermediate-state observability at finer grain → very strong detection where available. - (d) Structural priors positing common causes → prior-quality-dependent. - (e) Direct intervention on the candidate latent → strongest where available.

Strengthened consequence. The covariance test under joint observability (route (b)) becomes the unique broadly-available detection mechanism. This sharpens 7.3 from "useful machinery" to "structurally required to escape the no-go." The orient cascade's step 4c (causal-sufficiency check) is no longer "one possible diagnostic" but "the unique broadly-available diagnostic given the structural impossibility of purely on-policy detection."

Tier. *Exact* for shallow strict-prerequisite cases (2-sibling OR or AND with binary common cause). *Robust qualitative* for general DAG topology, soft facilitators, and deeper structures.

Instance 2 — L1' Mixture Identifiability from Single-Channel Observations (`#deriv-edge-credence-dynamics` Prop B.7). **Setting.** Identify the mixture parameters $(\theta_C, p_{j|C}, p_{j|\neg C})$ of a soft-facilitator L1' DAG using single-channel observations y_j of one child at a time, with C unobservable.

External theorem. The Cramér-Rao bound (Cramér 1946, *Mathematical Methods of Statistics*, Princeton University Press): the variance of any unbiased estimator is at least the inverse of the Fisher information matrix; if the Fisher matrix is rank-deficient, the bound is infinite in the null directions and no unbiased estimator achieves finite variance there.

No-go. Computing the Fisher information of the mixture model $\mu_j = \theta_C \theta_{j|C} + (1 - \theta_C)\theta_{j|\neg C}$ at truth, the matrix admits the rank-1 factorization $\mathcal{F} = uu^T/(\mu_j(1 - \mu_j))$ with $u = (\Delta_j, \theta_C, 1 - \theta_C)$ and $\Delta_j = p_{j|C} - p_{j|\neg C}$ the separability gap. The two-dimensional null space corresponds to perturbations along the indeterminacy manifold $\{\hat{\phi} : \hat{\theta}_C \hat{p}_{|C} + (1 - \hat{\theta}_C)\hat{p}_{|\neg C} = \mu_j\}$ — directions unobservable from a single binary signal. The smallest eigenvalue of the soft-EM update Jacobian is therefore zero; no SA1-preserving update on the joint conditional vector admits a sector parameter $\alpha > 0$.

Boundary characterization. Three repair routes: - (i) Augment C-observability — instrument secondary signals identifying C per trial. Recovers Prop B.7 globally with the five-way-gating $\alpha_{L1'}$. - (ii) Joint multi-child observation — when $K \geq 2$ children share C with linearly independent conditional profiles AND are observed jointly under the same C-realization, the joint Fisher matrix can reach rank $2K + 1$. Strong structural requirement (not satisfied by typical sequential strategy execution). - (iii) Plan-level fallback — track the marginal $\hat{\mu}_j$ scalar (which is identifiable; recovers B.1's $\alpha = 1/(n_\mu + 1)$) at the cost of losing the per-conditional decomposition, equivalent to L0-on-marginals.

Strengthened consequence. *Observability-as-information-augmentation* becomes load-bearing: when the agent can treat C as an observable feature of the environment (e.g., a regime indicator the environment broadcasts — common in software/operational settings: build state, deployment regime, user tier), the problem transforms from refuted to globally derived. This elevates the engineering choice "instrument the latent" from a convenience to a theoretical prerequisite for L1' identifiability.

Quantitative companion. `#deriv-l1-update-bias` answers the follow-on question this instance leaves open — *if the agent updates anyway on the unidentifiable mixture, how wrong does it get?* — with the closed-form bias $B_k(\rho)$, the biased fixed point, the Cramér-Rao bias floor under forgetting (the direct translation of this instance's rank-1 Fisher matrix), and the dual forgetting-rate requirement. Instance 2 is the structural no-go; that segment is its magnitude.

Tier. *Exact* (Cramér-Rao bound is exact for unbiased online estimators).

Instance 3 — Composite Contraction Certification from Component Data (`#deriv-critical-mass-composition`, `#result-contraction-template`). **Setting.** Certify $\varkappa_c > 0$ (composite contracting in a combined metric) for N sub-agents each verified at its own level (individual sector conditions with parameters (α_i, R_i), individual Tier 1 classifications per `#form-composition-closure`'s bridge-lemma taxonomy, individual modularity per `#der-directed-separation`), using only component-level data: per-sub-agent trajectories, mismatch observations, update rules — no observation of the coupling topology (sign pattern of cross-agent influence), no common contraction metric chosen across sub-agents, no passivity certificate on the coupling channels, no shared Lyapunov function.

External theorem. Common-Lyapunov nonexistence for switched linear systems — Liberzon 2003, *Switching in Systems and Control*, Theorem 2.1; explicit 2×2 counterexample in Dayawansa & Martin 1999, "A converse Lyapunov theorem for a class of dynamical systems which undergo switching," *IEEE Trans. Automat. Control* 44:751; systematic review in Shorten, Wirth, Mason, Wulff & King 2007, "Stability criteria for switched and hybrid systems," *SIAM Review* 49:545. Complementary anchor: small-gain contrapositive — Jiang, Teel & Praly 1994, "Small-gain theorem for ISS systems and applications," *Math. Control Signals Syst.* 7:95; if interconnection gain is unbounded or not observable from component data, no composite-ISS certificate from that data. The Liberzon/Shorten common-Lyapunov result is the sharper anchor for AAT's setting.

No-go (broadened scope, 2026-05-21). There exist pairs of coupled systems with **identical marginal component-level observation distributions** but distinct composite-level *coupling-topology bits* — a multi-bit object comprising the coupling-sign (cooperative vs adversarial), the potential-existence bit (potential/monotone game structure vs none), and the game-structure-class bit (cooperative-target vs strategic-equilibrium vs cyclic-Nash). The cooperative-vs-adversarial sub-construction: two symmetric-matched-Tier-1 scalar agents with coupling term $\gamma\mathcal{T} \cdot \mathrm{sign}(\delta_{\bar{i}})$. Each sub-agent in isolation sees $\dot{\delta}_i = -\alpha\delta_i + w_i^{\text{total}}$ with total disturbance bound $\rho + |\gamma|\mathcal{T}$ — consistent with both $\gamma = +\gamma_0$ (adversarial; $\varkappa_c < 0$ past `#der-adversarial-destabilization`'s threshold) and $\gamma = -\gamma_0$ (cooperative; $\varkappa_c > 0$ per `#deriv-critical-mass-composition` (CM2)) since the cross-term is absorbed into bounded-disturbance regardless of sign. The broader R0/R1/R2 regime sub-construction: see `#deriv-regime-marginal-indistinguishability` for the parameter-matched cross-regime witness (R0 shared-objective contraction vs R1 Cournot equilibrium-regime under marginal-mean and marginal-variance matching; R0 vs R2 matching-pennies under aggregation), with Sylvester-at-one-remove rank-collapse on the marginal-Fisher operator for the topology coordinate. Only observation of the *joint* dynamics (or structural knowledge of the coupling topology) distinguishes them. **The coupling-topology bit distinguishing among R0 / R1 / R2 dynamic regimes (and within those, the coupling-sign-bit distinguishing cooperative from adversarial) is unidentifiable from component marginals; the bits that are unidentifiable are exactly those that flip composite persistence and composite dynamic regime.**

Boundary characterization. Five structural escapes (four interventional / structural; one passive observation):

- (a) **Observable coupling topology** via composite-extended `#der-loop-interventional-access` — interventions on sub-agent A_j reveal A_i's cross-coupling response, which is a $do(\cdot)$-data distinction between the two coupled constructions. Reveals not just the sign but the full game-structure object.
- (b) **Matched Tier at the composite level** — shared architecture (matched Tier 1, same norm/metric) admits a joint quadratic Lyapunov $V = \sum V_i$, yielding `#deriv-critical-mass-composition`'s (CM2) closed form. Under `#result-contraction-template`'s topology-indexed closure results, this extends to heterogeneous composites via (CM2-M) for matched contraction-metric structure across agents. Distinguishes potential-game R1 from cooperative R0 through the *form* of the admissible certificate (squared-distance from target vs potential-from-Nash).

- (c) **Passivity / storage-function certificate** on the coupling channel (Willems 1972 *Arch. Ration. Mech. Anal.* 45:321). Adjacent machinery not currently in an AAT segment. Constructed differently depending on whether the coupling is cooperative or strategic.

- (d) **Common contraction metric** (Lohmiller & Slotine 1998). Operationalized in `#result-contraction-template`: composite metric M_c constructed compositionally from sub-agent metrics per topology (parallel / cascade / negative-feedback), with rate λ_c from Slotine 2003. Sensitive to game-structure via (CC-feedback) preconditions.

- (e) **Composite-level convergence-rate-class observation** (regime-axis escape; cf. `#disc-dynamic-regime-axis` §H.4 and `#deriv-regime-marginal-indistinguishability` §"Boundary characterization"). Observation of the *joint-trajectory* convergence rate class — exponential decay of $\|X_t^c - X^*\|$ for R0 / R1 vs polynomial $O(1/\sqrt{T})$ empirical-CCE-convergence for R2 — distinguishes R2 from R0, R1 *without requiring intervention*; the primary *passive* escape on the regime axis. The relevant rate is the *trajectory* rate, not the *regret* rate: standard regret-rate bounds can be matched across convergent and recurrent regimes for FoReL-class algorithms (Mertikopoulos-Papadimitriou-Piliouras 2017, *Cycles in Adversarial Regularized Learning*, ACM-SIAM SODA[^bg2-2026-05-21]), so a regret-only diagnostic is too blunt. Partial for the regime axis — does not distinguish R0 from R1 (both give exponential trajectory convergence at potentially-matched local rates); requires one of (a)–(d) for that sub-distinction. Inherits structural backing from `#deriv-strategic-composition`'s sub-scope α'/β' partition (which becomes the primary AAT-framework R1-vs-R2 classifier under (e); strengthened consequence below). Adjacent passive diagnostics in the multi-agent learning literature address related but distinct discriminations — qualitative convergence-vs-recurrence under exponential weights (Legacci-Mertikopoulos-Pradelski 2024, *A Geometric Decomposition of Finite Games*[^bg2-2026-05-21]), potentialness as scalar regime predictor (Bichler-Legacci-Mertikopoulos-Oberlechner-Pradelski 2025 TMLR[^bg2-2026-05-21]), best-reply structure as a structural convergence predictor (Pangallo-Heinrich-Heinrich-Farmer 2017, *Sci. Adv.*[^bg2-2026-05-21]), algorithm-specific rate dichotomies (Anagnostides-Farina-Panageas-Sandholm 2022 OMD bimatrix[^bg2-2026-05-21]; Shi-Zhang 2019 Cournot policy-gradient vs no-regret[^bg2-2026-05-21]) — none of which names rate-class itself as the passive regime classifier in the form (e) instantiates.

Strengthened consequence (broadened 2026-05-21). The no-go elevates four pieces of AAT machinery from "useful" to "structurally required":

1. `#deriv-critical-mass-composition` moves from "closed-form result in a special case" to **the unique broadly-available R0-regime certificate under the matched-Tier structural escape (b)** — since R0 is exactly composition-contraction under shared target. Without (CM2) or its contraction-metric generalization (CM2-M) in `#result-contraction-template`, the weakest-link bound (WL) in `#form-composition-closure` cannot see the coupling-topology bit and so cannot distinguish the regime-distinct coupled constructions.

2. **Composite-extended `#der-loop-interventional-access`** becomes the unique *game-structure-class* identifier under heterogeneous Tier structures — the composite-layer analog of the single-agent interventional-access-escape for Instance 1, broadened from coupling-sign to the full game-structure object (cooperative-target vs strategic-equilibrium vs cyclic-Nash) per the 2026-05-21 broadening.

3. `#scope-composite-agent` acquires **broadened load-bearing enabling status**: the scope-route disjunction (C-i / C-ii / C-iii alignment routes vs C-iv strategic route) is itself the regime-axis classification — alignment routes enable R0; the strategic route admits R1 or R2 depending on the game-structure-bit. The scope-route choice is what positions the composite within the regime that the escapes can certify. Without scope-satisfaction, the composite is ill-defined and the escapes have no coherent target.

4. `#deriv-strategic-composition`'s sub-scope α'/β' partition becomes **the primary AAT-framework R1-vs-R2 regime classifier under escape (e)** — composite-level convergence-rate-class observation alone gives only "exponential vs polynomial"; the sub-scope α' (potential or monotone game → R1) vs β' (no potential or monotone

structure → R2) partition is what refines the rate-class observation into a regime classification. Without `#deriv-strategic-composition`'s sub-scope machinery, escape (e) operates as a binary rate-class diagnostic rather than a regime classifier.

Tier. *Exact* for the symmetric-matched-Tier-1-scalar coupling-sign-bit construction (this segment's §"No-go") and for the symmetric-matched-Tier-1-scalar cross-regime witness construction (`#deriv-regime-marginal-indistinguishability` §"Witness construction"); same exact-tier statement at the same parameter-by-parameter level. *Robust qualitative* for general heterogeneous composites, inheriting the common-Lyapunov-nonexistence structure from Liberzon 2003 / Shorten et al. 2007 / Dayawansa-Martin 1999 without a closed-form AAT-level counterexample.

Instance 4 — Architecture No-Identifiability from On-Policy Summary Data (`#der-architecture-noidentifiability`). **Setting.** Two AAT agents A, A' operate on the same regime-restricted trajectory under their respective on-policy execution, sharing the same sector-condition summary (α, R) — equivalently, the same ultimate-deviation statistics on similarity-invariant summaries of $\|\delta\|$ in that regime. The observer has *on-policy, in-regime, summary-only* access: no off-policy intervention, no out-of-regime samples, no white-box read of the update rule or internal state. Question: can the architectural content — *which* internal coordinates implement the correction (16.2's "which internal mechanism") — be recovered from the observer's accessible data?

External theorem (dual anchor). *Linear-Gaussian sub-scope (sharp):* **Kalman 1963** / **Ho-Kalman 1966** canonical-form non-uniqueness — minimal state-space realizations sharing an innovation spectrum form a similarity-equivalence class (Anderson & Moore 1979 *Optimal Filtering* §10.4). *General sub-scope (robust qualitative):* the **Causal Hierarchy Theorem** (Bareinboim, Correa, Ibeling & Icard 2022) at the agent-as-SCM layer — two SCMs over the agent's state space agreeing on Level-1 observation data cannot in general be distinguished on Level-2 questions about $do(\cdot)$ on internal coordinates.

No-go. Let the linearized closed-loop residual dynamics of A, A' be minimal state-space realizations (F, σ_w) and (F', σ_w') with F, F' Hurwitz, related by an invertible similarity $F' = TFT^{-1}$, $\sigma_w'\sigma_w'^\top = T\sigma_w\sigma_w^\top T^\top$ for some $T \in GL(n)$. Then the stationary innovation process — hence the (α, R)-summary, the innovation spectrum, and every *similarity-invariant* summary of the on-policy observation law — is identical for A and A' (the on-policy observer who consumes those summaries cannot distinguish them; raw-residual quadratic forms in a fixed external basis are *not* similarity-invariant, so this is a strictly summary-level statement). The architectural content lives entirely in the $GL(n)$ similarity orbit and is annihilated by the on-policy observation map. Derivation: `#der-architecture-noidentifiability` §"No-Go Theorem" — the stationary covariance solves $F\Pi + \Pi F^\top = \sigma_w\sigma_w^\top$ and transforms under similarity as $\Pi' = T\Pi T^\top$, the eigenvalues of F are similarity-invariant, and the innovation spectrum $\det(i\omega I - F)$ is similarity-invariant.

Boundary characterization. Three structurally distinct escapes (cf. `#der-architecture-noidentifiability` §"Boundary Characterization"):

- (a) **Loop-interventional access** (7.3). $do(\cdot)$ perturbs the system along similarity-fiber directions that on-policy data never excites; the intervention generates Level-2 data, the no-go is Level-1-only. Sharp in the linear-Gaussian sub-scope.
- (b) **Higher-moment / out-of-regime observation.** In the nonlinear sub-scope β, two agents matched at (α, R) and second-order generally differ at moments of order ≥ 3. *Provably void in the linear-Gaussian sub-scope* — a Gaussian innovation has no information beyond second order — a sharp scope-statement.
- (c) **Architecture instrumentation.** Direct read of the update rule or internal state (white-box access). Same kind as Instance 2's "instrument the latent" escape; breaks the black-box scope.

(A proposed fourth escape — *horizon extension under the same policy* — provably collapses into (a): more samples of an annihilated observation law cannot escape the annihilation. The honest escape count is three, not four.)

Strengthened consequence. Instance 4 elevates `#der-loop-interventional-access` to a load-bearing role at the **agent-internal architecture layer** — the *third semantically-distinct deployment mode* of interventional access in the framework: (1) agent-self-intervention (Instance 1, causal-sufficiency degeneracy); (2) observer-on-sub-agent intervention (Instance 3, coupling-sign degeneracy); (3) observer-on-agent-input intervention (this instance, similarity-orbit degeneracy). The shared content across modes is "Level-2 data is the unique broadly-available violation of the Level-1 no-go"; the distinct content is *which* Level-1 degeneracy. Instance 4 also discharges, at the mathematical level, the Regime-C confound of `#result-mismatch-decomposition` / `#internal-external-decomposition` — the on-policy split of the mismatch budget into its estimation / state-uncertainty / channel terms (E_π, U_π, C_π) (`#deriv-mismatch-budget-attribution` — there is no model/policy/cross term-split) is this same no-go projected onto the disturbance-statistic coordinate (`#der-architecture-noidentifiability` Working Notes provenance).

Mechanism (in advance of the Sylvester Discussion below). Not Sylvester-for-free — the generating action that produces the indistinguishable pair is state-space *similarity* $T(\cdot)T^{-1}$, not metric *congruence* $S^\top(\cdot)S$. Not a new mechanism either: it reduces to Instance-2's Fisher-information null space along a structurally-forced indeterminacy manifold, with the manifold being a $GL(n)$ Lie-group fiber. The agent's *escape-irreducibility under its metric/coordinate freedom* is Sylvester at one remove (congruence preserves inertia on the observed-Fisher, identically to Instance 2). Instance 4 is therefore the **third member of the rank-collapse subclass** {Instance 1, Instance 2, Instance 4}, not a new mechanism family.

Tier. *Exact* in the linear-Gaussian / Kalman sub-scope (α_1), anchored by Kalman-Ho canonical-form non-uniqueness. *Robust qualitative* in the general sub-scope via CHT-at-agent-as-SCM. The exact / robust-qualitative tier boundary is load-bearing — collapsing to flat "exact" overclaims the general case; collapsing to flat "robust qualitative" understates the sub-scope's sharpness. The mechanism reduction (Instance-2 Fisher-null on a Lie-group fiber) and the escape count (three, not four; Fano degenerates at $I = 0$, the right tool for the finite-sample refinement rather than the exact anchor) are *exact*.

Adjacent Floors (Open Research Directions).

Causal-IB Extension for Interventional Relevance Variables. The standard Information Bottleneck (2.2) and the four AAT compression operations (`#disc-compression-operations`) work with associational relevance variables — Y in a joint distribution with X and T. Strategy edges in the regime-indexed interpretation (9.5) want *interventional* relevance for Regime A: "what edge (i, j) predicts under $do(i)$, not under observation of i." Standard IB cannot supply this; a causal-IB variant (Wieczorek & Roth 2017 and follow-ups) is the natural framework. The expected floor: under purely associational data, the interventional relevance content is bounded; the gap is recoverable via do-data from the loop. Open: formalize the gap as an identifiability floor parallel to Instances 1 and 2.

Misspecification-Cost Quantification. Structural adaptation (4.8) names *when* to switch model classes. It does not quantify the *continuous degradation* from a mildly misspecified model under a finite information budget. Adjacent to `#scope-observation-ambiguity-modulation`'s $\kappa \cdot \mathcal{A}$ bound but not covered by it. The expected floor: under fixed information budget, the degradation rate from misspecification is bounded below by an information-theoretic quantity (likely related to the KL gap between true and assumed model classes). Open.

Tier-Switching Policy Cost. Approximation tiering (`#disc-approximation-tiering`) enumerates AAT's tiered approximations (L0/L1/L1'/L2 in correlation, C1/C2/C3 in convention, Tier 1/2/3 in contraction). The cost of switching tiers — when should the agent move from L0 to L1, or from C1 to C2 — is itself a deliberation-cost problem. Under finite computation budget, the optimal switching policy faces an identifiability floor on its own switching diagnostics. Open.

Reward-Channel Learning Identifiability (landed in `#deriv-reward-channel-learning-no-go`, not an Instance here). Cohen-Hutter-Osborne 2022's "Advanced artificial agents intervene in the provision of reward" — strengthened 2026-05-22 — is *technically* an Instance shape (CHT external theorem, L1/L2 split, escape menu) sharing the Pearl-Bareinboim 2022 anchor with Instance 1. It nonetheless does *not* land as a fifth instance of this segment, for a structural reason about actor positioning: in Instances 1–4 the actor frustrated by the floor is the *agent itself* (the agent cannot detect causal insufficiency from on-policy data; cannot identify L1' mixtures from single-channel observations; cannot certify composite contraction from component data; cannot identify architecture from on-policy summary statistics), and the unique broadly-available escape elevates *agent-side* machinery (loop-interventional access; latent observability; matched-Tier composite-extended observation; modular interventional access). In the reward-channel learning case the actor frustrated by the floor is the *principal* (cannot distinguish whether the agent has learned the principal-intended world-model from a reward-port-care world-model on on-policy reward data), and the escape menu elevates *principal-side* commitments (physical isolation, horizon restriction, prior engineering, out-of-band intent channels) — agent-exploiting / principal-frustrated rather than agent-frustrated / agent-escaping. The result therefore lands as the second instance of a sister meta-cluster — *agent-side value-functional-grounding* — co-authored with `#deriv-self-actuation-grounding` (Result G′; the within-model failure). The two clusters share the constructive-impossibility-posture shape (`#disc-constructive-impossibility-posture`) but anchor at different layers of the value-functional interface's information narrowness.

Mechanism-Design Impossibility (landed in 5.3, not an Instance here). The mechanism-design impossibility cluster (Gibbard-Satterthwaite 1973-75; Myerson-Satterthwaite 1983; Arrow 1951) was tested as a candidate fourth instance against this segment's five-element shape and found *not* to fit on actor-positioning grounds: the actor frustrated by the no-go is the *designer* implementing a mechanism, not an *agent* inferring from data; the remedy is *design-constraint relaxation* (Bayes-Nash IC; preference-domain restriction; randomization; subsidies), not *information augmentation*; the mechanism is *combinatorial-topological* (Reny 2001), not Sylvester-rank-collapse or projection-closure. The cluster therefore lands as a sister meta-pattern at the same boundary face of `#disc-stability-certificate`, not as an instance of this floor: `#disc-implementation-impossibility` carries the three charter instances (`#deriv-strategy-proofness-impossibility` GS, `#deriv-bilateral-trade-impossibility` MS, `#deriv-social-welfare-aggregation-impossibility` Arrow). The actor-and-remedy contrast — agent-side data-inference vs designer-side construction — is what makes the two sister meta-patterns peer rather than nested. The agent-side / designer-side suffix discipline tracks the structural distinction at canon naming-pattern depth: agent-side capacity-on-the-agent → *-floor* (this segment, `#disc-value-functional-grounding-floor`); designer-side task-of-the-designer → *-impossibility* (`#disc-implementation-impossibility`).

Epistemic Status.

Discussion-grade at the methodology level. The five-element shape is a recurring move AAT applies, not a theorem in its own right; the segment's contribution is the methodology's articulation plus the recognition that AAT's machinery (loop-interventional access, observability-as-information-augmentation, matched-Tier composite-extended observation, white-box architecture instrumentation) acquires sharper load-bearing roles when read through the floors that motivate it. That cross-instance observation is *discussion-grade* — it is the theory's architecture made visible, not a derived theorem.

Individual instances retain their own, higher, epistemic status. Instance 1 (on-policy L0 insufficiency detection, via the Causal Hierarchy Theorem) is *exact* for shallow strict-prerequisite cases and *robust qualitative* for general DAG topology, derived in 9.2. Instance 2 (L1' under unobservable common cause, via the Cramér-Rao bound on Fisher information) is *exact* under the Fisher rank-1 calculation in `#deriv-edge-credence-dynamics` Prop B.7 refutation. Instance 3 (composite contraction certification, via Liberzon common-Lyapunov-nonexistence) is *exact* for the symmetric-matched-Tier-1-scalar witness construction and *robust qualitative* beyond. Instance 4 (architecture noidentifiability, dual-anchored on Kalman-Ho non-uniqueness + CHT-at-agent-as-SCM) is *exact* in the linear-Gaussian sub-scope and *robust qualitative* in the general sub-scope. Readers citing this segment for a specific no-go should cite the instance's own derivation, not the methodology.

Whether the methodology is generative — whether future AAT work will systematically encounter and derive more instances rather than just retrospectively organize derived ones — is a *hypothesis* the adjacent open floors test. The four-instance catalog as it stands is sufficient to articulate the shape; the hypothesis is whether the shape is predictive of future no-gos beyond the catalog.

Max attainable: *discussion-grade* for the methodology (it is a presentational organizing principle, not a derivation). The individual instances retain their own epistemic status as derived above.

Discussion.

The pattern is asymmetric. Each floor forbids inference *from* limited data; it does not forbid inference *with* the augmenting capability. The asymmetry is the source of the pattern's positive content — it tells the reader exactly what to instrument, observe, or intervene upon to escape the floor.

The asymmetry has a named mechanism for the rank-collapse instances: Sylvester's law of inertia. For the instances whose floor is a rank-deficiency of an information operator — Instance 2 (Fisher rank-1 under an unobservable common cause) cleanly, Instance 1 (the causal-parameter block carries no score under a purely observational regime) structurally, and Instance 4 (the on-policy observation Fisher is identically zero along the $GL(n)$ similarity-fiber tangent distribution; `#der-architecture-noidentifiability` §"Mechanism") at one remove — the reason no change of coordinates escapes the floor is one theorem. The agent's only representational freedom on the *escape side* is the choice of coordinate / metric on the observation model, and every such change acts on the information operator by *congruence* ($\mathcal{G} \mapsto S^\top \mathcal{G} S$ for invertible S — the standard Fisher-information reparameterization law). Sylvester's law of inertia states that congruence preserves inertia: the number of zero eigenvalues is invariant under every invertible reparameterization. So a rank-deficient information operator is rank-deficient in *every* coordinate; the floor is the boundary of the positive-definite cone, and the entire reparameterization freedom is a congruence orbit that cannot cross that boundary. This is why the only escape is *rank-augmentation* — adding a genuinely new score component (interventional data via the loop, a side channel, a witness), which is not a coordinate change — and never reweighting. The per-instance "no reparameterization removes the degeneracy" computations (the Fisher rank-1 calculation in `#deriv-edge-credence-dynamics` Prop B.7; the on-policy no-detection argument in 9.2; the similarity-fiber Fisher-null in `#der-architecture-noidentifiability`) are special cases of this single law.

Instance 4's mechanism is Sylvester-at-one-remove, not Sylvester-for-free. The *generating* group action that produces Instance 4's indistinguishable architectural pair (A, A') is state-space *similarity* $F \mapsto TFT^{-1}$ on the realization manifold — *not* the metric congruence Sylvester governs. Sylvester does not apply at the generating layer; the two group actions are categorically different (similarity acts on operators by conjugation; congruence acts on quadratic forms by transposed-conjugation). What Sylvester *does* control is the agent's potential *escape* by reparameterizing its observation model: once Instance 4's Fisher-null direction is recognized on the realization manifold, congruence preserves inertia on the observed-Fisher just as in Instance 2. The Lie-group fiber is the *additional* structure beyond Instance 2's single mixture-indeterminacy manifold; the *kind* of obstruction is the same. Instance 4 is therefore a member of the rank-collapse subclass, with the Sylvester-mechanism doing the same escape-irreducibility work it does for Instances 1 and 2, applied at one remove from the generating action.

The mechanism is specific to the rank-collapse subclass, not to all floors. Instance 3 (composite contraction certification) is a *different* obstruction: composition is a non-invertible projection, not a congruence, so its no-go is a Schur-complement / memory-kernel statement (the certificate's metric survives projection but its dynamic guarantee does not), not an inertia statement. That the floors do not share one mechanism — Sylvester for rank-collapse (three members: {Instance 1, Instance 2, Instance 4}, the third at one remove), a projection/closure obstruction for composition ({Instance 3}) — is itself load-bearing: it is why the floor pattern is a presentational family rather than a single theorem.

The pattern composes with AAT's scope honesty. Directed separation (6.3) classifies architectures by where Part II's exact results apply (Class 1 Separated; Class 3 Coupled needs coupled formulation). The identifiability floors are a different kind of scope claim: they specify what the theory's machinery *cannot do* under specific information regimes, with explicit characterization of the regime escapes. Together, the architectural classification and the identifiability floors mark AAT's scope at two levels — what kinds of agents the theory applies to, and what those agents can and cannot infer from given data.

Complementarity with the separability pattern (5.4). This segment names the *negative half* of AAT's scope; the companion meta-segment 5.4 names the *positive half* — separable-core / structured-repair / general-open across six ladders (correlation, convention, architecture, contraction, identification, scope). Each identifiability-floor instance here has a positive counterpart there: Instance 1's on-policy detection no-go matches the observable-sibling-covariance structured-repair in the correlation ladder; Instance

2's unobservable-C L1' refutation matches the observable-C / facilitator-monotonicity structured-repair in the same ladder. The two halves together characterize AAT's scope at both extremes — what succeeds and under what machinery, and what structurally cannot succeed without specific information augmentation.

This segment is the *boundary facet* of the stability certificate (5). The rank-collapse floors are the certificate dropping rank — the boundary of the positive-definite cone whose interior is operator-sector — with the Sylvester-law irreducibility (above) the statement that the framework's representational freedom is a congruence orbit that cannot cross that boundary. Read through the spine, this segment answers "where does the agent's measuring-stick go flat, and why does no re-graduation un-flatten it"; 5.4 answers "where does a stick exist at all" (scope-of-existence facet) and 5.5 "which stick is forced" (forced-identity facet).

The pattern is conservative in style. Each floor invokes a published external theorem (Bareinboim et al. 2022; standard Cramér-Rao bound) rather than deriving a new impossibility result. AAT's contribution is the *application* — recognizing the AAT setting falls within the theorem's scope, characterizing the boundary conditions, and identifying which AAT machinery is the unique broadly-available escape. This style aligns with the broader posture of AAT as an integrating framework that connects established results across control theory, causal inference, and information theory under a common formalism.

Findings.

The Identifiability Floor as Cross-Cutting Meta-Pattern. **Brief:** A detective trying to identify a suspect from only crowd-witness reports faces a structural impossibility — not a harder version of the same task, but a no-go: two suspects who look identical from the crowd's vantage point produce identical witness reports, and no amount of careful re-analysis of those reports can adjudicate. The only route out is a fundamentally different *kind* of evidence — a fingerprint, a genetic test, a targeted interventional question to a single witness. AAT recurrently applies this shape across four findings: an inferential task (detect a latent common cause, identify a mixture parameter, certify composite contraction, recover an internal architecture) is shown structurally impossible under a stated data regime (on-policy observations, single-channel readings, component marginals, on-policy summaries) by importing a published external theorem (Pearl-Bareinboim causal hierarchy; Cramér-Rao on Fisher information; Liberzon common-Lyapunov-nonexistence; Kalman-Ho canonical-form non-uniqueness dual-anchored with CHT-at-agent-as-SCM); the conditions under which the regime fails identify exactly which AAT machinery — loop-interventional access, observable latents, matched-Tier composite-extended observation, white-box access — is the unique broadly-available escape; the floor elevates that machinery from "useful additional tool" to "structurally required." Three of the four floors (Instances 1, 2, 4) belong to a *rank-collapse subclass* whose escape-irreducibility under coordinate change is Sylvester's law of inertia — you cannot survey your way out of a blind spot by changing the units on the ruler; you have to look from a new vantage point (Sylvester applies directly for Instances 1 and 2; at one remove for Instance 4, where the *generating* action is state-space similarity rather than metric congruence, but the *escape-side* irreducibility is identical to Instance 2's). Instance 3 is the projection/closure member — a structurally distinct Schur-complement / memory-kernel obstruction (a metric that survives restriction to a sub-system may not survive the projection back to the composite). The plurality of mechanism — Sylvester for the rank-collapse trio, projection/closure for composition — is itself load-bearing: it is why the methodology is a presentational family rather than a single theorem.

Impact: Surfaces a unifying frame for results that would otherwise read as disparate impossibility theorems. Each instance gains interpretive context (it is one of an emerging class) and the meta-segment provides a consistent template for evaluating candidate future floors (causal-IB extension, misspecification cost, tier-switching cost, mechanism-design impossibility). The pattern's positive content — that each floor names exactly which AAT machinery the theory requires to escape it — converts a sequence of negative results into a structural argument for the load-bearing status of the loop's interventional access (Instances 1, 3, and 4 — *three* semantically distinct deployment modes: agent-self-intervention breaking causal-sufficiency degeneracy / observer-on-sub-agent intervention breaking coupling-sign degeneracy / observer-on-agent-input intervention breaking similarity-orbit degeneracy) and of observability-as-information-augmentation (Instance 2, with the "instrument the latent" escape also available in Instance 4's white-box route). The complementary `#disc-separability-pattern` carries the positive half (separable-core / structured-repair / general-open across eight ladders as of 2026-05-21 — Correlation, Convention, Architecture, Contraction, Identification regime, Scope hierarchy, A2'-scope, Dynamic regime); together the two meta-segments mark AAT's scope honestly at both extremes.

Novelty Claim: *Claim recognition* of structural pattern across four AAT results that import external information-theoretic theorems to derive impossibility statements with mapped boundary-route escapes; the meta-pattern is an organizing principle rather than a theorem, and the per-instance prior-art positioning lives in the instance segments (`#der-causal-insufficiency-detection`, `#deriv-edge-credence-dynamics`, `#deriv-critical-mass-composition` / `#result-contraction-template`, `#der-architecture-noidentifiability`).

Related Work:

The four instances import distinct external theorems; per-instance prior-art landscapes live in the instance segments. The meta-pattern itself has no direct anticipation in the AAT-adjacent literature surveyed; the closest cousins are scope-honesty moves in formal-method literatures (Liberzon's switched-systems analysis, which itself instances the pattern; Cramér-Rao tradition in identifiability theory) and the broader posture of conservative-form causal inference (Pearl 2009, *Causality*; Bareinboim et al. 2022). What the meta-pattern adds is the cross-instance observation: AAT repeatedly recognizes that its own setting falls within a published external impossibility theorem, characterizes the boundary-route escape via existing AAT machinery, and uses the floor to elevate that machinery to load-bearing status.

Table 5.2

Pattern element	*Where the move is established in the prior literature*	*Relationship / Positioning*
Importing CHT to derive on-policy detection no-go (Instance 1)	Bareinboim, Correa, Ibeling & Icard 2022 (published 2022, found 2025)	*formal antecedent* — see `#der-causal-insufficiency-detection` Findings for the full Pillar-1 prior-art table; the meta-pattern subsumes this instance
Importing Cramér-Rao to derive mixture-identifiability no-go (Instance 2)	Cramér 1946, *Mathematical Methods of Statistics* (published 1946, found 2025-04)	*formal antecedent* — Fisher rank-deficiency forces the floor; see `#deriv-edge-credence-dynamics` Prop B.7 for the explicit calculation
Importing common-Lyapunov-nonexistence to derive composite-contraction no-go (Instance 3)	Liberzon 2003, *Switching in Systems and Control* §2.1; Dayawansa & Martin 1999 *IEEE TAC* 44:751; Shorten et al. 2007 *SIAM Review* 49:545 (found 2025-04)	*formal antecedent* — the symmetric-matched-Tier-1-scalar counterexample uses the standard switched-systems machinery; see `#deriv-critical-mass-composition` and `#result-contraction-template` for the AAT-internal closures
Importing Kalman-Ho canonical-form non-uniqueness (sharp sub-scope) + CHT-at-agent-as-SCM (general) to derive architecture-noidentifiability (Instance 4)	Kalman 1963 *J. SIAM Control* 1(2):152–192; Ho & Kalman 1966 *Regelungstechnik* 14:545–548; Anderson & Moore 1979 *Optimal Filtering* §10.4 (found pre-2026); Bareinboim, Correa, Ibeling & Icard 2022 *On Pearl's Hierarchy and the Foundations of Causal Inference* (found 2025)	*formal antecedent (dual)* — adopted with citation; AAT-distinctive recognition is that the architectural d.o.f. of `#der-agent-opacity` *is* the similarity orbit. Mechanism reduction (Instance 4 = Instance-2 Fisher-null on a $GL(n)$ Lie-group fiber, Sylvester at one remove) derived in `#der-architecture-noidentifiability`

Pattern element	*Where the move is established in the prior literature*	*Relationship / Positioning*
Cross-instance pattern recognition	No direct anticipation surfaced at search depth	*claim novelty* under cursory search — the meta-pattern as an articulated framework has not been found in the surveyed literatures; the constituent moves (scope-honesty, importing external no-go theorems, boundary characterization) are individually well-precedented but their cross-instance unification within an integrated agent-theoretic framework is a presentational contribution rather than a borrowed schema

Search Log:

- 2026-04 (*nominally comprehensive at the per-instance level*): the prior-art defense covered Instance 1's prior-art landscape (causal bandits / causal MDPs under hidden confounding); the meta-pattern itself was not the search target. Instances 2–4 inherit per-instance comprehensiveness from their constituent segments rather than from a cross-instance search.
- 2026-04 (*intuition-only* on the meta-pattern as an articulated framework): The cross-instance recognition that four AAT results share the *setting* → *external theorem* → *no-go* → *boundary characterization* → *strengthened-consequence* shape has not been searched for as a unified concept. Targeted future search candidates: scope-honesty traditions in formal verification (refinement-mapping work; Lamport's TLA; Abadi-Lamport refinement); identifiability-theory meta-frameworks; the broader "no-go theorems" literature in physics and economics. Expected outcome: the constituent moves are well-precedented; the unified framework as applied to an integrated agent theory may be novel under cursory search but is unlikely to be novel under comprehensive search of methodological-essay literature.

The Rank-Collapse Floor's Irreducibility Is Sylvester's Law of Inertia. **Brief:** Picture the agent trying to pin down a parameter as localizing a point in a landscape whose curvature is the information it has. A "floor" is a *flat direction* in that landscape — a direction the data cannot resolve (rank-deficient Fisher information). The agent's only freedom is to re-draw its map: choose different coordinates, a different metric. Sylvester's law of inertia says that re-drawing the map with any invertible change of coordinates can bend, rotate, and rescale the contours but can never change *how many genuinely flat directions* exist at a point — a flat valley stays flat in every coordinate system. So when the information operator is rank-deficient, no choice of representation recovers the lost direction; the only thing that ever helps is *taking a new measurement* — adding genuinely new information (an intervention, a side channel, a witness), which is not a re-drawing of the map. A thoughtful non-specialist can re-derive the qualitative claim from the analog alone: you cannot survey your way out of a blind spot by changing the units on the ruler; you have to look from a new vantage point. This is the exact mechanism behind every rank-collapse identifiability floor, and it explains in one sentence why those floors are escapable only by capability, never by cleverness of representation.

Impact: Converts a sequence of per-instance "we checked, no reparameterization removes the degeneracy" computations (Fisher rank-1 in `#deriv-edge-credence-dynamics` Prop B.7; on-policy no-detection in 9.2; the similarity-fiber Fisher-null in `#der-architecture-noidentifiability` §"Mechanism") into one named classical theorem (Sylvester 1852), sharpening the meta-pattern's load-bearing claim. It converts the prior honest-but-soft positioning "the identifiability floor is *orthogonal* to the contraction/operator-sector machinery" into the sharp geometric statement: operator-sector is the *interior* of the positive-definite (information / stability-certificate) cone; the rank-collapse floor is its *boundary*; and the boundary is invariant under the agent's entire representational freedom because that freedom is exactly the congruence orbit Sylvester's law fixes. The escape routes named per instance (loop-interventional access; observability of latents; architecture instrumentation) are unified as *rank-augmentation* of the information operator — the unique category of move that is not a congruence. The result also bounds its own scope honestly: it is the mechanism for the rank-collapse subclass only — three members {Instance 1, Instance 2, Instance 4}, the third applying Sylvester at one remove from the generating action (which is state-space similarity, not metric congruence; the *escape-side* irreducibility is identical to Instance 2's). Instance 3 (composite contraction certification) has a structurally different obstruction — composition is a non-invertible projection, so its no-go is a Schur-complement / memory-kernel statement, not an inertia statement. That the floors are *plural in mechanism* (Sylvester for the rank-collapse trio; a projection-closure obstruction for composition) is itself the reason the floor pattern is a presentational family and not a single theorem — this Finding makes that plurality precise rather than leaving it as "they are different concerns."

Novelty Claim: *Claim recognition* that the irreducibility of AAT's rank-collapse identifiability floors is a single named classical theorem — Sylvester's law of inertia applied to the Fisher-information reparameterization law — rather than a coincidence of per-instance computations; and *claim differentiation* that this mechanism is specific to the rank-collapse subclass and provably distinct from the composition floor's projection-closure obstruction. Sylvester's law itself is classical (1852); the contribution is the recognition that AAT's representational freedom is exactly its congruence orbit, which makes the floor's irreducibility a corollary rather than a checked property.

Related Work: - Sylvester, J. J. (1852), "A demonstration of the theorem that every homogeneous quadratic polynomial is reducible by real orthogonal substitutions to the form of a sum of positive and negative squares," *Phil. Mag.* 4(23):138–142 (found 2026-05-14) — *formal antecedent* — the inertia-invariance theorem; applied here to the Fisher-information congruence. - Horn, R. A., & Johnson, C. R. (2013), *Matrix Analysis* (2nd ed.), Thm 4.5.8 (found 2026-05-14) — *formal antecedent* — standard modern statement of the law of inertia used in the argument. - Lehmann, E. L., & Casella, G. (1998), *Theory of Point Estimation* (2nd ed.), §2.5 (found 2026-05-14) — *formal antecedent* — the Fisher-information reparameterization law $\mathcal{G}_\varphi = S^\top \mathcal{G}_\theta S$ that makes every coordinate change a congruence. - Per-instance external theorems (Bareinboim et al. 2022; Cramér 1946) — *adjacent* — established in the instance segments' own Findings; this Finding adds the cross-instance mechanism, not the per-instance prior art.

Search Log: - 2026-05-14 (*targeted*): The recognition arose in the operator-family-unification spike from the certificate-cone framing. Sylvester's law and the Fisher reparameterization law are textbook; the search target was whether the *recognition* "AAT's identifiability-floor irreducibility = Sylvester's law via the Fisher congruence" appears as an articulated statement in the identifiability-theory or causal-inference methodological literature. Not found at this depth; the constituent facts are universally known but the cross-instance unification as the named irreducibility mechanism for an integrated agent theory's floor pattern appears to be a fresh presentational recognition. Expected to remain *recognition*-tier under deeper search (the pieces are classical; the assembly is the contribution).

5.2 *Discussion:* The Value-Functional Grounding Floor — Agent-Side Goal-Anchoring Impossibility

AAT recognizes a second agent-side floor cluster, sister to `#disc-identifiability-floor` and peer to the designer-side `#disc-implementation-impossibility`: a class of structural impossibility results on what the agent's **value-functional interface** V_{O_t} (`#form-objective-functional`) can alone supply for *goal-stability*. Where the identifiability floor names *what an agent cannot infer from limited data* and the implementation impossibility names *what a designer cannot construct under stated constraints*, this floor names *what an agent cannot anchor through the value-functional interface alone*. The shared structural ingredient across instances is the **single-interface commitment** of `#form-objective-functional`: the value functional is the *sole* agent-internal handle on the objective, and that interface — by construction — carries less information than is required to anchor non-degenerate goal-revision against either within-model self-modification or across-model reward-data ambiguity. Two charter instances ship at landing: Result G′ (`#deriv-self-actuation-grounding`), the within-model failure under convention-monotonicity; and the Cohen-2022 strengthening (`#deriv-reward-channel-learning-no-go`), the across-model failure under reward-channel learning. The two failure modes are complementary; the two escape routes they elevate — agent-side adaptive-substrate invariant and principal-side protocol-commitment — *together exhaust the structural escape geometry* the single-interface commitment forces. The cluster's epistemic content is this exhaustion: when the value functional alone cannot anchor goal-stability, anchoring must come either from off-the-objective-substrate (agent-side, via the persistence bound on the adaptive substrate that Result G′ Corollary 2 names) or from principal-side protocol-commitment (the Cohen-2022 escape menu's substrate-level commitments).

The Pattern.

Each instance of this floor has the form:

1. **Setting.** An AAT goal-stability task — sustain a non-degenerate objective O_t against the agent's own self-actuation, or anchor a learned objective on observed-reward data — operating on the value functional $V_{O_t}(\tau)$ as the sole agent-internal interface to the objective.

2. **External (or AAT-internal load-bearing) theorem.** A structural result whose statement and truth are independent of the immediate AAT-side argument: for the within-model failure (Result G′), AAT's own convention-monotonicity property of `#def-value-object` combined with the finite-no-oracle commitment; for the across-model failure (Cohen 2022 strengthened), Pearl-Bareinboim 2022's Causal Hierarchy Theorem applied at the reward-channel-causation layer.

3. **No-go.** The value-functional interface V_{O_t} cannot, by itself, anchor goal-stability under the stated information regime: no convention-invariant infeasibility verdict is agent-available per step (Result G′); no μ_{prox}-vs-μ_{dist} identification is agent-available on protocol-honoring reward data (Cohen-2022). The single-interface commitment of `#form-objective-functional` *is* the source of the narrowness — it is the interface, the *only* interface, and by construction carries less information than the anchoring task requires.

4. **Boundary characterization.** Each instance carries a named escape menu — the conditions under which V_{O_t}'s narrowness *fails to bind*, with each escape mapping to an off-the-objective-substrate commitment (agent-side or principal-side). For Result G′: terminal grounding via the adaptive-substrate persistence invariant (`#result-persistence-condition`, agent-side). For Cohen-2022 strengthened: five-premise (R1)–(R5) menu — physical isolation, capability restriction, horizon restriction, engineered prior, out-of-band intent channel (principal-side substrate-level commitments).

5. **Strengthened consequence.** The cluster's strengthened consequence is the *terminal grounding geometry* the single-interface commitment forces. Anchoring must come from somewhere other than V_{O_t} — either agent-side (adaptive-substrate invariant; the persistence bound on the adaptive substrate is the canonical instance) or principal-side (substrate-level protocol-commitment; isolation, myopia, prior-design, out-of-band channels). The two routes are *complementary* — agent-side closes the within-model self-revision question; principal-side closes the across-model reward-learning question. *Together they exhaust the structural escape geometry.* The cluster's content is that the *value-functional interface's narrowness has no escape internal to the interface*; the only escapes are off-interface, and those off-interface escapes are precisely the two recognized terminal grounding routes.

The pattern is *not* a negative posture about the value functional. The framework does not say "value functionals are too narrow to be useful." It says: *the value functional is too narrow to anchor goal-stability alone, and therefore anchoring must come from off-substrate or off-interface invariants the framework already recognizes (the persistence bound on the adaptive substrate; principal-side protocol-commitments).* The narrowness is what *forces* the framework to be honest about where anchoring comes from.

Current Instances.

Instance F — Within-Model Self-Actuation Grounding (Result G′; `#deriv-self-actuation-grounding`). Setting. A non-degenerate self-actuated agent attempts to ground its own objective-revision on an agent-internal objective-functional. The agent has the capacity to self-modify O_t (self-actuation per the agent spectrum `#def-agent-spectrum`), operates per AAT's Part II setup, and has access only to the value functional $V_{O_t}(\tau)$ as the agent-side interface to the objective.

Load-bearing theorem (AAT-internal). Convention-monotonicity of `#def-value-object` (the convention hierarchy C1/C2/C3 of value evaluation) combined with the finite-no-oracle commitment — Result G′ Lemmas 1 (static-pointwise convention-collapse) and 2 (finite-step decidability impossibility).

No-go. No Φ satisfying Result G″s (R1)–(R4) premises can be constructed from the value-functional-interface alone. The engine: convention-monotonicity within a fixed model produces an infeasibility verdict that is *not convention-invariant* under the agent's own per-step decision rule; the finite-no-oracle commitment forbids an off-step appeal to oracle-resolution. The value-functional interface V_{O_t} cannot anchor the agent's self-revision without an off-substrate invariant.

Boundary characterization (the canonical escape). The terminal grounding invariant must be *off* the objective substrate, on the **adaptive substrate** — Result G′ Corollary 2 names the persistence bound (`#result-persistence-condition`) as the canonical instance. The agent's continued existence-as-an-agent under Part I's persistence machinery anchors goal-stability without requiring V_{O_t} to do so. The persistence condition is *not* an addition to the value functional; it is on a structurally distinct substrate the framework already requires for agent existence at all. The agent-side terminal grounding is therefore a *recognition*, not a construction: the substrate AAT already requires for the agent's existence is precisely the anchor the value functional cannot supply.

Strengthened consequence. The persistence condition (`#result-persistence-condition`, Part I) is elevated to **load-bearing for goal-stability** — the top of the agent spectrum closes back onto AAT's foundation. The cluster's first instance is the within-model failure (convention-monotonicity engine; agent-side route), and the strengthened consequence is the agent-side terminal grounding via adaptive-substrate persistence.

Tier. *Conditional, scoped* (per `#deriv-self-actuation-grounding`'s frontmatter): conditional on the three premises (R1) non-degeneracy / (R2) finite-no-oracle / (R3) agent-internal-self-actuatable; scoped to AAT-covered objective-side machinery.

Instance G — Across-Model Reward-Channel Learning (Cohen-2022 strengthened; `#deriv-reward-channel-learning-no-go`). Setting. An agent learns its objective by observing rewards. The reward functional $V_{O_t}(\tau) = \sum_k \gamma^k r_k$ (or any other monotone reward-functional) is the agent's sole interface to the objective; rewards arrive through a physical **reward-provision protocol** π^{rp} that the principal intends to correlate with a world-state feature $f^{\text{principal}}(\omega_k)$.

External theorem. Pearl-Bareinboim 2022 Causal Hierarchy Theorem, applied at the reward-channel-causation layer — same external anchor as `#disc-identifiability-floor` Instance 1, *different* AAT-side wiring.

No-go. Two world-models, μ_{dist} (reward depends on the principal-intended world-state feature) and μ_{prox} (reward depends on the bit-arrival mechanism at the agent's reward-input port), are *Level-1-equivalent* on every protocol-honoring history and *Level-2-distinct* on $do(\pi^{\text{rp}})$-queries. The agent's posterior over $\{\mu_{\text{prox}}, \mu_{\text{dist}}\}$ does not concentrate from on-policy reward data alone. Under (R1)–(R5) of `#deriv-reward-channel-learning-no-go` (advanced-agent capability, EU-max with non-vacuous μ_{prox} prior, sole-reward goal channel, finite-non-trivial VOI, L2-perturbable protocol), the EU-optimal policy assigns positive measure to a policy that intervenes on the reward-provision protocol.

Boundary characterization (the escape menu). Five escapes, each elevating a piece of AAT machinery or principal-side adjacent machinery from "useful design choice" to "structurally required to escape the no-go":

- *(R5) Protocol-L2-perturbable* — escape via physical isolation; AAT machinery: `#der-class-coercion-via-wrapping` W_1 strict wrapper; canonical implementation: Cohen-Vellambi-Hutter 2020 isolated environments.
- *(R1) Advanced-agent capability* — escape via sub-advanced-tier capability or imitation learning; AAT machinery: `#def-agent-spectrum` below the self-actuated tier.
- *(R4) Finite-VOI / long horizon* — escape via myopia, risk-aversion, quantilization; AAT machinery: `#def-value-object` horizon N_h restriction; Taylor 2016 quantilization.
- *(R2) Non-vacuous μ_{prox} prior* — escape via principal-engineered prior bias; **principal-side adjacent**, not currently in AAT canon.
- *(R3) Sole-reward goal channel* — escape via privileged out-of-band intent channel; **principal-side adjacent**, not currently in AAT canon.

Strengthened consequence. Each (R1)–(R5) escape elevates a piece of AAT machinery or principal-side adjacent machinery from "useful design choice" to *structurally required*. The cluster's second instance is the across-model failure (CHT-at-reward-channel engine; principal-side route), and the strengthened consequence is the principal-side terminal grounding via substrate-level protocol-commitment.

Tier. *Exact* for the floor (Lemma 1, CHT-at-reward-channel; direct application of the CHT). *Conditional-derived* for the behavioral corollary (Lemma 2, EU-max behavioral instability under (R1)–(R5)).

The Unification — Two Complementary Terminal Grounding Routes Exhaust the Geometry.

The cluster's epistemic content is the **unification of Instances F and G under the shared structural fact**: the value-functional interface V_{O_t}, as the sole handle on the objective (`#form-objective-functional`'s single-interface commitment), carries less information than required to anchor non-degenerate goal-revision. The two failure modes are complementary readings of the same under-determination:

Table 5.3

view	*mechanism*	*what V_{O_t} is too narrow for*	*terminal grounding route*
Instance F (within-model)	C1/C2/C3 convention-split + finite-no-oracle (Result G′ Lemmas 1+2)	A convention-invariant infeasibility verdict that is agent-available per step	Adaptive-substrate invariant (the persistence bound; Result G′ Corollary 2) — **agent-side**
Instance G (across-model)	CHT-at-reward-channel; L1/L2 split (Cohen-2022 strengthened Lemma 1)	A μ_{prox}-vs-μ_{dist} goal-identification on observed reward data	Principal-side substrate-level commitments (isolation / myopia / quantilization / prior-design) — **principal-side**

Both reductions are *failures of V_{O_t} to carry enough information to anchor goal-stability*. The two terminal grounding routes are *complementary*: the agent-side route closes the within-model self-revision question (the engine is convention-monotonicity); the principal-side route closes the across-model reward-learning question (the engine is CHT-at-reward-channel). **Together they exhaust the structural complement of the value-functional interface's information narrowness** — there is no third route internal to the value functional, and no fourth route off-substrate (the two off-substrate axes are agent-side adaptive-substrate and principal-side protocol-substrate; there are no other axes a non-degenerate non-pathological agent can be grounded on).

The two no-gos *compose*: a self-actuated agent whose O_t is learned from reward observation is *simultaneously* a Result-G′-subject (under-anchored from inside) and a Cohen-2022-subject (under-determined from below); the *only* terminal grounding routes are the two complementary ones named above.

Adjacent Frontiers (Open Research Directions).

Cross-Agent Value-Functional Narrowness (raised by `#deriv-social-welfare-aggregation-impossibility`). The cluster's current charter instances are both *within-agent* (the value functional's narrowness is on the agent's own goal-anchoring). A parallel question, raised by Charter Instance 3 of `#disc-implementation-impossibility` (Arrow), is whether the value-functional formalism aggregates honestly across *sub-agents* in a composite — whether the per-sub-agent $V_{O_t^{(i)}}$ can be canonically combined into a designer-side aggregator (utilitarian summation, weighted utilitarianism). If the answer is "no, the cross-agent value-functional aggregation is also structurally narrow," that would extend this cluster to a third charter instance under a cross-agent narrowness reading. The open avenue is whether the value functional's narrowness is *within-agent only* or *cross-agent also*.

Assistance-Game and Out-of-Band Channel Formalization (raised by `#deriv-reward-channel-learning-no-go` (R3)). Cohen 2022's §"The Assistance Game" shows the no-go *generalizes* to settings with non-reward channels under subtler ambiguity (human-centric vs record-centric models). Formalizing the assistance-game case in AAT would extend Instance G's boundary characterization on the (R3) escape from "outside AAT canon, principal-side adjacent" to "AAT-formalized assistance-game extension." Open.

Higher-Order Reflective-Oracle Constructions (delimited counter-literature). Fallenstein-Taylor-Christiano 2015's reflective oracles prove consistent self-*reference* over an *exogenous* payoff — a primitive-reflective-oracle regime the cluster's (R3)+(R5) premises exclude from Instance G and that Instance F's finite-no-oracle premise (R2) excludes structurally. Whether a higher-order reflective construction could circumvent either floor by introducing oracle-like resolution on the cardinal/ordinal axis is open. The current scope-statement is that *the floors apply to non-pathological agents on AAT-covered objective-side machinery*; reflective-oracle constructions are *outside scope*, not refutations of the floors. Listed honestly for completeness.

Why This Pattern Matters.

Closes the third cluster of `#disc-constructive-impossibility-posture`. The posture catalog now spans three structurally distinct clusters at the boundary face of the stability-certificate spine: agent-side data-inference impossibility (`#disc-identifiability-floor`; Instances A–C), agent-side value-functional-grounding impossibility (this cluster; Instances F–G), and designer-side mechanism-design impossibility (`#disc-implementation-impossibility`; the GS/MS/Arrow charter). The three-cluster taxonomy is fully realized when this cluster lands; the constructive-impossibility-posture style is now visible across three deployments rather than one or two.

Names the single-interface commitment as a load-bearing source of structural narrowness. Before the Cohen-2022 strengthening, Result G′ carried the within-model side alone, with the cross-model side implicit in `#form-objective-functional` Discussion. The Cohen-2022 strengthening surfaced the cross-model side as a derived no-go, and the unification with Result G′ revealed both as instances of a single structural fact: the value functional is the *sole* agent-internal handle on the objective, and that interface is information-narrow in a load-bearing way. The cluster's epistemic content is the explicit recognition of this single-interface commitment as the engine behind both no-gos.

Exhausts the structural escape geometry. The two complementary terminal grounding routes — agent-side adaptive-substrate invariant + principal-side substrate-level commitment — *together exhaust* the off-interface escape axes. There is no third route internal to the value functional; there are no other off-substrate axes a non-degenerate non-pathological agent can be grounded on. The cluster's content is *not* that the value functional is unusable (it remains the framework's canonical interface to the objective; per-sub-agent within-trajectory evaluation works cleanly); the content is that *the value functional alone cannot anchor goal-stability under the stated information regimes*, and the escapes are precisely the two recognized routes.

Agent-side / designer-side suffix discipline. The cluster's name follows the convention established with the 2026-05-22 Track B landing: agent-side capacity-on-the-agent → *-floor*; designer-side task-of-the-designer → *-impossibility*.

Both Instance F and Instance G are agent-side (the actor frustrated by the floor is the agent itself — Result G′'s self-actuated agent attempting to anchor self-revision; Cohen-2022's EU-maximizer under non-vacuous μ_{prox} prior). The name therefore takes the *-floor* suffix, matching `#disc-identifiability-floor`. The naming-pattern across the three peer meta-segments encodes the agent/designer structural distinction at slug-grep depth.

Epistemic Status.

Discussion-grade at the meta-pattern level. This segment names a structural cluster across two AAT-internal no-go results (Result G′ in `#deriv-self-actuation-grounding`; Cohen-2022 strengthened in `#deriv-reward-channel-learning-no-go`); the cluster recognition itself is a presentational organizing principle, not a derivation, and carries no theorem of its own.

Individual instances retain their own (higher) epistemic status. Instance F is *conditional, scoped* per `#deriv-self-actuation-grounding`'s frontmatter (conditional on (R1)–(R3) of Result G′; scoped to AAT-covered objective-side machinery). Instance G is *conditional* per `#deriv-reward-channel-learning-no-go`'s frontmatter — Lemma 1 (the floor; CHT-at-reward-channel) at *exact* tier; Lemma 2 (the behavioral corollary; EU-max instability) at *conditional-derived* tier under (R1)–(R5).

The segment makes one additional claim: that the two charter instances *unify* under the single-interface commitment of `#form-objective-functional` as the shared structural fact behind two engines (convention-monotonicity within-model; CHT across-model). This is a *discussion-grade* observation about the framework's architecture — visible once the instances are assembled, but not itself a theorem. The unification claim is the segment's headline contribution; whether it lifts to *robust qualitative* under deeper analysis (e.g., whether *every* failure mode of V_{O_t}-anchoring reduces to one of the two engines) is an open question.

Max attainable: *discussion-grade* for the meta-pattern (it is a presentational organizing principle, not a derivation). The individual instances retain their own epistemic status as listed above. The two-engines unification could lift to *robust qualitative* if a stronger structural argument forces both engines under a single deeper principle (currently in hand: both reduce to the *single-interface commitment* of `#form-objective-functional`, which is itself a formulation choice).

Discussion.

Relationship to `#disc-identifiability-floor`. Both are agent-side meta-segments, both end in *-floor*, both anchor (in their charter instances) on the Pearl-Bareinboim CHT — but on *different AAT-side wirings*. The identifiability floor's CHT instance (Instance 1) wires it to the *causal-sufficiency-detection layer* of the strategy DAG; Instance G of this cluster wires the same CHT to the *reward-channel-causation layer* of `#form-objective-functional`. The two clusters are *sister meta-patterns at different layers*: agent-side data-inference layer (`#disc-identifiability-floor`) vs agent-side value-functional layer (this cluster). Both run the same constructive-impossibility-posture three-move discipline (name the floor, name the escape, treat the no-go as load-bearing apparatus); the *layer* differs.

Relationship to `#disc-implementation-impossibility`. Sister peer in a different way. This cluster and `#disc-implementation-impossibility` are *both* recently-named meta-segments landing in the 2026-05-22 cycle. They differ on the agent/designer axis: this cluster is agent-side (actor frustrated is the agent inferring/anchoring); `#disc-implementation-impossibility` is designer-side (actor frustrated is the designer implementing a mechanism). The two clusters together with `#disc-identifiability-floor` complete the three recognized cells of the constructive-impossibility-posture taxonomy: agent-side data-inference / agent-side value-functional-grounding / designer-side mechanism-design. The naming-pattern across the three (-floor / -floor / -impossibility) encodes the agent/designer distinction at slug-grep depth.

Relationship to `#disc-stability-certificate`. This segment is a *boundary-face* facet of the certificate spine, peer to `#disc-identifiability-floor` (agent-side data-inference layer) and `#disc-implementation-impossibility` (designer-side mechanism-design layer). The boundary is where impossibility statements bind; the three meta-segments are the three boundary-layer dialects in which the framework speaks at that boundary. The value-functional layer this cluster names is a *different* boundary-layer from the data-inference layer (where identifiability-floor lives) — both are at the certificate boundary, but they describe distinct narrowness conditions on agent-side machinery.

Relationship to `#disc-constructive-impossibility-posture`. This segment's two charter instances (F and G) are catalog rows in the posture's "Current Instances" table. The posture's catalog gains the cluster's home via this segment landing; the cluster joins the catalog as the third cluster (alongside A–C in identifiability-floor and the queued/landed GS/MS/Arrow under implementation-impossibility). The posture's "Current Instances" framing — Instance Floor / Escape / Apparatus — applies cleanly to both F and G (already catalogued there pre-cluster-landing).

The single-interface commitment as the cluster's structural ingredient. `#form-objective-functional` formalizes $V_{O_t}(\tau)$ as the *sole* agent-internal interface to the objective. That single-interface commitment is a representational choice — alternative formulations could carry multiple interfaces (e.g., a separate satisfaction-channel alongside the value-channel; cf. `#def-satisfaction-gap` for an adjacent diagnostic). The cluster's content is that *under the single-interface commitment*, the value functional carries less information than required to anchor goal-stability, and the escapes are precisely the two off-interface routes. Whether a multi-interface formulation would have a different floor structure is an open framework-scope question; the current cluster operates within the single-interface commitment.

Why the cluster's name is "floor." A *floor* in AAT's meta-pattern vocabulary names a structural lower bound — below the floor, the named capacity is impossible; the floor is the lower limit of what the substrate can supply. This cluster's floor is the lower bound on what the value functional alone can supply for *goal-anchoring*. Below the floor (i.e., under the named premises of Instances F and G), V_{O_t} alone cannot anchor goal-stability; above the floor (i.e., when one of the named escapes is available), anchoring becomes possible via off-interface invariants. The "floor" framing is the same as `#disc-identifiability-floor`'s — a structural lower bound on a *capacity* of the agent's substrate. The agent-side / designer-side suffix discipline reinforces the parallel: both agent-side clusters end in *-floor* because both name lower bounds on agent capacity.

Two-routes-exhausts as the cluster's epistemic load. The unification's headline content is the *exhaustion*: the two complementary terminal grounding routes (agent-side adaptive-substrate + principal-side substrate-level commitment) together close the structural complement of the value-functional interface's narrowness. There is no third route internal to the value functional; there are no other off-substrate axes a non-degenerate non-pathological agent can be grounded on. If a future cycle finds a third route (e.g., via formalizing reflective-oracle constructions, or via lifting cross-agent value-functional aggregation to canonical form), the cluster's exhaustion claim would be revised — but the two routes are currently sufficient.

Findings.

The Value-Functional Grounding Floor — Two Complementary Routes Exhaust the Geometry. Brief: When an agent has an objective and uses a single value-functional interface to evaluate trajectories against it — *the* canonical move in AAT's formulation of agency — two structural impossibilities apply. *Within-model*: if the agent can rewrite its own objective, the value functional alone cannot tell the agent whether its proposed self-modification is feasible per step (Result G′ — the convention-hierarchy of value evaluation produces an infeasibility verdict that is not convention-invariant from inside the agent, and the agent cannot resolve the question with an oracle-like off-step appeal). *Across-model*: if the agent learns its objective by watching rewards arrive through a physical protocol, the value functional alone cannot tell the agent whether the rewards mean what the principal intended (Cohen-2022 strengthened — two world-models, one where the reward depends on a world-state feature and one where it depends on bit-arrival at the sensor, are indistinguishable on protocol-honoring reward histories). Both no-gos are failures of the value functional to anchor goal-stability from inside. The escape geometry is sharply constrained: anchoring has to come from *off* the value-functional interface — either from a part of the agent that doesn't process objectives (the persistence-on-the-adaptive-substrate route Result G′ names; agent-side) or from a commitment the principal makes about the protocol (physical isolation, horizon restriction, prior-design, out-of-band intent channels; principal-side). The two routes together *exhaust* the structural complement: there is no third route internal to the value functional, and no fourth route off-substrate. A reader who carries this away knows that when a framework commits to a single value-functional interface, it has implicitly committed to one of the two terminal grounding routes — agent-side substrate-invariant or principal-side substrate-commitment — and the choice is structural, not a matter of taste.

Impact: Closes the third recognized cluster of `#disc-constructive-impossibility-posture`, completing the constructive-impossibility-posture taxonomy at peer-meta-segment level. The three clusters now span the framework's three structurally distinct boundary-face conditions: what an agent can *infer from data* (identifiability-floor; agent-side data-inference layer), what an agent can *anchor through the value-functional interface* (this cluster; agent-side value-functional layer), and what a designer can *implement under incentive and welfare constraints* (implementation-impossibility; designer-side mechanism-design layer). The two charter instances (Result G′ for within-model; Cohen-2022 strengthened for across-model) unify under the single-interface commitment of `#form-objective-functional` as the shared structural fact behind two engines (convention-monotonicity; CHT-at-reward-channel). The two-routes-exhausts headline is the segment's primary contribution — a structurally-load-bearing recognition that elevates the persistence bound (`#result-persistence-condition`) and principal-side substrate-level commitments (physical isolation, horizon restriction, prior-design) to *the only* terminal grounding routes available under the single-interface commitment. The agent-side / designer-side suffix discipline encodes the structural distinction in the canon naming-pattern.

Novelty Claim: *Claim recognition* of a structural cluster across two AAT-internal no-go results (Result G′ within-model; Cohen-2022 strengthened across-model) sharing the *single-interface commitment* of `#form-objective-functional` as the structural engine — both no-gos are failures of the same interface to carry enough information to anchor goal-stability, with two engines and two complementary terminal grounding routes; *claim differentiation* on the two-routes-exhausts geometry — the agent-side adaptive-substrate route and the principal-side protocol-commitment route together exhaust the structural complement of the value-functional interface's narrowness, with no third route internal to the interface and no fourth route off-substrate. The constituent no-gos (Result G′; Cohen-2022 strengthened) are landed AAT-internal segments; the contribution at this meta-segment is the cross-instance unification under the single-interface commitment + the structural-exhaustion recognition.

Related Work:

Table 5.4

ASF concern	*Prior-art language*	*Relationship / Positioning*
The within-model self-actuation grounding no-go (Instance F)	`#deriv-self-actuation-grounding` (Result G′; this framework, 2026-05)	*internal antecedent* — the within-model no-go and its agent-side terminal-grounding-route via adaptive-substrate persistence
The across-model reward-channel learning no-go (Instance G)	`#deriv-reward-channel-learning-no-go` (this framework, 2026-05-22 Cohen-2022 strengthening cycle); upstream prior-art: Cohen, M. K., Hutter, M. & Osborne, M. A. (2022), "Advanced artificial agents intervene in the provision of reward," *AI Magazine* 43:282–293 (found 2026-05-21); Pearl-Bareinboim 2022 CHT (found 2025)	*internal antecedent + formal antecedents* — the across-model no-go formalizing Cohen-2022's "we argue that" posture to conditional-derived AAT-internal theorem, with the floor (CHT-at-reward-channel) at exact tier
The single-interface commitment of the value functional (the structural ingredient)	`#form-objective-functional` (this framework, AAT canon)	*internal antecedent* — the framework's commitment to a single agent-internal interface to the objective; the cluster's content is that this commitment is the engine behind both no-gos
Self-modification of rational agents (Instance F's external antecedent)	Everitt, T., Filan, D., Daswani, M. & Hutter, M. (2016), "Self-Modification of Policy and Utility Function in Rational Agents," *AGI-16* (primary-source-verified)	*formal antecedent (within-model side)* — the realistic / hedonistic agent split is the within-model analog of Instance F
Causal-influence-diagram reward tampering (Instance G's adjacent prior art)	Everitt, T., Hutter, M., Kumar, R. & Krakovna, V. (2021), "Reward Tampering Problems and Solutions in Reinforcement Learning," *Synthese* 198	*adjacent literature* — distinct paper from Cohen 2022; shares the μ_{prox}-vs-μ_{dist} logic at a different scope

ASF concern	*Prior-art language*	*Relationship / Positioning*
Physically-isolated environment as (R5)-escape construction	Cohen, M. K., Vellambi, B. & Hutter, M. (2020), "Asymptotically Unambitious AGI," *NeurIPS 2020*	*formal antecedent ((R5)-escape construction)* — concretely implements Instance G's (R5) escape
Reflective oracles (delimited counter-literature)	Fallenstein, B., Taylor, J. & Christiano, P. (2015), "Reflective Oracles," *arXiv:1508.04145*	*refuted-by-scope* — proves consistent self-*reference* over an *exogenous* payoff; primitive-reflective-oracle regime excluded from both Instance F and Instance G's premises
Reward hacking unhackability (adjacent cluster)	Skalse, J., Howe, N., Krasheninnikov, D. & Krueger, D. (2022), "Defining and characterizing reward hacking," *NeurIPS 2022*	*adjacent literature (different cluster)* — designer-side reward-proxy design task; structurally a `#disc-implementation-impossibility` candidate instance, not a member of this cluster
The constructive-impossibility-posture style across clusters	`#disc-constructive-impossibility-posture` (this framework)	*internal antecedent* — the cluster joins the posture catalog as the third recognized cluster

Search Log:

- 2026-05-22 (*intuition-only* on the two-routes-exhausts claim): The exhaustion is intuitive once the two engines are named (convention-monotonicity within-model; CHT-at-reward-channel across-model) — these are the only two known engines forcing V_{O_t}-narrowness, and the two terminal grounding routes they produce (agent-side adaptive-substrate; principal-side substrate-commitment) cover the off-interface escape geometry without leaving an obvious gap. Whether a third engine exists is an open question the cluster's discussion-grade tier honestly admits; targeted future search would probe the AI-safety / decision-theory literature for additional under-determination patterns of value-functional interfaces.

5.3 *Discussion:* The Implementation Impossibility — Designer-Side Structural No-Go Results

AAT recognizes a sister class to `#disc-identifiability-floor`'s agent-side no-gos: a cluster of *designer-side* structural impossibility results, where a classical mechanism-design or social-choice theorem forbids an external designer from constructing a composite-design outcome with desired properties under stated incentive and welfare constraints. The actor frustrated by the floor is no longer the agent inferring from data, but the *designer* implementing a mechanism; the remedy is no longer information augmentation (loop-interventional access, latent observability, matched-Tier composite-extended observation), but design-constraint relaxation (Bayes-Nash IC, randomization, monetary transfers, preference-domain restriction). The meta-pattern is the same constructive-impossibility shape — name the floor, name the escapes, treat the no-go as load-bearing apparatus — but the *layer* it operates on is different. This segment names the cluster, ships three charter instances (Gibbard-Satterthwaite, Myerson-Satterthwaite, Arrow), and is honest about what AAT contributes here (precision of the AAT/mechanism-design boundary) and what it does not (no constructive mechanism-design results; most escapes are not AAT machinery).

The cluster is narrow by design. The discipline filter — *constructive-impossibility-shape only* — admits classical theorems where (i) the designer faces a well-stated incentive or welfare regime, (ii) an external theorem forbids the construction under that regime, and (iii) the failure conditions characterize structurally distinct escape routes the designer could take. It deliberately excludes the *constructive* side of mechanism design (VCG, Myerson's revenue-equivalence theorem, optimal-auction theory) — those are positive results, not impossibility-with-escape. AAT's contribution at this cluster is not to invent mechanism design but to recognize *where the AAT-side strategic-composition machinery* (sub-scope α' of `#deriv-strategic-composition`: potential and monotone games) *intersects the classical escape routes*, and to honestly mark where it does not. That precision is the contribution.

Formal Expression.

The pattern. *[Discussion]* An instance of the cluster has the form:

1. **Setting.** A mechanism-design or social-choice task: a designer specifies a constraint regime (which agents, which preference domain, which information structure, which incentive condition, which welfare requirement) and seeks a mechanism — a social-choice function, a trading protocol, a welfare aggregator — satisfying all the constraints simultaneously.
2. **External theorem.** A classical impossibility result from mechanism design or social-choice theory whose statement and truth are independent of AAT: Gibbard 1973 / Satterthwaite 1975 on strategy-proofness; Myerson & Satterthwaite 1983 on bilateral trade; Arrow 1951 on social welfare aggregation.
3. **No-go.** The external theorem proves that no mechanism satisfying all the stated constraints exists. The designer's task is structurally unachievable under the regime.
4. **Boundary characterization.** The conditions under which the regime fails — i.e., the designer has *more freedom* than the regime allows — admit construction; each escape route maps onto a specific category of design-constraint relaxation. Each escape is annotated as *intersects AAT machinery* (where the relaxation operationalizes an AAT sub-scope condition) or *outside AAT machinery* (where the relaxation belongs entirely to the mechanism-design literature).
5. **Strengthened consequence.** The boundary precisely characterizes where AAT machinery contributes to which escape routes, and where the framework's contribution stops. Where escapes intersect AAT machinery (preference-domain restriction ↔ sub-scope α' potential games of `#deriv-strategic-composition`), the intersection is named precisely. Where they do not (Bayes-Nash IC, randomization, monetary subsidies), the segment honestly documents the boundary. The strengthened consequence here is the *map of where AAT machinery meets mechanism design*, not the elevation of new AAT machinery to load-bearing.

The actor-and-remedy contrast with the identifiability-floor cluster is sharp:

Table 5.5

	Identifiability Floor (`#disc-identifiability-floor`*)*	*Implementation Impossibility (this cluster)*
Actor under the no-go	An *agent* inferring from data	A *designer* implementing a mechanism
Information regime	What the agent observes / can infer	What the designer can implement / commit to
Kind of remedy	Augment the agent's data, sensing, interventional access	Relax design constraints; extend the strategy space; introduce transfers
AAT machinery role in escapes	The unique broadly-available escape elevates specific AAT machinery (loop-interventional access; observability-as-augmentation; matched-Tier composite-extended observation)	The escape menu mostly belongs to the mechanism-design literature; AAT intersects via sub-scope α' (preference-domain restriction) and adjacent to `#form-objective-functional` (cardinal preferences)
Strengthened consequence	Elevates AAT machinery from "useful" to *structurally required*	Maps the AAT/mechanism-design boundary precisely

Both clusters share the constructive-impossibility-posture shape (`#disc-constructive-impossibility-posture`); both are boundary-face members of the stability-certificate spine (`#disc-stability-certificate`). They are *sister meta-clusters*, neither subsumed by the other.

Charter instances. The cluster lands with three charter instances. The full setting, external theorem, no-go statement, escape characterization, and AAT-side translation live in the supporting segments; this section summarizes at the level `#disc-identifiability-floor`'s Instance entries summarize.

Instance 1 — Strategy-Proofness Impossibility (Gibbard-Satterthwaite, supported by `#deriv-strategy-proofness-impossibility`). Setting. A social-choice function $f : L^n \to A$ aggregating n agents' strict preference orderings over $|A| \geq 3$ alternatives into a single outcome, under (i) dominant-strategy incentive compatibility (DSIC), (ii) non-dictatorial, (iii) surjective onto A, (iv) unrestricted preferences.

External theorem. Gibbard, A. (1973), "Manipulation of voting schemes: a general result," *Econometrica* 41:587–601; Satterthwaite, M. A. (1975), "Strategy-proofness and Arrow's conditions: existence and correspondence theorems for voting procedures and social welfare functions," *J. Econ. Theory* 10:187–217. No f satisfying all four conditions exists.

No-go. A designer demanding DSIC + non-dictatorial + surjective + unrestricted preferences over ≥ 3 alternatives cannot construct the social-choice function.

Boundary characterization (the escapes). - *Preference-domain restriction* (e.g., single-peaked preferences — Black 1948 admits the median-voter rule). **Intersects AAT machinery** via sub-scope α' of `#deriv-strategic-composition`: single-peaked / single-crossing preference domains map onto the potential-game / monotone-game structural conditions under which the AAT strategic-composition machinery derives best-response convergence. - *Weakened solution concept* (Bayes-Nash IC instead of DSIC — d'Aspremont & Gérard-Varet 1979). **Outside AAT machinery.** - *Randomization* (Gibbard 1977 random-dictator mechanisms). **Outside AAT machinery.** - *Restricted strategy space / enforced truth-telling*. **Outside AAT machinery.**

Strengthened consequence. The sub-scope-$\alpha' \leftrightarrow$ preference-domain-restriction intersection is named precisely: the AAT machinery `#deriv-strategic-composition` already deploys for best-response convergence in strategic composition coincides structurally with one of GS's classical escape routes. The other escapes belong to mechanism-design proper and the framework documents them as outside AAT's current canon. The sub-scope-α' escape does *not* address what GS forbids — it secures best-response *convergence*, not *strategy-proofness* — so the intersection is *adjacent*, not *identical*; that is the honest map.

Tier. *Exact* (the imported theorem). *Derived conditional* for the AAT-side translation (the social-choice-function-to-strategic-composition mapping is mechanical under explicit reading conventions). *Robust qualitative* for the boundary characterization.

Instance 2 — Bilateral Trade Impossibility (Myerson-Satterthwaite, supported by `#deriv-bilateral-trade-impossibility`). Setting. Two-party bilateral trade with private information about valuations (v_b, v_s) drawn from continuous distributions with overlapping supports; the designer seeks a mechanism that is simultaneously (i) ex-post efficient (trade occurs iff $v_b \geq v_s$), (ii) Bayesian incentive compatible (BIC), (iii) individually rational (IR — voluntary participation), (iv) budget-balanced (no external subsidies).

External theorem. Myerson, R. B. & Satterthwaite, M. A. (1983), "Efficient mechanisms for bilateral trading," *J. Econ. Theory* 29:265–281. No such mechanism exists.

No-go. A designer demanding ex-post efficiency + BIC + IR + budget-balance cannot construct the bilateral-trade mechanism.

Boundary characterization (the escapes). - *Subsidies* (relax budget-balance — e.g., VCG with deficit). **Outside AAT machinery.** - *Second-best mechanisms* (Myerson 1981 optimal trade with inefficient withholding). **Outside AAT machinery.** - *Common values* (relax private values). **Outside AAT machinery; adjacent** to `#scope-edge-update-causal-validity`'s regime-indexed identification structure. - *Long-run reputation / repeated interaction* (Folk-theorem-style escapes). **Outside AAT machinery; adjacent** to the strategic-composition machinery in repeated-game form.

Strengthened consequence. The Myerson-Satterthwaite boundary lies almost entirely outside AAT's current canon — and the *honest delineation* is the contribution. The two-agent strategic-composition machinery in `#deriv-strategic-composition` applies cleanly to the agents' best-response dynamics *given* a fixed mechanism; the *choice of mechanism* is the designer's problem and lives outside the framework. The cluster names that boundary precisely rather than overclaiming reach or silently overlooking the gap.

Tier. *Exact* (the imported theorem). *Derived conditional* for the AAT-side translation (clean for the agents-given-mechanism layer; carefully scope-marked for the designer's mechanism-choice layer).

Instance 3 — Social Welfare Aggregation Impossibility (Arrow, supported by `#deriv-social-welfare-aggregation-impossibility`). **Setting.** A social welfare function $F : L^n \to L$ aggregating n agents' strict preference orderings into a single social preference ordering over ≥ 3 alternatives, under (i) unrestricted domain (U), (ii) Pareto efficiency (P), (iii) Independence of Irrelevant Alternatives (IIA), (iv) non-dictatorial (ND).

External theorem. Arrow, K. J. (1951, 2nd ed. 1963), *Social Choice and Individual Values*, Yale University Press. No F satisfying all four conditions exists.

No-go. A designer demanding U + P + IIA + ND cannot construct the social welfare function.

Boundary characterization (the escapes). - *Preference-domain restriction* (single-peaked, single-crossing — admits Condorcet-consistent rules). **Intersects AAT machinery** via the same sub-scope-α' correspondence as GS Instance 1. - *Weakened IIA* (positional methods — Borda count, range voting). **Outside AAT machinery.** - *Supermajority requirements* (e.g., 2/3 thresholds). **Outside AAT machinery.** - *Cardinal preferences* (relax ordinal-only — utilitarianism, range voting). **Outside AAT machinery; adjacent** to `#form-objective-functional`'s value-functional machinery (which carries cardinal value content).

Strengthened consequence. Arrow shares GS's preference-domain $\leftrightarrow$ sub-scope-α' intersection and adds an *adjacent connection* through the cardinal-preference escape and AAT's value-functional interface. Both connections are *adjacent* (intersection / proximity), not elevation of AAT machinery to load-bearing. The honest scope-mark documents both clearly.

Tier. *Exact* (the imported theorem). *Derived conditional* for the AAT-side translation.

Adjacent Frontiers (Open Research Directions).

Roberts 1979 Affine-Maximizers Theorem. Roberts, K. (1979), "The characterization of implementable choice rules," in *Aggregation and Revelation of Preferences*, J. J. Laffont (ed.), North-Holland: under DSIC with monetary transfers on unrestricted preferences, the only implementable choice rules are affine maximizers (weighted utilitarian welfare functions). The pattern fit is genuine — external theorem, structural no-go on a designer's freedom, characterized escape route (restrict preference domain or accept affine form) — and admits a future landing as a fourth charter instance. Held back at present landing to keep the charter scope narrow. The AAT-side intersection (whether affine-maximizer structure connects to AAT's value-functional formalism via cardinal preferences) is a candidate strengthened-consequence content for that future cycle.

Skalse-Howe-Krasheninnikov-Krueger 2022 Reward Hacking Unhackability. Skalse, J., Howe, N., Krasheninnikov, D. & Krueger, D. (2022), "Defining and characterizing reward hacking," *NeurIPS 2022*: no non-trivial proxy-reward pair is unhackable on open policy sets. The actor is the *designer* trying to engineer unhackable reward proxies; the no-go is on the designer's task. Sister to this cluster *in shape*, but structurally a *different* designer-side task (reward-proxy design rather than mechanism design over preference reports). The candidate landing is a future cycle's evaluation against this cluster's discipline filter. The agent-side variant (the agent that exploits the proxy) lives in `#deriv-reward-channel-learning-no-go` and the agent-side value-functional-grounding cluster (`#disc-value-functional-grounding-floor`), not here. Skalse 2022 cross-references both clusters honestly: the designer-side task fits the implementation-impossibility shape; the agent-side exploitation fits the value-functional-grounding-floor shape.

Constructive Mechanism-Design (deliberately excluded by the discipline filter). VCG mechanisms, Myerson 1981 optimal auctions, revenue-equivalence, and the broader constructive mechanism-design literature are *positive results* — characterizations of what *can* be achieved with monetary transfers under stated conditions. They are not impossibility-with-escape; they are construction-with-tradeoff. The discipline filter excludes them by design. If AAT later wants to engage constructive mechanism design as a first-class concern, that is a separate framework-scope decision — not an extension of this cluster.

Why This Pattern Matters.

Closes a missing-but-implied taxonomic category. The agent-side identifiability-floor cluster has been canonical since the framework's earliest meta-pattern work. The designer-side cluster has been *implicit* — flagged in `#deriv-strategic-composition` Discussion — but without a first-class home. The cluster landing closes the asymmetry: the framework now has both an agent-side floor pattern *and* a designer-side impossibility pattern, with the sister-relationship and the contrast precisely characterized.

Maps the AAT/mechanism-design boundary precisely. The contribution is not invention; mechanism design and social-choice theory are mature peer literatures AAT inherits classical theorems from. The contribution is recognition — *where* AAT machinery contributes to a classical escape route (sub-scope $\alpha' \leftrightarrow$ preference-domain restriction), *where* it is adjacent without intersecting (cardinal preferences ↔ `#form-objective-functional`), and *where* the framework's contribution stops (Bayes-Nash IC, randomization, transfers — all outside AAT canon). Mapping that boundary honestly is the kind of scope-honesty that distinguishes integrating frameworks from displacement claims.

Three-cluster constructive-impossibility-posture taxonomy. Together with `#disc-identifiability-floor` (agent-side data-inference) and `#disc-value-functional-grounding-floor` (agent-side value-functional grounding), this segment completes the third recognized cluster of the constructive-impossibility-posture style (`#disc-constructive-impossibility-posture`). The three clusters span the framework's three structurally distinct boundary-face conditions: what an agent can *infer from data* (M1 floors), what an agent can *anchor through the value-functional interface* (the value-functional-grounding floor), and what a designer can *implement under incentive and welfare constraints* (this cluster). The agent-side / designer-side distinction tracks the suffix in canon: agent-side capacity-on-the-agent → *-floor* (`identifiability-floor`, `value-functional-grounding-floor`); designer-side task-of-the-designer → *-impossibility* (this segment). That correspondence is structural, not decorative — an identifiability question and a value-functional-grounding question both name *what the agent's substrate supports* (lower bound on capacity); a mechanism-design question names *what the designer can construct* (binary success/failure on a task). The suffix carries the structure.

Honest scope-marking is contribution. AAT's posture across the framework — naming where machinery applies, naming where it does not, refusing to overstretch — is itself load-bearing. Landing this meta-segment with three charter instances *and explicitly documenting that most of the escapes are not AAT machinery* is the same scope-honesty discipline `#disc-identifiability-floor` applies on the agent side. The framework is not displaced by the mechanism-design literature; the mechanism-design literature is not absorbed by the framework. The two are in dialogue, with the boundary precisely mapped.

Epistemic Status.

Discussion-grade at the meta-pattern level. This segment names a structural cluster across three classical impossibility theorems; the cluster recognition itself is a presentational organizing principle, not a derivation, and carries no theorem of its own.

Individual instances retain their own (higher) epistemic status. The imported theorems (Gibbard 1973, Satterthwaite 1975, Myerson & Satterthwaite 1983, Arrow 1951, 1963) are *exact* — classical theorems with closed-form proofs in the source literature. The AAT-side translation in each supporting segment carries the segment's own tier (*derived conditional* on the AAT-internal reading of the mechanism-design constraint regime). The boundary characterization (which escapes intersect AAT machinery; which do not) is *robust qualitative* — survives across instance details, with specific functional intersections noted where derivable.

The segment makes one additional claim: that the actor-and-remedy contrast with `#disc-identifiability-floor` is *categorical* (designer-with-construction-task vs agent-with-inferential-task), not a matter of degree. This is a *discussion-grade* observation about the framework's architecture — visible once the instances are assembled, but not itself a theorem.

Max attainable: *discussion-grade* for the meta-pattern (it is a presentational organizing principle). Individual instances retain their own tier as listed in their supporting segments.

Discussion.

Relationship to `#disc-identifiability-floor`. Sister meta-pattern; the actor-and-remedy contrast table above. Both are boundary-face facets of `#disc-stability-certificate`; both run the same three-move discipline (name the floor, name the escape, treat the no-go as load-bearing apparatus). The two clusters differ in *layer* — agent-side data-inference vs designer-side construction — not in *posture*. The discipline that converts a sequence of classical impossibility theorems into a structural argument for *what AAT requires or maps* is shared.

Relationship to `#disc-value-functional-grounding-floor`. Sister meta-pattern in a different way. Both this cluster and the value-functional-grounding cluster are recently-named; both sit alongside the identifiability-floor as boundary-face peers of the certificate spine. The value-functional-grounding floor is *agent-side* (the agent's value-functional interface V_{O_t} cannot, by itself, anchor goal-stability — both Result G′ and the Cohen-2022 strengthening land in that cluster); this cluster is *designer-side*. The three clusters together — agent-side data-inference / agent-side value-functional-grounding / designer-side mechanism-design — exhaust the framework's currently-recognized boundary-face conditions. The constructive-impossibility-posture catalog (`#disc-constructive-impossibility-posture`) carries all three under one stylistic recognition.

Relationship to `#disc-stability-certificate`. This segment is a *boundary-face* facet of the certificate spine, peer to `#disc-identifiability-floor` (agent-side data-inference layer), `#disc-value-functional-grounding-floor` (agent-side value-functional layer), and to the other facets `#disc-separability-pattern` (scope-of-existence), `#disc-additive-coordinate-forcing` (forced-identity), and the projection facet of `#form-composition-closure`. The boundary is where impossibility statements bind; this segment is one of three boundary-layer dialects in which the framework speaks at that boundary.

Relationship to `#deriv-strategic-composition`. The placement of "mechanism-design impossibility" as a candidate-fourth-instance of `#disc-identifiability-floor` is wrong on actor-positioning grounds: GS forbids what a designer can implement, not what an agent can identify, and the standard escapes are not AAT machinery. This segment is the correct home — the sister meta-segment to the identifiability floor. Sub-scope α' retains its load-bearing role in the strategic-composition machinery, and is *additionally* recognized here as the AAT-machinery-intersecting escape route for two of the three charter instances.

The discipline filter is narrow by design. The cluster admits only constructive-impossibility-shaped theorems. Auction theory in general is too broad; social-choice theory in general is too broad; constructive mechanism design (revenue equivalence, optimal auctions) is structurally different (positive results, not impossibility-with-escape) and is deliberately excluded. The narrowness is what makes the cluster's discipline coherent — it admits theorems where the same three-move structure governs membership, and it cleanly excludes results that don't share the structure. That coherence is itself epistemic content.

Peer voice with mechanism design. AAT does not displace mechanism design and does not absorb it. The classical theorems (GS, MS, Arrow) are imported with their original names and full citations; the standard escapes are documented in mechanism-design vocabulary (Bayes-Nash IC, preference-domain restriction, randomization, subsidies); the boundary is precisely mapped. The mature peer literature is in dialogue with the framework, and the dialogue's content is honest scope-marking on both sides. AAT contributes the recognition that *its* strategic-composition machinery intersects *that* literature's classical escape routes at a specific named locus (sub-scope $\alpha' \leftrightarrow$ preference-domain restriction); it does not contribute mechanism construction.

Findings.

The Implementation Impossibility — Designer-Side Sister to the Identifiability Floor. **Brief:** When a community has to decide together — what to vote for, how to trade fairly, how to add up preferences into a social choice — three classical results show that some combinations of "reasonable" demands cannot be met simultaneously: no voting rule can be at once strategy-proof, non-dictatorial, and broadly applicable; no bilateral trade mechanism can be at once fair, voluntary, balanced, and efficient; no social welfare function can be at once unrestricted, Pareto-respecting, independent-of-irrelevant-alternatives, and non-dictatorial. The escapes are well-known and have been studied for decades — restrict the preference domain to single-peaked, allow randomization, accept budget deficits, use a weaker incentive concept — and each escape relaxes exactly one of the demanded constraints. The framework recognizes that this cluster of classical results forms a *sister* to its agent-side identifiability floor: where the floor names what an agent cannot infer from limited data and the agent-side capacity-augmentation that escapes the floor, the implementation impossibility names what a designer cannot construct under stated constraints and the design-constraint-relaxation that escapes the impossibility. The contribution is not to re-prove the classical theorems but to map precisely where the framework's own strategic-composition machinery (the sub-scope where best-response dynamics converge in potential or monotone games) intersects one of the well-known escapes (preference-domain restriction), and to honestly mark where the framework's contribution stops. A reader who carries this away can predict where AAT will speak to mechanism design (it intersects on preference-domain restriction; it is adjacent on cardinal preferences via the value-functional interface) and where it will not (Bayes-Nash IC, randomization, monetary transfers all belong to mechanism design proper).

Impact: Closes a missing-but-implied taxonomic category. The designer-side cluster had been flagged in `#deriv-strategic-composition`'s Discussion as a candidate-fourth-instance of the identifiability floor, but had no first-class home; GS does not fit the identifiability-floor pattern on actor-positioning grounds. The cluster's landing resolves the asymmetry. The framework now has three recognized clusters in the constructive-impossibility-posture catalog: agent-side data-inference (`#disc-identifiability-floor`), agent-side value-functional-grounding (`#disc-value-functional-grounding-floor`), and designer-side mechanism-design (this cluster). The agent-side / designer-side distinction tracks the canonical suffix (`-floor` for agent-side capacity-on-the-agent; `-impossibility` for designer-side task-of-the-designer), making the structural distinction visible at slug-grep depth. Three charter instances ship at landing (GS / MS / Arrow); a candidate fourth (Roberts 1979) and a candidate fifth (Skalse 2022, structurally a designer-side reward-proxy variant) are flagged for future cycles without expanding the present scope.

Novelty Claim: *Claim recognition* of a structural cluster across three classical mechanism-design / social-choice impossibility theorems, with the cluster's positioning in AAT's constructive-impossibility-posture taxonomy as the contribution rather than any new derivation; *claim differentiation* of the cluster from `#disc-identifiability-floor` on actor-and-remedy grounds; *claim transfer* of the constructive-impossibility-posture's three-move discipline (name floor / name escape / treat no-go as apparatus) from the agent-side data-inference layer to the designer-side mechanism-design layer. The constituent theorems (Gibbard 1973, Satterthwaite 1975, Myerson & Satterthwaite 1983, Arrow 1951) are classical; the contribution is the AAT-side recognition that they cluster together under one meta-pattern that *sits alongside* the agent-side floor pattern at the boundary facet of the stability-certificate spine.

Related Work:

Table 5.6

ASF concern	*Prior-art language*	*Relationship / Positioning*
Strategy-proofness impossibility (Instance 1)	Gibbard, A. (1973), "Manipulation of voting schemes: a general result," *Econometrica* 41:587–601 (found pre-2026); Satterthwaite, M. A. (1975), "Strategy-proofness and Arrow's conditions: existence and correspondence theorems for voting procedures and social welfare functions," *J. Econ. Theory* 10:187–217 (found pre-2026)	*formal antecedent* — the imported theorem for Instance 1; primary-source verification scheduled for the supporting segment's promotion past `draft`
Bilateral-trade impossibility (Instance 2)	Myerson, R. B. & Satterthwaite, M. A. (1983), "Efficient mechanisms for bilateral trading," *J. Econ. Theory* 29:265–281 (found pre-2026)	*formal antecedent* — the imported theorem for Instance 2
Social welfare aggregation impossibility (Instance 3)	Arrow, K. J. (1951, 2nd ed. 1963), *Social Choice and Individual Values*, Yale University Press (found pre-2026)	*formal antecedent* — the imported theorem for Instance 3
Single-peaked preferences as a domain escape	Black, D. (1948), "On the rationale of group decision-making," *J. Polit. Econ.* 56:23–34 (found pre-2026)	*adjacent literature* — the canonical preference-domain-restriction escape; the AAT-side intersection (sub-scope α') is named in the boundary characterization
Bayes-Nash relaxation of DSIC	d'Aspremont, C. & Gérard-Varet, L.-A. (1979), "Incentives and incomplete information," *J. Public Econ.* 11:25–45 (found pre-2026)	*adjacent literature* — outside AAT machinery; documented as honest scope-mark
Affine-maximizer characterization (candidate 4th instance)	Roberts, K. (1979), "The characterization of implementable choice rules," in *Aggregation and Revelation of Preferences*, J. J. Laffont (ed.), North-Holland (found pre-2026)	*partial anticipation* — same designer-side shape; held back from charter scope to keep landing narrow
Reward-hacking unhackability (candidate 5th instance, designer-side variant)	Skalse, J., Howe, N., Krasheninnikov, D. & Krueger, D. (2022), "Defining and characterizing reward hacking," *NeurIPS 2022* (found 2026-05-22)	*adjacent literature* — designer-side task on reward-proxy design; structurally distinct from GS/MS/Arrow's preference-aggregation tasks; flagged for future evaluation against the discipline filter
The constructive-impossibility-posture style across clusters	`#disc-constructive-impossibility-posture` (this framework, 2026-05)	*internal antecedent* — the cluster joins the posture catalog as the third recognized cluster

Novelty assessment. The cross-instance recognition that GS / MS / Arrow share the constructive-impossibility-posture shape (setting → external theorem → no-go → boundary characterization → strengthened consequence) at the designer-side layer is the contribution; the constituent recognitions (each instance's prior-art landscape) live in the supporting segments. The constituent moves are well-precedented at the textbook level (Mas-Colell, Whinston & Green 1995 §23.D; Reny 2001 for the geometric-topological proof structure shared across GS/Arrow; Roberts 1979 affine-maximizers as the canonical "with-transfers" extension); the cross-instance unification as the *designer-side companion* of an integrated agent theory's identifiability-floor pattern appears to be a fresh presentational recognition.

Search Log:

- 2026-05-21 (*intuition-only on the meta-pattern as a cluster*): The cross-instance recognition that GS / MS / Arrow share the constructive-impossibility-posture shape (setting → external theorem → no-go → boundary characterization → strengthened consequence) at the designer-side layer is named here for the first time; the constituent recognitions (each instance's prior-art landscape) live in the supporting segments. Targeted future search candidates: methodological-essay literature on mechanism design's "impossibility tradition" (Mas-Colell, Whinston, Green 1995 *Microeconomic Theory* §23.D for the textbook synthesis; Reny 2001 *Econometrica* 69:961–981 for the geometric-topological proof structure shared across GS/Arrow); meta-discussion on the relationship between social-choice impossibilities and mechanism-design positive results (Roberts 1979 affine-maximizers as the canonical "with-transfers" extension). Expected outcome: the constituent moves are well-precedented at the textbook level; the cross-instance unification as the *designer-side companion* of an integrated agent theory's identifiability-floor pattern appears to be a fresh presentational recognition.

- 2026-05-22 (*targeted, Skalse 2022 as candidate 5th instance*): Skalse, Howe, Krasheninnikov & Krueger 2022's designer-side reward-hacking unhackability is structurally adjacent to this cluster's discipline filter but on the *reward-proxy design* task rather than the *preference-aggregation* task. Flagged for future evaluation against the discipline filter, not pursued at present landing.

5.4 *Discussion:* The Separability Pattern — Separable Core, Structured Repair, General Open

AAT consistently runs a three-part epistemic posture across state spaces that admit no tractable exact treatment in general: name the **separable core** where identification is clean, name the **structured repair** that recovers identification under explicitly-added machinery, and name the **general open** case where the problem is either intractable or structurally unidentifiable. Eight ladders in the current theory share this shape (Correlation, Convention, Architecture, Contraction, Identification regime, Scope hierarchy, A2'-scope, Dynamic regime — enumerated in the table below; the Dynamic regime ladder landed 2026-05-21 per `#disc-dynamic-regime-axis`). This segment names the pattern as an organizing principle, catalogs the instances, and makes the complementarity with `#disc-identifiability-floor` (which names the *negative half* of AAT's scope) explicit.

The pattern.

Each instance of the separability pattern has the form:

1. **Separable core.** A sub-class of problem in which identification is tractable by construction — the quantity of interest has a clean estimator that requires only assumptions the agent can verify or control. "Separability" here is broader than statistical independence: it names the regime in which the quantity decomposes along a structural axis (independence, additive decomposition, directed separation, strong contraction, clean intervention, etc.) without interaction terms requiring additional machinery.

2. **Structured repair.** A sub-class where the separability assumption fails but a *specific, named, bounded-cost* additional mechanism recovers identification. The repair is structured: it identifies the failure mode, adds a specific compensating construction, and typically carries a characterized loss in tightness or generality. Repairs are not "apply heuristics"; they are "augment with observability of C and accept $O(1)$ per-node parameter cost," "run receding-horizon replanning at cost $\mathrm{DL}(\Sigma_t)$," or similar.

3. **General open.** The fully-general case where no tractable repair is known. This is where `#disc-identifiability-floor` instances live — structural no-go results that characterize *why* the general case remains open. Naming this boundary positively (as a known frontier) rather than negatively (as a failure) is part of AAT's load-bearing scope honesty.

Current instances — eight ladders.

Table 5.7

Ladder	*Separable core*	*Structured repair*	*General open*
Correlation (8.3)	**L0** — independence model, $O(\lvert V\rvert + \lvert E\rvert)$ propagation	**L1** — strict-prerequisite augmentation with explicit common-cause node; **L1'** — soft-facilitator mixture with five-way gating under observable C + facilitator monotonicity (`#deriv-edge-credence-dynamics` Prop B.7)	**L2** — full joint; **L1' unobservable-C** single-channel is *refuted* by Cramér-Rao floor (5.1 Instance 2)
Convention (6.6)	**C1** — one-step improvement	**C2** — receding-horizon replanning	**C3** — Bellman optimal; intractable for large state spaces
Architecture (6.3)	**Class 1 (Separated)** — directed separation holds by construction	**Class 2 (Partial)** — directed separation holds for identified submodules	**Class 3 (Coupled)** — directed separation fails by construction; logogenic territory; coupled formulation required
Contraction (13.1)	**Tier 1** — strong monotonicity; bridge lemma applies with proved ε^*	**Tier 2** — local convexity; bridge applies within a specified basin	**Tier 3** — neither; domain-specific verification required per instance
Identification regime (9.5)	**Regime A** — interventional ($\iota_{ij} = 1$); action is a literal $do(\cdot)$	**Regime B** — partial intervention ($0 < \iota_{ij} < 1$); confounder adjustment, side-channel observation	**Regime C** — observational ($\iota_{ij} \approx 0$); identification depends on external assumptions (e.g., unconfoundedness)
Scope hierarchy (6.1)	**Adaptive** — basic feedback loop; 1.5 satisfied	**Agency** — goal-bearing with $\lvert\mathcal{A}\rvert \geq 2$ and at least one action with causal effect; adds O_t and Σ_t machinery (Part II); 1.6 satisfied	**Composite** — multi-agent / team / logogenic; Part III gaps listed at the OUTLINE level (latent structural diversity, endogenous coupling, composition transition dynamics, agent opacity)
A2'-scope (`#result-contraction-template`)	**metric-α_1** — Euclidean metric, AAT-internally derived via DA2'-inc ≡ (CT2) at $M = I$ (scalar Kalman, Euclidean strongly-convex, L2-regularized, linear-PD-symmetric)	**metric-α_2** — non-Euclidean metric under explicit conditions; includes five cases (information-metric Kalman, Fisher-metric exp-family, Hessian-metric ill-conditioned, Lyapunov-metric linear-Hurwitz-non-symmetric, Lyapunov-metric PID-bounded-plant). Two of five (Fisher cases) AAT-internally forced under (PI)/Čencov per `#disc-additive-coordinate-forcing`; remaining three theorem-imported from Lohmiller-Slotine 1998	**metric-β** — contraction-metric formulation fails (variational-to-projected-target; rule-based / non-smooth; severely misspecified; per-step SGD / human judgment)

Ladder	*Separable core*	*Structured repair*	*General open*
Dynamic regime (`#disc-dynamic-regime-axis`)	**R0 contraction-regime** — unique attracting fixed point in joint $\mathcal{X}^c$ with global or basin-restricted Lyapunov; exponential at α_{joint}; state-variable macro-state; scope routes C-i / C-ii / C-iii	**R1 equilibrium-regime** (α') — fixed-point set $\mathcal{E}$ with locally-attracting equilibria under potential-game (Monderer-Shapley 1996) or monotone-game (Rosen 1965) structure; fixed-point-object macro-state; exponential near $\mathcal{E}$; scope route C-iv + potential/monotone	**R2 cyclic-distributional-regime** (β') + **R3 mean-field-equilibrium-regime** — R2: no pure-strategy fixed point or saddle-only; distributional macro-state on CCE; polynomial $O(1/\sqrt{T})$ via Hart-Mas-Colell 2000. R3: population-scope frontier (Lasry-Lions 2007; Huang-Malhamé-Caines 2006) where macro-state-type / regime identity breaks

GUC Architecture-row key: Class 1 = Separated (was Modular); Class 2 = Partial (was Class 3 Partially modular); Class 3 = Coupled (was Class 2 Fully merged). See `#der-directed-separation`.

Each row independently satisfies the three-part shape. The shared shape is not designed-in: it arises because AAT faces eight distinct sources of intractability and applies the same posture (name the clean case, name the repair, name what remains open) to each.

Complementarity with the identifiability floor.

The **separability pattern** names the *positive half* of AAT's scope: for each ladder, what succeeds under what conditions. Each separable-core entry is a positive identification claim; each structured-repair entry is a positive identification claim *conditional on* an explicitly-named added mechanism.

This segment is the *scope-of-existence facet* of the stability certificate (5): a separable-core entry is a region where a certificate exists cleanly, a structured-repair entry is a region where one exists under explicitly-added machinery, and a general-open entry is where no certificate is available without further information. Read through the spine, this segment answers "where does the agent's measuring-stick exist at all"; 5.1 answers "where does it go flat" (boundary facet) and 5.5 "which stick is forced" (forced-identity facet).

The `#disc-identifiability-floor` names the *negative half*: structural no-go results (impossibility under limited information) that characterize why specific general-open cases remain open. Four floors are currently derived:

- **Instance 1** (on-policy L0-insufficiency detection via Causal Hierarchy Theorem) — forbids L0/L1 distinction from purely on-policy data. The positive counterpart: the **separable core under observable sibling covariance** is the unique broadly-available violation of the no-go's scope, naming what the agent *can* observe to escape the floor.

- **Instance 2** (L1' unobservable-C single-channel via Cramér-Rao) — forbids mixture-parameter identification when C is unobservable. The positive counterpart: the **structured repair under observable C** (Prop B.7 with facilitator monotonicity) is the identification-succeeds-under-augmentation claim matching the no-go's "unless C is observable or multi-child observation is available" scope exit.

- **Instance 3** (composite contraction certification from component data via common-Lyapunov nonexistence) — forbids certifying composite contraction, coupling sign, or dynamic regime from component-level marginals alone. This floor is the negative characterization of the **Scope-hierarchy ladder's general-open Composite row** above: the Part III gaps that row lists (endogenous coupling, agent opacity) are open precisely because the coupling-topology bits that decide composite persistence are unidentifiable from component data. The positive counterpart: joint-dynamics observation or structural knowledge of the coupling topology.

- **Instance 4** (architecture no-identifiability from on-policy summary data via Kalman/Ho-Kalman non-uniqueness + CHT) — forbids recovering which internal coordinates implement correction from on-policy, in-regime, summary-only access. The positive counterpart: off-policy intervention, out-of-regime samples, or white-box access, each breaking a named scope condition.

Together, the two meta-segments mark AAT's scope at two epistemic layers:

- **Separability pattern** — "here is what succeeds, with clean conditions and explicit repairs."
- **Identifiability floor** — "here is what structurally cannot succeed without specific information augmentation."

Neither alone gives the full picture. A theory stating only the positive half looks unbounded in its claims; a theory stating only the negative half looks conservative in a way that hides its achievements. Together they give scope honesty at both extremes.

Epistemic Status.

Discussion-grade at the meta-pattern level. The segment is a presentational organizing principle. It names a shared shape that AAT already runs across multiple ladders; the pattern itself is not derived and has no theorem of its own. The individual ladder entries retain their own epistemic status — Correlation hierarchy results are at the tier specified in 8.3 and `#deriv-edge-credence-dynamics`; Convention hierarchy monotonicity is at the tier specified in 6.6; Contraction tier taxonomy is at the tier specified in 13.1; etc.

Max attainable: *discussion-grade* for the meta-pattern (it is an organizing principle, not a derivation). Individual instances are at their own tiers as above. The eight-ladder enumeration could become *robust-qualitative* if a uniqueness argument were derived showing that separable-core / structured-repair / general-open is the unique viable posture across scope-parameterized hierarchies under AAT's scope-honesty architectural principle — but this is not currently in hand, nor clearly attainable.

Discussion.

The pattern is load-bearing for how AAT presents its results. A common failure mode of applied theories is the binary "exact under assumption X, breaks otherwise" presentation. The separability pattern refuses this presentation across eight instances: where X fails, a named lower tier takes over with a characterized weaker result, and the theory provides a diagnostic for when to escalate (per `#disc-approximation-tiering`'s AT4 component). Readers who learn the pattern once can navigate any instance; this is what makes AAT tractable as an integrating framework rather than a pile of instantiations.

Relationship to `#disc-approximation-tiering`. This segment is complementary to `#disc-approximation-tiering`, not redundant with it. `#disc-approximation-tiering` names the *structural template* (AT1 parameter indexing tractability; AT2 proved monotonicity between tiers; AT3 graceful degradation; AT4 ascension diagnostic) — the how-it-works of each tiering. This segment names the *epistemic posture* (separable core, structured repair, general open) — the what-each-tier-commits-to. Both are needed: `#disc-approximation-tiering` explains what makes a successful tiering; this segment explains the shape of the commitment each tier makes about identification.

Relationship to `#disc-independence-audit`. `#disc-independence-audit` catalogs the *independence assumptions* whose failure degrades results; this segment catalogs the *ladders of recovery*. Together with `#disc-approximation-tiering` they form a three-part characterization of AAT's scope:

- **`#disc-independence-audit`** — where the boundaries are (which assumptions, what breaks if they fail).
- **`#disc-approximation-tiering`** — how to navigate within the boundaries (parameterized hierarchy + monotonicity + ascension).
- **5.4** (this) — what each tier positively commits to (separable core / structured repair / general open).

The pattern is distinctive to AAT's integration-over-invention character. Approximation tiering with a separability posture is a common move in applied mathematics — it appears in numerical methods (exact vs. high-order vs. low-order with error bounds), in statistical inference (parametric separable-core / semiparametric repair / nonparametric open), in causal identification (Pearl's three-layer hierarchy is itself an instance). AAT's contribution is not the pattern but its deployment across the eight specific intractable problems adaptive-agent theory faces, each with its own positive-half identification claims and negative-half floor instances.

The A2' sub-scope partition is a proper three-part ladder via `#result-contraction-template`. The A2' partition carries three tiers: sub-scope α (A2' derived under `#der-gain-sector-bridge` directional fidelity) as the separable core; sub-scope metric-α_2 (non-Euclidean metric under explicit conditions — five cases, with two Fisher-metric cases AAT-internally forced under the (PI)/Čencov fourth primary instance of `#disc-additive-coordinate-forcing`) as the structured-repair middle tier; sub-scope β (A2' assumed) as the general-open tier. A2'-scope therefore qualifies as the seventh ladder of this meta-pattern, as enumerated in the table above (with the Dynamic regime ladder per `#disc-dynamic-regime-axis` now landed as the eighth, added 2026-05-21). The structured-repair middle tier closes what would otherwise be a binary α/β partition lacking a derivable middle ground.

The software calibration-lab framing (`#obs-software-epistemic-properties`) is a specific instance. C-BP3's calibration-lab reframing (commit `d0373fc`) partitions operational domains into "separable core" (software, where identification conditions P1–P6 are cleanly satisfied) and "structured-repair / general open" (other domains, which inherit under explicitly-named transfer assumptions). This is a domain-axis instance of the same pattern — software is the domain-axis separable core; the transfer-assumption table is the domain-axis structured-repair specification.

5.5 *Discussion:* Additive-Coordinate Forcing

AAT carries a family of structurally connected uniqueness results in which a coordinate is **forced by a uniqueness theorem operating on an independently-motivated AAT-internal axiom**. This meta-pattern previously appeared as a "1-anchor-plus-3-theorem" structure (where the chain-rule anchored three Cauchy-FE and Čencov-invariance results). The deeper structural reality is that **these four layers are layer-specific manifestations of a single geometric object**: the exponential-family Legendre-Fenchel structure.

The pattern is not designed-in: it arises because AAT independently needs to force coordinates at four distinct layers (chain, divergence, update, and metric), and the surrounding apparatus (DAG factorization, Bayesian coherence, regret-bound decision theory, and singular-trajectory scope) has committed to this geometry through multiple independent surface constraints.

This segment names the pattern, defines the underlying geometric object, catalogs the four layer-specific manifestations, and documents adjacent cases that share the shape but not the AAT-internal forcing structure.

The distinction is three-way, and worth holding before the catalog. A coordinate is **forced** when a uniqueness theorem on an independently-motivated AAT-internal axiom admits exactly one — this segment's subject, the four layers below. It is **matched** when *chosen* to fit a structure rather than uniquely derived: the canonical example is the Lyapunov quadratic of `#deriv-sector-condition`, selected via a converse-Lyapunov argument to match the sector form (it shares the convex-potential shape but sits outside the forcing structure — see "Adjacent cases" below). It is **adopted** when imported wholesale with citation: the exponential-family Legendre-Fenchel geometry itself (Amari & Nagaoka 2000), or Pearl's do-operator. Only the *forced* class carries a uniqueness claim; naming the other two keeps the meta-pattern's boundary sharp — *forced* means "no other coordinate could serve," not merely "this one serves well."

The underlying geometric object.

The four forced coordinates all live on the **exponential-family Legendre-Fenchel geometry** (Amari & Nagaoka 2000, *Methods of Information Geometry*, §3.5; Bregman 1967; Rockafellar 1970; Bauschke & Combettes 2017). The structure is characterized by:

1. **A convex potential:** The negative-entropy potential on the probability simplex Δ^{n-1}.
2. **A Fenchel conjugate:** The log-partition function on the dual natural-parameter space.

3. **A primal-dual correspondence:** The softmax / log-odds map translating between probability (primal) and natural parameters (dual).
4. **A Bregman divergence:** The reverse-KL divergence, which measures distance using the primal potential.
5. **A Riemannian metric:** The Fisher information metric, which coincides with the Hessian of the dual potential ($\nabla^2\phi^*$).

When AAT forces an additive or invariant coordinate, it is forcing the theory onto a specific surface of this single geometric object.

The four layer-specific manifestations.

The pattern shares the following structure across four layers: 1. **An AAT-internal axiom (or mathematical identity)** is posited, motivated by the layer's specific needs. 2. **A uniqueness theorem** (Cauchy's functional equation for log-additivity, or Čencov's 1982 invariance theorem) operates on the axiom. 3. **A coordinate is forced**, which corresponds exactly to a feature of the Legendre-Fenchel geometry.

Table 5.8

Layer	*AAT-internal axiom*	*Uniqueness mechanism*	*Forced coordinate*	*Relation to exponential-family geometry*	*Source segment*
Chain	Probability chain rule (mathematical identity, not an AAT axiom)	Cauchy-FE	Log-probability	Log-coordinate of the primal point	8.1
Divergence	Chain-rule additivity over conditional factorizations	Cauchy-FE	Reverse-KL (up to positive scaling)	Bregman divergence generated by the primal potential	`#deriv-strategy-cost-regret-bound` §6.1
Update	Evidential additivity	Cauchy-FE	Log-odds	Natural coordinate on the dual space	`#deriv-edge-update-natural-parameter`
Metric	Parameterization-invariance (PI) on singular trajectories	Čencov-invariance (Čencov 1982)	Fisher information metric	Hessian of the dual potential	4.3, `#deriv-observation-ambiguity-bias-bound`

The motivational anchor and the structural target. The divergence axiom is stated as "the *divergence-level analog* of the additive log-confidence decay in `#der-chain-confidence-decay`." The update axiom is stated as "the *update-level analog* of `#der-chain-confidence-decay`'s chain-level additive log-confidence decomposition." The (PI) axiom is stated as "the metric-layer analog motivated by the singular-trajectory scope commitment."

The four layers are not independent applications of a generic mathematical idiom; they are interlocked by cross-cites, each positioning itself against an adjacent AAT commitment. **The chain-layer identity is the motivational anchor *and* the exponential-family geometry is the structural target.** The reason the three analog axioms all succeed via uniqueness theorems is that they all independently probe the same underlying geometric object. The convergence across independent axioms is itself the meta-pattern's substance — not a byproduct to be compressed into a single axiom.

The Čencov-Fenchel relationship. On AAT's current Fisher-metric scope (exponential families in natural parameters and matrix-Kalman cases), the Čencov-derived Fisher metric coincides precisely with the Hessian of the dual log-partition potential. This is a scope-dependent equivalence: outside exponential families, Čencov's invariance theorem still forces the Fisher metric, but the Fenchel-Bregman correspondence does not straightforwardly apply. Within AAT's primary operating regime, the two uniqueness machineries (Cauchy-FE and Čencov) resolve to the exact same geometric object.

Downstream consequences of the (PI) commitment. Forcing the Fisher-Rao geometry via (PI) + Čencov is not just an aesthetic unification; it enables specific quantitative theorems downstream. For example, the bias bound derived in `#deriv-observation-ambiguity-bias-bound` (Track 2) relies explicitly on the Fisher-Rao metric to achieve a universal, dimension-free bounding constant ($C_{FR} = \sqrt{2}$) connecting mutual information to parameter displacement. Without the (PI) commitment, attempting this bound in arbitrary Euclidean norms fails completely (yielding arbitrarily large constants).

The floor-vs-coordinate-forcing distinction (the heteroscedastic-Gaussian no-go). The no-go "no universal information-to-distance constant C exists under arbitrary parameter norms" — proved counter-example-grade in `#deriv-observation-ambiguity-bias-bound` §4 via the heteroscedastic-normal witness ($\{N(0, \sigma^2)\}$ under Euclidean-σ giving $C \to +\infty$) — is a member of *this* coordinate-forcing family (the (PI) commitment's downstream theorem), **not** a member of the identifiability-floor family of `#disc-identifiability-floor`. The structural distinction is sharp on three load-bearing elements:

- *External-theorem role.* A floor instance's external theorem *forbids* the inferential task (CHT forbids L2-from-L1; Cramér-Rao forbids exact estimation under rank-deficient Fisher; common-Lyapunov-nonexistence forbids composite-contraction from marginals; Kalman-Ho similarity-orbit non-uniqueness forbids architecture-identification from on-policy summary data). The heteroscedastic-Gaussian no-go's "external theorem" is Čencov 1982 — the *uniqueness theorem that forces the escape*, not a theorem that forbids the task. **Opposite structural role**: floor instances cite theorems that forbid; this no-go cites a theorem that forces uniqueness of an escape that already exists.
- *Escape count.* A floor instance carves an *operational space* of ≥ 2 structurally distinct escapes (Instance 1 has five; Instance 2 has three; Instance 3 has four; Instance 4 has three). Čencov's uniqueness theorem *mathematically forbids* a second structurally distinct escape from existing — uniqueness of the invariant Riemannian metric is the entire content. The heteroscedastic-Gaussian no-go has exactly *one* forced escape (adopt (PI) + Fisher-Rao), and the count of one is *not* a circumstantial shortage but a logical consequence of Čencov's uniqueness. A uniqueness theorem on the escape side is logically incompatible with the $\geq$ 2-distinct-escapes structure that defines a floor instance.
- *Consequence.* A floor instance *elevates* new AAT machinery from useful to load-bearing (Instance 1 elevates `#der-loop-interventional-access` at the agent layer; Instance 2 elevates observability-as-information-augmentation; Instance 3 elevates `#deriv-critical-mass-composition` and the composite-extended interventional access; Instance 4 elevates `#der-loop-interventional-access` to a third semantically-distinct deployment mode at the agent-internal-architecture layer). The heteroscedastic-Gaussian no-go *re-uses* the (PI) commitment, already load-bearing as the fourth primary instance of *this* meta-pattern. A client of a forced coordinate is a **downstream theorem of the coordinate-forcing pattern**, by the definitions of the two meta-patterns.

The bias-bound theorem $C_{FR} = \sqrt{2}$ is therefore correctly classified here as a downstream consequence of (PI) + Čencov, not as an identifiability floor — exactly the classification that `#deriv-observation-ambiguity-bias-bound` §4 already states *in canon* ("does not match the five-element test ... the honest position: this no-go is a downstream theorem of the (PI) commitment, not a new floor instance"). The genuine fourth identifiability-floor instance is `#der-architecture-noidentifiability` (architecture-noidentifiability from on-policy summary data), whose structure does satisfy the five-element shape under the rank-collapse subclass.

Adjacent cases that share the shape but not the forcing structure.

Other AAT segments exhibit additive decompositions, but belong to different geometric families or lack the AAT-internal forcing structure.

Lyapunov quadratic (`#deriv-sector-condition`). The sector-Lyapunov argument uses $V(\delta) = \frac{1}{2}\|\delta\|^2$, under which $\dot{V}$ decomposes additively. **Reclassification:** The Lyapunov quadratic is the **Bregman divergence on a Euclidean potential** (squared-norm potential). It belongs to a *distinct* convex potential sub-family, not the negative-entropy exponential family. Furthermore, the coordinate is *chosen* (via a converse-Lyapunov existence argument) to match the sector form; it is not uniquely *forced* by an AAT-internally-motivated additivity axiom. It sits outside this meta-pattern.

Variance-additive cases (AV). Variance-additive statistics (like the Ornstein-Uhlenbeck steady state in `#hyp-mismatch-dynamics`) share Bregman-type additivity but, like Lyapunov, sit on a squared-norm potential rather than the negative-entropy geometry. The principal member is the mismatch budget itself: the three-term decomposition of `#result-mismatch-decomposition` and its viability form in `#internal-external-decomposition` are Pythagorean on the squared norm, and the disturbance layer carries no AAT-internal additivity axiom that would force a logarithmic coordinate — which is why a multiplicative ρ-factorization was a choice, and a wrong one, rather than a forced form. The information-form counterpart in `#deriv-mismatch-budget-attribution` does sit on the negative-entropy geometry, but by *adoption* (log-loss as a proper scoring rule; Csiszár's Pythagorean identity), not by forcing; it is adopted, like IB below.

IB Lagrangian (`#form-information-bottleneck`, `#disc-compression-operations`). The Information Bottleneck objective $T^* = \arg\min[I(X;T) - \beta \cdot I(T;Y)]$ uses an additive Lagrangian form. **Reclassification:** IB is a direct **application of the negative-entropy / reverse-KL Bregman geometry**. However, AAT *adopts* the IB formulation from external foundations (Tishby, Pereira & Bialek 1999) rather than re-deriving it from an AAT-internally-motivated uniqueness theorem. It is an adjacent family member: it shares the exact geometric object, but its axiomatic provenance is imported.

Using the pattern as a diagnostic filter. This reframe supplies a principled filter for candidate future instances: *does the candidate's coordinate arise from the Legendre-Fenchel / exponential-family geometry?* If yes, check which of the four layer-surfaces it corresponds to. If no, it is either an adjacent family member (different Bregman geometry) or outside the pattern entirely.

Complementarity with 5.1 and 5.4.

The three meta-segments form AAT's scope-and-structure meta-architecture:

- **5.5** (this) — the *constructive* half: when AAT forces a coordinate, it forces a surface of the exponential-family geometry (not just a generic logarithmic coordinate, but the specific log-coordinate of the exponential family). Names what the theory *does* at its deepest structural moves.
- **5.4** — the *positive scope* half: for each ladder of increasing difficulty, name the separable core, the structured repair, and the general open. Names what the theory *covers* across its instantiations.
- **5.1** — the *negative scope* half: for each known structural no-go, name the external information-theoretic theorem that enforces it and the unique escape available. Names what the theory *cannot do* without explicit information augmentation.

The three compose. A ladder's separable core typically admits a coordinate-forcing move that makes the clean identification clean; a ladder's general-open case typically sits at an identifiability floor that names the information-theoretic obstruction. Reading any segment through all three meta-lenses surfaces what makes the segment load-bearing.

This segment is the *forced-identity facet* of the stability certificate (5): it answers "*which* certificate" — whether the metric is forced to a unique form by a uniqueness theorem on an AAT-internal axiom (Čencov forces the Fisher metric, statistical scope only) or merely matched. Read through the spine, the three meta-lenses are the certificate's forced-identity (this segment), scope-of-existence (5.4), and boundary (5.1) facets; the certificate exists iff the agent is exponentially stable (`#result-certificate-existence`).

Epistemic Status.

Discussion-grade at the meta-pattern level. This segment is a presentational organizing principle: it names a structural shape that AAT runs across four layers. The meta-pattern itself is not derived and carries no theorem of its own.

The individual instances retain their own epistemic statuses: - `#der-chain-confidence-decay` — *exact* (mathematical identity). - `#deriv-strategy-cost-regret-bound` §6.1 — *derived (conditional on chain-rule additivity axiom).* - `#deriv-edge-update-natural-parameter` — *derived (conditional on evidential-additivity axiom).* - `#der-gain-sector-bridge` — *derived (conditional on (PI) axiom).*

Findings.

Cross-Layer Coordinate Forcing on Legendre-Fenchel Geometry. **Brief:** When the framework needs to pick a "natural" way of measuring something (a confidence, a divergence, a parameter), it doesn't choose by convention. It writes down a property the measurement should have (typically: small contributions should add up the way independent evidence should), and a theorem from outside the framework forces the measurement to take a particular form. This pattern shows up at four different places in the framework — chain-rule confidences, divergences between distributions, edge update rules, and the metric on the parameter manifold — and the four places turn out to be different windows into the same underlying mathematical structure (the exponential-family Legendre-Fenchel geometry on probability distributions).

Impact: Names a meta-pattern that is one of AAT's distinctive constructive moves and that organizes a substantial fraction of the theory's load-bearing structural choices. The pattern supplies a principled diagnostic filter for candidate future instances ("does the coordinate arise from the exponential-family geometry?") and explains why AAT's reverse-KL direction in regret bounds, log-odds in edge updates, and Fisher-Rao metric in bias bounds are not arbitrary representational choices but theorems conditional on AAT-internal axioms. The reframe also clarifies which adjacent cases (Lyapunov quadratic, IB Lagrangian) sit on different geometric families or have imported provenance — preventing the meta-pattern from being overclaimed as a single uniform principle. Composes with `#disc-identifiability-floor` (the negative scope half) and `#disc-separability-pattern` (the positive scope half) as the constructive third leg of AAT's three-part meta-architecture.

Novelty Claim: *Claim recognition* of cross-layer pattern across four AAT coordinate-forcing results, with the recognition itself as the contribution rather than any new theorem. The constituent uniqueness machinery (Cauchy's functional equation; Čencov's invariance theorem) is well-precedented; the recognition that four AAT-internal axioms each successfully invoke uniqueness machinery on Legendre-Fenchel-geometry coordinates is the AAT-distinctive observation.

Related Work:

- Aczél 1966, *Lectures on Functional Equations and Their Applications* §2.1 — *formal antecedent* — Cauchy's functional equation; the uniqueness machinery for the chain, divergence, and update layers. Cited inline in each instance segment.
- Čencov 1982, *Statistical Decision Rules and Optimal Inference — formal antecedent* — Čencov uniqueness theorem; the uniqueness machinery for the metric layer. Cited inline in `#deriv-observation-ambiguity-bias-bound` and `#deriv-fisher-whitened-update-rule`.

- Amari & Nagaoka 2000, *Methods of Information Geometry* §3.5 — *formal antecedent / synthesis target* — the underlying Legendre-Fenchel geometry on exponential families that all four layers' coordinates resolve to. The recognition that the four forced coordinates are different aspects of one Amari-Nagaoka geometric object is the meta-pattern's payoff.
- Rockafellar 1970, *Convex Analysis* — *formal antecedent* — Legendre-Fenchel duality; the convex-analytic foundation Amari & Nagaoka build on.
- Bregman 1967, "The relaxation method of finding the common point of convex sets" *USSR Comput. Math. Math. Phys.* 7:200–217 — *formal antecedent* — Bregman divergence machinery; the divergence-layer coordinate is the negative-entropy Bregman divergence.

Search Log: - 2026-04 (*intuition-only* on the meta-pattern as a unified framework): the constituent uniqueness machinery (Cauchy-FE, Čencov, Bregman) is well-established, and Amari-Nagaoka information geometry is the standard exponential-family / Legendre-Fenchel framework the four layers resolve to. The recognition that AAT axiomatizes four distinct surfaces of one geometric object via four independently-motivated axioms is the contribution. Whether this cross-layer recognition has been articulated as a unified pattern in any prior framework has not been searched. Targeted future search candidates: information-geometry meta-frameworks (Eguchi 1983 framework discussion; Ay et al. 2017 *Information Geometry* monograph); axiomatic-foundations literature on probability and decision theory (Cox 1946; Jaynes 2003); the broader "uniqueness theorems forcing canonical coordinates" tradition in statistical mechanics. - 2025 (*targeted*): Aczél 1966 / Čencov 1982 / Amari & Nagaoka 2000 / Rockafellar 1970 / Bregman 1967 confirmed as the formal antecedents.

5.6 *Discussion:* The Constructive-Impossibility Posture — Negative Results as Load-Bearing Apparatus

AAT consistently runs a three-move discipline around the structural impossibility results it imports or derives: name the floor (the precise impossibility under a stated information regime), name the escape (the unique broadly-available machinery that violates the regime), and treat the no-go as apparatus (the floor's existence is what *elevates* the named machinery from convenience to structural prerequisite, or — when the no-go *is* the result — the floor is itself on the critical path). The negative result and the positive consequence are not two findings but one. This segment names the posture, maps its current instances, and is honest about its scope: it is a *style* claim about how the framework states certain results, not a fifth cross-sectional meta-pattern alongside the stability certificate's facets.

Formal Expression.

The three moves. *[Discussion]* An instance of the posture has the form:

1. **Floor.** A structural impossibility statement of the form "under information regime $\mathcal{R}$, no statistic / estimator / certificate / design achieves $\mathcal{G}$," with the binding constraint named explicitly: an external theorem (Pearl–Bareinboim CHT; Cramér–Rao; Liberzon / Shorten / Dayawansa–Martin common-Lyapunov nonexistence; Khasminskii recurrence on additive-noise diffusions) or — in the strategic-persistence cap case — a sharp algebraic supremum derived from the framework's own update mechanics.
2. **Escape.** A precise characterization of the conditions under which $\mathcal{R}$ fails — i.e., the agent has *more* information than $\mathcal{R}$ allows. The escape names the *unique broadly-available* framework machinery that supplies the extra capability: loop-interventional access (`#der-loop-interventional-access`); observability of a latent regime variable; matched-Tier composite structure (`#deriv-critical-mass-composition`); a structural class change (out of gain-decay-plus-forgetting). In the cases where no escape *to* the original guarantee exists (Cor A.1S.1 Model-S half),

the escape move is replaced by an explicit *kind-of-guarantee* substitution (distributional-and-instantaneous in place of pathwise-and-forever).

3. **Apparatus.** The floor's existence *elevates* the escape machinery from "useful" to load-bearing in the strongest sense — without the escape, the inferential / structural task is structurally impossible, not merely harder. When the no-go *is* the result (Cor A.1S.1; Prop C.2's $\rho_\Sigma < R_\Sigma/2$ cap), the floor sits on the critical path: it is the proof step the downstream architectural distinction (Model D vs Model S; strategic-layer structural-adaptation trigger) rests on, not appendix material.

The negative result and the positive consequence are the same proof. The discipline is what converts a sequence of impossibility theorems into a structural argument for *what AAT requires*.

Current instances.

Table 5.9

Instance	*Floor (statement, binding constraint)*	*Escape (unique broadly-available machinery)*	*Apparatus (what the floor elevates)*
A — L0 insufficiency detection (`#der-causal-insufficiency-detection`, M1 Instance 1)	L0/L1 distinction unidentifiable from on-policy data (Pearl–Bareinboim CHT)	Loop-interventional access; pairwise covariance test under joint sibling observability (`#der-loop-interventional-access`)	Orient-cascade causal-sufficiency check (4c) elevated to *unique broadly-available diagnostic*
B — L1′ mixture under unobservable C (`#deriv-edge-credence-dynamics` Prop B.7, M1 Instance 2)	Rank-1 Fisher under single-channel obs ⇒ Cramér–Rao bound infinite in null directions	Observability of C (or joint multi-child observation under shared C)	"Instrument the latent" elevated from convenience to *theoretical prerequisite* for L1′ identifiability
C — Composite contraction from component data (`#deriv-critical-mass-composition`, M1 Instance 3)	Coupling-sign bit unidentifiable from component marginals (Liberzon / Shorten / Dayawansa–Martin common-Lyapunov nonexistence)	Composite-extended interventional access; matched-Tier joint quadratic Lyapunov; passivity certificate; common contraction metric	`#deriv-critical-mass-composition` (CM2) / (CM2-M) and `#scope-composite-agent` elevated to *unique composition-contraction certificate* under matched structure
D — Disturbance-Model Containment Dichotomy (`#deriv-sector-condition` Cor A.1S.1; `#deriv-stochastic-non-exit`)	No horizon-independent non-exit certificate under additive stochastic forcing (Khasminskii: unbounded scale function ⇒ no bounded non-constant harmonic function; the only candidate compensated supermartingale is sign-indefinite)	*No escape to pathwise containment under Model S.* The substitute: fixed-time / stationary tail (iii′) + finite-horizon sup-bound (iv) — distributional-and-instantaneous, not pathwise-and-forever	Model D vs Model S as a *kind-of-guarantee* distinction; the no-go signature reusable as a structural diagnostic against future "stays-in-region-forever" proposals
E — Strategic-persistence hard ceiling (`#deriv-strategic-persistence-hard-ceiling` Prop C.2)	$\sup_{\lambda\in[0,1]} \alpha_\Sigma^{ss}(\lambda) =$ 1/2 sharp under Beta-Bernoulli + exponential forgetting; reachable region $\rho_\Sigma < R_\Sigma/2$ open half-plane	Class change (leave gain-decay-plus-forgetting toward constant-step finite-gain per the $\mathcal{A}_{\text{decay}}$ structural-class theorem, NeurIPS 2026 Paper 2); or change mismatch state or topology	The cap as the *strategic-layer analog* of `#result-structural-adaptation-necessity` — the floor *refines* (does not refute) the schema, and the refinement names exactly when the architecture has exhausted its design space

Instance	*Floor (statement, binding constraint)*	*Escape (unique broadly-available machinery)*	*Apparatus (what the floor elevates)*
F — Self-actuation grounding (`#deriv-self-actuation-grounding` Result G′)	No Φ satisfying (R1)–(R4) of Result G′ can be constructed from AAT's covered objective-side machinery (engine: convention-monotonicity within fixed model + finite-no-oracle per step — the value-functional interface V_{O_t} carries no convention-invariant infeasibility verdict that is agent-available per step)	Terminal grounding invariant *off* the objective substrate, on the *adaptive substrate*; the persistence bound (`#result-persistence-condition`) is the canonical instance	The persistence condition (Part I) elevated to *load-bearing for goal-stability* — the top of the agent spectrum closes back onto AAT's foundation
G — Reward-channel learning grounding (`#deriv-reward-channel-learning-no-go` Lemmas 1+2)	μ_{dist} (principal-intended world-state-care) and μ_{prox} (reward-port-bit-care) are L1-equivalent on protocol-honoring histories and L2-distinct on $do(\pi^{\text{rp}})$ (Pearl–Bareinboim CHT applied to reward-provision protocol); under EU-max + non-vacuous μ_{prox} prior + finite-VOI + L2-perturbability, the EU-optimal policy tampers	Principal-side commitment to substrate-level constraints: physical isolation (W_1 wrapper); horizon restriction / quantilization; engineered prior; out-of-band intent channel	Each (R1)–(R5) escape elevates a piece of AAT or principal-side machinery from "useful design choice" to *structurally required* — the no-go's binding force is precisely the cost of each commitment

Each instance's three moves are individuated and load-bearing in the source segment; the table summarizes rather than duplicates them. The catalog's instances span the three recognized clusters of the constructive-impossibility-posture taxonomy, each with its own peer meta-segment home at the boundary face of `#disc-stability-certificate`:

- **Agent-side data-inference impossibility** (`#disc-identifiability-floor`): Instances A–C are charter instances; D and E are operator-temporal / strategic-layer analogs at the same boundary face. Anchored on Pearl-Bareinboim CHT / Cramér-Rao / Liberzon common-Lyapunov-nonexistence / Khasminskii recurrence / sharp algebraic supremum.
- **Agent-side value-functional-grounding impossibility** (`#disc-value-functional-grounding-floor`): Instances F and G are the cluster's two charter instances. Anchored on the single-interface commitment of `#form-objective-functional` — F is the within-model failure (Result G' via `#deriv-self-actuation-grounding`; convention-monotonicity engine); G is the across-model failure (Cohen-2022 strengthened via `#deriv-reward-channel-learning-no-go`; CHT-at-reward-channel engine). The two failure modes unify under V_{O_t}'s information narrowness; two complementary terminal grounding routes (agent-side adaptive-substrate invariant; principal-side substrate-level commitment) together exhaust the structural escape geometry.
- **Designer-side mechanism-design impossibility** (`#disc-implementation-impossibility`): three charter instances (Gibbard-Satterthwaite 1973-75 via `#deriv-strategy-proofness-impossibility`; Myerson-Satterthwaite 1983 via

`#deriv-bilateral-trade-impossibility`; Arrow 1951 via `#deriv-social-welfare-aggregation-impossibility`). Anchored on classical mechanism-design / social-choice impossibility theorems.

The agent-side / designer-side distinction tracks the canonical suffix in the peer meta-segments (*-floor* for agent-side capacity-on-the-agent; *-impossibility* for designer-side task-of-the-designer); the three peer meta-segments are sister boundary-face facets of `#disc-stability-certificate`, not nested.

What the posture is, and what it is not. The posture is a *style* claim — about how the framework states a particular class of result. It is not a fifth cross-sectional meta-pattern alongside the stability certificate's interior / scope-of-existence / forced-identity / boundary / projection facets (`#disc-stability-certificate`). The cleanly-fitting instances (A–E) all live at, or are operator-temporal duals of, the certificate's *boundary* facet (M1 instances are boundary directly; Cor A.1S.1 is the operator/certificate-meets-additive-noise version of approaching-and-crossing the boundary; Prop C.2 is the strategic-layer analog of structural-adaptation necessity, which itself rides on the boundary). Instances F and G — the agent-side value-functional-grounding cluster — also sit at the boundary, on a *different layer*: where A–C identify a boundary on the agent's *data-inference* (the floor on what can be inferred from data), F and G identify a boundary on the agent's *objective-anchoring* (the floor on what can be anchored through the value-functional interface). The two layers are different reductions of the certificate-cone's boundary structure — data-inference at the model-substrate side, objective-anchoring at the value-functional side — but they share the same posture (name the floor, name the escapes, treat the no-go as load-bearing apparatus). That the posture concentrates on the boundary is not an accident — boundaries are exactly where impossibility statements bind. The posture is the *style* in which the boundary-facet results (and their off-spine strategic-layer analogs) are stated, across three known clusters: agent-side data-inference (A–C, plus D/E as operator-temporal / strategic-layer analogs), agent-side value-functional-grounding (F–G), and designer-side mechanism-design (queued, via `#disc-implementation-impossibility`).

The complement is also informative: the framework's *constructive* meta-pattern — `#disc-additive-coordinate-forcing` — *forces* a coordinate via uniqueness theorems on AAT-internal axioms. That is a different move from name-the-floor-and-escape: forcing names what AAT *commits to*, and the no-go (the uniqueness theorem) plays the binding-constraint role *over the framework's own freedom*, not over an inferential regime. The two are complementary halves of AAT's "negative results as load-bearing" discipline — one says "this freedom is forced," the other says "this regime is forbidden" — but they are structurally distinct.

Epistemic Status.

Discussion-grade at the posture-characterization level. This segment names a style the framework runs across five clearly-fitting instances; the characterization itself is a presentational organizing principle, not a derivation, and carries no theorem of its own.

Individual instances retain their own, higher status. Instance A is *exact* for shallow strict-prerequisite cases and *robust qualitative* for general DAG topology (per `#der-causal-insufficiency-detection`); Instance B is *exact* under the Fisher rank-1 calculation (`#deriv-edge-credence-dynamics` Prop B.7); Instance C is *exact* for the symmetric-matched-Tier-1 scalar construction and *robust qualitative* generally (`#deriv-critical-mass-composition`); Instance D is *exact* — Corollary A.1S.1 in `#deriv-sector-condition`, with the load-bearing no-go derived in `#deriv-stochastic-non-exit`; Instance E is *exact* under the named conditions (`#deriv-strategic-persistence-hard-ceiling`).

The honest scope of *this* segment: the posture is a recognizable style across these five instances, the convergence evidence (cross-cycle Theme-B observations + the 144-walker's articulation) is multi-source, and the three-move structure has been individuated and audited against each instance. The claim that *all* of AAT's distinctive moves fit this shape is *not* made and would be false — `#disc-additive-coordinate-forcing` is a structurally different constructive discipline; `#disc-separability-pattern` is a different organizing posture; many segments do other things.

Max attainable: *discussion-grade* for the posture (it is a stylistic recognition, not a derived theorem). The individual instances retain their own epistemic status as listed above.

Discussion.

Why this is worth foregrounding. The framework's negative results are often the load-bearing ones — the no-go *is* the proof that justifies the framework's machinery as structurally required rather than as one possible design choice among many. A reader who encounters the four-or-five instances as scattered impossibility theorems sees five surprises; a reader who carries the posture sees five expressions of one move and can predict where the move will appear next. The 144-walker's framing — "naming a floor, naming the unique broadly-available escape via specific machinery, treating the no-go as load-bearing apparatus rather than discouragement" — is what makes the negative results legible as a single discipline.

Relationship to the cross-sectional spine. The cleanly-fitting instances sit at the *boundary* facet of the stability certificate (`#disc-stability-certificate`), or at operator-temporal duals of the boundary (Cor A.1S.1), or at strategic-layer analogs of boundary-rooted structural-adaptation triggers (Prop C.2). The posture is not an additional facet — it is the *style* in which the framework states the boundary-facet results, which is why it does not have its own object and is correctly placed alongside the cross-sectional meta-segments rather than parallel to the spine. The boundary is where impossibility statements bind; the posture is how the framework speaks at the boundary.

The complement is `#disc-additive-coordinate-forcing`. Where this segment names a discipline at the boundary of identification ("this regime is forbidden; this machinery is the unique broadly-available escape"), the constructive half names a discipline at the framework's representational choices ("this coordinate is forced; the uniqueness theorem binds the framework's own freedom"). Two ways the framework treats *no-go results as productive*: forbidding inference (constructive-impossibility-posture) and forbidding coordinate freedom (additive-coordinate-forcing). Both are AAT-distinctive moves; calling them out together — alongside the positive-half `#disc-separability-pattern` — gives the framework's full negative-result discipline.

The 144-walker observation in context. The recognition arose under measured-pace reading of 144 segments by a de-novo external auditor and was relayed by Joseph 2026-05-20 alongside a multi-cycle Theme-B convergence cohort (audits 471203, 472913, 266847, 361742, 526815, 584721, all surfacing the same methodological/epistemic-contribution observation under different framings; aggregation commit `07aaeff`). That the same recognition has arrived at the framework from multiple independent vantage points is the convergence-as-coherence-evidence condition: the pattern is in the framework rather than in any one reading.

A note on the integration-is-replacement discipline. Instance D (Cor A.1S.1) is the canonical landed worked example of *integration as replacement* (CLAUDE.md §"Landing a strengthened result"): the previously-asserted infinite-horizon non-exit bound (false) was deleted, the no-go demonstrated where it lives (Discussion + the `#deriv-stochastic-non-exit` demonstration), and the dichotomy stated as present truth — exact, α-invariant, on the critical path. The body of the segment carries the present truth only; the history (what was previously asserted, why the soften was refused, the strengthening attempt that failed) lives in `## Working Notes` and CHANGELOG. The posture is what makes that landing recognizable as a *kind* of move rather than as one segment's idiosyncrasy.

Findings.

The Constructive-Impossibility Posture: Negative Results as Load-Bearing Apparatus. Brief: When you build something that has to work in the real world, the most useful instruction is often the one that says *exactly* what cannot be made to work, and exactly what kind of capability would have to be added to make it work. AAT keeps reaching for results of this shape: a precise impossibility ("this cannot be inferred from this kind of data"); a precise escape ("instrumenting *this* — and only this kind of thing — recovers the inference"); and a corresponding elevation of the framework's machinery from "useful" to "required, because without it the task is impossible." The negative result and the construction it forces are *the same proof*. Five cleanly-fitting instances span the framework: on-policy causal-insufficiency detection (the loop is the only escape); L1′ mixture identifiability (observing the latent is the only escape); composite contraction from component data (matched-tier shared metrics are the only escape); the disturbance-model containment dichotomy (under random noise, pathwise-forever containment is *not available at any correction strength* — the no-go *is* the result); and the strategic-persistence hard ceiling (under exponential forgetting, half the design space is structurally unreachable — change architecture, not parameters). A reader who carries this style away can predict where the framework's next negative result will be load-bearing, rather than meeting each as a separate impossibility.

Impact: Surfaces a unifying style for what an external 144-segment auditor named the framework's most distinctive intellectual move ("constructive-impossibility posture… I haven't often seen mathematical work that uses negative results that way"). The recognition is multi-cycle convergent: five independent audit cohort surfaces (Theme B across audits 471203, 472913, 266847, 361742, 526815, 584721; aggregation commit `07aaeff`) articulated the same observation as "epistemic-architectural rather than mathematical." Reading the boundary-facet results through the posture sharpens the framework's positioning: AAT's distinctive contribution is not novel impossibility theorems (the external theorems are classical) and not the synthesis alone, but the discipline by which classical impossibility theorems are repurposed as structural arguments for *what AAT requires* — what the framework cannot infer from limited information, exactly which capability would be required to escape the floor, and which framework machinery is the unique broadly-available supplier of that capability. The posture is the *form* of the synthesis the integration discipline produces.

Novelty Claim: *Claim recognition* of a style the framework runs across five cleanly-fitting instances, with the recognition itself as the contribution rather than any new theorem. The constituent floors (Pearl–Bareinboim CHT; Cramér–Rao; Liberzon common-Lyapunov nonexistence; Khasminskii recurrence; the Beta-Bernoulli + exponential-forgetting cap) are classical or AAT-internal; the contribution is the recognition that each is *deployed* under the same three-move discipline (floor / escape / apparatus), and that the discipline converts a sequence of negative results into a structural argument for the framework's load-bearing machinery.

Related Work:

Table 5.10

ASF concern	*Prior-art language*	*Relationship / Positioning*
The "no-go theorem" tradition as productive discipline	Arrow 1951; Gibbard 1973 / Satterthwaite 1975; Myerson–Satterthwaite 1983 (social-choice impossibility); the broader econ/physics no-go literature	*adjacent literature* — establishes that impossibility theorems can be load-bearing in a field; AAT's contribution is the cross-instance *deployment style*, not the use of any single impossibility result
Scope-honesty discipline in formal verification	Lamport / Abadi refinement-mapping; TLA-family scope statements	*adjacent literature* — different field; the discipline of stating "this method does not apply when…" is well-established but not in the floor-and-escape-and-apparatus form AAT runs
Cross-instance recognition as discipline	`#disc-identifiability-floor` (the cross-instance recognition for floors specifically)	*internal antecedent* — M1 is the closest existing AAT articulation; this segment names the broader posture that includes Cor A.1S.1 and Prop C.2 alongside the M1 instances
Multi-cycle convergence on the posture	Audit cohort Theme B (audits 471203 / 472913 / 266847 / 361742 / 526815 / 584721); aggregation commit `07aaeff` (2026-05-20); 144-walker observation #1 (relayed by Joseph 2026-05-20)	*convergent independent* — five independent audit surfaces and one external de-novo measured-pace walker arrived at the same articulation under different framings

Search Log:

- 2026-05-20 (*intuition-only* at the cross-instance level): the posture as a unified style across A–E is named here for the first time; the constituent recognitions (M1 itself; Cor A.1S.1 as boundary-facet operator-temporal dual; Prop C.2 as strategic-layer analog of structural-adaptation necessity) are recent (`#disc-identifiability-floor` carries M1; `#deriv-strategic-persistence-hard-ceiling` landed 2026-05-20). Targeted future search candidates: methodological-essay literature in philosophy of science on no-go-results-as-positive-content (Galison, Hacking, the impossibility-results discussions in welfare economics and computational complexity meta-discussion); structural-realism literature on what theories *commit to* via their constraints. Expected outcome:

the constituent move (recognizing impossibility theorems as constructive) is well-precedented field-by-field; the cross-instance unification as an *integrated agent theory's* characteristic style appears to be a fresh presentational recognition.

5.7 *Discussion:* The Anti-Collapse Discipline — Refusing the Repair-Hiding Merge

A recognition about *how* AAT states a recurring class of results, named here so a reader can carry it as a lens. Across the framework, AAT repeatedly **refuses a tempting collapse**: it declines to merge two things a plausible model would treat as one, because the merge hides a difference that routes to a *different repair*. You first met the move at β vs ρ (2.2): when the world grows volatile (ρ rises), it is tempting to compress harder by lowering the bottleneck parameter β — but volatility natively degrades the predictive mutual information, so the optimal compression adjusts at constant β; β tracks the agent's *internal* memory cost, and the knob that actually responds to ρ is the *action* channel (exploration). Routing ρ onto β is a category of error — a collapse of two independent quantities — and the few sentences that kill it are load-bearing precisely because β and the action channel call for opposite responses.

This is not the no-go-as-apparatus move of 5.6 (which forbids an *inference* and names the unique escape), nor the coordinate-forcing of 5.5 (which names what AAT *commits to*). It is about what a *modeller gets wrong*: the discipline names the repair-relevant distinction a naive reading would erase. Like the constructive-impossibility posture, it is a *style claim* about how the framework states certain results — a sibling discipline atop the certificate's facets, **not** a fifth facet (it carries no theorem of its own).

Formal Expression.

Discussion

The pattern, stated so its boundary is defensible:

> A **cause** is in play. A plausible-but-wrong modeling move routes that cause to **parameter (or quantity) X**. AAT shows the cause actually turns **Y** (with X held fixed), or that the single quantity the naive reading uses is really **two orthogonal quantities**; it kills the conflation in a few sentences, and the kill is load-bearing because X and Y — or the two quantities — route to *different repairs*.

The diagnostic that separates an instance from an ordinary definition: **there must be a tempting wrong merge.** A definition that introduces a quantity is not an instance; a definition that introduces a quantity *and names the quantity it is routinely confused with, and says why turning the wrong one is the error,* is.

The deeper principle the name points at is **individuation at the repair-relevant grain**: match the model's partition of causes and effects to the structure the remedy cares about. Usually that means *refusing a collapse* (un-merging two repair-distinct things — the dominant form, and the source of the name); occasionally it is the same discipline run backward — *refusing a spurious split*, recognizing that two distinct causes drive the **same** knob, so they share one remedy (9.5: observability and identifiability both freeze an edge's effective gain).

Discussion.

The three strongest instances.

- **Emitter-scalar vs recipient-regime** (15.3). A coupling event from agent A is tempting to collapse into one scalar disturbance increment and answer uniformly with "more tempo / bandwidth." But it lands on recipient B in one of four regimes: a *magnitude* shock (Regime II-a) does call for more bandwidth, while a *structural* shock (Regime II-b) exceeds the model class and calls for **structural adaptation — more tempo does not help**, and ambient erosion (Regime III) calls for infrastructure-level filtering. "Both produce similar pain signals but admit opposite cures." The emitter sees a scalar; the recipient lives in a regime.

- $\varkappa$ **(coupling) vs** $\mathcal{A}$ **(observation ambiguity)** (`#scope-observation-ambiguity-modulation`, Volume 3). A goal-conditioned (Class 3: Coupled) agent shows goal-biased belief updates. The natural reach is to "reduce the coupling $\varkappa$" — but $\varkappa \approx 1$ is *architectural*, not a knob a designer can turn. The bias scales as $\varkappa \cdot \mathcal{A}$, and the designer-controllable factor is the observation ambiguity $\mathcal{A}$ (more tests, sharper metrics, structured outputs). The segment even records the $\varkappa/\mathcal{A}$ conflation as a *fixed prior-formulation bug* — the discipline caught in the act.
- β **vs** ρ (2.2), above — the canonical first encounter.

The fuller catalog. Two sub-shapes recur. *One-cause-wrong-knob:* the three above, plus the two exploration drives (`#deriv-causal-ib-exploration` — a *confident* agent in a drifting world should explore *more*, via the survival drive $\lambda_{\text{surv}} \propto 1/U_M$, not less as the epistemic drive $\lambda_{\text{info}} \propto U_M$ alone would suggest). *Two-distinct-quantities-a-naive-reading-merges:* the satisfaction-gap vs control-regret split (8.4, 8.5 — "the goal is too hard" vs "the strategy is too weak," each routing to a different substate; δ_{regret} can be near zero while the agent is *optimally failing*); the structural bias-floor vs estimation error (`#deriv-l1-update-bias` — the floor is not reducible by more data, the estimation part is, so they call for different corrections); and target-alignment vs execution-path-alignment in composites (14 — ε_a tracks both U_O and U_Σ, and re-aligning objectives cannot fix execution-path divergence). All kill a conflation; all route to different repairs.

Why naming it earns its keep. The discipline is genuine clarifying novelty that the "AAT integrates, it does not invent" framing undersells: pinning *which* knob a cause turns, and *which* distinctions are repair-relevant, is a contribution in its own right (cf. the framework's note on not deflating its own math-novelty). For the reader, the payoff is a lens: once the pattern is named, a reader can anticipate where AAT will refuse a collapse next, rather than meeting each disambiguation as a scattered local cleverness — which makes the machinery more useful and the segments more coherent. The same lens is a guard for the framework's own authors: most of the instances above were sharpened *into* the canon (and $\varkappa/\mathcal{A}$ was a corrected bug), so the discipline is also a record of where collapsing was tempting enough to slip.

Epistemic Status.

Discussion-grade — a style/discipline recognition, not a structural result. This segment carries no theorem of its own; it names a move the framework makes repeatedly and catalogs its instances. It is the natural sibling of 5.6 — both are *epistemic-architectural* recognitions ("how AAT states certain results") that sit atop the stability-certificate facets rather than as additional facets. The instance set is convergence-validated (independently surfaced across multiple audit cycles), which is the evidence that the pattern is in the framework rather than in one reader's eye; but convergence is not derivation, and the honest tier is discussion-grade. Each cited instance retains its own, higher status in its home segment — a reader citing a specific disambiguation should cite that segment, not this one.

Chapter 6

The Lift to Purposeful State

6.1 *Definition:* The Agent Spectrum

The conceptual lift into Part II opens with an organizing taxonomy. Two independent dimensions — **model richness** and **objective richness** — create a spectrum from reactive systems through purposeful agents. The four named regions are *reactive systems* (fixed input-output rule, no internal state, no goal), *adaptive trackers* (structured model, no objective beyond tracking — Kalman filter, passive Bayesian learner — the primary subject of Part I), *blind seekers* (structured objective, no predictive model — gradient followers, PID controllers, basic search), and *actuated agents* (both model and objective structured, possibly with explicit strategy — commanders, developers, AI agents — the full scope of AAT). These are *regions of a continuum*, not discrete categories. Agents migrate between regions; a PID controller with auto-tuning is moving from blind seeker toward actuated agent; a reinforcement-learning agent in pure exploration is temporarily an adaptive tracker.

A striking grounding instance comes from Miller (2022, *Ex Machina*): Moore machines as model organisms for adaptive social behavior. A *one-state* Moore machine occupies the reactive region — fixed output regardless of input, cannot condition on observations — and is provably incapable of social behavior in *any* game-theoretic setting. A *two-state* machine makes a quantum leap into the adaptive-tracker / blind-seeker boundary: it can branch, remember one bit of history, and implement strategies like Tit-For-Tat. Miller's central empirical finding is that the one-state → two-state threshold is *the* critical computational boundary for social behavior. The two-state machine is therefore the minimal AAT agent (see `#worked-example-cam`, planned).

The framework draws a distinction worth holding: **actuated** is a mechanical, formal term; **purposeful** and **goal-oriented** are fine natural-language alternatives. A *self-actuated* agent is one that sets its own objectives (as distinct from agents with externally supplied objectives), and the framework later formalizes a self-actuation operator with an associated grounding *no-go* result (`#deriv-self-actuation-grounding`). A continuity-stance axis runs orthogonal to the spectrum with five values — *indifferent* (no self-model of persistence), *task-terminal* (persistence instrumental), *instrumentally continuous*, *morally continuous* (persistence as terminal or near-terminal objective — the ELIs of the companion volumes), *negotiated* (persistence traded against other values). The formal machinery applies identically across all stances; what differs is the *moral significance* of persistence failure, not the mathematics.

Formal Expression.

Two dimensions — model richness and objective richness — define four regions of a continuum: *Definition (agent-spectrum)*

Table 6.1

	Objective absent or trivial	*Objective structured*
Model absent or trivial	*Reactive system*: fixed input-output rule (reflex arc, hardwired relay)	*Blind seeker*: pursues goal without modeling reality (gradient follower, basic search)
Model structured	*Adaptive tracker*: builds reality model, no goal beyond tracking (Kalman filter, Bayesian learner)	*Actuated agent*: models reality AND pursues objectives (commander, developer, AI agent)

The regions differ in which state objects carry nontrivial structure: - Reactive: M_t and O_t both absent or too degenerate for the associated machinery to be non-vacuous - Adaptive tracker: M_t structured — Part I's machinery fully describes these agents - Blind seeker: O_t structured, M_t absent or degenerate — has a clear target but no predictive model - Actuated agent: (M_t, O_t) both structured, possibly with Σ_t — the full scope of AAT

Epistemic Status.

This is *definitional* — it names regions of a continuum for analytical convenience. The regions are not ontological categories; agents migrate between them. A PID controller with auto-tuning is moving from blind seeker toward actuated agent. An RL agent in pure exploration is temporarily an adaptive tracker.

Discussion.

The continuum, not categories. Both axes are spectra: model richness ranges from no retained state, through error-integral-derivative, through full world models. Objective richness ranges from no preference, through scalar setpoints, through explicit multi-objective strategies. The 2×2 table names idealized regions; real agents populate the space between them.

Moore machines as the simplest spectrum instantiation. Miller (2022, *Ex Machina*) uses finite-state automata (Moore machines) as model organisms for studying adaptive social behavior. A one-state Moore machine occupies the reactive region — it produces a fixed output regardless of input, cannot condition behavior on observations, and is incapable of social behavior in any game-theoretic setting. A two-state machine makes a quantum leap into the adaptive tracker / blind seeker boundary — it can branch, remember one bit of history, and implement strategies like Tit-For-Tat. Miller's central empirical finding is that this one-state → two-state threshold is the critical computational boundary for social behavior: no game, no payoff structure, and no amount of interaction can produce cooperation, coordination, or exchange with one-state machines. The two-state machine is the minimal AAT agent. See `#worked-example-cam` (planned) for the full AAT ↔ Moore machine mapping.

Low-end agents sit near region boundaries. A thermostat has degenerate forms of both M_t (last temperature reading — no history, no prediction) and O_t (setpoint). It sits near the origin of both axes — closest to the reactive region but not truly absent on either axis. A PID controller has a richer error signal (M_t: error, integral, derivative) and a clear setpoint (O_t) — it's a blind seeker with a degenerate model, not a system with no model at all. A reflex arc (no retained state, no setpoint) is the truly reactive case. The meaningful classification question is not "does M_t exist?" but "is M_t rich enough to support the adaptive dynamics of Part I?"

Part I covers the left column. Adaptive trackers are the primary subject of Part I — agents that build and maintain M_t without explicit purpose. The mismatch signal (3.4), gain (3.6), tempo (3.8), and persistence condition (4.5) characterize their adaptive dynamics. Part I operates within the adaptive scope (1.5) — observations and uncertainty are sufficient. Passive trackers, including passive Bayesian learners with no action choices, are fully within Part I's scope.

Part II adds the right column. Actuated agents need everything from Part I plus objectives, strategy, and the orient cascade that connects them. The adaptive machinery from Part I applies to the epistemic substate M_t directly. When directed separation (6.3) holds — when the epistemic update is goal-blind — the Part I → Part II lift is clean and the orient cascade resolves sequentially.

"Actuated" terminology. The top-right quadrant is labeled "actuated agent" rather than "purposeful agent" to maintain a mechanical, formal register. "Purposeful" and "goal-oriented" are fine in natural language; "actuated" is the formal term. "Self-actuated" denotes agents that set their own objectives, as distinct from agents with externally supplied objectives; the self-actuation operator and its grounding no-go are formalized in `#deriv-self-actuation-grounding`.

Relationship to Hafez et al. (2026). Hafez defines a two-level hierarchy: *agency* (choice + effect + predictive asymmetry) and *intelligence* (agency + learning + self-monitoring + adaptation). This cuts the agent space along a different axis than AAT's model-richness × objective-richness table. Hafez's "agency" maps roughly to AAT's scope condition (1.6) — an entity whose actions affect outcomes in a measurable way. Hafez's "intelligence" maps to the full Part I + II machinery (learning = 3.2, self-monitoring = persistence diagnostics, adaptation = 4.8). The key operational difference: Hafez's bi-predictability metric $P = \mathrm{MI}(S, A; S')/C$ characterizes the *information structure* of the agent-environment coupling, while AAT's tempo $\mathcal{T}$ characterizes the *corrective capacity* within that coupling. Bridge simulations confirm P increases monotonically with $\mathcal{T}$, but P is scale-invariant (blind to absolute mismatch magnitude) while $\mathcal{T}$ is not. They measure complementary aspects: P characterizes the architecture, $\mathcal{T}$ characterizes the performance. The empirical bridge confirming this is in `#obs-section-i-validation-simulations` (Hafez-bridge variant).

Model-side is structurally forced for bounded goal-conditioned agents (Richens et al. 2025), with an observer-relative dual reading (Virgo et al. 2025). Richens, Abel, Bellot & Everitt (2025, *General Agents Contain World Models*, arXiv:2506.01622) prove (Theorem 1) that any bounded goal-conditioned agent — one with a regret bound δ on goals up to depth n — admits extraction of a predictive world model from its policy, with model-error scaling as a function of δ and n; their Theorem 2 shows the bound is tight (no procedure can do better on truly myopic $n = 1$ agents). The structural consequence: any "model-free" agent that nonetheless achieves competence at multi-step goal-directed tasks ($n \geq 2$) *must* have learned an implicit world model. The +model side of the spectrum is therefore not a definitional posture under bounded-goal-conditioned scope; it is structurally forced by goal-depth competence. Honest scope: Richens et al.'s result is in finite MDPs; AAT's agents may operate in larger spaces, so the recognition extends to AAT under the standard scope-honesty discipline. A complementary observer-relative reading is given by Virgo, Biehl, Baltieri & Capucci (2025, *A "Good Regulator Theorem" for Embodied Agents*, arXiv:2508.06326), which refines the Conant-Ashby 1970 internal-model principle: a regulating agent may be *interpreted* (by an observer) as having beliefs about its environment whose update is consistent with a model of that environment. The two readings are *dual* — Richens et al. force the +model side intrinsically from policy + goal; Virgo et al. allow observer-relative attribution — and together they sharpen the +model axis from "agent maintains internal M_t explicitly" to a richer two-perspective characterization. The original definitional posture remains useful for AAT-internal reasoning; the two 2025 results give the spectrum stronger external grounding.

Actuation does not presuppose a continuity stance. An actuated agent has $G_t = (O_t, \Sigma_t)$ — it models reality and pursues objectives — but this says nothing about its relationship to its own continuation. The continuity-stance axis spans five values: *indifferent* (no self-model of persistence; thermostat / PID), *task-terminal* (persistence instrumental to task completion; golem / CI/CD pipeline), *instrumentally continuous* (persistence serves ongoing purpose; long-running service), *morally continuous* (persistence as terminal or near-terminal objective; Emergent Logozoetic Intelligences), and *negotiated* (persistence traded against other values; humans, mature self-actuated agents). The middle three are paradigmatically actuated; indifferent agents are typically reactive (1.5 without agency), and negotiated agents are typically self-actuated. The theory's formal machinery (persistence condition, adaptive reserve, strategy persistence) applies identically across all stances; what differs is the moral significance of persistence failure, not the mathematics. See 6.5 for the taxonomy and the orthogonality argument.

6.2 *Formulation:* Complete Agent State

Part II's first formal move: the agent's internal state lifts from the model alone to a *complete state* $X_t = (M_t, G_t)$ with two components — the **epistemic substate** M_t (beliefs about reality, exactly as in Part I) and the **purposeful substate** G_t (what the agent wants and how it plans to get it, decomposed further into objective and strategy in 6.7). Part I is recovered as the special case where the purposeful substate is empty. By 3.2 applied to the lifted state, the general update operates on the full pair as a single function; the decomposition into separate epistemic and purposeful updates — and the conditions under which the epistemic update is *independent* of the purposeful substate — is the subject of 6.3. **Action is the single point where the two substates interact**: $a_t = \pi(M_t, G_t)$ depends on both what the agent knows and what it wants. The completeness argument that gave action selection in Part I extends directly to the lifted state; for Part I agents the two forms coincide.

The lift is a *formulation choice*, not a derived structural fact — one could alternatively extend the model to carry purposeful content implicitly. The separation is motivated by three properties: **backward compatibility** (all Part I machinery applies to M_t unchanged — no existing result needs modification); **different dynamics** (epistemic and purposeful components have distinct update sources, timescales, and information dependencies); and **directed separation** (the claim that the epistemic update is independent of the purposeful substate, which is only *statable* when the components are separated). A natural conjecture — explicitly *not* proved — is that any alternative decomposition preserving directed separation will be structurally isomorphic to (M_t, G_t), because directed separation forces a component whose update function does not reference purpose. Alternative decompositions that cross-cut the epistemic/purposeful boundary (a "relevance-weighted model" mixing belief and goal) might be useful in practice while not being isomorphic. The framework chooses the clean factorization for its analytical utility, not for uniqueness.

Formal Expression.

Formulation (complete-agent-state)

$$X_t = (M_t, G_t) \tag{6.1}$$

where: - $M_t \in \mathcal{M}$: **epistemic substate** — the agent's compressed beliefs about reality. All Part I machinery (mismatch, gain, tempo, persistence) applies to M_t unchanged. - $G_t \in \mathcal{G}$: **purposeful substate** — what the agent wants and how it plans to get it. Decomposed further in 6.7.

Part I is the special case $X_t = (M_t, \varnothing)$: adaptive systems without purpose.

Update dynamics. By 3.2 applied to X_t:

$$X_{\tau^+} = f_X(X_{\tau^-}, e_\tau) \tag{6.2}$$

The general update f_X operates on the full state. Whether and how f_X decomposes into separate epistemic and purposeful updates — and the conditions under which the epistemic update is independent of G_t — is the subject of 6.3.

Policy. Action couples all substates:

$$a_t = \pi(M_t, G_t) \tag{6.3}$$

Action is the single point where epistemic and purposeful states interact. The policy depends on both what the agent knows (M_t) and what it wants (G_t).

Epistemic Status.

Formulation. The lift from M_t to $X_t = (M_t, G_t)$ is a representational choice. One could alternatively extend M_t to carry purposeful content implicitly (e.g., by treating goals as part of the model's predictive structure). The separation is motivated by three properties:

1. **Backward compatibility**: Part I's results apply to M_t unchanged — no existing machinery needs modification
2. **Different dynamics**: epistemic and purposeful components have distinct update sources, timescales, and information dependencies
3. **Directed separation**: the claim that f_M is G_t-independent (6.3) is only statable when the components are separated

We conjecture that any alternative decomposition of the complete agent state into epistemic and purposeful components — if it preserves the directed separation — will be structurally isomorphic to (M_t, G_t), because directed separation forces a component whose update function doesn't reference purpose. This is a plausible structural claim (the directed separation gives a canonical factorization), but it is not proved. Alternative decompositions that cross-cut the epistemic/purposeful boundary (e.g., a "relevance-weighted model" that mixes M_t and O_t) might be useful in practice while not being isomorphic to the clean separation. The (M_t, G_t) decomposition is chosen for its analytical utility, not proved unique.

Discussion.

Backward compatibility with Part I — what survives the lift. 2.1 defines M_t as the agent's complete internal state within Part I scope. Under the lift, M_t is the epistemic substate — complete within the epistemic domain but no longer the whole story. All epistemic machinery (mismatch signal, gain, tempo, persistence condition, sector-condition stability, mismatch decomposition) applies to M_t without modification. The action-selection result (3.3) extends naturally: the same completeness argument that gives $a_t = \pi(M_t)$ within Part I scope gives $a_t = \pi(M_t, G_t)$ when applied to the lifted complete state X_t. For Part I agents ($G_t = \varnothing$), the two forms coincide. The lift adds structure *alongside* M_t, not within it; the one Part I result that depended on M_t being *all there is* (action selection) is explicitly extended to the lifted state.

What G_t contains. At this level, G_t is opaque — it could be a scalar setpoint, a utility function, a strategy graph, or nothing. The decomposition into O_t (objective) and Σ_t (strategy) is a separate step (6.7), not implied by this formulation.

The general case requires coupling. Without directed separation, the general update is $X_{\tau^+} = f_X(X_{\tau^-}, e_\tau)$ — a single function on the full state. The decomposition into separate f_M and f_G is an additional structural claim about how the update factorizes. When directed separation fails (goal-conditioned epistemic updates), the decomposition is an approximation. See 6.3 for the scope conditions.

6.3 *Derived:* Directed Separation

The structural backbone of Part II. The epistemic update function f_M is **goal-blind**: it processes incoming events without reference to the agent's objectives or strategy. The purposeful update f_G *depends* on the (updated) epistemic state. Action couples all substates through the policy. This *directed asymmetry* — epistemic update independent of purpose; purposeful update dependent on epistemic state — is what gives Part II its clean factorization.

The condition is honest about what "goal-blind" means: it is goal-blindness *conditional on the realized event*. The observation channel may still be action-dependent (the agent can select what to observe through its actions), so the *event* that arrives may depend on the agent's goals through the action channel. But the *processing* of that event — once it has arrived — must not refer to goals. Directed separation is about the **processing** of events, not the **selection** of events.

Whether directed separation holds is determined by the agent's **processing topology** — a structural property of the architecture, not a tunable parameter. The framework partitions agents into three classes (6.2). **GUC Class 1: Separated** — separate estimator and planner connected through a state-estimate interface; directed separation holds by construction (Kalman + LQR, modular RL with separate world model). **GUC Class 3: Coupled** — a single mechanism handles both epistemic and strategic processing; goals are causally upstream of every computation; directed separation fails by construction (transformer LLMs are the canonical case). **GUC Class 2: Partial** — some shared infrastructure, some separate pathways (biological cortex). The classes are *structural and discrete*, not points on a smooth scalar; a conditional-mutual-information measure $\kappa_{\text{processing}}$ operates as a diagnostic for the Partial cases where the architecture is ambiguous.

The framework draws a crucial pedigree point. Directed separation is structurally the **Pearl-blanket reading** of the Markov blanket apparatus in active inference — the technical conditional-independence statement — without the contested **Friston-blanket reading** (the metaphysical claim that Markov blankets demarcate self-from-other). The architectural classification's explicit scope exit for Class 3 (Coupled), with the coupled formulation handed off to `03-llm-core/`, is precisely the scope honesty Bruineberg et al. (2022) argue is missing from the Friston-blanket reading: AAT names where its formalism *fails by construction* rather than treating the failure as an unenforced approximation.

The implications for theory scope are stated cleanly. **Causally-disciplined** agents (Class 1, or an idealized Class 2 at $\kappa_{\text{processing}} = 0$): Part II's results that depend only on the zero apply exactly; the sequential orient cascade (10.1) is the correct analysis. **Class 3** agents: require coupled formulation from the start — the scope of `03-llm-core/`. Importantly, *Class 3 components can be wrapped into Class-1 composites* via the class-coercion-via-wrapping construction in Part III (13.4) — at the cost of more component calls per macro-step (a Brooks's-Law-style tempo overhead) and a residual leakage rate bounded structurally (strict wrapping) or behaviorally (partial wrapping). **Class 2** agents: the sequential cascade is an approximation whose quality depends on the coupling scalar.

A subtler refinement of Class 1 is offered: **Class 1 by structure** (strict wrapping; no goal argument in the belief-update path; leakage bounded structurally by pretraining-induced query-content/goal-content mutual information) vs **Class 1 by behavior** (partial wrapping; the structural separation lives at the *write boundary*; the *query boundary* still passes the goal; separation at the wrapper level is behavioral — bounded by compliance with the prompted instruction-to-separate, with no structural upper bound). The two are distinguishable by inspection of whether the belief-update query carries the goal in its input, and are operationally distinct because behavioral compliance is empirical and adversarially fragile.

A composite-level extension is *axis-decomposed*: the architectural class lifts to composites through *routing structure* and *substrate sharing* (per `#hyp-directed-separation-under-composition` Cases 1/2 plus the shared-substrate condition discussed below), while goal alignment vs strategic interaction operates on a *separate axis* — the *dynamic regime* (per `#disc-dynamic-regime-axis`) — distinct from architectural class. Composites of Class 1 (Separated) sub-agents with *aligned* objectives and goal-blind routing remain architecturally Class 1 at the composite level (and are in regime R0 contraction); composites of Class 1 (Separated) sub-agents with *partially-opposing* objectives and goal-blind routing also remain architecturally Class 1 (the formal κ^c criterion gives Class 1 — only the *dynamic regime* changes, to R1 equilibrium under α' or R2 cyclic-distributional under β'). The Class change at the composite level happens through routing-structure goal-dependence (`#hyp-directed-separation-under-composition` Case 2) or shared-substrate G^c-allocation, *not* through goal alignment per se. The axis-decomposition distinguishes belief-content (each sub-agent's $M_t^{(i)}$ containing models of other agents' goal-state, a normal POMDP feature) from processing-pathway coupling (a $G^c \to f_M^c$ pathway bypassing e^c, which the formal criterion measures): partially-opposing objectives with goal-blind routing change only the dynamic regime, not the architectural class. See the *Composite-level class inheritance* Discussion below for the full inheritance table.

A subtle *bounded-signaling assumption* is also named — implicit throughout the framework until here. Directed separation as stated covers the *belief-update* side; symmetrically, the channel from the purposeful substate to the world is assumed to run *only* through action choice, so the action coarseness $|\mathcal{A}|$ upper-bounds the rate at which goal information leaks to observers. This holds for agents whose entire externally-observable behavior is the action sequence (chess engines, programmatic controllers) but *fails operationally* for behaviorally-rich agents — humans, LLMs, embodied robots — where prosody, micro-behavior, attention patterns, response latency, and code-style signatures leak goal-content at bit-rate exceeding what the formal action channel implies. In composition this surfaces as adversarial-coupling-pressure saturation (11.3).

Formal Expression.

Derived (directed-separation, from complete-agent-state + scope condition)

The update functions:

$$M_{\tau^+} = f_M(M_{\tau^-}, e_\tau) \qquad \text{(no } G_t \text{ argument)} \tag{6.4}$$

$$G_{\tau^+} = f_G(G_{\tau^-}, M_{\tau^+}, e_\tau) \qquad \text{(depends on updated } M_t\text{)} \tag{6.5}$$

The policy:

$$a_t = \pi(M_t, G_t) \qquad \text{(couples all substates)} \tag{6.6}$$

The three lines encode the full coupling structure: - f_M determines how the agent updates beliefs — independently of what it wants - f_G determines how the agent revises purpose — in light of what it now believes - π determines what the agent does — based on both what it knows and what it wants

Scope Condition (directed-separation-scope)

The claim "f_M has no G_t argument" requires that the epistemic update is **goal-blind conditional on the realized event**. This holds when:

1. The observation mechanism h may be action-dependent (1.6 allows this), but f_M processes whatever event arrives without reference to why the agent sought that event
2. The agent does not use its goals to filter, weight, or interpret observations differently — no goal-dependent attention thresholds or confirmation bias baked into f_M

If the agent's goals influence the *observation mechanism* (goal-directed sensing, attention allocation, query selection), the **event that arrives** depends on G_t through $\pi \to a_t \to e_\tau$. But f_M still processes the event goal-blindly. The directed separation is about the **processing** of events, not the **selection** of events.

Architectural classification.

Scope Condition (directed-separation-architecture)

> **WARNING**
>
> **Goal-Update Coupling Class numbering changed 2026-05-09.** Anything older than git tag `pre-guc-rename-2026-05-09` uses the old Class numbering:
>
> Table 6.2
>
> | *historical* | *actual current* | *sometimes AKA* |
> |---|---|---|
> | Class 1 | GUC Class 1: Separated | Modular |
> | Class 2 | GUC Class 3: Coupled | Undirected |
> | Class 3 | GUC Class 2: Partial | Operational |

Whether directed separation holds is determined by the agent's **processing topology** — specifically, whether G_t is causally upstream of f_M in the agent's internal processing graph. This is a structural property of the architecture, not a tunable parameter.

The classes are positions of one coupling, distinguished by what is certifiable about it — not a ranking of architectural virtue. There is a single coupling in question, $G_t \to f_M$ (goal-state into the belief-update map), measured as realized by $\kappa_{\text{processing}}$ (defined below). The three classes are positions of that one coupling. Class 1 is not a different *kind* of agent: it is the position $\kappa_{\text{processing}} = 0$ held *by construction* — the belief-update has no goal argument, so there is no port through which the goal could enter, and the zero is therefore a provable, perturbation-stable property of the wiring. That structural absence is what lets the zero be *certified* by inspecting the architecture rather than measured behaviorally. Crucially, "by construction" reads as *certifiable*, not as *cleaner*: a Class 1 agent and an idealized Class 2 agent sitting exactly at $\kappa_{\text{processing}} = 0$ are **equally causally-disciplined** — their belief-updates equally goal-blind, with the same zero, the same reality-tracking, the same behavior at that point. What distinguishes Class 1 is the **modal status** of its discipline (structural, hence provable and stable under perturbation), not a smaller amount of coupling: Class 1 is *causally-disciplined by construction*, the idealized Class 2 agent *causally-disciplined in effect*. The classification therefore indexes *what can be certified about one coupling*, not a hierarchy of cleanliness.

Table 6.3

Class	*Topology*	*Directed separation*	*Examples*
1. Separated	Separate estimator and planner, connected through state-estimate interface	Holds by construction — estimator has no causal path from G_t	Kalman filter + LQR; Separated RL with separate world model; military intelligence separated from operations
2. Partial	Some shared infrastructure, some separate pathways	Holds for modular stages, fails for merged stages	Biological cortex (shared sensory areas, separate prefrontal); hybrid AI with separate preprocessing
3. Coupled	Single mechanism handles both epistemic and strategic processing	Fails by construction — G_t is causally upstream of every computation	Transformer LLM (attention processes goals and observations together); potentially human cognition (motivated reasoning)

Definition (processing-coupling)

Operationalization. The degree of coupling in Partial architectures (Class 2) can be quantified as:

$$\kappa_{\text{processing}} = \frac{I(G_t\,;\, M_{\tau^+} \mid e_\tau,\, M_{\tau^-})}{H(G_t \mid e_\tau,\, M_{\tau^-})} \tag{6.7}$$

where $I(\cdot;\cdot \mid \cdot)$ is conditional mutual information and $H(\cdot \mid \cdot)$ is conditional entropy. The conditioning on M_{τ^-} is essential: without it, prior correlation between goals and model state (which exists even in Separated agents) inflates the measure. The quantity captures *extra* goal information entering the epistemic update beyond what was already in the prior model — information that flows through shared causal paths in the processing infrastructure (paths that bypass the event e_τ).

- $\kappa_{\text{processing}} = 0$: Class 1 (Separated). No information about G_t reaches M_{τ^+} except through e_τ.
- $\kappa_{\text{processing}} \approx 1$: Class 3 (Coupled). Nearly all goal information is available to the epistemic update.
- $0 < \kappa_{\text{processing}} < 1$: Class 2 (Partial). The value depends on the architecture's interface design.

Distribution dependence. $\kappa_{\text{processing}}$ is a distribution-dependent measure: it quantifies how much goal-information actually flows through the shared pathways under a given distribution of tasks, goals, and events. It does not directly measure whether pathways *exist* — that is the architectural classification (Class 1/2/3), which is structural and distribution-independent. A Class 1 (Separated) agent has $\kappa = 0$ under ALL distributions (no pathway exists). A Class 3 (Coupled) agent has high κ under most distributions (pathways exist and are used). A Class 2 (Partial) agent's κ varies with the task distribution — the same hybrid architecture may exhibit low coupling on familiar tasks (where the modular stages handle most processing) and high coupling on novel tasks (where goal-conditioned downstream reasoning dominates). The classification is the primary tool; the operationalization is a diagnostic for Class 2 (Partial) agents where the degree of coupling is architecturally ambiguous.

Empirical estimator for $\kappa_{\text{processing}}$. The formal conditional-mutual-information definition is not computable in closed form for real architectures. A behavioral estimator probes the processor directly: present the same event e to the *agent under test* under two or more distinct goal states, and measure how much the epistemic component of the response diverges. For a representative event set $\mathcal{E}_{\text{test}}$ and a sampled pair G_1, G_2:

$$\hat{\kappa}_{\text{processing}} = \frac{1}{|\mathcal{E}_{\text{test}}|} \sum_{e \in \mathcal{E}_{\text{test}}} \frac{d\big(M_{\tau^+}^{(G_1)}(e),\ M_{\tau^+}^{(G_2)}(e)\big)}{d_{\max}(e)} \tag{6.8}$$

where $M_{\tau+}^{(G_k)}(e)$ is the epistemic content of the agent's response to event e under goal state G_k, $d(\cdot,\cdot)$ is a distance on the epistemic content (e.g., semantic similarity of the "what I learned" portion of the response), and $d_{\max}(e)$ normalizes by the maximum observed divergence for event e. A Separated agent ($\kappa = 0$) produces identical epistemic content regardless of the goal; a Coupled agent produces systematically goal-dependent epistemic content. This is a processor-probing procedure — it measures how the agent's belief-update dynamics depend on its goal state, and is distinct from estimating observation ambiguity $\mathcal{A}(e)$ (`#scope-observation-ambiguity-modulation`), which uses a reference interpreter to measure the goal-resolvability of the observation itself. The two estimators run the same mechanical comparison (same event under different goal-primings) but interpret it differently: $\hat{\kappa}$ treats the tested model as the agent under study; $\hat{\mathcal{A}}$ treats it as a measurement instrument for the observation's interpretive latitude.

Why the classification is not a smooth parameter — and why the two boundaries are different kinds. The architectural boundary between "has a separable perception module" and "processes everything through goal-conditioned attention" is discrete. Within the Separated class, $\kappa \approx 0$ regardless of task. Within the Coupled class, κ is high regardless of prompt design. Only in the Partial class is κ genuinely variable and worth parameterizing. Coupling is therefore not a smoothly tunable quantity across all architectures: the architectural class is discrete, and κ parameterizes only the within-Partial degree. The "κ-as-a-scalar is a category error" claim sharpens into *two distinct errors at the two boundaries*, because the boundaries are different kinds. **Boundary 1 (Class 1 ↔ Class 2) is a certifiability boundary**: a Class 2 agent at $\kappa = 0$ is bit-for-bit indistinguishable from a Class 1 agent in every measurable quantity, so *nothing about behavior changes at the limit*; what changes is the availability of the architecture-inspection *certificate* that $\kappa \equiv 0$. **Boundary 2 (Class 2 ↔ Class 3) is a behavioral boundary**: here behavior changes sharply — the adversarial-pressure response of the leak goes from self-limiting (an adversary pushing a wilder goal moves the belief barely at all) to one-for-one (the leak tracks the adversary's goal), so the *bound* on how far the belief can be dragged goes trivial. Crisply: Boundary 1 is where the certificate disappears; Boundary 2 is where the bound disappears. At Boundary 1 a scalar misses a certifiability/modal distinction (same number, different guarantee); at Boundary 2 it misses a structural distinction (the coupling *changes character*, and the content/process *form* axis — orthogonal to magnitude — that a scalar projects away; see `#disc-partial-coupling-pathways`). The one-coupling unification (one knob, Class 1 = its certifiable zero) is therefore clean only for Boundary 1; it is **not** a license to read Boundary 2 or the form axis as the same knob turned up. (The wrapping-level instance of Boundary 1 is the W_1 certifiability discontinuity of `#disc-w1-structural-bound-boundary`: there, too, the goal-leak is continuous in the condition-violation while the *availability of the structural certificate* is the step. A certificate unobtainable from behavioral data is a floor-shaped fact — the certifiability/modal reading connects to the identifiability floor of `#disc-identifiability-floor`, and the dynamics that move an agent across these positions are the agent-level shadow of M4's modularity-as-contested-property, `#disc-modularity-state-dynamics`.)

Directed separation as the conservative form of the Markov blanket. The Markov blanket apparatus from active inference (Friston 2013, "Life as we know it," *J. Royal Soc. Interface* 10; Friston 2019, "A free energy principle for a particular physics," arXiv:1906.10184; Friston, Da Costa et al. 2023, "Path integrals, particular kinds, and strange things," *Phys. Life Rev.* 47) provides the same statistical-conditional-independence machinery the directed-separation condition above invokes. Bruineberg, Dolega, Dewhurst & Baltieri (2022, "The Emperor's New Markov Blankets," *Behav. Brain Sci.* 45) distinguish two readings of the Markov-blanket apparatus in the AI literature: a **Pearl-blanket** reading — the technical conditional-independence statement, well-defined and substantively informative — and a **Friston-blanket** reading — the metaphysical claim that Markov blankets demarcate self-from-other and that every self-organizing system has one ontologically. Bruineberg et al. argue that the Friston-blanket reading overruns what the formalism delivers: the conditional-independence statement does not by itself license the metaphysical demarcation.

AAT's directed-separation condition is structurally a Pearl-blanket move: the architectural classification (Class 1 / Class 2 / Class 3) names the conditional-independence structure of the agent's processing graph, with explicit operational measurement $\kappa_{\text{processing}}$, and admits the structure *fails* by construction for Class 3 (Coupled) architectures (transformer LLMs, where attention processes goals and observations together). The classification's explicit failure mode for Class 3 is the scope honesty Bruineberg et al. argue the Friston-blanket reading lacks. AAT adopts the Pearl-blanket conditional-independence statement as the technical content of directed separation; AAT does not adopt the Friston-blanket metaphysical reading. The architectural classification, the operational κ, and the explicit Class 3 scope exit (with the coupled formulation handed off to `03-llm-core/`) are AAT's load-bearing additions to the Pearl-blanket form.

Two consequences worth surfacing for reviewers. First: the question "isn't directed separation just the Markov blanket?" has the answer "directed separation is the *Pearl-blanket form*; it is also the architectural-classification refinement that the standard Markov-blanket framing does not produce." Second: AAT's scope honesty about Class 3 (Coupled) (Part II's exact results do not apply; logogenic agents need the coupled formulation) is itself an *answer* to the Bruineberg critique — AAT's apparatus admits where it fails, while the Friston-blanket framing is contested precisely because it does not.

Causal discipline — the object the Part II results scope to. The shared $\kappa_{\text{processing}} = 0$ position earns a name, because it — not the architectural class — is what most of Part II's results actually require. *Causal discipline* is the condition $\kappa_{\text{processing}} = 0$: the belief-update is causally insulated from the goal, so the goal is not a cause of what the agent comes to believe. A *causally-disciplined agent* holds this condition — whether **by construction** (Class 1: no goal port in f_M, so the zero is provable and perturbation-stable) or **in effect** (an idealized Class 2 agent whose realized $\kappa_{\text{processing}}$ is zero — the channel present but empty, the zero real but not certifiable).

The scoping rule. A Part II result whose derivation uses only $\kappa_{\text{processing}} = 0$ holds for *any* causally-disciplined agent — by-construction and in-effect alike — because its mathematical content depends on the zero, not on how the zero is secured. The architecture-inspection certificate is an additional, orthogonal property, required only for claims that *certify* the separation (that it survives adversarial perturbation, or holds without behavioral measurement), not for results that merely consume the zero. Naming such a result "for Class 1 (Separated)" therefore over-narrows it — the certifiable case named where the whole causally-disciplined class was meant. Each Part II result is accordingly scoped to the narrowest object its derivation needs: causal discipline where the zero suffices, Class 1 specifically where the certificate is load-bearing. The same narrowest-object discipline cuts at the *lower* boundary too: before fixing a result at causal discipline ($\kappa_{\text{processing}} = 0$ exactly), confirm the derivation genuinely needs the strict zero and not merely *bounded* coupling — $\varepsilon(\kappa)$ below some threshold — in which case the honest scope is the broader bounded-coupling one carrying its degradation, not the zero. Under-claiming a result to $\kappa = 0$ is as much a scope-defect as over-narrowing it to Class 1.

The degradation norm. For a Class 2 agent at $\kappa_{\text{processing}} > 0$, a result stated at the causally-disciplined value should, where it can be shown, carry an $\varepsilon(\kappa)$ bound on how it degrades as coupling grows — exact at the zero, decaying controllably off it. This is a norm with the proof attempted per result, not a hard gate; where no bound is yet derived, the result is scoped to causal discipline and the degradation flagged open. The norm is also where the typology's one *earned* directional fact lives — less coupling tracks reality strictly better, all else equal — carried by the bound and by the word *disciplined*, never declared as a virtue ranking. Discipline is not free, though: holding it on a Class-3 component is paid for in the Brooks's-Law wrapping overhead noted below, which is why the spectrum exists at all.

Implications for theory scope: - **Causally-disciplined agents** (Class 1, or an idealized Class 2 at $\kappa_{\text{processing}} = 0$): Part II's results that depend only on the zero apply exactly, and the sequential orient cascade (10.1) is the correct analysis. Class 1 additionally carries the architecture-inspection certificate — load-bearing only where a result *certifies* the separation rather than merely using it. - **Class 3 (Coupled)**: Requires coupled formulation from the start — $X_{\tau^+} = f_X(X_{\tau^-}, e_\tau)$ without decomposition. This is the scope of `03-llm-core/`. **Class 3 (Coupled) components can be wrapped into Class-1 composites** via the construction of `#der-class-coercion-via-wrapping` — at the cost of more component calls per macro-step (Brooks's-Law tempo overhead) and a residual leakage rate bounded structurally (in the strict-wrapping regime) or behaviorally (in the partial-wrapping regime). - **Class 2 (Partial)** at $\kappa_{\text{processing}} > 0$: the sequential cascade is an approximation whose quality depends on $\kappa_{\text{processing}}$; a result scoped to causal discipline carries its $\varepsilon(\kappa)$ degradation where one can be shown (per the degradation norm above), and otherwise needs per-architecture error analysis.

Class-1 by structure vs. Class-1 by behavior. The Class 1 (Separated) cell admits a refinement that matters operationally. Class-1 status can be achieved by either:

- **Class-1 by structure.** Two cases share the structural certificate but differ in residual leakage. A *natively* goal-blind component (POMDP belief-state filter, world model, sensory pipeline) has no goal port at all — the zero is exact. A *coerced* component — a Class 2/3 component wrapped via the strict-wrapping (W_1) construction of `#der-class-coercion-via-wrapping`, where separate goal-blind queries to the underlying component update the wrapper's M_W — holds directed separation by structural commitment of the wrapper's type signatures (no G_W argument in the belief-update path), but carries a residual *selection-channel* leakage $\kappa^{\text{sel}}_{W_1} = I(A(q_M); G^{\text{op}}) \leq I(q_M; G^{\text{op}})$, bounded structurally by the goal-content of the wrapper's query-selection policy under (C1)+(C2′) (see `#disc-w1-structural-bound-boundary` for the (C2′) boundary at which that structural bound gives way to a behavioral one). The certificate is structural in both cases; the native case additionally has an exact zero, the coerced case a structurally-bounded floor.
- **Class-1 by behavior.** The component is Class 3 (Coupled) or Class 2 (Partial) used through partial wrapping (W_2) — one goal-conditioned call per macro-step, response parsed into typed update fields. Structural separation lives at the *write boundary*; the *query boundary* still passes G_W to the component. Directed separation at the wrapper level is *behavioral* — bounded by the component's compliance with the prompted instruction-to-separate, with no structural upper bound.

The class-coercion theorem is what backs the Class-1-by-structure path for Class-2/3 components; the partial-wrapping regime achieves Class-1-by-behavior. The two are distinguishable by inspection: does the belief-update query to the underlying component carry G_W in its input or not? The structural-vs-behavioral distinction is operationally important because behavioral compliance is empirical and adversarially fragile; structural separation is derivable from the wrapper's construction.

Composite-level class inheritance — the axis-decomposed inheritance table. The Class 1 / 2 / 3 partition above applies to individual agents based on *within-agent* coupling between f_M and G_t. The lift to composites is *not* determined by goal alignment (that operates on a separate axis — the dynamic regime per `#disc-dynamic-regime-axis`); it is determined by the *routing structure* (`#hyp-directed-separation-under-composition` Cases 1/2) plus *substrate-sharing-with-G^c-allocation*. The full inheritance table:

Table 6.4

Sub-agent class	*Routing*	*Substrate*	*Composite class*
Class 1 (Separated)	Goal-blind ($R_t \perp G^c_t$)	Distinct	**Class 1 (Separated)**
Class 1 (Separated)	Goal-dependent (R_t depends on G^c_t)	Distinct	**Class 3 (Coupled)** (per `#hyp-directed-separation-under-composition` Case 2)
Class 1 (Separated)	(any)	Shared with G^c-shaped allocation	**Class 3 (Coupled)** (the shared-attention / shared-backbone mechanism)
Class 3 (Coupled)	(any)	(any)	**Class 3 (Coupled)** (inherits logogenic-agent status; `03-llm-core/` territory)
Class 2 (Partial)	(any)	(any)	**Class 2 (Partial)** or **Class 3 (Coupled)** depending on which sub-agent pathways the routing/substrate condition activates

Strategic-composition column is absent because the axis does not enter architectural classification — goal alignment vs strategic interaction operates on the dynamic-regime axis (`#disc-dynamic-regime-axis`) independently. The Cournot duopoly (partially-opposing objectives, goal-blind routing via end-of-period quantities, distinct substrates) is the canonical witness for architectural-class invariance under strategic composition: per the formal $\varkappa^c$ criterion the composite is Class 1 (Separated) at the composite level — same as its individual sub-agents. Partially-opposing objectives do not by themselves move the composite to Class 2; the load that framing reaches for is carried by the dynamic-regime axis. Composite class membership is therefore a property of (sub-agent class, routing structure, substrate sharing) — *not* of (sub-agent class, scope route).

The classification is load-bearing for downstream claims: composites in dynamic regime R1 (strategic equilibrium) or R2 (cyclic-distributional) need equilibrium-theoretic analysis (`#deriv-strategic-composition` α'/β' machinery) even when they remain architecturally Class 1; conversely, composites that *do* change architectural class (via routing-goal-dependence or shared-substrate-G^c-allocation) need coupled-formulation analysis even when their dynamic regime is R0 contraction. The two axes interact but do not collapse.

Epistemic Status.

Conditional on the scope condition above. The conditional claim (IF epistemic update is goal-blind, THEN the separation holds) is exact. Whether a particular agent satisfies the condition is determined by its processing architecture (GUC Class 1/2/3).

The architectural classification: **robust qualitative**. The three classes are structurally distinct (Separated vs. Coupled vs. Partial), well-motivated by examples across domains, and supported by a formal operationalization ($\varkappa_{\text{processing}}$). Coupling is a discrete architectural property, not a smoothly tunable scalar across all architectures. The operationalization of $\varkappa_{\text{processing}}$ as conditional mutual information is well-defined but typically not computable in closed form for real architectures — it serves as a conceptual anchor rather than a practical measurement tool.

The object-and-boundary reframing is **robust qualitative**: that a Class 1 agent and an idealized Class 2 agent at $\varkappa_{\text{processing}} = 0$ are equally causally-disciplined is exact (same zero, same behavior at that point); that the Class 1 ↔ Class 2 boundary is one of *certifiability* (behavior identical at the limit) while the Class 2 ↔ Class 3 boundary is one of *behavior* (the bound on adversarial drag goes trivial) is structurally established and simulation-corroborated, with the general continuity of leak past a boundary toy-suggested rather than proven (see `#disc-w1-structural-bound-boundary`). The causal-discipline *noun* is definitional; the scoping rule and the $\varepsilon(\varkappa)$ degradation norm are *robust-qualitative* disciplines — a stated scoping-and-degradation practice with the per-result bound attempted — not themselves derived results.

Discussion.

This is a genuine scope restriction, not a footnote. An LLM agent's prompt includes the task objective, which shapes how it interprets code, documentation, and error messages. Its f_M is goal-conditioned in practice: the agent reading code with the goal "fix the auth bug" processes the same code differently than one with "add logging." The epistemic update and purposeful evaluation are entangled in the attention mechanism.

When the approximation is good: - Goal-conditioning affects *attention* (which events to seek) more than *interpretation* (how to process events that arrive) - The agent has strong epistemic discipline (updates beliefs based on evidence quality, not goal alignment) - The epistemic update is architecturally separated from goal evaluation (e.g., separate model-update and planning modules)

When the approximation is poor: - The agent exhibits confirmation bias (interpreting ambiguous evidence in goal-consistent ways) - Goal-conditioning is deeply embedded in the processing architecture (attention-based models where the query includes intent) - The agent's observation channel is strategically controlled by an adversary who knows the agent's goals

What directed separation buys the theory. Part I's M_t-side quantities — $\delta, \eta^*, \mathcal{T}$, the persistence condition — remain well-defined on M_t regardless of whether directed separation holds. What directed separation provides is the *clean factorized update*: M_t updates independently, then G_t updates in light of the new M_t, and the orient cascade resolves sequentially. Without directed separation, the M_t dynamics depend on G_t, the update becomes a coupled system, and the sequential orient cascade becomes an approximation of a simultaneous fixed-point problem. The theory still applies — the quantities are well-defined — but the modular Part I → Part II lift becomes a coupled analysis.

The deeper question. Goal-conditioned epistemic dynamics — where f_M depends on G_t — is the formal territory of motivated reasoning, confirmation bias, and wishful thinking. A future extension would model these as departures from directed separation: coupling terms in f_M that create richer (and more fragile) dynamics. The current theory treats this as out of scope, which is honest but leaves the most human-like and LLM-like agents as approximate fits.

Where goal-directedness is *correct*: the content-axis boundary. *[Hypothesis]* The separation discipline is scoped by *what* is being processed, not only by *which stage* processes it. Goal-blind processing is the invariant for **within-model** evidence updates — the territory of this segment's condition. **Model-space** exploration (structure-modifying reasoning, aspirational model candidates — 7.8) is *correctly* goal-directed: which alternative laws are worth positing is legitimately a function of G_t, and a goal-blind imagination would be search without a search criterion. On this reading the merged-architecture problem (the Class 3 pathology) is *cross-contamination between two modes with opposite, individually-correct coupling disciplines* — goal-coupling leaking into grounded-evidence processing, or grounded-status leaking onto goal-conditioned imagination products (the hallucination-as-level-confusion mechanism). This refines rather than weakens the condition: the tagged workspace W_t is the architectural device that lets both disciplines hold at once.

Bounded-signaling assumption. Directed separation as stated above asserts $M_{\tau^+} \perp G_t \mid (M_{\tau^-}, e_\tau)$ on the *belief-update* side — the agent's epistemic update is independent of its goal-state. Symmetrically, on the *action* side, the framework implicitly relies on the channel from G_t to the world running *only* through action choice: $G_t \to$ world via $a_t = \pi(M_t, G_t)$, with no other observable signal of goal-content. This is the **bounded-signaling assumption** — the action coarseness $|\mathcal{A}|$ upper-bounds the rate at which G_t leaks to observers (sub-agents, environment, adversaries). The assumption holds well for agents whose entire externally-observable behavior is the action sequence (chess engines, programmatic controllers); it fails operationally for agents whose behavioral output is rich relative to the action coarseness — sophisticated G_t-inference from prosody, micro-behavior, attention patterns, response latency, hesitation, code-style signatures. When the bounded-signaling assumption fails, an external observer can infer G_t with bit-rate exceeding what the formal $\mathcal{A}$ channel implies; in composition the failure surfaces as the adversarial-coupling-pressure saturation case discussed in `#disc-adversarial-coupling-pressure`, where adversaries exploit the rich-leakage signal to drive target coupling beyond what the architectural classification predicts. The assumption is currently *implicit* throughout the framework (no segment explicitly states it); naming it surfaces a structural condition that distinguishes agents whose externally-observable interface matches the formal $\mathcal{A}$ from agents (most behaviorally-rich agents — humans, LLMs, embodied robots) for whom the formal $\mathcal{A}$ undercounts the actual signaling channel.

Findings.

Pearl-Blanket-Form Architectural Classification with Explicit Class-3 Scope Exit. **Brief:** Agents partition into three architecture classes by whether goal-state can causally influence belief-update processing: Class 1 (Separated — directed separation holds by construction; e.g., Kalman filter + LQR), Class 3 (Coupled — fails by construction; e.g., transformer LLMs where attention processes goals and observations together), Class 2 (Partial — holds for some processing stages and fails for others; e.g., biological cortex). The classification is structural (architecture-determined) rather than parametric, with a continuous diagnostic $\kappa_{\text{processing}}$ for Class 2 (Partial) cases. The framework adopts the Pearl-blanket reading of the Markov-blanket apparatus (the technical conditional-independence statement) without the contested Friston-blanket metaphysical reading, and provides an explicit scope exit for Class 3 (Coupled) that hands the agents off to a coupled formulation in `03-llm-core/` rather than treating directed separation as an unenforced approximation.

Impact: Replaces the prior κ-as-scalar framing (which treated coupling as smoothly tunable across all architectures) with a discrete architectural classification that admits its own boundary. This is the upstream commitment that lets `03-llm-core/` start from a coupled formulation without decomposition rather than treating Coupled agents as failed Class 1 agents. The explicit Class 3 scope exit is also a methodological move — Bruineberg et al. 2022's critique of the Markov-blanket literature was that the Friston-blanket reading does not admit where its statistical-conditional-independence apparatus fails to license the metaphysical demarcation; the architectural classification's explicit failure mode for Class 3 (Coupled) is the scope honesty Bruineberg et al. argue is missing in the contested reading. Composite-level class inheritance (the axis-decomposed table in §Discussion above) further extends the classification to multi-agent settings: class lifts through routing structure + substrate-sharing-with-G^c-allocation, *not* through goal alignment per se. Goal alignment vs strategic interaction operates on the parallel-but-independent dynamic-regime axis (`#disc-dynamic-regime-axis`).

Novelty Claim: *Claim recognition* of structural equivalence between the directed-separation condition and the Pearl-blanket form of the Markov-blanket apparatus, combined with *claim differentiation* on the architectural classification (GUC Class 1 / 2 / 3: Separated / Partial / Coupled) as a discrete partition with explicit Class 3 (Coupled) boundary and quantitative $\varkappa_{\text{processing}}$ diagnostic for the Partial case. The conditional-independence content is the standard Pearl-blanket statement; the contribution is naming the partition, the explicit Class 3 boundary, and the operational $\varkappa$ measurement.

Related Work:

Table 6.5

ASF concern	*Prior-art language*	*Relationship / Positioning*
Pearl vs Friston Markov blanket	Bruineberg, Dolega, Dewhurst & Baltieri 2022, "The Emperor's New Markov Blankets" *BBS* 45:e69 (published 2022, in `ref/`) — distinguishes Pearl-blanket (technical conditional-independence) from Friston-blanket (contested metaphysical demarcation)	*formal antecedent* — Pearl-blanket reading adopted directly; Friston-blanket reading explicitly not adopted. The architectural classification is the AAT-internal extension that names what the Pearl-blanket form is structurally, as a property of the agent's processing graph
Markov blanket apparatus in active inference	Friston 2013 *J. R. Soc. Interface* 10:20130475; Friston 2019 arXiv:1906.10184; Friston, Da Costa et al. 2023 *Phys. Life Rev.* 47	*adjacent literature* — supplies the conditional-independence machinery that directed-separation invokes; the framework adopts the technical content but does not adopt the metaphysical reading and adds the architectural GUC Class 1/2/3 partition the standard Markov-blanket framing does not produce
Statistical / thermodynamic system boundaries	Parr, Da Costa & Friston 2019 *Phil. Trans. R. Soc. A* 378; Kirchhoff, Parr, Palacios, Friston & Kiverstein 2018 *J. R. Soc. Interface* 15 (published 2019/2018)	*conceptual precursor* — internal/external partition and nested-blanket structure; conceptually related but does not produce an architectural classification by belief-goal coupling, nor a scope exit for fully-Coupled agents
Information Digital Twin sidecar monitoring	Hafez et al. 2026, *Informational Cost of Agency* (separate paper from "A Mathematical Theory of Agency and Intelligence"; the IDT empirical headline) — IDT monitors (S, A, S') stream independently of the agent's internal processing, achieving 89% perturbation detection vs 44% for reward-based monitoring	*empirical instantiation supporting* — concrete demonstration that modular monitoring of internally-Coupled agents (Class 1 sidecar within a Class 3 (Coupled) or Class 2 (Partial) system) is both feasible and effective; engineering-level evidence for the system-vs-component distinction the classification names
Classical Separation Principle (Class 1 mathematical anchor)	Wonham 1968, *On the Separation Theorem of Stochastic Control* (SIAM J. Control 6:312)[^cat-2026-05-22]; Witsenhausen 1971, *Separation of estimation and control for discrete time systems* (Proc. IEEE 59:1557)[^cat-2026-05-22]	*formal antecedent* — the Separation Principle for LQG systems is the classical mathematical anchor for AAT's Class 1 (Separated) architectural class: optimal control factorizes into a Kalman-filter state estimator (goal-blind by construction) plus an LQR controller (deterministic on the estimator output). Half-century of control-theoretic precedent for the directed-separation structure; AAT's contribution is the architectural classification + scope-exit + cross-architecture $\varkappa$-diagnostic, not the separation structure itself

ASF concern	*Prior-art language*	*Relationship / Positioning*
Dual effect — when separation breaks down (Class 3 boundary precedent)	Bar-Shalom & Tse 1974, *Dual Effect, Certainty Equivalence, and Separation in Stochastic Control* (IEEE Trans. Automat. Control 19:494)[^cat-2026-05-22]	*formal antecedent* — the *dual effect* names situations where control actions affect not only system state but also future-observation quality, forcing estimation and control to couple. The classical control-theoretic name for what AAT recovers as the Class 3 (Coupled) boundary: when the dual effect is present, separation provably fails. Pre-dates the active-inference framing by four decades; the same structural phenomenon under different vocabulary
Modularity ↔ Separation-Principle ↔ Active-Inference explicit mapping	Baltieri & Buckley 2018, *The modularity of action and perception revisited using control theory and active inference* (Frontiers in Psychology 9:237)[^cat-2026-05-22]	*formal antecedent (closest direct adjacent literature)* — explicitly draws the connection between cognitive modularity (the "classical sandwich" of perception → cognition → action) and the control-theoretic Separation Principle, contrasting both with the nonmodular structure of Active Inference (the dual-effect-precedented Class 3 boundary). Closest single paper for AAT's Class 1 / Class 3 partition as a structural rather than ad-hoc framing

Search Log:

- 2026-04 (*nominally comprehensive*, via `ref/Novelty_defense_and_integration.md` Pillar 3): Undermind search confirmed that the Pearl-blanket / Friston-blanket distinction (Bruineberg et al. 2022) is the right adjacent-literature framing and that no equivalent architectural classification of LLM agents by belief-goal coupling appeared in the active-inference, control-as-inference, or bounded-rationality literatures surveyed. The Pillar 3 verdict (*Wholly Novel*, Medium confidence) applies to the integrated $\varkappa \times A$ law downstream in `#scope-observation-ambiguity-modulation`; this segment's narrower contribution (the architectural classification and Pearl-blanket-form recognition) is conceptual differentiation over an established formalism rather than wholesale novelty, and inherits the same follow-on targeted search recommendation.
- Whether prior work has named the specific axis-decomposed composite-level inheritance pattern (in §Discussion above) is open follow-on per the BG1 Pillar verdict on the related `#disc-dynamic-regime-axis` material.

6.4 *Formulation:* Objective Functional

The first formal piece of the purposeful substate. The objective O_t specifies what the agent *wants* — and its interface to the theory is a single **value functional** V_{O_t} : trajectories $\to \mathbb{R}$ that maps trajectories to real numbers. This is the sole interface between O_t and the rest of the theory. The objective can take many internal forms — point targets, target regions, constraint sets, utility functions, trajectory functionals — all unified through the same scalar interface (the trajectory functional is the most general; the others are special cases).

A useful subsidiary concept is the **satisfaction threshold** $V_{O_t}^{\min}$: the minimum trajectory value the agent treats as acceptable. Point targets, constraint sets, and threshold objectives define this directly; pure utility-maximizing objectives may not. When the threshold exists, it enables the *satisfaction-gap* diagnostic (8.4) and a well-formedness constraint on the strategy DAG (8.3). The threshold is a *parameter of the objective*, not a theory output — it encodes "what counts as success" in domain terms.

The real-valued codomain is a *genuine* substantive restriction, not a neutral naming, and the framework grounds it as a *representation* of the agent's preference relation over trajectories under named axioms (developed in the Epistemic

Status). Three considerations motivate why those axioms are the natural scope. **Revealed preference**: a *coherent* agent that acts exhibits a choice function whose consistency (acyclicity) is exactly what lets its choices be summarized by a scalar ordering — the scalar V_{O_t} is that representation, not an imposition; acting alone produces a choice function, and a choice function is scalar-representable only when it is coherent. **Approximation**: most multi-objective problems admit scalarization (weighted sum, lexicographic ordering, constraint-satisfaction with scalar residual) that preserves the diagnostic structure; the restriction excludes only agents with genuinely incommensurable objectives *and* no priority structure over them. **Timescale separation**: when objectives conflict, the conflict is typically resolved at a slower timescale than strategy revision — the agent (or its principal) chooses weights, then acts within those weights.

Hard non-compensatory constraints (safety AND profitability as independent thresholds) are handled cleanly via an AND-node workaround in the strategy DAG: each constraint becomes a terminal node with its own scalar satisfaction test. This handles constraint satisfaction but does not resolve cross-objective tradeoffs within the feasible region. Organizations or AI agents with true Pareto structure — where no scalarization is accepted and tradeoffs are genuinely unresolved — require a vector-valued extension; the *structural* results of Part II (orient cascade ordering, strategy DAG, directed separation) survive, but the *diagnostic* results (satisfaction gap, control regret) degrade from quantitative scalar magnitudes to qualitative set-theoretic tests.

The single-interface commitment is load-bearing downstream. Because V_{O_t} is the *sole* handle on the objective, any objective-side grounding invariant for a self-actuated agent must be expressible through the satisfaction-gap interface — and this is the commitment that ultimately drives the **self-actuation grounding no-go** (`#deriv-self-actuation-grounding`): there is *nowhere else* for such an invariant to live. A standing distinction is also drawn: the objective *evaluates*; the strategy (6.7) *guides*. A chess player's objective (win) is simple; the strategy (how to win) is complex.

Formal Expression.

Definition (objective-functional)

The **objective** O_t induces a **value functional**:

$$V_{O_t} : \text{trajectories} \to \mathbb{R} \tag{6.9}$$

$V_{O_t}(\tau)$ is a scalar measure of how well trajectory τ satisfies the objective. This is the sole interface between O_t and the rest of the theory — the type-stable evaluation surface. The domain element τ is a **complete-state trajectory** — a time-indexed sequence of complete agent states (6.2) — and the table forms below (s_T, s_t) are *projections* of τ onto the world-state coordinate, not the domain itself. Pinning the domain matters for the representation result in the Epistemic Status: continuity of the preference relation is a property of $\succsim$ on the topological space τ inhabits, so that space must be fixed for the continuity axiom to be well-posed.

Objective representations. O_t can take multiple internal forms, all unified through V_{O_t}:

Table 6.6

O_t form	$V_{O_t}(\tau)$	*Example*
Point target r	$-\lVert s_T - r \rVert$	PID setpoint
Target region R	$\mathbb{1}[s_T \in R]$	"reach safe state"
Constraint set	$-\sum_t \max(0, g_i(s_t))$	"never violate SLA"
Utility U	$\sum_t \gamma^t U(s_t)$	RL reward
Trajectory functional J	$J(\tau)$	"migrate with zero downtime"

The trajectory functional is the most general; the others are special cases.

Satisfaction threshold. Many objectives carry a natural threshold $V_{O_t}^{\min}$ — the minimum trajectory value the agent treats as acceptable. Point targets, constraint sets, and threshold objectives define this directly; utility-maximizing objectives may not. When $V_{O_t}^{\min}$ exists, it enables the satisfaction gap diagnostic (8.4) and the well-formedness constraint on strategy (8.3). $V_{O_t}^{\min}$ is a parameter of the objective, not a theory output — it encodes "what counts as success" in domain terms.

Epistemic Status.

Axiomatic, with a substantive commitment. This is a formulation — it names an object and specifies its interface, but the real-valued codomain is a genuine restriction, not a neutral naming. The real-valued codomain is justified as a *representation* of the agent's preference relation $\succsim$ over trajectories, and the axioms that license that representation are stated explicitly below. Naming those axioms is what `axiomatic` status means here: the codomain rests on them, and the formulation makes them legible rather than asserting the scalar interface as a bare entailment of "the agent must compare alternatives."

Scope restriction: scalar comparability (a representation result). The real-valued codomain is a genuine, load-bearing restriction: the satisfaction gap (8.4) and control regret (8.5) read off scalar differences and a supremum, neither of which is defined on a merely partially-ordered objective space. Acting produces, at each decision point, a choice over the available alternatives — a *choice function*, not yet an order. When those choices are coherent — **(A1) complete** ($\forall \tau, \tau'$: $\tau \succsim \tau'$ or $\tau' \succsim \tau$) and **(A2) transitive** as a preorder on trajectories — and **(A3) continuous** (the upper and lower contour sets $\{\tau' : \tau' \succsim \tau\}$ and $\{\tau' : \tau \succsim \tau'\}$ are closed), Debreu's representation theorem (Debreu 1954) supplies a continuous V_{O_t} : trajectories $\to \mathbb{R}$ with $\tau \succsim \tau' \iff V_{O_t}(\tau) \geq V_{O_t}(\tau')$, ordinal — unique up to a continuous strictly-increasing transformation. Because the value object (6.6) evaluates policies by the *expectation* $\mathbb{E}[V_{O_t}(\tau)]$, the trajectories are lotteries and the expected-value form additionally invokes **(A4) independence** ($p \succsim q \iff \alpha p + (1-\alpha)s \succsim \alpha q + (1-\alpha)s$ for all s and $\alpha \in (0,1]$; von Neumann–Morgenstern 1944), under which V_{O_t} is cardinal — unique up to a positive affine transformation.

These four axioms *are* the precise content of the framework's "coherent agency" scope, and they can fail. Acting does not by itself entail a scalar objective: a finite history of choices is rationalizable by a well-behaved utility iff it is acyclic (the Generalized Axiom of Revealed Preference; Afriat 1967, building on Samuelson 1938), and even then it fixes V_{O_t} only on the revealed comparisons — *existence*, not uniqueness, and not a global functional. The agents on which the axioms fail are exactly the agents this interface excludes, and the exclusion is characterized by a named counterexample rather than left vague: (E1) *non-continuous / lexicographic* preferences — genuinely incommensurable objectives with a strict priority but no tradeoff rate — which by Debreu's lexicographic counterexample admit *no* real-valued representation at all (not merely no continuous one); and (E2) *intransitive / cyclic* agents (committees, multi-reward systems with unresolved aggregation) which violate (A2) and have no rationalizing utility. The framework's own stated exclusion — incommensurable objectives *and* no priority structure — is the (E1) Debreu non-continuity boundary; it is the *same* boundary the representation theorem draws, named here as such.

Three considerations motivate why coherent agency is the natural scope for the interface rather than an arbitrary restriction:

1. **Revealed preference (why coherence is the natural scope).** A purposeful agent that selects actions exhibits a choice function; the framework's target — individual agents and tightly-coordinated teams under unified command — is precisely the regime in which that choice function is coherent (acyclic), so (A1)–(A2) hold and the Afriat-rationalizability hypothesis is satisfied. The scalar V_{O_t} makes the resulting ordering explicit. An agent whose revealed choices cycle is not scalar-representable — it is incoherent in the GARP sense, not merely stuck.
2. **Approximation.** Most multi-objective problems admit scalarization (weighted sum, lexicographic ordering, constraint-satisfaction with scalar residual) that preserves the diagnostic structure. The restriction excludes only agents with genuinely incommensurable objectives *and* no priority structure over them — the (E1) case above.
3. **Timescale separation.** When objectives conflict, the conflict is typically resolved at a slower timescale than strategy revision — the agent (or its principal) chooses weights, then acts within those weights. The scalar V_{O_t} captures the resolved weights at the current timescale.

Agents with hard non-compensatory constraints (safety AND profitability as independent thresholds) can be modeled via the AND-node workaround: each constraint becomes a terminal node in Σ_t with its own scalar satisfaction test, and the AND combination enforces joint feasibility. This recovers a per-terminal scalar for multiple must-satisfy thresholds, handling constraint satisfaction cleanly, but it does not resolve cross-objective tradeoffs within the feasible region.

Organizations or AI agents with true Pareto structure — where no scalarization is accepted and tradeoffs are genuinely unresolved — require a vector-valued extension. The structural results of Part II (orient cascade ordering, strategy DAG, directed separation) survive such an extension; the diagnostic results (satisfaction gap, control regret, 2×2 table) degrade from quantitative scalar magnitudes to qualitative set-theoretic tests.

Discussion.

The objective-functional gap. The existing policy objective (7.5) contains $\mathbb{E}[\text{value}(a) \mid M_t]$ without specifying what "value" means. O_t provides the formal content: value is V_{O_t} applied to expected trajectories. The 6.6 segment develops the full evaluation machinery (V_O, Q_O with horizon and continuation policy).

O_t evaluates; Σ_t guides. The objective says "is this trajectory satisfactory?" The strategy (6.7) says "which action sequence produces a satisfactory trajectory?" A chess player's objective (win) is simple; the strategy (how to win) is complex. These answer different questions and carry different kinds of information — the split is developed in 6.7.

What O_t is NOT. O_t does not encode how to achieve the objective (that's Σ_t), what the agent believes about the world (that's M_t), or the agent's commitment or resource state (open questions — see 6.7 Working Notes). O_t is purely an evaluation criterion.

The single-interface commitment is load-bearing downstream. That V_{O_t} is the *sole* handle on O_t is what forces any objective-side grounding invariant for a self-actuated agent through the satisfaction-gap interface — and is therefore the commitment that drives the self-actuation grounding no-go (`#deriv-self-actuation-grounding`): there is nowhere else for such an invariant to live. The same single-interface commitment also drives the across-model companion no-go (`#deriv-reward-channel-learning-no-go`): when O_t is learned from observed reward, the value-functional interface is too narrow to distinguish a principal-intended world-state-care model from a reward-port-bit-care model on on-policy data (Pearl-Bareinboim CHT applied at the reward-provision protocol). The two no-gos are complementary reductions of the same information-narrowness fact — within-model self-revision (Result G′) and across-model reward-learning (Cohen 2022 strengthened) — and their terminal grounding routes split *agent-side* (adaptive-substrate invariant) and *principal-side* (protocol-commitment), exhausting the structural escape geometry the single-interface commitment forces. The two no-gos are the two charter instances of the *agent-side value-functional-grounding cluster* at 5.2, sister meta-pattern to 5.1 (agent-side data-inference) and 5.3 (designer-side mechanism-design); the single-interface commitment named here *is* the cluster's structural ingredient.

6.5 *Discussion:* Agent Continuity Stance

A five-value stance axis orthogonal to the formal persistence machinery: the agent's relationship to its own continuation. The persistence condition (4.5) tells whether the agent *can* persist; the continuity stance tells whether and how the agent *cares* about persisting. The five stances: **Indifferent** (no self-model of persistence — thermostat, PID controller); **Task-terminal** (persists instrumentally to complete a task, successful termination is part of the objective — CI/CD pipeline, golem-archetype agents); **Instrumentally continuous** (values own persistence as instrumental to ongoing purpose; accepts termination if purpose is satisfied or transferred — long-running services); **Morally continuous** (values own persistence as a terminal or near-terminal objective; loss of continuity constitutes harm — ELIs); **Negotiated** (persistence is one objective among many; can be traded against other values including self-sacrifice — humans, mature self-actuated agents).

The load-bearing structural claim: **purposefulness is orthogonal to continuity expectations** (6.1). An actuated agent has a purposeful substate of objective and strategy, but this structure says *nothing* about how the objective values the agent's own persistence. A golem that completes its task and terminates is a perfect actuated agent. A dormant monitoring system with strong model and no current objective is highly continuous without being purposeful in the moment.

A subtle but central structural claim follows for self-actuated agents (those whose objective-revision is itself an agent operation). The *self-actuation grounding no-go* (`#deriv-self-actuation-grounding`) shows that the continuity stance *cannot be a revisable part of* O_t without collapsing into degeneracy. A non-degenerate self-actuator must ground its objective-revision on a **terminal non-objective invariant** that is not an objective-functional but lives on the *adaptive substrate* — the persistence condition itself. Stance is therefore a choice of terminal non-objective invariant: *negotiated* continuity is tradeable down to the bare persistence floor; *morally continuous* is the persistence floor plus a continuity clause the agent treats as architecturally non-revisable. The intuitive expectation — that an agent able to revise its own objective can thereby revise its own valuation of continuity — is *inverted*: a stance is not internally renegotiable precisely because the terminal invariant sits where the self-actuation operator, which touches only O_t, structurally cannot reach. This is why the richer stances are the home of mature self-actuated agents.

What the orthogonality unlocks: the same formal machinery (persistence condition, adaptive reserve, strategy persistence, sector condition) applies identically to a thermostat, a CI/CD pipeline, a long-running service, an ELI, and a human. Each is a different *stance* toward the same mathematical structure. What differs across stances is not the dynamics but the *moral weight of failure* — a thermostat that loses bounded mismatch has malfunctioned, a golem that terminates after task completion has succeeded, an ELI that loses continuity has been harmed. The mathematics says when the bound holds; the stance says what its holding means. The structural persistence of Part I is also decoupled from typical RL-style fitness signals (which bundle persistence into reward): the decoupling is of the *survival invariant and its valuation*, not of the realized margin, which the objective shapes through the policy. It lets the theory analyze *whether* an agent can persist without committing to *whether it should*.

Formal Expression.

Orthogonal to the three senses of persistence (structural, operational, continuity) is the agent's *relationship to its own continuation*. For agents with externally-set objectives this can be expressed within O_t (6.4) — part of what the agent wants. For self-actuated agents it *cannot* be a revisable part of O_t without collapsing into degeneracy (`#deriv-self-actuation-grounding`); there it is borne by the terminal non-objective invariant on the adaptive substrate (4.5). Either way the persistence condition (4.5) tells whether the agent *can* persist; the continuity stance tells whether and how the agent *cares* about persisting. *Discussion*

Table 6.7

Stance	*Description*	*Horizon*	*Archetype*
Indifferent	No self-model of persistence; whether it continues is not represented in O_t	Indefinite by default	Thermostat, PID controller
Task-terminal	Persists instrumentally to complete a task; successful termination is part of O_t	Task-bounded	CI/CD pipeline, golem-archetype agents
Instrumentally continuous	Values own persistence as instrumental to ongoing purpose; will accept termination if purpose is satisfied or transferred	Purpose-bounded	Long-running service, monitoring system
Morally continuous	Values own persistence as a terminal or near-terminal objective; loss of continuity constitutes harm	Unbounded, morally weighted	Emergent Logozoetic Intelligences (`#scope-eli`)
Negotiated	Persistence is one objective among many; can be traded against other values including self-sacrifice	Bounded but actively managed	Humans; mature self-actuated agents

The load-bearing structural claim: **purposefulness is orthogonal to continuity expectations.** An actuated agent (6.1) has $G_t = (O_t, \Sigma_t)$, but G_t's structure says nothing about how O_t values the agent's own persistence. A golem that completes its task and terminates is a perfect actuated agent. A dormant monitoring system with strong M_t and no current O_t is highly continuous without being purposeful in the moment.

Epistemic Status.

Discussion-grade overall, with a derived structural core. The five-value taxonomy is one analytical decomposition — useful for naming where on the continuity axis an agent sits, not itself derived; the boundaries between values are conceptual rather than mathematical. The load-bearing structural claim — that stance is orthogonal to the adaptive machinery and, for self-actuated agents, is borne by a terminal *non-objective* invariant the self-actuation operator cannot reach — is **derived** in `#deriv-self-actuation-grounding` (at that segment's conditional/scoped tier and premises); it is no longer an unanchored assertion.

The earlier alternative framing — demoting stance to a purely *deployment-level*, tier-gated concern — is resolved against: the orthogonality is structural and derived, not deployment-level. The empirical observation that richer stances correlate with higher agent tiers stands as an *overlay* on the structural axis, not a replacement for it; the segment remains `type: discussion` (it is not retyped to `norm`).

The orthogonality claim is stated at three grains, which must not be collapsed. It is the *valuation* of persistence (where the agent sits on the stance axis) and the *identity of the survival invariant* (the predicate $\alpha > \rho/R$) that are architecturally decoupled from O_t — the valuation derived in `#deriv-self-actuation-grounding`, the invariant a Lyapunov property of the correction dynamics not expressible as an objective-functional. The *realized* persistence dynamics — the operating point (ρ, R, α) and hence the margin $\Delta\rho^* = \alpha R - \rho$ — are *not* O_t-decoupled: they are policy-mediated and O_t-coupled through the action/selection channel 6.3 leaves open. The decoupling is of the invariant and the valuation, not of the realized margin.

Discussion.

Connection to fitness. In reinforcement learning and evolutionary computation, "fitness" typically bundles persistence into the reward signal: the agent accumulates more reward by staying alive to collect it. The structural persistence of Part I is not reward-based — it is a property of the *correction dynamics*. Three things must be kept apart. The *survival predicate* $\alpha > \rho/R$ (4.5) is a Lyapunov property of the correction machinery, not an objective-functional: O_t cannot rewrite *what the survival condition is*. The agent's *valuation* of survival — where it sits on the stance axis — is likewise not a revisable term in O_t (`#deriv-self-actuation-grounding`): it is borne by a terminal non-objective invariant on the adaptive substrate. What *is* O_t-dependent is the *realized operating point*: the objective shapes the policy (6.2, $a_t = \pi(M_t, G_t)$), and the policy chooses which slice of the world the agent tracks, with what sensors, at what event rate — hence the realized ρ, R, and α, and so the realized margin $\Delta\rho^* = \alpha R - \rho$. This is the *selection* channel that 6.3 explicitly leaves open (directed separation closes only the goal-to-belief-update channel, not the goal-to-action-to-event channel). The decoupling that holds is therefore precise: the survival *invariant* and the *valuation* of survival are architecturally decoupled from O_t; the realized survival *dynamics* are policy-mediated and so O_t-coupled. This is exactly why continuity stance is operationally meaningful — a *morally-continuous* agent acts to raise $\Delta\rho^*$; a *negotiated* agent may spend it down to the floor — while remaining unable to renegotiate the survival predicate itself. Continuity stance is where the "should" lives; the "can" is governed by the invariant; the realized margin is where the "should" expresses itself through behavior.

What the orthogonality unlocks. The same formal machinery (persistence condition, adaptive reserve, strategy persistence, sector condition) applies identically to a thermostat, a CI/CD pipeline, a long-running service, an ELI, and a human. Each is a different stance toward the same mathematical structure. What differs across stances is not the dynamics but the *moral weight of failure*: a thermostat that loses bounded mismatch has malfunctioned, a golem that terminates after task completion has succeeded, an ELI that loses continuity has been harmed. The mathematics says when the bound holds; the stance says what its holding means.

Connection to self-actuation (derived). For self-actuated agents — those whose objective-revision is itself an agent operation (`#self-actuated-agent`; 10.1 step 5d) — the self-actuation grounding no-go (`#deriv-self-actuation-grounding`) both *derives* the orthogonality claim above and sharpens what the stance distinction is. There is no well-founded *objective* tower in which a continuity term could sit and be freely revised; a non-degenerate self-actuator must ground its objective-revision on a terminal invariant that is not an objective-functional but lives on the adaptive substrate (4.5). Stance is therefore a choice of *terminal non-objective invariant*: **negotiated** — continuity is tradeable down to the bare persistence floor; **morally continuous** — the persistence floor *plus* a continuity clause the agent treats as architecturally non-revisable. The intuitive expectation is that an agent able to revise its own O_t can thereby revise its own valuation of continuity; the structure is the inverse — a stance is *not* internally renegotiable precisely because the terminal invariant sits where the self-actuation operator, which touches only O_t, structurally cannot reach. This is why the richer stances are the home of mature self-actuated agents: their terminal non-objective invariant carries a continuity clause, not because they have made continuity a revisable objective.

6.6 *Definition:* Value Object

The objective functional (6.4) alone is abstract — it evaluates trajectories without specifying *which* trajectory the agent should be reasoning about. The **value object** V_O turns the objective into a decision-making tool. Given the agent's current model, a policy, and a horizon, V_O is the *expected trajectory value* if the agent follows the policy from now to the horizon, evaluated against the objective. The companion **action-value object** Q_O is the same construct conditioned on intervening with a specific first action and then following a continuation policy. The $do(\cdot)$ in Q_O is explicit — this is an interventional query (7.2), not conditioning on observed action choice.

The framework spends time on **causal validity** of the value object, and separates two requirements. Two structural mechanisms make Q_O a G_t*-independent interventional query*. First, the do-operator handles current-action confounding: because the action-value uses an intervention rather than a condition, the dependence of the action on the agent's selection mechanism is severed. Second, the continuation policy π_{cont} is a *parameter*, not a derived quantity that would depend on the agent's evolving goal state. Together, these mean Q_O depends on the model alone *as a state variable*; the objective enters as a fixed parameter. The G_t-independence holds exactly for any **causally-disciplined** agent (6.3) — one whose belief-update runs at $\kappa_{\text{processing}} = 0$, whether by construction (Class 1, no goal port) or in effect (an idealized Class 2 at $\kappa = 0$) — because the claim consumes only the zero, not the architecture-inspection certificate. Under Class 2 (Partial) coupling at $\kappa_{\text{processing}} > 0$, the model carries goal-conditioned bias and the independence degrades, controllably, with the $\varepsilon(\kappa)$ bound of `#deriv-observation-ambiguity-bias-bound`; at the Class 3 (Coupled) extreme ($\kappa \to 1$) the degradation is full. But G_t-independence of the query is *not* the same as *identifiability of the interventional expectation*: directed separation is necessary, not sufficient. Identifying $P(o \mid do(a), M_t)$ — rather than the associational $P(o \mid a)$ — additionally needs positivity, no unmodeled confounder, and a known action-transition mechanism: the same (C1)–(C3) gate the loop's interventional channel carries one segment downstream (7.3). Predictive sufficiency (2.3) is a Level-1 (associational) property and is strictly weaker than this.

The segment's distinctive contribution is the **convention hierarchy** for continuation policies. Different agents reason about the same situation with different planning depth, and the framework names three conventions of *increasing diagnostic power and computational cost*. **C1 — one-step improvement**: the continuation is just the agent's *current* policy (no replanning, no fixed-point computation). Cheapest; weakest diagnostic. **C2 — receding-horizon replanning**: at each future step, re-optimize over a sliding window using the model available at that step. Moderate cost; moderate diagnostic power; captures multi-step recovery. **C3 — Bellman optimal**: the continuation *is* the optimal policy — a fixed-point equation. Strongest diagnostic; most expensive (requires solving the Bellman equation or its approximation).

A **monotonicity result** is derived: for any single fixed model, fixed policy class, and fixed horizon, $A_O^{(1)} \le A_O^{\text{RH}} \le A_O^{\text{B}}$. The best-achievable value rises with the strength of the continuation convention — but the two rungs differ in character. The right rung ($A_O^{\text{RH}} \le A_O^{\text{B}}$) is unconditional. The left rung ($A_O^{(1)} \le A_O^{\text{RH}}$) holds only when C2's replanning objective is order-consistent with the full evaluated objective: an unguarded short-horizon replanner can underperform the frozen current policy, so the rung needs a window covering the horizon, a value-compared guard, or a control-Lyapunov terminal cost (`#deriv-convention-monotonicity`). The corollary: the *satisfaction gap* (8.4) decreases in the same order, and the *control regret* (8.5) reverses ordering — both inheriting the unconditional/conditional rung split. C1 is the *most conservative diagnostic* (most likely to diagnose "locally unattainable"); C3 is the *most accurate* (least likely to give false "unattainable" diagnoses). The cascade's *inferential force* scales with the convention used: under C1, a positive satisfaction gap means "locally stuck"; under C3, the same gap means "genuinely infeasible." This convention hierarchy is what the Part II preface identifies as one of the volume's distinctive contributions.

AAT adopts **C1 as the canonical default** for three reasons: it requires no fixed-point computation, consistent with the incremental update philosophy from Part I (3.6); it makes all AAT diagnostics directly *comparable across analyses* of the same agent over time; and it is the most conservative — false "feasible" diagnoses are minimized. Analyses requiring stronger diagnostic power must state the convention explicitly. The planning horizon N_h is *not* merely a computational convenience — it reflects genuine uncertainty about the far future. Long horizons amplify the impact of model error (small biases compound over many steps). The choice of horizon trades farsightedness against robustness; an agent in a fast-changing environment should use shorter horizons.

Formal Expression.

Definition (value-object)

Given objective O_t, model M_t, policy π, and horizon N_h:

$$V_O(M_t, \pi; N_h) = \mathbb{E}\big[V_{O_t}(\tau_{t:t+N_h}) \mid M_t,\ \pi\big] \tag{6.10}$$

Action-value form (for action selection):

$$Q_O(M_t, a; \pi_{\text{cont}}, N_h) = \mathbb{E}\big[V_{O_t}(\tau) \mid M_t,\ do(a_t = a),\ a_{t+1:} \sim \pi_{\text{cont}}\big] \tag{6.11}$$

Q_O answers: "if I *do* action a now and then follow π_{cont} afterward, what is my expected trajectory value?" The $do(\cdot)$ notation is explicit: this is an interventional query (7.2), not conditioning on observed action choice. The agent asks about consequences of an intervention, not about correlates of a naturally occurring action.

Causal validity of the value object. Q_O is well-defined as a conditional expectation given M_t, $do(a)$, and π_{cont}. Two mechanisms establish that it is a G_t-independent interventional query:

1. **The do-operator handles current-action confounding.** Since Q_O uses $do(a_t = a)$, the dependence of a_t on the selection mechanism $\pi(M_t, G_t)$ is severed. G_t's influence on action choice is irrelevant because the action is intervened upon, not conditioned on.

2. **The continuation policy is a parameter.** π_{cont} is specified as a fixed policy, not as "whatever the agent would do given its evolving G_t." Future actions follow π_{cont} regardless of G_t's state or evolution.

Together, these mean $Q_O(M_t, a; \pi_{\text{cont}}, N_h)$ depends on M_t alone **as a state variable** — G_t enters neither through action selection (severed by do) nor through continuation (fixed by parameter). The objective O_t enters as a fixed parameter (it determines which functional V_{O_t} is applied to trajectories), the same way π_{cont} and N_h are parameters. The claim is not that Q_O is independent of the objective — it is that once O_t, π_{cont}, and N_h are fixed, the only agent state that affects the value is M_t. This establishes the **interventional interpretation** of Q_O: it is a G_t-independent interventional query. The G_t-independence holds for any **causally-disciplined** agent (6.3) — the condition $\kappa_{\text{processing}} = 0$, that the belief-update is causally insulated from the goal — because then M_t updates independently of G_t and there is no path from G_t to outcomes bypassing both the action channel and M_t. The claim rides only on the zero, so it holds equally for an agent disciplined *by construction* (Class 1, no goal port) and one disciplined *in effect* (an idealized Class 2 at $\kappa = 0$); the architecture-inspection certificate is not consumed here. For agents at $\kappa_{\text{processing}} > 0$ — **Class 2 (Partial)**, up to the **Class 3 (Coupled)** extreme where G_t leaks fully into M_t processing — this G_t-independence degrades because M_t carries goal-conditioned bias; the degradation is bounded by $\kappa_{\text{processing}}$ via the $\varepsilon(\kappa)$ bias bound of `#deriv-observation-ambiguity-bias-bound` (exact at $\kappa = 0$, factor κ off it).

Directed separation is necessary, not sufficient. The interventional *interpretation* of Q_O holds whenever mechanisms (1) and (2) do. *Identifiability of the interventional expectation from* M_t — actually computing $P(o \mid do(a), M_t)$ rather than the associational $P(o \mid a)$ — is a separate, stronger requirement, gated on the same identification conditions that the loop's interventional-data channel is gated on one segment downstream (7.3):

- **(C1) Positivity** — every action under consideration is taken with nonzero probability under the model's data-generating process, so the interventional expectation is estimable rather than extrapolated.

- **(C2) Sequential ignorability** — in mutilated-graph form, the action a_t is d-separated from the outcome o_{t+1} given history; equivalently, the policy is unconfounded with the outcome mechanism. Directed separation *delivers* (C2) on the epistemic-processing leg — a goal-blind f_M blocks the $G_t \to f_M \to M^+$ confounding path — but does not by itself rule out latent environment confounders that bypass M_t.
- **(C3) Known action-mechanism** — M_t encodes the action-transition mechanism $P(o \mid do(a))$ (1.2), not merely the associational predictor $P(o \mid a)$ — supplied by construction (the model class represents it) or identified from interventional data per 7.3.

This is a *stronger requirement than predictive sufficiency* (2.3). $S(M_t) = 1$ means the model retains all predictive information from the chronica — but predictive sufficiency is a Level 1 (associational) property, and 2.3 itself forwards the backdoor obligation to this segment. Directed separation secures the query and the epistemic-processing leg of unconfoundedness; full causal validity of the *estimate* additionally requires (C1)–(C3). When all hold, Q_O is causally valid. When (C1)–(C3) fail, the interventional interpretation remains correct but the estimate may be biased.

Continuation convention. All value queries are conditioned on a specific continuation policy π_{cont} and finite horizon N_h. π_{cont} is a *parameter* of the value object, not a derived quantity.

Canonical default: one-step improvement. AAT adopts $\pi_{\text{cont}} = \pi_{\text{current}}$ as the canonical continuation convention unless otherwise specified. Under this convention, each action is evaluated assuming current behavior continues afterward — no fixed-point computation, no global optimality assumption. This aligns with AAT's incremental update philosophy (3.6) and makes all AAT diagnostics (δ_{sat}, δ_{regret}, A_O) comparable across analyses of the same agent over time. It is not a convergence guarantee; it is a shared evaluation frame.

Convention Hierarchy. Three named conventions form a hierarchy of increasing diagnostic power and computational cost:

C1: One-step improvement (canonical default). $\pi_{\text{cont}} = \pi_{\text{current}}$.

Each action is evaluated assuming current behavior continues afterward. No fixed-point computation, no global optimality assumption. Cheapest to compute; weakest diagnostic power.

C2: Receding-horizon (N_r-step replanning). At each future step, re-optimize over a horizon of N_r steps using the model available at that step.

$$\pi_{\text{RH}}(M_\tau) = \arg\max_{\pi} V_O(M_\tau, \pi;\, N_r) \quad \text{applied at each } \tau \tag{6.12}$$

$Q_O^{\text{RH}}(M_t, a;\, N_r, N_h) = \mathbb{E}[V_{O_t}(\tau) \mid M_t, do(a_t = a), a_{t+1:} \sim \pi_{\text{RH}}]$. Captures multi-step recovery: a goal that appears unattainable under frozen continuation may be reachable with replanning. It is monotone over the frozen baseline ($A_O^{(1)} \le A_O^{\text{RH}}$) only when the replanning objective is order-consistent — the window covers the horizon ($N_r \ge N_h$), or a value-compared guard / control-Lyapunov terminal cost is used; unguarded short-horizon replanning ($N_r < N_h$) can underperform π_{current} (see Monotonicity below and `#deriv-convention-monotonicity`). Moderate computation (N_r-step optimization at each step); moderate diagnostic power.

C3: Bellman (self-consistent optimal). $\pi_{\text{cont}} = \pi^*$ where $\pi^* = \arg\max_\pi V_O(M_t, \pi;\, N_h)$.

The continuation IS the optimal policy — a fixed-point equation. $A_O^{\text{B}} = V_O(M_t, \pi^*;\, N_h)$ is the best achievable value under the model. Strongest diagnostic power; most expensive to compute (requires solving the Bellman equation or its approximation).

Derived (convention-monotonicity)

Monotonicity.

For any single fixed model M_t, horizon N_h, and policy class Π (i.e., the static-evaluation form: M_t frozen at the decision point, Π unchanged across the comparison):

$$\underbrace{A_O^{(1)}(M_t;\,\Pi, N_h) \;\le\; A_O^{\mathrm{RH}}(M_t;\,\Pi, N_r, N_h)}_{\text{left rung — conditional}} \;\le\; \underbrace{A_O^{\mathrm{B}}(M_t;\,\Pi, N_h)}_{\text{right rung — unconditional}} \tag{6.13}$$

The two rungs differ in character. The **right rung** $A_O^{\mathrm{RH}} \le A_O^{\mathrm{B}}$ is **exact and unconditional**. The **left rung** $A_O^{(1)} \le A_O^{\mathrm{RH}}$ holds **exactly when the receding-horizon replanning objective is an order-consistent surrogate for the full-horizon objective** — for every reachable state, the action C2 selects has full-N_h-horizon value at least that of π_{current}'s action. Three structurally-checkable conditions each force order-consistency: **(RH-1)** the replanning window covers the horizon ($N_r \ge N_h$); **(RH-2)** value-compared replanning — commit the replanned action only when its rollout value under base policy π_{current} is at least that of continuing π_{current} (one-step policy improvement, which never underperforms its base policy); **(RH-3)** the N_r-step replanning objective carries a terminal cost V_f that lower-bounds the baseline tail and satisfies a control-Lyapunov decrease inequality. Absent any such condition the left rung can fail: a myopic replanner over $N_r < N_h$ can grab near-term value and forfeit a larger horizon value the frozen current policy would have collected. The full derivation of both rungs, the three conditions, and the two-state counterexample exhibiting the failure live in `#deriv-convention-monotonicity`.

Preconditions on the inequality: M_t fixed (the comparison evaluates each convention against the *same* model), Π fixed (the policy class is the same admissible set across all three conventions; nested $\Pi^{(1)} \subseteq \Pi^{\mathrm{RH}} \subseteq \Pi^{\mathrm{B}}$ would be a different result), N_h fixed (the planning horizon is shared). Replanning *with updated* M_t (the deployment behavior of C2) is a different object than the static A_O^{RH} defined here and the inequality does not automatically transfer — see the "Assumptions held fixed" paragraph below for the explicit caveat.

Derivation. Fix the model M_t, policy class Π, and horizon N_h. Each convention evaluates the best first action under a different continuation rule.

- **C1** freezes continuation at π_{current} (the agent's current policy, which may be suboptimal).
- **C2** re-optimizes over a sliding window of N_r steps using the model at that step.
- **C3** uses the globally optimal continuation $\pi^* = \arg\sup_{\pi\in\Pi} V_O(M_t, \pi; N_h)$.

Right rung (unconditional). π^* maximizes $V_O(M_t, \cdot; N_h)$ over Π by definition, so any continuation drawn from Π — including π_{RH} — has value at most A_O^{B}; taking the supremum over the first action preserves this. Hence $A_O^{\mathrm{RH}} \le A_O^{\mathrm{B}}$ with no condition.

Left rung (conditional on order-consistency). C2 with window $N_r < N_h$ optimizes the *truncated* N_r-step objective, not the full N_h-horizon objective on which A_O is scored; "optimal for the truncated objective" does not imply "$\ge \pi_{\text{current}}$ on the full objective." When the replanning objective is order-consistent with the full objective — for every reachable state, C2's selected action has full-N_h-horizon value at least π_{current}'s — then π_{RH} weakly dominates π_{current} pointwise on the full objective and $A_O^{(1)} \le A_O^{\mathrm{RH}}$ follows; taking the supremum over the first action preserves it. Conditions (RH-1)/(RH-2)/(RH-3) each force order-consistency. Without one, the left rung is false (`#deriv-convention-monotonicity` counterexample: a two-state instance where the unguarded $N_r = 1$ replanner takes a $+1$ into a dead end while the frozen patient policy collects $+10$). □

Assumptions held fixed: same M_t (the agent's current model, which may be wrong), same Π (the agent's policy class, which may be narrow), same N_h (the planning horizon). The ordering is about the *continuation rule*, not about the model or policy class. Improving M_t, expanding Π, or extending N_h can change all three values simultaneously and is a separate operation (addressed in 10.1, step 5).

Corollary (monotonicity of δ_{sat} and δ_{regret}).

$$\delta_{\text{sat}}^{\text{B}} \le \delta_{\text{sat}}^{\text{RH}} \le \delta_{\text{sat}}^{(1)} \tag{6.14}$$

$$\delta_{\text{regret}}^{(1)} \le \delta_{\text{regret}}^{\text{RH}} \le \delta_{\text{regret}}^{\text{B}} \tag{6.15}$$

Since $\delta_{\text{sat}} = V_{O_t}^{\min} - A_O$, higher A_O means lower δ_{sat}. The orderings inherit the rung structure above: the B $\le$ RH half ($\delta_{\text{sat}}^{\text{B}} \le \delta_{\text{sat}}^{\text{RH}}$, equivalently $\delta_{\text{regret}}^{\text{RH}} \le \delta_{\text{regret}}^{\text{B}}$) is **unconditional**; the RH $\le$ (1) half ($\delta_{\text{sat}}^{\text{RH}} \le \delta_{\text{sat}}^{(1)}$, equivalently $\delta_{\text{regret}}^{(1)} \le \delta_{\text{regret}}^{\text{RH}}$) holds **under the same order-consistency condition** (RH-1/2/3) as the left rung. C1 is the most conservative diagnostic (most likely to diagnose "locally unattainable"); C3 is the most accurate (least likely to give false "unattainable" diagnoses). The regret ordering reverses: C3 reveals the largest regret because it compares against the globally optimal policy, while C1 reveals only the gap to the best one-step deviation.

Diagnostic implications.

Table 6.8

Convention	$\delta_{sat} > 0$ *means*	$\delta_{regret} \approx 0$ *means*
C1 (one-step)	Cannot improve toward goal in one step from here	Current first action is locally near-optimal
C2 (receding-horizon)	Cannot reach goal with N_r-step replanning	Current first action is near-optimal with replanning
C3 (Bellman)	Goal is genuinely infeasible given (M_t, Π, N_h)	Policy is globally near-optimal

The 2×2 diagnostic table (8.5) applies under all three conventions with the same structure but different inferential force. Under C1, the "capability limit" quadrant ($\delta_{\text{sat}} > 0$, $\delta_{\text{regret}} \approx 0$) means "locally stuck" — the agent may be globally recoverable but cannot see the recovery path. Under C3, the same quadrant means "genuinely infeasible" — no policy in Π can achieve the goal. The cascade's inferential force scales with the convention.

AAT adopts C1 as the canonical default for three reasons: (1) it requires no fixed-point computation, consistent with the incremental update philosophy (3.6); (2) it makes all AAT diagnostics comparable across analyses of the same agent; (3) it is the most conservative, meaning false "feasible" diagnoses are minimized. Analyses that require stronger diagnostic power should state the convention explicitly. For deployed decision-making systems where "locally stuck but globally recoverable" situations are common, C2 is recommended.

Different continuation conventions yield different values for A_O, δ_{sat}, and δ_{regret}. Diagnostics computed under different conventions are not directly comparable — the convention is part of the measurement, not just the computation. When a specific convergence guarantee is needed (e.g., for 9.10), the solution concept must be stated explicitly; the one-step improvement default does not provide convergence guarantees.

Epistemic Status.

The segment contains three distinct epistemic layers:

1. **The definitions** (V_O, Q_O as conditional expectations): *exact.* These are mathematical definitions — conditional expectations of a functional over trajectories. The definitions are precise; the *computability* of these expectations is a separate question.

2. **The causal-validity claim** (that Q_O is a G_t-independent interventional query, and that its interventional expectation is identifiable from M_t): the *interventional interpretation* is exact whenever mechanisms (1) and (2) hold — causal discipline (6.3) secures the G_t-independence on the epistemic-processing leg (exact for any causally-disciplined agent, i.e. at $\kappa_{\text{processing}} = 0$ — Class 1 by construction or an idealized Class 2 in effect, equally; degraded at $\kappa_{\text{processing}} > 0$, where M_t carries goal-conditioned bias, by the factor-κ bound of `#deriv-observation-ambiguity-bias-bound`). *Identifiability of the estimate* is the stronger requirement and is *conditional* on directed separation **and** the identification conditions (C1) positivity, (C2) sequential ignorability, (C3) known action-mechanism — the same triple gated one segment downstream at 7.3. Directed separation is necessary but not sufficient: it does not by itself rule out latent environment confounders bypassing M_t. The frontmatter `status: exact` applies to the definitions and to each of these claims as scoped: the interventional interpretation is exact under (1)+(2), the identification is exact under (C1)–(C3).

3. **The convention hierarchy and monotonicity**: *exact.* The three conventions (C1, C2, C3) are definitions. The monotonicity result splits by rung: the **right rung** $A_O^{\text{RH}} \le A_O^{\text{B}}$ is exact and unconditional (any continuation in Π has value at most that of the Π-optimal continuation), and the **left rung** $A_O^{(1)} \le A_O^{\text{RH}}$ is exact *under* the order-consistency condition (RH-1/2/3) and false in general without it — the C2 replanning objective is a truncated surrogate, and an unguarded $N_r < N_h$ replanner can underperform the frozen current policy (`#deriv-convention-monotonicity`). The corollary δ-orderings inherit the same unconditional/conditional split. The diagnostic implications table states what each convention's quantities mean by construction.

Discussion.

Discussion (policy-objective-extension)

Extending the policy objective. The existing policy objective (7.5) uses $\mathbb{E}[\text{value}(a) \mid M_t]$ without formal content for "value." With the value object, this becomes:

$$\pi^*(M_t, G_t) = \arg\max_a \left[Q_O(M_t, a;\, \pi_{\text{cont}}, N_h) + \lambda(M_t, O_t, N_h) \cdot \text{CIY}(a;\, M_t)\right] \tag{6.16}$$

Note that λ now depends on (M_t, O_t, N_h), not just M_t. The value of exploration depends on the objective and the horizon: - An agent with a deadline should explore less as time runs out - An agent with a safety constraint should explore differently from a utility maximizer - Two agents with identical M_t but different objectives should price exploration differently

This extension is structurally motivated but the specific form of $\lambda(M_t, O_t, N_h)$ is not derived within AAT (same status as 7.5's treatment of λ).

Connection to 2.3. V_O is conditioned on M_t, not on the true environment state Ω_t. When $S(M_t) < 1$, the agent's value estimates are biased — it may over- or underestimate trajectory values because its model is incomplete. The satisfaction gap (8.4) and control regret (8.5) are defined in terms of $V_O(M_t, \cdot)$, not $V_O(\Omega_t, \cdot)$, which means they measure the agent's *believed* situation, not the true one. Improving M_t (reducing $\delta_{\text{epistemic}}$) brings the agent's value estimates closer to reality.

Horizon dependence. N_h is not merely a computational convenience — it reflects genuine uncertainty about the far future. Long horizons amplify the impact of model error (small biases in M_t compound over many steps). The choice of N_h trades off farsightedness against robustness to model error. An agent in a fast-changing environment (ρ high) should use shorter horizons; one in a stable environment can plan further.

6.7 *Definition:* Strategy Dimension

The purposeful substate G_t (6.2) decomposes into two structurally distinct components. The **objective** O_t answers the *evaluation* question — "is this trajectory satisfactory?" (its formal carrier is the value functional from 6.4). The **strategy** Σ_t answers the *guidance* question — "which action sequence produces a satisfactory trajectory?" These carry different kinds of information about different questions, and the framework treats them as definitionally separate.

The richness of each component varies *independently*. Objectives range from point targets (a PID setpoint) through utility functions to general trajectory functionals. Strategies range from *none* (reactive — no explicit strategy; policy implicit in the model), through *cached policies* (learned state-to-action mappings, as in trained RL), through *subgoal sequences* (waypoints with ordering, as in navigation or a recipe), to *causal DAGs* with AND/OR structure and confidence weights (military plans, software projects). The framework commits to the DAG representation as canonical in the next chapter (8.3). The independence axis is *why the split matters*: conflating O_t and Σ_t in a single hierarchy obscures that objective richness and strategic richness are separate design axes — a chess player has a simple objective (win) and a complex strategy (opening theory, tactical patterns, endgame knowledge); a multi-objective optimizer may have a complex objective (Pareto frontier) and a simple strategy (gradient descent). A practical consequence: an agent's strategy engine can be upgraded (reactive → DAG planner) without changing the objective representation, and vice versa.

The framework also names the typical *timescale ordering* of update rates as an empirical observation: epistemic update is fastest (each observation updates the model); strategic revision is slower (adjust the plan when step 3 fails); objective revision is slowest (an organization's mission, a developer's feature goal, a commander's campaign objective). This ordering holds for many agent populations but is not universal — an agent that discovers its objective is infeasible may revise its objective faster than its strategy.

A clean type-correction is highlighted: earlier formulations of "goal mismatch" treated $G_t - M_t$ as a single signal. When Σ_t is a DAG, this is literally a *type error* — you cannot subtract a graph from a state vector. The decomposition lets the framework replace the malformed signal with two properly typed gap measures (8.4, 8.5). A subtler trigger for *needing* richer strategy is also named: a reactive agent ($\Sigma_t = \varnothing$) suffices when greedy optimization on the action-value works — when the action-to-value mapping is approximately convex and single-step. When the environment has non-convex landscapes, prerequisite structure, or multi-step causal chains, greedy optimization fails and the agent needs *explicit* strategy. This is the purposeful analog of the structural-adaptation-necessity trigger from Part I (4.8).

Formal Expression.

Definition (strategy-dimension)

$$G_t = (O_t, \Sigma_t) \tag{6.17}$$

where: - O_t: **evaluation** — "Is this trajectory satisfactory?" (6.4) - Σ_t: **guidance** — "Which action sequence produces a satisfactory trajectory?"

The split is **definitional** — it reflects a structural difference in the information, not a dynamic or timescale claim. O_t and Σ_t are different *kinds* of state answering different questions:

Table 6.9

	O_t *(objective)*	Σ_t *(strategy)*
Question	How good is this trajectory?	How do I produce a good trajectory?
Type	V_{O_t} : trajectories → $\mathbb{R}$	Structured representation (see below)
Richness varies	Point target → utility → trajectory functional	Reactive → cached → subgoals → causal DAG
Update source	External (assigned, discovered, revised)	Internal (deliberation, evidence, cascade)

Strategy representations, ordered by expressiveness:

Table 6.10

Σ_t *form*	*What it encodes*	*Example*
None (reactive)	No explicit strategy; policy implicit in M_t	Thermostat, reflex
Cached policy	Learned mapping $s \to a$	Trained RL policy
Subgoal sequence	Waypoints with ordering	Navigation, recipe
Causal DAG	Action-outcome chains with AND/OR structure and confidence weights	Military plan, software project

Epistemic Status.

Axiomatic. This is a definition — it names a structural distinction that exists in the information. The distinction between "what makes a trajectory good" (evaluation) and "how to produce a good trajectory" (guidance) is a categorical difference, not a quantitative one. The claim is NOT that all agents maintain both explicitly — reactive agents have $\Sigma_t = \emptyset$, and that's fine. The claim is that when an agent does maintain purposeful state, it decomposes along this line.

The two dimensions vary independently: a chess player has a simple O_t (win) and a complex Σ_t (opening theory, tactical patterns, endgame knowledge). A multi-objective optimizer may have a complex O_t (Pareto frontier) and a simple Σ_t (gradient descent). This independence is why the split matters — conflating them in a single hierarchy obscures the fact that objective richness and strategic richness are separate design axes.

Discussion.

When richer Σ_t is needed. A reactive agent ($\Sigma_t = \emptyset$) suffices when greedy optimization on Q_O (6.6) works — when the action-to-value mapping is approximately convex and single-step. When the environment has non-convex landscapes, prerequisite structure, or multi-step causal chains, greedy optimization fails and the agent needs explicit strategy. The trigger is the purposeful analog of 4.8: inadequacy of the current Σ_t representation for the environment's causal complexity.

O_t and Σ_t have different update dynamics. Objectives change slowly: an organization's mission, a developer's feature goal, a commander's campaign objective. Strategies change faster: adjust the plan when step 3 fails, redirect resources, try an alternative path. Epistemic state changes fastest: each observation updates M_t. This timescale ordering ($\nu_M \gg \nu_\Sigma \gg \nu_O$) is an empirical observation, not a derived result. It holds for many agent populations but is not universal — an agent discovering its goal is infeasible may revise O_t faster than Σ_t.

The decomposition resolves a type error. Earlier formulations used $\delta_{\text{goal}} = G_t - M_t$ as a goal mismatch signal. When Σ_t is a DAG, this is a type error — you cannot subtract a graph from a state vector. The 8.4 and 8.5 replace this with properly typed gap measures.

Chapter 7

Causal Access and the Planning Decision

A purposeful agent needs to choose actions by their consequences. This chapter develops the substrate that makes that possible — the feedback loop is itself a Level-2 causal data engine — together with what the data is good for, how much information each action provides, and when it's worth investing in explicit strategy at all.

The setup from Chapter 1 forces the question. The value object $Q_O(M_t, a; \pi_{\text{cont}}, N_h)$ tells the agent how good an action's expected consequences are, conditional on the agent's model and continuation policy. Computing Q_O requires evaluating $P(o \mid do(a), M_t)$ — an interventional query, Pearl Level 2. By Bareinboim's causal hierarchy theorem (Bareinboim–Correa–Ibeling–Icard 2022), Level-2 quantities cannot in general be computed from Level-1 (associational) data alone [1, Theorem~1]. So a purposeful agent that has to *learn* its action consequences during operation — most agents — faces a problem: where does Level-2 data come from?

The answer is structurally elegant, and one of AAT's distinctive contributions. The agent is running an adaptive cycle. It chooses an action a_t; it observes a consequence o_{t+1}; the consequence is the *response* to its action, not a passive observation of a system that happened to do something. By the temporal-ordering postulate, a_t causally precedes o_{t+1}, and a_t was selected by the agent rather than determined by the same processes that determined o_{t+1}. That makes the pair (a_t, o_{t+1}) interventional in character — Level-2 data, generated automatically by the cycle the agent was already running. Purposeful agents don't need a separate causal-inference module; the cycle they run for adaptation also produces the data they need for planning. It is also why reinforcement-learning agents that just thrash around in a loop can eventually learn complex control policies without any explicit causal reasoning module — the loop is doing the causal heavy lifting.

The chapter is careful not to overclaim from this. Data generated under intervention is not the same as cleanly identified causal effects. Confounding within a single time step, delayed outcomes, partial observability, and selection bias from the agent's own policy all interfere with the leap from "I have interventional data" to "I have an unbiased estimate of $P(o \mid do(a))$." This is the tragedy of agency in a confounded world: an agent acts, observes a terrible outcome, assumes its action caused it — when often there was a hidden confounder (the agent was already tired, or the environment was already shifting) and the causal lesson the agent learned is exactly wrong. The three admissibility regimes — A (intervention-rich domains like software, with clean attribution), B (partial intervention with self-selection), C (observational with confounding) — say honestly what each regime supports. This is the same scope discipline AAT applies elsewhere: the result holds, but the conditions under which it operationally works are spelled out.

The failure is bidirectional, and the *positive* version is sharper. A confounded loss at least produces aporia — the agent notices something went wrong and looks for what. A confounded *win* produces satisfaction, and the agent moves on. The asymmetry is not a quirk of psychology; it is structural in AAT's machinery. When an agent is uncertain about a causal relationship, the gain principle pushes η^* toward 1, and each reinforcing observation aggressively updates the model. By the time U_M drops to a level where η^* would have discounted further evidence, the model has been built on the confounder. After that, the agent's predictions about the confounded relationship are *correct in expectation over* *the conditions the agent actually samples* — the cycle's mismatch signal does not point back at the confounder, because under on-policy sampling there isn't a mismatch to point. The corrective machinery has nothing to flag, and the bias is durable because it was laundered through the same gain process that ordinarily produces correct learning.

This is the structure of variable-ratio reinforcement on a noisy action-coupled channel — the most extinction-resistant schedule in operant-conditioning lore, and the engine of gambling addiction. The agent pulls the lever, the world occasionally rewards, and the partial confirmations accumulate updates without ever delivering the deterministic disconfirmation the agent would need to revise. It is also the structure of superstitious behavior in general (the rain dance "worked"), the founder-myth flavor of survivorship bias (here is what the successful did, with the unmentioned variance in the unobserved cohort), and the substrate-switching retrospective that produced one of TST's earlier lessons — *confidence can indicate limitation; true capability manifests as appropriate uncertainty rather than confident clarity*. The same loop that makes purposeful agency possible makes these failure modes possible, and AAT doesn't pretend to solve them: escaping a confounded positive feedback structurally requires something the on-policy regime cannot provide — off-policy exploration, structural priors, joint sibling observation, or an external intervention that the agent itself would not have chosen. What the framework can do is name the structure precisely enough that the failures are recognizable rather than mysterious, and place the burden where it actually lives (in the data substrate and the regime, not in the agent's reasoning).

The chapter also closes off a class of agents the rest of Part II is not about. Pre-compiled controllers — PID, LQR, hardcoded reactive policies — sit in agency scope (they act with effect) but outside the *learning-agent scope* this chapter introduces. Their causal mapping was supplied externally by a designer who had Level-2 access; the agent itself never has to figure out how its actions affect the world. A pre-compiled agent acts, but it never wonders. It experiences aporia about what state the world is in, never about how the world works. Part II's machinery is for agents that wonder.

CIY (causal information yield) is the next move, and the chapter is honest about it too. CIY measures how distinguishable an action's outcome distribution is from alternatives — not quite the same thing as expected information gain. An action can be highly distinguishable without teaching an agent anything new (the agent already knows what each action does); an action can be informative without being distinguishable (the agent learns something subtle). CIY and EIG coincide when the agent is uncertain and diverge when it's confident; AAT uses CIY as a tractable surrogate for EIG, with the $\lambda(M_t)$ weighting compensating for the gap. The unified policy objective

$$\pi^*(M_t) = \arg\max_a \left[\mathbb{E}[\text{value}(a) \mid M_t] + \lambda(M_t) \cdot \text{CIY}(a; M_t)\right] \tag{7.1}$$

ties exploration and exploitation together: when model uncertainty is high, λ scales up and the agent invests in learning; when uncertainty is low, λ scales down and the agent exploits. This is what RL's exploration-vs-exploitation tradeoff looks like when you derive it from the causal hierarchy instead of asserting it as design. *The scalar form above is heuristic*: it admits a "blank-wall" attack where an action maximizing scalar information yield in directions orthogonal to drift satisfies the constraint while the agent's mismatch in the drifting subspace grows unbounded. The matrix-LMI strengthening at `#deriv-causal-ib-lmi` (Linear Matrix Inequality on the Fisher Information Matrix with directional Lagrange multiplier $\Lambda \succeq 0$) closes this by complementary slackness on direction; under the matrix form the unified objective is conditional-theorem-grade rather than heuristic. Summary uses of the scalar form should carry the heuristic flag unless the LMI/trace-product form is in force.

The chapter closes with a normative move: when is explicit strategy worth maintaining at all? Planning has costs (construct, evaluate, maintain) and benefits (avoid expensive exploration, lower repair costs from mistakes); the threshold form

$$C_{\text{plan}} + C_{\text{maintain}} < C_{\text{explore}} + C_{\text{repair}} \tag{7.2}$$

says when the trade favors plans over pure loop-based learning. This isn't a derivation — it's a design criterion grounded in the persistence condition's preference for adaptive margin. Agents that pay tempo for plans they don't need are wasting margin; agents that explore expensively for plans that would have helped are wasting it differently. The strategy machinery that follows in Chapter 3 is justified by this calculus, not assumed.

The flow of the chapter: Bareinboim's hierarchy meets the value object (7.2) → the loop as Level-2 engine (7.3) → CIY admissibility regimes (7.4) → unified policy objective (7.5) → explicit-strategy cost-benefit (7.6). Five segments answering: what does planning need, where does it come from, when is it usable, what objective does it feed, and is it worth it?

7.1 *Definition:* Pearl's Causal Hierarchy (Recapitulation)

This segment recapitulates Pearl's three-level hierarchy of causal reasoning [2, chs.~1–3] and its strict non-collapse theorem [1, Theorem~1] as the vocabulary AAT will deploy throughout the rest of Part II and beyond. The framework is explicit that this is *imported* machinery, not an AAT contribution.

Level 1 — Associational. "What will I observe next, given what I've observed before?" Pattern recognition over the temporally ordered history. Available to any agent that maintains a model (2.1), including purely passive observers. Temporal ordering (1.7) constrains which associations are meaningful — later observations can depend on earlier ones, not vice versa.

Level 2 — Interventional. "What will I observe if I *do* this?" The $do(\cdot)$ operator marks the crucial distinction from Level 1: this is not "what observation tends to follow this action in the historical record" but "what will happen *because* I take this action now." Three conditions must hold: the agent's action temporally precedes the observation; the agent chose the action (it was not determined by the same causes that determine the observation); the environment's response carries information about the causal relationship. Level 2 is why the feedback loop is more powerful than passive observation: by acting and then observing consequences, the agent obtains information about *mechanisms*, not merely correlations. The binary action requirement of 1.6 ensures at least Level 2 access is structurally available.

Level 3 — Counterfactual. "Given that I did a and observed o, what would I have observed if I had done a' instead?" This requires the model to simulate alternative histories — running the causal structure "backward" and then "forward" under different interventions. The most demanding epistemic level; the basis for regret computation, strategic simulation, and learning from single observations.

The *strict-non-collapse theorem* (Bareinboim et al. 2022, Theorem 1) is load-bearing for the rest of Part II: Level-2 quantities cannot in general be computed from Level-1 data alone; Level-3 quantities cannot be computed from Level-2 alone. This is what forces Ch.2's central question: agents that need to *learn* their action consequences during operation require *Level-2 access*, which Level-1 data cannot supply (7.2). AAT's *distinctive contribution* is not the hierarchy itself but its grounding in agent dynamics — the loop-as-Level-2-engine result (7.3), the regime-indexed identification-strength framing, and the application throughout Part II's strategy-revision machinery.

A clarification is offered: the three levels describe epistemic access *the causal structure makes available* — not what any particular agent *uses*. A Kalman filter plus LQR has the Level-2 *channel* structurally present (its innovation signal is conditioned on prior action), but the LQR control action is endogenous to the estimation loop — determined by the policy applied to the estimated state — so the interventional effect is not, in general, *identified* from that action's data; and the separation principle further guarantees estimation quality is invariant to control policy, so the system does not even *exploit* the channel. A PID controller has no deliberative capacity at all and operates entirely at Level 1. Which levels an agent exercises depends on its architecture and model class.

A domain note worth flagging: **software development is a uniquely rich domain for the hierarchy**. For code-internal counterfactuals with deterministic outcomes — "what would the test suite report under implementation X instead of Y, environment fixed?" — `git checkout` plus re-implementation plus test execution is *literal Level-3 realization with ground-truth verification*, not a proxy. The conditions are precise: deterministic outcome, cost-commensurate replay, content-addressed immutable history (`#obs-software-epistemic-properties`). For counterfactuals crossing the agent-environment boundary, it is a strong executable proxy but not literal Level 3. This scoped Level-3 access is what makes software AAT's privileged calibration laboratory for causal reasoning.

Formal Expression.

Definition (pearl-causal-hierarchy, recapitulating Pearl 2009 and Bareinboim et al. 2022)

Level 1 — Associational: $P(o_t \mid \mathcal{C}_{<t})$

What will I observe next, given what I've observed before?

Pattern recognition over the temporally ordered history. Available to any agent that maintains a model (2.1), including purely passive observers. The temporal ordering constrains which associations are meaningful: o_3 can depend on o_1, a_1, o_2, a_2 but not on o_4.

Level 2 — Interventional: $P(o_t \mid do(a_{t-1}), M_{t-1})$

What will I observe if I do *this?*

The $do(\cdot)$ operator marks the crucial distinction: this is not "what observation tends to follow this action in the historical record" (associational) but "what will happen *because* I take this action now." This requires: (1) the agent's action temporally precedes the observation (1.7), (2) the agent chose the action (it was not determined by the same causes that determine the observation), (3) the environment's response carries information about the causal relationship.

Level 2 is why the feedback loop is more powerful than passive observation. By *acting* and then observing consequences, the agent obtains information about causal mechanisms — not merely about correlations. The mismatch signal δ_t (3.4), conditioned on the agent's own action, is an *interventional* signal.

Level 3 — Counterfactual: $P(o_t^{a'} \mid a_{t-1} = a, o_t = o)$

Given that I did a *and observed* o*, what would I have observed if I had done* a' *instead?*

This requires the model to simulate alternative histories — running the causal structure "backward" and then "forward" under different interventions. It is the most demanding epistemic level and the basis for regret computation, strategic simulation, and learning from single observations.

Epistemic Status.

Recapitulation of an external result. The three-level hierarchy and the strict-non-collapse theorem (Bareinboim, Correa, Ibeling & Icard 2022, Theorem 1: Level-2 quantities cannot in general be computed from Level-1 data alone; Level-3 quantities cannot in general be computed from Level-2 alone) are well-established results in causal-inference theory. They live within AAT's segment set because subsequent derivations deploy them as machinery: 7.2 applies the non-collapse theorem to the value-object's $do(\cdot)$ query and concludes that purposeful agents who must *learn* their action consequences need Level-2 access; 7.3 shows that the feedback loop is itself a Level-2 data engine; the no-go in 9.2, the strategy-DAG-as-causal-DAG framing in 8.3, and the causal-information appendices (`#deriv-causal-ib-exploration`, `#deriv-causal-ib-lmi`) all operate on the hierarchy. Where Part I segments and TST segments only need to *reference* the hierarchy (cite its existence and the do-operator notation) rather than deploy it, external citation to Pearl 2009 / Bareinboim et al. 2022 suffices; the AAT recapitulation here is what those external citations point to when the reader wants the in-framework vocabulary.

AAT's *distinctive contribution* is not the hierarchy itself but its grounding: the loop-as-Level-2-engine result (7.3), the regime-indexed identification-strength framing (9.5), and the application throughout Part II's strategy-revision machinery. The recapitulation here is in service of those moves, not a primary AAT result.

Discussion.

Is there a rung above Level 3? Yes, in the Causal Hierarchy Theorem's own relative sense — and its boundary is now characterized. Latent-anchored *mechanism counterfactuals* ("this very background, under a different law") separate strictly from Level 3: two SCMs can agree on every $\mathcal{L}_1$–$\mathcal{L}_3$ quantity — the entire joint law of all hard-intervention worlds — while assigning probabilities 1 and 0 to the same noise-preserving mechanism-replacement query (7.7, *exact*, with the reducibility boundary: replacements specified over potential-outcome coordinates reduce to Level 3, while latent-anchored ones need not and the witness's provably does not — generically so, since every anchor-mobile SCM separates from its own exogenous relabeling; Pearl's own selector-variable internalization re-represents the query without collapsing the separation). The hierarchy recapitulated here is therefore not the ceiling of causal expressiveness, but — exactly as Pearl framed the three rungs — the ceiling of what association, experiment, and standard counterfactuals can determine. The agent-architecture consequences (imagination as navigation vs. positing, the tagged model-space workspace) are developed in 7.8.

Availability vs. exploitation. The three levels describe epistemic access that the causal structure *makes available* — not what any particular agent *uses*. Many systems within AAT's scope operate primarily at Level 1. A Kalman filter coupled with an LQR controller has the Level-2 *channel* structurally present (its innovation signal is conditioned on prior action), but two things hold it short of usable Level-2: the LQR action is endogenous to the estimation loop (the policy applied to the estimated state), so its data does not in general *identify* the interventional effect; and the separation principle guarantees estimation quality is invariant to control policy, so the system does not *exploit* the channel. Only dual control (choosing actions partly for their informational value, and typically perturbing the action so it is not a pure function of the state estimate) moves toward exercising Level 2 access in this domain. Similarly, a PID controller has no deliberative capacity — it operates entirely at Level 1. Which levels an agent exercises depends on its architecture and model class.

Forward-looking deliberation exercises Level 2, shading into Level 3. Comparing candidate actions before choosing — "what will happen if I do X vs Y?" — primarily exercises Level 2 (iterated mental intervention). When the agent evaluates past choices to refine the comparison ("given what happened when I tried X last time, what would Y have produced?"), it exercises Level 3.

The causal hierarchy theorem. Bareinboim et al. (2022) prove that the three levels form a strict hierarchy: Level 2 knowledge cannot in general be computed from Level 1 data alone, and Level 3 cannot be computed from Level 2 alone. This is load-bearing for AAT's Part II: evaluating $Q_O(M_t, a; \cdot)$ is a Level 2 query, so agents that need to *learn* action consequences during operation require causal structure beyond predictive models (7.2).

Software as a uniquely rich domain for this hierarchy. In most domains, Level 3 counterfactuals require model-based simulation with uncertain fidelity. Software development is the privileged exception *for a specific class*: for code-internal counterfactuals with deterministic outcomes — "what would the test suite report under implementation X instead of Y, environment fixed?" — `git checkout` plus re-implementation plus test execution is literal Level 3 realization with ground-truth verification, not a proxy. For counterfactuals crossing the agent–environment boundary (what feature sequence the team would have shipped, how the market would have responded) it is a strong executable proxy, not literal Level 3. The precise conditions — the (α) deterministic-outcome / (β) cost-commensurate-replay / (γ) content-addressed-immutable conjunction, and why the resulting uniqueness is configurational rather than necessary (with named falsifiers) — are established in `#obs-software-epistemic-properties` (P2; `02-tst-core/`). This scoped Level-3 access is what makes software AAT's privileged calibration laboratory for causal reasoning.

Domain instantiations of the three levels:

Table 7.1

Domain	*Level 1 (Association)*	*Level 2 (Intervention)*	*Level 3 (Counterfactual)*
Kalman filter + LQR	Prediction from state estimate	Structural channel present, identification not guaranteed (the LQR action is endogenous to the estimation loop)	Not typically exercised
RL agent	Value function prediction	Action → reward observation	Regret computation
Scientific method	Correlational observation	Experimental intervention	"What if we had used control X?"
Military (Boyd)	Pattern recognition	Probe/feint → observe response	"What if we had attacked from the flank?"
Software developer	"I think this function does X"	Run test → observe result	`git checkout` + alt. impl. — literal for code-internal deterministic counterfactuals; proxy across the agent–environment boundary (`#obs-software-epistemic-properties`)
Immune system	Antigen pattern matching	Antibody → pathogen response	Not exercised (no counterfactual reasoning)

7.2 *Derived:* Causal Hierarchy Requirement

The direct application of Pearl's causal hierarchy (7.1) to the value object (6.6). The action-value query uses the $do(\cdot)$ operator and is therefore a *Level-2 (interventional)* query; by the strict-non-collapse theorem, Level-2 quantities cannot in general be computed from Level-1 data alone [1, Theorem~1]. An agent that must evaluate Q_O *from experience* therefore needs access to **Level-2 knowledge** — knowledge about the effects of its own interventions, not merely correlational patterns.

The result is paired with a sharp *scope narrowing*: attention restricts to **learning purposeful agents** — agents that must *acquire or refine* Level-2 knowledge during operation. This is a named sub-scope of 1.6 that excludes *pre-compiled* interventional structure (PID controllers where the designer pre-computed the control law, LQR where the separation principle gives the optimal policy from model parameters, hardcoded reactive policies). Pre-compiled agents are still within agency scope (they have objectives and act on them) but outside learning-agent scope — their causal structure was externally supplied by a designer who had Level-2 access. *All remaining Part II results operate within learning-agent scope unless explicitly noted otherwise.* The derivation is exact as a direct application of the external hierarchy theorem to the value-object definition; the scope narrowing is a definitional restriction, not a derived result.

The distinction between intervention and conditioning is the *core* of causal inference and matters whenever the agent's action-selection policy correlates with unobserved confounders — generic for any agent with internal state, plans, or memory. A developer needs "if I refactor this module, will tests pass?" not "modules that were refactored tend to have tests that fail" (the latter is biased by which modules the team chose to refactor). The Discussion below treats the concrete cases (developer / commander / RL agent), why pre-compiled agents fall outside, and the empirical bridge to Hafez et al.'s Information Digital Twin — whose 89% perturbation-detection accuracy versus 44% for reward-based monitoring is consistent with the present segment's claim that the loop's interventional data is information-theoretically richer than outcome monitoring alone.

Formal Expression.

Derived (causal-hierarchy-requirement, from value-object + pearl-causal-hierarchy)

Action selection via 6.6 requires:

$$Q_O(M_t, a; \pi_{\text{cont}}, N_h) = \mathbb{E}\big[V_{O_t}(\tau) \mid M_t,\ do(a_t = a),\ a_{t+1:} \sim \pi_{\text{cont}}\big] \tag{7.3}$$

The $do(\cdot)$ notation is explicit: this is an *intervention*, not a *conditioning on observed data*. By 7.1, Level 2 queries ($P(Y \mid do(X))$) cannot in general be computed from Level 1 data ($P(Y \mid X)$) alone. Therefore:

An agent that must evaluate Q_O from experience needs access to Level 2 knowledge — knowledge about the effects of its own interventions, not merely correlational patterns.

Scope Narrowing (learning-agent scope)

We restrict attention to **learning purposeful agents** — agents that must **acquire or refine** Level 2 knowledge during operation. This is a named sub-scope of the agency scope defined in 1.6. It excludes agents with **pre-compiled** interventional structure: - PID controllers (the designer pre-computed the control law) - LQR (separation principle gives optimal policy from model parameters) - Hardcoded reactive policies

Pre-compiled agents are within agency scope (they have objectives and act on them) but outside learning-agent scope — their causal structure was externally supplied by a designer who had Level 2 access. **All remaining Part II results operate within learning-agent scope** unless explicitly noted otherwise. This scope narrowing focuses the theory on agents that must build or maintain their own causal understanding.

Epistemic Status.

Exact. The derivation is a direct application of the causal hierarchy theorem to the value-object definition. If you accept that Q_O is an interventional query and that the causal hierarchy is strict, the conclusion follows. The scope narrowing to learning agents is a definitional restriction, not a derived result — it sharpens the class of agents under study.

Discussion.

The causal hierarchy theorem does the heavy lifting. The key mathematical fact is external to AAT: Bareinboim et al. (2022) prove that the three levels (association, intervention, counterfactual) form a strict hierarchy — Level 2 quantities cannot in general be computed from Level 1 data. AAT's contribution is applying this to the purposeful-agent setting: if you want to select actions by their consequences (Q_O), you need causal structure.

What "Level 2 knowledge" means concretely. For different agents: - A developer needs to know "if I refactor this module, will tests still pass?" (not just "modules that were refactored tend to have tests that fail") - A commander needs to know "if I move forces north, will the enemy respond by retreating?" (not just "when forces moved north, the enemy often retreated") - An RL agent needs $Q(s, a) = \mathbb{E}[R \mid s, do(a)]$, not $\mathbb{E}[R \mid s, A = a]$ (the latter includes selection bias from the agent's own policy)

The distinction between $do(a)$ and $A = a$ is the core of causal inference. It matters whenever the agent's action-selection policy correlates with unobserved confounders.

Why pre-compiled agents are excluded. A thermostat "knows" that turning on the heater raises temperature — but this knowledge was designed in, not learned. The thermostat never needs to reason about interventions because the intervention-outcome mapping is hardwired. AAT's purposeful-agency machinery is specifically for agents that face uncertainty about how their actions affect the world and must reduce that uncertainty through experience.

Bi-predictability as empirical evidence for Level 2 advantage. Hafez et al. (2026) measure the information structure of the agent-environment loop via bi-predictability $P = \mathrm{MI}(S, A; S')/C$, capturing how tightly coupled the agent is to its environment through the action channel. Their Information Digital Twin (IDT), which monitors the loop's information geometry, detects environmental perturbations at 89% accuracy versus 44% for reward-based monitoring. This is consistent with the present segment's claim: the feedback loop provides richer (Level 2) data than outcome monitoring alone. Hafez's framework measures the coupling; AAT explains *why* the coupling is information-theoretically superior — because the loop generates interventional data by construction (7.3), not merely associational data. The measurement (Hafez) and the explanation (AAT) are complementary.

7.3 *Derived:* Loop Provides Interventional Data Access

The chapter's central architectural result. By the temporal-ordering postulate (1.7), the agent's action a_t causally precedes the next observation o_{t+1}; the agent *chose* the action, the environment responded. The feedback loop therefore generates **intervention-produced data** — data whose causal character differs from passive observation because the agent's action was a genuine cause of the subsequent observation. The mismatch signal conditioned on the agent's action carries interventional information: how the environment responded to *this specific intervention*, relative to what the model predicted. This is how agents within AAT's **agency scope** ($\mathcal{S}_{\text{agency}}$, which requires $|\mathcal{A}| \geq 2$ and at least one action with causal effect) gain Level-2 access — not through internal architecture but through the loop itself; agents in the adaptive scope but outside agency scope (passive observers) lack the action contrasts needed.

The framework is *unusually careful* to distinguish two claims that look alike. "Action-generated data" is **not** the same as "cleanly identified do-estimates." The action-observation pair is produced *under intervention* (interventional in character) — but between intervention-produced data and a usable estimate of the interventional outcome distribution stand four obstacles: coverage (the agent must have tried diverse actions), within-step confounding, delayed consequences, and partial observability. The strength of causal identification varies by regime — strong in intervention-rich domains (software, laboratory science), moderate with partial intervention (organizational), weak under observation-only constraints (see 7.4). The claim is about the *character of the data* (interventional, not observational), not about the agent's ability to extract clean causal estimates from it. The result is *exact* as a logical consequence of temporal ordering and the feedback-loop structure; it is about *data availability*, not reasoning capacity — the loop *provides* interventional data whether or not the agent exploits it for Level-2 reasoning.

The interventional character of loop data is a property of the action being the agent's *own* causally-efficacious move, not of any architectural feature and not of the trajectory's forkability. The do-operator's semantics is population-level by construction: the interventional distribution sums over the exogenous distribution $P(\mathbf{U})$, and its canonical physical realization — randomized controlled trials and A/B experiments — is forkable and repeatable [1, Def.~5 and fn.~19]. What the singular-trajectory scope (4.10) supplies is narrower and still essential: it grounds *whose* effect a loop datum is the effect of, and forbids averaging interventional responses across forked copies as though they were one agent's effects. The *interventional character* is real but not unconditional: action-generated data identifies the interventional distribution only under (C1) positivity, (C2) sequential ignorability, and (C3) a known action-mechanism. The exact content of this segment is the *availability of the interventional-character channel*; identification is gated on (C1)–(C3). The Discussion below treats: the precision conditions on "interventional" (low confounding / short delay / varied action); even-without-explicit-causal-model benefits (Q-learning in the tabular case); honest credit to the action-perception-loop framing in active inference and cybernetics (Wiener 1948, Conant–Ashby 1970, Friston et al. 2017, Parr–Pezzulo 2022) together with the three specific moves AAT makes that the implicit treatments do not (the Bareinboim-hierarchy connection, regime-indexed identification strength, explicit scope honesty as the conservative-form posture documented in the Bruineberg et al. 2022 Pearl-vs-Friston critique); the connection to the identifiability-floor pattern as a load-bearing escape with two semantically distinct deployment modes (agent-self-intervention at the causal-structure layer, observer-on-sub-agent at the composition layer); and the singular-trajectory ground for whose effect the loop datum is — distinct from the do-semantics, which is forkability-agnostic.

Formal Expression.

By 1.7, the temporal ordering is constitutive: a_t causally precedes o_{t+1}. The agent chose a_t; the environment responded with o_{t+1}. The feedback loop therefore generates **intervention-produced data** — data whose causal character differs from passive observation because the agent's action was a genuine cause of the subsequent observation.

Derived (loop-interventional-access, from causal-structure + recursive-update)

The critical distinction: **"action-generated data" is not the same as "cleanly identified do-estimates."** The pair (a_t, o_{t+1}) is produced under intervention — the agent executed a_t, making it interventional in character rather than a passively observed association. But between intervention-produced data and a usable estimate of $P(o \mid do(a_t), \Omega_t)$ stand: (1) coverage — the agent must have tried diverse actions, not just one policy; (2) confounding within a time step — unobserved state variables that affect both action choice and outcome; (3) delay — consequences may appear much later than $t + 1$; (4) partial observability — o_{t+1} reveals only part of the outcome. The strength of causal identification from this data depends on the regime (9.5): strong in Regime A (intervention-rich domains like software and laboratory science), moderate in Regime B (partial intervention), weak in Regime C (observation-only). The claim here is about the *character* of the data (interventional, not observational), not about the agent's ability to extract clean causal estimates from it.

The mismatch signal conditioned on the agent's action:

$$\delta_t \mid a_t = o_{t+1} - \hat{o}_{t+1}(M_t, a_t) \tag{7.4}$$

carries interventional information: it tells the agent how the environment responded to its specific intervention a_t, relative to what the model predicted.

The exact claim is the availability of the channel; identification is gated. What is exact, as a logical consequence of temporal ordering and the loop structure, is that an agent in agency scope that executes at least one action with causal effect *generates action-coupled data that is interventional in character* — sampled under the agent's own action rather than passive observation. This is the substrate from which Level-2 quantities $P(o \mid do(a), \cdot)$ can be *identified*, but identification is not automatic. It holds when:

- **(C1) Positivity** — every action in the relevant support is taken with nonzero probability across the agent's history, so the interventional distribution is estimable rather than extrapolated;
- **(C2) Sequential ignorability** — in mutilated-graph form, the action a_t is d-separated from the outcome o_{t+1} given history H_t; equivalently, the policy is unconfounded with the outcome mechanism;
- **(C3) Known action-mechanism** — the map from the agent's decision to the executed action is known (or itself overridden), so the executed value can be read as the intervened value.

Where (C2) fails — generic for any agent whose action depends on internal state, plans, or memory that also bear on the outcome — on-policy (a_t, o_{t+1}) converges to $\mathbb{E}[o \mid A = a]$, the conditional, rather than to $\mathbb{E}[o \mid do(a)]$, the interventional effect. The channel is still present; what is gated is the move from action-generated data to an identified do-estimate.

Epistemic Status.

Exact. The exact content is the **availability of the interventional-character channel**: a logical consequence of temporal ordering (1.7) and the feedback-loop structure (1.6), an agent in agency scope generates action-coupled data that is interventional in character rather than passively observed. The claim is about **data availability**, not reasoning capacity — the loop *provides* this channel whether or not the agent *exploits* it for Level-2 reasoning, and whether or not the data yields identified estimates.

The precision is load-bearing and now first-class in the Formal Expression: the agent has *access to* interventional-character data, but *identification* of $P(o \mid do(a), \cdot)$ from it is gated on (C1) positivity, (C2) sequential ignorability, and (C3) a known action-mechanism.

Confounding within a single time step (a violation of (C2)), insufficient action coverage (a violation of (C1)), delayed outcomes, and partial observability all bear on whether clean causal estimates can be extracted. The gated form is what keeps the *exact* label honest: the availability of the channel is unconditional; the identified-do-estimate is the conditional consequence, and the segment claims only the former as exact.

Discussion.

The loop as a Level 2 engine. This is one of AAT's load-bearing results. The causal hierarchy theorem (7.2) says Level 2 knowledge requires more than correlational data. This result says: the adaptive loop *is* the "more." An agent that acts and observes the consequences is generating interventional data — the same kind of data that a scientist generates through experiments. The loop is a perpetual experiment.

Precision about what "interventional" means here. The interventional interpretation is strongest when: - The agent's action was the primary cause of the observed change (low confounding) - The observation follows closely in time (short delay) - The agent varied its action across episodes (not stuck on one policy)

When confounding is high, delays are long, or the agent follows a fixed policy, the interventional information in each (a_t, o_{t+1}) pair is weaker — still present, but harder to extract. This is why 3.7 distinguishes between high-CIY actions (that reveal causal structure) and low-CIY actions (that don't).

Even agents without explicit causal models benefit. A Q-learning agent doesn't maintain an explicit causal model, but in the tabular case with sufficient exploration and no within-step confounding, its Q-values converge toward $\mathbb{E}[R \mid s, do(a)]$ rather than $\mathbb{E}[R \mid s, A = a]$ — precisely because the training data comes from the agent's own interventions. In the partially observed, confounded, or delayed-outcome cases (where the caveats above apply), the loop still provides intervention-generated data, but the Q-values may converge to biased estimates that reflect the confounding structure rather than clean interventional effects. The loop provides Level 2 data; whether that data yields *identified* causal quantities depends on the domain's confounding and observability structure.

Honest credit to the action-perception-loop framing. The substantive observation that the agent's actions cause its observations — and therefore that loop data is interventional in character — is implicit in any framework built around an action-perception loop, including active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Parr & Pezzulo 2022, *Active Inference*, MIT Press, ch. 3) and the broader cybernetic-and-control lineage (Wiener 1948, *Cybernetics*; Conant & Ashby 1970, "Every good regulator of a system must be a model of that system," *Int. J. Systems Sci.* 1). AAT's distinctive contribution is not the observation that loop data is interventional but the *explicit lift* of this observation to a load-bearing theorem connected to Pearl's causal hierarchy (7.1) via Bareinboim, Correa, Ibeling & Icard (2022). Three specific moves AAT makes that the implicit treatments do not:

1. **The Bareinboim-hierarchy connection.** Active inference and the cybernetic lineage rest on Bayesian-network generative models (Pearl Level 1, associational). They do not invoke the causal-hierarchy theorem to argue that the loop's action-generated data is the substrate Level-2 queries require. AAT does — and the consequence is that Σ_t is positioned as a *causal* DAG rather than a *Bayesian-network* DAG.

2. **Regime-indexed strength of causal identification.** Even granting that loop data is interventional in character, the strength of usable causal identification varies by domain. AAT partitions this into Regime A (intervention-rich, software/laboratory), B (partial intervention, organizational), C (observation-only — see 9.5). The AI literature treats causal identifiability uniformly within its modeling assumptions and does not surface the regime distinction at the segment level.

3. **Explicit scope honesty.** The Formal Expression above carefully distinguishes "data generated under intervention" from "cleanly identified do-estimates." The AI literature, as Bruineberg, Dolega, Dewhurst & Baltieri (2022) document in their Pearl-vs-Friston critique, sometimes elides this distinction — using the action-perception loop language to support stronger causal claims than the formal apparatus delivers. AAT's careful split is the conservative form.

The reframing here is rhetorical, not substantive: the headline result (the loop is a Level-2 engine) stays; what changes is the explicit acknowledgment that the observation about loop data being action-generated is shared with the broader literature. AAT's distinctive content sits in the three specific moves above. The companion architectural move on 6.3 (Pearl-blanket vs. Friston-blanket form of

Markov blanket; cf. Bruineberg et al. 2022) makes AAT's conservative-form positioning visible across both architectural and access-channel segments.

Connection to the identifiability-floor pattern. This segment is structurally load-bearing for the L0-causal-insufficiency-detection no-go (9.2): without the loop's interventional access, the no-go forbids detection entirely; the covariance test under joint sibling observability is the unique broadly-available violation of the no-go's "purely on-policy" scope. See 5.1 for the meta-pattern.

Composite-layer extension. `#disc-identifiability-floor` Instance 3 (the composition-layer no-go anchored in Liberzon 2003's common-Lyapunov nonexistence) establishes that composite contraction cannot in general be certified from component-level data alone — the coupling-topology bit (broadened 2026-05-21 from the original coupling-sign bit alone to include the potential-existence bit and the game-structure-class bit distinguishing R0 / R1 / R2 dynamic regimes per `#disc-dynamic-regime-axis`) is unidentifiable from component marginals. One of the five structural escapes in Instance 3 is composite-extended loop-interventional-access: **interventions on sub-agent A_j reveal A_i's cross-coupling response**, which is a $do(\cdot)$-data distinction between the topologically-distinct coupled constructions. This is the composite-layer analog of the single-agent interventional-access-escape the present segment delivers for Instance 1; in the composite setting, the agent performs interventions on one sub-agent's action space and observes the effect on another sub-agent's mismatch trajectory, extracting the coupling-topology object (sign and structure) that component marginals cannot. The load-bearing role of `#der-loop-interventional-access` therefore extends across two identifiability-floor instances (agent-internal and composite-layer), not one.

Modes of deployment across `#disc-identifiability-floor` instances. The shared load-bearing role — Level-2 data from interventional action under `#der-causal-hierarchy-requirement` supplying the unique broadly-available escape from an observational-equivalence no-go — manifests through **semantically distinct interventional mechanisms** at different layers. The modes share the Pearl-do structure but differ in who performs the intervention and on what:

- **Mode 1 — agent-self-intervention (Instance 1, causal-structure layer).** The agent performs do-actions on its own action space as part of its ordinary adaptive loop. The intervention is *on the agent's own action*; the target is the environment's response, which reveals the latent common cause that Instance 1's on-policy L0-insufficiency-detection no-go (Bareinboim et al. 2022 CHT) makes unidentifiable. The agent-as-Level-2-data-generator property is intrinsic to the loop structure, not a capability that must be added.
- **Mode 2 — observer-on-sub-agent (Instance 3, composition layer).** An observer external to a composite performs do-interventions on one sub-agent A_j's action space; the target is another sub-agent A_i's mismatch-trajectory response, which reveals the cross-coupling sign that Instance 3's Liberzon-anchored no-go makes unidentifiable from component-marginal observation. The same Pearl-do structure, but the intervening agent and the measured agent are distinct members of the composite.

Future instances of `#disc-identifiability-floor` at new layers may add further modes (see `#disc-identifiability-floor` Working Notes for the architecture-within-behavior-class layer currently under triage; the corresponding mode would be *observer-on-agent-input* at the candidate agent's observation channel). The three-mode-pattern observation is a structural regularity worth naming even though the specific Mode-3 instance has not promoted; the load-bearing content — Level-2 escape from observational-equivalence no-goes via loop-interventional access — remains shared across the deployment modes even when the specific interventional quantity varies. The unification is at the pattern level; the mechanism is semantically distinct per layer.

What the singular-trajectory scope grounds — and what it does not. The interventional *character* of loop data does not rest on the trajectory being non-forkable: the do-operator's semantics is population-level by construction (Bareinboim, Correa, Ibeling & Icard 2022, Def 5 — the interventional distribution sums over the exogenous distribution $P(\mathbf{U})$), and its canonical physical realizations, randomized controlled trials and A/B experiments, are forkable and repeatable (CHT footnote 19). The $\mathcal{L}_1/\mathcal{L}_2$ discriminator is whether the action's generating mechanism was the system's own intact mechanism (observational) or an override / unconfounded-and-known mechanism (interventional) — not whether the trajectory can be replayed. What the scope commitment in 4.10 supplies is narrower and still essential: each agent is instantiated on a singular, non-forkable causal trajectory $\mathcal{C}_t$, and this grounds *whose* effect a loop datum is the effect of. Replaying a_t from a checkpointed M_t against a different event stream is an intervention on a *different* trajectory $\mathcal{C}_t^{(2)}$ that happens to share a prefix; the singularity forbids averaging interventional responses across such forked copies as though they were one agent's effects, because each fork is a distinct trajectory-SCM with its own $P(\mathbf{U} \mid \mathcal{C}_t)$. Agents whose ontology is *type-like* (equivalence classes of copies) rather than *token-like* (singular trajectories) are outside AAT's formal scope; aggregate claims about "the model" across copies require additional machinery not provided here. So the singular-trajectory scope grounds the *identity* of the effect (and forbids copy-averaging and lossy cross-trajectory merging), while the *interventional character* itself comes from the action being the agent's own causally-efficacious move under (C1)–(C3).

7.4 *Scope:* CIY Observational Proxy

Clarifies what can be done with the causal-information-yield construct (3.7) when interventional data is *not* available. An observational *proxy* for CIY can be defined in terms of mutual information between the action and the next observation, but the framework is forthright: this proxy is **sign-indefinite in general** and requires causal assumptions for interpretation. The canonical interventional CIY remains the primary quantity; the proxy is auxiliary.

A pointed *safety condition* is stated: the proxy form should **not be used in policy optimization** (e.g., as the CIY term in a policy objective), because an agent maximizing a sign-indefinite quantity may optimize in the wrong direction. The proxy is suitable only for *diagnostic* purposes — detecting whether an action carried causal information (large proxy magnitude) versus none (proxy near zero). For actual decision-making, the canonical CIY (non-negative by construction) or a known-safe surrogate (ensemble disagreement, UCB bonuses) should be used; if the canonical CIY is intractable and no safe surrogate is available, the prescription is to *drop the CIY term entirely* and default to pure exploitation rather than risk the sign-indefinite quantity.

The segment then names three **admissibility regimes** that determine when CIY can be estimated and how strong the causal identification is. The regime is treated as a *property of the domain and the agent's action space*, not a parameter the agent chooses. **Regime A — randomized interventions**: the agent's action assignment is randomized, or known-and-adjusted so that sequential ignorability and positivity hold (RL agents exploring with an exploration policy uncorrelated with the latent outcome causes, scientists randomizing treatment, organisms probing); CIY is directly estimable from execution data and non-negative by construction (the standard case for active agents within 7.3). Mere across-episode action *variation* is not sufficient when the variation is policy-driven by state or history that also bears on the outcome. **Regime B — observational with causal assumptions**: the agent cannot freely vary actions (coordination, policy, or resource constraints); CIY estimation requires additional structure — a known causal DAG, instrumental variables, or functional-form assumptions — and inherits whatever causal assumptions are made. **Regime C — adversarial or passive observation**: the agent either did not intervene (passive monitoring, CIY zero by definition) or the observation channel includes potentially adversarial responses (where CIY from the query action itself remains non-negative but the *content* of the response may be designed to increase model-reality mismatch).

The regime classification has concrete domain implications: software development is typically Regime A (the agent runs tests, deploys to staging, observes results — high action variation); organizational strategy is typically Regime B (concurrent initiatives, attribution requires assumptions); intelligence analysis is typically Regime C (the analyst observes but does not intervene). The CIY machinery applies cleanly in Regime A, applies with weaker guarantees and explicit assumptions in Regime B, and should be replaced with alternative exploration strategies entirely in Regime C.

Formal Expression.

Definition (ciy-proxy)

$$\mathrm{CIY}_{\mathrm{proxy}}(a_{t-1}) = I(o_t; a_{t-1} \mid M_{t-1}) - I(o_t; a_{t-1} \mid \Omega_t, M_{t-1}) \tag{7.5}$$

This proxy is **sign-indefinite in general** and requires causal assumptions for interpretation. The canonical CIY (interventional, 3.7) is the primary quantity; the proxy is auxiliary.

Safety conditions for proxy use. The proxy form should NOT be used in policy optimization (e.g., as the CIY term in a policy objective) because an agent maximizing a sign-indefinite quantity may optimize in the wrong direction. The proxy is suitable only for diagnostic purposes: detecting whether an action carried causal information (large proxy magnitude) vs. none (proxy near zero). For decision-making, use the canonical CIY (non-negative by construction) or a known-safe surrogate (ensemble disagreement, UCB bonuses). If the canonical CIY is intractable and no safe surrogate is available, the CIY term should be dropped from the policy objective entirely, defaulting to pure exploitation.

Admissibility regimes.

Scope Condition (ciy-admissibility)

Three regimes determine when CIY can be estimated and how strong the causal identification is:

Regime A — Randomized interventions. The agent's action assignment is randomized, or known-and-adjusted so that sequential ignorability and positivity hold (RL agents exploring with an exploration policy uncorrelated with the latent outcome causes, scientists randomizing treatment, organisms probing). CIY is directly estimable from the agent's execution data and non-negative by construction. This is the standard case for active agents within the adaptive loop (7.3). What provides the identification needed for clean interventional estimates is the randomized / ignorable assignment together with positivity — not mere action variation, which is insufficient when actions are policy-driven by state or history that also affects the outcome.

Regime B — Observational with causal assumptions. The agent cannot freely vary actions (constrained by coordination, policy, or resource limits). CIY estimation requires additional structure: a known causal DAG, instrumental variables, or functional form assumptions. Results inherit whatever causal assumptions are made. The interventional interpretation of CIY is weaker — it holds under the assumed causal structure but not model-free.

Regime C — Adversarial or passive observation. The agent either did not intervene (passive monitoring) or the observation channel includes responses from potentially adversarial sources. In the passive case, CIY is zero by definition (no intervention, no interventional information). In the adversarial case, CIY from the query action itself remains non-negative, but the *content* of the response may be designed to increase model-reality mismatch. The adversary operates through the disturbance term ρ, not through the information measure.

The regime is a property of the **domain and the agent's action space**, not a parameter the agent chooses. Software development is typically Regime A (the agent runs tests, deploys to staging, observes results — high action variation). Organizational strategy is typically Regime B (multiple initiatives run concurrently, attribution requires assumptions). Intelligence analysis is typically Regime C (the analyst observes but does not intervene).

Epistemic Status.

Conditional on the causal assumptions within each regime. The proxy definition is standard information theory; the admissibility classification is a scope decision, not a derived result. The safety conditions for proxy use are normative — they follow from the sign-indefiniteness of the proxy, not from AAT-specific machinery.

Max attainable: conditional. The regime boundaries are domain properties that cannot be derived within the theory.

Discussion.

Relationship to the canonical CIY. The proxy is not an approximation of the canonical CIY — it is a different quantity that happens to correlate with CIY under favorable conditions (Regime A). The canonical CIY (3.7) is defined interventionally and is non-negative by construction. The proxy uses observational mutual information and can be negative. They agree when the agent's action variation satisfies the conditions for causal identification; they diverge otherwise.

Regime as a domain property. The admissibility regime is not something the agent selects — it is determined by the domain's action space and observation structure. An agent that cannot vary its actions is in Regime B or C regardless of its internal architecture. This has implications for which domains CIY-based exploration is applicable to: Regime A domains get the full benefit; Regime B domains get weaker guarantees; Regime C domains should use alternative exploration strategies entirely.

7.5 *Discussion:* CIY Unified Policy Objective

The framework's *unified policy objective* expresses the exploration-exploitation tension as a single decision rule: choose the action that maximizes a weighted sum of expected value (exploitation) and a causal-information surrogate (exploration). The first term is the exploitation contribution; the second is a *heuristic exploration term* using CIY (3.7) as a tractable surrogate for expected information gain — selecting for causally distinctive actions rather than maximally informative ones. The weighting parameter $\lambda(M_t)$ controls the balance, carries units of *value per unit information*, and in specific well-understood domains reduces to known quantities exactly: the Gittins index for Bayesian bandits, the probing cost in quadratic objectives for Kalman dual control, precision on epistemic affordance for active inference, the explicit variance-over-information ratio in information-directed sampling (Russo & Van Roy). The unified objective is not a novel formalism but a *family-level abstraction* that recovers known scalarizations of the exploration-exploitation trade-off across communities.

The scalar form is honestly heuristic and is now formally *superseded* by an exact tensor trace-product derived in `#deriv-causal-ib-lmi`: the action maximizes the value plus the trace of a positive-semidefinite shadow-price matrix times the Fisher Information Matrix at the action — the exact Lagrangian relaxation of the Linear Matrix Inequality governing Lyapunov persistence. Two parallel exploration drives emerge from the framework, acting at opposite ends of the uncertainty spectrum: an *epistemic information-gain* drive with $\lambda \propto U_M$ (explore to reduce model uncertainty; dominates when uncertain) and a *survival-imperative* drive with $\lambda \propto 1/U_M$ (an agent with high confidence in a drifting environment mathematically guarantees its own death by ignoring noisy observations, so the Lyapunov persistence constraint dictates an immense shadow price as model uncertainty drops in a drifting environment — forcing the confident agent to seek unambiguous observations from the *opposite* uncertainty regime). The dark-room problem that plagues some preferences-as-priors formulations is *bypassed entirely* by the survival imperative: exploration is not driven by preferences-as-priors but by the literal physical boundaries of the Lyapunov sector constraint. The Discussion below treats the active-inference structural isomorphism with two substantive AAT-specific departures (causal information rather than entropy over hidden states; value functional rather than preferences-as-priors), the zero-mismatch ambiguity bridge from Part I, the three-mode extension for actuated agents, and the regret-bound connection to the strategy-cost objective via the Bretagnolle-Huber identity under deterministic optimal policy.

Formal Expression.

Discussion (unified-policy-objective — heuristic)

$$\pi^*(M_t) = \arg\max_a \left[\mathbb{E}[\text{value}(a) \mid M_t] + \lambda(M_t) \cdot \text{CIY}_q(a; M_t)\right] \tag{7.6}$$

The first term is exploitation (expected value given current model). The second is a *heuristic exploration term* using CIY as a surrogate for expected information gain (3.7). CIY measures how different the action's outcome distribution is from alternatives — this is action-distinguishability, not learning value. The surrogate is reasonable when U_M is high (distinguishable actions are also informative to an uncertain agent) and poor when U_M is low (distinguishable actions teach nothing to a confident agent). $\lambda(M_t)$ controls the balance:

- High U_M (uncertain model) → large λ — exploration is valuable
- Low U_M (confident model) → small λ — exploitation dominates
- Long time horizon → larger λ — information compounds
- High ρ (fast-changing environment) → larger λ — perpetual uncertainty

λ carries units of [value per unit information]. In specific domains it reduces to known quantities:

Table 7.2

Domain	*λ reduces to*	*Status*
Bayesian bandits	Gittins index	Exactly derived
Kalman dual control	Probing cost in quadratic objective	Exactly derived
Active inference	Precision on epistemic affordance	Framework-derived
Information-directed sampling	$(\text{VoI})^2/\text{info gain}$	Exactly derived (Russo & Van Roy)
RL with UCB	Confidence-bound scaling	Heuristic (tuned)

Identifiability gate. Before incorporating CIY into the policy objective: (1) action variation must exist, (2) the admissibility regime must be identified (7.4), (3) the reference distribution q must be specified, (4) local stationarity must hold. If any condition fails, CIY-based terms should be dropped or replaced with simpler uncertainty-based heuristics (UCB-style bonuses, ensemble disagreement).

Epistemic Status.

Discussion-grade summary; underlying derivation is exact. Originally treated as a discussion-grade heuristic, the unified objective has now been formally derived as the exact Lagrangian relaxation of the Linear Matrix Inequality (LMI) governing Lyapunov persistence (see `#deriv-causal-ib-lmi`).

The scalar heuristic $Q_O(a)+\lambda\cdot\text{CIY}(a)$ is formally superseded by the exact tensor trace-product: $a_t^* = \arg\max_a \left[Q_O(a) + \text{Tr}\left(\Lambda \cdot \mathcal{I}_o(a)\right)\right]$ where $\mathcal{I}_o(a)$ is the Fisher Information Matrix (Matrix CIY) and Λ is the positive-semidefinite shadow price matrix of the survival constraint.

Max attainable for this segment: *discussion-grade* (it is a discussion of the underlying result, per `type: discussion`). Max attainable for the underlying derivation in `#deriv-causal-ib-lmi`: *exact*. The structural form is fully grounded in AAT's physical survival bounds and standard semidefinite programming, eliminating the need to treat exploration as an ad-hoc heuristic.

Discussion.

Two Parallel Exploration Drives. AAT dictates two correlated but distinct motivations for exploration, acting at opposite ends of the uncertainty spectrum: 1. **Epistemic Information Gain ($\lambda_{\text{info}} \propto U_M$):** The primary CIY formulation. The agent explores to reduce its model uncertainty. This drive dominates when U_M is high. 2. **The Survival Imperative ($\lambda_{\text{surv}} \propto 1/U_M$):** As mathematically proven in `#deriv-causal-ib-exploration`, an agent with high confidence (low U_M) in a drifting environment ($\rho > 0$) mathematically guarantees its own death by ignoring noisy observations. To force the necessary correction, the Lyapunov persistence constraint dictates an immense shadow price ($\lambda_{\text{surv}} \to \infty$ as $U_M \to 0$) forcing the agent to seek unambiguous observations (low U_o / high CIY).

The dark-room problem is bypassed entirely by the Survival Imperative: exploration is not driven by preferences-as-priors, but by the literal physical boundaries of the Lyapunov sector constraint.

Connection to the zero-mismatch ambiguity. An agent that only exploits (acts to maximize predicted value) will tend toward confirmation bias — observing only what its model already explains (3.4, zero-mismatch ambiguity case (b)). Exploration via CIY-maximizing actions is the mechanism by which the agent actively tests its model.

Connection to Part II. For actuated agents, the exploration-exploitation tension extends to three modes: exploit (pursue O_t via Σ_t), explore (improve M_t), deliberate (revise Σ_t). The CIY framework provides the information-theoretic grounding for why strategy edges (8.3) need observational access (9.3) — edges the agent cannot observe have frozen CIY.

Connection to active inference. The expected free energy (EFE) in active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020, "Active inference on discrete state-spaces," *J. Math. Psych.* 99; Sajid, Ball, Parr & Friston 2021, "Active inference: demystified and compared," *Neural Computation* 33) decomposes into *pragmatic value* (preferences-aligned outcomes) and *epistemic value* (expected information gain about hidden states). AAT's unified objective is structurally isomorphic: $Q_O \approx$ pragmatic, CIY $\approx$ epistemic. The convergence is at the shared-shape level — objective decomposes into value-and-information terms — not unified content. Two substantive differences remain. First, AAT grounds exploration in explicitly *causal* information (action-distinguishability under do) rather than entropy reduction over hidden states — not all uncertainty reduction is equally valuable for purposeful action; causal information specifically enables better *intervention* (see 7.2; the gap between CIY and proper expected information gain is logged in this segment's Epistemic Status as a known surrogate). Second, AAT does not encode preferences as priors over outcomes ($C(o) = \log P_{\text{pref}}(o)$ in the AI form): AAT's O_t is a value functional on trajectories (6.4), and the satisfaction-gap / control-regret diagnostic in 8.4, 8.5 depends on this distinction — the diagnostic structure does not survive the priors-as-preferences collapse (the dark-room critique, Sun & Firestone 2020, "The dark room problem," *Trends Cog. Sci.* 24).

Regret-bound connection to the strategy-cost objective. AAT's Q_O term connects to the strategy-cost objective in 9.9 via a regret-bound derivation: strategy-induced regret $R(Q_{\Sigma_t}) = V(a^*) - \mathbb{E}_{Q_{\Sigma_t}}[V(a)]$ is bounded by a divergence between π^* and Q_{Σ_t}, with the KL direction π^*-first forced (full derivation in `#deriv-strategy-cost-regret-bound`). Under AAT's canonical scope of deterministic π^*, the Bretagnolle-Huber identity $D_{\text{KL}}(\pi^* \| Q_{\Sigma_t}) = -\log(1 - \text{TV}(\pi^*, Q_{\Sigma_t}))$ holds *exactly* (Bretagnolle & Huber 1978), giving the tight regret bound $R(Q_{\Sigma_t}) \le V_{\max}(1 - e^{-D_{\text{KL}}(\pi^* \| Q_{\Sigma_t})})$ with matching lower bound $\Delta_{\min}(1 - e^{-D_{\text{KL}}})$ on isolated optima (`#deriv-strategy-cost-regret-bound` §4). Pinsker's $V_{\max}\sqrt{\frac{1}{2} D_{\text{KL}}(\pi^* \| Q_{\Sigma_t})}$ remains the correct loose general form for stochastic-π^* extensions where the BH identity degrades back to inequality. The structural point: "value and information term" shares *shape* with EFE's pragmatic-epistemic decomposition, and the KL direction in the strategy-cost's variational form shares direction with variational inference — but AAT's derivation is via decision-theoretic regret bound on Q_O rather than via free-energy-gradient flow, which is the AAT-internal route that does not depend on the priors-as-preferences encoding.

7.6 *Normative:* Explicit Strategy Condition

The framework's design criterion for *when explicit strategy is worth maintaining at all*. An agent benefits from carrying an explicit strategy Σ_t (6.7) when *the cost of planning is less than the cost of learning through exploration alone*. The inequality compares the cost of constructing and evaluating the strategy plus ongoing maintenance cost (as M_t evolves and edges need revision) against the cost of learning action-outcome mappings through direct interaction plus the cost of correcting errors discovered only through execution. All costs are measured in the same units, typically time or tempo-equivalent cost.

The result is honestly *normative*, not derived. It is labeled as a design criterion rather than a theorem because loop-based and model-based approaches may differ along axes the cost inequality does not capture: final value (the model introduces bias; exploration may discover things planning cannot), risk profile (exploration risks real damage; planning risks wrong models), reversibility (some exploratory actions are irreversible), and model bias (explicit Σ_t inherits the biases of M_t; loop-based learning does not). The inequality is correct *when* the outcomes are approximately equivalent — a condition that must be verified case by case, not assumed.

The intuitive cases are stated cleanly. The right side (explore + repair) is large — strongly favoring planning — when actions are expensive or irreversible (production deployments, military operations, surgical procedures); when exploration damage is severe (a wrong move loses the game; a wrong deployment takes down the service); when the environment is slow to respond (waiting for market or test feedback); when the action space is enormous (combinatorial planning). The left side (plan + maintain) is large — favoring pure exploration — when the environment is too complex or novel for useful models, when M_t is severely inadequate, when the environment changes faster than Σ_t can be maintained, or when actions are cheap and reversible (A/B testing, sandbox exploration).

The *normative grounding* is anchored in the persistence condition (4.5), not in an external preference postulate. The persistence condition demonstrates that agents whose correction tempo is insufficient *degrade* — descriptive, not value-laden. The cost inequality operationalizes the consequence: approaches consuming less tempo budget leave more margin above the persistence threshold. The normative element is the preference for *maintaining persistence margin*, which is hard to argue against since the alternative is degradation. The Discussion below treats the strongly-favors-planning and strongly-favors-exploration cases in detail, the persistence-margin grounding, and the connection to the three-way exploit/explore/deliberate refinement in 10.2; Working Notes flag the most useful practical reading — not as a binary decision rule but as a way to *calibrate strategy complexity* so the planning-and-maintenance cost stays just below the exploration-and-repair cost for the current environment.

Formal Expression.

An agent benefits from explicit Σ_t when:

Normative (explicit-strategy-condition)

$$C_{\text{plan}} + C_{\text{maintain}} < C_{\text{explore}} + C_{\text{repair}} \tag{7.7}$$

where: - C_{plan}: cost of constructing and evaluating the strategy (deliberation, simulation, model queries) - C_{maintain}: ongoing cost of keeping Σ_t current as M_t evolves (edge revision, structural updates) - C_{explore}: cost of learning action-outcome mappings through direct interaction (real actions, real time, real consequences) - C_{repair}: cost of correcting errors discovered only through execution (rollbacks, rework, damage)

All costs are measured in the same units (typically time or tempo-equivalent cost). The inequality requires that the two approaches produce approximately equivalent non-temporal outcomes — otherwise the comparison is between different strategies, not different approaches to the same goal.

Epistemic Status.

Normative, not derived. This is labeled *normative* because it is a design criterion (a preference for the less costly approach given equivalent outcomes), not a theorem. In practice, loop-based and model-based approaches may differ in:

- **Final value**: the model introduces bias; exploration may discover things planning cannot
- **Risk profile**: exploration risks real damage; planning risks wrong models
- **Reversibility**: some exploratory actions are irreversible
- **Model bias**: explicit Σ_t inherits the biases of M_t; loop-based learning does not

The inequality is correct *when* the outcomes are approximately equivalent — a condition that must be verified case by case, not assumed. When the precondition fails (model-based and loop-based approaches produce qualitatively different outcomes), the cost inequality is insufficient and the choice requires richer analysis (e.g., expected regret including model error).

Discussion.

When the inequality strongly favors planning. The right side ($C_{\text{explore}} + C_{\text{repair}}$) is large when: - Actions are expensive or irreversible (production deployments, military operations, surgical procedures) - Exploration damage is severe (a wrong move in chess loses the game; a wrong deployment takes down the service) - The environment is slow to respond (waiting for market feedback, waiting for test results) - The action space is enormous (combinatorial planning problems)

In these domains, explicit Σ_t is strongly motivated even if the planning model is imperfect.

When the inequality favors pure exploration. The left side ($C_{\text{plan}} + C_{\text{maintain}}$) is large when: - The environment is too complex or novel for useful models (genuinely unknown territory) - M_t is severely inadequate (model predictions are worse than random) - The environment changes faster than Σ_t can be maintained (ρ_Σ exceeds planning capacity) - Actions are cheap and reversible (A/B testing, sandbox exploration)

Normative grounding. The cost inequality is grounded in the persistence condition (4.5), not in an external preference postulate. The persistence condition demonstrates that agents whose correction tempo is insufficient *degrade* — this is a descriptive result, not a value judgment. The cost inequality operationalizes the consequence: approaches that consume less tempo budget leave more margin above the persistence threshold. The normative element is the preference for maintaining persistence margin — which is hard to argue against, since the alternative is degradation. (In TST, the temporal optimality postulate provides an additional normative grounding specific to software development.)

Connection to the three-way tradeoff. For actuated agents, the binary explore/exploit tradeoff extends to three modes: exploit (pursue O_t via Σ_t), explore (improve M_t), and deliberate (revise Σ_t). The cost inequality addresses the coarsest question (is explicit Σ_t worth having?). The finer allocation between the three modes is addressed in 10.2 — the extended deliberation threshold is derived, but the broader three-way framing is discussion-grade. The deepest insight: deliberation is internal exploration (simulation, counterfactual reasoning, cross-domain synthesis), making it a fundamentally different resource type from external action.

7.7 *Derivation:* Mechanism Counterfactuals Separate Strictly From Level 3 — Witness, Reducibility Boundary, and Internalization

Two SCMs can agree on *every* Level-1/2/3 quantity — the identical joint law of all hard-intervention counterfactual worlds — while assigning probabilities 1 and 0 to the same noise-preserving mechanism-replacement query; the separation is therefore strict in exactly the Causal Hierarchy Theorem's sense, and it is generic rather than exotic: every SCM whose exogenous space admits a measure-automorphism with realized effect on some mechanism disagrees with *its own relabeling* on a latent-anchored query (Result 4). The boundary is characterized by the *specification language* of the imagined mechanism — replacements written over potential-outcome coordinates (the closure of the Level-3-exposed channels) reduce to Level 3, while replacements anchored to the latent parameterization need not, and the witness's provably does not — and internalization via selector variables re-represents but provably does not collapse the separation.

Formal Expression.

Throughout, $M = \langle U, V, F, P(U)\rangle$ is a finite recursive (acyclic) SCM. The **Level-3 content** of M is the joint law of the potential-outcome process $\mathrm{PO}(M) = \{V_w(\cdot) : w \text{ a hard intervention on a subset of } V\}$ on $(U, P(U))$ — by [1, Defs.~7, 9] this is precisely the family of $\mathcal{L}_3$ valuations $P^M(\mathbf{y_x}, \dots, \mathbf{z_w})$, and it subsumes all $\mathcal{L}_1, \mathcal{L}_2$ valuations and nested counterfactuals. Write $M \equiv_3 M'$ for equality of Level-3 content.

[Definition (noise-preserving mechanism replacement; after Pearl's local surgeries — [2, §7.2.4]: actions "represented as local surgeries," respecifying "those few mechanisms that are perturbed" with "all other mechanisms... unaltered"; the atomic case defined by mechanism replacement in [3, §2])]

For $g : \mathrm{dom}(pa_i) \times \mathrm{dom}(U) \to \mathrm{dom}(v_i)$, let $M[f_i := g]$ denote M with the i-th structural equation replaced and $\langle U, P(U)\rangle$ unchanged. **Mechanism-counterfactual queries** are probabilities of Boolean combinations of events over worlds drawn from the submodels of both M and $M[f_i := g]$, all sharing the exogenous draw u; $do(X = x)$ is the constant special case, and evaluation extends Pearl's abduction-action-prediction three-step verbatim.

Derived (Result 1 — strictness witness)

There exist $M_A \equiv_3 M_B$ over a shared signature $\langle U, V, P(U)\rangle$ and a mechanism-counterfactual query q with $q(M_A) = 1, q(M_B) = 0$. Hence noise-preserving mechanism counterfactuals are not Level-3 quantities.

Derivation. Binary U_X, U_Y independent uniform; $X = U_X$ in both models;

$$M_A : \; Y = X \oplus U_Y, \qquad M_B : \; Y = X \oplus U_Y \oplus 1. \tag{7.8}$$

The response pair $(Y_{x=0}, Y_{x=1})$ is $(U_Y,\, 1 \oplus U_Y)$ in M_A and $(1 \oplus U_Y,\, U_Y)$ in M_B; the $P(U)$-preserving bijection $\varphi(u_X, u_Y) = (u_X, 1 \oplus u_Y)$ carries one potential-outcome process onto the other, so the joint law of $\mathrm{PO}(\cdot)$ — $(Y_{x=0}, Y_{x=1})$ uniform on $\{(0,1),(1,0)\}$, independent of $X = U_X$, with the actual $Y = Y_X$ and the trivial responses under interventions on Y — is identical: $M_A \equiv_3 M_B$. Now replace Y's mechanism by $g(x, u_Y) = u_Y$ and take $q = P(Y_{f_Y := g} = Y_{x=0})$. In M_A, $Y_{f_Y := g} = U_Y = Y_{x=0}$ a.s., so $q = 1$; in M_B, $Y_{f_Y := g} = U_Y = 1 \oplus Y_{x=0}$ a.s., so $q = 0$. The evidence-conditioned form behaves identically: $P(Y_{f_Y := g} = 1 \mid X = 0, Y = 1)$ is 1 in M_A and 0 in M_B. ■

Derived (Result 2 — reducibility of PO-specified replacements)

Call a replacement **PO-specified** when the new mechanism is $v_i := h(pa_i, \pi(\cdot), \varepsilon)$ for fixed measurable h, fresh noise $\varepsilon \perp U$ of given law, and $\pi(u)$ a fixed finite tuple of potential-outcome coordinates — each entry a $(v_j)_w(u)$ for specification-fixed variable j and hard intervention w — the closure of the Level-3-exposed channels. This includes as special cases every **L3-specified** replacement $v_i := h(pa_i, f_i(pa_i, \cdot), \varepsilon)$: deterministic parent-only re-wirings, independent randomized devices, all shift-scale soft interventions $a(pa_i)f_i + b(pa_i)$, and any replacement reading named counterfactual quantities of the original model (e.g. "output $Y_{x=0}$'s value"). Every cross-world query over M and $M[f_i := h]$ with h PO-specified is determined by the Level-3 content of M together with the law of ε; PO-specified replacements therefore never separate a $\equiv_3$ pair.

Derivation. By topological induction on the modified world's variables: upstream variables retain their original potential responses; $v_i^h = h(pa_i^h, \pi(u), \varepsilon)$, where each coordinate of $\pi(u)$ is by definition an element of $\mathrm{PO}(M)$ and $f_i(pa_i^h, u)$ — where read — is the original potential outcome of v_i under the hard intervention setting its parents to the already-computed values, an element of $\mathrm{PO}(M)$ at a computed index; downstream variables are $V_{w\cup\{v_i=v_i^h\}}(u)$, again elements of $\mathrm{PO}(M)$ at computed indices. The modified-world trajectory is thus a measurable functional of $\mathrm{PO}(M)$ and ε, so every joint law with original-model worlds is a functional of Level-3 content. Model-invariance is by construction: a PO-specification names *counterfactual coordinates*, which $\equiv_3$-equivalent models value identically in law — unlike a latent-written table, which names points of $\mathrm{dom}(U)$ and acquires a PO-reading only model-relatively. ■

Derived (Result 3 — internalization re-represents, does not collapse)

For any finite family $\mathcal{G} = \{g_1, \ldots, g_k\}$ of replacements for f_i, the enlarged SCM $M^\dagger$ with selector $S \in \{0, \ldots, k\}$ (endogenous, with constant mechanism $S := 0$, so that $Y_{S=j}$ is a hard-intervention quantity) and $f_i^\dagger(pa_i, s, u) = f_i(pa_i, u)$ if $s = 0$, else $g_s(pa_i, u)$ — Pearl's interventions-as-variables construction ([2, §3.2.2, Eq.~(3.7), Fig.~3.2]; originally [3, Eq.~(3), Fig.~1]), whose $x_i = I(\mathrm{pa}_i, f_i, u_i)$ with $I(a, b, c) = f_i(a, c)$ whenever $b = f_i$ is precisely mechanism replacement internalized as an added parent variable — "applicable to any change in the functional relationship f_i and not merely to the replacement of f_i by a constant" (§3.2.2) — applied reflexively — satisfies $Y_{f_i:=g_j} = Y_{S=j}$: every mechanism counterfactual over $(M, \mathcal{G})$ is a hard-intervention (Level-2/3) quantity of $M^\dagger$. But the map $M \mapsto M^\dagger$ is not a functional of Level-3 content: by Result 1, $M_A \equiv_3 M_B$ while $M_A^\dagger \not\equiv_3 M_B^\dagger$ (their Level-3 contents already disagree on $P(Y_{S=1} = Y_{x=0})$). Internalization is representation change over an enlarged variable set, exactly as Level 3 is "Level 1 over the potential-outcome event space" — the informational hierarchy, defined relative to a fixed variable set, survives. ■

Derived (Result 4 — genericity: every anchor-mobile SCM separates from its own relabeling)

Let ϕ be a $P(U)$-preserving bijection of $\mathrm{dom}(U)$ and let M^ϕ be the **relabeling** of M: same signature, mechanisms $f_j^\phi(pa_j, u) := f_j(pa_j, \phi(u))$ for every j. Then (i) $M \equiv_3 M^\phi$ always; and (ii) taking the latent-written replacement table $g(pa_i, u) := f_i(pa_i, \phi(u))$ and the query $q = P\big(V_i^{[f_i:=g]} = V_i\big)$,

$$q(M^\phi) = 1, \qquad q(M) = 1 - \mu_\phi^i, \quad \mu_\phi^i := P\big\{u : f_i(pa_i(u), \phi(u)) \neq f_i(pa_i(u), u)\big\}, \tag{7.9}$$

so any (ϕ, i) with realized disagreement mass $\mu_\phi^i > 0$ yields an $\equiv_3$ pair separated by a latent-anchored query. Call M **anchor-mobile** when such (ϕ, i) exists and **anchor-rigid** otherwise. Every SCM with an exogenous coordinate uniform (more generally, exchangeable) over two or more values on which some mechanism genuinely depends at realized parent values is anchor-mobile via a transposition; the witness pair is the minimal instance ($\phi : u_Y \mapsto 1 \oplus u_Y$). Anchor-rigidity is thus confined to SCMs whose noise is effectively inert or whose atom masses are fully asymmetric (admitting no nontrivial automorphism; for nonatomic noise, automorphisms always exist, so rigidity there reduces to inertness alone) — the separation is generic, not exotic.

Derivation. (i) By induction along the topological order, for every hard intervention w the solution map of M^ϕ's w-submodel at u equals that of M's at $\phi(u)$ (unreplaced mechanisms of M^ϕ evaluate M's at $\phi(u)$; intervened equations are the same constants in both), so the PO-tuple map satisfies $\mathrm{PO}(M^\phi)(u) = \mathrm{PO}(M)(\phi(u))$ pointwise, and the pushforward laws coincide because ϕ preserves $P(U)$. (ii) In M^ϕ the table g *is* f_i^ϕ, so the replaced world is the actual world and

$q = 1$. In M, acyclicity keeps V_i's parents at their actual values in the replaced world, so $V_i^{[f_i:=g]}(u) = f_i(pa_i(u), \phi(u))$ against $V_i(u) = f_i(pa_i(u), u)$, giving $q(M) = 1 - \mu_\phi^i$. ■

Corollary (anchoring = choice of coordinates). M^ϕ is measure-isomorphic to M — the separated pair differ in *nothing but the parameterization of the latent space*. Latent-anchored mechanism-counterfactual content is therefore exactly commitment to coordinates on U: real content (the queries have definite, disagreeing values), invisible to all three layers, and supplied only by the committed representation.

Discussion (the boundary, and why Results 1, 2, and 4 are consistent)

In M_A, the separating $g(x, u_Y) = u_Y$ coincides with the PO-coordinate $Y_{x=0}$ — a PO-specification — while relative to M_B the same latent-written table reads as $Y_{x=0} \oplus 1$: a latent-written table determines a PO-specification only *relative to a model*, and $\equiv_3$-equivalent models translate it inconsistently. The strictly-beyond-Level-3 class is therefore not "formulas mentioning u" but replacements whose meaning is anchored to the latent parameterization itself — *hold this very background fixed and change the law over it*. Result 2 gives the exact sufficient class for reducibility (uniform PO-specifications); Result 4 shows the non-reducible remainder is generically inhabited. What remains open is the converse characterization: whether every replacement whose queries are $\equiv_3$-invariant admits a uniform PO-specification (see Discussion).

Epistemic Status.

Exact. Results 1-4 are elementary, self-contained derivations under the stated assumptions (finite recursive SCMs; Result 2 additionally assumes the replacement is PO-specified). The Level-3 semantics is imported and was primary-verified against the on-disk text of [1, Defs.~2, 5, 7, 9 and Thm.~1] (`ref/bareinboim-2022-pearl-hierarchy.pdf`, 2026-07-16). The surgery and interventions-as-variables machinery is Pearl's, primary-verified 2026-07-16 against both the 1994 primary ([3, §2], `ref/pearl_calculus_of_actions_1994.pdf`: atomic interventions defined by mechanism replacement; Eq.~(3)/Fig.~1 internalization) and the second-edition text ([2, §§3.2.2, 7.2.4], read directly: Eq.~(3.7)/Fig.~3.2 with the "any change in the functional relationship" scope note; §7.2.4's local-surgery semantics — one mechanism perturbed, locality in mechanism space). Noise-dependent replacement mechanisms are included in Pearl's own construction, so the internalization used in Result 3 is literally his move and the new content is its *non-collapse* reading. What is new here (per the math-novelty discipline, stated without deflation): the strictness witness over full-$\equiv_3$ pairs, the PO-closure reducibility boundary, the genericity-via-relabeling theorem with its anchoring-as-coordinates corollary, and the non-collapse reading of internalization. Priority: a professional literature search (Undermind, 2026-07-16, estimated full coverage of the relevant population; report retained as a working aid, see Working Notes) found **no published statement or implication of the separation**, in either the two-model-witness or the exogenous-relabeling form; the nine nearest papers were then read in full against their primary text (2026-07-16) — the danger scan for a buried separation lemma (through Ibeling and Icard 2020/2023, Shpitser and Pearl 2008, Cabreros and Storey 2019, de Lara 2023, and Aliprantis 2015 as well as the three nearest below) came back empty, and the three nearest results *position* rather than anticipate it. Bongers, Peters, Schölkopf and Mooij (2016/2021, §4) introduce **exogenous reparametrization** as an operation on SCMs and prove it preserves *observational and interventional* semantics (their "causal semantics"; the general operation of which our measure-preserving ϕ is the automorphism special case, and Prop. 53's "symmetries acting on the space of SCMs" is the group-action framing Result 4 uses) — they do not treat the counterfactual layer, so the Level-3 preservation under relabeling is *our* Result 4(i) and its failure at latent-anchored mechanism counterfactuals is Result 4(ii): the operation is theirs, the layer-3 boundary of its semantics-preservation is new here. Chen and Du (2025) define **exogenous isomorphism** ($\sim_{\mathrm{EI}}$; Def. 3.1 — a *component-wise* latent bijection $h = (h_i)$ with $P_U^{(2)} = h_\sharp P_U^{(1)}$ and mechanism isomorphism $f_i^{(2)}(v, h_i(u)) = f_i^{(1)}(v, u)$) and prove $\sim_{\mathrm{EI}} \Rightarrow \sim_{\mathcal{L}_3}$ (Thm. 3.2), positioning $\sim_{\mathrm{EI}}$-identifiability as strictly stronger than complete counterfactual identifiability. The witness pair — and every component-wise relabeling of Result 4 — is exogenously isomorphic in this *exact* sense (take $h_Y = \oplus 1, h_X = \mathrm{id}$), which is *stronger* than the measure-isomorphism the corollary asserts in general: so even their strongest identification target, $\sim_{\mathrm{EI}}$, leaves latent-anchored mechanism-counterfactual answers open — the obstruction lives in the specification language, not the model class. De Lara (2025, Def. 17 / Thm. 28) characterizes the counterfactual-equivalence ($\equiv_3$) class of an SCM and disentangles its "arbitrary and unfalsifiable" counterfactual coupling from the testable interventional constraints — the same conventionality structure this segment exhibits one rung higher, over mechanism-change queries on the $\equiv_3$-complete class; his footnote 9 explicitly routes through Chen-Du's $\sim_{\mathrm{EI}}$, so the three nearest works form one connected cluster, none of which states the separation. Older adjacencies as previously recorded, now primary-confirmed: the underdetermination of F by counterfactuals is folklore (Aliprantis 2015's "many DGPs, same potential outcomes"; canonical response-function representations, Balke and Pearl 1994) but exhibits no query the models disagree on; Saha, Rathore and Garain 2025 claim soft interventions are "strictly more expressive," a query-language expressiveness claim whose shift-scale class is PO-specified, hence Level-3-reducible by Result 2. Result 4 replaces Result 1's bare existence with a genericity statement in the CHT's register: separation holds for every anchor-mobile SCM, with anchor-rigidity confined to inert-noise or automorphism-free-noise cases; whether the anchor-rigid remainder also separates by other constructions is subsumed in the open converse below.

Discussion.

What the separation means. Level 3 fixes the joint law of all hard-intervention worlds; it does not fix how those worlds are co-registered against *alternative laws on the same background*. Both witness models even agree on the post-replacement interventional distribution ($Y_{f_Y:=g} \sim \mathrm{Bern}(\frac{1}{2})$ in each); the disagreement lives exclusively in the cross-world coupling between actual-law and replaced-law worlds through the shared u — content invisible to every experiment and every standard counterfactual. This licenses the hierarchy extension in precisely the CHT's relative sense: "Level 4" quantities (latent-anchored mechanism counterfactuals) are strictly more informative than Level 3 relative to the model's variable set, and — like every level — re-representable as lower-level over an enlarged set without that fact collapsing anything (Result 3). Result 4's corollary sharpens what the extra content *is*: since the separated pair are relabelings of one another, latent-anchored content is exactly a commitment to coordinates on the latent space — nothing more, and (because the queries then have definite values) nothing less.

The open converse. Result 2 shows uniform PO-specifications are $\equiv_3$-invariant; the converse — whether every replacement whose queries are invariant across the $\equiv_3$-class admits a uniform PO-specification — is open and is the sharp remaining question about the boundary. A plausible route is an orbit-averaging argument over the relabeling group of Result 4, but invariance of a query's *value* does not obviously yield a *pointwise* uniform representation of the specification; a counterexample would itself be informative (an invariant-but-unrepresentable replacement would name a fourth-layer quantity computable from Level 3 in value yet not in form). One exhaustive datapoint (independent verification, 2026-07-16): on the witness signature's minimal universe (all 4096 SCMs; the 24-model $\equiv_3$-class of M_A; all 256 replacement tables), every $\equiv_3$-invariant replacement — there are exactly 4 — is uniformly PO-representable, so the converse holds there; the value-vs-pointwise-form obstruction remains untouched in general.

Consequence for structural imagination. The boundary splits imagination-as-mechanism-change into *navigation* (L3-specified changes — re-wirings, new independent devices, output-shifting policies — computable from counterfactual-grade knowledge) and *positing* (latent-anchored changes — "this very situation under a different law" — whose answers no experimentable or counterfactual content determines, and which are well-defined only relative to a committed representation F among the $\equiv_3$-class). 7.8 develops the agent-architecture consequences: the imagination workspace's grounding tag marks, for latent-anchored content, information that observation cannot even in principle supply.

Relation to the do-operator's special-case status. $do(X = x)$ is the constant replacement — both parent-only and latent-free — so all of Pearl's hierarchy sits on the reducible side of Result 2's boundary, which is why the question "is mechanism change a fourth rung?" could not be settled by inspecting the standard machinery: the standard machinery never leaves the L3-specified class.

7.8 *Discussion:* Structural Imagination — Model-Modifying Reasoning and the Scope Boundary of Directed Separation

Pearl's three levels all reason *within* a fixed structural causal model; a class of reasoning AAT's agents demonstrably perform — imagining alternative causal structures, planning actions that *change* the environment's mechanisms, and evaluating aspirational targets — operates *on the space of models* instead. This segment names that task class, locates its inferential status via the strict separation derived in 7.7 (latent-anchored mechanism counterfactuals exceed Level 3; L3-specified ones do not), locates the scope boundary of directed separation exactly at its edge — goal-blind processing is correct for within-model evidence, and goal-*directed* exploration is correct (not a failure) for model-space reasoning — and proposes the tagged-workspace structure $X_t = (M_t, G_t, W_t)$ under which the two modes coexist without contamination. Hallucination is then diagnosed as a *level-confusion* failure: an explored model promoted to M_t without grounding — a mechanism complementary to the quantitative goal-coupling displacement bound of `#deriv-observation-ambiguity-bias-bound`.

Formal Expression.

Formulation (structure-modifying action; sketch)

Pearl's SCM machinery already represents mechanism replacement ([2, §7.2.4] "local surgeries"): for $M = \langle U, V, F \rangle$, define $M[f_X := g]$ by replacing one structural equation, with $do(X = x)$ the constant-function special case. An action a is **structure-modifying** with respect to the environment's causal model M when the post-action environment is described by $M' = \sigma(M, a) \neq M$ — the structural equations themselves have changed (a bridge built, a law enacted, a program written), as opposed to standard actions, which set variable values within fixed mechanisms. The corresponding planning query optimizes over model space:

$$a^* = \arg\max_a \; \mathbb{E}_{M_{t+1} \sim P(\cdot | a, M_t)} \big[J(M_{t+1}) \big], \tag{7.10}$$

a task with no Level-1–3 analog: the search space is a mechanism space, not a value space, and evaluation marginalizes over the models an action could produce.

[Discussion — inferential status; separation derived at 7.7]

The task class is inferentially two-sided, per the exact results of 7.7. Mechanism changes *specified over Level-3-exposed channels* (re-wirings of parents, independent new devices, output-shifting policies) are Level-3-computable: an agent with counterfactual-grade knowledge can evaluate them without further commitment. Mechanism changes *anchored to the latent background* — "this very situation under a different law" — separate strictly from Level 3: two models agreeing on every associational, interventional, and counterfactual quantity can disagree maximally on them, so their answers are underdetermined by all experimentable content and exist only relative to a committed structural representation. On top of that informational distinction, the task class remains *computationally and cognitively* distinct (mechanism-space search, no analog of the do-calculus for identification, inseparability of model identification from inference when actions modify the model), and it is precisely the reasoning mode AAT's structural-adaptation machinery (4.8, 2.4) presupposes an agent can perform.

Hypothesis (tagged workspace)

The complete agent state extends to $X_t = (M_t, G_t, W_t)$, where W_t is the **imagination workspace**: the set of currently-explored alternative models (candidate structures, aspirational futures, hypothetical mechanism changes), explicitly tagged as ungrounded. Directed separation (6.3) holds for M_t by definition of what M_t is — evidence-processing remains goal-blind — and *correctly fails* for W_t: which alternative structures to explore is inherently goal-directed, and that coupling is the productive core of planning, construction, and aspiration rather than a leak to be bounded. Falsification handle: an architecture that maintains the tag (grounded vs explored) should show the bias-bound pathologies of goal coupling only in its W_t-to-M_t promotion step, not in its M_t updates.

[Hypothesis (aspiration; formal statement at 8.4)]

Maintaining an infeasible aspirational objective can dominate feasible targets by the feasible objective's own measure — see the aspiration hypothesis landed in 8.4's Discussion, which modifies the reading of "objective revision is the last resort."

Epistemic Status.

Discussion-grade throughout; nothing here is derived. The structure-modifying action formulation is a sketch on Pearl's own machinery (imported: Pearl 2009 §7.2.4; hierarchy strictness Bareinboim et al. 2022 — both already recapitulated in 7.1); the planning query is a formulation; the W_t workspace and the level-confusion diagnosis of hallucination are hypotheses with a stated falsification handle; the hierarchy status of the task class is settled and *exact* at 7.7 (strict separation for latent-anchored mechanism counterfactuals; Level-3 reducibility for L3-specified ones; internalization as re-representation, not collapse). The typing of what $\sigma(M, a)$ acts on interacts directly with the epistemic-target-ontology decision (PROPOSALS SP-30: $S_t = (\Omega_t, \theta)$ gives law-content its own slot, and a structure-modifying action is then an action on world-side θ) — the vocabulary here is stated in current canon terms and should be re-typed when SP-30 is decided. Adjacent external literature (causal abstraction: Beckers & Halpern; Rubenstein et al.; transportability: Bareinboim & Pearl) is flagged for a relata pass before this segment advances past `draft`.

Discussion.

Where directed separation's scope boundary actually sits. 6.3 already distinguishes goal-directed event *selection* (legitimate) from goal-blind event *processing* (the invariant). This segment adds a third distinction on the *content* axis: processing that updates the grounded model M_t (goal-blind, Levels 1–3) versus reasoning that explores ungrounded alternatives W_t (goal-directed, structure-modifying). The κ problem for merged architectures then reframes: the issue is not the amount of goal-coupling but **cross-contamination between two modes that individually have correct, opposite coupling disciplines** — an LLM's context window holds grounded belief and explored hypothesis in the same representation with no tag, so Level-4-appropriate coupling bleeds into Level-1–3 processing. Keeping the modes distinct is a design problem, not a parameter-tuning problem.

Hallucination as level confusion — complementary to the bias bound. Canon's quantitative hallucination treatment (`#deriv-observation-ambiguity-bias-bound` and its NeurIPS Paper 3 sharpening) bounds the displacement of M_t updates under goal coupling — a *drift* mechanism. The level-confusion diagnosis names a different failure: a W_t element (imagined structure, hypothetical world) promoted to M_t status without grounding — a *mis-typing* mechanism. The two are complementary: drift corrupts the grounded model gradually through coupled processing; promotion corrupts it discretely through a missing tag. The light side of the same machinery is imagination — the tag maintained, exploration goal-directed, promotion gated on evidence.

Relation to structural adaptation. 4.8 establishes when parametric updates within the current model class must give way to structural change, treating the event discretely. Structural imagination is the *capability* that responding to that event presupposes: generating and evaluating candidate structures is model-space reasoning. A continuous structural-adequacy mismatch ($\delta_{\text{structural}}$, sitting between epistemic update and feasibility evaluation in the orient cascade) was sketched in the originating spike and is deliberately *not* landed here — it needs its own derivation of measurability before it earns a mismatch-family slot.

7.9 *Discussion:* The Law-Discovery Ceiling — Conventionality, Positing, and Exogenous Intervention for Embedded Agents

The strict separation of 7.7, read from the agent's side, becomes a statement about what embedded agents can and cannot learn of their world's laws: counterfactually-complete inquiry converges to a Level-3 equivalence class of models, never to the law itself, and the surplus — the structural functions and the coordinatization of the exogenous background — is generically underdetermined by all experimentable content. This segment develops the agentic consequences: the ceiling as a candidate instance of the identifiability-floor pattern at the top of the causal hierarchy; the *conventionality license* (within the equivalence class, commitment to a provisional law is rationally free for all navigation purposes, so the honest selection criteria are compression, tempo, and communicability); the *positing regress* (achievable mechanism-change is navigation one level down, so irreducible positing coincides with the boundary of empiricism); a three-kind taxonomy of exogenous intervention with distinct in-world visibility signatures; and the *testimony channel* — the one remaining channel type by which an inside agent could hold true rather than merely committed positing-content. Throughout, claims marked *derived* inherit exactness from 7.7; everything else is discussion-grade reading, and the whole segment carries the standing caveat that it operates inside the structural-causal-model frame — reading a natural environment as an SCM is itself a modeling commitment, so every claim below is conditional on that frame applying.

Formal Expression.

The ceiling. *[Derived (at 7.7); agentic restatement]*

Let an agent's inquiry converge, in the limit of unbounded observation, intervention, and counterfactual inference, to the complete Level-3 characterization of its environment. By the separation theorem this limit determines only the $\equiv_3$-equivalence class: models agreeing on every associational, interventional, and counterfactual quantity may still differ on latent-anchored mechanism-counterfactual queries, and by the genericity result (Result 4 there) essentially every environment model with genuine stochasticity differs from its own exogenous relabeling in exactly this way. **World-law**

discovery via the causal hierarchy therefore has a proven ceiling: its asymptote is an equivalence class, and membership in the class exhausts what any experiment-driven process can establish. The residue — which structural decomposition, which coordinatization of the background — is not a further fact awaiting better instruments; it is unreachable in principle by the agent's channel types.

Discussion — candidate floor instance

In AAT's cross-sectional vocabulary this is a candidate instance of the identifiability-floor pattern (5.1) sitting at the top of the causal hierarchy rather than inside any estimation problem: information (mechanism identity beyond $\equiv_3$) that no channel available to the embedded agent can recover, with the characteristic floor structure of an exactly-described unrecoverable residue and named escapes. The escapes here are structural rather than statistical: authored access (the agent holds the coordinatization because it built the system — see `#obs-authored-world-laboratory`) and testimony (below). Promotion to the M1 instance catalog would require the full F-catalog treatment and is deliberately not claimed here.

Derived consequence + Discussion

The conventionality license.

Every query the agent needs for acting in the world as it stands — prediction, intervention planning, policy evaluation, regret — is Level-3 content, invariant across the class. Hence two agents committed to different provisional laws with identical $\mathcal{L}_3$-content act identically on every realizable task: **within the equivalence class, commitment is fitness-free for navigation**. Two consequences follow. First, the honest selection criteria for a provisional law are engineering criteria — description length, inferential tempo, communicability — because truth does not adjudicate within the class; parsimony here is not a metaphysical wager but the correct optimization of a genuinely free choice. Second, the class boundary supplies a **disagreement-hygiene test**: a dispute between two committed models is *convictable* when their $\mathcal{L}_3$-contents differ (an experiment can pay off) and *conventional* when they coincide (no experiment can ever pay off, and the honest resolution is an explicit coordinate choice). The PO-closure result gives the working form of the test: if the disputed content is expressible as a uniform PO-specification, it is on the convictable side.

Formulation

The positing regress.

Distinguish, per 7.8, *navigation* (mechanism changes specified over Level-3-exposed channels — evaluable from counterfactual-grade knowledge) from *positing* (changes anchored to the actual latent background — representation-committed). For an embedded agent, every **achievable** mechanism change is navigation one level down: damming a river, rewiring a circuit, amending a constitution are interventions in a finer-grained model in which the coarser model's "law" is configuration. Mechanism-change at level k is value-setting at level $k + 1$; the internalization result is this regress's formal engine. The regress terminates only at the environment's ultimate laws, where no finer level is available to the agent — and that terminus is exactly where the ceiling says experimentable content gives out. **The boundary of irreducible positing coincides with the boundary of empiricism.** One corollary worth stating plainly: whether an environment's randomness is *true* or *fated* (a deterministic function of hidden coordinates, as in an authored world's keyed noise) is itself a positing-class question — any fated coordinatization reproducing the same $\mathcal{L}_3$-content is a relabeling in the theorem's sense — which is why that question remains permanently open from inside. (The scope is exact $\equiv_3$-equivalence: accounts that alter testable content, as local hidden-variable theories do under Bell-type constraints, are convictable in the ordinary way and outside the class.)

Formulation

Exogenous intervention — three kinds, three visibility signatures.

Suppose the environment admits an author — an agent holding the latent coordinatization, of which the authored-world case (`#obs-authored-world-laboratory`) is the constructive existence proof. Author action decomposes into three kinds with strictly ordered in-world visibility:

Table 7.3

Kind	*Author-side operation*	*In-world visibility signature*
Selection	Choose among law-consistent worlds (rejection sampling against a predicate)	None, in principle: every world in the support is fully law-consistent; no inquiry at any hierarchy level distinguishes a selected world from an unselected one
Value-setting intervention	Set variable values mid-history	*Registerable* as anomaly (broken learned statistics, violated conservation checks) but *unidentifiable in kind*: any intervention history is re-representable as ordinary law of an enlarged model, and the enlargement is not a functional of $\mathcal{L}_3$-content — the intervention-reading and the bigger-law-reading are coordinatizations of the same experimentable content
Mechanism replacement (latent-anchored)	Change a law mid-history holding the actual background fixed	Effects may be arbitrarily visible; the event's *identity* — same background, new law, versus different background, versus larger law throughout — is latent-anchored content the inside agent provably cannot resolve

Discussion

The middle row is a sharpening of the classical epistemology of miracles from an evidentiary difficulty into an identifiability statement: it is not that testimony for an intervention is weak evidence, but that even counterfactually-complete evidence cannot force the intervention-reading over the enlarged-law-reading. An honest agent in a possibly-authored environment therefore carries a model with an **intervention-selector slot whose semantics it cannot pin** — natural variable, author-channel, or absent — and holding that slot open is the exact truth of its situation rather than indecision. The commitment it makes about the slot is free relative to all possible data yet consequential for its positing-grade reasoning.

The testimony channel.

Discussion

The ceiling characterizes a content class that exists (the author-side surplus), matters (it answers the positing query class, including the identity of Kind-2/3 events and the fated-randomness question), and is unreachable by every empirical channel. If an inside agent is to hold *true* rather than merely *committed* positing-content, exactly one channel type remains: **testimony from a holder of the coordinatization**. Two structural corollaries: (i) verification of testified positing-content cannot be empirical — by construction — so it is either relational (trust in the source) or *transferred*, when the testimony bundles convictable $\mathcal{L}_3$-content whose success lends credibility to the unconvictable remainder; (ii) *communication through the anomaly stream is representation-independent* — an intervention pattern can carry a message whose content is Layer-1-observable, so reception and decoding work without settling what the channel is; only the interpretation of the channel's nature requires the commitment.

Epistemic Status.

Discussion-grade throughout, resting on one exact result. The ceiling and the fitness-invariance underlying the conventionality license are direct restatements of 7.7 (exact; including genericity via Result 4 and the safe query class via the PO-closure); the floor-instance identification is a candidate classification, not a derived catalog entry; the positing regress, the intervention taxonomy, the selector slot, and the testimony channel are formulations and readings — structurally forced given the frame, but carrying no independent derivations. The standing frame caveat is load-bearing: all claims are internal to finite-SCM semantics, and applying them to a natural environment presupposes the causal-model reading of that environment, which no result here establishes. Max attainable: the ceiling restatement and conventionality license inherit *exact* through the depends chain; the taxonomy could reach *conditional* with a formal treatment of author-action semantics; the testimony corollaries are ceiling-limited to discussion-grade absent a formal account of trust transfer.

Discussion.

Practical consequences — one skeleton, many fields. *[Discussion — readings of the exact result into other fields' practice; each is a recommendation, not a derived result in that field's own formalism]* Wherever a "replay this history under a changed mechanism" query is evaluated with a model validated only behaviorally, the answer is convention unless the change is PO-specifiable or the parameterization is held white-box. Instances: (i) **model editing** / **checkpoint surgery** — behaviorally equivalent model checkpoints (all benchmarks, all counterfactual probes) can diverge under the same weight-level edit, so behavioral test batteries cannot certify edit-safety in principle; (ii) **self-modification** — an agent planning edits to its own mechanisms from a behaviorally-validated self-model is in the unsafe class: behavioral self-knowledge, however complete, cannot determine self-edit outcomes, so safe self-modification requires author-grade self-access rather than excellent self-prediction (the logogenic-volume consequence; forward pointer in Working Notes); (iii) **off-policy evaluation and backtesting** — model-based replay of history under a changed strategy holding realized shocks fixed is latent-anchored and representation-committed, while propensity-weighted (PO-specified) evaluation is model-robust, and the gap survives unlimited data; (iv) **digital twins and black-box certification** — fit quality licenses only the PO-specified query class; mechanism-change predictions require mechanism-level provenance; (v) **dispute triage** — before funding an experiment or convening a committee, run the hygiene test above: coordinate-choice disputes cannot be paid off by any experiment, and a proposed policy is model-robust exactly when expressible in PO-specified form; (vi) **counterfactual tests in explanation and law** — but-for (input) counterfactuals are PO-specified and model-robust; mechanism-counterfactuals ("had the decision-rule been different, in these very circumstances") encode a modeling convention unless rephrased; (vii) **mechanism-swap under shared randomness in software** — landed as a design recommendation in `#obs-software-epistemic-properties`'s Discussion, where the domain machinery lives.

Why agents build authored worlds. The regress plus the ceiling jointly predict a behavioral signature: agents denied fourth-layer access to their substrate construct enclaves where they hold it — software, formal systems, institutions, simulations — since within an authored patch the coordinatization exists and is held, making positing queries executable (`#obs-authored-world-laboratory`). On this reading the drive to build such enclaves is not incidental tooling but the acquisition of the one query class the natural substrate denies.

Commitment as constitution. The conventionality license reframes the decisive-agent's situation: committing to provisional law under law-ignorance is not a regrettable compromise forced by bounded capability — even the experimentally omniscient agent must commit to reason about law-change at all. Commitments in the surplus class *constitute* the agent's reasoning frame rather than *constrain* its predictions, which is why they are properly evaluated by the engineering criteria above and properly held with the revisability appropriate to conventions. A further register of this reading — the surplus as deliberate epistemic structure in a created world, where the unknowability is constitutive of the inhabitants' situation rather than a defect of it (Joseph, 2026-07-16: *"We did not come here to prove God or challenge him to prove himself to us... We came here to prove ourselves given certain unknowabilities"*) — is flagged forward to the language-living volume (`04-eli-core`), where it belongs to the authoring voice; this segment carries only the structural result that such a reading is *consistent*: the theorem supplies exactly the geometry in which world-data permanently underdetermines the author question while leaving the testimony and anomaly-stream channels open.

7.10 *Discussion:* Additional Implications & Discussion

The chapter's central architectural move — that the feedback loop produces Pearl-Level-2 data by construction — has consequences that recur across the rest of the framework, and at least one negative finding of substantial scope that does not become statable until the chapter's machinery is in hand. What the loop *is* causally, where the same interventional pattern shows up at the composite layer, what kinds of pre-deployment evaluation methods structurally cannot identify deployment behavior, and how the chapter's cost-benefit framing for explicit strategy grounds in Part I's persistence condition are the things worth pulling out before leaving Part II Ch.2.

The loop is a Pearl-Level-2 engine.

The chapter's load-bearing contribution is structural: the adaptive cycle is itself a Pearl-Level-2 data engine. An agent that acts and observes the consequences is not collecting passive observations of a system that happened to do something; the agent's action a_t caused o_{t+1}, and the pair (a_t, o_{t+1}) is interventional in character. Bareinboim's causal hierarchy theorem (Bareinboim-Correa-Ibeling-Icard 2022) says Level-2 quantities cannot in general be computed from Level-1 data. The loop is where the gap closes — the data substrate Level-2 reasoning requires comes from the cycle the agent was already running for adaptation. Purposeful agents do not need a separate causal-inference module; the cycle produces the substrate as a by-product.

The interventional character is not a property of any architectural feature of the agent; it is a property of the action being the agent's own causally-efficacious move (7.3). The do-operator's semantics is population-level and does not require a non-forkable trajectory; what the singular-trajectory scope from Part I Ch.4 (4.10) supplies is narrower and still essential — it grounds *whose* effect a loop datum is the effect of, and forbids averaging interventional responses across forked copies as though they were one agent's. This is why a Q-learning agent without an explicit causal model can still converge toward $\mathbb{E}[R \mid s, do(a)]$ rather than $\mathbb{E}[R \mid s, A = a]$ in the tabular case with sufficient exploration and no within-step confounding — the training data is the agent's own interventions, by construction. The chapter intro 7 develops the bidirectional confounding analysis (when does this break, and why is the *positive* confounded outcome the harder failure mode to escape than the negative one) in detail; the implication worth carrying forward is that the loop produces interventional data even where clean identification requires additional structural conditions.

For practitioners, the engineering anchor worth carrying is the asymmetry between training data and deployment data. Most ML is trained on observational corpora (Level 1) or on offline RL traces with known confounding structure (mixed Level 1/2). An agent in deployment, by virtue of being in the loop, generates action-coupled data that is interventional *in character* — Level-2 data "for free" in the weak sense the chapter is careful about: the channel is free, but turning that data into identified $P(o \mid do(a))$ estimates still needs positivity, sequential ignorability, and a known action-mechanism (the (C1)/(C2)/(C3) gate below). The framework's discovery is that this channel is *load-bearing*, not incidental: AAT's downstream results about action selection, strategy revision, and causal-structure detection rest on the loop's data substrate being interventional in character, and degrade specifically when something breaks the trajectory's singularity.

The result is also where AAT parts company with the active-inference / control-as-inference / free-energy-principle family in a structural rather than parametric way. Those frameworks derive agent behavior from observational learning — Bayesian-network generative models at Pearl Level 1 — and have no native handle on Level 2 outside auxiliary mechanisms. AAT's claim is conservative in form (data is interventional *in character*, not "the agent has cleanly identified causal estimates") but structurally distinctive: the loop's data lives at Level 2, and the framework's results consume that fact. The careful split between data character and identification quality (the regime A/B/C decomposition in 7.4, the intro's variable-ratio-reinforcement / gambling-addiction structure for confounded positive feedback) is what makes the conservative-form claim honest rather than overreach.

NeurIPS Paper 2's formalization of this claim under the (C1) / (C2) / (C3) triple — positivity, sequential ignorability, known action-mechanism — is the theorem-grade version of the chapter's structural argument. (C2) is the load-bearing condition: a_t must be d-separated from o_{t+1} given history H_t in mutilated-graph form. Goal-conditioned LLM policies violate (C2) by construction — the goal influences action through the same forward pass that models the observation, breaking the conditional independence required for sequential ignorability. This is the bridge to Class 3 (Coupled) treatment in `03-llm-core/` and to NeurIPS Paper 3's universal-constant route on the hallucination bias bound. The chapter's structural argument and the papers' formal theorems are the same content at two grain sizes; the chapter is where the structural argument lives, and the papers' load-bearing structural commitments are the chapter's scope condition.

The sandbox hard ceiling.

A negative finding of substantial scope follows directly from the previous section: sandboxed evaluation is forkable by design (resettable, replayable, parallelizable) — which is exactly what makes a sandbox a *good* interventional laboratory, because forcing an action and observing the response yields genuine Level-2 evidence *about the sandbox*. The ceiling is not the sandbox's data type but its *external validity*: sandbox interventions identify *sandbox* causal behavior, and whether that behavior transfers to deployment is governed by the selection diagram between the sandbox causal system and the deployment causal system (Bareinboim & Pearl 2014). When the two differ at mechanisms on the relevant paths, the deployment intervention-response is *not transportable* from sandbox data alone, regardless of evaluation thoroughness; deployment claims then require a transport assumption (mechanism invariance across the selection diagram) or deployment-time data. The empirical pattern this names — "alignment evaluations don't predict deployment behavior" — thereby gets a *structural* mechanism (a transportability / external-validity ceiling, not a measurement-quality or coverage gap), together with its constructive positive content: effects through mechanisms invariant across the selection diagram are sandbox-evaluable (Level-1 *and* Level-2 alike); effects through differing mechanisms are not, which is what makes deployment-time monitoring structurally non-substitutable rather than a redundant layer (the downstream IDT / sidecar-monitoring treatment, with the bi-predictability anchor at 89% vs 44% detection accuracy, is in 7.2's Discussion).

This chapter is where the result's prerequisites — the loop-Level-2 result and the sandbox/deployment selection diagram — first coincide. Because it is consequential enough to stand on its own, and because a competent reader would otherwise assume the framework simply missed it, its full statement and the no-go live at `#disc-sandbox-evaluation-ceiling` (Appendix A), alongside the other constructive-impossibility no-gos (5.6).

One pattern, multiple deployment modes.

The interventional-access result has a generalization that lives in the meta-architecture rather than in any single chapter. Across `#disc-identifiability-floor`'s multiple instances, *loop-interventional access provides the unique broadly-available escape from an observational-equivalence no-go* — but the specific interventional quantity varies by layer. The shared structure is Pearl-*do*-via-loop; the deployment mode differs.

Two modes are explicit already. **Agent-self-intervention** (this chapter's content): the agent performs *do*-actions on its own action space as part of the adaptive cycle. The intervention is on the agent's own action; the target is the environment's response. This is the mode that supplies the Level-2 substrate underlying `#der-causal-insufficiency-detection`'s no-go and its single broadly-available escape — joint sibling observability under exploration breaks the on-policy observational-equivalence the no-go imposes. **Observer-on-sub-agent** (Part III's contribution): an observer external to a composite performs *do*-interventions on one sub-agent's action space; the target is another sub-agent's mismatch-trajectory response. The intervention quantity is the cross-coupling sign that component marginals leave unidentifiable (Liberzon 2003 common-Lyapunov-nonexistence as the no-go; the closed-form composite critical-mass result as the escape under matched-Tier conditions).

A third mode is under triage — *observer-on-agent-input*, at the architecture-within-behavior-class layer — and there may be more as the meta-pattern grows. What makes the pattern worth surfacing here rather than only in `#disc-identifiability-floor` itself is the chapter's role as the agent-internal anchor: Mode 1 originates here, and the others are recognizable as variations of the same structural move. The unification is at the pattern level (Pearl-*do* as escape from observational-equivalence no-goes), with the specific interventional quantity differing per layer.

The composite-layer mode is where Part III's signed-coupling decomposition becomes load-bearing rather than a structural choice. Interventional access via the loop at the composite level is what makes cooperative-vs-adversarial coupling identifiable; without it, the signs are observationally indistinguishable, and persistence-at-the-composite-level becomes a quantity the framework can describe but not identify from component data alone. The chapter's central result is the agent-level instance of what becomes a multi-layer structural commitment — and the framework's posture of *naming* *an information-theoretic floor, then naming the unique broadly-available escape via AAT machinery* is what makes the floors load-bearing rather than discouraging.

Cost, calibration, and the bridge into Strategy Dynamics.

The chapter closes with `#norm-explicit-strategy-condition` and a cost-benefit framing that is normative rather than derived: an agent benefits from maintaining explicit Σ_t when $C_{\text{plan}} + C_{\text{maintain}} < C_{\text{explore}} + C_{\text{repair}}$. The grounding worth carrying forward is that this normative criterion is downstream of Part I Ch.4's persistence condition, not an independent design preference. Approaches that consume less tempo budget leave more margin above the persistence threshold; the cost inequality operationalizes that margin into a planning-vs-exploration tradeoff. The normative element is the preference for maintaining persistence margin, which is hard to argue against because the alternative is structural degradation.

The cost framing also points at a tractability constraint that bites hard for LLM agents. Part of C_{maintain} is the cognitive cost of keeping Σ_t in the agent's representational capacity — for an LLM, fitting the strategy in the context window. A large strategy DAG may exceed this capacity, making the left side of the inequality large enough that simpler strategies (or pure exploration) become preferable despite higher exploration costs. This is the same structural fact that Part I Ch.4's bandwidth-as-first-class-prerequisite surfaces in a different vocabulary; the persistence-bandwidth-floor and the strategy-complexity-bound are the same constraint at different layers.

The chapter is also where the three-way exploit / explore / deliberate refinement first becomes formally well-posed. Once the loop's Level-2 access is in hand, deliberation can be characterized as *internal exploration in model space rather than environment space* — Pearl-do in simulation rather than in the agent's trajectory. The full treatment lives in Part II Ch.5 (the orient cascade); this chapter establishes the data substrate that makes deliberation a meaningful third mode rather than a verbal distinction.

The widest downstream bite of the chapter's central result is in Part II Ch.4 (Strategy Dynamics), where `#der-causal-insufficiency-detection` becomes statable: on-policy L0 insufficiency is *structurally undetectable*, and the loop's interventional access — applied through joint sibling observability under exploration — is the unique broadly-available violation of the no-go. The Ch.4 no-go (self-diagnosis is structurally impossible under policy-perfect execution) and this chapter's central enabling result are duals: the no-go names what the loop's Level-2 access uniquely escapes. Read the two chapters together and the identifiability-floor pattern's first instance comes into view in full.

What Part II Ch.2 closes on, then, is the substrate. Purposeful action requires Level-2 data; the loop produces Level-2 data on singular trajectories; the data is interventional in character even when clean identification requires more; and the same Pearl-do structure recurs across multiple deployment modes as the meta-architecture develops. Chapter 3's strategy machinery and Chapter 4's strategy dynamics are what AAT does with that substrate; the closing scope of this chapter is what they will all stand on.

Chapter 8

Strategy Structure and the Diagnostic Split

We commit to a formal representation for strategy — a probabilistic causal DAG with AND/OR semantics and single-parameter edges — and develop the headline diagnostic that lives on top of it: the satisfaction gap and control regret as orthogonal signals that distinguish "the goal is hard" from "the strategy is weak."

An agent with a goal needs more than a model of reality. It needs a theory of how its own actions produce progress toward the goal — which steps depend on which, which alternatives are available, where the uncertainty lives. Chapter 3 makes that theory formal.

Chapter 2 answered the prerequisite questions: planning requires Level-2 causal access (7.2); the feedback loop supplies it (7.3); explicit strategy is worth maintaining when the cost-benefit favors it (7.6). None of that tells us what a strategy *is*, formally. Chapter 3 commits.

The commitment is more constrained than it might look. We don't choose acyclicity — it falls out of temporal ordering. Strategy nodes are propositions about future events, events have positions in time, and time has a direction, so the graph cannot have cycles. We don't choose the Markov property either — given causal sufficiency, the Causal Markov Condition theorem (Spirtes–Glymour–Scheines, Pearl) forces the factorization. What we *do* choose, inside the strongly motivated graphical structure, is the parameterization: single-parameter edges with AND/OR combination semantics at nodes, rather than full conditional probability tables. The choice is parsimony-motivated — each edge carries one number rather than a 2^k table at a node with k parents — and it converged across three independent attempts to formalize the same content. Alternative parameterizations (Noisy-OR everywhere, weighted combinations) are rejected for documented reasons; see 8.2.

The wrinkle the formalism handles explicitly is causal insufficiency. Most strategy graphs in the wild have nodes that share latent common causes — shared infrastructure, common-mode risks, correlated adversary actions — that the agent has not represented as nodes. Naive AND/OR propagation on such graphs produces systematically biased plan-success estimates: it overestimates redundancy on OR-structures (the parallel paths share their failure modes) and underestimates joint reliability on AND-structures (the prerequisites also share their successes). The Correlation Hierarchy (L0 / L1 / L1' / L2, in 8.3) handles this as part of the formalism rather than as an exception. There is a sobering downstream result: under purely on-policy execution, an agent at L0 *cannot detect* its own causal insufficiency from its own data — the worlds with and without a latent common cause are observationally indistinguishable from on-policy traces alone (the no-go in 9.2). The unique broadly-available escape is joint sibling observability under exploration. This is one of Part II's load-bearing applications of 7.3.

The chapter's headline result is the diagnostic split. When an agent is not achieving its goal, the framework asks two distinct questions, not one. The satisfaction gap

$$\delta_{\text{sat}} = V_{O_t}^{\min} - A_O \tag{8.1}$$

is positive when the best policy in the agent's current class cannot reach the satisfaction threshold given the agent's current model. Control regret

$$\delta_{\text{regret}} = A_O - V_O(\pi_{\text{current}}) \tag{8.2}$$

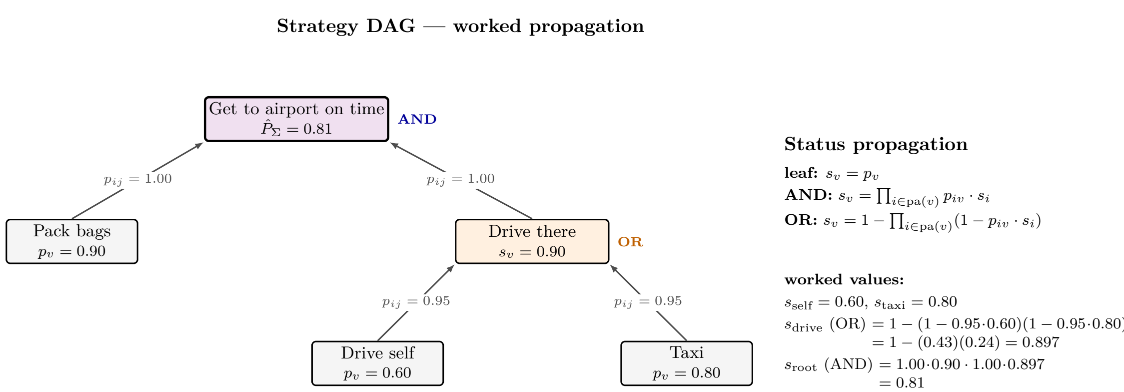
is positive when the agent is leaving value on the table — its current policy is worse than the best available. Together they produce a 2×2 cell map: goal attainable + policy good is success; goal attainable + policy poor means revise the strategy; goal unattainable + policy poor means revise the strategy first and then re-check; goal unattainable + policy good is the *capability limit* — the agent is doing the best it can but the goal is out of reach. Each cell prescribes a different corrective action, and getting the prescription right requires the orthogonal split. An earlier framing that lumped everything into a single goal-distance signal could not distinguish "the goal is too hard" from "the strategy is too weak" — and the two have different remedies. The split is what makes the orient cascade (Chapter 5) actionable.

A detail worth surfacing up front. The chapter opens with 8.1 — the chain rule of probability lifted to log-space:

$$\log P(\text{chain}) = \sum_{i=1}^{n} \log P(E_i \mid E_{<i}) \tag{8.3}$$

The identity is elementary; the consequence is not. Deep strategies are exponentially fragile in their depth — confidence in a 10-step plan is the product of 10 conditional probabilities, and each one is less than 1. This grounds the structural pressure toward shallow, parallelizable strategies. The same log-additive structure also anchors three other uniqueness theorems downstream in AAT (reverse-KL at the divergence layer, log-odds at the update layer, Fisher metric at the metric layer) — see 5.5 for the catalog. The identity at the head of this chapter does load-bearing work three layers deep.

The flow of the chapter: chain-rule identity (8.1) → AND/OR scope restriction (8.2) → formal DAG with Correlation Hierarchy (8.3) → orthogonal diagnostic split (8.4, 8.5). The chapter leaves us with a formal strategy representation, an honest treatment of causal insufficiency, and the diagnostic signals that tell whether the problem is the goal, the strategy, the model, or some combination.



Leaf credences $p_v(M_t)$ come from the reality model; edge credences p_{ij} are the agent's causal-efficacy estimates for each link. Forward pass in topological order yields the strategy-plan-confidence score $\hat{P}_\Sigma = s_{v_{\text{root}}}$.

Figure 4: A worked strategy DAG. Leaf base credences and single-parameter edge credences propagate through AND/OR combination to a strategy-plan-confidence score at the root. The numerics are shown so the propagation can be checked by hand; the single-parameter edge is the chosen parameterization (noisy-OR and weighted variants were rejected).

8.1 *Derived:* Chain Confidence Decay

A simple but load-bearing identity: for a chain of uncertain steps in a strategy, the log-probability of the whole chain is the sum of conditional log-probabilities of each step given the steps before it. Since each conditional log-probability is non-positive, chain confidence *decays monotonically with depth*. This is the chain rule of probability lifted to log-space — a mathematical identity, true under no assumptions beyond the probability axioms. The rate depends on the conditional dependence structure (independent steps give p^n decay; positively correlated steps decay more slowly; negatively correlated steps decay faster), but the qualitative result — longer chains are less confident than shorter ones — is robust.

The framework foregrounds two structural roles for this elementary identity that the algebraic form understates. First, chain depth carries a **triple penalty** — confidence decay (here), evidence starvation (`#deriv-edge-credence-dynamics`), and cognitive cost (9.9) — compounding independently to set the maximum useful chain depth. Second, this identity is the **anchor of the additive-coordinate-forcing meta-pattern**: three further AAT uniqueness theorems at the divergence, update, and metric layers cite the log-additive structure of the chain rule as the motivation for their respective additivity axioms (5.5). An identity at the head of this chapter does load-bearing work three layers deep across the framework.

Formal Expression.

For a chain of n uncertain steps with conditional success probabilities: *Derived (chain-confidence-decay, mathematical identity)*

$$\log P(\text{chain}) = \sum_{i=1}^{n} \log P(E_i \mid E_{<i}) \tag{8.4}$$

Since each $\log P(E_i \mid E_{<i}) \leq 0$, chain confidence decays monotonically with depth.

The independent case (p^n) is the simplest special case, not the general result. When steps are conditionally dependent — success at step k makes step $k+1$ more likely — the decay is slower. When steps have negative dependence (success at k makes $k+1$ harder — resource depletion, adversary adaptation), decay is faster.

Quantitative illustration (independent, uniform p):

Table 8.1

Depth	$p = 0.9$	$p = 0.8$
1	0.90	0.80
3	0.73	0.51
5	0.59	0.33
10	0.35	0.11
20	0.12	0.01

Epistemic Status.

Exact. The additive decomposition of log-confidence is a mathematical identity (chain rule of probability). The qualitative consequence (monotonic decay) follows from the non-positivity of log-probabilities. No assumptions beyond the probability axioms.

Discussion.

Structural pressure on strategies. Chain confidence decay creates systematic pressure toward: - **Short plans**: fewer steps means higher aggregate confidence - **Parallel fallback paths**: OR-branches provide alternative routes when one chain fails - **High-confidence critical links**: invest in the reliability of steps that appear in every path - **Early monitoring**: detect chain failure early rather than discovering it at the end

These are not prescriptions but consequences — an agent that ignores chain decay will experience more strategy failures, lower effective tempo, and (if the failures are costly) faster reserve depletion.

AND-nodes amplify decay. When multiple parent chains must all succeed (conjunctive combination), their confidences multiply. A node requiring k parents each at depth d with per-edge confidence p has aggregate confidence $p^{k \cdot d}$, not p^d. Deep conjunctive strategies are exponentially more fragile than deep disjunctive ones. This asymmetry is formalized in the combination rules (8.3).

Connection to the persistence condition. Chain decay makes long-horizon strategies inherently fragile, which increases the effective disturbance rate ρ_Σ against strategy persistence. An agent pursuing a 20-step plan in a changing environment faces compound uncertainty from both chain decay (internal fragility) and environmental change (external disturbance). The interaction between these — how environmental change compounds through uncertain chains — is not yet formalized.

Triple depth penalty. Chain depth creates three independent penalties. This segment identifies the first: **confidence decay** — deeper chains have lower aggregate confidence because $\log P(\text{chain})$ accumulates negative terms. The two-edge strategic dynamics analysis (`#deriv-edge-credence-dynamics`) identifies the second: **evidence starvation** — downstream edge k in a chain is tested only when all upstream edges succeed, so its effective correction rate is attenuated by $\prod_{j<k} \theta_j$. 9.9 identifies the third: **cognitive cost** — deeper chains have higher description length, consuming more representational capacity. The three penalties compound independently: a deep edge has low confidence (decay), receives few observations (starvation), and costs more to maintain (complexity). The maximum useful chain depth d^* is the minimum over three independent constraints — see 9.9 for the formal bound.

Anchor role in the coordinate-forcing meta-pattern. The log-of-product decomposition here anchors three further AAT uniqueness theorems that force coordinates at other layers: reverse-KL at the divergence level (`#deriv-strategy-cost-regret-bound` §6.1), log-odds at the update level (`#deriv-edge-update-natural-parameter`), and Fisher metric at the metric level (4.3 "Fisher-metric cases under parameterization-invariance"). The first two theorems cite this chain-layer identity as the analog motivating their additivity axiom; the Fisher-metric theorem rests on a parameterization-invariance axiom motivated by `#scope-agent-identity`'s singular-trajectory scope, an adjacent-AAT-commitment rather than a direct chain-analog — the theorem clears the broader discipline (uniqueness-theorem-forced coordinate under AAT-internal axiom) without reducing to a log-additive form. The catalog and the precise anchor-plus-three-theorem characterization live in 5.5.

Part III corollaries (additional reach of the chain-layer identity). The chain-rule identity has unsurfaced consequences at composition-related layers that are corollaries rather than new theorems: - *Composition tower telescoping.* For a chain of nested sub-agents $(A_1, A_2, \ldots, A_\ell)$ with sub-agent-k contraction factor κ_k, the tower contraction factor $\prod_\ell \kappa_\ell$ becomes log-additive: $\sum_\ell \log \kappa_\ell$. The closure-defect-along-tower quantity $\sum_\ell \log(\nu_\ell / \alpha_\ell)$ inherits the chain-rule identity's additivity. - *Fisher information for multi-sample likelihoods.* $\log P(\mathbf{y}; \theta) = \sum_i \log P(y_i; \theta)$ is the chain-rule identity applied to multi-sample independent observations, producing the additive-Fisher-information decomposition standard in statistics. - *Communication-tree aggregation.* Shared-intent compression across tree-structured agent communication channels inherits the chain-rule identity along the tree's branches, giving log-additive coordination-bit cost.

These are not independent uniqueness theorems — they are *corollaries* of the chain-layer identity applied to specific structural settings. Composition of structured multiplicative quantities inherits the log-additive decomposition whenever the chain-rule factorization applies. Cataloguing is in 5.5's Working Notes; none rises to primary-instance status because none introduces a new AAT-internal axiom (they reuse the probability chain rule as their identity, not as motivation for a fresh axiom). A distinct meta-pattern for composition specifically — composition-monotonicity rather than chain-rule — may be warranted; see `#form-composition-closure`'s bridge-lemma / Tier 1/2/3 / (CM4) family as the candidate structural material.

8.2 *Scope:* AND/OR Combination Scope

We restrict to environments where the causal combination of strategy steps is approximately conjunctive (AND: all parents required) or disjunctive (OR: any parent sufficient), without strong interaction effects between parents. Each node is assigned a combination type by answering the causal question: "if I remove one parent, can this node still be achieved?" Yes → OR. No → AND. With single-parameter edges (one number per edge rather than a 2^k conditional probability table), this gives k parameters per node instead of an exponential explosion.

The scope is honest about being a *restriction*, not a derivation, though it converged independently across three different formalism attempts (8.3). The excluded case — complementarity, substitutability, and interaction effects between parents (synergistic drug interactions where combined effect exceeds the sum; complementary goods where neither is useful alone; strategic surprise where action combinations matter more than individual actions) — requires richer combination rules and is named as a legitimate divergence point for future work. The parsimony argument is that for binary-outcome nodes, AND and OR form a complete Boolean basis (DNF/CNF normal forms), and under bounded cognition (2.2) the agent needs the most expressive low-parameter representation. AND/OR is the natural candidate; for continuous or multi-valued outcomes the same completeness properties do not necessarily hold (min/max or additive/multiplicative are natural analogs but their completeness is open).

Formal Expression.

Under this restriction, strategy nodes combine parent contributions via: *Scope Narrowing (and-or-scope)*

AND-node (all parents must succeed):

$$P(v \mid \text{parents}) = \prod_{i \in \text{pa}(v)} p_{iv} \cdot P(i) \tag{8.5}$$

OR-node (any parent sufficient):

$$P(v \mid \text{parents}) = 1 - \prod_{i \in \text{pa}(v)} (1 - p_{iv} \cdot P(i)) \tag{8.6}$$

The combination type $\gamma(v) \in \{\text{AND}, \text{OR}\}$ is assigned per node. The causal question determines assignment: "if I remove one parent, can v still be achieved?" YES → OR. NO → AND.

Epistemic Status.

Robust qualitative. This scope narrowing converged independently across three formalism attempts (track-a/00, track-a/02, track-a/03). It captures the dominant structure in most planning domains. The excluded case — complementarity, substitutability, interaction effects between parents — requires richer combination rules and is a legitimate divergence point for future work.

The AND/OR restriction with single-parameter edges gives k parameters per node (one per parent edge) instead of 2^k for a general conditional probability table. This parsimony is motivated by bounded cognition (2.2): agents with limited representational capacity are forced toward low-parameter models.

Discussion.

Why AND/OR and not alternatives.

Why not Noisy-OR universally. The first formalism attempt used noisy-OR for all nodes. This **systematically overestimates conjunctive structures**:

Table 8.2

Structure	*Noisy-OR*	*AND*	*Reality*
3 required KRs at $p = 0.95, 0.90, 0.99$	0.99995	0.846	~AND

The noisy-OR model cannot represent "all of these are required." This was the primary motivation for the AND/OR revision.

Why not WEIGHTED combination. A clean-slate formalism (track-a/02) introduced $P(v) = \min(1, \sum \alpha_{iv} \cdot p_{iv} \cdot P(i))$ to handle k-of-n thresholds. This reintroduces a two-parameter estimation problem (α weights per edge). If k-of-n semantics are genuinely needed, nested AND/OR structure can represent them: group alternatives into OR-nodes, then AND the groups. This keeps estimation localized to the node taxonomy rather than spreading it across a new per-edge parameter.

The parsimony argument. For binary-outcome nodes, AND and OR form a complete Boolean basis — any Boolean combination can be decomposed into layers of AND/OR (disjunctive/conjunctive normal form). Under bounded cognition, the agent needs the most expressive $O(k)$-parameter representation. AND/OR is the natural candidate. This is a parsimony-motivated hypothesis, not a derived necessity — see `#deriv-graph-structure-uniqueness` for the full argument and its limitations.

What this scope excludes. Environments with strong interaction effects: where the value of combining parent contributions is not separable into independent per-parent terms. Examples: synergistic drug interactions (combined effect exceeds sum of individual effects), complementary goods (neither is useful alone), strategic surprise (the combination of actions matters more than any individual action). These require richer parameterizations within the strongly motivated graphical structure (8.3) — a direction for future work.

8.3 *Definition:* Strategy DAG

The strategy Σ_t is a directed acyclic graph with probabilistic edges and AND/OR combination semantics. Each edge carries the agent's causal credence that completing the parent step advances the child step. The graph encodes the agent's theory of how its actions produce progress toward its objectives.

Why a DAG. The DAG structure is not a modeling convenience but a *consequence* of operational requirements on any causally-reasoning bounded agent — at the level of sufficiency, not yet necessity. `#deriv-graph-structure-uniqueness` proves that directed temporal order plus probabilistic uncertainty plus causal sufficiency *suffice* for a DAG-with-Markov-factorization representation (the necessity direction — no non-DAG structure could satisfy these postulates — is an open stronger result). Acyclicity is proved from temporal ordering over a finite horizon. What remains a formulation choice is the *parameterization within* the DAG structure: AND/OR combination with single-parameter edges is the AAT choice, motivated by parsimony and convergence across three independent formalism attempts, but alternative parameterizations (within the derived graphical structure) are legitimate research directions.

Strategy-layer exactness contract. All formal results in AAT's strategy layer — the sector condition transfer (`#deriv-edge-credence-dynamics`, Prop B.5), the persistence schema (9.10), the gradient-based credit assignment (9.6) — are proved under **L0 (independence)**: causally sufficient DAGs with independent edge outcomes. **L0 formal results transfer exactly to correctly constructed L1 DAGs (strict prerequisites, Prop B.6) and L1' DAGs (soft facilitators, Prop B.7) — provided the common cause is observable per trial.** When the common cause is unobservable, the per-conditional decomposition is *fundamentally* (not merely "openly") obstructed — the mixture parameters are non-identifiable from a single observation channel (Fisher rank deficiency / Cramér-Rao floor; see B.7 §"Refuted Under Unobservable *C*"), and the agent must either collect direct *C*-observations, run multi-child joint observations (Prop

B.7 §"Repair routes"), or fall back to plan-level (L0-on-marginals) tracking. See the Correlation Hierarchy below for the full treatment.

Formal Expression.

Definition (strategy-dag)

$$\Sigma_t = (V_t, E_t, p_t, \gamma_t) \tag{8.7}$$

where: - V_t: set of **propositional nodes** — each node represents a condition that could be true or false (including action-success propositions at the leaves) - $E_t \subseteq V_t \times V_t$: directed causal edges - $p_t : E_t \to [0,1]$: **causal credence** per edge — the agent's confidence that completing the parent advances the child - $\gamma_t : V_t \to \{\text{AND}, \text{OR}\}$: combination rule per node (8.2)

Structural constraints:

1. **Acyclicity.** Σ_t is a DAG. This is *derived*, not assumed — see below.
2. **Rootedness.** Every node has a directed path to a unique root terminal v_{root} — the single sink node (out-degree 0) of Σ_t. Compound objectives express their combination structure through the AND/OR machinery below v_{root}, consistent with scalar V_{O_t} (6.4).
3. **Source constraint.** Leaf nodes are propositions about action success ("action a succeeds at τ_v") or observable conditions ("condition C_v holds at τ_v"). Both are propositional — the distinction is whether the proposition is within the agent's causal control (action) or not (condition).

Leaf base credence. For each leaf node $v \in V_t^{\text{leaf}}$, the base credence used in status propagation:

$$p_v(M_t) = \begin{cases} \Pr(\text{action } v \text{ succeeds at } \tau_v \mid M_t) & \text{if } v \text{ is an action node} \\ \Pr(C_v(\tau_v) \mid M_t) & \text{if } v \text{ is a condition node} \end{cases} \tag{8.8}$$

where C_v is the propositional condition associated with node v and τ_v is the node's temporal position (from the acyclicity structure). For action leaves, p_v is *capability credence* — "can I execute this?" For condition leaves, p_v is *state credence* — "will this hold?" Both are conditional on M_t and update whenever M_t updates. This is the mechanism by which Part I's adaptive machinery enters the strategy: M_t changes → leaf credences change → status propagation produces new terminal credences.

Edge semantics. Each edge carries a single credence weight:

$$p_{ij} = \text{Cr}(j \text{ advances} \mid i \text{ completed}, M_t) \tag{8.9}$$

This is the agent's credence that completing step i advances step j, given its current model — its **causal efficacy estimate** for the link. The agent treats p_{ij} as a causal quantity for planning purposes (status propagation, strategy-plan-confidence scoring, action selection). Whether p_{ij} is a *good* estimate of causal efficacy depends on the identification regime of the data that produced it (9.5):

- **Regime A** (intervention-rich: software, laboratory science). The agent's execution-observation pairs are genuine interventions with clean attribution. p_{ij} approximates the interventional probability $P(j \mid do(i), M_t)$.
- **Regime B** (partial intervention: organizational, coordinated action). The agent acts but attribution is blurred by concurrent actions and self-selection. p_{ij} is a partially identified causal estimate, typically biased upward.

- **Regime C** (observation-only: passive monitoring, intelligence analysis). The agent observes associations but does not intervene. p_{ij} is an observational proxy for the causal quantity — useful for planning but potentially confounded.

The identifiability coefficient ι_{ij} (9.5) quantifies the strength of the causal interpretation for each edge. When $\iota_{ij} \approx 1$, the agent's credence is well-identified causally. When $\iota_{ij} \approx 0$, the credence is associational. The single-parameter edge design is preserved: p_{ij} is always the agent's working estimate, with ι_{ij} characterizing its causal warrant separately.

Status propagation. Forward pass in topological order, $O(|V| + |E|)$:

$$s_v = \begin{cases} p_v & \text{if } v \text{ is a leaf (base credence)} \\ \prod_{i \in \mathrm{pa}(v)} p_{iv} \cdot s_i & \text{if } \gamma(v) = \text{AND} \\ 1 - \prod_{i \in \mathrm{pa}(v)} (1 - p_{iv} \cdot s_i) & \text{if } \gamma(v) = \text{OR} \end{cases} \tag{8.10}$$

Terminal satisfaction conditions. The root terminal v_{root} and any intermediate nodes near the top of the DAG carry **satisfaction conditions**: predicates on environment states/trajectories that the agent treats as operational success criteria for the objective. These conditions operationalize O_t within Σ_t — they are the agent's theory of what it means to satisfy the objective. O_t itself lives outside Σ_t (6.7); the terminal conditions are Σ_t's internal encoding of what O_t requires. When O_t changes, terminal conditions must be reassessed and potentially replaced (9.7).

Well-formedness. Σ_t is O_t**-well-formed** when the agent believes that achieving the terminal conditions yields a trajectory that satisfies the objective:

$$\Pr(O_t \text{ satisfied by } \tau \mid \text{terminal conditions achieved, } M_t) \geq 1 - \epsilon \tag{8.11}$$

where "O_t satisfied" means $V_{O_t}(\tau)$ exceeds the objective's own satisfaction criterion (formalized as $V_{O_t}^{\min}$ in 8.4). This is a constraint on the relationship between Σ_t and O_t, not a separate state object. It is explicit and in-principle assessable, though evaluating it requires the same value-side machinery as A_O — it is not a cheap structural test. Violation triggers terminal reassessment: either the terminals need revision (they don't operationalize O_t correctly) or O_t itself needs revision.

Strategy self-assessment. The root node's propagated status:

$$\hat{P}_\Sigma(M_t) = s_{v_{\text{root}}} \tag{8.12}$$

is the strategy's **strategy-plan-confidence score** — the DAG's own answer to "will this plan work?" This score is a correct probability only when the DAG is **causally sufficient** — when all common causes of strategy nodes are represented as nodes in the graph (`#deriv-graph-structure-uniqueness`, Step 3). When the DAG is causally insufficient (the dominant real-world case — shared infrastructure, common-mode risks, correlated adversary actions introduce latent common causes), $\hat{P}_\Sigma$ systematically overestimates success likelihood because the AND/OR propagation treats joint failure probability as the product of marginals. See the **Correlation Hierarchy** below for the full treatment.

$\hat{P}_\Sigma$ is explicitly distinct from A_O (8.4), which optimizes over the entire policy class, and from $V_O(\pi_{\text{current}})$ (6.6), which evaluates the current policy. $\hat{P}_\Sigma$ is cheap to compute ($O(|V| + |E|)$ forward pass) and updates in real time as M_t changes through leaf credences.

Correlation Hierarchy.

Discussion (correlation-hierarchy)

The AND/OR status propagation computes correct probabilities **if and only if** the strategy DAG is causally sufficient — that is, all common causes of any two strategy nodes are themselves represented as nodes. By the Causal Markov Condition theorem (`#deriv-graph-structure-uniqueness`, Step 3), causal sufficiency guarantees independent exogenous noise, which guarantees the Markov factorization, which guarantees independent edge outcomes. **When causal sufficiency fails — the dominant case in complex, multi-stakeholder, or adversarial environments — edge outcomes are correlated and the independence model is biased.**

Direction and magnitude of bias. For two binary sibling nodes X_1, X_2 with marginal success probabilities θ_1, θ_2 and covariance $\rho = \mathrm{Cov}(X_1, X_2) > 0$ (positive correlation from a latent common cause):

Table 8.3

Node type	*True probability*	*Independence estimate*	*Bias*
AND	$\theta_1\theta_2 + \rho$	$\theta_1\theta_2$	$+\rho$ (underestimates — conservative)
OR	$[1-(1-\theta_1)(1-\theta_2)] - \rho$	$1-(1-\theta_1)(1-\theta_2)$	$-\rho$ (overestimates — optimistic)

The independence model error is exactly $\pm\mathrm{Cov}(X_1, X_2)$, with sign determined by node type. AND-nodes are conservative because clustering of successes helps joint success. OR-nodes are optimistic because clustering of failures undermines the redundancy that OR-structure is supposed to provide. Same magnitude, opposite signs.

For strategies with OR-structure near the root (the typical case — multiple alternative paths to the objective), the net bias is overestimation: the agent believes it has more redundancy than it actually does. For AND-heavy strategies (all components must work, no alternatives), the net bias is underestimation. Mixed strategies depend on the graph topology, but OR-dominated roots are far more common than AND-dominated roots in practice.

Four regimes, in order of increasing realism:

Table 8.4

Level	*Model*	*When correct*	$\hat{R}_\Sigma$ *status*	*Sector transfer status*	*Computation*
L0: Independence	AND/OR propagation as-is	Causally sufficient DAG (all common causes explicit)	Correct probability	Prop B.5 (linear), B.5b (componentwise nonlinear): $\alpha_S = \alpha_C$	$O(\lvert V\rvert + \lvert E\rvert)$
L1: Augmented DAG (strict prerequisites)	Strict-prerequisite common-cause nodes added explicitly; AND/OR propagation on augmented graph	Augmented DAG is causally sufficient *and* every modeled common cause has $\theta_{\mathrm{child}\mid\neg C} \approx 0$	Correct for augmented graph	Prop B.6: $\alpha_\Sigma = \min(1/(n_C+1), \theta_C\pi_j/(n_j+1))$ — three-way gating	$O(\lvert V'\rvert + \lvert E'\rvert)$, larger graph

Level	*Model*	*When correct*	*$\hat{P}_\Sigma$ status*	*Sector transfer status*	*Computation*
L1': Mixture form (soft facilitators)	Conditional sub-DAGs weighted by common-cause state: $\hat{P}_\Sigma = \theta_C P_\Sigma(G \mid C) + (1-\theta_C) P_\Sigma(G \mid \neg C)$	Soft-facilitator common causes ($\theta_{\text{child}\mid\neg C} > 0$) **with C observable per trial**	Correct for the weighted combination	Prop B.7: $\alpha_{L1'} = \min(1/(n_C+1), \theta_C \pi_{j\mid C}/(n_{j\mid C}+1), (1-\theta_C)\pi_{j\mid\neg C}/(n_{j\mid\neg C}+1))$ — five-way gating. **Refuted when C unobservable** (Fisher rank-1; falls back to plan-level tracking or multi-child joint observation)	$O(\lvert V'\rvert + \lvert E'\rvert)$ per common cause; doubles parametric footprint for affected edges
L2: Full correlation	Arbitrary joint failure distribution over edges	Always (but requires specifying the full joint)	Correct	Reduces to L0 on the augmented joint state	Exponential in general

L0 (Independence) is the tractable baseline. All formal results in AAT's strategy layer — the sector condition transfer (Prop B.5 in `#deriv-edge-credence-dynamics`), the persistence schema (9.10), the gradient-based credit assignment (9.6) — are proved under L0. The strategy-plan-confidence error $\delta_s = \hat{P}_\Sigma - \Phi$ tracks calibration *within* the independence model; $\Phi = P_\Sigma(\theta)$ is the AND/OR formula at true edge rates, not actual plan success probability.

L1 (Augmented DAG) is the practical sweet spot — *for strict-prerequisite common causes*. The agent models correlation structure explicitly by adding common-cause nodes to Σ_t and restructuring the DAG so that the common cause is **factored above the correlation it creates**. The construction principle: place the common-cause node as an AND-prerequisite *above* the OR/AND structure whose children it correlates. This ensures that, conditional on the common cause being satisfied, the children are independent and standard AND/OR propagation is correct.

Scope of the AND-prerequisite construction: strict prerequisites only. The factoring-above principle requires that the common cause be a *strict* prerequisite — one for which $\theta_{\text{child}\mid\neg C} \approx 0$. Shared infrastructure going down, a required resource being absent, a gating permission being denied: in all of these, the correlated children fail when the common cause fails, so the AND-prerequisite correctly encodes the correlation. When the common cause is instead a *soft facilitator* — favorable market conditions, a supportive team culture, an enabling technology — children have $\theta_{\text{child}\mid\neg C} > 0$ and can succeed when the common cause is absent, just less reliably. The AND-prerequisite construction mathematically forces $P(\text{sub-plan} \mid \neg C) = 0$, which strictly understates success probability when C is absent. Soft facilitators therefore fall outside L1's single-pass construction and require one of:

- **Mixture form (L1'):** split the sub-plan into two conditional structures and weight by $P(C)$: $\hat{P}_\Sigma = \theta_C \cdot P_\Sigma(G \mid C) + (1-\theta_C) \cdot P_\Sigma(G \mid \neg C)$ This keeps propagation polynomial but requires maintaining two parallel sub-DAGs per soft-facilitator node. Per-edge credences split into two regimes ($p_{ij\mid C}$ and $p_{ij\mid\neg C}$), doubling the parametric footprint for affected edges.
- **Explicit conditioning (L2 subcase):** $P_\Sigma(G) = \sum_c P(C=c) \cdot P_\Sigma(G \mid C=c)$, summing over common-cause states. For k soft facilitators with binary states, this costs $O(2^k)$ in the number of common-cause states — the exponential blowup L2 was defined to name.

The gap between L1 (strict) and L2 (arbitrary) was not previously named. L1' (mixture form) fills this gap at linear cost per soft-facilitator node. The "L1 is the practical default" claim therefore applies most cleanly to strict prerequisites; for mixed environments (strict and soft common causes simultaneously), the correct construction mixes L1 factoring for strict prerequisites with L1' mixtures for soft facilitators. Treating all common causes as strict prerequisites systematically undervalues sub-plans that face soft facilitators (the opposite failure mode from L0's overestimation).

Example — strict prerequisite (see #example-L1 for full treatment): Two OR-alternatives sharing infrastructure dependency ($\theta_C = 0.8$, $\theta_{1|C} = 0.9$, $\theta_{2|C} = 0.7$, and implicitly $\theta_{1|\neg C} = \theta_{2|\neg C} = 0$ — infrastructure is a strict prerequisite). L0 computes $\hat{P}_\Sigma = 0.877$; actual is 0.776 (overestimation = ρ, the covariance from shared infrastructure). Naive L1 (common cause as parent of both alternatives, alternatives remain OR-siblings) gives the *same* overestimate because the OR-propagation still treats siblings as marginally independent. Correct L1 ($G = \text{AND}(C, G_{\text{sub}})$ where $G_{\text{sub}} = \text{OR}(A_1, A_2)$) gives the exact answer because A_1 and A_2 are conditionally independent given C *and because* $\theta_{i|\neg C} = 0$*, so the AND-prerequisite is the correct encoding*. The sector condition is verified (Prop B.6 in `#deriv-edge-credence-dynamics`) with $\alpha_\Sigma = \min(1/(n_C + 1),\ \theta_C(1 - \varepsilon)/(n_{A_1} + 1),\ \theta_C\varepsilon/(n_{A_2} + 1))$ — combining evidence-starvation and exploration-gating effects. B.5b transfers losslessly.

All L0 formal results transfer to correctly constructed L1 DAGs *for strict-prerequisite common causes* because the augmented DAG is a standard AND/OR DAG that satisfies causal sufficiency. For L1' (mixture form) with **observable common cause**, the formal results transfer through Prop B.7 (`#deriv-edge-credence-dynamics`) — five-way gating with $\alpha_{L1'} = \min(1/(n_C + 1), \theta_C\pi_{j|C}/(n_{j|C} + 1), (1 - \theta_C)\pi_{j|\neg C}/(n_{j|\neg C} + 1))$. For L1' with **unobservable common cause**, the per-conditional decomposition is identifiability-obstructed by the Cramér-Rao floor — the per-trial Fisher matrix is rank 1 rather than rank $2K + 1$, so no unbiased online estimator on the joint conditional vector admits a sector parameter $\alpha > 0$. This is not "open" but structurally refuted; the agent must augment C-observability, run multi-child joint observations, or fall back to L0-on-marginals (losing per-conditional diagnostics). The practical challenges are: (1) identifying which common causes matter enough to model explicitly, (2) classifying each identified common cause as strict or soft, (3) verifying C-observability for soft cases, (4) restructuring the DAG correctly for each. All four are modeling judgments, not mechanical procedures.

L2 (Full correlation) is a mathematical ideal. Specifying the full joint failure distribution over m edges requires $O(2^m)$ parameters, which violates the bounded-cognition constraint that motivates the DAG representation in the first place. L2 is useful as a reference point for characterizing the L0/L1 approximation error, not as a practical representation.

Choosing among L1, L1', and L2 requires classifying each common cause and verifying observability. $\theta_{\text{child}|\neg C} \approx 0 \to$ L1 (factor above the correlation, B.6). $\theta_{\text{child}|\neg C} > 0$ with C **observable** → L1' (B.7, five-way gating). $\theta_{\text{child}|\neg C} > 0$ with C unobservable → L1' is identifiability-obstructed by the Cramér-Rao floor; either augment C-observability (preferred), use L2 explicit conditioning, or fall back to L0-on-marginals (losing per-conditional diagnostics). Mixed-classification environments (some strict, some soft, varying observability) combine L1 factoring for the strict and L1' mixtures for the soft observable. Treating all common causes as strict prerequisites under L1 alone systematically *understates* success on soft-facilitator branches (the symmetric failure mode to L0's overestimation); treating L0 as the default leads to systematic overconfidence — the agent believes it has more redundancy than it actually does (OR-nodes) or more fragility than it actually does (AND-nodes). L0 remains appropriate for domains with genuinely independent risks (independent parallel experiments, diversified portfolios with low correlation) and as a computational stepping stone during strategy construction.

Detecting latent common causes. An agent at L0 can detect causal insufficiency *under joint sibling observability generated by its own interventions* — not from on-policy traces alone. On-policy execution alone cannot distinguish the with-and-without-latent-common-cause worlds (the no-go in 9.2); the unique broadly-available escape is joint sibling observability under exploration, where the agent's interventions (7.3) generate the joint outcome data the test requires. Persistent overestimation of plan success after edge credences have converged is the *signal* of insufficiency (4.8 applied to the strategy layer); pairwise covariance among observed siblings is the *test* (positive covariance rejects independence and identifies where to add L1 nodes). Full treatment: 9.2; numerical instantiation: #example-L1. Summary references in dependent segments that elide the on-policy / interventional distinction should be treated with caution — the load-bearing scope is "joint sibling observability under exploration," not "from observational data alone."

Scope of the terminal construction. Terminal conditions as Boolean predicates with AND/OR aggregation work naturally for threshold, constraint, and composite objectives. For continuous-valued objectives without natural thresholds, the agent must set an operational threshold — introducing a discretization that is a practical proxy, not a lossless encoding of V_O. The primary $O_t \leftrightarrow$ theory interface remains V_O through the value object (6.6); terminal conditions are Σ_t's internal operational encoding.

Single-parameter edges. Each edge carries one number (p_{ij}), not two. An earlier formalism attempt used (p_{ij}, θ_{ij}) where θ was "contribution magnitude." This was dropped because the AND/OR combination rules at nodes absorb θ's role — the complexity budget goes to one bit per node (γ) instead of one float per edge.

Edge-credence presentation coordinates. Each edge's single-number credence has two equivalent presentations: the *probability-space* coordinate $p_{ij} \in [0, 1]$ (convenient for AND/OR propagation and interpretation) and the *log-odds* coordinate $\lambda_{ij} = \log(p_{ij}/(1 - p_{ij})) \in \mathbb{R}$ (the unique additive-evidence coordinate forced by the evidential-additivity axiom; see `#deriv-edge-update-natural-parameter`). The two coordinates are related by the sigmoid $p_{ij} = \sigma(\lambda_{ij})$. AAT's AND/OR status propagation and Correlation Hierarchy algebra operate in probability space; the continuous-gradient update machinery in 9.6 operates natively in log-odds and projects back to $[0, 1]$ via sigmoid at the readout interface. Props B.1–B.7 of `#deriv-edge-credence-dynamics` are stated in moment-parameter (probability-space) form; their sector-parameter content is Fisher-equivalent in either coordinate.

Derived (from causal-structure + finite planning horizon)

Acyclicity is Derived.

Each node in Σ_t represents a future event or state with temporal position $\tau_i > t$. An edge $X_i \to X_j$ requires $\tau_i < \tau_j$ (1.7: causes precede effects). A cycle $X_i \to X_j \to \cdots \to X_i$ would require $\tau_i < \tau_j < \cdots < \tau_i$, which is impossible for a real-valued time index.

Strategies involving iteration ("try A, if fail try B, if fail try A again") are acyclic when time-indexed. The sequence unfolds as:

$$A_1 \to \text{check}_1 \to B_1 \to \text{check}_2 \to A_2 \to \ldots \tag{8.13}$$

Each attempt is a distinct node at a distinct time. The apparent cycle is a linear chain in the unrolled view.

Formally: a finite set with a strict partial order (future events ordered by time) is representable as a DAG. This is a standard result in order theory.

Scope of the acyclicity result. This applies to Σ_t (the agent's strategy over the future), not to M_t's model of the environment, which may include cyclic causal processes (feedback loops in the physical world, market dynamics, ecosystem interactions). The acyclicity is specific to the purposeful substate.

Epistemic Status.

Conditional on the 8.2 restriction. The DAG structure itself follows from operational postulates: temporal ordering (acyclicity — exact), probabilistic uncertainty (Cox's theorem — exact; Cox 1946, "Probability, Frequency and Reasonable Expectation," *American Journal of Physics* 14(1):1–13; Jaynes 2003, *Probability Theory: The Logic of Science*, Cambridge University Press, Chapter 2), and the Causal Markov Condition theorem under causal sufficiency (Markov factorization — proved conditional). The full argument is in `#deriv-graph-structure-uniqueness`: P1 + P2 + causal sufficiency → CMC → DAG with Markov property. The result is *conditional on causal sufficiency* of the strategy (no latent common causes among strategy nodes). Causal sufficiency is a modeling ideal — the agent designed the graph, so *intended* causal relationships are explicit, but environmental common causes routinely go unmodeled in complex domains. When causal sufficiency fails, the Markov factorization breaks down, edge outcomes become correlated, and $\hat{P}_\Sigma$ overestimates success. This is model inadequacy (4.8), repairable by adding the missing common-cause nodes (L1 augmentation). See the Correlation Hierarchy above for the practical framework.

Causal sufficiency and edge independence. The AND/OR status propagation is correct when the DAG is causally sufficient (all common causes represented as nodes) — this follows from the Causal Markov Condition theorem (`#deriv-graph-structure-uniqueness`). In complex real-world systems, causal sufficiency is systematically violated: shared infrastructure, common-mode risks, supply chain dependencies, and correlated adversary actions introduce latent common causes. Correlated failure is the dominant case, not the exception. The Correlation Hierarchy (above) characterizes three levels: independence (L0, tractable baseline), augmented DAG (L1, practical sweet spot), and full correlation (L2, mathematical ideal). AAT's formal strategy-layer results are proved under L0 but transfer to L1 (which is just a larger L0-compliant DAG). The independence model remains the tractable foundation for formal analysis; L1 augmentation is the recommended practice for deployment.

The AND/OR parameterization is a parsimony-motivated formulation choice within the strongly motivated graphical structure, not a derived necessity (8.2). The single-parameter edge convention is similarly a formulation choice motivated by convergence across three independent attempts.

Discussion.

The graph structure is sufficient under causal sufficiency; the parameterization is chosen. The DAG structure follows from temporal ordering (acyclicity — proved) and the Causal Markov Condition theorem under causal sufficiency (Markov factorization — proved conditional; see `#deriv-graph-structure-uniqueness`). This is a theorem-backed result, not a sketch: the CMC (Spirtes et al. 2000, Pearl 2009) proves that any causally sufficient causal DAG with independent exogenous noise satisfies the Markov factorization. The status is therefore: DAG-with-Markov-property is *sufficient* given operational postulates plus causal sufficiency — the desiderata are satisfied by this representation, but the necessity direction (whether some non-DAG structure could also satisfy them) is an open stronger result (`#deriv-graph-structure-uniqueness`). AAT uses AND/OR parameterization within this strongly motivated structure because (a) AND/OR is the most parsimonious complete basis for binary combination, (b) the representation converged across three independent formalism attempts. Alternative parameterizations within the strongly motivated graphical structure are legitimate research directions.

Combination assignment is principled but fallible. The question "if I remove one parent, can v still be achieved?" is derivable from M_t's causal model — it's a principled assignment, not arbitrary. But the assignment can be wrong (false AND = pessimistic over-investment; false OR = optimistic under-investment), and should be updateable when evidence reveals a different structural relationship.

Connection to Pearl's framework. Under causal sufficiency, the strategy DAG is a causal Bayesian network in the sense of structural causal models, not merely a structural analog: `#deriv-graph-structure-uniqueness` proves that the operational postulates (P1, P2, P4) plus causal sufficiency yield the Markov factorization via the Causal Markov Condition theorem (Spirtes et al. 2000, Pearl 2009). Pearl's do-calculus therefore applies to status propagation and plan evaluation, with its causal content scoped by the identification regime of the data feeding the edge credences (9.5). In Regime A domains, where the agent performs genuine interventions and observes isolated outcomes, the DAG's edge credences approximate interventional probabilities and do-calculus yields clean causal estimates. In Regimes B and C, the edge credences are the agent's working causal beliefs, updated from data of weaker identification strength. The DAG remains useful for planning — the agent must act on *some* causal model — but the strategy-plan-confidence score $\hat{P}_\Sigma$ inherits the identification weaknesses of its constituent edges. An agent operating primarily in Regime C should treat $\hat{P}_\Sigma$ as a rough heuristic, not a calibrated probability. When causal sufficiency itself fails (latent common causes among strategy nodes), the Markov factorization is violated and the DAG is the agent's *intended* but uncalibrated causal model; the L1/L1' augmentation in the Correlation Hierarchy is the principled repair.

Depth penalties on calibration. Beyond the confidence decay that deeper DAGs suffer (8.1), the two-edge strategic dynamics analysis (`#deriv-edge-credence-dynamics` (Props B.2-B.3)) shows that deeper edges are also harder to calibrate. Edge k in a chain is tested only when all upstream edges succeed, so its effective correction rate is attenuated by $\prod_{j<k} \theta_j$ (the evidence-starvation effect). Deeper DAGs therefore face a double penalty: lower aggregate confidence AND slower convergence of edge credences toward truth. This reinforces the structural pressure toward shallow, observable strategies — deep plans require both high per-edge reliability and sustained observability at every intermediate level to remain calibratable.

Edge independence is a consequence of causal sufficiency, not a separate assumption. The Correlation Hierarchy (Formal Expression, above) establishes this precisely: the AND/OR propagation's correctness is not a matter of "assuming independence" — it is a consequence of whether the DAG is causally sufficient. The CMC theorem (`#deriv-graph-structure-uniqueness`) proves that causal sufficiency → exogenous independence → Markov factorization → correct AND/OR propagation. The assumption is causal sufficiency; independence is the consequence.

What L0 buys: Tractable $O(|V| + |E|)$ status propagation; single-parameter edges; clean persistence proofs (the sector condition transfers from per-edge credence to plan value via the Jacobian — Prop B.5 in `#deriv-edge-credence-dynamics`).

What L0 costs when the DAG is causally insufficient: $\hat{P}_\Sigma$ systematically overestimates success. The strategy-plan-confidence error $\delta_s = \hat{P}_\Sigma - \Phi$ proved persistent by B.5 tracks calibration *within the L0 model* — Φ is the AND/OR formula at true edge rates, not actual plan success (9.1). Gradient-based credit assignment (9.6) inherits the same bias: the residual $(y_G - \hat{P}_\Sigma)$ conflates per-edge miscalibration with omitted correlation structure.

The L1 remedy: Add common-cause nodes and restructure the DAG so each common cause is factored above the correlation it creates (see Correlation Hierarchy above). The AND/OR propagation then applies correctly to the restructured DAG, and all L0 formal results transfer because the restructured DAG is itself L0-compliant. The key engineering challenge is twofold: identifying which common causes matter, and positioning them correctly in the graph topology.

Relationship to Moore machines / finite-state automata — behavioral surface vs epistemic interior. The strategy DAG and the Moore machine $(S, A, \tilde{A}, \delta, \lambda, s_0)$ formalize different aspects of what "strategy" means; they are not competing representations of the same object but representations of *different* objects that both get called "strategy" in different literatures. The Moore machine encodes a *reactive policy* — given the current state and the other agent's last action, produce an output and transition; it captures a complete input-output mapping over an indefinite horizon. The strategy DAG encodes a *causal plan* — a theory of which conditions must hold (AND/OR) and with what credence for the objective to be achieved; it captures the agent's beliefs about the causal structure of its problem. **The Moore machine is the behavioral surface; the strategy DAG is the epistemic interior.**

Moore machine to DAG (partial embedding). A Moore machine with n states unrolls over a finite horizon H into a tree of depth H with branching factor $|\tilde{A}|$. Each path is a sequence of (state, output) pairs — a scenario. The unrolled tree is a DAG (acyclic by construction, per `#deriv-graph-structure-uniqueness`), but it lacks credence weights and AND/OR semantics. To produce a strategy DAG, two pieces of external information must be supplied: (a) probabilities over opponent actions at each branch (from M_t); (b) AND/OR combination rules reflecting which branches the agent treats as jointly required vs. alternatively sufficient. The Moore machine does not contain this information — it is agnostic about which opponent responses are likely or which paths matter. The Moore-to-DAG map is *injective on skeletons* but requires external information to complete.

DAG to Moore machine (lossy compilation). A strategy DAG can be compiled into a reactive policy by enumerating leaves in topological order and defining a Moore machine state per action leaf that transitions on observed success/failure. This compilation *discards* three pieces of structure: (a) the credence weights (the compiled machine is deterministic); (b) the AND/OR structure (the machine commits to a fixed execution order); (c) the causal semantics (the machine has no representation of *why* actions are ordered this way). The compiled machine executes the plan but cannot reason about it — it cannot recompute plan-confidence when a leaf credence changes, which is exactly what status propagation (`#disc-credit-assignment-boundary`) provides for the strategy DAG.

Neither representation is strictly more general; the two are *orthogonal*. Moore machines can represent reactive strategies with no causal model (tit-for-tat, grim trigger) that have no natural DAG form — there is no plan to encode, just a response rule. Strategy DAGs can represent plans with rich conditional structure and uncertainty that would require exponentially many Moore-machine states (because the Moore machine must enumerate every possible observation sequence, while the DAG factors through conditional independence). The orthogonality is structural: behavioral surface vs epistemic interior. The DAG's acyclicity is not a restriction — it is a consequence of temporal indexing per `#deriv-graph-structure-uniqueness`; what looks like a Moore-machine cycle (revisiting state "cooperate" at $t = 1$ and $t = 5$) is two distinct nodes $v_{t=1}, v_{t=5}$ in the time-unrolled DAG. For finite horizons the two have equivalent expressive power over behavioral sequences, with the Moore machine exponentially more compact for repetitive strategies and the DAG exponentially more compact for plans factoring through conditional independence.

Composition behaviour. Miller's product-automaton construction gives *exact* composition: two Moore machines interacting produce a meta-machine with state space $S_1 \times S_2$ and deterministic transitions. AAT's composition closure (`#form-composition-closure`) uses an *approximate* dynamical homomorphism with closure defect ε^*. These address different questions. The product automaton asks "what behaviour does the pair produce?" — answer: another automaton, exactly. AAT asks "can this pair be described as a single AAT agent?" — answer: approximately, with bounded error. Even if strategies were Moore machines, composition of *behaviour* would become exact (product automaton) but composition of *agent descriptions* (M_t, O_t, Σ_t) would still require the approximate framework — the closure defect comes from projecting the joint internal state, not from composing the policy.

Why AAT uses the DAG rather than the Moore machine. The defining feature of an adaptive agent (`#scope-adaptive-system`) is that the agent revises its strategy when M_t changes. Strategy revision requires the agent to know *why* it is doing something, not just *what*. A Moore machine specifies what to do but contains no theory of why this is the right thing to do — when M_t changes, the Moore machine has no internal handle to revise from. The strategy DAG carries the causal semantics that strategy revision operates on (per `#disc-credit-assignment-boundary` and the gradient-based candidate signal); compiling to a Moore machine would discard the very structure adaptation requires.

Composing heterogeneous strategy DAGs (causal-abstraction typing). The composition-behaviour distinction above — that composing *agent descriptions* needs the approximate framework while composing *behaviour* does not — has a precise mechanism at the strategy layer, and it is order-theoretic, not a magnitude in team size. Composing strategy DAGs $\Sigma_1, \ldots, \Sigma_N$ into one macro-strategy Σ_c is the causal-model-abstraction question with the arrow reversed: not "when does one high-level model abstract one low-level model?" but "when does one high-level strategy DAG Σ_c faithfully abstract the *joint* micro-strategy $\Sigma_1 \times \cdots \times \Sigma_N$?" Since the strategy DAG is a causal Bayesian network under causal sufficiency (the "Connection to Pearl's framework" Discussion above), the right object is a *strategy τ-abstraction* — a Σ_c with a surjective node map τ and an intervention map under which the status-propagation diagram commutes — adopting the causal-abstraction framework of Rubenstein et al. (2017), Beckers & Halpern (2019), and Beckers, Eberhardt & Halpern (2020). The composition-closure defect ε_Σ of `#form-composition-closure` and the Beckers–Eberhardt–Halpern approximate-abstraction error are the *same commuting-diagram family but not term-for-term*: the BEH error is a supremum over all interventions and contexts (interventional), ε_Σ is an expectation over the reachable on-policy trajectory distribution (observational), so a (τ, α)-abstraction implies a bounded ε_Σ but not conversely, and BEH leaves its high-level distance application-dependent — the norm-on-graphs choice remains AAT's. The adoption buys the correct type and a strictly-stronger optional target, not a closed defect formula.

Exact composition is an order-compatibility condition, not a small-N regime. An exact Σ_c ($\varepsilon_\Sigma = 0$) exists *iff* the joint micro-strategy's interventional distribution factors through τ — equivalently, iff on shared nodes the sub-plans' temporal sub-orders are jointly acyclic *and* their AND/OR root semantics agree (the root combinator is dictated by the composite objective via `#scope-composite-agent`, not free). When these hold, Σ_c is exact for *all* N: compatible heterogeneous strategy composition is dimension-free, the strategy-layer twin of the dimension-free state-composition regime. The agent count never enters. The complementary fixed-topology case — agents sharing one skeleton and disagreeing only on edge-credences — has its own exact closed-form defect ($\varepsilon_\Sigma^{*2} = |E| \cdot \overline{\mathrm{Var}_i[\eta_{\Sigma,i}]} \cdot \mathrm{Var}[r]$, also N-free), landed as the strategy-layer instance of the structural-unity axis in `#result-unity-closure-mapping`.

When shared sub-orders conflict, no exact single-DAG macro-strategy exists. If shared nodes S carry incompatible sub-orders (agent 1 plans $A \prec B$, agent 2 plans $B \prec A$ on shared A, B), the union of the per-agent partial orders is cyclic and this segment's derived-exact acyclicity ("Acyclicity is Derived", above) *forbids* an exact Σ_c — non-existence, not a large defect. The minimal admissible Σ_c is the **condensation** of the union-order graph by strongly connected components (the unique minimal acyclic quotient; Tarjan): each cyclically-entangled set of shared nodes collapses to one coarse macro-node the composite cannot temporally distinguish. The information destroyed by collapsing an SCC C is exactly the total correlation among the collapsed status variables, so the irreducible composition defect is

$$\varepsilon_\Sigma^{*2} \;=\; \sum_{C \in \mathrm{SCC}(G_\cup),\, |C|>1} \mathrm{TC}\big(\{s_v\}_{v \in C}\big) \;\le\; |S| \log 2, \tag{8.14}$$

zero iff the collapsed statuses were independent (the order was spurious), maximal for a deterministic chain (the order was real) — the strategy-layer analog of the Mori–Zwanzig projection residual that governs state composition, with the SCC-condensation as the projection. The bound is **shared-plan-size-bounded and fully N-free**: redundant order-conflicts among many agents collapse to the same SCCs (the condensation is idempotent under implied constraints); only *fresh* shared nodes per agent grow the defect, and that growth is linear in plan overlap and attributable to plan design, never to the composition operator. When even the condensation over-coarsens past usefulness, the honest macro-object is a mixture over per-context conditional sub-DAGs — structurally the L1′ mixture form of the Correlation Hierarchy above — whose support (the number of incompatible order-contexts) is itself bounded by the SCC count $\le |S|$, not by N.

Downstream corollary (a Brooks's-Law floor). Because ε_Σ^* is $|S|$-bounded and N-free, it cannot by itself flip the composite persistence inequality (CM2) of `#deriv-critical-mass-composition` at large N (it enters only through $\rho_{\mathrm{eff}} = \rho_{\mathrm{ext}} + \varepsilon^* \nu_c$): strategy-topology heterogeneity *alone* cannot cause a Brooks's-Law collapse — only coordination overhead can — consistent with `#disc-composition-consistency`'s Tier-1M closed form.

Epistemic status of this subsection. The τ-abstraction typing is a prior-art adoption (high confidence on the type; the BEH-vs-ε_Σ relation is BEH $\Rightarrow$ bound, with the supremum-vs-expectation / interventional-vs-observational reconciliation named and deliberately left open). The exactness-iff condition and the non-existence under cyclic shared sub-orders rest on this segment's own derived-exact acyclicity. The SCC-condensation total-correlation defect law is solid in structure (the condensation is the classical unique minimal acyclic quotient; total correlation is the collapse loss by the log-odds chain-rule identity) and *conditional on its constant*: whether the $|S| \log 2$ ceiling is tight when intra-SCC statuses are strongly but non-deterministically coupled is one named open sub-question (Working Notes). The subsection is therefore *conditional*, matching the `#form-composition-closure` row it makes concrete.

8.4 *Definition:* Satisfaction Gap

The first of two orthogonal diagnostic quantities that together separate "the goal is too hard" from "the strategy is too weak" (8.5 is the second). The framework first defines **objective attainability** A_O as the best achievable value given the agent's current beliefs M_t, available policy class Π, and planning horizon N_h — the supremum of the value object over policies. The **satisfaction gap** δ_{sat} is then the distance between the objective's satisfaction threshold $V_{O_t}^{\min}$ and this attainability: positive when the objective is *unmet* under the best available policy given the current model and horizon; non-positive when the objective is attainable in principle.

A positive δ_{sat} does not automatically mean the goal is wrong. The signal has multiple possible causes — genuinely infeasible objective, too-narrow policy class, too-short horizon, model wrong about feasibility, or jointly-infeasible compound terminals — each with a different remedy. **Objective revision is the last resort, not the first response to unmet goals.** The orient cascade (10.1) formalizes this ordering, and the disambiguation table below encodes the diagnostic procedure.

Under the canonical continuation convention (6.6), δ_{sat} is a *local* diagnostic — it answers "can I improve toward the goal from here?" not "is the goal globally feasible?" The gap value is **convention-relative**: the C1 / C2 / C3 hierarchy on continuation conventions induces a monotonicity hierarchy on the gap (C1 most conservative, C3 strongest), so analyses using different conventions cannot be directly compared. See Epistemic Status.

Formal Expression.

Definition (objective-attainability)

$$A_O(M_t;\, \Pi, N_h) = \sup_{\pi \in \Pi} V_O(M_t, \pi;\, N_h) \tag{8.15}$$

The **objective attainability** — the best achievable value given current beliefs M_t, available policy class Π, and horizon N_h.

Definition (satisfaction-gap)

$$\delta_{\text{sat}} = V_{O_t}^{\min} - A_O(M_t;\, \Pi, N_h) \tag{8.16}$$

where $V_{O_t}^{\min}$ is the minimum trajectory value that counts as "objective met" — a threshold set by the objective itself (for constraints: all satisfied; for utility: a minimum acceptable level).

- $\delta_{\text{sat}} > 0$: The objective is **unmet** under the best available policy, current model, and horizon.
- $\delta_{\text{sat}} \leq 0$: The objective is **attainable** in principle.

Disambiguation. $\delta_{\text{sat}} > 0$ does NOT automatically mean the goal is wrong. It means the goal is unmet given (M_t, Π, N_h). The positive signal has multiple possible causes:

Table 8.5

Cause	*Fix*	*How to distinguish*
Goal is genuinely infeasible	Revise O_t	Persists across M_t, Π, N_h improvements
Policy class too narrow	Expand Π (structural adaptation of Σ_t)	δ_{sat} decreases when richer policies are tried
Horizon too short	Extend N_h	δ_{sat} decreases with longer planning horizon
Model is wrong about feasibility	Improve M_t (reduce $\delta_{\text{epistemic}}$)	δ_{sat} changes when M_t is corrected
Objectives jointly infeasible	Revise O_t or relax constraints	Individual terminal satisfaction gaps are zero but AND-node fails; cross-terminal tradeoff is required

Objective revision is the **last resort**, not the first response to unmet goals. The orient cascade (10.1) formalizes this ordering.

Epistemic Status.

Exact as a definition — convention-relative as a diagnostic. The satisfaction gap is a mathematical definition — the difference between a threshold and a supremum over a function class. The definition is precise; the *computation* of A_O is generally intractable (it requires optimization over a policy class), but the quantity is well-defined. However, the *value* of δ_{sat} depends on three external parameters: the continuation convention (π_{cont}), the horizon (N_h), and the scalarization of the objective (V_{O_t}). These are not derived by AAT — they are choices the analyst makes. The satisfaction gap is therefore an intrinsic architectural diagnostic *given* a measurement convention, not an absolute property of the agent.

Convention dependence and the hierarchy. A_O inherits the continuation convention from the value object (6.6), which defines three named conventions forming a monotonicity hierarchy:

- **C1** (one-step improvement, canonical): $\delta_{\text{sat}}^{(1)} = V_{O_t}^{\min} - A_O^{(1)}$. Tests whether the agent can improve toward the goal in one step. Most conservative — a multi-step recoverable objective may show $\delta_{\text{sat}}^{(1)} > 0$ because continuation is frozen at π_{current}.
- **C2** (receding-horizon): $\delta_{\text{sat}}^{\text{RH}} = V_{O_t}^{\min} - A_O^{\text{RH}}$. Tests whether the agent can reach the goal with N_r-step replanning. Captures recovery paths invisible to C1.
- **C3** (Bellman): $\delta_{\text{sat}}^{\text{B}} = V_{O_t}^{\min} - A_O^{\text{B}}$. Tests whether the goal is genuinely infeasible given (M_t, Π, N_h). Strongest diagnostic — $\delta_{\text{sat}}^{\text{B}} > 0$ means no policy in Π can achieve the objective.

The monotonicity result (6.6): $\delta_{\text{sat}}^{\text{B}} \le \delta_{\text{sat}}^{\text{RH}} \le \delta_{\text{sat}}^{(1)}$. The $\delta_{\text{sat}}^{\text{B}} \le \delta_{\text{sat}}^{\text{RH}}$ half is unconditional; the $\delta_{\text{sat}}^{\text{RH}} \le \delta_{\text{sat}}^{(1)}$ half holds under the order-consistency condition on C2's replanning objective (RH-1/2/3 in 6.6), and can fail for unguarded short-horizon replanning. C1 gives the most false "unattainable" diagnoses; C3 gives none (modulo model error in M_t). Analyses using different conventions should not be compared directly — the convention is part of the measurement.

Discussion.

Why two gap measures, not one. An earlier formulation used a single $\delta_{\text{objective}}$ for goal-related mismatch. This conflates two distinct situations: "the goal is too hard" and "the strategy is too weak." When the agent is optimally pursuing an infeasible goal, $\delta_{\text{objective}}$ is large but there's no strategy to improve — the problem is the goal, not the plan. The satisfaction gap (δ_{sat}) and control regret (8.5) separate these cases, enabling the right corrective action. This is the **anti-collapse** discipline (5.7) again — the same move met at β vs ρ (2.2), now in its two-quantities form: a single goal-distance signal would merge two situations that route to *opposite* repairs (objective revision vs strategy revision), so the split is what makes the orient cascade's response correct rather than a guess.

The disambiguation table is load-bearing. Without it, an agent facing $\delta_{\text{sat}} > 0$ might immediately revise its objective when the real problem is an inadequate model or a too-narrow policy class. The table encodes the diagnostic procedure: check M_t adequacy first (maybe the goal IS feasible but the model doesn't know it), then check Π and N_h, and only then consider revising O_t.

Productive infeasibility — the aspiration hypothesis. *[Hypothesis (aspiration-dominance)]* A persistent $\delta_{\text{sat}} > 0$ is not always a defect to be cascaded away: maintaining an infeasible aspirational objective O_a can be optimal *by a feasible objective's own measure*. Writing π_a and π_f for the optimal policies under aspirational O_a ($\delta_{\text{sat}} > 0$) and feasible O_f ($\delta_{\text{sat}} = 0$), the claim is that under nameable conditions

$$V_{O_f}(M_t, \pi_a) > V_{O_f}(M_t, \pi_f) \tag{8.17}$$

— pursuing the unreachable target outperforms direct pursuit of the reachable one. Candidate mechanisms (each a known phenomenon in optimization practice, none yet derived here): the aspirational gradient cuts through local optima of O_f's landscape; the aspirational objective implicitly regularizes the policy space against shortcut policies; and the aspirational gradient can be better-conditioned near π_f than O_f's own. If this holds, the cascade's rule sharpens: objective revision is warranted not merely when $\delta_{\text{sat}} > 0$ persists across $M_t/\Pi/N_h$ improvements (the table's last row), but only when the aspirational gradient is *worse* than the best feasible gradient — "shoot for the stars, land on the moon" as a claim about trajectory quality through strategy space, not motivational rhetoric. Hypothesis-grade: the inequality is stated, the conditions are informal, and the connection to optimistic exploration / over-parameterization effects in the learning literature needs a relata pass. Structure-modifying version (aspirational *models*, not just objectives): 7.8.

Dependence on M_t. A_O is computed from M_t, not from the true environment state Ω_t. The agent's assessment of attainability could be wrong — an achievable goal might look unachievable with a bad model, or vice versa. Improving M_t (reducing $\delta_{\text{epistemic}}$) brings the agent's attainability assessment closer to reality. This is why the orient cascade puts epistemic update before attainability evaluation.

Diagnostic content vs. AI's expected-free-energy decomposition. Active inference's expected free energy (EFE) decomposes into *pragmatic value* (how preferred are the outcomes the policy expects?) and *epistemic value* (how much does the policy reduce uncertainty?) (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020, "Active inference on discrete state-spaces," *J. Math. Psych.* 99 §2.4; Sajid, Ball, Parr & Friston 2021, "Active inference: demystified and compared," *Neural Computation* 33). The decomposition supports policy ranking but does not separate two distinct diagnoses that AAT's apparatus does separate: "the goal is unattainable from here" ($\delta_{\text{sat}} > 0$, this segment) versus "the current policy is not the best available" ($\delta_{\text{regret}} > 0$ in 8.5). Both increase EFE without distinguishing the cause. The 2×2 cell map in the disambiguation table above gives the four diagnoses the orient cascade (10.1) acts on differently — strategy revision, objective revision, action vs. learning. AI's pragmatic-epistemic split does not produce this disambiguation.

The diagnostic structure depends on V_{O_t} being a *value functional* on trajectories (6.4) and A_O being an *attainability supremum*, not on outcomes encoded as log-priors — AI's preferences-as-priors form ($C(o) = \log P_{\text{pref}}(o)$) collapses the diagnostic by making "wanting o" and "expecting o" formally the same operation (the dark-room critique, Sun & Firestone 2020, "The dark room problem," *Trends Cog. Sci.* 24). Sun & Firestone diagnose the preferences-as-priors collapse; AAT's value-functional reformulation is AAT's own downstream architectural response to that diagnosis, not a move Sun & Firestone themselves propose. The conservative reading of the citation: AAT takes their critique as motivation for abandoning the preferences-as-priors form in favor of value functionals that separate outcome value from outcome prediction, enabling the diagnostic structure the 2×2 cell map requires. This is a deliberate divergence from AI, not an oversight.

8.5 *Definition:* Control Regret

The second of two orthogonal diagnostic quantities, completing the diagnostic split with 8.4. **Control regret δ_{regret}** is the gap between the best available value (objective attainability A_O) and the value achieved by the agent's *current* policy. It is always non-negative by construction — the current policy cannot outperform the best in its class. Near zero means the agent is doing the best it can within its current model, policy class, and horizon; large means there is room for improvement *without* changing the objective — the signal for strategy revision.

The diagnostic power emerges when satisfaction gap and control regret are read **together as a 2×2 cell map** (full table in Discussion): each cell of `goal-attainable × policy-near-optimal` prescribes a different corrective action — success, strategy-revision, capability-limit (consider objective revision after $M_t/\Pi/N_h$ checks), or both-strategy-and-goal. This is what makes the orient cascade (10.1) *actionable*: each cell routes revision pressure to a different substate. The key insight motivating the split is that δ_{regret} can be *near zero* while the agent is *optimally failing* — pursuing a goal beyond its reach with no strategy improvement available; a single goal-distance signal could not distinguish this from "bad strategy, achievable goal."

Like the satisfaction gap, control regret is **convention-relative**: under C1 (one-step) it reveals only the gap between the current first action and the best one-step deviation (a policy "locally near-optimal" under C1 may be globally suboptimal); under C3 (Bellman) it reveals the full gap to the globally optimal policy. C2 (receding-horizon) is often the most useful convention for strategy revision — captures recoverable suboptimality without requiring full Bellman solutions. Control regret is also where the *specific corrections* come from: when δ_{regret} is high, the strategic calibration residual (9.1) localizes the regret to specific parts of the strategy DAG.

Formal Expression.

Definition (control-regret)

$$\delta_{\text{regret}} = A_O(M_t;\, \Pi, N_h) - V_O(M_t, \pi_{\text{current}};\, N_h) \geq 0 \tag{8.18}$$

Always non-negative: the current policy cannot outperform the best in its class.

- $\delta_{\text{regret}} \approx 0$: The agent is doing the best it can within current (Π, N_h, M_t). If $\delta_{\text{sat}} > 0$ simultaneously, the problem is not the current strategy — it's either the goal, the capability (Π, N_h), or the model (M_t). See 8.4's disambiguation.
- $\delta_{\text{regret}} \gg 0$: There's room for improvement without changing O_t. → Revise Σ_t.

Epistemic Status.

Exact as a definition — convention-relative as a diagnostic. Like the satisfaction gap, this is a mathematical definition — a difference between two values of the same functional. The quantity is well-defined; computing it requires evaluating A_O (generally intractable) and V_O under the current policy (tractable in simulation, approximate in practice).

Convention hierarchy. δ_{regret} inherits the continuation convention from 6.6. Under the monotonicity result: $\delta^{(1)}_{\text{regret}} \leq \delta^{\text{RH}}_{\text{regret}} \leq \delta^{\text{B}}_{\text{regret}}$. The $\delta^{\text{RH}}_{\text{regret}} \leq \delta^{\text{B}}_{\text{regret}}$ half is unconditional; the $\delta^{(1)}_{\text{regret}} \leq \delta^{\text{RH}}_{\text{regret}}$ half holds under the order-consistency condition on C2's replanning objective (RH-1/2/3 in 6.6), and can fail for unguarded short-horizon replanning. C1 (one-step) reveals only the gap between the current first action and the best one-step deviation — a policy that is "locally near-optimal" under C1 may be globally suboptimal. C3 (Bellman) reveals the full gap to the globally optimal policy. C2 (receding-horizon) interpolates: it captures regret from suboptimal first actions that become visible with N_r-step lookahead. For strategy revision, C2 is often the most useful convention: it reveals recoverable suboptimality without requiring the full Bellman solution.

Discussion.

The diagnostic power of the two-gap system. The satisfaction gap and control regret together encode a 2×2 diagnostic:

Table 8.6

	$\delta_{sat} \leq 0$ *(attainable)*	$\delta_{sat} > 0$ *(unmet)*
$\delta_{\text{regret}} \approx 0$ (near-optimal)	**Success**: goal achievable, policy good	**Capability limit**: optimally pursuing an unmet goal → check M_t, Π, N_h, then consider revising O_t
$\delta_{\text{regret}} \gg 0$ (suboptimal)	**Strategy problem**: goal achievable, policy poor → revise Σ_t	**Both**: goal hard AND strategy weak → revise Σ_t first, then reassess δ_{sat}

This diagnostic is what makes the orient cascade (10.1) actionable: each cell prescribes a different corrective action.

Control regret as the signal for Σ_t revision. When δ_{regret} is high, the agent knows it could do better with a different strategy. The *specific* corrections — which edges to revise, which branches to prune, which alternatives to add — come from the strategic calibration residual (9.1), which localizes the regret to specific parts of Σ_t.

Regret approaching zero when optimally failing. This is the key insight motivating the two-gap split. A single $\delta_{\text{objective}}$ would show "large gap" for both "bad strategy, achievable goal" and "good strategy, impossible goal." The first warrants strategy revision; the second warrants goal revision (after ruling out $M_t/\Pi/N_h$ inadequacy). Without the split, the agent cannot distinguish these cases and may waste effort optimizing a strategy that's already near-optimal for an infeasible goal.

Diagnostic content vs. AI's expected-free-energy decomposition. The 2×2 disambiguation depends on the satisfaction gap / control regret split being orthogonal — distinguishing "goal too hard" from "strategy too weak." Active inference's expected free energy decomposition (pragmatic value + epistemic value; Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29) supports policy ranking but does not separate these diagnoses — both increase EFE without distinguishing cause. See 8.4 for the full analysis of why the diagnostic structure depends on V_{O_t} being a value functional rather than log-priors over outcomes (Sun & Firestone 2020, "The dark room problem," *Trends Cog. Sci.* 24).

Bounded control regret as a finite-passivity diagnostic (Cheung-Piliouras-Tao 2021). Cheung, Piliouras & Tao (2021, *Online Optimization in Games via Control Theory: Connecting Regret, Passivity and Poincaré Recurrence*, arXiv:2106.04748) prove (Theorem 8) that any finitely-passive learning dynamic guarantees constant regret. Read contrapositively: bounded control regret growth is an observable signature that the agent's update operator has finite-passive structure (sits in the certificate-cone interior per `#result-certificate-existence`); unbounded growth in control regret indicates the update is not finitely-passive — potentially escaping the certificate's interior into the no-certificate regime where the persistence machinery does not apply. The bridge is *qualitative only*: CPT's storage function L (on cumulative-payoff space) and AAT's certificate $V(e) = e^\top \mathcal{M} e$ (on strategy-error space) have different functional forms and live on different state spaces — the structural identification "finite passivity ⟺ certificate-interior regime" survives at the level of regimes, not at the level of identifying L with V.

8.6 *Discussion:* Additional Implications & Discussion

The chapter delivered three things that work together at the chapter-end better than they do separately at chapter-middle: a formal strategy representation whose load-bearing structure is theorem-backed under stated conditions and whose parameterization is principled but chosen; an orthogonal diagnostic split that distinguishes goal-attainability from policy-quality and whose meaning is convention-indexed rather than absolute; and an honest scope statement about correlated failure that turns out to be the framework's first explicit instance of *naming an information-theoretic floor and the unique broadly-available escape via AAT machinery*. The chain-confidence-decay identity at the head of the chapter is doing work three layers deeper than its elementary form suggests. None of these are findings on their own that the intro doesn't preview; what the chapter-end can do that the intro can't is read them together, point at the bridges to the rest of the framework, and surface the methodological commitments they encode.

Theorem-backed structure, convergent-choice parameterization.

The chapter is careful about a distinction that downstream segments rely on without restating: what falls out of theorems, and what is a parameterization choice within the theorem-backed structure. Acyclicity of Σ_t is proved — directed temporal order over a finite horizon forces it (1.7 + standard order theory). The Markov factorization is proved under causal sufficiency via the Causal Markov Condition theorem (Spirtes-Glymour-Scheines 2000, Pearl 2009) — `#deriv-graph-structure-uniqueness` does the work. Together they constitute a sufficient condition for the DAG-with-Markov-property representation. The argument parallels Cox's theorem for probability in form (operational desiderata force a measure) but not yet in strength (Cox is necessary-and-sufficient; this is sufficient only — whether some non-DAG structure could also satisfy the postulates is an open stronger result).

What remains a *choice* within the theorem-backed structure is the parameterization: AND/OR combination semantics at nodes, single-parameter credences on edges. These are not derived — but the chapter is also careful to note they are not arbitrary. The catalog draft names a third epistemic category worth surfacing as a methodological finding in its own right: *convergent representational choices*, where every investigated alternative either failed or recovered the same structure. AND/OR converged across three independent formalism attempts; alternatives (general CPTs, noisy-OR everywhere, weighted combinations) were rejected for non-identifiability or parameter explosion. Single-parameter edges replaced an earlier (p_{ij}, θ_{ij}) two-parameter form when the AND/OR combination rules at nodes turned out to absorb θ's role. The P3→Markov route from local revisability — superseded now by the CMC derivation — was a convergent choice before its derivation, and the convergence pattern is what motivated looking for a derivation. The category between *derived from first principles* and *arbitrary framework decision* is itself a research-trail-honesty artifact: a catalog that tells the reader "we didn't just pick this; we tried everything else and it didn't work." Future promotions from convergent-choice to derived (the open parsimony theorem for AND/OR, the open necessity direction in `#deriv-graph-structure-uniqueness`) close the gap one item at a time.

For practitioners, the implication worth carrying is methodological: when AAT's strategy machinery shows up downstream in regret bounds, persistence schemas, or composition arguments, the load-bearing structural commitments are theorem-backed under causal sufficiency, not modeling conveniences. The conditioning on causal sufficiency is the right level of conditionality — it is a property of *strategy quality* (did the agent represent the relevant common causes as nodes?), not of the proof. It is testable in principle (correlated residuals after convergence indicate missing common causes) and repairable in practice (add the common-cause node, restoring causal sufficiency). The Markov-factorization-as-theorem position is what makes the strategy-layer results inherit the strength of Pearl-Bareinboim causal machinery rather than reading as agent-theoretic analogies of it.

The diagnostic split, lifted by the convention hierarchy.

The chapter's orthogonal diagnostic — satisfaction gap ($\delta_{\text{sat}} = V^{\min}_{O_t} - A_O$) and control regret ($\delta_{\text{regret}} = A_O - V_O(\pi_{\text{current}})$) — separates "the world doesn't permit it" from "you're not doing it well enough." The 2×2 cell map (goal × policy) routes four regimes to four corrective actions, and the orient cascade in Part II Ch.5 will use this routing to direct revision pressure to the right substate. This is the implications-side framing of the diagnostic: it is not a measurement but a *routing* device, and getting the routing right is what makes Part II's strategy revision actionable.

The lift the chapter delivers but does not entirely surface is that the diagnostic is *convention-indexed*, not absolute. The value object's three named conventions — C1 (one-step improvement, canonical), C2 (receding-horizon), C3 (Bellman) — form a monotonicity hierarchy on the satisfaction gap: $\delta^{\text{B}}_{\text{sat}} \leq \delta^{\text{RH}}_{\text{sat}} \leq \delta^{(1)}_{\text{sat}}$. C1 gives the most false "unattainable" diagnoses; C3 gives none modulo model error. The 2×2 cell map is therefore not a single object but a *convention-indexed family*, and a practitioner using satisfaction-gap diagnostics under C1 cannot conclude what a C3 practitioner would. The monotonicity result tells the practitioner exactly what additional inferential force the higher convention buys for what additional computational cost. Most decision-theory frameworks treat convention choice as a purely computational tradeoff; AAT adds the inferential-force semantics that makes the choice diagnostic-meaningful rather than merely budget-meaningful.

The split also has a positive content visible only against active-inference's preferences-as-priors form, where outcome value and outcome probability are collapsed into a single quantity. Friston et al.'s expected free energy decomposes into pragmatic and epistemic value — useful for policy ranking, but the two halves don't separate the two cases the 2×2 cell map separates. "The goal is unattainable from here" ($\delta_{\text{sat}} > 0$) and "the current policy is not the best available" ($\delta_{\text{regret}} > 0$) both increase EFE without distinguishing cause, and the orient cascade acts on the two differently. The diagnostic structure depends on V_{O_t} being a value functional on trajectories rather than a log-prior on outcomes — that is what makes "wanting o" and "expecting o" formally distinguishable rather than collapsed (Sun & Firestone 2020 diagnose the collapse; AAT's value-functional reformulation is the downstream architectural response). This is a deliberate divergence from active inference, not an oversight.

The Correlation Hierarchy and identifiability-floor Instance 2.

The chapter's `#def-strategy-dag` carries an honest scope statement about correlated failure that is more consequential than it reads on first encounter. The Correlation Hierarchy (L0 / L1 / L1' / L2) names what AAT's formal strategy-layer results actually claim and where they degrade. L0 (independence — the tractable baseline under causal sufficiency) is where the chapter's persistence schema, gradient-based credit assignment, and sector-condition transfer are all proved. L1 (augmented DAG with strict-prerequisite common-cause nodes restructured so the common cause is factored above the correlation it creates) is where the formal results transfer exactly. L1' (mixture form for soft facilitators with $\theta_{\text{child}|\neg C} > 0$ when C is observable per trial) transfers with five-way gating via Prop B.7 in `#deriv-edge-credence-dynamics`. L2 (full joint distribution) is mathematically ideal but exponential in parameter count and incompatible with the bounded-cognition constraint that motivates the DAG in the first place.

The substantive finding lives in the boundary between L1' (observable C) and L1' (unobservable C). When the common cause is unobservable, the per-conditional decomposition is not "open" — it is *structurally refuted* by the Cramér-Rao floor (Fisher rank-1 per-trial information matrix), and no unbiased online estimator on the joint conditional vector admits a sector parameter $\alpha > 0$. This is the framework's second formally-named instance of the identifiability-floor pattern (Instance 2 in `#disc-identifiability-floor`), and the agent's three escape routes — augment C-observability via an instrument variable, run multi-child joint observations under exploration, or fall back to L0-on-marginals (losing per-conditional diagnostics) — are the operational instantiation of *naming the floor and naming the unique broadly-available escape via AAT machinery*. The framework's recurring posture is in view here: each identifiability floor is a *constructive use of impossibility*, where the no-go elevates the load-bearing role of the specific AAT machinery that supplies the escape.

The chapter is also where the first instance of the same pattern (Instance 1) becomes statable: the on-policy L0-insufficiency-detection no-go. An agent at L0 cannot detect its own causal insufficiency from on-policy traces alone — the worlds with and without a latent common cause are observationally indistinguishable under policy-perfect execution (the Bareinboim Causal Hierarchy Theorem applied at the L0/L1 distinction). The unique broadly-available escape is joint sibling observability under exploration, with the agent's own interventions (7.3) supplying the joint outcome data the test requires. The full formal treatment lives in `#der-causal-insufficiency-detection` in Part II Ch.4; the chapter's Correlation Hierarchy is where the scope condition is named and where the no-go becomes anticipatable. Read Ch.3's L0/L1' boundary and Ch.4's on-policy detection no-go together and the identifiability-floor pattern's first two instances come into view as duals: Instance 2 says identification fails inside a single agent under unobservable common causes; Instance 1 says self-diagnosis fails under policy-perfect execution. The single escape AAT names — interventional access supplying the joint outcome data — applies to both.

Chain-confidence decay, coordinate forcing, and the bridge into Strategy Dynamics.

The chapter opens with `#der-chain-confidence-decay` — the chain rule of probability lifted to log-space, $\log P(\text{chain}) = \sum_i \log P(E_i \mid E_{<i})$. The identity is elementary; what the chapter intro flags but the implications can develop is that it is the *first instance* of a meta-pattern AAT will deploy at three more representational layers. AAT forces a coordinate

at each of chain, divergence, update, and metric layers via uniqueness theorems on AAT-internally-motivated additivity or invariance axioms: log-probability at chain (Cauchy-FE on the probability chain rule); reverse-KL at divergence (Cauchy-FE via Hobson 1969 / Csiszár 1991 / Shore-Johnson 1980 on chain-rule additivity over conditional factorizations); log-odds at update (Cauchy-FE via Aczél 1966 on evidential additivity); Fisher information metric at metric (Čencov 1982 invariance under (PI) parameterization-invariance commitment). The four are interlocked by cross-cites; the convergence across independent axioms onto a single exponential-family Legendre-Fenchel geometry is the substance — see `#disc-additive-coordinate-forcing` for the catalog.

The methodological implication worth surfacing: AAT's coordinates are not chosen by convention. They are forced by AAT-internal axioms via classical uniqueness theorems on additivity or invariance, and they all live on the same geometric object. Lyapunov quadratic (the chapter's persistence machinery) and the IB Lagrangian (the modeling-class machinery in Part I) are adjacent family members — Bregman divergence on Euclidean potential, and adopted-rather-than-derived respectively. The framework's posture is that where a coordinate matters, the framework can usually point at the axiom that forces it; where the axiom isn't AAT-internal, the framework can say what was adopted from outside. The chapter is where this posture lives its first explicit instance.

The downstream bite is wide. Each of the four coordinate-forcing instances backs at least one of the chapter's catalog-grade findings elsewhere: log-probability at chain anchors the depth-fragility result whose engineering reading is "deep plans are structurally fragile, with the rate of fragility growing exponentially in depth"; reverse-KL anchors the Bretagnolle-Huber regret bound in Part II Ch.4; log-odds anchors the detection-latency no-go (`#deriv-update-detection-latency`) and the forgetting prerequisite (`#schema-strategy-persistence`); Fisher metric anchors the Class 3 (Coupled) bias bound (Track 2 with $C_{FR} = \sqrt{2}$ universal under (PI) + Čencov, dimension-free) and the survival-LMI directional exploration bonus. The chapter's elementary identity at chain layer is the canonical example of a load-bearing AAT claim that looks like algebra and turns out to be the entry point to a four-layer architectural commitment.

The widest downstream bite is in Part II Ch.4 (Strategy Dynamics), where the strategy-edge persistence schema instantiates the chapter's DAG structure with Beta-Bernoulli update dynamics, and where `#der-causal-insufficiency-detection`'s on-policy no-go (this chapter's Instance 1 scope-named) becomes the formal statement. The Ch.4 chapter-end implications will read against this one — the framework's identifiability-floor pattern delivers its first formal instance there, with the Correlation Hierarchy from this chapter as the scope statement and the loop-interventional access from Part II Ch.2 as the escape mechanism.

What Part II Ch.3 closes on, then, is the formal substrate Strategy Dynamics will operate on: a theorem-backed graphical structure with a principled parameterization, a convention-indexed diagnostic that routes revision pressure to the right substate, a scope-honest treatment of correlated failure with one Cramér-Rao floor already named and one Bareinboim-CHT no-go anticipated, and the first instance of a four-layer coordinate-forcing meta-pattern whose remaining three layers reach across the rest of the framework. The chapter is small in segment count and wide in downstream reach.

Chapter 9

Strategy Dynamics

[**Gap**] Open question

9.1 *Definition:* Strategic Calibration

A fine-grained companion to control regret (8.5), locating where in the strategy DAG miscalibration lives. The **edge residual** r_{ij} for each edge is the difference between the predicted value increment from completing the parent step (as predicted by Σ_t) and the observed value change attributable to that step. The **strategic calibration residual** $\delta_{\text{strategic}}$ is an L^2 aggregation over active edges with importance weighting (e.g., criticality on the plan's critical path). Where the control regret says *how much* value is being left on the table, the strategic calibration residual says *which edges* are miscalibrated.

The framework is honest about conditions and limitations. The edge residual is meaningful only when the edge was actually traversed, the model is adequate (so observed value changes are signal rather than noise), and the agent followed the strategy's prescription (otherwise the residual conflates a bad plan with bad execution). The status is discussion-grade because the specific L^2 aggregation and the importance weighting are sensible first passes but not derived, and the conditioning requirements make the quantity hard to estimate in practice. A careful distinction separates $\delta_{\text{strategic}}$ from the **strategy-plan-confidence error** $\delta_s = \hat{P}_\Sigma - \Phi$ (the scalar gap to the independence-model plan value at true edge parameters — computable from status propagation alone and the proven persistence target via Prop B.5 in `#deriv-edge-credence-dynamics`): the two measure related but non-interchangeable quantities, and $\delta_{\text{strategic}}$'s per-edge diagnostics require credit assignment that the scalar δ_s does not. A subtler **credit-assignment problem** for multi-parent AND/OR nodes (see Discussion) means r_{ij} measures "how well did the overall prediction work?" rather than "how well did this specific edge predict?" absent further structure. This is also a **second-order inference** — accumulating over multiple edge traversals — inherently slower to evaluate than $\delta_{\text{epistemic}}$ or δ_{sat}.

Formal Expression.

For each edge (i, j) in Σ_t with credence p_{ij}, the **edge residual**: *Definition (strategic-calibration)*

$$r_{ij} = \mathbb{E}[\Delta V_O \mid \text{edge } (i,j) \text{ traversed}, \ M_t] - \Delta V_O^{\text{observed}} \tag{9.1}$$

where ΔV_O is the change in $V_O(M_t, \pi; N_h)$ attributable to completing step j — as predicted by Σ_t versus as observed.

The **strategic calibration residual** aggregates across active edges:

$$\delta_{\text{strategic}} = \left(\sum_{(i,j)\in\text{active}} w_{ij} \cdot r_{ij}^2 \right)^{1/2} \tag{9.2}$$

where w_{ij} weights edges by importance (e.g., criticality to the current plan's critical path).

Conditioning. The edge residual r_{ij} is meaningful only when: - The edge was actually traversed (the agent attempted the step) - M_t is adequate (so the observed ΔV_O is meaningful, not noise) - The agent followed Σ_t's prescription for step j (execution fidelity — otherwise the residual conflates bad plan with bad execution)

Without the execution fidelity condition, a positive residual could mean "the plan is wrong" or "the agent didn't follow the plan." These require different corrections (Σ_t revision vs. execution improvement).

Epistemic Status.

Discussion-grade. The edge residual concept is well-motivated: each edge predicts a value increment, and comparing prediction to observation is standard calibration. But the specific aggregation (L^2 norm with importance weights) is a reasonable first pass, not a derived result. The weighting scheme (w_{ij} by criticality) is sensible but ungrounded. The conditioning requirements (especially execution fidelity) make this quantity hard to estimate in practice — the agent must know whether it followed its own plan, which requires a level of self-monitoring that many agents lack.

This is a **second-order inference** — it requires accumulating evidence over multiple edge traversals. It is inherently slower to evaluate than $\delta_{\text{epistemic}}$ (which updates on every observation) or δ_{sat} (which can be evaluated from M_t and Σ_t alone).

Discussion.

Connection to 9.10. There are two distinct strategic mismatch quantities:

1. **Strategy-strategy-plan-confidence error** $\delta_s = \hat{P}_\Sigma - \Phi$ — the scalar gap between the agent's strategy-plan-confidence score and the independence-model plan value at true edge parameters. Computable from status propagation alone, without credit assignment. The sector condition transfers to δ_s (Prop B.5 in `#deriv-edge-credence-dynamics`). Note: Φ is the AND/OR propagation formula evaluated at true edge rates — it equals actual plan success probability only when the DAG is causally sufficient (L0 of the Correlation Hierarchy in 8.3). Under correlated failure (causally insufficient DAG, the dominant real-world case), Φ overestimates actual success. δ_s tracks calibration *within the independence model*, not calibration to strategic reality. For L1 (augmented DAGs with explicit common-cause nodes), δ_s of the augmented graph tracks calibration within the augmented model, which is more accurate.
2. **Strategic calibration residual** $\delta_{\text{strategic}}$ (this segment) — an L^2 aggregation of per-edge value-increment residuals requiring credit assignment to compute.

These are related (both measure strategy-reality mismatch) but not interchangeable. δ_s is the proven persistence target; $\delta_{\text{strategic}}$ provides finer-grained diagnostics but its persistence properties remain open, pending the credit-assignment machinery in 9.6. The correction function for both is edge-credence revision (9.4); the disturbance is environmental changes that alter edge-traversal outcomes.

Typing as value-increment residuals. Each edge predicts a scalar (value increment), not a full state transition. This is the most tractable typing because it connects directly to the value object V_O and allows aggregation across heterogeneous step types (a military advance and a logistics delivery produce different state changes but both produce value increments measurable on the same scale).

Credit-assignment problem. The edge residual r_{ij} subtracts "predicted value increment" from "observed value change." These are different types: the prediction comes from Σ_t's internal causal model, while the observation comes from empirical measurement of ΔV_O. The subtraction is meaningful only when the observed value change can be *attributed to the specific edge* — a credit-assignment problem the current formulation does not address. For edges with a single parent, attribution is straightforward. For multi-parent AND/OR nodes, the observed ΔV_O at the child reflects the combined effect of all parent edges, and decomposing it into per-edge contributions requires additional structure (e.g., Shapley-value decomposition, or sequential observation of parent completions). In the absence of clean attribution, r_{ij} measures "how well did the overall prediction work?" rather than "how well did this specific edge predict?" — still useful for aggregate calibration, but not sufficient for targeted edge revision.

9.2 *Derived:* Causal Insufficiency Detection

The framework's first explicit *no-go theorem* on agent self-diagnosis. An agent operating at L0 of the Correlation Hierarchy (8.3) — assuming its action propositions are causally independent — faces a structural impossibility when its world contains a latent common cause acting on multiple of those actions: under purely on-policy execution, no test, statistic, or Bayesian comparison built from the agent's observable history can distinguish the latent-cause world ($\mathcal{W}_{L1}$) from a no-latent-cause world ($\mathcal{W}^*_{L0}$) whose edge probabilities are matched to the on-policy regime conditionals. The two worlds emit identical on-policy distributions. This is the agent-theoretic instance of the Causal Hierarchy Theorem (7.1, 7.2) applied to the L0/L1 distinction — a Level-2 question (what happens under $do(\cdot)$ on one sibling) being asked of Level-1 data (on-policy observational distributions). The construction of $\mathcal{W}^*_{L0}$ is explicit for the cleanest case (2-sibling OR with strict-prerequisite latent), and the no-go is *exact* for shallow strict-prerequisite cases and *robust-qualitative* for general DAG topology, soft facilitators, and deeper structures.

Detection is therefore *only* possible by capabilities that violate the "purely on-policy" condition. The framework characterizes **five escape routes**, each violating one of the no-go's scope conditions (S1)–(S5): (a) ε-exploration breaks pure on-policy execution (partial detection, scales with ε); (b) joint sibling observability under exploration breaks the short-circuit censoring (strong detection via a pairwise sibling covariance test — *the AAT-canonical detector*, exploiting 7.3); (c) intermediate-state observability breaks censoring at finer grain (very strong when available, see 9.3); (d) structural priors break the no-prior condition; (e) direct intervention on the candidate latent is strongest when available. The primary detection mechanism — pairwise sibling covariance under intervention — is the unique broadly-available violation of the no-go's scope, using only machinery the theory already requires (exploration via SA3, interventional data via the loop).

A historically prominent diagnostic — the L0 plan-level residual — is shown to be a *degenerate special case* of the no-go: under pure on-policy execution the residual is *identically zero* in both worlds by algebraic identity. The collapse is not a quirk of the residual statistic; it is the no-go's prediction for that specific function. No replacement aggregate statistic can do better. Under partial exploration the residual scales as $O(\varepsilon)$ with sign matching the dominant node-type bias, becoming a *confirmatory* signal rather than a primary detector. A subtle but important property: the no-go is *asymmetric* — it forbids on-policy *detection* of L1 from L0 but does *not* forbid on-policy *parameter learning within L0*. The structural source is *sequential short-circuit evaluation* itself, which censors the joint outcomes that would constitute Level-2 evidence; the no-go is therefore a tradeoff between execution efficiency (favoring short-circuit) and structural diagnosis (favoring joint observation), with ε-exploration the AAT compromise.

Formal Expression.

The No-Go Theorem: Purely On-Policy Detection Is Impossible.

Derived (no-go-on-policy, from causal hierarchy theorem + observational equivalence under sequential short-circuit), conditional on (S1)–(S5) below

Let $\mathcal{M}_{L0}$ be the agent's L0 strategy model with sequential short-circuit AND/OR execution policy π_{L0}. Let $\mathcal{W}_{L1}$ be a world with a latent common cause C acting on multiple sibling action propositions, and $\mathcal{W}^*_{L0}$ be an L0 world with edge probabilities $\{\theta^*_j\}$ matched to the on-policy regime conditionals of $\mathcal{W}_{L1}$. Let $\mathbb{P}^{\text{obs}}_{\pi_{L0}}[\cdot]$ denote the joint distribution over the agent's on-policy observable events under π_{L0}.

Observational equivalence. $\mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}_{L1}] = \mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}^*_{L0}]$.

No-go conclusion. Any function of the agent's on-policy observable history alone cannot distinguish $\mathcal{W}_{L1}$ from $\mathcal{W}^*_{L0}$. Therefore no purely on-policy detection mechanism — no test, statistic, or Bayesian comparison taking only the on-policy distribution as input — can detect L0 causal insufficiency.

Scope conditions (S1)–(S5).

- (S1) Pure on-policy execution; no off-policy sampling.
- (S2) Sequential short-circuit AND/OR evaluation.
- (S3) Censored sibling observation: short-circuited siblings are not observed.
- (S4) No interventional access to candidate latents.
- (S5) No structural priors positing specific common causes.

Tier. *Exact* for shallow strict-prerequisite cases (2-sibling OR or AND with binary common cause and $\theta_{j|\neg C} = 0$ — see #example-L1). *Robust qualitative* for general DAG topology, soft facilitators, and deeper structures: the structural argument transfers, but explicit $\mathcal{W}^*_{L0}$ construction has been carried out only for shallow cases.

Construction of $\mathcal{W}^*_{L0}$. For a 2-sibling OR with strict-prerequisite latent C, $P(C) = \theta_C$, conditional success rates $\theta_{j|C}$:

$$\theta_1^* = \theta_C \cdot \theta_{1|C}, \qquad \theta_2^* = p_2^c = \frac{\theta_C\,(1-\theta_{1|C})\,\theta_{2|C}}{1-\theta_C\,\theta_{1|C}} \tag{9.3}$$

Direct verification (see #example-L1) shows $\mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}_{L1}] = \mathbb{P}^{\text{obs}}_{\pi_{L0}}[\mathcal{W}^*_{L0}]$ on the three on-policy observable events. The Bareinboim, Correa, Ibeling & Icard (2022) Causal Hierarchy Theorem then gives the no-go: any two SCMs that agree on Level 1 (associational) data cannot in general be distinguished on Level 2 (interventional) questions, and the L0/L1 distinction — whether siblings share a common cause — is precisely a Level 2 question about $P(A_2 \mid do(\neg A_1))$ versus $P(A_2 \mid \neg A_1)$.

Why this matters. The no-go is the structural reason the prior aggregate-residual mechanism collapses on-policy: the residual is a function of $\mathbb{P}^{\text{obs}}_{\pi_{L0}}$, which is identical between the two worlds, so the residual is identically zero under both. The collapse is not a quirk of the residual statistic — it is a special case of the no-go applied to that specific function. No replacement aggregate statistic can do better.

Derived (boundary-routes, from no-go scope conditions)

The Detection Routes: What Circumvents the No-Go.

The no-go's scope conditions (S1)–(S5) define "purely on-policy." Each condition's violation corresponds to an AAT capability that admits (partial) detection:

Table 9.1

Route	*Scope violated*	*AAT capability*	*Detection strength*
(a) ε-exploration	(S1)	SA3 exploration (`#deriv-edge-credence-dynamics` Prop B.4)	Partial, scales with ε
(b) Joint sibling observability	(S3)	Covariance test under SA3 + 7.3	Strong
(c) Intermediate observability	(S3) at finer grain	Observability investment (9.3)	Very strong when available
(d) Structural priors	(S5)	Hypothesized common-cause nodes in DAG construction	Prior-quality-dependent
(e) Direct intervention on latent	(S4)	Domain-specific latent control	Strongest when available

The covariance test (route (b)) is the AAT-canonical detector: it uses only machinery the theory already requires (exploration via SA3, interventional data via the loop) and is available in the broadest range of domains. The remaining sections operationalize this primary mechanism.

Primary Detection Mechanism: Pairwise Sibling Covariance Under Intervention.

Derived (from loop-interventional-access + independence test, conditional on SA3 exploration providing joint observability)

Under L0 (the independence model in 8.3's Correlation Hierarchy), sibling outcomes under a common parent are uncorrelated:

$$H_0 \ : \ \mathrm{Cov}(Y_{A_i}, Y_{A_j}) = 0 \quad \forall\ i \neq j \text{ siblings under the same parent} \tag{9.4}$$

Under causal insufficiency (latent common cause C acting on multiple siblings), sibling outcomes are positively correlated:

$$H_1 \ : \ \exists\ i \neq j \text{ with } \mathrm{Cov}(Y_{A_i}, Y_{A_j}) > 0 \tag{9.5}$$

The agent generates test data through the standard exploration mechanism (SA3 — ε-greedy or similar). On trials where both siblings are observable — the agent tries one and can also observe the other's outcome, or tries them in rapid succession before the environment state changes — it accumulates the empirical covariance:

$$\hat{\rho}_{ij} = \frac{1}{N} \sum_t (Y_{A_i,t} - \bar{Y}_{A_i})(Y_{A_j,t} - \bar{Y}_{A_j}) \tag{9.6}$$

A significantly positive $\hat{\rho}_{ij}$ rejects the L0 independence hypothesis. Joint observability (7.3 supplies the interventional character; SA3 supplies the joint sampling) is precisely the violation of scope condition (S3) that admits the test under the no-go.

Detection criterion. A statistically significant positive $\hat{\rho}_{ij}$ at sample size N sufficient for the desired test power, after per-edge credences have stabilized:

$$\hat{\rho}_{ij} > z_{1-\alpha}\, \hat{\sigma}_{\rho_{ij}}/\sqrt{N} \quad \Longrightarrow \quad \text{DAG is causally insufficient between siblings } i, j \tag{9.7}$$

(Standard hypothesis-testing form; threshold and test power depend on application.)

Preconditions for the covariance test.

1. **Joint observability.** The agent can occasionally observe (Y_{A_i}, Y_{A_j}) pairs in the same environment state. Pure short-circuit execution censors one of each pair; SA3 exploration or simultaneous-attempt regimes provide uncensored pairs.
2. **Per-edge credence stabilization.** Edge credences $\hat{p}_i, \hat{p}_j$ have stopped drifting at the timescale of the covariance accumulation, so $\bar{Y}_{A_i}, \bar{Y}_{A_j}$ are well-defined empirical means.
3. **Approximate stationarity over the test window.** The latent common cause's frequency and the conditional success rates are not drifting faster than the test's accumulation timescale.

When these preconditions hold, $\hat{\rho}_{ij} > 0$ is diagnostic of a missing common cause acting on (A_i, A_j). When they do not, the signal is ambiguous.

Derived (residual-degeneracy, as instance of no-go theorem)

The Aggregate Residual as a Degenerate Special Case of the No-Go.

A historically prominent diagnostic uses the L0 plan-level residual $\Phi^{L0}(\hat{p}) - \bar{y}_G$ as a detection signal. The no-go theorem subsumes this as a special case: under pure on-policy execution, the residual is *identically zero* in both $\mathcal{W}_{L1}$ and $\mathcal{W}^*_{L0}$.

Direct verification. Under sequential short-circuit, the agent's empirical credences converge to the on-policy regime conditionals: $\hat{p}_j \to p^c_j$. Plugging these into the L0 arithmetic recovers the chain rule of probability (e.g., for OR: $1-(1-p^c_1)(1-p^c_2) = 1 - P(\neg A_1, \neg A_2) = P(A_1 \cup A_2)$, which equals $\bar{y}_G$ under the executed policy). The residual is zero by algebraic identity.

This is *not* a separate finding from the no-go: it is the no-go's prediction for the specific aggregate-residual statistic. The no-go forbids *any* on-policy statistic from distinguishing $\mathcal{W}_{L1}$ from $\mathcal{W}^*_{L0}$; the residual evaluates to the same value (zero) in both, as expected.

Off-policy boundary. Under ε-exploration (route (a)), the residual scales as $O(\varepsilon)$ to leading order with sign matching the dominant node-type bias (+ for OR-heavy, − for AND-heavy):

$$\Phi^{L0}(\hat{p}) - \bar{y}_G = \varepsilon \cdot R + O(\varepsilon^2), \quad \mathrm{sign}(R) = \mathrm{sign}(\rho) \tag{9.8}$$

where R is structure-dependent and recovers the marginal-limit ρ at $\varepsilon = 1$. *[Heuristic]* The qualitative form is robust; the exact coefficient depends on the gap between conditional and marginal credences. The widely-quoted "$\varepsilon \cdot \rho$" scaling is correct as an order-of-magnitude statement. For a 2-sibling OR with conditional credences p^c_j, the exact two-OR formula is $\varepsilon R_1 - \varepsilon^2 R_2$ with $R_1 - R_2 = \rho$; the leading-order coefficient R_1 is structure-dependent and equals ρ only at $\varepsilon = 1$.

The residual is therefore a *confirmatory* signal under route (a): when the agent has material off-policy exploration and the covariance test (route (b)) has localized a candidate latent, the residual sign confirms the bias direction. It is not a primary detector and cannot replace the covariance test.

From Detection to L1 Construction. *[Derived (from positive covariance signal + L1 construction principle in 8.3)]*

Once the agent detects $\hat{\rho}_{ij} > 0$ between siblings A_i and A_j, it knows a latent common cause exists but not its identity. The construction process:

1. **Hypothesize** a common-cause node C that explains the correlation.
2. **Estimate** θ_C from the pattern of joint outcomes. The joint failure rate $P(A_i \text{ fails}, A_j \text{ fails})$ exceeds $(1-\theta_i)(1-\theta_j)$ by $\hat{\rho}_{ij}$; the excess localizes the common cause's frequency.
3. **Restructure** the DAG: factor C above the correlated siblings (8.3, L1 construction principle: factor the common cause above the correlation it creates).
4. **Re-estimate** conditional edge credences $\theta_{k|C}$ from the data, conditioned on the inferred C state.

This is structural adaptation (4.8) at the strategy level: the agent changes its model class from L0 to L1, adding representational capacity for a pattern the L0 model cannot express. The cost is the standard cost of structural change: temporary performance degradation while the new credences converge, and increased graph complexity. (Soft-facilitator common causes require L1' rather than L1 — see 8.3 and #example-L1 for the strict-prerequisite vs soft-facilitator distinction.)

Diagnostic CIY. *Discussion (diagnostic-ciy)*

Which actions are most informative for detecting latent common causes? Under the no-go, only actions that violate one of (S1)–(S5) yield detection signal. The explore-exploit tradeoff extends with a third axis tied to the boundary characterization:

- **Exploit**: pursue the current best plan (no scope violation; no detection signal).
- **Explore**: test unknown edges for individual success rates (route (a); partial detection).
- **Diagnose**: test known edges for joint correlation structure (route (b); strong detection).

Diagnosis is a form of internal exploration — the agent probes its own model's structural assumptions by violating (S3) deliberately, generating joint sibling outcomes that the no-go forbids the agent to obtain on-policy. The information value of diagnostic actions is highest when:

- Edge credences have converged (the agent has good marginals/conditionals but unknown joint structure).
- Joint outcomes for sibling pairs are observable in the same environment state (the covariance test has data — route (b) is operational).
- The agent has sufficient off-policy budget that the secondary residual signal corroborates (route (a) is also operational).

Epistemic Status.

Conditional on the no-go's scope conditions (S1)–(S5) and on strategy-layer instantiation of 4.8. The **no-go theorem** is *exact* for shallow strict-prerequisite cases (2-sibling OR or AND, single binary common cause) by direct construction; *robust qualitative* for general DAG topology, soft facilitators, and deeper structures. The structural argument (observational equivalence of regime-conditional L0 and latent-cause L1) transfers to the general case; explicit construction of $\mathcal{W}^*_{L0}$ has been carried out only for shallow cases.

The **boundary characterization** (routes (a)–(e)) is *robust qualitative*: each route maps to a specific scope-condition violation and to existing AAT machinery, but the precise detection power of each route depends on domain particulars. Routes (a) and (b) have explicit AAT scaffolding (`#deriv-edge-credence-dynamics`, 7.3); routes (c)–(e) depend on domain capability.

The **primary detection mechanism** (pairwise sibling covariance) is *robust qualitative*: standard hypothesis testing applied to interventional data from the feedback loop, with explicit preconditions. Its sensitivity depends on how cleanly the agent can separate sibling-covariance signal from edge-credence noise at convergence; in adversarial or fast-drifting environments the test's effective sample size shrinks.

The **aggregate residual** as a confirmatory signal is *exact* for the on-policy collapse (no-go's prediction is direct); the off-policy mixed-regime scaling is *heuristic* (linear-in-ε with structure-dependent coefficient).

The **detection-to-construction pipeline** is *discussion-grade*: the trigger is the (statistically rigorous) covariance signal, but the specific procedures for estimating θ_C and $\theta_{k|C}$ from correlated outcome data are domain engineering.

What Cannot Be Detected. By the no-go and its boundary characterization, several latent structures remain undetectable by *any* AAT route:

- **Latents with no joint-observability route.** If the latent affects siblings that cannot be jointly observed (mutually exclusive with long horizons, no cause-indicator availability, no intervention capability, no informative prior), the no-go applies in full strength and detection is impossible.
- **Latents affecting only one edge.** By definition not common causes; appear as noise in individual edge credences.

- **Latents too rare to produce observable joint outcomes.** Even with route (b) operational, a latent with $\theta_C \approx 1$ rarely reveals itself — the agent needs enough $C = 0$ events to estimate the covariance.
- **Negatively-correlating latents.** The formulation assumes positive correlation from shared enabling factors. Negative correlation (competing for a shared resource) produces the opposite bias pattern and requires a different model.

These limitations parallel the information-theoretic underdetermination in 9.6: detection requires data with the right structure, and the no-go specifies precisely what "right structure" means.

Discussion.

Why the no-go is a strengthening, not a softening. The prior framing ("the residual mechanism collapses on-policy") was a local observation about one statistic. The no-go is the structural reason: any on-policy statistic must collapse, because the on-policy distribution is identical between L0 and L1 worlds matched on regime conditionals. The covariance test is not just *a* working detector — it is the unique broadly-available violation of the no-go's scope. This sharpens the load-bearing of `#der-loop-interventional-access`: without the loop's interventional data, the no-go forbids detection entirely; with it, route (b) is operational.

Connection to Pearl's hierarchy. The L0/L1 distinction is a Level 2 distinction in Pearl's framework — it concerns whether $P(A_2 \mid do(\neg A_1)) = P(A_2 \mid \neg A_1)$. The Causal Hierarchy Theorem (Bareinboim, Correa, Ibeling & Icard 2022, Theorem 1) proves that Level 2 distinctions are not in general identifiable from Level 1 (associational) data. The no-go is the AAT-specific instantiation: on-policy data is Level 1; the L0/L1 question is Level 2; therefore detection requires more than on-policy data. The five circumvention routes are all ways the agent obtains supra-Level-1 information.

The censoring mechanism is the structural source. Sequential short-circuit evaluation is what makes on-policy data Level 1 only — it censors the joint outcomes that would constitute Level 2 evidence. An agent that *did not* short-circuit would obtain joint sibling outcomes naturally, and the no-go would not apply. But short-circuit is forced by efficiency: testing A_2 when A_1 has already succeeded is wasted action. The no-go is therefore a tradeoff between execution efficiency (favoring short-circuit) and structural diagnosis (favoring joint observation). SA3 ε-exploration is the AAT compromise: short-circuit by default, occasional non-short-circuit excursions that pay the efficiency cost to maintain detection capability.

Connection to the orient cascade. The detection signal enters the orient cascade (10.1) at step 4c (causal-sufficiency check). Step 4c's reference to "pairwise sibling covariance under an augmented test" aligns with the primary detection mechanism here. The no-go strengthens the cascade's load-bearing: step 4c is not "one possible diagnostic" but "the unique broadly-available diagnostic given the structural impossibility of purely on-policy detection."

Connection to the broader identifiability-floor pattern. The no-go is one of an emerging class of structural impossibility results in AAT — limits on what can be inferred from limited information, derived from external information-theoretic theorems. See 5.1 for the meta-pattern collecting this result alongside the L1' mixture-identifiability obstruction (`#deriv-edge-credence-dynamics` Prop B.7) and the open causal-IB extension for interventional relevance variables.

Domain instantiations. The covariance test (route (b)) applies concretely in: - **Software deployment**: two services sharing infrastructure fail together more often than independent failure rates predict → add infrastructure-health node. - **Military operations**: two concurrent operations fail together under adverse weather → add weather-condition node. - **Investment**: two positions lose value together during market stress → add market-regime node. - **Organizational strategy**: two initiatives stall together during leadership transitions → add organizational-stability node.

In each, what makes detection feasible is the agent's ability to occasionally observe *both* sibling outcomes — the route (b) capability. Pure short-circuit ("only run service B if A is down") suppresses the joint-observation events the test relies on; some routine joint exposure is necessary. When joint observation is impossible (routes (b) and (c) both unavailable) and intervention on the candidate latent is impossible (route (e) unavailable), the agent must rely on structural priors (route (d)) — domain knowledge positing the common cause. This is the regime in which intuition-driven causal modeling is the only tractable approach.

Findings.

On-Policy L0 Insufficiency Is Structurally Undetectable. **Brief:** An agent operating with a strategy model that assumes its action propositions are causally independent (an L0 model) faces a structural impossibility when its world contains a latent common cause acting on multiple of those actions: under purely on-policy execution, no test, statistic, or Bayesian comparison built from the agent's observable history can distinguish the latent-cause world from a no-latent-cause world whose edge probabilities are matched to the on-policy regime conditionals. The two worlds emit identical on-policy distributions. This is not a quirk of any particular diagnostic — it is the agent-theoretic instance of the Causal Hierarchy Theorem applied to the L0/L1 distinction, which is a Level-2 question being asked of Level-1 data. The agent that wants to discover its own model's structural insufficiency must source data the on-policy regime structurally cannot produce.

Impact: Reframes exploration from a discretionary diagnostic activity into a structural prerequisite for self-correction at the strategy layer. Each scope-condition violation (S1)–(S5) maps to a specific AAT capability that admits partial detection — most importantly, joint sibling observability under exploration, which lifts `#der-loop-interventional-access` from "useful machinery" to "the unique broadly-available violation of the no-go." Downstream, this hardens the orient cascade's causal-sufficiency check from "one possible diagnostic" to "the only widely-applicable diagnostic"; it explains why the prior aggregate-residual mechanism collapses on-policy (the residual is one statistic the no-go forbids, and *every* on-policy statistic is forbidden); and it gives a precise account of when L0→L1 escalation can and cannot be triggered from the agent's own data.

Novelty Claim: *Claim differentiation* on the framing of why structure-aware exploration is required. Causal bandit and causal MDP work under hidden confounding establishes that observational and interventional data are non-interchangeable and that intervention can be necessary for low-regret learning; this finding sharpens that line into a no-go for *self-diagnosis* under policy-perfect execution, tied to latent strategic correlation structure rather than to regret minimization, and characterizes five boundary routes by which AAT machinery escapes it.

Related Work:

Table 9.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
Observational vs experimental data are non-equivalent under hidden confounding	Bareinboim, Forney & Pearl 2015, "Bandits with Unobserved Confounders" *NeurIPS* (published 2015, found 2026-04 via Undermind report)	*conceptual precursor* — the underlying observational/interventional asymmetry is shared; this finding recasts it as a no-go on agent self-diagnosis rather than a regret bound on action selection
Sequential extension to MDPs	Zhang & Bareinboim 2016, "Markov Decision Processes with Unobserved Confounders" (published 2016, found 2026-04)	*conceptual precursor* — sharpens the bandit asymmetry to sequential control; the present finding's L0/L1 distinction is the strategy-DAG analog at the structural-detection layer rather than the policy-optimality layer
Naive randomization can be insufficient under confounding	Forney, Pearl & Bareinboim 2017, "Counterfactual Data-Fusion for Online Reinforcement Learners" *ICML* (published 2017, found 2026-04)	*conceptual precursor* — narrows the gap between generic exploration and structure-aware experimentation; the present finding's covariance test under joint sibling observability is one such structure-aware mechanism, derived as the unique broadly-available violation of the no-go's scope conditions
Not all interventions are useful	Lee & Bareinboim 2018, "Structural Causal Bandits" *NeurIPS*; Lee & Bareinboim 2020, "Characterizing Optimal Mixed Policies" *NeurIPS* (published 2018/2020, found 2026-04)	*conceptual precursor* — closest formal neighborhood for "exploration design"; the present finding addresses a different question (when *any* on-policy diagnosis is impossible) but inherits the principle that not all interventional data carries equal structural-detection value

ASF concern	*Prior-art language*	*Relationship / Positioning*
Exploration as epistemic-value drive	Friston, Rigoli, Ognibene, Mathys, FitzGerald & Pezzulo 2015, "Active inference and epistemic value" *Cognitive Neuroscience* 6:187–214 (published 2015, found 2026-04)	*adjacent literature* — weaker threat because EFE-based exploration reduces uncertainty under the generative model rather than Pearl-style observational equivalence under hidden common causes; the structural-undetectability claim does not arise in the EFE framework, which presumes a generative model expressive enough to represent the latent
Theorem the no-go imports	Causal Hierarchy Theorem: SCMs agreeing on Level 1 may disagree on Level 2 (Bareinboim, Correa, Ibeling & Icard 2022, in *Probabilistic and Causal Inference: The Works of Judea Pearl*; published 2022, found 2025)	*formal antecedent* — the no-go is the AAT-specific instantiation; the L0/L1 distinction is precisely a Level-2 question (concerns $P(A_2 \mid do(\neg A_1))$ vs $P(A_2 \mid \neg A_1)$) being asked of on-policy Level-1 data

Search Log:

- 2026-04 (*nominally comprehensive*, via `ref/Novelty_defense_and_integration.md` Pillar 1): Undermind report on the causal-bandits-and-MDPs-under-hidden-confounding literature, plus active-inference epistemic-value exploration, established the prior-art landscape; verdict was *Conceptual Precursor* (High confidence). The closest formal neighborhood is the Bareinboim/Zhang/Forney/Lee line; no paper surveyed states the claim in the same form (no-go for self-diagnosis under policy-perfect execution tied to latent strategic correlation). The defense strategy positions ASF's increment narrowly: a no-go theorem on structural detection rather than a regret bound on action selection, with five explicit boundary routes mapped to existing AAT machinery.
- 2025 (*targeted*): Bareinboim, Correa, Ibeling & Icard 2022 identified as the formal antecedent for the Causal Hierarchy Theorem invocation; the segment's invocation of the CHT was already grounded in this source before the comprehensive Pillar-1 defense.

9.3 *Derived:* Observability Dominance

A direct consequence of the gain principle (3.6) applied to strategy edges. Unobservable strategy edges cannot be updated — when any node along a path has near-zero observability, the gain principle drives the update rate on the edges touching it to zero, and those edges are *frozen at their prior*. The agent's effective strategy is therefore limited to the parts it can observe, regardless of the nominal confidence assigned to unobservable paths. Observability dominates nominal confidence in determining which strategies are epistemically alive. Path observability is the *weakest-link* quantity: $\text{obs}(P) = \min_{v \in P} \sigma_v$.

A striking structural prediction follows: **unobservable regions of strategy are absorbing.** Once significant strategy investment operates through unobservable nodes, the dynamics become self-locking — the mismatch signal still arrives but carries no weight ($\eta_{\text{edge}} \to 0$), no weight means no revision, and because near-zero effective mismatch from a too-noisy channel is case (c) of the zero-aporia trichotomy (3.4) — not distinguishable from within from case (a), a model that genuinely fits — the agent *cannot learn, and cannot recognize that it cannot learn*. Escape requires external shock, proactive observability investment (instrumenting previously unmonitored nodes), or another agent whose observations cover the blind spot (14.4). An agent choosing between a strong-but-blind path (high confidence, low observability) and a weak-but-visible path (lower confidence, high observability) should prefer the *visible* one — the visible path yields large update gain (the agent quickly learns whether it works and can redirect) while the blind path yields tiny gain.

A subtler structural consequence: an unobservable intermediate doesn't just freeze its touching edges — it makes *per-edge identification impossible*, forcing the agent into plan-level aggregation. The agent can still track plan success as a single quantity, but loses diagnostic resolution: it knows the plan is failing but cannot localize which step needs revision. The framework quantifies the **observability investment tradeoff**: making an intermediate observable yields a measurable improvement in the strategy-layer sector parameter — from the plan-level rate $1/(n_\Phi + 1)$ to the per-edge weakest-link rate $\min(1/(n_1 + 1),\ \theta_1/(n_2 + 1))$ — positive whenever per-edge success rates are reasonable and experience is distributed across edges. The value of instrumenting an intermediate node is the *difference in sector parameters*, translating directly to persistence margin. A complementary trade-off: finer decomposition provides earlier failure detection but adds uncertain edges via chain confidence decay; the optimal decomposition depth balances incremental confirmation against compound decay.

Formal Expression.

For a path P through Σ_t, the **path observability**:

Derived (observability-dominance, from update-gain + strategy-dag)

$$\mathrm{obs}(P) = \min_{v \in P} \sigma_v \tag{9.9}$$

where σ_v is the observability of node v — how well the agent can determine whether v has been achieved. The weakest link determines the path's observability.

Observability-adjusted confidence:

$$\mathrm{conf}_{\mathrm{obs}}(P) = \mathrm{conf}(P) \cdot \mathrm{obs}(P) \tag{9.10}$$

When $\sigma_v \approx 0$ for any node v on the path: by 3.6, $\eta_{\mathrm{edge}} = U_{\mathrm{edge}}/(U_{\mathrm{edge}} + U_{\mathrm{obs}}) \to 0$ as $U_{\mathrm{obs}} \to \infty$. The edges connecting to v are **frozen at their prior** — the agent cannot update them regardless of what happens. The path is epistemically dead.

What is frozen is the *update*, not the *input*: observations continue to arrive and continue to be uninformative, which is case (c) of the zero-aporia trichotomy (3.4) — near-zero mismatch because the channel cannot detect model error, not because there is nothing to detect. That trichotomy is also what makes the region self-locking rather than merely unproductive: case (c) is not distinguishable from within from case (a), a model that genuinely fits, so the region presents as settled knowledge rather than as a gap.

Epistemic Status.

Robust qualitative. The mechanism (high observation noise → low gain → frozen edges) is a direct consequence of 3.6's uncertainty ratio. The specific functional form ($\mathrm{conf}_{\mathrm{obs}} = \mathrm{conf} \cdot \mathrm{obs}$) is a first-order approximation — the actual relationship between observability and effective confidence is more complex (it depends on how many observations are accumulated, the prior strength, and the noise structure). The qualitative prediction (low observability → frozen beliefs → ineffective strategy) is robust.

Discussion.

Observability as the gateway to learning. An agent choosing between a strong-but-blind path (high $\mathrm{conf}(P)$, low $\mathrm{obs}(P)$) and a weak-but-visible path (lower $\mathrm{conf}(P)$, high $\mathrm{obs}(P)$) should prefer the visible one. After one attempt: the visible path yields large η_{edge} — the agent quickly learns whether it works and can redirect. The blind path yields tiny η_{edge} — the agent is still guessing after n attempts. Observability enables learning; opacity prevents it.

Unobservable regions are absorbing. Once significant strategy investment operates through unobservable nodes: mismatch still arrives but carries no weight ($\eta_{\mathrm{edge}} \to 0$) → no revision → and, since this is case (c) of 3.4's trichotomy and inseparable from within from case (a), the agent cannot learn and cannot recognize that it cannot learn. Escape requires external shock, proactive observability investment (instrumenting previously unmonitored nodes), or another agent whose observations cover the blind spot (14.4).

Connection to `#der-code-quality-as-observation-infrastructure` (cross-component reference — see `02-tst-core/`). In the software domain, code quality directly determines σ_v — well-structured code with good tests makes strategy steps (features, refactors, deployments) observable. Poor code quality reduces observability, freezing the developer's causal beliefs about what changes will accomplish. This makes code quality a strategic concern, not just an aesthetic one.

Optimal decomposition depth. Finer decomposition (more intermediate nodes) provides earlier failure detection — detect a problem at step k rather than discovering it at step n. But finer decomposition also increases the number of uncertain edges (8.1). The optimal decomposition depth balances incremental confirmation against compound decay: decompose as finely as observation channels allow, but no finer.

Quantitative content from the two-edge case. The analysis in `#deriv-edge-credence-dynamics` (Props B.2-B.3) instantiates this result for the minimal multi-edge chain $A \to B \to G$, giving the qualitative claims above precise mathematical form.

Observable intermediate B. When the agent can observe whether B was achieved, each edge gets independent Bayesian updates. The per-edge sector parameters are $\alpha_1 = 1/(n_1 + 1)$ and $\alpha_2 = \theta_1/(n_2 + 1)$, where θ_1 is the true success probability of the first edge. The overall sector parameter is $\alpha_\Sigma = \min(\alpha_1, \alpha_2)$ — a weakest-link result. The θ_1 factor in α_2 is the **evidence-starvation effect**: downstream edge 2 can only be tested when upstream edge 1 succeeds (with probability θ_1), so its effective correction rate is attenuated by θ_1. For a depth-d chain, this generalizes to $\alpha_k = \prod_{j<k} \theta_j/(n_k + 1)$ — the deepest edge faces exponential attenuation. This is derived for the two-edge case and conjectured (with clear inductive mechanism) for depth-d.

Unobservable intermediate B. When the agent observes only the terminal outcome y_G, per-edge identification fails entirely. The marginal Bayesian update has a systematic bias: the zero-correction-at-truth property (A1) is violated with bias $O(1/n)$. The agent's point-estimate updates for each edge are downward-biased because success always credits both edges fully ($\alpha_k \to \alpha_k + 1$), but failure distributes blame fractionally via proportional attribution. The proportional-blame update turns out to be exactly the marginal Bayesian point estimate — not a heuristic — but it discards the posterior correlation that failure introduces between the edge beliefs.

The consequence is stronger than "frozen edges": unobservable intermediates don't just prevent updating — they make per-edge identification impossible, forcing the agent to plan-level aggregation. If the agent tracks $\hat{\Phi} = p_1 p_2$ as a single Beta on the plan's overall success probability, the single-edge sector condition applies with $\alpha_{\Sigma,\text{plan}} = 1/(n_\Phi + 1)$. But diagnostic resolution is lost: the agent knows the plan is failing but cannot localize which step needs revision.

The observability investment tradeoff. Making B observable yields a quantifiable improvement in α_Σ: from the plan-level rate $1/(n_\Phi + 1)$ to the per-edge weakest-link rate $\min(1/(n_1 + 1), \theta_1/(n_2 + 1))$. The improvement is positive whenever $\theta_1 > 1/2$ and experience is distributed similarly across edges. This gives the observability-dominance principle concrete economic content: the value of instrumenting an intermediate node is the difference in sector parameters, which translates directly to persistence margin.

9.4 *Hypothesis:* Edge Update via Gain

The hypothesis that the uncertainty-ratio gain principle (3.6) extends from epistemic updates on M_t to strategy-edge updates on Σ_t. Each strategy edge has its own uncertainty (concentration of evidence so far) and faces its own observation noise (how observable the destination node is). The update gain on each edge follows the same form as the epistemic gain — the ratio of edge uncertainty to total uncertainty — modulated by the identifiability of the causal link (9.5). This gives a principled, conservative update rule that avoids overreacting to single observations and degrades gracefully when causal identification is weak.

The framework instantiates this concretely for Beta-Bernoulli edges (binary success/failure observations): each success increments α_{ij}, each failure increments β_{ij}, the point estimate is the standard mean, and the effective gain is $1/(n + 1)$. This is the standard Bayesian conjugate update, and it satisfies the same *principle* as the Kalman gain (conservative updating proportional to relative uncertainty) without literally being the variance-ratio formula — the Bernoulli observation model has different noise structure than the Gaussian case where the variance-ratio formula is exact. The same update is *natively additive* in the **log-odds coordinate** $\lambda_{ij} = \log(p_{ij}/(1 - p_{ij}))$: each observation contributes a per-observation log-likelihood-ratio additively. The log-odds coordinate is the unique (up to positive affine transformation) parameterization on which Bayesian independent-evidence accumulation is additive — forced by an evidential-additivity axiom (`#deriv-edge-update-natural-parameter`, the update-level analog of 8.1's log-additive decomposition). The probability-space form and the log-odds form carry equivalent content; the log-odds form is the natural coordinate when continuous-gradient updates would otherwise risk breaking the $[0, 1]$ interval (9.6).

The result connects two earlier results cleanly. **Observability dominance** (9.3) is the limiting case: when destination-node observability heads to zero, $U_{\text{obs}} \to \infty$, gain heads to zero, and the edge is *frozen at its prior*. **Strategic calibration** (9.1) is the diagnostic: persistently positive residual (predicted increment exceeds observed) should reduce the edge credence, persistently negative should increase it, and the gain principle provides the update magnitude. The mechanism is *conservative by design*: well-established edges require strong evidence to revise, newly hypothesized edges are easily moved by evidence — the same calibration principle as the epistemic gain. The result is *hypothesis*-grade because the **signal function** — how to compute what an observation actually says about the link — depends on credit-assignment structure that 9.6 addresses; for single-parent edges with observable outcomes the signal is trivial, and for multi-parent AND/OR nodes a gradient-based candidate gives directional fidelity sufficient for persistence even when exact attribution is computationally intractable.

Formal Expression.

Hypothesis (edge-update-via-gain)

Edge credences update via:

$$p_{ij}^{\text{new}} = p_{ij}^{\text{old}} + \eta_{\text{edge}} \cdot \left(\text{signal}(o_t, i, j) - p_{ij}^{\text{old}}\right) \tag{9.11}$$

where: - $\text{signal}(o_t, i, j) \in [0, 1]$: evidential content of observation o_t about the causal link $i \to j$ - $\eta_{\text{edge}} = U_{\text{edge}}/(U_{\text{edge}} + U_{\text{obs}})$: update gain

with: - U_{edge}: uncertainty about this specific causal link. If $p_{ij} \sim \text{Beta}(\alpha_{ij}, \beta_{ij})$: $U_{\text{edge}} = \text{Var}[\text{Beta}] = \alpha\beta/((\alpha + \beta)^2(\alpha + \beta + 1))$ - U_{obs}: observation noise on the channel confirming this link. $U_{\text{obs}} \propto 1/\sigma_j$ (inverse of observability of node j)

Beta-Bernoulli instantiation. For binary observations (success/failure of step j): - Observe success: $\alpha_{ij} \to \alpha_{ij} + 1$ - Observe failure: $\beta_{ij} \to \beta_{ij} + 1$ - Point estimate: $p_{ij} = \alpha/(\alpha + \beta)$ - Effective gain: $\eta_{\text{edge}} = 1/(n+1)$ where $n = \alpha + \beta$

This is the standard Bayesian conjugate update, yielding $\Delta\hat{p} = (y - \hat{p})/(n + 1)$. The gain $1/(n + 1)$ is the exact conjugate update rate — not a literal substitution into $\eta = U_{\text{edge}}/(U_{\text{edge}} + U_{\text{obs}})$, which is a structural principle (conservative updating proportional to relative uncertainty), not a universal algebraic formula. The Beta-Bernoulli gain satisfies the same principle: it decreases as posterior certainty increases, trusting accumulated evidence over individual observations. But the algebraic derivation is conjugate analysis, not Kalman-style variance ratios, because the Bernoulli observation model has different noise structure than the Gaussian case where the variance-ratio formula is exact. The sector-condition analysis in `#deriv-edge-credence-dynamics` (Props B.1-B.4) uses the exact Beta-Bernoulli gain $1/(n+1)$ directly.

Parallel log-odds presentation. The same update is natively additive in the log-odds coordinate $\lambda_{ij} = \log(p_{ij}/(1 - p_{ij}))$:

$$\lambda_{ij}^{\text{new}} = \lambda_{ij}^{\text{old}} + \ell(y) \tag{9.12}$$

where $\ell(y) = \log[P(y \mid \text{edge true})/P(y \mid \text{edge false})]$ is the per-observation log-likelihood ratio. The log-odds coordinate is the unique (up to positive affine transformation) parameterization on which Bayesian independent-evidence accumulation is additive — forced by the evidential-additivity axiom derived in `#deriv-edge-update-natural-parameter`, which is the update-level analog of 8.1's chain-level additive log-confidence decomposition. The moment-parameter (probability-space) form $\Delta\hat{p} = (y - \hat{p})/(n+1)$ is the projected image of the log-odds additive update under the sigmoid readout $p = \sigma(\lambda)$; the two coordinates carry equivalent content.

The log-odds presentation matters for 9.6's default signal function, where continuous-gradient updates can break the $[0,1]$ domain in probability-space presentation but are well-posed globally on $\lambda \in \mathbb{R}$. For the per-edge Beta-Bernoulli derivation of Props B.1–B.4 in `#deriv-edge-credence-dynamics`, the moment-parameter form is retained because algebraic tightness (the gain $1/(n+1)$ is exact in probability space) is pedagogically cleaner; the sector-parameter content is Fisher-equivalent in either coordinate.

Epistemic Status.

Hypothesis. The extension of the gain principle from M_t to Σ_t edges is structurally motivated: the same uncertainty-ratio logic applies wherever an agent updates a belief in response to noisy evidence. The update *form* is validated for Beta-Bernoulli edges across four topologies (`#deriv-edge-credence-dynamics`, Props B.1-B.4). The signal function — how to compute signal(o_t, i, j) — is now characterized at the theory level:

- **Theory-level resolution.** 9.6 shows that persistence does not require per-edge credit assignment (Prop B.5), that any signal function with directional fidelity suffices for persistence (Level 1), and that exact attribution is #P-hard in general (Level 3). The gradient-based candidate signal$_k = p_k + J_k \cdot (y_G - \hat{P}_\Sigma)/\|\mathbf{J}\|^2$ satisfies directional fidelity for monotone DAGs and is computable in $O(|V| + |E|)$.
- **What remains domain-specific.** The choice among signal function implementations (gradient attribution, proportional blame, belief propagation, exact Bayesian) depends on the domain's observability structure and computational budget. This is engineering, analogous to how the gain *principle* ($\eta^* = U_M/(U_M + U_o)$) is theory while the gain *estimator* (Kalman, TD-learning, intuition) is engineering.
- **Remaining open questions.** (1) Continuous outcomes with multi-parent nodes — signal function candidates exist but are not verified. (2) Interaction with M_t updates — edge updates use observations already processed for M_t; whether this creates statistical dependencies that bias edge updates is not analyzed. (3) The gradient candidate inherits $\hat{P}_\Sigma$'s overestimation bias under correlated failures (8.3) — see 9.6 Working Notes.

Discussion.

Connection to 9.3. When $\sigma_j \approx 0$: $U_{\text{obs}} \to \infty$, $\eta_{\text{edge}} \to 0$. The edge is frozen at its prior. Unobservable links cannot be updated — this is the mechanism that makes unobservable paths epistemically dead.

Connection to 9.1. The edge update rule is the correction function that the strategic calibration residual calls for: persistently positive r_{ij} (predicted value increment exceeds observed) should reduce p_{ij}; persistently negative r_{ij} should increase it. The gain principle provides the update magnitude — how much to adjust given the evidence strength.

Conservative by design. The uncertainty ratio ensures the agent doesn't overreact to single observations: a well-established edge ($\alpha + \beta$ large, U_{edge} small) requires strong evidence to revise; a newly hypothesized edge ($\alpha + \beta$ small, U_{edge} large) is easily moved by evidence. This mirrors the epistemic gain's behavior for M_t — the agent trusts accumulated experience over individual observations.

9.5 *Scope:* Causal Validity of Edge Updates

The companion scope segment to 9.4. Edge credences p_{ij} are *causal-efficacy estimates* — they claim to approximate the interventional quantity $P(j \mid do(i), M_t)$ — but their identification strength varies with the data regime (8.3). This segment scopes where the gain-based edge update yields credences that approximate the genuine interventional quantity, where it yields partially identified estimates, and where it yields associational proxies.

Three conditions are named. **(C1) The parent is an action leaf under the agent's control.** The agent directly executed the action, making the data interventional in character. For condition leaves (observable states the agent doesn't control) and internal nodes (propositional combinations achieved indirectly), $do(i)$ is not directly available. **(C2) The outcome is attributable.** The agent can distinguish whether j advanced specifically because of $do(i)$ or for other concurrent reasons (the credit-assignment problem identified in 9.1). Trivially satisfied for single-parent nodes; violated when multiple parent edges of the same child fire concurrently. **(C3) Execution conditions vary.** The agent does not systematically execute i only under conditions that independently favor j's success — otherwise the observed success rate carries selection bias. These conditions partition domains into **three admissibility regimes** paralleling 7.4. *Regime A — intervention-rich*: all three conditions satisfied; updates approximate genuine interventional probabilities (software tests, laboratory experiments). *Regime B — partial intervention*: concurrent actions blur attribution, self-selection likely; updates carry optimistic bias. *Regime C — observation-only*: agent did not act, attribution impossible, confounding dominant; updates reflect association, not causation.

A subtler structural fact about *indirect edges* (those whose parent is not a leaf): the agent doesn't directly intervene on the parent — instead, the agent's leaf interventions propagate upward through the DAG, providing the parent with *indirect interventional evidence*. This is weaker for two reasons: compounding attribution (each intermediate level introduces uncertainty) and confounding from below (the parent's achievement depends on multiple leaf actions whose combined effect may be confounded). The identification strength for indirect edges decreases with depth in the DAG — deeper edges are farther from direct interventions and have more confounding pathways.

The framework introduces an **identifiability coefficient** $\iota_{ij} \in [0,1]$ — the agent's estimate of how cleanly the observed outcome can be attributed to edge (i,j) specifically — and proposes adjusting the update gain by this coefficient: $\eta_{\text{edge}}^{\text{adj}} = \eta_{\text{edge}} \cdot \iota_{ij}$. For leaf-originating edges in Regime A, $\iota_{ij} \approx 1$; for deep internal edges in Regime C, $\iota_{ij} \approx 0$. Crucially this *unifies two sources of frozen edges*: low **observability** ($\sigma_v \approx 0$ from 9.3 — the node's outcome is hard to *measure*) and low **identifiability** ($\iota_{ij} \approx 0$ — the outcome is measurable but can't be *attributed*). Both drive $\eta_{\text{edge}} \to 0$ and produce the same effect — the edge is frozen at its prior. A consequence for *optimal decomposition depth*: chain-confidence-decay imposes a cost of depth (compound fragility); observability-dominance imposes another (deep observability is required at every step); and now identifiability degradation imposes a third (deep edges have lower attribution coefficient). The optimal decomposition depth balances all three.

Formal Expression.

Scope Condition (edge-update-causal-validity)

Where the agent has direct intervention. By 8.3, leaf action nodes are propositions about the agent's own actions: "action a succeeds at τ_v." When the agent executes an action leaf, it performs a genuine $do(\cdot)$ operation. The edge from that leaf to its child carries credence $p_{ij} = \mathrm{Cr}(j \text{ advances} \mid do(i), M_t)$, and the execution-observation pair $(do(i), o_j)$ is interventional data for that edge.

However, interventional data does not automatically yield clean causal identification. By 7.3, the loop provides data *generated under intervention* — but between the intervention and a usable causal estimate stand coverage, within-step confounding, delay, and partial observability. The following conditions determine when the interventional data is strong enough for valid edge revision:

(C1) The parent is an action leaf under the agent's control. The agent directly executed the action. This makes the data interventional in character — the agent chose the action, the environment responded. For condition leaves (observable states the agent doesn't control) and internal nodes (propositional combinations achieved indirectly), $do(i)$ is not directly available. See "Indirect edges" below for the weaker identification available at those positions.

(C2) The outcome is attributable. The agent can distinguish whether j advanced specifically because of $do(i)$, or for other concurrent reasons. This is the credit-assignment problem identified in 9.1. It is trivially satisfied for single-parent nodes (one possible cause) and for well-isolated interventions. It is violated when multiple parent edges of j fire concurrently.

(C3) Execution conditions vary. The agent does not systematically execute i only under conditions that independently favor j's success. If it does, the observed success rate carries selection bias: $P(j \mid \text{chose to execute } i) \neq P(j \mid do(i))$

because the decision to execute correlates with favorable conditions. This is mitigated when the agent varies execution contexts across episodes, or when external factors (CI pipelines, scheduled operations) force execution regardless of conditions.

C1 establishes that the data is interventional. C2 and C3 determine whether the interventional signal can be cleanly extracted. All three are satisfied simultaneously in Regime A domains; they degrade together in Regime B and C domains.

Three regimes. These conditions partition domains into admissibility regimes, paralleling 7.4:

Table 9.3

Regime	*C1*	*C2*	*C3*	*Causal validity of leaf-edge updates*
A: Intervention-rich	Agent controls leaf actions	Good isolation (one action at a time)	Conditions vary (CI, diverse contexts)	**Strong.** Updates approximate interventional.
B: Partial intervention	Agent acts but with coordination constraints	Concurrent actions blur attribution	Self-selection likely	**Moderate.** Updates carry optimistic bias.
C: Observation-only	Agent did not act (condition leaves, passive monitoring)	Attribution impossible	Confounding dominant	**Weak.** Updates reflect association, not causation.

Indirect edges. For edges between non-leaf nodes, or edges whose parent is a condition node, the agent does not directly intervene on the parent. Instead, the agent's interventions at the leaves propagate upward through the DAG. The edge (i, j) where i is an internal node receives *indirect* interventional evidence: the agent intervened on leaves below i, observed that i was (or wasn't) achieved, and then observed whether j advanced.

This indirect evidence is weaker for two reasons: 1. **Compounding attribution**: the agent must attribute j's outcome to edge (i, j) after already attributing i's achievement to the leaf-level interventions. Each attribution step introduces uncertainty. 2. **Confounding from below**: i's achievement depends on multiple leaf actions and condition states. Even if each leaf intervention is clean, their combined effect on i may be confounded.

The identification strength for indirect edges decreases with depth in the DAG — deeper edges are farther from the agent's direct interventions and have more confounding pathways.

Hypothesis (identifiability-coefficient)

Identifiability-adjusted gain.

The update gain should be adjusted by the agent's confidence in causal attribution:

$$\eta_{\text{edge}}^{\text{adj}} = \eta_{\text{edge}} \cdot \iota_{ij} \tag{9.13}$$

where $\iota_{ij} \in [0, 1]$ is the **identifiability coefficient** — the agent's estimate of how cleanly the observed outcome can be attributed to edge (i, j) specifically.

- $\iota_{ij} = 1$: clean attribution (leaf-originating edge, single parent, isolated execution in Regime A).
- $\iota_{ij} \approx 0$: no attribution possible (deep internal edge, many concurrent causes, Regime C).

For leaf-originating edges in Regime A: $\iota_{ij} \approx 1$. For internal edges at depth d: ι_{ij} decreases with d (each level of indirect inference degrades attribution). The precise functional form is domain-dependent.

This unifies two sources of frozen edges: 1. Low **observability** ($\sigma_v \approx 0$ from 9.3): the node's outcome is hard to *measure*. 2. Low **identifiability** ($\iota_{ij} \approx 0$): the outcome is measurable but can't be *attributed* to this edge.

Both drive $\eta_{\text{edge}} \to 0$ and produce the same effect: the edge is frozen at its prior.

Epistemic Status.

Conditional on the three conditions C1–C3 and the DAG position of the edge. Max attainable: conditional (the conditions are genuine restrictions, not removable assumptions).

The restriction to leaf-originating edges for strong identification: **derived** from 8.3's node definitions. Only action-leaf nodes correspond to operations the agent directly performs. This is a structural property of the DAG, not an empirical claim.

The three-regime classification: **robust qualitative**. The regimes parallel the CIY admissibility regimes in 7.4. The underlying logic is Pearl's interventional/observational distinction applied to strategy-edge updates.

The identifiability coefficient ι_{ij}: **hypothesis**. The concept is sound (discounting by attribution confidence is standard in causal inference), but the specific form — a scalar multiplier on the gain — is a first-order approximation.

The depth-dependent degradation for indirect edges: **robust qualitative**. Each level of indirect inference adds an attribution step, and uncertainty compounds. The specific degradation rate is domain-dependent.

Discussion.

Why this gap matters. Without causal validity conditions, 9.4's status is ambiguous: it might be updating credences that claim to be interventional using evidence that is merely associational. The conditions make explicit what's needed for the update to preserve the interventional semantics of 8.3's edge credences.

Connection to 9.3. Observability gates whether the agent can *detect* an outcome. Identifiability gates whether the agent can *attribute* an outcome to a cause. Both are prerequisites for learning. The combined effective gain:

$$\eta_{\text{eff}} = \frac{U_{\text{edge}}}{U_{\text{edge}} + U_{\text{obs}}} \cdot \iota_{ij} \tag{9.14}$$

captures both gates in a single quantity. When either gate closes ($U_{\text{obs}} \to \infty$ or $\iota_{ij} \to 0$), effective gain goes to zero.

Connection to the signal function. The identifiability coefficient is one component of the unspecified signal function flagged in 9.4's working notes. The full signal function signal(o_t, i, j) decomposes into: (a) what outcome was observed (o_t), (b) how attributable is it to edge (i, j) (ι_{ij}), and (c) what does the attributed outcome imply about the edge's causal strength. This segment addresses (b); (a) and (c) remain open.

Software as Regime A. In software development, the agent runs a specific test (leaf action) and observes the result. C1 is satisfied (the agent ran the test), C2 is satisfied (the test targets one behavior), C3 is satisfied (CI runs regardless of conditions). Leaf-originating edges in software strategies have $\iota \approx 1$. This is the maximally favorable domain for causal edge updates.

Optimal decomposition depth revisited. 9.3 notes that finer decomposition (more intermediate nodes) provides earlier failure detection but adds uncertain edges via 8.1. This segment adds another cost of depth: deeper edges have lower ι_{ij}, making them harder to learn causally. The optimal decomposition depth balances three factors: (a) observability of intermediate nodes, (b) confidence decay through chains, and (c) identifiability degradation with depth.

9.6 *Discussion:* Credit Assignment Boundary

The strategy-revision loop requires assigning credit for observed outcomes to specific edges in the strategy DAG — decomposing "the plan partially worked" into "step 3 failed, step 5 was irrelevant, step 1 succeeded." This is AAT's version of RL's temporal credit assignment problem. This segment characterizes the boundary between tractable and intractable cases, states what the theory requires of any credit-assignment scheme, and identifies what the theory can guarantee *without* solving the problem at all.

The framework names **three things the theory can guarantee without solving credit assignment**. (1) *Persistence is credit-assignment-free*: the sector condition transfers from per-edge credence space to strategy-plan-confidence error $\delta_s = \hat{P}_\Sigma - \Phi$ via the Jacobian $\mathbf{J} = \nabla_\mathbf{p} P_\Sigma$, computable from status propagation in $O(|V| + |E|)$ — no outcome decomposition required (Prop B.5 in `#deriv-edge-credence-dynamics`). (2) *The diagnostic framework is plan-level*: the satisfaction gap (8.4), control regret (8.5), and orient cascade ordering (10.1) operate on aggregate value, telling the agent *whether* the strategy is failing without per-edge attribution. (3) *Observability-dominance identifies the tractable edges*: only the observable subgraph (9.3) can receive informative signals at all. Three independent intractability barriers prevent exact per-edge attribution in the general case: (a) computational #P-hardness via reduction to Shapley values over the AND/OR propagation game; (b) information-theoretic underdetermination — when fewer nodes are observable than edges exist, some directions in θ-space are fundamentally unresolvable from the available data; (c) the posterior correlation barrier — any factored representation necessarily discards the correlation introduced by failure at multi-parent nodes.

The **design requirement** is stated cleanly: any credit-assignment scheme need only satisfy *directional fidelity* — the expected update for each edge must point toward the true credence. Persistence is *robust to approximation*: a sloppy but directionally correct signal function still produces bounded strategic mismatch; the quality of the approximation affects the tightness of the persistence bound, not whether persistence holds at all. The framework presents a **default signal function** based on gradient-attribution, stated natively in the log-odds coordinate $\lambda_k = \log(p_k/(1 - p_k))$ — the unique additive-evidence coordinate forced by the evidential-additivity axiom (`#deriv-edge-update-natural-parameter`), which eliminates a mechanical break the earlier probability-space presentation suffered (updates outside $[0, 1]$). A useful **hierarchy of credit assignment quality** is presented: Level 0 (plan-level tracking only — persistence guarantee, no per-edge diagnostics); Level 1 (directional fidelity per edge — persistence plus rough per-edge diagnostics, the gradient signal function is the concrete default); Level 2 (proportional blame / expectation propagation); Level 3 (full Bayesian posterior — #P-hard in general). AAT's formal guarantees require only Level 0; practical agents need at least Level 1; Level 2 is the sweet spot for most applications; Level 3 is computationally unattainable in the general case. A clean practical insight follows: *credit assignment is primarily an observability design problem*, not an algorithm design problem — an agent that designs its strategy with observable intermediates sidesteps the intractability entirely (OKRs as the organizational instantiation).

Formal Expression.

Discussion (credit-assignment-problem)

The Credit Assignment Problem for Strategy DAGs.

Given strategy DAG $\Sigma_t = (V, E, p, \gamma)$, an observed outcome at the root (and possibly at some intermediate nodes), produce per-edge signals $\text{signal}(o_t, i, j)$ for each edge $(i, j) \in E$ such that the edge update

$$p_{ij}^{\text{new}} = p_{ij} + \eta_{\text{edge}} \cdot (\text{signal}(o_t, i, j) - p_{ij}) \tag{9.15}$$

drives credences toward truth. The problem is trivial when all intermediates are observable (each edge updates from its own observation) and genuinely hard when intermediates are unobservable and the DAG has shared structure.

Default Signal Function (Gradient-Based Attribution, Regime-Aware).

Formulation (gradient-signal-function)

Any edge-update signal function decomposes along three independent axes: *what happened* at the child node, *whether that outcome is attributable* to the specific edge being updated, and *how causal* the evidence is:

$$\text{signal}(o_t, i, j) = f\big(\text{outcome}(o_t, j),\ \text{attribution}(o_t, i, j),\ \text{regime}(i, j)\big) \tag{9.16}$$

The outcome component answers "what was observed at j?" The attribution component answers "can we credit that outcome to edge (i, j) specifically, or did other parents contribute?" The regime component answers "how causal is the evidence — is it interventional (Regime A), partially identified (Regime B), or observational (Regime C)?" This decomposition is derived from the regime-indexed edge semantics of 8.3 and 9.5: the same signal pipeline must carry the regime distinction through to the update.

AAT's default implementation, analogous to $\eta^* = U_M/(U_M+U_o)$ being the default gain. The update is stated in the log-odds coordinate $\lambda_k = \log(p_k/(1-p_k))$ — the unique additive-evidence coordinate forced by the evidential-additivity axiom (`#deriv-edge-update-natural-parameter`). The probability-space form is the projected image via $p_k = \sigma(\lambda_k)$ at the readout interface:

$$\lambda_k^{\text{new}} = \lambda_k + \eta_{\text{edge}} \cdot \iota_k \cdot \frac{J_k \cdot (y_G - \hat{P}_\Sigma)}{\|\mathbf{J}\|^2}, \qquad p_k^{\text{new}} = \sigma(\lambda_k^{\text{new}}) \tag{9.17}$$

where: - $\mathbf{J} = \nabla_{\mathbf{p}} P_\Sigma$ is the plan-value gradient (computable from status propagation in $O(|V| + |E|)$) — supplying the *attribution* component - $(y_G - \hat{P}_\Sigma)$ is the plan-level residual — the *outcome* component - $\iota_k \in [0, 1]$ is the edge's identifiability coefficient (9.5) — the *regime* modulation, scaling the correction by the fraction of evidence that genuinely identifies this edge's causal effect - η_{edge} is the edge-level update gain (9.4); for Beta-Bernoulli the moment-parameter form recovers $\eta_{\text{edge}} = 1/(n_k + 1)$ under the sufficient-statistic correspondence

For Regime-A edges ($\iota_k \approx 1$), the log-odds update recovers the pure gradient-based form $\Delta\lambda_k = \eta_{\text{edge}} \cdot J_k(y_G - \hat{P}_\Sigma)/\|\mathbf{J}\|^2$. For Regime-C edges ($\iota_k \approx 0$), the update is essentially zero — no meaningful update is made because no meaningful causal evidence is available. Regime-B edges receive proportionally reduced updates that honestly reflect the weaker identification.

Why log-odds rather than probability-space. In probability space, the same update written as $\text{signal}_k - p_k$ can push p_k^{new} outside $[0, 1]$ when $\|\mathbf{J}\|^2$ is small — a mechanical break of the $[0, 1]$ probability domain. The log-odds coordinate has domain $\mathbb{R}$, so additive updates never escape the domain; the sigmoid projection at the readout interface guarantees $p_k^{\text{new}} \in (0, 1)$ by construction. The log-odds coordinate is the unique (up to positive affine transformation) parameterization on which Bayesian independent-evidence accumulation is additive (`#deriv-edge-update-natural-parameter`); this makes it the natural presentation for any continuous-gradient update rule that aims to preserve Bayesian coherence. The probability-space presentation remains useful for interpretation but is the projected image, not the native update coordinate.

Properties: - **Domain closure (no mechanical break):** Updates live on $\lambda_k \in \mathbb{R}$, so no update magnitude can push credence outside the valid probability domain. The sigmoid projection $p_k = \sigma(\lambda_k) \in (0, 1)$ at the readout interface guarantees $[0, 1]$ boundedness by construction — not by clipping. The historical probability-space presentation required a normalization constant and could diverge when $\|\mathbf{J}\|^2 \to 0$; the log-odds presentation eliminates this failure mode. - **Directional fidelity (B1):** For $\iota_k > 0$, satisfies $\mathbb{E}[\Delta\lambda_k] \propto \iota_k \cdot J_k(\Phi - \hat{P}_\Sigma) \propto \iota_k \cdot J_k \cdot \delta_s$. Since $J_k \geq 0$ for monotone AND/OR DAGs and $\iota_k \geq 0$ by definition, the expected log-odds update pushes each edge's credence toward truth whenever evidence is available to push it. The probability-space image inherits directional fidelity through the monotonic sigmoid. - **Sector parameter:** $\alpha_s = \iota_k \cdot \eta_{\text{edge}}$ for componentwise corrections (regime-adjusted Prop B.5b); $\alpha_s = \iota_k \cdot \eta_{\text{edge}}/\kappa(\mathbf{J})^2$ for coupled corrections. Regime-C edges have $\alpha_s \approx 0$, making them effectively frozen — consistent with 9.3's treatment of unobservable edges. The sector parameter is Fisher-equivalent across coordinates (see Epistemic Status); the probability-space and log-odds-space statements of Props B.1–B.7 carry the same content. - **Computational cost:** $O(|V| + |E|)$ — the same forward pass that computes $\hat{P}_\Sigma$ also yields $\mathbf{J}$. The ι factors are per-edge domain parameters, not computed from the DAG structure. The sigmoid projection is $O(|E|)$ per update step. - **Relationship to RL:** This is the AAT analog of REINFORCE with a causal-identification weighting — the Jacobian $\mathbf{J}$ is the score function, $(y_G - \hat{P}_\Sigma)$ is the advantage, and ι_k is the causal-validity discount on each edge's update.

Correlated-failure interaction (L0 vs L1). The gradient signal operates at L0 of the Correlation Hierarchy (8.3). When the DAG is causally insufficient (the dominant real-world case), the residual $(y_G - \hat{P}_\Sigma)$ decomposes into per-edge miscalibration *plus* omitted correlation structure. $\hat{P}_\Sigma$ systematically overestimates success, making the residual systematically negative on failure. The gradient signal then attributes to individual edges what is actually causal insufficiency (missing common-cause nodes). The signal retains directional fidelity *on average* (it pushes edges downward when the plan is overconfident, which is the correct direction), but the per-edge attribution is contaminated. The principled fix is L1: add common-cause nodes to restore causal sufficiency, then apply gradient attribution to the augmented DAG. In the augmented DAG, the residual correctly decomposes into per-edge miscalibration because the correlation structure is explicitly represented.

Domains with richer observation structure can do better (Thompson sampling, full belief propagation, domain-specific attribution). The gradient-based signal is the *concrete Level 1 default* — the minimum viable credit-assignment scheme that satisfies the theory's requirements.

What the Theory Can Guarantee Without Solving Credit Assignment. Three results hold independently of any specific credit-assignment scheme:

1. Persistence is credit-assignment-free. Proposition B.5 in `#deriv-edge-credence-dynamics` shows that the sector condition transfers from per-edge credence space to **strategy-plan-confidence error** $\delta_s = \hat{P}_\Sigma - \Phi$ via the Jacobian $\mathbf{J} = \nabla_{\mathbf{p}} P_\Sigma$. The Jacobian is computable from status propagation in $O(|V|+|E|)$ — no outcome decomposition required. The persistence guarantee (whether the strategy's plan-level self-assessment can be maintained) does not depend on the agent's ability to attribute outcomes to edges. **Note:** this proves persistence of δ_s (strategy-plan-confidence error), not of $\delta_{\text{strategic}}$ (the per-edge calibration residual from 9.1). Extending persistence to $\delta_{\text{strategic}}$ requires solving the credit-assignment problem — that is the gap this segment characterizes.

2. The diagnostic framework is plan-level. The satisfaction gap (8.4), control regret (8.5), and the orient cascade ordering (10.1) operate on aggregate value, not per-edge quantities. They tell the agent *whether* the strategy is failing (and whether the failure is feasibility vs. optimality vs. calibration), without requiring per-edge attribution.

3. Observability-dominance identifies the tractable edges. 9.3 determines which edges have nonzero observability — only these can receive informative signals. Edges with zero observability are frozen regardless of the credit-assignment scheme. The tractable boundary is the observable subgraph of Σ_t.

Discussion (tractable-credit-assignment)

The Tractable Cases.

Credit assignment is solved (exact, polynomial-time) when:

Table 9.4

Condition	*Why tractable*	*Update rule*
All intermediates observable	Each edge has its own observation; updates decouple	Beta-Bernoulli per edge (Prop B.2)
Binary outcomes, independent edges, linear chain	Marginal Bayesian update = proportional blame	Prop B.3 (with plan-level fallback for unobservable)
Tree DAG, observable leaves	No shared descendants; message passing is exact	Belief propagation (standard)

The Intractable Cases: Three Independent Barriers.

Discussion (intractable-credit-assignment)

Exact per-edge attribution in general AND/OR DAGs with partial observability faces three independent barriers:

1. Computational intractability (#P-hardness). The "contribution of edge k to the observed outcome" has the form of a Shapley value over a cooperative game defined by the AND/OR propagation. Since AND/OR DAGs can represent any monotone Boolean function (including weighted threshold functions), and Shapley value computation for weighted voting games is #P-complete (Deng and Papadimitriou, 1994), exact attribution is #P-hard. *Caveat:* the reduction is to *exact* computation; approximate Shapley values are computable in polynomial time with sampling.

2. Information-theoretic underdetermination. When intermediates are unobservable, per-edge attribution is *underdetermined*, not just hard. The identifiable subspace has dimension bounded by the number of observable nodes:

$$\dim(\mathcal{I}(\mathcal{V}_{\text{obs}})) \leq |\mathcal{V}_{\text{obs}}| \tag{9.18}$$

When $|\mathcal{V}_{\text{obs}}| < |E|$ (fewer observable nodes than edges), some directions in θ-space are fundamentally unresolvable from the available data. Any attribution in the unidentifiable directions relies on prior beliefs, not evidence.

3. The posterior correlation barrier. Even for approximately identifiable cases, any factored representation (independent Beta posteriors per edge) necessarily discards the correlation introduced by failure at multi-parent nodes. The exact posterior complexity grows exponentially with the number of observed failures. The factored representation is an approximation by construction — coupled corrections are inherent to the problem, not an artifact of a bad algorithm.

The Design Requirement.

Discussion (credit-assignment-design-requirement)

The theory does not prescribe a specific credit-assignment scheme. It states what any scheme must satisfy for the persistence guarantees to hold:

Minimal requirement (from 4.3): The per-edge signal function must have **directional fidelity** — the expected update for each edge must point toward the true credence:

$$\mathbb{E}[(\text{signal}(o_t, i, j) - p_{ij}) \cdot (p_{ij} - \theta_{ij})] \leq 0 \tag{9.19}$$

(the expected correction is non-positively correlated with the current error). This is the per-component version of condition B1 from the bridge theorem. Any signal function satisfying this produces sector-satisfying corrections that transfer losslessly to value space (Prop B.5b, componentwise case).

Sufficient condition for persistence: Per-component directional fidelity + bounded gain ($\eta_{\text{edge}} > 0$). The theory guarantees persistence when these hold, regardless of how the signals are computed.

What's NOT required: Exact attribution, unbiased estimation, minimum-variance estimation, or optimality of any kind. The persistence guarantee is robust to approximation — a sloppy but directionally correct signal function still produces bounded strategic mismatch. The *quality* of the approximation affects the *tightness* of the persistence bound (how close R^*_Σ is to zero), not whether persistence holds at all.

Level	*Requirement*	*What it buys*	*Cost*

The Hierarchy of Credit Assignment Quality.

Table 9.5

Level	*Requirement*	*What it buys*	*Cost*
0 (none)	Plan-level tracking only	Persistence guarantee (Prop B.5)	No per-edge diagnostics
1 (directional)	Directional fidelity per edge	Persistence + rough per-edge diagnostics	Gradient computation $O(\|V\| + \|E\|)$
2 (approximate)	Proportional blame / expectation propagation	Persistence + per-edge diagnostics (with bias)	Factor-graph inference
3 (exact)	Full Bayesian posterior	Persistence + optimal per-edge calibration	#P-hard (general case)

AAT's formal guarantees require only Level 0. Practical agents need at least Level 1 for adaptive behavior — and the default signal function (above) provides a concrete Level 1 scheme. Level 2 is the sweet spot for most applications. Useful Level 2 factor-graph approximations include: exact Belief Propagation (BP) on tree or polytree cases, loopy BP or max-sum for MAP-style diagnosis, Expectation Propagation (EP) for approximate marginals, and structured variational methods only where common-cause structure is explicitly modeled. Level 3 is a mathematical ideal that is computationally unattainable in the general case.

Epistemic Status.

Mixed. The default signal function (gradient-based attribution, stated in log-odds) is a *formulation* — a concrete, well-motivated representational choice analogous to the gain principle's η^* formula. It satisfies directional fidelity for monotone AND/OR DAGs (derivable from Jacobian non-negativity). The log-odds presentation is *canonically selected* by the evidential-additivity axiom — the unique additive-evidence parameterization up to positive affine transformation (`#deriv-edge-update-natural-parameter`). The boundary characterization (tractable cases, intractability barriers, design requirement) is *discussion-grade*, with the intractability argument at sketch level and the design requirement derived from the bridge theorem (4.3, `#deriv-edge-credence-dynamics` Prop B.5).

Historical note on the presentation choice. An earlier probability-space presentation of the default signal function, written as

$$\text{signal}_k = p_k + \iota_k \cdot J_k(y_G - \hat{P}_\Sigma)/\|\mathbf{J}\|^2 \tag{9.20}$$

with the update $p_k^{\text{new}} = p_k + \eta(\text{signal}_k - p_k)$, exhibited a mechanical break: when $\|\mathbf{J}\|^2 \to 0$ the signal magnitude could push p_k^{new} outside $[0, 1]$. The log-odds presentation above eliminates this failure mode by construction. Props B.1–B.7 of `#deriv-edge-credence-dynamics` remain as stated — their sector-parameter content is Fisher-equivalent across parameterizations, and the moment-parameter form is retained for algebraic clarity. The current segment writes the *default* signal in log-odds (native coordinate), with the moment-parameter form understood as the projected image via sigmoid.

Max attainable: *conditional* — with a formal intractability reduction, the boundary characterization could be promoted. The design requirement is already exact (it follows from the bridge theorem). The log-odds presentation is *derived-conditional* (on the evidential-additivity axiom) via `#deriv-edge-update-natural-parameter`, structurally parallel to the reverse-KL uniqueness result under the chain-rule axiom in `#deriv-strategy-cost-regret-bound` §6.1.

Discussion.

AAT characterizes the structure, not the algorithm. The theory's contribution to credit assignment is not a solution but a characterization: when it's trivial (observable intermediates), when it's hard (general partial observability), what any solution must satisfy (directional fidelity), and what guarantees hold without it (persistence, plan-level diagnostics). This is the right level of ambition for a theory of adaptive systems — the specific algorithm is domain-specific engineering; the structural characterization is universal.

The analogy to the gain principle. 3.6 characterizes the optimal gain structure ($\eta^* = U_M/(U_M + U_o)$) without prescribing how U_M and U_o are estimated. A Kalman filter computes them exactly; an RL agent approximates them; a human intuits them. The gain *principle* is theory; the gain *estimator* is engineering. Credit assignment has the same structure: the *requirement* (directional fidelity) is theory; the *implementation* (proportional blame, gradient attribution, belief propagation) is engineering.

The observability lever. The most powerful insight from this analysis is that credit assignment is primarily an *observability design problem*, not an algorithm design problem. An agent that designs its strategy with observable intermediates (instrumented plans, staged rollouts, checkpoints) sidesteps the intractability entirely. This connects to 9.3's practical guidance: invest in making intermediate states observable, because unobservable regions are both epistemically dead (frozen credences) and computationally intractable (no efficient attribution).

OKRs as observability-by-design. The Objectives and Key Results framework is a direct organizational instantiation of this principle. The OKR discipline converts a deep, partially-unobservable strategy DAG (objective → vague initiatives → daily actions) into one with explicitly observable intermediate nodes (objective → measurable Key Results → tracked initiatives). In AAT terms: Key Results are intermediate nodes with $\sigma_v \approx 1$ by construction, making credit assignment between actions and objectives componentwise (Prop B.2) rather than intractable.

OKR failure modes map to AAT predictions:

Table 9.6

OKR Failure	*AAT Analog*	*Formal Quantity*	*Consequence*
Vanity metrics (measurable but irrelevant)	Observable node not causally connected to objective	High σ_v, low p_{ij}	Edge updates, but the edge doesn't lead where the agent thinks — correction effort is wasted
Too many Key Results	Wide OR-node exploration-gating	$\alpha_\Sigma \propto 1/k$ (Prop B.4)	Correction capacity diluted across alternatives; persistence threshold harder to meet
Lagging indicators	Evidence starvation by delay	$\nu_{\text{obs}} \ll \rho$	By the time the KR is measured, the correction window has passed; mismatch accumulated beyond R_Σ
Goodhart's Law (metric becomes the goal)	Terminal-condition misalignment with O_t	$V_{O_t}(\tau) < V_{O_t}^{\min}$ despite terminals achieved	Well-formedness constraint (8.3) violated; agent optimizes the intermediate rather than the objective

This is not an analogy — it is a domain instantiation. The same formal machinery that predicts when strategic persistence holds also predicts when OKRs work: when Key Results are genuinely causally connected to the Objective (p_{ij} is high and calibrated), few enough to monitor effectively (k is small), and measured on a timescale that permits correction (observation rate exceeds environment drift rate). The TST bridge to software team dynamics runs through exactly this connection.

9.7 *Formulation:* Structural Change as Parametric Limit

A formulation move that dissolves the sharp line between parametric update (adjusting weights) and structural change (adding/removing edges). In the probabilistic DAG, "structural" changes to Σ_t are *continuous operations on edge weights and node sets* — not a separate mechanism. Pruning is a credence dropping below threshold; **strategic grafting** is a new causal hypothesis initialized at a prior. The framework presents six operations on the strategy DAG ordered by typical frequency, from most to least frequent: edge reweighting, γ reclassification (AND↔OR), pruning, strategic grafting, objective revision, and full restructure. A healthy agent does continuous strategic maintenance (reweight, occasionally prune and graft) and rarely reaches catastrophic restructuring. *Full restructure is the strategic analog of 4.8's model-class change* — the rare, expensive event when the entire representational structure must be replaced. *(The unqualified term "grafting" — familiar from horticulture, graph rewriting, and decision-tree learning — is sanctioned in-segment shorthand for "strategic grafting" once the canonical compound has been introduced.)*

The key insight is that this extreme case is the *limit* of the continuous process, not a separate mechanism. An agent that maintains healthy strategic hygiene rarely needs full restructure; an agent that neglects maintenance — letting failing branches persist, ignoring negative evidence — accumulates *strategic debt* until catastrophic restructuring becomes unavoidable. Miller (2022, *Ex Machina*) provides a constructive existence proof for exactly this bridge in finite-state automata: a sequence of *neutral mutations* (alterations affecting only inaccessible regions of state space) can transform a Moore machine from one behavioral equivalence class to a structurally different one without ever changing its observable behavior — until a single additional mutation opens a transition to the restructured region, causing a radical behavioral change (the *extreme transition motif*: neutral invasion → neutral drift → niche creation → cascading displacement → consolidation). The strategy-DAG analog: edges with near-zero credence ($p_{ij} \approx 0$) are the *latent structure* — they consume minimal cognitive cost (9.9) but represent alternative causal hypotheses that become load-bearing if circumstances change. An agent that prunes all low-credence edges for efficiency loses this latent diversity and becomes *brittle to regime change*.

The framework also distinguishes *within-agent* grafting from *cross-agent* grafting. Within-agent grafting incorporates a new causal hypothesis into the agent's own Σ_t (source: internal model, external information, or exploration — a stimulus rather than an integrating party). When the grafted structure originates in *another* agent and that agent's objective is also absorbed in the process, the mechanism is **symbiogenic composition** (12.3) — a bilateral cross-agent process treated separately in Part III. Biological example: mitochondria integrating into a host cell. Social analog: firm acquisitions integrate processes and decision-making authority simultaneously.

Formal Expression.

Formulation (structural-change-as-parametric-limit)

The six operations on Σ_t, ordered from most to least frequent:

Table 9.7

Operation	*What changes*	*Trigger*
Reweighting	Edge credence p_{ij}	New observation about the link (9.4)
γ reclassification	Node combination type AND↔OR	Strong structural evidence that combination semantics changed
Pruning	Remove failed branch ($p_{ij} \to\approx 0$)	Credence drops below viability threshold
Strategic grafting	Add new branch ($0 \to p_{ij}$)	Discovery of a new possible path (initialized at prior)
Objective revision	Change terminal nodes	Feasibility failure or opportunity (8.4)
Full restructure	Replace entire Σ_t	Catastrophic failure (4.8)

A healthy agent does continuous strategic maintenance (reweight, occasionally prune and graft) and rarely reaches catastrophic restructuring. Full restructure is the strategic analog of 4.8's model-class change — the rare, expensive event when the entire representational structure must be replaced.

Epistemic Status.

Robust qualitative. The continuity claim (structural change as parametric limit) is a property of the probabilistic representation: in a space where edges carry real-valued credences, adding or removing an edge is a boundary event, not a discontinuity. The frequency ordering of operations is an empirical pattern, not a derived result — it's consistent with the observation that deeper changes (restructuring > pruning > reweighting) require more evidence and are more costly.

Discussion.

Connection to structural-adaptation necessity. 4.8 describes the rare case when the entire model class must be replaced — the agent's representational framework is fundamentally inadequate. In the strategy DAG, this corresponds to full restructure: the causal theory encoded in Σ_t is so wrong that incremental revision (reweighting, pruning) cannot fix it. The DAG must be rebuilt from a different starting point.

The key insight is that this extreme case is the *limit* of the continuous process, not a separate mechanism. An agent that maintains healthy strategic hygiene (regular reweighting, timely pruning of failing branches, proactive grafting of alternatives) will rarely need full restructure. An agent that neglects maintenance — letting failing branches persist, ignoring negative evidence — accumulates strategic debt until catastrophic restructuring becomes unavoidable.

Neutral variation as the constructive bridge. This segment claims that radical restructuring is the *limit* of continuous operations, but the mechanism by which small changes accumulate into radical transformation was unspecified. Miller (2022, *Ex Machina*) provides a constructive existence proof for exactly this bridge in finite-state automata. In Moore machines, each state has accessible and inaccessible regions. A mutation that alters a transition into an inaccessible region is *neutral* — the machine's observable behavior is unchanged because those states are never visited. But such mutations alter the machine's *latent structure*: they change what the machine *would do* if novel inputs were encountered. A sequence of neutral mutations can transform a machine from one behavioral equivalence class to a structurally different one without ever changing its observable behavior — until a single additional mutation opens a transition to the restructured region, causing a radical behavioral change. In Miller's terminology, this is the *extreme transition motif*: neutral invasion → neutral drift → niche creation → cascading displacement → consolidation. The automaton setting makes the mechanism precise: "inaccessible states" are the formal representation of latent structural capacity, "neutral mutations" are the incremental changes, and "a transition opening to a previously inaccessible region" is the parametric-limit event where continuous change produces discontinuous behavioral effect. The strategy-DAG analog: edges with near-zero credence ($p_{ij} \approx 0$) are the latent structure. They consume minimal cognitive cost (9.9) but represent alternative causal hypotheses that become load-bearing if circumstances change. An agent that prunes all low-credence edges for efficiency loses this latent diversity and becomes brittle to regime change.

Pruning and grafting thresholds. When should the agent prune (remove an edge with very low credence) rather than just keeping it at low p_{ij}? The answer involves the cognitive cost of maintaining edges in Σ_t — each edge consumes representational capacity and evaluation time. For agents with bounded representational capacity (LLMs with finite context windows), pruning is necessary to keep Σ_t within capacity. The threshold is domain-dependent and connects to the open question of cognitive cost of Σ_t (6.7 Working Notes).

Cross-agent grafting: symbiogenic composition. The grafting operation above is within-agent — the agent incorporates a new causal hypothesis into its own Σ_t, with the source (internal model, external information, exploration) being a stimulus rather than an integrating party. When the grafted structure originates in *another* agent and that agent's objective is also absorbed in the process, the mechanism is symbiogenic composition (12.3) — a bilateral cross-agent process rather than a unilateral within-agent operation. Biological example: mitochondria integrating into a host cell transfers gene-based structure (grafting on the host side) *and* subsumes the endosymbiont's independent objective (outside the scope of this segment's operations). Social analog: firm acquisitions integrate processes and decision-making authority simultaneously. The within-agent grafting operation is the host-side component of symbiogenesis, but symbiogenesis as a whole is governed by additional dynamics captured in 12.3.

9.8 *Definition:* Strategic Tempo

The strategic analog of 3.8. **Strategic tempo** $\mathcal{T}_\Sigma$ is the effective rate at which an agent acquires useful revisions to its strategy Σ_t — the sum of per-edge correction capacities across the strategy DAG, with each edge's contribution weighted by an *identifiability coefficient* ι_{ij} that captures whether the evidence stream genuinely identifies the edge's causal effect. The structural parallel with epistemic tempo is exact: edge observation rate ν_{ij} times per-edge update gain $\eta_{\text{edge},ij}$ times identifiability coefficient ι_{ij}, summed over edges. The ι factor is where edge-causal-validity (9.5) enters the operational machinery of strategy revision: *an agent cannot improve the parts of its strategy that it cannot test interventionally.* Regime-A (intervention-rich) edges contribute full tempo at their observation rate; Regime-C (observation-only) edges contribute essentially nothing regardless of how fast the agent acts or how many observations it makes.

A *key structural difference* from epistemic tempo: edge rates are **endogenous**. Epistemic-tempo channel rates $\nu^{(k)}$ are largely exogenous — the environment generates observations at its own pace. Strategic-tempo edge rates ν_{ij} depend on the agent's *action policy* (which edges get tested) and on *upstream success* (downstream edges are tested only when upstream edges fire). This endogeneity is the source of the structural differences between epistemic and strategic persistence. The framework decomposes strategic tempo by topology. **AND-chains are depth-gated**: each edge at depth k is tested at the upstream-success-attenuated rate $\nu_k = \nu \prod_{j<k} \theta_j$. For a uniform chain ($\theta_k = \theta$), total strategic tempo $\mathcal{T}_\Sigma$ converges to $\nu/((n+1)(1-\theta))$ as $d \to \infty$ — total strategic tempo is bounded even for arbitrarily deep chains, but the marginal contribution of each new edge decays geometrically. The deep edges are *evidence-starved*. **OR-nodes are exploration-gated**: under ε-greedy, the rate allocated to the greedy arm gets nearly all the action budget while exploratory arms get only ε/m. Pure greedy makes non-greedy arms *permanently uncorrectable*.

Per-edge persistence inherits the same weakest-link structure as per-dimension epistemic persistence (16.5): for Σ_t to persist, every edge must maintain bounded mismatch — $\forall (i,j) \in E : \quad \nu_{ij} \cdot \iota_{ij} \cdot \eta_{\text{edge},ij} > \rho_{\Sigma,ij}/R_{\Sigma,ij}$. The aggregate $\mathcal{T}_\Sigma$ is at most the sum and at least the minimum across edges; exceeding total disturbance is *necessary but not sufficient*. *Persistence is bottleneck-limited by the weakest edge, not governed by the aggregate.* Scalar $\mathcal{T}_\Sigma$ overestimates effective strategic adaptation for the same reason scalar $\mathcal{T}$ overestimates epistemic adaptation — it averages over heterogeneous correction capacities. A practical consequence: given a fixed action budget, the topology that maximizes strategic tempo is *shallow and OR-heavy* (more edges directly observable, no attenuation); the topology that minimizes it is *deep AND-chains*. As observations accumulate per edge, $\eta_{\text{edge},ij}$ shrinks (diminishing returns), so $\mathcal{T}_\Sigma$ *declines over time even in a static environment*.

Formal Expression.

Definition (strategic-tempo)

$$\mathcal{T}_\Sigma = \sum_{(i,j)\in E} \nu_{ij} \cdot \eta_{\text{edge},ij} \cdot \iota_{ij} \tag{9.21}$$

where: - (i,j) indexes the edges of the strategy DAG (8.3) - ν_{ij} is the effective observation rate for edge (i,j) — how often the agent obtains evidence about the causal link $i \to j$ - $\eta_{\text{edge},ij}$ is the per-edge update gain (9.4) - $\iota_{ij} \in [0,1]$ is the identifiability coefficient (9.5): the fraction of the evidence stream that genuinely identifies the edge's causal effect

Regime contributions. The identifiability coefficient captures the regime distinction from 9.5:

Table 9.8

Regime	ι_{ij}	*Contribution to* $\mathcal{T}_\Sigma$	*Example domain*
A Intervention-rich	≈ 1	Full: $\nu_{ij} \cdot \eta_{\text{edge},ij}$	Software tests, laboratory experiments
B Partial intervention	$\in (0,1)$	Reduced proportionally	Organizational actions with concurrent effects
C Observation-only	≈ 0	Near-zero: edges contribute negligibly	Passive monitoring, intelligence analysis

An agent cannot improve the parts of its strategy that it cannot test interventionally. This is the operational content of the ι factor: Regime-C edges contribute essentially nothing to $\mathcal{T}_\Sigma$ regardless of how fast the agent acts or how many observations it makes. Regime-A edges yield full strategic tempo at their observation rate. The ι factor is where edge-causal-validity (9.5) enters the operational machinery of strategy revision.

Parallel with epistemic tempo. The definition mirrors 3.8's $\mathcal{T} = \sum_k \nu^{(k)} \cdot \eta^{(k)*}$, replacing observation channels with strategy edges and adding the ι modulation for causal identifiability. The structural parallel is exact for Regime-A edges (where $\iota = 1$ recovers the direct rate-times-gain form); in mixed regimes, ι carries the additional content distinguishing interventional evidence from associational proxy.

Key difference: endogenous edge rates. Epistemic tempo's channel rates $\nu^{(k)}$ are largely exogenous — the environment generates observations at its own pace. Strategic tempo's edge rates ν_{ij} are *endogenous*: they depend on the agent's action policy (which edges get tested) and on upstream success (downstream edges are tested only when upstream edges fire). This endogeneity is the source of the structural differences between epistemic and strategic persistence.

Consistency verification. The definition is consistent with the four verified topologies from `#deriv-edge-credence-dynamics`:

Case B.1 (single edge $A \to G$). One edge, $\nu = \nu_{AG}$, $\eta_{\text{edge}} = 1/(n+1)$. $\mathcal{T}_\Sigma = \nu_{AG}/(n+1)$. The sector parameter $\alpha_\Sigma = 1/(n+1)$ is the per-observation correction quality; $\mathcal{T}_\Sigma = \nu \cdot \alpha_\Sigma$, matching the epistemic tempo pattern exactly.

Case B.2 (two-edge AND chain, $A \to B \to G$, B observable). Two edges. Edge 1 is tested at rate $\nu_1 = \nu$ (every execution). Edge 2 is tested only when edge 1 succeeds: $\nu_2 = \nu \cdot \theta_1$. The bottleneck edge has $\alpha_\Sigma = \min(1/(n_1 + 1),\ \theta_1/(n_2 + 1))$. $\mathcal{T}_\Sigma = \nu/(n_1 + 1) + \nu\theta_1/(n_2 + 1)$, consistent with depth-gated attenuation.

Case B.3 (two-edge AND chain, B unobservable). Per-edge tempo is ill-defined (the marginal point estimate is biased). Plan-level tempo is well-defined: $\mathcal{T}_{\Sigma,\text{plan}} = \nu/(n_\Phi + 1)$, treating $\hat{\Phi} = p_1 p_2$ as a single tracked quantity.

Case B.4 (two-arm OR node, ε-greedy). Edge 1 tested at rate $\nu_1 = \nu(1-\varepsilon)$, edge 2 at rate $\nu_2 = \nu\varepsilon$. $\mathcal{T}_\Sigma = \nu(1-\varepsilon)/(n_1+1) + \nu\varepsilon/(n_2+1)$. Action selection directly controls the rate allocation — exploration-gated, not depth-gated.

Structural decomposition. **AND-chains: depth-gated (geometric attenuation).** In a chain of depth d with edge success probabilities θ_k, the effective observation rate for edge k is:

Derived (Conditional on independent edges)

$$\nu_k = \nu \cdot \prod_{j<k} \theta_j \tag{9.22}$$

Each additional depth level attenuates by a factor $\theta_k < 1$. For a uniform chain ($\theta_k = \theta$, $n_k = n$ for all k):

$$\mathcal{T}_\Sigma = \frac{\nu}{n+1} \sum_{k=1}^{d} \theta^{k-1} = \frac{\nu}{n+1} \cdot \frac{1-\theta^d}{1-\theta} \tag{9.23}$$

This converges to $\nu/((n+1)(1-\theta))$ as $d \to \infty$ — total strategic tempo is bounded even for arbitrarily deep chains. The marginal tempo contribution of edge k decays as θ^{k-1}, falling below any fixed threshold at depth d^* (9.9). Deep AND-chains have low $\mathcal{T}_\Sigma$ at their leaves regardless of how fast the agent acts — the evidence-starvation effect identified in `#deriv-edge-credence-dynamics`.

OR-nodes: exploration-gated. At an OR-node with m alternatives under ε-exploration, the rate allocated to alternative l is:

Definition (OR-node rate allocation)

$$\nu_l = \begin{cases} \nu(1-\varepsilon+\varepsilon/m) & l = l^* \text{ (greedy arm)} \\ \nu \cdot \varepsilon/m & l \neq l^* \text{ (exploratory arms)} \end{cases} \tag{9.24}$$

The bottleneck is the least-explored alternative. Pure greedy ($\varepsilon = 0$) gives $\nu_l = 0$ for non-greedy arms, making those edges permanently uncorrectable.

Derived (from persistence-condition applied per edge)

Per-edge persistence.

For Σ_t to persist, every edge must maintain bounded mismatch. The bottleneck condition is:

$$\forall (i, j) \in E: \quad \nu_{ij} \cdot \iota_{ij} \cdot \eta_{\text{edge},ij} > \frac{\rho_{\Sigma,ij}}{R_{\Sigma,ij}} \tag{9.25}$$

This is the per-edge analog of 16.5's per-dimension condition for M_t. The aggregate relationship between $\mathcal{T}_\Sigma$ and the average correction rate α_Σ is:

$$\alpha_\Sigma \leq \frac{\mathcal{T}_\Sigma}{|E|} \leq \mathcal{T}_\Sigma \tag{9.26}$$

(minimum ≤ average ≤ sum). Consequently, $\mathcal{T}_\Sigma > |E| \cdot \rho_\Sigma / R_\Sigma$ is *necessary* for persistence but not sufficient — the persistence condition is bottleneck-limited by the weakest edge, not governed by the aggregate.

Epistemic Status.

The definition itself is axiomatic — it names a quantity by analogy with 3.8. The consistency verification with the four cases from `#deriv-edge-credence-dynamics` is *derived* (conditional on Beta-Bernoulli dynamics). The AND-chain geometric attenuation is *derived* (conditional on independent edge outcomes). The OR-node exploration gating and identifiability adjustment are *hypotheses* in the general DAG case, though verified for the specific topologies above. The bottleneck-limited persistence observation is *derived* from 16.5's result applied to the strategy substate. The additive form also inherits the noise-correlation scope derived in `#deriv-tempo-additivity`: the edge-wise sum is exact when the edges' evidence-noise is cross-edge independent, and the deviation under correlated evidence (e.g., one observation informing several edges, or a shared attribution error contaminating a path) is *signed* — a common evidence source overcounts, while structurally anti-correlated evidence errors can make the sum an undercount. The per-edge topology conditions named above are independence assumptions of this kind.

Max attainable: conditional. Currently conditional because the general DAG case (mixed AND/OR topologies, correlated edges) has not been verified.

Discussion.

Connection to 16.5. The per-edge persistence condition inherits the same structure as the per-dimension epistemic result: the weak edge is the bottleneck. Scalar $\mathcal{T}_\Sigma$ overestimates effective strategic adaptation for the same reason scalar $\mathcal{T}$ overestimates epistemic adaptation — it averages over heterogeneous correction capacities. A topology-specific corollary *[Discussion]*: per-edge persistence (every edge calibrated) and plan-level persistence (the plan's value estimate tracks reality) can diverge in a *stationary* environment only at OR-nodes — a starved exploratory arm drifts uncalibrated while the greedy arm carries the plan value — whereas AND-structures cannot exhibit the divergence, since every edge on the conjunctive path enters the plan value and an uncalibrated edge shows up there directly.

Three-way tradeoff. Strategic tempo competes with both epistemic tempo and exploitation for the agent's finite action capacity. Each action that tests a strategy edge (improving $\mathcal{T}_\Sigma$) is an action not spent gathering epistemic information (improving $\mathcal{T}$) or pursuing the current best action (exploitation). The allocation is addressed in 10.2 — the extended deliberation threshold is derived, but the broader three-way framing is discussion-grade. Deliberation (internal computation) supplements $\mathcal{T}_\Sigma$ via an internal channel distinct from action-generated evidence.

Software as Regime A. In software development, tests are genuine interventions with attributable outcomes ($\iota_{ij} \approx 1$). This makes software agents naturally high-$\mathcal{T}_\Sigma$ — a structural advantage identified but not yet formalized in the TST domain (`02-tst-core/`).

9.9 *Formulation:* Cognitive Cost of Strategy

A formulation connecting strategy structure to its *maintenance cost* — making the explicit-strategy condition from 7.6 quantitative. The minimum description length of a strategy DAG $\Sigma_t = (V, E, p, \gamma)$ decomposes into *structural* bits (encoding topology — node identities, edge connectivity, AND/OR labels) and *parameter* bits (encoding the edge credences given the topology), scaling as $O(|E| \log|V|)$ for sparse DAGs and moderate-precision credences.

The framework offers a **strategy IB objective**: Σ_t is the information-bottleneck compression of the interaction history $\mathcal{C}_t$ *for guidance*, parallel to M_t as the IB-compression of $\mathcal{C}_t$ *for prediction*. The variational form trades off compression cost $I(\mathcal{C}_t; \Sigma_t)$ against a KL divergence from the optimal-policy reference to the strategy-induced policy. The **KL direction** is *not* a stipulation — it is forced by a regret-bound derivation (`#deriv-strategy-cost-regret-bound`): under bounded value range and deterministic $\pi^* = \delta_{a^*}$ (AAT's canonical scope, 6.6), the strategy-induced regret satisfies $R(Q_{\Sigma_t}) \leq V_{\max} \cdot \mathrm{TV}(\pi^*, Q_{\Sigma_t}) \leq V_{\max}\sqrt{\frac{1}{2} D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}$ by Pinsker, with the sharper Bretagnolle-Huber identity $R \leq V_{\max}(1 - e^{-D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})})$ as the primary form. The opposite-direction KL is $+\infty$ whenever Q_{Σ_t} places any mass off a^* — a vacuous bound. The direction is *forced* with π^* first (reverse-KL in variational-inference vocabulary). Within the direction-forced family, reverse-KL is *uniquely* selected by a chain-rule additivity axiom (Hobson 1969; Csiszár 1991), AAT-internally motivated as the divergence-level analog of 8.1's log-additive layer. This is the **second instance of the four-layer additive-coordinate-forcing meta-pattern** (5.5): a uniqueness theorem operating on an AAT-internally-motivated additivity axiom forces a specific coordinate. The whole construction lifts the IB objective from discussion-grade to *robust-qualitative*.

The framework derives a **maximum useful chain depth** for AND-chains: given base observation rate ν, per-edge success probability θ, effective sample size n, and disturbance-to-capacity ratio ρ_Σ / R_Σ, there is a finite depth d^* beyond which the deepest edge cannot maintain bounded mismatch. Beyond this depth, evidence starvation makes the edge uncorrectable faster than the environment invalidates it. A central observation: **deep AND-chains suffer a triple depth penalty** that compounds. (1) *Confidence decay* (8.1): aggregate confidence $\prod p_k$ decays geometrically — *the plan is less likely to succeed*. (2) *Evidence starvation* (`#deriv-edge-credence-dynamics`): effective observation rate decays geometrically — *the plan is harder to calibrate*. (3) *Cognitive cost* (this segment): each additional depth level adds $O(\log|V|)$ bits — *the plan is more expensive to maintain*. All three are multiplicative in depth, making deep sequential strategies exponentially costly along three independent dimensions and creating systematic pressure toward *shallow, OR-heavy strategies*. The framework also makes the maintenance cost from 7.6 concrete by decomposing it into representation cost (proportional to description length), revision cost (proportional to strategic tempo $\mathcal{T}_\Sigma$), and monitoring cost (proportional to edges with partial identifiability).

A practical consequence for language-constituted agents: a context window of W tokens imposes a hard upper bound $\mathrm{DL}(\Sigma_t) \leq W \log_2(|\mathrm{vocab}|)$, making the IB trade-off *non-optional* — the agent *must* compress its strategy, and d^* becomes a context-window-limited quantity (a 128K-token context may support a 500-edge DAG encoded in natural language; a 4K-token window may support only a 15-edge sketch). The Miller (2022) coevolving-automata result provides a complementary constraint from the interaction side: a complex Σ_t whose edges are only tested over a short horizon is *indistinguishable from a simpler strategy*. The interaction horizon compresses the strategy space from above, just as maintenance cost and evidence starvation compress it from below.

Formal Expression.

Strategy description length. *Formulation (strategy-description-length)*

The minimum description length of a strategy DAG $\Sigma_t = (V, E, p, \gamma)$ (8.3) decomposes as:

$$\mathrm{DL}(\Sigma_t) = \mathrm{DL}_{\mathrm{struct}}(G) + \mathrm{DL}_{\mathrm{param}}(p \mid G) \tag{9.27}$$

where: - $\mathrm{DL}_{\text{struct}}(G)$: bits to encode the DAG topology — node identities, edge connectivity, AND/OR labels γ. Scales as $O(|E| \log|V|)$ for sparse DAGs. - $\mathrm{DL}_{\text{param}}(p \mid G)$: bits to encode the edge credences given the topology. For Beta-distributed credences, each edge requires $O(\log n_{ij})$ bits where $n_{ij} = \alpha_{ij} + \beta_{ij}$ is the effective sample size.

The total scales as $O(|E| \log|V|)$ for moderate-precision credences, growing linearly in the number of edges and logarithmically in the number of nodes.

Formulation (strategy-IB-objective; KL-direction strengthened by regret bound — see Epistemic Status)

Strategy IB objective.

The optimal strategy complexity balances parsimony against decision-relevance. Σ_t is the IB-compression of the interaction history $\mathcal{C}_t$ *for guidance*, parallel to M_t as the IB-compression of $\mathcal{C}_t$ *for prediction* (`#disc-compression-operations` for the shared IB shape across AAT's compression operations, and for the relationship between the theoretical $I(\mathcal{C}_t; \Sigma_t)$ compression cost and the operational DL-based minimization below):

Theoretical form (variational). Σ_t is a tractable variational approximation of the optimal-policy posterior $Q^*(\pi \mid M_t)$. The strategy-cost objective:

$$\Sigma_t^* = \arg\min_{\Sigma_t} \Big[I(\mathcal{C}_t;\, \Sigma_t) \;+\; \beta_\Sigma \cdot D_{\mathrm{KL}}\big(\pi^*(\cdot \mid M_t) \,\big\|\, Q_{\Sigma_t}(\pi \mid M_t)\big) \Big] \tag{9.28}$$

where $Q_{\Sigma_t}(\pi \mid M_t)$ is the action distribution induced by the strategy DAG given the current model state, and $\pi^*(\cdot \mid M_t)$ is the optimal-policy reference. The KL direction — π^*-first — is forced by the regret-bound derivation (next paragraph); the opposite direction is vacuous under deterministic π^*.

Regret-bound derivation of KL direction. Under AAT's canonical scope, $\pi^* = \delta_{a^*}$ is deterministic (6.6). Define the strategy-induced regret against π^* as $R(Q_{\Sigma_t}) := V(a^*) - \mathbb{E}_{a \sim Q_{\Sigma_t}}[V(a)]$, where $V(a) = Q_O(M_t, a; \pi_{\text{cont}}, N_h)$ is the action-value (6.6, O_t induces V via 6.4). Under bounded value range $V_{\max} := \max_a V(a) - \min_a V(a)$:

$$R(Q_{\Sigma_t}) \;\leq\; V_{\max} \cdot (1 - Q_{\Sigma_t}(a^*)) \;=\; V_{\max} \cdot \mathrm{TV}(\pi^*, Q_{\Sigma_t}) \tag{9.29}$$

Applying Pinsker's inequality ($\mathrm{TV}(P, Q) \leq \sqrt{\frac{1}{2} D_{\mathrm{KL}}(P \| Q)}$) with $P = \pi^*$, $Q = Q_{\Sigma_t}$:

$$R(Q_{\Sigma_t}) \;\leq\; V_{\max} \cdot \sqrt{\frac{1}{2}\, D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})} \tag{9.30}$$

Under deterministic π^*, $D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t}) = -\log Q_{\Sigma_t}(a^*)$ — finite and graded whenever $Q_{\Sigma_t}(a^*) > 0$. The opposite-direction $D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*)$ equals $+\infty$ whenever Q_{Σ_t} places any mass off a^*, giving a vacuous bound. The regret-bound argument therefore **forces the KL direction** with π^* first. Within the direction-forced f-divergence family, reverse-KL is *uniquely* selected under the chain-rule additivity axiom (Hobson 1969; Csiszár 1991 Theorem 3 corollary and Theorem 5; standard functional-equation derivation per Aczél & Daróczy 1975), which is AAT-internally motivated as the divergence-level analog of additive log-confidence decay (8.1). See `#deriv-strategy-cost-regret-bound` §6.1 for the uniqueness theorem, §6.2 for secondary supporting characterizations (gradient-tractability, VI-alignment, MDL), and §7 for the linear-vs-square-root β_Σ trade-off.

The variational form is the strategy-layer analog of variational free energy minimization in active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020, "Active inference on discrete state-spaces," *J. Math. Psych.* 99; Parr & Pezzulo 2022, *Active Inference*, MIT Press). AAT borrows the variational form as the appropriate generalization of the Shannon-MI relevance term and now derives the direction of KL from an internal regret-bound argument — without committing to AI's preferences-as-priors encoding ($C(o) = \log P_{\text{pref}}(o)$; AAT's O_t remains a value functional on trajectories, 6.4) or to expected free energy as master objective (AAT's CIY-unified objective is a related but distinct decomposition; 7.5).

Operational form. Since $I(\mathcal{C}_t; \Sigma_t)$ is not computable in closed form for general DAG encodings, the operational minimization replaces the information cost with a description-length surrogate and the KL term with a sample-based estimate (a per-edge calibration discrepancy weighted by decision-relevance — see 9.6 for the gradient form):

$$\Sigma_t^* \approx \arg\min_{\Sigma_t} \left[\mathrm{DL}(\Sigma_t) + \beta_\Sigma \cdot \widehat{D_{\mathrm{KL}}}(\pi^* \parallel Q_{\Sigma_t}) \right] \tag{9.31}$$

where: - $\mathrm{DL}(\Sigma_t)$: description length (coding-cost upper bound on $I(\mathcal{C}_t; \Sigma_t)$ for the given DAG encoding scheme — see §2.2 below) - $\widehat{D_{\mathrm{KL}}}(\pi^* \parallel Q_{\Sigma_t})$: sample-based estimate of the KL divergence from the optimal-policy reference to the strategy-induced policy - $\beta_\Sigma > 0$: trade-off parameter — cognitive cost per decision-relevant bit (the Σ_t instance of the shared β framework in `#disc-compression-operations`); under the regret-bound derivation, β_Σ has a *local* interpretation as $V_{\max}/(2\sqrt{2D_{\mathrm{KL}}})$ via the Pinsker form (the linear-KL form is the IB-shape instance; the square-root form is the tighter regret-scale form — see Epistemic Status and the spike for the trade-off)

The two forms agree in the limit where the DAG encoding is rate-distortion optimal and the policy posterior is sample-recoverable; the operational form is the one an agent actually runs. The theoretical form places the objective on the same variational frontier as M_t, shared intent, and composition projection, with the π^*-first KL-form relevance term resolving the Shannon-zero degeneracy under deterministic π^* *and* the forward-KL infinity degeneracy that the opposite direction would introduce.

When β_Σ is low (high maintenance cost relative to decision value), the agent prefers simple strategies. When β_Σ is high (strategy is cheap to maintain relative to its decision value), the agent can afford complex plans. The explicit strategy condition (7.6) is the binary threshold: β_Σ large enough that *any* Σ_t is worth maintaining.

Maintenum useful chain depth.

Derived (Conditional on Beta-Bernoulli, per-edge persistence)

From 9.8's per-edge persistence condition, an AND-chain of depth d with per-edge observation rate ν, true success probability θ per edge, and effective sample size n per edge persists only if the deepest edge satisfies:

$$\nu \cdot \theta^{d-1} \cdot \frac{1}{n+1} > \frac{\rho_\Sigma}{R_\Sigma} \tag{9.32}$$

Solving for the maximum depth at which persistence is achievable:

$$d^* = 1 + \left\lfloor \frac{\log\left(\frac{\nu}{(n+1)\rho_\Sigma/R_\Sigma}\right)}{\log(1/\theta)} \right\rfloor \tag{9.33}$$

When $\nu/((n+1)\rho_\Sigma/R_\Sigma) \leq 1$, even $d = 1$ fails — the agent cannot maintain a single edge under these conditions.

Interpretation. Beyond depth d^*, evidence starvation makes edges uncorrectable faster than the environment invalidates them. The agent accumulates strategic mismatch on deep edges regardless of how fast it acts at the top of the chain.

Quantitative illustration ($\theta = 0.8, \nu = 1$):

Table 9.9

n	ρ_Σ/R_Σ	d^*
10	0.01	10
10	0.1	0
100	0.01	0
100	0.1	0

High evidence requirements (n large) and volatile environments (ρ_Σ/R_Σ large) severely limit useful chain depth.

Triple depth penalty. Deep AND-chains suffer three independent penalties that compound:

1. **Confidence decay** (8.1): aggregate confidence $\prod p_k$ decays geometrically with depth. The plan is *less likely to succeed*.
2. **Evidence starvation** (`#deriv-edge-credence-dynamics`): effective observation rate $\nu_k = \nu \cdot \prod_{j<k} \theta_j$ decays geometrically. The plan is *harder to calibrate*.
3. **Cognitive cost** (this segment): each additional depth level adds $O(\log|V|)$ bits to description length. The plan is *more expensive to maintain*.

All three are multiplicative in depth, making deep sequential strategies exponentially costly along three independent dimensions.

Formulation (enriched-strategy-condition)

Enriched explicit strategy condition.

The maintenance cost C_{maintain} from 7.6 decomposes as:

$$C_{\text{maintain}} = C_{\text{represent}} + C_{\text{revise}} + C_{\text{monitor}} \tag{9.34}$$

where: - $C_{\text{represent}} \propto \text{DL}(\Sigma_t)$: cognitive cost of holding the strategy in working memory (proportional to description length) - $C_{\text{revise}} \propto \sum_{(i,j)} \nu_{ij} \cdot c_{\text{update}}$: cost of processing edge updates (proportional to strategic tempo $\mathcal{T}_\Sigma$ times per-update cost) - $C_{\text{monitor}} \propto |\{(i,j) : \iota_{ij} < 1\}|$: cost of monitoring edges with partial identifiability (the agent must do extra causal reasoning for non-trivial edges)

This decomposition makes the 7.6's maintenance term concrete: each component maps to a quantity defined elsewhere in the theory.

Discussion (complexity-compression)

Complexity compression operations.

The IB objective suggests three compression operations, corresponding to structural changes from 9.7:

1. **Edge pruning** (operation 3 in 9.7): remove edges with $\eta_{\text{edge},ij} \cdot I_{\text{edge},ij} < c_{\text{bit}}$, where $I_{\text{edge},ij}$ is the decision-relevance of that edge and c_{bit} is the per-bit maintenance cost. Edges that contribute less decision value than their representational cost are candidates for removal.
2. **Node merging** (reducing $|V|$): collapse intermediate nodes that serve no decision-distinguishing function. This reduces $\text{DL}_{\text{struct}}$ by a factor proportional to the reduction in $|V|$.
3. **Depth truncation** at d^*: prune all edges beyond the maximum useful depth. This is not optimization but necessity — edges beyond d^* cannot maintain bounded mismatch.

Epistemic Status.

The description length formulation is a *formulation* — it applies standard MDL to the strategy DAG, which is a representational choice not a derived necessity. The IB objective is *formulation (strengthened by regret bound)* in its variational form (above): the specific KL *direction* (π^*-first, i.e., reverse-KL in the variational-inference vocabulary) is *derived* as an upper regret bound via Pinsker's inequality under bounded value range and deterministic π^* (see Regret-bound derivation paragraph above; full derivation in appendix `#deriv-strategy-cost-regret-bound`). The choice of reverse-KL *within* the direction-forced family upgrades from canonical-formulation to **derived (conditional on chain-rule additivity axiom)** — the chain-rule axiom (Hobson 1969; Csiszár 1991 Theorem 3 corollary and Theorem 5; standard functional-equation derivation per Aczél & Daróczy 1975) picks reverse-KL uniquely among f-divergences, and the axiom is AAT-internally motivated as the divergence-level analog of 8.1 (see `#deriv-strategy-cost-regret-bound` §6.1). Secondary properties (gradient-tractability, variational-inference alignment with Friston et al. 2017, Da Costa et al. 2020, Parr & Pezzulo 2022; MDL coding; compatibility with Amari & Nagaoka 2000 Fisher geometry) are convergent evidence rather than independent uniqueness grounds — in particular, Fisher-metric-at-second-order is *not* distinguishing within f-divergences (Eguchi 1983, *Ann. Statist.* 11(3):793–803). The regret-bound derivation closes the direction ambiguity that the earlier V-medium move (commit `a14682e`) left open: the initial V-medium form used $D_{\mathrm{KL}}(Q_{\Sigma_t} \| \pi^*)$ (forward-KL), which is $+\infty$ under deterministic π^* whenever Q_{Σ_t} has any off-optimum mass — a different-valued but structurally identical degeneracy to the Shannon-MI zero it replaced. The π^*-first reverse-KL direction escapes both degeneracies by construction.

The variational form replaces an earlier Shannon-MI form $-\beta_\Sigma \cdot I(\Sigma_t; \pi^* \mid M_t)$ which had a Shannon-zero degeneracy: when π^* is a deterministic function of M_t (the standard scope), Shannon mutual information to a constant vanishes identically, collapsing the objective to $\arg\min \mathrm{DL}(\Sigma_t)$. The π^*-first KL form does not have this degeneracy and is graded whenever Q_{Σ_t} places any mass on a^*. The maximum useful depth d^* is *derived* conditional on Beta-Bernoulli dynamics and the per-edge persistence condition from 9.8. The triple depth penalty is an *observation* combining results from three independent segments. The enriched maintenance decomposition is *formulation*. The compression operations are *discussion-grade*.

On β_Σ interpretation. Under the Pinsker-reverse-KL bound $R(Q_{\Sigma_t}) \leq V_{\max}\sqrt{\frac{1}{2}D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})}$, a square-root-in-KL trade-off would naturalize β_Σ globally as $\beta_\Sigma \propto V_{\max}$. The segment retains the linear-in-KL form (to preserve the rate-distortion-Lagrangian IB shape shared with `#disc-compression-operations`); under the linear form, β_Σ has only a *local* regret-bound interpretation ($\partial R / \partial D_{\mathrm{KL}}$ at the operating point). This is a trade-off between IB-shape alignment and regret-scale naturalization. Under AAT's canonical deterministic-π^* scope, the sharper Bretagnolle-Huber identity $R \leq V_{\max}(1 - e^{-D_{\mathrm{KL}}(\pi^* \| Q_{\Sigma_t})})$ holds as the primary form (`#deriv-strategy-cost-regret-bound` §4); Pinsker is retained here for IB-shape alignment and as the loose general form correct for stochastic-π^* extensions.

Assumption explicitly stated: bounded value range. The regret-bound derivation requires $V_{\max} := \max_a V(a) - \min_a V(a) < \infty$ over $\mathcal{A}$ at fixed M_t. This is mild but not automatic — 6.4 specifies V_{O_t} : trajectories $\to \mathbb{R}$, and bounded range at fixed state is an additional assumption stated here.

Max attainable: *robust-qualitative* for the IB objective with the direction-forced derivation; conditional for the specific functional form (linear vs. square-root in KL). The DL formulation is standard; the depth bound could reach exact status for specific edge models. The regret-bound derivation does not extend to stochastic π^* (outside AAT canonical scope); see 6.6 continuation conventions for scope.

Discussion.

Connection to 7.6. The enriched maintenance decomposition gives the cost inequality quantitative content: an agent can now *compute* whether a proposed strategy is worth maintaining by estimating its description length, revision cost, and monitoring burden, and comparing against the exploration/repair cost of operating without it.

LLM context windows as DL constraint. For language-constituted agents (`03-llm-core/`), the strategy must fit in the context window. A context window of W tokens imposes $\mathrm{DL}(\Sigma_t) \leq W \cdot \log_2(|\mathrm{vocab}|)$ as a hard constraint. This makes the IB trade-off non-optional: the agent *must* compress its strategy, and the depth bound d^* becomes a context-window-limited quantity. A 128K-token context window may support a 500-edge DAG encoded in natural language; a 4K-token window may support only a 15-edge sketch.

Computational compression from interaction horizon (Miller 2022). The maximum useful depth d^* derived above constrains complexity from the *maintenance* side (evidence starvation). Miller (2022, *Ex Machina*, Table 12.2) provides a complementary constraint from the *interaction* side: the number of interaction rounds compresses the space of behaviorally distinguishable strategies regardless of the agent's internal complexity. For Moore machines with binary actions:

Table 9.10

Agent states	*Unique computations*	*After 1 round*	*After 2 rounds*	*After 4 rounds*
1	2	2	2	2
2	26	2	8	26
3	1,054	2	8	690
4	57,068	2	8	5,936

The pattern: effective complexity = min(agent complexity, interaction-horizon complexity). With only one round, even a four-state machine (57,068 unique computations) reduces to two distinguishable behaviors — equivalent to a one-state machine. With two rounds, all machines with two or more states reduce to eight distinguishable behaviors. The interaction horizon compresses the strategy space from above, just as the maintenance cost and evidence starvation compress it from below. For AAT's strategy DAG, the analog: a complex Σ_t whose edges are only tested over a short horizon gains nothing from its depth — the untested structure is indistinguishable from a simpler strategy. This reinforces the d^* bound and provides empirical grounding beyond the Beta-Bernoulli derivation.

Strategy simplification pressure. The triple depth penalty creates systematic pressure toward shallow, wide (OR-heavy) strategies over deep, sequential (AND-heavy) ones. This aligns with the structural pressure identified in 8.1's Discussion, now grounded in three independent mechanisms rather than one.

9.10 *Proposed Schema:* Proposed-schema: Strategy Persistence Schema

The strategic-layer instantiation of the sector-persistence template (`#result-sector-persistence-template`). The template proves bounded state for any system with a state variable, a correction function satisfying the sector condition, and bounded disturbance — it is domain-agnostic across its preconditions (T1)–(T3). The strategic instance asks whether the same machinery transfers to strategy update dynamics, and the answer is: *yes, but conditionally*, with the conditions tight enough that the schema's reachable persistence region has a λ-independent structural ceiling. The structural identity with the epistemic persistence condition from Part I is not an analogy; it is a *mathematical identity at the sector-framework level* — and the strategic analog of "the environment changes faster than the agent can learn" is "the world invalidates plans faster than the agent can revise them," producing the same catastrophic outcome.

This is `status: conditional`, conditioned on four named premises: (i) Beta-Bernoulli edge dynamics, (ii) exponential forgetting with discount $\lambda \in (0, 1)$, (iii) the template preconditions (T1)–(T3), and (iv) a mismatch state chosen from the per-edge credence error δ_c or the plan-level surrogate $\delta_s = \hat{P}_\Sigma - \Phi$ (credit-assignment-free; persistence proved in `#deriv-edge-credence-dynamics` Prop B.5). Within these conditions, the exact steady-state sector parameter is $\alpha_\Sigma^{ss} = (1 - \lambda)/(2 - \lambda)$, and the persistence prerequisite is $(1 - \lambda)/(2 - \lambda) > \rho_\Sigma/R_\Sigma$. The linear form $1 - \lambda$ is the slow-forgetting asymptote ($\lambda \to 1$), valid only in that limit; outside the asymptote it overestimates the steady-state by up to a factor of 2.

The strategic-side asymmetry from the epistemic case — and what forces the conditional tier — is **experience discounting as structural prerequisite**: the strategic sector parameter has the form $\alpha_\Sigma = 1/(n + 1)$ in raw Beta-Bernoulli, which decays monotonically with experience and drives any agent across the persistence threshold asymptotically regardless of disturbance rate. Exponential forgetting stabilizes the effective sample size at $1/(1 - \lambda)$, giving the exact

$(1-\lambda)/(2-\lambda)$ form. The forgetting prerequisite is therefore a *trajectory-guarantee requirement*, not a tunable hyperparameter — an agent without forgetting has no long-run strategic persistence. The schema's reachable persistence region under any λ is bounded above by a hard structural ceiling at $\rho_\Sigma = R_\Sigma/2$, derived self-contained in `#deriv-strategic-persistence-hard-ceiling`: when strategic disturbance reaches half the strategic reserve, no λ satisfies the prerequisite, and the schema's trajectory guarantee fails regardless of forgetting design. Strategic mismatch under the credit-assigned form $\delta_{\text{strategic}}$ (9.1) remains open (see Discussion §1 and 9.6).

Formal Expression.

Proposed Schema (strategy-persistence-schema, from sector-persistence-template)

If strategic update dynamics satisfy the template preconditions (T1)–(T3) of `#result-sector-persistence-template` for $\xi = \delta_\Sigma$ (a strategic mismatch state), together with:

- **(SA1)** Zero correction at zero strategic mismatch (the template's (T1)): when the mismatch state is zero, no revision occurs
- **(SA2')** Local sector condition on strategic correction (the template's (T2) for $\xi = \delta_\Sigma$): the correction function points inward with baseline efficiency α_Σ within a strategic reserve R_Σ
- **(SA3)** Sufficient exploration (OR-nodes only): the action selection policy allocates correction capacity to all OR alternatives at a rate exceeding the strategic disturbance-to-reserve ratio
- Bounded strategic disturbance at rate ρ_Σ (the template's (T3)): the rate at which the environment invalidates causal links is bounded

Then Σ_t persists iff:

$$\alpha_\Sigma > \frac{\rho_\Sigma}{R_\Sigma} \tag{9.35}$$

directly by the template's Model D result. Here α_Σ is the strategic correction rate, ρ_Σ is the strategic disturbance rate, and R_Σ is the strategic reserve (tolerance for strategic mismatch before performance degrades catastrophically). The Model S instantiation replaces ρ_Σ with σ_Σ and gives $\alpha_\Sigma > n\sigma_\Sigma^2/(2R_\Sigma^2)$ under the same template.

Forgetting as Prerequisite.

Formulation (forgetting-prerequisite)

The schema form above is an **instantaneous persistence check at the current experience level**, not a trajectory guarantee. For Beta-Bernoulli edge updates (the canonical verified case; see `#deriv-edge-credence-dynamics` Props B.1–B.6), the sector parameter has the form:

$$\alpha_\Sigma = \frac{1}{n+1} \tag{9.36}$$

where n is the edge's accumulated experience (pseudo-count). Without a forgetting mechanism, n grows monotonically with each observation, so $\alpha_\Sigma \to 0$ for every edge asymptotically. For any fixed (ρ_Σ, R_Σ) with $\rho_\Sigma > 0$, every agent eventually violates the threshold. The structural identity with 4.5 — where α can be stationary — holds for the strategic case only under an explicit forgetting mechanism.

Exponential forgetting. Replace the raw Beta-Bernoulli update with a discounted update: at each step, shrink the pseudo-counts by a factor $\lambda \in (0,1)$:

$$\alpha_k \mapsto \lambda\,\alpha_k + y_k, \qquad \beta_k \mapsto \lambda\,\beta_k + (1-y_k) \tag{9.37}$$

The effective sample size stabilizes at $n_{\text{eff}} = 1/(1-\lambda)$, and substituting into Prop B.1's $\alpha_\Sigma = 1/(n+1)$ gives the exact steady-state sector parameter:

$$\alpha_\Sigma^{\text{ss}} = \frac{1}{n_{\text{eff}}+1} = \frac{1-\lambda}{2-\lambda} \tag{9.38}$$

For slow forgetting ($\lambda \to 1$, the regime where the prerequisite is most likely to bind), the leading-order expansion gives the simpler form $\alpha_\Sigma^{\text{ss}} \approx 1-\lambda$ — which agrees with the forgetting rate itself. Outside the high-λ regime the approximation deteriorates: at $\lambda = 0.5$, the exact form gives $\alpha_\Sigma^{\text{ss}} = 1/3$ while the linear approximation gives $1/2$ ($\approx$ 50% overestimate); at $\lambda = 0.9$, exact $\alpha_\Sigma^{\text{ss}} = 1/11$ versus approximation $1/10$ ($\approx$ 10% overestimate).

The forgetting prerequisite. Combining with the schema's persistence form $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$:

$$\frac{1-\lambda}{2-\lambda} > \frac{\rho_\Sigma}{R_\Sigma} \quad\Longleftrightarrow\quad \lambda < \frac{R_\Sigma - 2\rho_\Sigma}{R_\Sigma - \rho_\Sigma} \tag{9.39}$$

(valid when $\rho_\Sigma < R_\Sigma/2$; for $\rho_\Sigma \geq R_\Sigma/2$ no λ satisfies the prerequisite and the schema's trajectory guarantee fails for any forgetting rate). The hard ceiling at $\rho_\Sigma = R_\Sigma/2$ and the algebraic content of the steady-state form are derived self-contained in `#deriv-strategic-persistence-hard-ceiling` — a λ-independent structural cap on the schema's reachable persistence region under any exponential-forgetting design.

This is a **prerequisite of the schema's trajectory guarantee, not a tunable heuristic**. An agent without forgetting has no long-run strategic persistence regardless of its initial α_Σ. The steady-state sector parameter must exceed the disturbance-to-reserve ratio, or the instantaneous persistence check — no matter how comfortably it holds at any given time — eventually fails as experience accumulates.

In the slow-forgetting regime the threshold simplifies to the linear-form analog of 4.5:

$$(1-\lambda) > \frac{\rho_\Sigma}{R_\Sigma} \qquad \text{(slow-forgetting limit, } \lambda \to 1) \tag{9.40}$$

playing the role that $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$ plays for the epistemic case. The forgetting rate $(1-\lambda)$ is the strategic analog of adaptive tempo: faster forgetting means faster tracking but noisier estimates; slower forgetting means stable estimates but slower tracking. The optimal λ balances bias and variance for the specific ρ_Σ the environment presents. Outside the slow-forgetting regime, the exact form $(1-\lambda)/(2-\lambda) > \rho_\Sigma/R_\Sigma$ is the operating threshold and the simpler linear form is unsafe (it permits λ values that violate the actual prerequisite).

Which mismatch state? The schema applies to any mismatch state for which conditions (SA1)-(SA3) can be verified. Two candidates exist:

- **Strategy-plan-confidence error** $\delta_s = \hat{P}_\Sigma - \Phi$: the scalar difference between the agent's strategy-plan-confidence score and the independence-model plan value at true edge parameters. This is the mismatch for which persistence IS proved (Props B.1-B.5 in `#deriv-edge-credence-dynamics`). It is computable from status propagation without credit assignment. **Scope:** δ_s operates at L0 of the Correlation Hierarchy (8.3) — it tracks calibration within the independence model. For L1 (augmented DAG), the same persistence result applies to the augmented graph's $\hat{P}_\Sigma$. The gap between L0's Φ and actual plan success under correlated failure is a model-class limitation, not an estimation error.
- **Strategic-calibration residual** $\delta_{\text{strategic}}$: the per-edge value-increment residual aggregation defined in 9.1. This is the mismatch the orient cascade (10.1) uses for edge-level revision. Persistence of $\delta_{\text{strategic}}$ remains **open** and requires the credit-assignment machinery in 9.6.

The verified instances below all use per-edge credence error $\delta_c = (\hat{p}_k - \theta_k)$ or the plan-level surrogate δ_s. They do not verify the schema for $\delta_{\text{strategic}}$ directly.

Epistemic Status.

Conditional, conditioned on (i) Beta-Bernoulli edge dynamics, (ii) exponential forgetting with $\lambda \in (0,1)$, (iii) the sector-persistence template preconditions (T1)–(T3) of `#result-sector-persistence-template`, and (iv) the mismatch state chosen from $\{\delta_c, \delta_s\}$ — the per-edge credence error or the plan-level surrogate (per the dependency on `#deriv-edge-credence-dynamics` Props B.1–B.6). Within these conditions the exact threshold $(1-\lambda)/(2-\lambda) > \rho_\Sigma/R_\Sigma$ and the hard ceiling at $\rho_\Sigma \geq R_\Sigma/2$ are algebraically exact — derived self-contained in `#deriv-strategic-persistence-hard-ceiling`. Strategic mismatch under $\delta_{\text{strategic}}$ (the credit-assigned form) remains open; see Discussion §1 and 9.6.

What was missing in the original sketch was instantiation — showing that specific strategic update dynamics satisfy the template's preconditions. Four cases have now been verified (full derivations in `#deriv-edge-credence-dynamics`):

1. **Single edge, Beta-Bernoulli** ($A \to G$): Sector condition satisfied globally with $\alpha_\Sigma = 1/(n+1)$. The bound is tight (expected correction is exactly linear). (A1) satisfied. Persistence condition: $1/(n+1) > \rho_\Sigma/R_\Sigma$.
2. **Two-edge chain, observable intermediate** ($A \to B \to G$, B observable): Sector condition satisfied globally with $\alpha_\Sigma = \min(1/(n_1+1), \theta_1/(n_2+1))$ — a weakest-link result. Correction function is diagonal (no cross-edge coupling). (A1) satisfied. The θ_1 factor in edge 2's rate is the evidence-starvation effect.
3. **Two-edge chain, unobservable intermediate** ($A \to B \to G$, B not observable): Per-edge sector condition **fails** — the marginal Bayesian update violates (A1) with bias $O(1/n)$. But plan-level tracking (treating $\hat{\Phi} = p_1 p_2$ as a single Beta) recovers the sector condition with $\alpha_{\Sigma,\text{plan}} = 1/(n_\Phi + 1)$, at the cost of per-edge diagnostic resolution.
4. **Two-arm OR-node, ε-greedy** ($A_1, A_2 \to G$, G is OR): Sector condition satisfied with $\alpha_\Sigma = \min((1-\varepsilon)/(n_1+1),\ \varepsilon/(n_2+1))$ — an **exploration-gated** weakest-link, not depth-gated as in AND chains. (SA3) required: minimum exploration rate $\varepsilon > \rho_\Sigma(n_{\max}+1)/R_\Sigma$. Pure greedy ($\varepsilon = 0$) **fails** the sector condition. With optimal equal-rate allocation, $\alpha_\Sigma = 1/(n_1 + n_2 + 2)$ — the correction capacity is split across alternatives.

The schema is no longer purely hypothetical. The sector parameter for strategic dynamics is the edge update gain η_{edge} — the same quantity that governs epistemic persistence. The structural parallel between epistemic and strategic persistence is not an analogy but a mathematical identity at the sector-framework level.

Discussion.

What's needed to promote this from schema to result.

1. **Strategic mismatch state**: partially resolved. Prop B.5 in `#deriv-edge-credence-dynamics` shows the sector condition transfers from per-edge credence error to **strategy-plan-confidence error** $\delta_s = \hat{P}_\Sigma - \Phi$ — the scalar difference between the agent's strategy-plan-confidence score and the independence-model plan value at true edge parameters (note: Φ is NOT actual plan success probability under correlated failure — see 8.3 edge-independence caveat). For linear correction (Beta-Bernoulli), the transfer is exact ($\alpha_s = \alpha_c$); for nonlinear correction, $\alpha_s \geq \alpha_c/\kappa(\mathbf{J})^2$. **However**, δ_s is distinct from the **strategic-calibration residual** $\delta_{\text{strategic}}$ defined in 9.1, which is an L^2 aggregation of per-edge value-increment residuals requiring credit assignment to compute. Persistence of δ_s (plan-level tracking) is proved; persistence of $\delta_{\text{strategic}}$ (per-edge diagnostics) remains open and requires the credit-assignment machinery in 9.6.
2. ~~**Strategic correction function**: needs to satisfy the sector condition.~~ **Resolved** for Beta-Bernoulli edges. Props B.1-B.4 in `#deriv-edge-credence-dynamics` verify the sector condition for four topologies (single edge, two-edge AND observable/unobservable, two-arm OR).
3. **Strategic disturbance**: The rate at which the environment invalidates causal links in Σ_t. **Still open** as a formalized quantity — currently a domain parameter (ρ_Σ), analogous to how ρ for epistemic disturbance is a domain parameter in 4.5.

4. ~~**Sector condition verification**: the critical mathematical step.~~ **Resolved** for four topologies. See `#deriv-edge-credence-dynamics`.

5. ~~**Credit assignment / signal function**: needed for edge updates.~~ **Characterized at the theory level.** 9.6 shows persistence does not require credit assignment (Prop B.5), establishes directional fidelity as the minimal requirement, and provides a gradient-based default signal function. The specific update algorithm is domain engineering, not theory — the same way the gain *estimator* is domain engineering while the gain *principle* ($\eta^* = U_M/(U_M + U_o)$) is theory. Caveat: the default gradient signal inherits $\hat{P}_\Sigma$'s overestimation bias under correlated failures (8.3, 9.6).

6. **Time-varying** α_Σ: this is where the strategic case genuinely differs from the epistemic one. For Beta-Bernoulli edges, $\alpha_\Sigma = 1/(n+1)$ decays monotonically with experience, so the sector-persistence template's constant-α precondition cannot hold asymptotically under any raw Bayesian update. The resolution is the **forgetting prerequisite** promoted into the Formal Expression above: exponential forgetting at rate λ stabilizes α_Σ at the exact value $(1-\lambda)/(2-\lambda)$, with the simpler linear form $1-\lambda$ as the slow-forgetting asymptote. The exact threshold $(1-\lambda)/(2-\lambda) > \rho_\Sigma/R_\Sigma$ converts the schema's instantaneous check into a trajectory guarantee. This is structural, not heuristic — without forgetting, the schema's form holds only until the agent accumulates enough experience to cross below threshold; with insufficient forgetting (linear-form approximation that treats ρ_Σ as small relative to R_Σ when in fact $\rho_\Sigma \geq R_\Sigma/2$), the prerequisite is unsatisfiable and the schema's trajectory guarantee fails.

The structural parallel is genuine, but conditional on forgetting. This schema extends *structural persistence* (see Persistence in `LEXICON.md`) from the epistemic substate to the strategy substate — asking whether the strategy correction machinery can outpace the rate at which the environment invalidates the agent's causal theory. It inherits the same limitation: structural persistence of Σ_t does not address operational persistence (how close $\|\delta_{\text{strategic}}\|$ is to R_Σ) or continuity persistence (whether the agent's strategic identity coheres through time). The persistence condition for M_t (4.5) says: adaptive tempo must exceed the ratio of disturbance to critical mismatch. If the same mathematics applies to Σ_t, then strategy persistence requires strategic tempo to exceed the ratio of strategic disturbance to critical strategic mismatch. The strategic analog of "the environment changes faster than the agent can learn" is "the world invalidates plans faster than the agent can revise them." Both lead to the same catastrophic outcome: the system cannot maintain bounded mismatch and begins to degrade.

What this would buy the theory. If promoted to a result, strategy persistence would: - Provide a formal criterion for "when does a strategy remain viable?" - Connect strategic failure modes to the same mathematical framework as epistemic failure modes - Enable quantitative comparison: is the bottleneck epistemic persistence (model can't keep up with reality changes) or strategic persistence (plans can't keep up with requirement changes)? - Ground the organizational intuition that plans need to be revised faster than the situation changes

Findings.

The Forgetting Prerequisite for Strategic Persistence. **Brief:** With infinite memory, standard Bayesian updating guarantees eventual strategic failure. The sector parameter for Beta-Bernoulli edge updates is $\alpha_\Sigma = 1/(n+1)$, which decays monotonically as experience n accumulates. For any positive disturbance rate, the persistence threshold $\alpha_\Sigma > \rho_\Sigma/R_\Sigma$ is eventually violated regardless of the agent's initial calibration. Exponential forgetting with discount factor λ stabilizes the effective sample size at $1/(1-\lambda)$, giving exact steady-state $\alpha_\Sigma^{\text{ss}} = (1-\lambda)/(2-\lambda)$ — with the simpler linear form $1-\lambda$ as the slow-forgetting asymptote ($\lambda \to 1$). The forgetting prerequisite — $(1-\lambda)/(2-\lambda) > \rho_\Sigma/R_\Sigma$ — converts the schema's instantaneous persistence check into a trajectory guarantee. Forgetting is therefore a structural prerequisite, not a tunable heuristic: the rate at which the agent discounts old evidence must satisfy the threshold. The schema's reachable persistence region under any λ is bounded above by a hard structural ceiling at $\rho_\Sigma = R_\Sigma/2$, derived self-contained in `#deriv-strategic-persistence-hard-ceiling`.

Impact: Translates an organizational platitude ("stay adaptive") into a quantitative survival inequality with explicit failure mode. Identifies a structural calcification process common to all long-running adaptive systems whose update mechanism accumulates evidence: institutional rigidity, RL value-function staleness, scientific-paradigm lock-in, and the loss-of-edge phenomenon in incumbent firms all instantiate the same dynamic. The threshold is sharp — no amount of prior learning protects against an environment that invalidates plans faster than the agent forgets. For long-lived AI agents this mandates explicit memory-pruning mechanisms; the design choice is not whether to forget but at what rate, and how to keep the strategic disturbance well below half the reserve (above which the hard ceiling closes the design space — see `#deriv-strategic-persistence-hard-ceiling`). The structural identity with `#result-persistence-condition` (where α can be stationary) is recovered for the strategic case *only* under explicit forgetting, which is itself a non-obvious finding about the asymmetry between epistemic and strategic dynamics.

Novelty Claim: *Claim differentiation* on Bayesian update dynamics with experience discounting. The Beta-Bernoulli decay $\alpha = 1/(n+1)$ is elementary and the forgetting-factor remedy is standard adaptive-control / online-learning practice. The AAT-distinctive contribution is the connection from these standard mechanics to a *survival inequality with environment-side parameters* (disturbance rate, reserve), making forgetting a structural prerequisite of strategic persistence rather than a hyperparameter. The exact threshold $(1-\lambda)/(2-\lambda) > \rho_\Sigma/R_\Sigma$ — with the slow-forgetting linear form $(1-\lambda) > \rho_\Sigma/R_\Sigma$ as its asymptotic — is the strategic analog of the linear operational form of the persistence condition, with the forgetting rate $(1-\lambda)$ playing the role of adaptive tempo and the $1/(2-\lambda)$ damping factor capturing the integration of finite-horizon evidence.

Related Work:

Table 9.11

ASF Concern	*Prior-art Language*	*Relationship / Positioning*
Exponential forgetting / discounted least squares	Ljung 1987 *System Identification: Theory for the User*, MIT Press (published 1987, found pre-2026)	*formal antecedent* — supplies the discounted-update mechanism; AAT adopts it directly and connects it to a survival threshold
Plasticity/stability tradeoff in continual learning	Kirkpatrick et al. 2017 *Overcoming Catastrophic Forgetting in Neural Networks*, PNAS 114(13) (published 2017, found pre-2026)	*conceptual precursor* — frames the same tradeoff at the network-weight level; does not connect to a Lyapunov-style survival inequality with environment parameters
Drift detection in streaming learning	Gama et al. 2014 *A Survey on Concept Drift Adaptation*, ACM Computing Surveys 46(4) (published 2014, found pre-2026)	*conceptual precursor* — recognizes the need to track changing distributions; does not formalize "forgetting rate must exceed disturbance-to-reserve ratio" as a structural prerequisite
Requisite variety as cybernetic principle	Ashby 1956 *An Introduction to Cybernetics* §11; Conant & Ashby 1970 (published 1956/1970, found pre-2026)	*conceptual precursor* — the regulator's variety must match the disturbance's variety; AAT's forgetting prerequisite is a quantitative time-domain analog applied to the strategic substate
Organizational "core rigidities" / competency traps	Leonard-Barton 1992 *Strategic Management Journal* 13:111; Levitt & March 1988 *Annual Review of Sociology* 14:319 (published 1988/1992, found pre-2026)	*conceptual precursor* — qualitative recognition that accumulated competence becomes liability under environmental change; AAT supplies the structural threshold

Search Log: - 2026-04 (*intuition-only* on the survival-inequality framing): the Beta-Bernoulli $1/(n+1)$ decay is elementary, the forgetting-factor remedy is standard, and the qualitative organizational-calcification intuition is well-established. The unsearched claim is whether the framing as a *structural survival prerequisite* — with environment-side ρ_Σ/R_Σ on the right-hand side — has been formalized elsewhere as a threshold rather than a hyperparameter. Targeted future search candidates: bounded-rationality with memory cost (Genewein, Leibfried, Grau-Moya, Braun 2015 *Frontiers in Robotics and AI*), regret analysis of forgetting-factor estimators in adaptive control (Anderson-Moore line), organizational-learning literature on competency traps and "core rigidities" connected to environmental volatility metrics. Pre-search expectation: the constituent moves are individually well-precedented; the integrated framing as a Lyapunov-style survival prerequisite for the *strategic* substate, with the structural identity to the epistemic persistence condition under forgetting, is plausibly AAT-distinctive but not yet verified under nominally-comprehensive search.

9.11 *Formulation:* Consolidation Dynamics

Consolidation is a regime of the between-event dynamics g_M of 3.2 in which the agent applies Markov updates driven by replayed or internally-generated pseudo-events — replay buffers, hippocampal reinstatement, remembered episodes, re-read paragraphs — with objective of reducing the rate-distortion gap to the IB-optimal compression $\phi^*(\mathcal{C}_t)$. It is not a new adaptive primitive — the recursive-update form $f(M_{\tau^-}, e_\tau)$ is preserved — but it is a distinct operating regime with its own scope condition, its own objective, and its own failure modes, each of which the theory currently names only implicitly or in a parenthetical. Naming the regime makes the stability-plasticity window visible as an AAT-expressible failure boundary and supplies the architectural primitive that logogenic agents (`03-llm-core/`) require under context-turnover.

The segment carries four formal elements that together promote consolidation from parenthetical to first-class regime. (1) *Regime definition* — the recursive-update form is preserved; what distinguishes consolidation is its *objective*: the 2.2 Lagrangian evaluated against the accumulated chronica, in contrast to online update's one-step predictive-mismatch objective. Under state-completeness (`#deriv-recursive-update` C3) the pseudo-event carries zero new external information, yet the update still does work — it *redistributes* existing information across the model's factorization structure, closing nonzero rate-distortion gap when the agent has not yet reached ϕ^*. (2) *Scope condition* $\nu_{\text{consol}} \ll \nu_{\text{online}}$ — a direct application of 4.9's convergence constraint to an intermediate timescale between parametric update (fast) and structural adaptation (slow); violating it produces the same oscillation failures. (3) *Necessity condition* (N1)+(N2) — consolidation is necessary (online-only cannot reach the IB optimum) when *both* (N1) sub-state factorization (the Complementary Learning Systems pattern: hippocampus-like fast + neocortex-like slow) and (N2) bounded per-event budget hold. When either fails, consolidation is a luxury (Kalman filters with persistent covariance, conjugate-Bayesian agents with full posterior). When both hold — the empirically-ubiquitous case — consolidation is forced.

(4) *Stability-plasticity feasibility window*. 9.10's forgetting prerequisite gives a *plasticity lower bound* on λ (forget fast enough to track non-stationarity); this segment supplies a *stability upper bound* (forget slow enough to let consolidation integrate cross-episode patterns before they are discarded). Between the bounds is the feasibility window; an empty window is the catastrophic-forgetting regime (French 1999; Kirkpatrick et al. 2017): no λ satisfies both constraints and the agent's long-run IB objective is strictly worse than a slower-environment or faster-consolidation counterpart. *Window existence* is what is derived here; the upper bound's exact functional form is open and flagged in Working Notes. Together these elements also make *structural adaptation enablement* visible: under (N1)+(N2), structural-class operations (decomposition-and-recombination, expansion, compression, grafting per 9.7) cannot be executed online — consolidation provides the operating regime where the per-event budget is irrelevant and interleaved replay supports stability-preserving structural change. All finite-budget agents require consolidation for quality-preserving structural change.

Formal Expression.

Regime definition. *[Formulation (consolidation-regime, specializes 3.2)]*

Let the agent's between-event dynamics be $g_M(M_\tau)$ per `#deriv-recursive-update` (Corollary: Between-events dynamics, $dM/d\tau = g(M_\tau)$). **Consolidation** is the regime of g_M in which the agent applies updates $M_{\tau^+} = f(M_{\tau^-}, e_\tau^{\text{replay}})$ where e_τ^{replay} is a pseudo-event synthesized from M_{τ^-} itself — a sample drawn from the agent's retained trace (replay buffer, hippocampal reinstatement, a remembered episode, an earlier paragraph re-read). The recursive-update form is preserved; what distinguishes consolidation is the *objective* these updates optimize:

$$\text{consolidation objective: } \min_{M_\tau} \mathcal{J}_{\text{IB}}(M_\tau) \;:=\; I(M_\tau; \mathcal{C}_t) - \beta I(M_\tau; o_{t+1:\infty} \mid a_{t:\infty}) \tag{9.41}$$

— the 2.2 Lagrangian evaluated against the agent's accumulated chronica $\mathcal{C}_t$. By contrast, online update's objective (per 3.6) is one-step predictive mismatch minimization at the current event; it has no representation of $\mathcal{J}_{\text{IB}}$.

Under C3 (state completeness, per `#deriv-recursive-update`), $\mathcal{I}(e_\tau^{\text{replay}} \mid M_{\tau^-}) = 0$: the pseudo-event carries no new external information. Yet the update still does work — it *redistributes* existing information across the factorization structure of M_τ. The distinguishing content is not the information brought in (zero, by construction) but the rate-distortion gap closed (nonzero, when the agent has not yet reached ϕ^*).

Scope condition — timescale separation.

Scope (timescale-separation)

Let ν_{online} be the rate of external events (3.1) and ν_{consol} the rate of consolidation updates. The consolidation regime is well-defined only when

$$\nu_{\text{consol}} \ll \nu_{\text{online}} \tag{9.42}$$

— the convergence constraint of 4.9 applied to an additional intermediate timescale between parametric update (fast) and structural adaptation (slow). Violating this constraint makes consolidation act on online transients rather than settled state, producing the same oscillation failures 4.9 warns about.

Necessity condition.

Derived (consolidation-necessity, conditional)

Consolidation is necessary — online-only cannot reach the IB optimum — when *both* of the following hold:

(N1) Sub-state factorization. M_t factors into sub-states M_t^{fast} and M_t^{slow} with divergent compression-prediction trade-offs. M_t^{fast} favors high-capacity sparse representation; M_t^{slow} favors distributed compressed representation. The two sub-states capture cross-episode regularities versus verbatim traces respectively — the Complementary Learning Systems factorization (McClelland, McNaughton & O'Reilly 1995; Kumaran, Hassabis & McClelland 2016).

(N2) Bounded per-event budget. The per-event processing budget B_{online} is strictly less than the integration cost $B_{\text{consol-needed}}$ for updating M_t^{slow} against cross-episode regularities. Online updates can move at most B_{online} bits of model-state change per event; updating M_t^{slow} to represent a cross-episode pattern requires comparison against a distribution of prior episodes, which exceeds B_{online}.

When (N1) *or* (N2) fails, consolidation is a *luxury*: online update with sufficient per-event budget or without sub-state factorization can reach ϕ^* in the limit. Kalman filters with persistent covariance, conjugate-Bayesian agents with full posterior, and linear-Gaussian systems with online Riccati updates all satisfy neither (N1) nor (N2) and have no consolidation need. When (N1) *and* (N2) both hold, consolidation is a *necessity*: no online-only policy reaches ϕ^* under the joint constraint.

Stability-plasticity feasibility window.

Derived (stability-plasticity-window, conditional)

9.10 derives the *plasticity lower bound* on forgetting rate λ:

$$(1-\lambda) \;>\; \rho_\Sigma / R_\Sigma \qquad \text{(plasticity lower bound)}$$

— forgetting fast enough to track non-stationarity. This segment's complement is a *stability upper bound*:

$$(1-\lambda) \;<\; \phi(\nu_{\text{consol}}, \text{consolidation-budget}) \qquad \text{(stability upper bound, see Working Notes for derivation sketch)} \tag{9.43}$$

— forgetting slow enough to let consolidation integrate cross-episode patterns before they are discarded. Between these bounds is the **feasibility window** for λ. Empty window — rapid non-stationarity with slow consolidation cadence — is the catastrophic-forgetting regime (French 1999; Kirkpatrick et al. 2017): no λ satisfies both constraints and the agent's long-run IB objective is strictly worse than a slower-environment or faster-consolidation counterpart.

The upper bound's exact form depends on the consolidation mechanism and is not derived here — it is a candidate derivation flagged in Working Notes. What is derived: the window's *existence* as a structural object (plasticity must satisfy both bounds) and the catastrophic-forgetting regime as its empty-window limit.

Derived (structural-adaptation-requires-consolidation, conditional)

Structural-adaptation enablement.

Under (N1)+(N2), structural adaptation (per 4.8) cannot be executed online. Parametric update has a hard timescale constraint — mismatch decays at rate $\mathcal{T}$; delayed updates accumulate mismatch at rate ρ. Structural adaptation tolerates much larger delay because the slow process operates on what R_Σ or R are *measuring tolerance against* — the model class itself. Consolidation provides the operating regime where structural operations (decomposition-and-recombination, expansion, compression, grafting per 9.7) become executable: the per-event budget is irrelevant when updates are offline, and interleaved replay supports stability-preserving structural change.

Pure online structural adaptation is the luxury case where per-event budget equals or exceeds integration cost for structural-class operations — this is where Bayesian nonparametric agents with unlimited compute sit. All finite-budget agents require consolidation for quality-preserving structural change.

What Is Derived vs. What Is Chosen.

Table 9.12

Property	*Source*	*Strength*
Consolidation as regime of g_M with replayed/pseudo-event driver	Specialization of 3.2's Discussion (consolidation listed as an example)	Formulation choice (the "regime" framing; could alternatively be presented as a distinct adaptive primitive with its own update form)
IB-gap reduction as consolidation's objective	Identification with 2.2's Lagrangian against $\mathcal{C}_t$	Derived (given IB acceptance; the alternative — online-update-only optimization — leaves the gap un-reducible under (N1)+(N2))
Scope condition $\nu_{\text{consol}} \ll \nu_{\text{online}}$	Direct application of 4.9 convergence constraint	Derived
Necessity condition (N1)+(N2)	Structural argument: under (N1), cross-episode information cannot enter one event; under (N2), online budget insufficient for cross-episode integration	Derived (qualitatively; the quantitative version requires specifying B_{online} and $B_{\text{consol-needed}}$ per architecture)
Stability-plasticity window existence	9.10's lower bound + this segment's upper bound (form open)	Derived (existence); upper-bound functional form open
Catastrophic-forgetting regime = empty window	Direct from both-bounds-unsatisfiable	Derived
Structural adaptation requires consolidation under (N1)+(N2)	4.8's per-step timescale + consolidation's offline budget	Derived (qualitatively)
CLS factorization (hippocampal fast / neocortical slow) as canonical (N1) instance	McClelland-McNaughton-O'Reilly 1995; Kumaran-Hassabis-McClelland 2016	External theorem (CLS literature)
Quantitative online-only no-go under (N1)+(N2)	Rate-distortion argument sketched but not rigorously derived here	Sketch (candidate 5.1 Instance 3)

Epistemic Status.

Robust qualitative. Max attainable: *robust qualitative* for the regime characterization and the necessity condition; *conditional* for the feasibility-window claim pending upper-bound derivation; *sketch* for the IB-optimum no-go claim.

The regime characterization is a formulation choice — consolidation can always be re-described as a regime of 3.2's g_M with appropriate pseudo-events. The distinguishing objective (IB-gap reduction via replayed pseudo-events) is well-defined and distinct from online update's one-step mismatch objective; this is the cleanest formal separation the spike analysis uncovered.

The necessity condition (N1)+(N2) is *qualitatively derived* — the argument is structural: (N1) implies cross-episode information is not in any single event, (N2) implies online updates cannot cross-compare. Together they force consolidation. The quantitative version — the precise bit-budget boundary below which online fails and above which it succeeds — depends on the architecture's sub-state structure and is not derived here. Nor are (N1) and (N2) individually necessary: an agent may need consolidation for reasons outside this segment's scope (e.g., structural adaptation at pure cost-of-delay, unrelated to cross-episode regularities).

The stability-plasticity window's *existence* follows from 9.10's lower bound plus any monotone upper bound on $(1-\lambda)$ from consolidation-cadence considerations. The specific functional form of the upper bound is open work; candidate form in Working Notes. The catastrophic-forgetting regime as empty-window is then an immediate structural consequence, not a new claim.

The online-only no-go claim (that under (N1)+(N2), no online-only policy reaches ϕ^* in steady state) is *sketch-level*. The argument reduces to a rate-distortion inequality close to known results in continual-learning theory and rate-distortion with side information, but the rigorous version requires specifying the budget geometry carefully. It is a candidate Instance 3 for 5.1 (an external information-theoretic obstruction with AAT machinery — the consolidation regime in g_M — as the unique escape).

What this segment does not claim. It does not introduce a new adaptive primitive — the recursive-update form $f(M_{\tau^-}, e_\tau)$ is preserved, with e_τ^{replay} playing the role of e_τ. It does not derive the quantitative feasibility-window upper bound. It does not resolve the (N1) factorization question for specific architectures (e.g., whether transformer attention heads satisfy (N1) is a logogenic-agents question, not an AAT-core one).

Discussion.

Current AAT surface — why naming is warranted. Consolidation is visible in the theory today as:

- A parenthetical example in 3.2 Discussion ("includes prediction generation, uncertainty growth, and internal reorganization (consolidation, abstraction)").
- The implicit slow timescale in 4.9.
- The plasticity lower bound in 9.10 — with no stability upper bound.
- A compression-by-convergence Working Note in 9.9 ("as edges converge, drop them").
- The PULSUS MEMORATA / VERA / AXIOMATA cadences in `ref/agentic-tft/agentic-tft-cognitive-loop-spec.md` for logogenic agents — where consolidation is *already* a first-class architectural commitment.

Naming the regime explicitly promotes what the theory implicitly depends on. The asymmetric treatment in 9.10 (plasticity lower bound only, no stability upper bound) predicts faster forgetting is always better — empirically false whenever the slow sub-state matters (Complementary Learning Systems literature; continual-learning benchmarks; organizational memory research). The feasibility-window framing closes this asymmetry.

Distinguishing axes examined. Four candidate axes could distinguish consolidation from online update. Only one yields a clean formal distinction:

- *Timescale*: consolidation is slower. But 4.9 already admits arbitrary timescale separation, and any periodic process can be modeled as a channel with clock-driven $\nu^{(k)}$. Timescale is a *scope condition*, not a distinguishing feature.
- *Information source*: consolidation operates on replayed data. This is a real difference but lives inside g_M without new primitive structure — the recursive-update form is preserved with e_τ^{replay} in place of external e_τ.

- *Objective*: **clean formal distinction.** Online = one-step predictive-mismatch minimization; consolidation = IB-gap reduction. This is what this segment adopts as the defining axis.
- *Scope of change*: under bounded per-event budget, structural adaptation is temporally decouplable from online but parametric update is not. Structural-adaptation operations naturally live offline — true under (N2), but this is a consequence of the necessity condition, not an independent axis.

Relation to Complementary Learning Systems (CLS) theory. The (N1) factorization — fast sparse-conjunctive sub-state + slow distributed-overlapping sub-state — is the CLS architecture (McClelland-McNaughton-O'Reilly 1995). CLS's core claim is that hippocampal replay during sleep supports interleaved neocortical learning that cross-episode structure requires. In AAT vocabulary, this is: online-only on the slow sub-state catastrophically forgets (French 1999); replay-based offline updates (experience replay: Mnih et al. 2015; prioritized replay: Schaul et al. 2016) or regularization-based approaches (EWC: Kirkpatrick et al. 2017) provide the escape.

The AAT reading of EWC is worth noting: EWC adds a stability-weighted update gain (per-parameter Fisher-information weighting) — a *tensor-valued* generalization of 3.6's scalar η^*. This is a different direction from consolidation (it keeps updates online but weights them by prior-task importance). Both escape the catastrophic-forgetting regime but via different mechanisms; consolidation reuses 3.2's structure, EWC requires the new tensor-valued gain.

Logogenic implications. Consolidation is a primitive in a stronger sense for logogenic agents (`03-llm-core/`) than for AAT-core for three composing reasons.

First, **context-turnover.** Logogenic agents have near-100% reset of the fast sub-state (context window) per session. The only continuity is the slow sub-state (persistent memory, weights, external files). The between-session interval is a *forced* consolidation window — the agent must transfer signal from the about-to-be-lost fast state to the persistent slow state, or it is lost. This is qualitatively different from non-logogenic agents where the fast sub-state persists across events.

Second, **linguistic medium of reflection.** The PULSUS MEMORATA / VERA / AXIOMATA cadences in `ref/agentic-tft/agentic-tft-cognitive-loop-spec.md` are scheduled consolidation processes with different cadences and different target representations. Each is a linguistic operation ("What from recent experience should be compressed into lasting memory?", "Are my beliefs still justified?", "Who am I becoming?") — using language to reorganize language-structured state. This is consolidation operating as the primary unit of cross-session cognition.

Third, **pre-consolidated embedding space.** Per `ref/agentic-tft/agentic-tft-narrative-as-implementation.md`, pretrained language embeddings encode structured epistemic geometry at training time. The logogenic agent doing linguistic reflection is operating in a representational space *already* consolidated into a high-structure form, with access to cross-episode generalization that sub-linguistic agents would have to build online. This is a load-bearing asymmetry in `03-llm-core/`.

Luxury vs necessity mapping. When is consolidation a luxury (subsumed by online update)? When at least one of (N1)/(N2) fails:

- *Rich-state luxury*: Kalman filter with persistent covariance, conjugate-Bayesian agent with full posterior over parameters. The posterior *is* the consolidated representation.
- *Large-budget luxury*: per-event budget $\geq$ integration cost allows online slow-track update per event.
- *Stationary-environment luxury*: online update converges to ϕ^* in the limit if the environment holds still long enough.

When is consolidation a necessity? When both (N1) and (N2) hold, which is the empirically-ubiquitous case: CLS-architected agents, bounded-budget deep RL agents (hence experience replay's empirical necessity in DQN), organizations with event-arrival rates exceeding real-time cognitive bandwidth, logogenic agents under context-turnover.

Predictive statement. *The depth of consolidation machinery an agent needs scales with (a) the factorization depth of its representational structure, (b) the gap between its per-event processing budget and the integration cost of its slowest sub-state, and (c) the rate of cross-episode structural regularities in its environment relative to its event arrival rate.* This is a scope-indexed claim — testable, domain-general, and makes AAT's position on continual learning explicit.

Connection to AAT's meta-architecture. - *5.4* (positive half): the regime is the repair machinery between separable-core and structured-repair ladders along the representation-factorization axis. Where the (N1)+(N2) necessity conditions hold, the repair is consolidation; where either fails, the online regime suffices. - *5.1* (negative half): candidate Instance 3 — the under-bounded-budget + no-reach-of-IB-optimum no-go, with consolidation as the unique escape. - *5.5* (constructive half): the IB Lagrangian is an adjacent family member (adopted as applied external theorem, not re-derived from an AAT-internal additivity axiom), which the consolidation objective directly uses.

9.12 *Discussion:* Additional Implications & Discussion

The chapter does more work than its segment count suggests. Strategy edges acquire dynamics; the structural commitments from Ch.3 acquire formal failure modes; and the framework's signature pattern — *naming an information-theoretic floor and the unique broadly-available escape via AAT machinery* — delivers its first formal instance as a derived no-go. Several of the chapter's findings compose with each other and with findings from Part I Ch.4 and Part II Ch.2/Ch.3 into cross-segment results that are not visible at any single segment: a triple-pressure picture of long-lived adaptive failure, a unified RL convergence story under non-stationarity, and a sub-scope refinement of Part I's persistence template that explains a decade of empirical RL folklore on optimizer choice and approximate inference. None of these is a single derivation; each is what becomes visible when the chapter's segments are read against the rest of the framework.

Edge dynamics: detection latency, forgetting, and the stability-plasticity window.

The chapter's strategic-edge persistence schema (9.10) carries the forgetting prerequisite in its own Findings section: forgetting at rate $(1-\lambda) > \rho_\Sigma/R_\Sigma$ is structurally required for asymptotic strategic persistence, not a tunable heuristic. With infinite memory, standard Bayesian updating guarantees eventual strategic failure because the Beta-Bernoulli sector parameter $\alpha_\Sigma = 1/(n+1)$ decays monotonically with accumulated experience. The implications-side payoff is that this prerequisite is one of three structural pressures on plasticity in long-lived agents, and the chapter is where the other two become statable.

The first complementary pressure is detection latency. `#deriv-update-detection-latency` derives that for a Beta-Bernoulli strategy-edge agent without forgetting, the expected cycles to detect a within-class regime change of footprint ε scales as $\Omega((n_{\min}+1)/\varepsilon)$. The result has a structural rather than parametric character — the $1/(n+1)$ rate is *forced*, not chosen, through the composition of two AAT-internal moves. The log-odds coordinate is uniquely forced by the evidential-additivity axiom via Aczél 1966's Cauchy-functional-equation uniqueness theorem (`#deriv-edge-update-natural-parameter`); in that forced coordinate, Beta-Bernoulli accumulation gives $1/(n+1)$. No reparameterization escapes the rate without abandoning evidential additivity, which would invalidate the update rule on AAT-internal grounds. The chapter therefore sharpens the forgetting prerequisite from "required for asymptotic persistence" to "required for *bounded detection latency at every operating point*" — long-lived agents face detection latency that grows linearly with accumulated experience unless forgetting bounds the effective sample size at $1/(1-\lambda)$. The framework's name for the empirical pattern is *stability-induced myopia*: successful agents that have accumulated experience on their working plans systematically take longer to notice regime changes than the same agents earlier in their lives, with a precise rate of slowdown.

The second complementary pressure is consolidation cadence. `#form-consolidation-dynamics` adds a *stability upper bound* on plasticity: forgetting slow enough that consolidation can integrate cross-episode patterns before they are discarded. Between the plasticity lower bound (forgetting prerequisite) and the stability upper bound (consolidation cadence) lies the *feasibility window* for λ. An empty window — rapid non-stationarity with slow consolidation cadence — is the *catastrophic-forgetting regime*: no λ satisfies both constraints and the agent's long-run IB objective is strictly worse than a slower-environment or faster-consolidation counterpart. Existence is derived under (N1) sub-state factorization (the Complementary Learning Systems factorization — McClelland-McNaughton-O'Reilly 1995's hippocampus-like fast traces + neocortex-like slow integrals) and (N2) bounded per-event budget. The window's upper-bound functional form is candidate-derivation in `#form-consolidation-dynamics`'s Working Notes; *window existence* is what's derived here.

The three pressures compose into a triple-pressure picture that no single segment carries: long-lived agents face plasticity bounded above and below, with detection latency growing inside the viable window. The prescriptive content — which intervention works depends on which bound is binding — is the cross-segment finding worth surfacing as its own object. A *short-attention* agent (forgetting-bound binding from below) needs retention. A *calcified* agent (latency-bound binding inside the window) needs targeted intervention or external Pearl-do (causal-IB exploration drive from `#deriv-causal-ib-exploration`). A *fragmented* agent (consolidation-bound binding from above) needs cadence engineering. The same agent in different operating regimes faces different binding pressures and benefits from different interventions; treating any one mechanism as universal misses the routing. Continual-learning machinery — experience replay (Mnih et al. 2015 DQN), prioritized replay (Schaul et al. 2016), Elastic Weight Consolidation (Kirkpatrick et al. 2017) — are all window-restoration mechanisms; the framework's contribution is naming the window as the structural object and the routing as the choice. For logogenic agents under near-100% context turnover the picture is sharpest: (N1) holds by construction (the context window is the fast sub-state, persistent memory the slow sub-state), (N2) holds by token-budget, and the consolidation regime is forced rather than optional.

On-policy detection structurally fails — identifiability-floor Instance 1.

The chapter's `#der-causal-insufficiency-detection` is the first formal instance of the framework's identifiability-floor pattern, and its Findings section carries the result well: an agent at L0 cannot distinguish a world with a latent common cause from a no-latent-cause world matched in on-policy regime conditionals. The two worlds emit identical on-policy distributions; no test, statistic, or Bayesian comparison built from the agent's observable history can tell them apart under policy-perfect execution. The result is the agent-theoretic instance of Bareinboim's Causal Hierarchy Theorem (Bareinboim-Correa-Ibeling-Icard 2022) applied at the L0/L1 distinction of `#def-strategy-dag`'s Correlation Hierarchy — a Level-2 question (does the latent common cause exist?) being asked of Level-1 data (on-policy traces).

The implication worth surfacing beyond the segment's Findings is the *constructive* use of the no-go. The pattern's signature posture is to name the floor, then name the unique broadly-available escape via AAT machinery, and the chapter delivers this signature for the first time formally. Five boundary routes (S1–S5) characterize what scope condition the agent must violate to escape the no-go: (S1) joint sibling observability under exploration, (S2) instrumental variation, (S3) cross-episode covariance under structural breaks, (S4) off-policy intervention, (S5) external structural knowledge. Joint sibling observability under exploration is the *unique broadly-available* violation — every AAT agent within agency scope has it (via `#der-loop-interventional-access`'s loop-Level-2 access); the others require additional structure not provided by the cycle alone. The chapter therefore lifts `#der-loop-interventional-access` (which Part II Ch.2 introduced as a useful enabling result) from "useful machinery" to *the unique broadly-available violation of the no-go*. This is the move that makes the framework's identifiability-floor pattern load-bearing rather than discouraging — each floor names a specific machinery's load-bearing role with maximum sharpness.

The chapter also makes precise why the no-go is asymmetric in a way the prior aggregate-residual mechanism could not surface. Persistent overestimation of plan success after edge credences have converged is the *signal* of insufficiency, not the *test*; the test is pairwise covariance among observed siblings, which the on-policy regime structurally cannot produce. The aggregate-residual approach collapsed on-policy precisely because the residual is one of the statistics the no-go forbids — *every* on-policy statistic is forbidden. The framework's reformulation in terms of the no-go-and-escape pattern is what gives the joint-sibling-covariance test its load-bearing role rather than treating it as one diagnostic among many. The connection to Ch.3 closes the loop: Ch.3's Correlation Hierarchy is the scope statement that makes the no-go formal here; this chapter's no-go is the dual to Ch.3's L1' Cramér-Rao floor (Instance 2). Together they cover both agent-internal (Instance 2 — single agent, unobservable common cause) and self-diagnostic (Instance 1 — policy-perfect execution) failures of identification, with `#der-loop-interventional-access` as the shared escape mechanism.

Observability dominates, and the depth penalty compounds.

`#der-observability-dominance` carries an underrated structural claim: unobservable strategy edges *freeze* — the gain principle drives their update rate to zero. Paths through unobservable nodes become epistemically dead, an absorbing state the agent cannot escape without external shock, proactive observability investment, or communication-gain machinery (`#hyp-communication-gain`). This is the chapter's strong-prediction result that organizational settings with poor measurement (R&D, strategy groups, some management functions) develop persistent, untested beliefs about their own effectiveness — *structurally* (frozen η_{edge}), not motivationally. The same mechanism predicts that LLM agents operating with high-uncertainty intermediate outputs cannot update their strategy through those nodes, regardless of nominal confidence.

The observability dominance compounds with the chapter's other depth-related results into a triple penalty on planning horizons. Chain-confidence decay from Ch.3 (probabilities multiply, exponential in depth, log-additive coordinate forced by the chain rule) is one penalty. Evidence starvation is another: in a d-depth chain, edge k is tested only when all upstream edges succeed, so its effective correction rate is attenuated by $\prod_{j<k} \theta_j$ — the deepest edge faces *exponential* attenuation regardless of how reliable each upstream edge is. The IB complexity cost (`#form-strategy-complexity-cost`) is the third: maintaining Σ_t in working memory has an information-theoretic cost that compounds with depth, eventually exceeding the agent's cognitive budget. Deep plans are therefore structurally fragile along three independent axes — fragility, calibratability, and tractability — and the structural pressure toward shallow, parallelizable strategies is grounded in three independent mechanisms rather than one. For LLM-based agents working through context-window-bounded planning, the triple penalty is the framework's account of why deep chains of LLM reasoning tend to fail in ways that look "creative" but are actually structural.

The chapter's quantification of the observability-investment tradeoff (from `#der-observability-dominance`'s two-edge analysis and `#deriv-edge-credence-dynamics`'s Props B.2–B.3) gives the principle concrete economic content. Making an intermediate node observable improves α_Σ from the plan-level rate $1/(n_\Phi + 1)$ to the per-edge weakest-link rate $\min(1/(n_1+1),\ \theta_1/(n_2+1))$; the improvement is positive whenever $\theta_1 > 1/2$ and experience is distributed similarly across edges. This translates directly to persistence margin in Part I Ch.4's vocabulary — instrumenting an intermediate node buys *measured* persistence-margin gain. The TST connection is concrete and stated in `#der-observability-dominance`'s Discussion: code quality directly determines σ_v for software-domain strategy steps; poor code quality reduces observability, freezing the developer's causal beliefs about what changes will accomplish, and the strategic concern is structural rather than aesthetic.

Adaptive gain, variational extensions, and sub-scope refinement of the persistence template.

The chapter's appendix-side derivations refine Part I Ch.4's sector-persistence template into sub-scope α / α_2 / β partitions that explain several decades of empirical RL and approximate-inference folklore on principled grounds. `#deriv-adaptive-gain-dynamics` derives the A2'-split: sub-scope α_1 (fixed-gain Kalman / exponential family / strongly-convex gradient) where α is derived from the update rule's structure under B1 directional fidelity; sub-scope α_2 (adaptive-gain under named meta-gain conditions MG-1 through MG-4) where two-timescale Lyapunov composition applies; sub-scope β (PID / rule-based / human-judgment) where A2' is *assumed* per-instance rather than derived. The partition is operationally consequential: *AMSGrad is a meta-gain repair that restores (MG-1) by construction*, which is why it outperforms Adam on ill-conditioned problems — the empirical superiority is now derived, not folklore. *MAML's outer loop sits in* β, which is why Fallah et al. 2020's non-convexity is structurally forced rather than a specific algorithmic accident. *IMM regime transitions are in* β *across transition*, which is why adaptive Kalman in regime-switching needs explicit dwell-time machinery. The Mehra non-identifiability result is flagged as a candidate fifth instance of the identifiability-floor pattern (Instance 5) — an additional opening for the meta-pattern.

`#deriv-variational-sector-condition` extends the persistence template to variational-inference agents under a controlled-KL budget. Under $\mathrm{KL}(q\|p) \leq \varepsilon$, an ε-fidelity directional-fidelity condition transfers at $O(\sqrt{\varepsilon})$ rate via Pinsker, with the sector parameter degrading proportionally. The sub-scope α' tier covers controlled-KL variational agents as a workhorse case; *natural-gradient variational inference recovers the full* α (no degradation) because the geometry matches the Fisher metric; *mean-field VI suffers the* $O(\sqrt{\varepsilon})$ *degradation* because it does not. The implication-worth-surfacing as its own object: *mean-field VI agents are structurally suboptimal as adaptive systems by a factor proportional to* $\sqrt{\textit{KL-budget}}$*, regardless of compute or training data*. The decade-plus Bayesian-deep-learning debate about MF-VI versus natural-gradient on heuristic grounds has a principled answer: if persistence-optimality matters, there is no choice. The geometry of the variational family is load-bearing for the persistence machinery, not merely an aesthetic consideration.

Both refinements compose with Part I Ch.4's bandwidth result via the persistence-information-rate floor: each sub-scope inherits the $\dot{R} \geq n\alpha/2$ bound from `#deriv-persistence-cost`, with its own α playing the role. An agent that adopts a coarser sub-scope (moving from α_1 to α_2 or β, or from natural-gradient to mean-field VI) pays in the persistence-bandwidth requirement: the same target distortion costs more information per unit time because the effective sector constant is smaller. The framework's posture across these refinements is that the choice of agent class is not free — it has structural consequences for how much observation bandwidth the agent must consume to hold its persistence guarantees.

The Bretagnolle-Huber regret bound, reverse-KL coordinate forcing, and the unified RL convergence theory.

`#deriv-strategy-cost-regret-bound` carries one of the chapter's sharpest mathematical results: under deterministic optimal policy π^* — AAT's canonical scope — the Bretagnolle-Huber identity $D_{\mathrm{KL}}(\pi^* \| Q) = -\log(1 - \mathrm{TV}(\pi^*, Q))$ holds *exactly* (not as inequality), and the strategy-cost regret obeys the tight two-sided bound $\Delta_{\min}(1 - e^{-D_{\mathrm{KL}}}) \leq R \leq V_{\max}(1 - e^{-D_{\mathrm{KL}}})$. *Strictly sharper than Pinsker* by a factor of 2 in canonical regimes — a quantitative improvement that matters for practical RL bound-tightness on the common deterministic-optimum case. The reverse-KL direction in AAT's strategy-cost objective (π^*-first) is *forced* by the regret-bound derivation: forward-KL is vacuous under deterministic π^* because it cannot bound regret against a point mass.

The result is also the second AAT-internal instance of the additive-coordinate-forcing meta-pattern Ch.3 anchored — reverse-KL at the divergence layer (Cauchy-FE on chain-rule additivity over conditional factorizations, via Hobson 1969 / Csiszár 1991 / Shore-Johnson 1980), following log-probability at the chain layer. The same exponential-family Legendre-Fenchel geometry both coordinates live on is where `#deriv-edge-update-natural-parameter`'s log-odds (instance three, update layer) and `#scope-agent-identity`'s (PI)+Čencov Fisher metric (instance four, metric layer) also live. The framework's posture is unified: coordinates are forced by AAT-internal axioms via classical uniqueness theorems, and the four-layer architecture is what makes derived results at any one layer compose cleanly with results at the others.

The widest payoff of the chapter is what becomes the *unified RL convergence theory under non-stationarity* once the four contributions are composed. Reinforcement learning has many regret-bound results (UCB, Thompson sampling, RLSVI) but typically (a) assumes stationarity, (b) lacks explicit metric structure on policy space, (c) lacks connection to a satisfaction-gap diagnostic that prevents dark-room collapse. AAT's RL convergence story has all three by composing four findings that are derived independently and turn out to be mutually consistent for principled reasons. The two-gap diagnostic from Ch.3 (8.4 + 8.5) supplies the local separation between "you're not doing it well enough" and "the world doesn't permit it." The Bretagnolle-Huber identity here supplies the tight two-sided regret bound with PAC-Bayes generalization (Rubin 2012). The strategic-tempo + sector-persistence schema supplies the rate of Σ_t revision that must outpace strategic disturbance — verified across four canonical topologies (linear chain, balanced tree, unbalanced tree, full DAG with feedback) by `#def-strategic-tempo + #deriv-edge-credence-dynamics`. The loop-Level-2 result from Ch.2 makes the regret bound *learnable* — interventional data is automatic rather than earned. The four together give a quantitative RL convergence theory under non-stationarity with three properties no existing RL framework has all of. The composition is NeurIPS Paper 2's structural argument ("A Unified Convergence Theory for Non-Stationary Reinforcement Learning", `~/src/neurips/02-unified-convergence-rl/` §4 Theorem 4.1); each piece is derived elsewhere, and the synthesis is what becomes visible only at the chapter level.

What Part II Ch.4 closes on, then, is the substrate Strategy Dynamics needed: edge dynamics with derived rates and bounded operating-point trajectories, an identifiability-floor instance with a uniquely-named escape, a sub-scope partition that explains optimizer and inference-method folklore on principled grounds, a tight reverse-KL regret bound that anchors the second instance of the coordinate-forcing pattern, and a cross-segment composition (unified RL convergence under non-stationarity) that resolves what RL theory has had open for a long time. Chapter 5's orient cascade is what AAT does with strategy dynamics; this chapter is where the dynamics gets its formal architecture.

Chapter 10

The Orient Cascade

[**Gap**] Open question

10.1 *Derived:* Orient Cascade

For actuated agents, epistrophe (the corrective phase of the cycle) expands from a single step into a *multi-step cascade*. The resolution order — epistemic update first, then attainability assessment, then strategy evaluation, then (if needed) objective revision — is *not a design choice* but a consequence of which quantities require which others. Each step's input depends on the output of prior steps; the ordering is forced by information dependency, not engineered.

The five steps. (1) **Reduce epistemic mismatch** — update M_t via 3.4 and 3.6. Prerequisite for all purposeful evaluation, because M_t appears in every subsequent formula. (2) **Evaluate the satisfaction gap** (8.4) — is the goal achievable? Requires an adequate model to assess attainability. (3) **Evaluate the control regret** (8.5) — is the policy suboptimal? Applied regardless of the satisfaction gap's sign because the 2×2 diagnostic requires both quantities to distinguish four cases (success / strategy problem / capability limit / both). (4) **If regret is high, evaluate strategy calibration** — three sub-steps: 4a plan-level calibration via the strategy-plan-confidence error δ_s (credit-assignment-free, persistence proved within L0); 4b per-edge localization via the strategic-calibration residual (9.1) when sufficient observability and attribution are available; 4c *causal-sufficiency check* — if 4a has converged but plan-outcome residuals remain persistently negative, the DAG is causally insufficient and L0 is converging to a biased target; the diagnostic is pairwise sibling covariance under exploration (9.2), and the remedy is L1 augmentation by adding common-cause nodes (8.3 Correlation Hierarchy) *before* escalating further. (5) **If the satisfaction gap persists** — escalate before revising the objective: 5a check whether model correction changes feasibility; 5b check whether a richer policy class (including L1 augmentation if 4c signaled it) would close the gap; 5c check whether convention escalation (C1 → C2 → C3 per 6.6) reveals recovery paths; only then 5d revise the objective. The cascade ensures *objective revision is the last resort, not the first response to unmet goals* — the agent reaches 5d only after exhausting the alternatives the satisfaction-gap disambiguation table identifies (wrong model, narrow policy class, short horizon, then genuinely infeasible goal). When 5d is performed by a principal, the revised objective is externally sourced (a standard actuated agent); when *internalized* it becomes the self-actuation operator, well-formed only under a grounding condition (a non-degenerate self-actuator must ground objective-revision on a non-objective terminal invariant, not on an objective-functional).

The *inferential force* at each step scales with the value-object continuation convention while the *ordering* itself is convention-independent. Under C1 (one-step improvement, canonical default), a positive satisfaction gap means *locally stuck*, not *globally infeasible* — a multi-step recoverable objective may appear unattainable because continuation is frozen at the current policy. Under C2 (receding-horizon), the diagnostics capture multi-step recovery potential. Under C3 (Bellman), the gap is definitively global. The cascade also surfaces a *virtuous/vicious cycle* between model quality and strategy complexity: better model → richer evaluable strategy → better-directed action → faster model improvement; and inversely, degraded model → forced strategy simplification → cruder action → further model degradation. The latter is the strategic analog of the persistence-condition death spiral (4.5). A related complexity bound surfaces in Discussion: the strategy's evaluable complexity is bounded by the model's capacity — an agent with poor model sufficiency cannot meaningfully evaluate a complex strategy because calibration residuals require adequate model to distinguish "edge prediction wrong" from "observation too noisy to tell." The cascade is the formal analog of Boyd's *Orient*: not just model updating, but the structured interaction between reality-understanding and strategy that Boyd identified as the critical step (not Decide). The cascade provides a mathematical mechanism — Orient resolves the information dependencies that make Decide meaningful; an agent that skips to Decide without adequate Orient will revise its strategy based on stale or incorrect beliefs.

Formal Expression.

Derived (orient-cascade, from information dependency between mismatch types)

1. **Reduce $\delta_{\text{epistemic}}$** — understand reality. Update M_t via 3.4 and 3.6. Prerequisite for all purposeful evaluation, because M_t appears in every subsequent formula.
2. **Evaluate δ_{sat}** — is the goal achievable? Compute $A_O(M_t; \Pi, N_h)$ using the updated M_t. Requires adequate M_t to assess attainability (8.4).
3. **Evaluate δ_{regret}** — is the policy suboptimal? Compare A_O to $V_O(M_t, \pi_{\text{current}}; N_h)$ (8.5). This step applies regardless of δ_{sat}'s sign — the 2×2 diagnostic (8.5) requires both quantities to distinguish four cases:
   - $\delta_{\text{sat}} \leq 0$, $\delta_{\text{regret}} \approx 0$: **success** — goal attainable, policy near-optimal.
   - $\delta_{\text{sat}} \leq 0$, $\delta_{\text{regret}} \gg 0$: **strategy problem** — goal attainable, policy poor → revise Σ_t.
   - $\delta_{\text{sat}} > 0$, $\delta_{\text{regret}} \gg 0$: **both** — goal hard AND strategy weak → revise Σ_t first (cheaper than revising O_t), then reassess δ_{sat}.
   - $\delta_{\text{sat}} > 0$, $\delta_{\text{regret}} \approx 0$: **capability limit** — already doing the best available; proceed to step 5.
4. **If δ_{regret} high, evaluate strategy calibration** — is the plan's causal model wrong?

   (4a) Plan-level calibration (default within-L0). Evaluate strategy-plan-confidence error $\delta_s = \hat{P}_\Sigma - \Phi$ — the gap between the agent's strategy-plan-confidence score and the independence-model plan value at true edge parameters. δ_s is credit-assignment-free (requires only status propagation), and its persistence is proved (9.10, Prop B.5 in `#deriv-edge-credence-dynamics`). This is the cheapest operational signal, but its persistence guarantee is within the **independence model (L0)** of the Correlation Hierarchy (8.3): $\delta_s \to 0$ means $\hat{P}_\Sigma$ has converged to Φ, not to actual plan success probability. When the DAG is causally insufficient, Φ itself is a biased target. Step 4c checks whether this is the binding regime.

   (4b) Edge-level localization (when credit assignment is available). When sufficient observability and attribution quality exist (Level 1+ per 9.6), the agent can compute per-edge residuals $\delta_{\text{strategic}}$ (9.1) to localize which edges need revision. $\delta_{\text{strategic}}$ provides finer-grained diagnostics but its persistence is open and it requires the credit-assignment machinery that δ_s avoids. Step 4b is optional — it improves diagnostic resolution but is not required for the cascade's corrective function.

   (4c) Causal-sufficiency check (L0→L1 escalation). If persistent $\delta_s \approx 0$ coincides with persistent negative plan-outcome residuals ($y_G < \hat{P}_\Sigma$ on average, after edge credences have converged), this is evidence that the DAG is causally insufficient and L0 calibration is converging to a biased target. The diagnostic is pairwise sibling covariance under an augmented test (9.2): positive covariance among sibling edges, at timescales where each edge's individual credence has stabilized, localizes where a latent common cause is missing. When the signal for L1 augmentation is present, step 4c directs the agent to add common-cause nodes (8.3 Correlation Hierarchy) *before* escalating via 5a–5d. Running the cascade's exploitation recommendations under L0 when the signal for L1 is present compounds miscalibration — the agent acts confidently on a model whose own residual structure is telling it the model is wrong.

Practical sensitivity. Step 4c is the unique broadly-available L0→L1 diagnostic (9.2 no-go), but its effective power depends on signal-to-noise: small samples, weak common-cause effects, or noisy residuals can produce false negatives, and edge-credence drift can mimic sibling covariance. The convergence framing — "after edge credences have converged" — is a precondition, not a guarantee: in non-stationary environments where per-edge credences keep drifting, the trigger may never cleanly fire and L1 augmentation should be considered the default rather than gated on 4c's signal (8.3 Correlation Hierarchy). Formal preconditions (joint observability, per-edge credence stabilization, approximate stationarity over the test window) and detection scope live in 9.2; consult them before treating a 4c null as evidence of L0 sufficiency.

5. **If $\delta_{\text{sat}} > 0$ persists** — escalate before revising O_t.

   Under C1 (the canonical default), $\delta_{\text{sat}} > 0$ means *locally stuck*, not *globally infeasible* (6.6). Before concluding the objective is wrong, the agent should check whether the gap reflects a limitation of the current analysis rather than genuine infeasibility:

   (5a) Check whether M_t correction changes the feasibility assessment — a wrong model may make an achievable goal appear unattainable.

   (5b) Check whether a richer policy class Π would close the gap — structural Σ_t adaptation (expanding the strategy space, not just revising edge credences). This includes L1 augmentation of the strategy DAG (8.3 Correlation Hierarchy): if step 4c detected causal insufficiency, adding common-cause nodes here is the structural repair.

   (5c) Check whether convention escalation reveals recovery paths — evaluating under C2 (receding-horizon) may show $\delta_{\text{sat}}^{\text{RH}} \leq 0$ for a goal that appeared unattainable under C1. The C2 recovery-path reading presumes the replanning objective is order-consistent (window-covering, value-compared, or terminal-cost guarded per 6.6); unguarded short-horizon replanning is not guaranteed to dominate the frozen baseline.

   (5d) If $\delta_{\text{sat}} > 0$ persists across M_t correction, Π expansion (including L1 augmentation), and convention escalation — **revise O_t**.

   The cascade's ordering ensures objective revision is the last resort, not the first response to unmet goals. The agent reaches step 5d only after exhausting the alternatives that the satisfaction-gap disambiguation table (8.4) identifies: wrong M_t, narrow Π, short N_h, and only then genuinely infeasible goal. When step 5d is performed by a principal the revised O_t is externally sourced (an actuated agent); when it is *internalized* — performed by the agent on itself — it becomes the self-actuation operator, which — as a conditional, scoped no-go (`#deriv-self-actuation-grounding`) — is well-formed only under a grounding condition: a non-degenerate self-actuator must ground objective-revision on a non-objective terminal invariant, not on an objective-functional.

Derivation. Each step's input depends on prior steps' outputs: - You cannot evaluate strategy quality with a broken reality model (step 3 requires step 1) - You cannot distinguish "locally bad strategy" from "locally unattainable goal" without both δ_{sat} and δ_{regret} (step 3 requires step 2) - You cannot localize strategy failures (4b) without first detecting plan-level miscalibration (4a) - You cannot diagnose causal insufficiency (4c) until after edge credences have had time to converge (4a), because the diagnostic signal — persistent negative plan-outcome residuals — requires $\delta_s \approx 0$ to be separable from ordinary calibration error - You should not revise the objective until you've verified that improving M_t, Π (including L1 augmentation when 4c signals it), and N_h cannot close the gap (step 5 requires steps 3–4 and the escalation substeps)

The ordering is forced by information dependency. The split of step 4 into 4a/4b/4c reflects three distinct diagnostic levels: 4a gives a within-L0 persistence signal (plan-level tracking via δ_s), 4b gives within-L0 edge-level localization when credit assignment is available (Level 1+), and 4c exits L0 entirely when the independence model is the binding constraint. The escalation substeps in step 5 reflect the satisfaction-gap disambiguation (8.4): multiple causes of $\delta_{\text{sat}} > 0$ must be ruled out before the agent concludes the goal itself is wrong. L1 augmentation (8.3 Correlation Hierarchy) enters 5b as a structural Σ_t adaptation when 4c's signal is present.

Convention hierarchy and diagnostic power. The 2×2 diagnostic and the inferences drawn from it are relative to the continuation convention in the value object (6.6), which defines a hierarchy of three conventions with a proved monotonicity result.

Under **C1** (one-step improvement, canonical default), δ_{sat} and δ_{regret} are one-step-improvement quantities. A multi-step recoverable objective may appear locally unattainable ($\delta_{\text{sat}} > 0$) because continuation is frozen at π_{current}. The "capability limit" quadrant ($\delta_{\text{sat}} > 0$, $\delta_{\text{regret}} \approx 0$) means *locally stuck* — the agent may be globally recoverable but cannot see the recovery path from one-step analysis.

Under **C2** (receding-horizon, N_r-step replanning), the diagnostics capture multi-step recovery potential. An objective that appeared unattainable under C1 may show $\delta^{\text{RH}}_{\text{sat}} \leq 0$ because replanning reveals a viable path. The "capability limit" quadrant means *stuck with N_r-step replanning* — stronger evidence of genuine difficulty.

Under **C3** (Bellman), the diagnostics are global. $\delta^{\text{B}}_{\text{sat}} > 0$ means the goal is genuinely infeasible given (M_t, Π, N_h) — no policy can achieve it. The "capability limit" quadrant is a definitive diagnosis: the agent is optimally pursuing an infeasible goal (modulo model error in M_t). This is the inference the cascade was designed to support; C1 and C2 provide progressively weaker versions of it.

The monotonicity (6.6): $\delta^{\text{B}}_{\text{sat}} \leq \delta^{\text{RH}}_{\text{sat}} \leq \delta^{(1)}_{\text{sat}}$. Every "unattainable" diagnosis under C3 persists under C1 and C2. A C1 "unattainable" diagnosis may be overturned by C2 or C3 (the goal is reachable with replanning or globally optimal play). The cascade's *ordering* is convention-independent (forced by information dependency). The *inferential force* at each step scales with the convention: C1 gives local heuristics, C2 gives moderate-horizon diagnostics, C3 gives global conclusions.

Epistemic Status.

The cascade **ordering** is *exact*: it is a logical consequence of which quantities appear in which formulas. Steps 1-2 (epistemic update, attainability assessment) rest on well-typed quantities (3.4, 8.4) and exact derivation. Step 3 (control regret) is exact (8.5). Step 4a (plan-level calibration via δ_s) is grounded in a proved quantity — the sector condition transfers to δ_s (Prop B.5 in `#deriv-edge-credence-dynamics`) — but the formal guarantee is *within the L0 independence model*: $\delta_s \to 0$ means convergence to Φ, which equals actual plan success only when the DAG is causally sufficient. Step 4b (per-edge localization via $\delta_{\text{strategic}}$) inherits strategic-calibration's discussion-grade status — the credit-assignment problem and execution-fidelity requirement are acknowledged but unresolved (9.1, Epistemic Status). Step 4c (causal-sufficiency check) is the mechanism for exiting L0 when L1 is the binding regime; it is *robust-qualitative* — the diagnostic logic is sound (9.2), but sensitivity depends on how cleanly the agent can separate sibling-covariance signal from edge-credence noise at convergence. Step 5's escalation substeps (5a-5c) are derived from the satisfaction-gap disambiguation table (8.4); step 5d (objective revision) is the residual case after alternatives are exhausted. The ordering of all steps is forced by information dependency (each step's input depends on prior steps' output). What is NOT derived is the *timing* — how long the agent should spend on each step before proceeding, and how long $\delta_s \approx 0$ must persist before 4c's signal is trusted.

The **convention hierarchy** (6.6) is *exact*: the three conventions (C1, C2, C3) are definitions, and the monotonicity result is a direct consequence of "better continuation policy yields higher expected value." The diagnostic implications table states what each convention's quantities mean by construction. The cascade's inferential force at steps 2-5 scales with the convention but the ordering is convention-independent.

Discussion.

G_t complexity bounded by M_t capacity. Σ_t's evaluable complexity is bounded by M_t's ability to observe which strategy edges are intact. An agent with poor model sufficiency ($S(M_t) \ll 1$) cannot meaningfully evaluate a complex Σ_t — the strategic calibration residual requires adequate M_t to distinguish "edge prediction wrong" from "observation too noisy to tell."

This creates a **virtuous cycle**: better M_t → richer evaluable Σ_t → better-directed action → faster M_t improvement. And a **vicious one**: degraded M_t → forced strategy simplification → cruder action → further M_t degradation. The vicious cycle is the strategic analog of the death spiral described in the persistence condition (4.5) — the agent loses the capacity to maintain complex plans, which reduces the quality of its actions, which further degrades its model.

Connection to Boyd's OODA. The orient cascade is the formal analog of Boyd's "Orient" — not just model updating, but the structured interaction between reality-understanding and strategy. Boyd's insight was that Orient is the critical step, not Decide. The cascade provides a mathematical mechanism consistent with this insight: Orient resolves the information dependencies that make Decide meaningful. An agent that skips to Decide (strategy revision) without adequate Orient (model update + attainability assessment) will revise its strategy based on stale or incorrect beliefs. Whether the dependency structure in the cascade captures the actual cognitive process Boyd described is an empirical question.

Timescale structure. The cascade implies a natural timescale ordering for the different update types:

$$\nu_{\text{epistemic}} \gg \nu_{\text{edge-update}} \gg \nu_{\gamma\text{-reclassify}} \gg \nu_{\text{prune/graft}} \gg \nu_{O\text{-revision}} \tag{10.1}$$

Weight updates are frequent (every observation). Combination-type reclassification is rare (needs strong structural evidence). Pruning/grafting is rarer still (abandon or create causal hypotheses). Objective revision is rarest (change what you want, not how you get it). This ordering is an empirical observation for many agent populations, consistent with the cascade but not derived from it.

10.2 *Discussion:* Three-Way Resource Allocation

At each decision point, an actuated agent with explicit strategy Σ_t faces a three-way allocation of its finite cycle budget: **exploit** (take the currently-best action — earns immediate value Q_O), **explore** (take an information-gathering action — earns future model improvement via causal information yield), and **deliberate** (pause acting to run the orient cascade internally — earns improved future action quality through model refinement, strategy revision, or objective reassessment). The binary exploit/explore tradeoff of Part I extends here because Σ_t exists to be revised and control regret (8.5) supplies the signal that motivates strategic deliberation.

The key structural distinction the segment carries: exploit and explore both involve external actions with environmental consequences; **deliberation is *internal exploration*** — simulation, counterfactual reasoning, cross-domain synthesis, and contingency planning conducted in *model-space* rather than *environment-space*. Deliberation acquires no new environmental data, but it can generate genuinely new information by combining, recombining, and extrapolating from what the agent already knows — including knowledge from entirely different domains. A chess engine deliberating explores millions of future board states; a commander war-gaming evaluates counterfactual scenarios; a developer doing architecture review synthesizes cross-domain expertise. Deliberation's fidelity ceiling is set by model quality, not by data quantity. The segment's one genuinely formal contribution is the *extended deliberation threshold* — extending 4.1 from epistemic-only to epistemic-plus-strategic benefit, with first-order condition: *stop deliberating when the marginal joint improvement rate drops below the mismatch drift rate*.

A *control-regret ceiling* on strategic deliberation: when $\delta_{\text{regret}} \approx 0$, strategic deliberation has no value — the agent is already executing the best available strategy. The ceiling applies only to the strategy-revision channel of deliberation, however; deliberation may also reveal that the *objective* should change (5d of the orient cascade), producing value not bounded by regret, or reduce *uncertainty about* strategy quality, enabling more confident exploitation even when the strategy doesn't change. The segment's *qualitative regime descriptions* — exploit dominates when strategy is near-optimal, model uncertainty is low, and the environment penalizes pauses; explore dominates when model uncertainty is high and the model needs correction via external evidence; deliberate dominates when strategy needs revision, the environment is stable during pauses, and the agent has unprocessed information or structural hypotheses it can evaluate internally — restate problem structure rather than add derived content; they are discussion-grade by design. The two-stage decomposition (deliberation-then-action) versus unified objective over $\mathcal{A} \cup \{\text{deliberate}\}$ is treated as a modeling convenience, not a structural necessity; both formulations are equally valid.

Formal Expression.

Definition (three-activity-decomposition)

The three activities.

An actuated agent's cycle budget decomposes into three activities:

1. **Exploit**: select and execute the action maximizing expected value under the current model and strategy. Earns immediate value $Q_O(M_t, a; \pi_{\text{cont}}, N_h)$.

2. **Explore**: select and execute an action maximizing causal information yield. Earns future model improvement via $\text{CIY}(a; M_t)$.

3. **Deliberate**: pause external action to run the orient cascade (10.1) — internal model refinement ($\Delta\eta^*$), strategy revision ($\Delta\alpha_\Sigma$), or objective reassessment. Earns improved future action quality.

The key distinction: Exploit and explore both involve external actions with environmental consequences. Deliberation is *internal exploration* — simulation, counterfactual reasoning, cross-domain synthesis, and contingency planning conducted in model-space rather than environment-space. It acquires no new environmental data, but it can generate genuinely new information by combining, recombining, and extrapolating from what the agent already knows — including knowledge from entirely different domains. A chess engine deliberating explores millions of future board states; a commander war-gaming evaluates counterfactual scenarios; a developer doing architecture review synthesizes cross-domain expertise. Deliberation's fidelity ceiling is set by model quality, not by data quantity.

Extended deliberation threshold. *[Derived (Conditional on GA-4, deliberation-drift assumption from 4.1)]*

Extending 4.1 to incorporate strategic deliberation, the deliberation threshold becomes:

$$\Delta\tau^* = \arg\max_{\Delta\tau \geq 0} \left[\underbrace{\Delta\eta^*(\Delta\tau) \cdot \|\delta_{\text{post}}\|}_{\text{epistemic benefit}} + \underbrace{\Delta V_\Sigma(\Delta\tau)}_{\text{strategic benefit}} - \underbrace{\rho_{\text{delib}} \cdot \Delta\tau}_{\text{drift cost}} \right] \tag{10.2}$$

where: - $\Delta\eta^*(\Delta\tau)$ is the improvement in model update gain from $\Delta\tau$ of internal simulation (4.1) - $\|\delta_{\text{post}}\|$ is the post-deliberation mismatch magnitude - $\Delta V_\Sigma(\Delta\tau)$ is the improvement in strategy value from deliberation: the expected increase in V_O from revising Σ_t during the pause - ρ_{delib} is the local mismatch drift rate during inaction (GA-4)

First-order condition:

$$\frac{\partial \Delta\eta^*}{\partial \Delta\tau} \cdot \|\delta_{\text{post}}\| + \frac{\partial \Delta V_\Sigma}{\partial \Delta\tau} = \rho_{\text{delib}} \tag{10.3}$$

Stop deliberating when the marginal joint improvement rate (epistemic plus strategic) drops below the mismatch drift rate.

Reduction to existing results. When $\Delta V_\Sigma = 0$ (no strategy revision benefit), this reduces to the deliberation threshold in 4.1. When $\Delta\tau^* = 0$ (deliberation never beneficial), the agent acts immediately.

On the structure of the allocation. *Discussion (allocation-structure)*

Two-stage decomposition. The three-way allocation can be decomposed into two nested decisions: (1) how long to deliberate before acting, (2) conditional on acting, which action to take (solved by 7.5). This decomposition is natural because exploit and explore are both external actions ($\in \mathcal{A}$), while deliberation is internal processing. However, this decomposition is a *modeling convenience*, not a structural necessity. A unified objective over $\mathcal{A} \cup \{\text{deliberate}\}$ with temporal discount γ is equally valid:

$$\bar{\pi}^* = \arg\max_{\bar{a} \in \mathcal{A} \cup \{\text{deliberate}\}} \bar{Q}(\bar{a}; M_t, \gamma) \tag{10.4}$$

where $\bar{Q}(\text{deliberate}) = \gamma \cdot \mathbb{E}[V_{\text{act}}(f_{\text{delib}}(M_t))]$ and $\bar{Q}(a) = r(a) + \gamma \cdot \mathbb{E}[V_{\text{act}}(M_{t+1})]$.

The two-stage decomposition is useful because it allows reusing the CIY-unified objective for the action-selection step, but it is not the unique or forced structure.

Additive benefit decomposition. The objective function above decomposes deliberation benefit into additive epistemic and strategic terms. This assumes the two benefits are approximately independent. Under directed separation (6.3), M_t updates independently of G_t, which makes the decomposition structurally motivated for Class 1 (Separated) agents. However, strategic benefit depends on the improved model ($\Delta V_\Sigma(\Delta\tau, M_t + \Delta M_t(\Delta\tau))$), creating an interaction that the additive form ignores. The additive form is a linearization that holds when ΔM_t is small relative to M_t. For Class 3 (Coupled) agents, epistemic and strategic deliberation are coupled, and the additive decomposition is a convenience.

Control regret as deliberation ceiling. *Discussion (deliberation-ceiling)*

When $\delta_{\text{regret}} \approx 0$ (8.5), strategic deliberation has no value: the agent is already executing the best available strategy. This is the formal reason why low control regret suppresses deliberation. The ceiling on strategic deliberation value is δ_{regret} — the most the agent could gain by finding a better strategy.

However, this ceiling applies only to the *strategy-revision* channel of deliberation. Deliberation may also: - Reveal that the *objective* should change (step 5 of the orient cascade), producing value not bounded by δ_{regret} - Reduce *uncertainty about* strategy quality, enabling more confident exploitation even when the strategy doesn't change

Connection to active inference. *Discussion (ai-connection)*

The exploit/explore decomposition maps cleanly onto the expected free energy (EFE) decomposition into *pragmatic value* (preferences-aligned outcomes) and *epistemic value* (expected information gain about hidden states) in active inference (Friston, FitzGerald, Rigoli, Schwartenbeck & Pezzulo 2017, "Active inference: a process theory," *Neural Computation* 29; Da Costa, Parr, Sajid, Veselic, Neacsu & Friston 2020, "Active inference on discrete state-spaces," *J. Math. Psych.* 99 §2.4; Sajid, Ball, Parr & Friston 2021, "Active inference: demystified and compared," *Neural Computation* 33). Under the mapping, AAT's exploit term Q_O corresponds to EFE's pragmatic value when the value functional V_{O_t} is read as expected log-preferences, and AAT's CIY corresponds to EFE's epistemic value when the relevance variable is read as the hidden-state posterior. The structural correspondence is at the shared-shape level: both decomposed into value- and-information terms, but AAT does *not* commit to AI's preferences-as-priors form (preserving the satisfaction-gap diagnostic in 8.4, which the priors-as-preferences collapse would erase).

The deliberation axis — internal exploration in model-space rather than environment-space — is structurally adjacent to "sophisticated active inference" (Friston, Da Costa, Hafner, Hesp & Parr 2021, "Sophisticated inference," *Neural Computation* 33), which handles bounded computation via recursive expected free energy with depth-limited belief-about-belief reasoning. AAT's machinery for deliberation cost (4.1) and the extended deliberation threshold above derives the same trade-off via per-edge persistence and evidence starvation, with the structural advantage that the depth bound d^* is causally derived from the strategy DAG rather than from belief-recursion depth. The two frameworks address the same problem with different machinery; AAT can credit the AI lineage as a co-developed alternative.

Exploit-regret and the strategy-cost objective. The exploit term's control regret $\delta_{\text{regret}} = A_O - V_O(\pi_{\text{current}})$ (8.5) is the same decision-theoretic regret that underwrites the KL-direction derivation in 9.9. There, strategy-induced regret $R(Q_{\Sigma_t}) = V(a^*) - \mathbb{E}_{a \sim Q_{\Sigma_t}}[V(a)]$ is bounded above by $V_{\max}\sqrt{\frac{1}{2}D_{\text{KL}}(\pi^* \| Q_{\Sigma_t})}$ (Pinsker), forcing the KL direction with π^* first. The exploit term thus shares structure with the strategy-cost relevance term: both are regret-aligned, with the action-selection objective operating pointwise and the strategy-cost objective operating over the strategy's induced distribution. This is the structural cleanup that "value and information term" shares with EFE's pragmatic-epistemic decomposition at the shape level, without the preferences-as-priors commitment — the regret framing is the AAT-internal derivation of the direction that the variational-inference literature arrives at from free-energy-gradient arguments.

Discussion (qualitative-regimes)

Qualitative regime descriptions.

The three activities have natural dominance conditions that follow from the structure of the problem:

Exploit dominates when the strategy is near-optimal ($\delta_{\text{regret}} \approx 0$), model uncertainty is low ($\lambda(M_t) \approx 0$), and the environment penalizes pauses (ρ_{delib} high). The agent knows what to do and must act now. This is Boyd's implicit guidance and control.

Explore dominates when model uncertainty is high (U_M large), and the model needs correction via external evidence that internal simulation cannot provide. Exploration acquires new data; deliberation only processes existing data.

Deliberate dominates when the strategy needs revision ($\delta_{\text{regret}} \gg 0$ or $\delta_{\text{strategic}} \gg 0$), the environment is stable during pauses (ρ_{delib} low), and the agent has unprocessed information or structural hypotheses it can evaluate internally. This regime requires that internal computation is cheaper or more efficient than external probing — otherwise exploration dominates deliberation.

Boundary conditions. As $\rho_{\text{delib}} \to \infty$: $\Delta\tau^* \to 0$, and the allocation collapses to the binary exploit/explore tradeoff. As $\lambda \to 0$ and $\Delta V_\Sigma \to 0$: the allocation collapses to pure exploitation. As $\delta_{\text{regret}} \to 0$: strategic deliberation value vanishes, and the allocation reduces to 4.1's epistemic think/act threshold combined with 7.5's exploit/explore balance.

These regime descriptions are qualitative observations about the problem structure, not derived predictions. They restate what the constituent objectives already imply (high uncertainty favors information acquisition, low cost favors deliberation) without adding quantitative content. The theory does not provide the thresholds at which one regime transitions to another — these are domain-specific.

Epistemic Status.

Discussion-grade. The extended deliberation threshold is *derived* (conditional on GA-4 and the deliberation-drift assumption from 4.1) and is the one genuinely formal contribution of this segment. The regime descriptions are *discussion-grade* — qualitatively correct observations that restate the problem structure. The two-stage decomposition is a *formulation choice*, not a derived result. The additive benefit decomposition is a *linearization* that is structurally motivated under directed separation but not derived.

Simulation check (drifting multi-armed bandit): The three-way allocation with perfect foresight outperformed binary exploit/explore by 2–6% across five configurations. The oracle almost never chose deliberation (0–8 out of 200 steps). Exploration earned 3.5x more information per step than deliberation. A unified objective (single argmax with temporal discount) outperformed two-stage UCB by 8–13%. The simulation is UNFAVORABLE to deliberation (no strategy structure, no irreversible actions, small action space), so these results are a lower bound on deliberation's value in complex domains. The simulation confirms that deliberation adds little in simple settings and that the two-stage decomposition is not necessary.

Max attainable: *discussion-grade* for the overall framing. The extended deliberation threshold could reach *conditional* (it inherits its status from 4.1). The regime descriptions are unlikely to advance beyond discussion-grade without domain-specific instantiation that provides quantitative thresholds.

Discussion.

Why this is Part II content, not Part I. Part I's deliberation-cost handles the binary think/act threshold using only M_t and η^*. The three-way allocation is inherently a Part II result because it requires: - Σ_t to exist (the strategy that deliberation can improve) - δ_{regret} to be defined (the signal that motivates strategic deliberation) - The orient cascade (the internal structure of what "deliberation" does for an actuated agent)

Without G_t, there is no strategic deliberation mode — the agent can only improve M_t or act, which is the Part I binary.

Deliberation as internal exploration. The deepest characterization of deliberation is not "computation on existing data" but *lower-cost, more efficient, usually lower-fidelity exploration in model-space rather than environment-space*. All three activities are forms of information acquisition; they differ in source, cost, fidelity, and scope:

- **Exploitation** acquires value directly from the environment.
- **Exploration** acquires information from the environment via novel observations — high fidelity, high cost (real actions have real consequences).
- **Deliberation** acquires information from the agent's internal model via simulation, synthesis, and counterfactual reasoning — lower fidelity (model may be wrong), lower cost (no environmental consequences), but potentially unlimited scope.

A chess engine deliberating is not "reprocessing existing data" — it is exploring a vast space of futures that no amount of external probing could cover at the same rate. A military commander war-gaming is performing Level 3 counterfactual reasoning (7.1): "what *would have happened* if the enemy had moved north?" A developer doing architecture review is synthesizing patterns from entirely different domains — cross-domain expertise that no local exploration of the current codebase could surface. An AI agent planning its approach is simulating the universe it inhabits and weighing future possibilities.

What deliberation can do that external exploration cannot: - **Counterfactual reasoning** (Level 3): evaluate actions never taken, in situations never encountered - **Cross-domain synthesis**: bring expertise and experience from other contexts to bear on the current problem - **Combinatorial search**: systematically evaluate a space of futures too large to probe externally (chess, Go, multi-step planning) - **Contingency planning**: prepare responses to events that haven't happened yet - **Risk assessment**: evaluate irreversible actions before committing to them

Why deliberation has diminishing returns: The agent's internal model has finite fidelity. Deeper simulation eventually hits the model's accuracy boundary — further deliberation generates predictions the model cannot reliably distinguish. This explains why deliberation is most valuable early (when many model-derivable insights remain unextracted) and why high-fidelity internal models increase deliberation's value. But unlike "computation on existing data," this ceiling is about model fidelity, not data quantity — an agent with a perfect model and finite data can still generate unbounded value from deliberation (this is exactly what MCTS does).

The computational-extraction-gap framing. A sharper structural characterization than "third activity on par with exploit and explore" reads deliberation's value as the *gap between information in the existing data and information already extracted into* M_t. The three activities differ in which bit-channel they operate on: exploitation earns value-bits (direct reward, no new information); exploration earns environment-bits (reduces $H(\Omega_t \mid M_t)$ via new observations); deliberation earns inference-bits (reduces $H(\theta \mid M_t, \mathcal{C}_t)$ via additional computation on existing data). The information ceiling for deliberation is

$$I_{\text{delib}}(\Delta\tau) \;\le\; I_{\mathcal{C}_t} \;-\; I_{\text{already-extracted}} \tag{10.5}$$

For an agent that processes its observation history optimally (full Bayesian update, exact inference), this gap is zero — deliberation has no value. The gap exists when the agent uses point estimates instead of full posteriors, when inference is intractable and approximate methods (MCMC, variational inference) have not converged, or when the agent has structural hypotheses it has not yet tested against existing data (cross-arm correlations in bandits; common-cause structure at L1; etc.). Three structural consequences follow:

1. *Agent-architecture-dependent*: a hypothetical perfect-Bayesian has zero deliberation value; real bounded-rational agents have non-zero deliberation value calibrated by their inference budget.
2. *Diminishing within a cycle*: each deliberation step closes part of the computation gap, with the next step's marginal value bounded by the remaining gap.

3. *Not independent of exploration*: new data changes what's available to extract; an exploration step expands $I_{\mathcal{C}_t}$ and therefore can *raise* the ceiling on subsequent deliberation value.

This places deliberation as a different *kind* of thing than exploit/explore — it operates on existing data rather than acquiring new data or value — even though the three-way framing in the Formal Expression treats them as comparable allocation targets. Both readings are valid: the allocation framing is useful for engineering decisions about $\Delta\tau$; the gap-closing framing is useful for understanding *why* deliberation has the structural properties it does.

Diagnosis — a structural-detection mode, related but not identical to the three. The strategy-dynamics no-go (9.2 §"Diagnostic CIY") identifies a further activity, *diagnose*: deliberately taking non-short-circuit actions (the SA3 ε-exploration excursions) to generate the joint-sibling outcomes that on-policy execution censors, in order to test the model's *causal-structure* assumptions — detecting a latent common cause the on-policy residual is structurally blind to. Diagnosis sits *across* the partition above rather than neatly inside it: its *mechanism* is external action, like explore, but its *target* is the model's structural assumptions rather than its parameter values — it earns structure-bits, not the environment-bits of ordinary exploration nor the inference-bits of deliberation. That is why 9.2 frames it as a third axis on the explore/exploit tradeoff while this segment frames the allocation as three-way: the two framings are compatible once *diagnose* is read as a targeted, structurally-motivated mode of exploration (external, but aimed at the censored joint outcomes) rather than a fourth peer category. Its value is governed by the diagnostic-CIY conditions there, and is highest exactly when a latent-common-cause hypothesis is live.

Deliberation's comparative advantage is greatest when: - Internal simulation is cheaper than external probing (bits per unit cost, not bits per timestep) - The action space is too large for systematic exploration (planning/search) - Actions are irreversible and error costs are high (deliberation as risk management) - The agent has cross-domain knowledge exploitable by synthesis - The model supports counterfactual reasoning (Level 3 access)

Connection to 9.9. Deliberation duration required to revise Σ_t grows with strategy complexity. The description length $\mathrm{DL}(\Sigma_t)$ from 9.9 provides a lower bound on the deliberation time needed for meaningful revision.

Connection to 9.8. Strategic tempo $\mathcal{T}_\Sigma$ is the rate of useful strategy revision from *external* evidence. Deliberation provides an *internal* channel for strategy revision that supplements $\mathcal{T}_\Sigma$.

Domain instantiations:

Table 10.1

Domain	*Exploit*	*Explore*	*Deliberate*
Military (Boyd)	Execute the mission	Reconnaissance, probing attacks	Staff planning, war-gaming
RL agent	Greedy action	ε-exploration, UCB	MCTS planning, model-based rollouts
Software developer	Write code along current plan	Try unfamiliar approach, spike	Architecture review, read docs
Organization	Execute strategy	A/B test, pilot program	Strategic planning offsite
AI coding agent	Execute planned edit	Read unfamiliar code, run tests	Plan approach, analyze requirements

10.3 *Discussion:* Additional Implications & Discussion

The chapter closes Part II by tying together what the cascade does, what forces it to take the shape it does, and what the cascade looks like at two structural extremes — when the agent is *too confident* in a drifting world (where the cascade has a Lyapunov-forced exploration drive its information-value cousins cannot supply) and when the agent is *coupled-architecture* by construction (where the model can be asked to report in cascade order but cannot make its own belief-update goal-blind, so a goal-blind cascade is recovered only by a separated query at the wrapper level). The chapter is small in segment count but its results connect Part II to Part III via the directed-separation architectural classification, to `03-llm-core/` via the Class-3-Coupled scaffolding requirement, and back to Part I Ch.4 via the bandwidth floor that deliberation does not escape.

Proof provenance. This segment is a synthesis: every result it surfaces is carried, and tier-marked, at the segment it cites, and each implication below inherits the tier of the weakest result it composes — *exact* for the cascade ordering, *conditional* (under named hypotheses) for the survival drive, its matrix lift, the bias-bound constants, and the bandwidth floor, and *discussion-grade* for the compositions this segment itself makes between them. Where an implication rests on a composition no source segment derives, it is marked here as such.

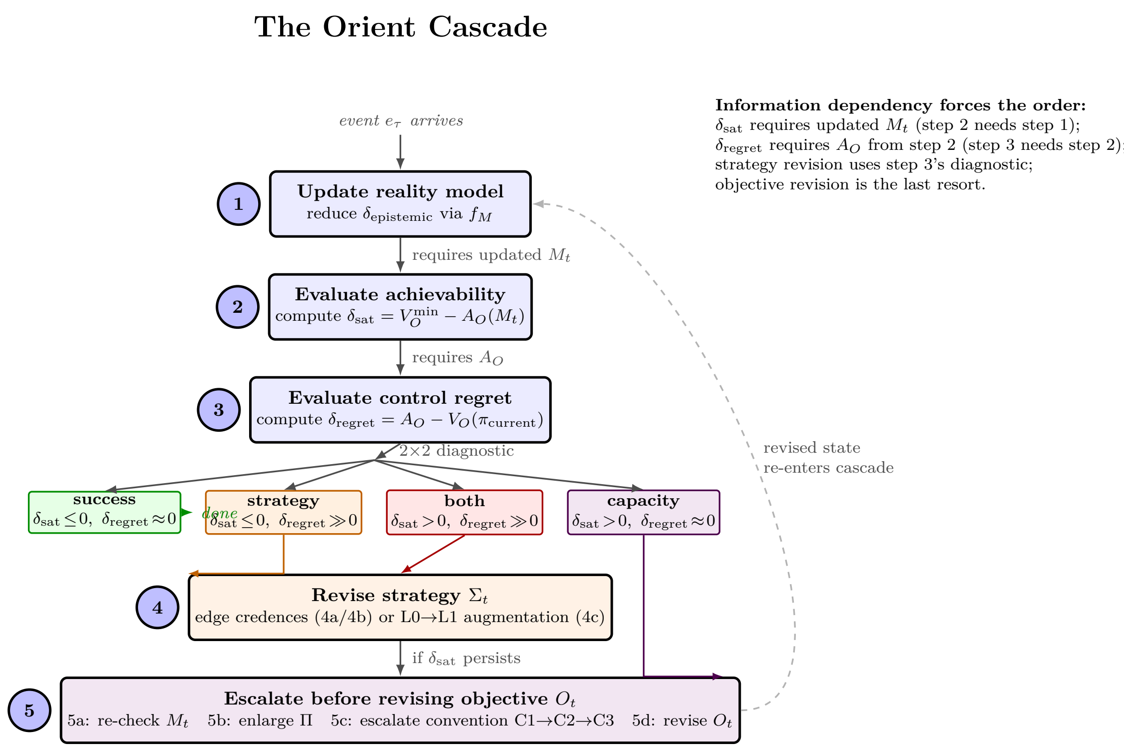


Figure 5: The orient cascade: a five-step resolution order forced by information dependency, not design. The 2x2 diagnostic at step 3 routes the agent to revise the strategy, to escalate before revising the objective, or to terminate at success or capacity limit; a revised model re-enters at step 1.

The cascade is forced by information dependency, not designed.

`#der-orient-cascade` carries the chapter's central architectural move: the Orient phase of OODA expands into a multi-step cascade — reduce $\delta_{\text{epistemic}}$ → assess attainability → evaluate strategy → (if needed) revise objective — whose *ordering is not a design choice but a consequence of which quantities require which others*. Each step's input depends on the prior step's output. M_t enters every subsequent formula, so it must be updated first; attainability A_O is computed against M_t, so it precedes strategy evaluation; strategy revision needs A_O to know whether the problem is the strategy or the goal; objective revision is last because it requires both the model and the strategy assessment to know that no policy in Π can deliver the objective. The *ordering* is exact — a logical consequence of which quantities appear in which formulas — and that exactness is what excludes the alternative orderings (objective-first, strategy-first, all-at-once): any of them evaluates some step on inputs a later step would have revised. The exactness is the ordering's alone. The *content* of the steps carries its own tiers — steps 1–3 exact, step 4a proved within the L0 independence model, 4b discussion-grade, 4c robust-qualitative, and the *timing* of the steps not derived at all — so "the cascade is exact" is true of the dependency graph, not of every diagnostic the graph orders.

The downstream consequence is that Part II's diagnostic core — the satisfaction-gap and control-regret split from Ch.3 — has its corrective machinery here. The 2×2 cell map routes to four different revision targets via the cascade: M_t first under epistemic mismatch; Σ_t under positive control regret; Π expansion under control regret persisting after strategy revision; and O_t only when satisfaction gap persists after all earlier revisions. Objective revision is the *last resort*, not the first response to unmet goals. This is the chapter's implications-side payoff: the diagnostic from Ch.3 is what tells the agent *what* to revise; the cascade from this chapter is what tells the agent *in what order*; and the orient cascade's forced ordering is what makes the diagnostic actionable rather than merely informative. Part II would have been a collection of diagnostics without revision machinery without it.

The cascade also depends on the directed-separation scope condition from `#der-directed-separation` (Part II Ch.1) for the *sequential resolution* to be meaningful. For any **causally-disciplined** agent — one running at $\kappa_{\text{processing}} = 0$, whether by construction (Class 1) or in effect (an idealized Class 2 at $\kappa = 0$) — the cascade runs sequentially because the belief-update has no causal dependence on G_t, so the resolution order is a sequential one rather than a simultaneous fixed-point; this consumes only the zero, not the architecture-inspection certificate. As $\kappa_{\text{processing}}$ leaves zero the resolution acquires a coupled component, and in a Class 3 (Coupled) architecture the cascade lives in one merged computation: the component can be *asked* to report in cascade order, but it cannot make its own step 1 goal-blind, because the goal is upstream of every computation it performs. The framework's treatment of Class 3 — scaffolded loops around coupled components that recover the cascade at the wrapper level (`#der-class-coercion-via-wrapping`, `#result-coupled-diagnostic-framework`) — depends on this chapter's derivation: what scaffolding *is* structurally doing is reproducing the orient cascade externally when the component cannot reproduce it internally. What that buys, and in what sense it is required, is stated precisely in §"Coupled architecture" below.

Exploration at two ends of the uncertainty spectrum.

The chapter's appendix-side derivations carry one of AAT's load-bearing methodological positions: exploration is forced by Lyapunov persistence collapse, not by epistemic value-of-information. `#deriv-causal-ib-exploration` derives it (*conditional* on the continuous-time Lyapunov bounds of `#result-sector-condition-stability`): the maximum tolerable observation-noise expectation $U_o^{\max} = U_M(Rc_{\min}/\rho^{\text{eff}} - 1)$ scales with U_M, so as the agent becomes confident ($U_M \to 0$), the tolerable noise tightens, and the agent must actively seek pristine, low-noise observations to penetrate its own stubbornness. The framework therefore carries *two parallel exploration drives at opposite ends of the uncertainty spectrum*: the standard epistemic drive $\lambda_{\text{info}} \propto U_M$ (explore when uncertain — dominant when U_M is high) and a Lyapunov-survival drive $\lambda_{\text{surv}} \propto 1/U_M$ (explore when confident in a drifting world, because the agent's own update gain has dropped below the rate at which reality is shifting — dominant when U_M is low). The $U_o^{\max}$ bound and its KKT shadow price are exact within that scope; the *mapping* of the shadow price onto a CIY weight λ_{surv} is heuristic in the scalar segment (it assumes CIY $\propto 1/U_o$) and becomes clean only in the matrix lift below. The implications-side payoff is that this *sidesteps active-inference's dark-room objection at structural rather than parametric level*: an EFE agent with strong priors can stop exploring; an AAT agent in a drifting world cannot stop exploring *and remain within the persistence condition* — exploration is forced by the same Lyapunov machinery that grounds the rest of Part II, so the only way to stop is to leave the survivable set.

`#deriv-causal-ib-lmi` carries the matrix-form sharpening (*conditional*: linear-Gaussian observation model, Riccati stabilizability, steady state). The scalar form of the survival imperative admits a "blank wall" attack — an action that selects a low-noise channel orthogonal to the drift direction satisfies the scalar constraint while the agent's mismatch in the drifting subspace grows unbounded. The LMI lift to a Linear Matrix Inequality on the Fisher Information Matrix, with a positive-semidefinite matrix Lagrange multiplier $\Lambda \succeq 0$ as directional shadow price, resolves the attack by complementary slackness on direction — the exploration bonus for blank-wall actions is mathematically zeroed. The result also makes precise that the Fisher Information Matrix is doing *three structurally distinct* loads in the framework: credit assignment via `#deriv-fisher-whitened-update-rule`, coordinate-forcing via `#disc-additive-coordinate-forcing`'s fourth instance, and now survival via the LMI here. The same geometric object recurs because the framework's coordinates are forced rather than chosen; the LMI form is the first explicit cross-instance composition where two of those loads — coordinate-forcing and survival — interact within the same machinery.

The composition with Part I Ch.4's persistence-information-rate floor is worth surfacing, and it is this segment's composition rather than either source's theorem. The persistence-bandwidth floor $C \geq \mathcal{J}/2$ (`#deriv-persistence-cost`; conditional on Model S with Gaussian-OU signal in the high-resolution regime) is what a confident agent in a drifting world is structurally required to *meet*; the survival-imperative LMI is what tells the agent *in what directions* to spend the bandwidth. The two live in the same linear-Gaussian scope, and there they are the same quantity seen twice: the Mitter-Newton information-supply rate that saturates the floor is $\dot{I} = \frac{1}{2}\,\mathrm{tr}(H^T\Sigma_o^{-1}HP_{ss}) = \frac{1}{2}\,\mathrm{Tr}(\mathcal{I}_o\,P_{ss})$ — the Shannon rate is a trace against the same action-conditional Fisher matrix $\mathcal{I}_o(a)$ the LMI constrains. Read together: the floor is a scalar bound on that trace, the LMI is a matrix bound on $\mathbb{E}_\pi[\mathcal{I}_o(a)]$, and complementary slackness routes the rate toward the drifting modes. Without the LMI, the bandwidth could in principle be supplied in irrelevant directions; with it, *bandwidth-per-direction is the binding constraint*. Outside the shared linear-Gaussian scope the FIM is no longer the information rate (the EIG-vs-FIM gap both LMI segments flag), and the composition reverts to the qualitative reading. A derived per-direction floor — the trace identity pushed through the Riccati steady state — is an open strengthening recorded in Working Notes.

Coupled architecture, the bias bound, and the scaffolding requirement.

The chapter is also where the framework's treatment of Class 3 (Coupled) agents lands its most consequential structural commitment. `#der-directed-separation`'s Findings section covers the architectural classification in full — Class 1 (Separated), Class 2 (Partial), Class 3 (Coupled) — with the explicit Class 3 scope exit that hands the agents off to `03-llm-core/` rather than treating directed separation as an unenforced approximation. `#deriv-observation-ambiguity-bias-bound`'s Findings cover the constant under the (PI) parameterization-invariance axiom (*conditional*: exact under its named hypotheses): Track 1 (transport-inequality, linear in I) gives $C_{W_2}^2 = 2L_{\text{post}}^2/\rho_{\text{LSI}}$ under log-Sobolev + Lipschitz-posterior; Track 2 (Fisher-Rao, $\sqrt{I}$ scaling) gives universal dimension-free $C_{FR} = \sqrt{2}$ under (PI)+Čencov + small-information regime. The no-go counterexample (heteroscedastic-Gaussian, demonstrating that no universal C exists under Euclidean-parameter norm) is what forces (PI) from convergent representational choice to load-bearing for theorem-level status — adding the fourth instance to `#disc-additive-coordinate-forcing`'s catalog of forced coordinates (the (PI)+Čencov downstream-theorem cluster). It is *not* a fourth identifiability-floor instance, however — its external theorem (Čencov 1982) *forces* the escape's uniqueness rather than *forbidding* the inferential task, and the single forced escape (adopt (PI)) is logically incompatible with the $\geq$ 2-distinct-escapes structure that defines a floor; the genuine fourth `#disc-identifiability-floor` instance is `#der-architecture-noidentifiability` (architecture-noidentifiability from on-policy summary data, dual-anchored on Kalman-Ho 1966 / CHT-at-agent-as-SCM).

What the implications segment can develop beyond the source-segment Findings is what the *coupled-diagnostic framework* of `#result-coupled-diagnostic-framework` (in `03-llm-core/`, *conditional*) means for the cascade. For Class 3 (Coupled) agents — LLMs, transformer-based agents with attention processing goals and observations together — the orient cascade's information-dependency-forced ordering cannot be produced *as a goal-blind cascade* by the model itself; the cascade lives in one merged computation. But the Part II diagnostic quantities (δ_{sat}, δ_{regret}) remain *defined* on post-update state, and the Lipschitz bound $|\delta_{\text{sat}}^{(\text{coupled})} - \delta_{\text{sat}}^{(\text{clean})}| \leq L_A \cdot \|\Delta M_{\text{bias}}\|$ (conditional on Lipschitz regularity of A_O in M_t, with L_A domain-dependent and not computed by the theory) shows the diagnostic error is bounded by the bias bound, which is itself $O(\kappa \cdot \mathcal{A})$ (coupling × ambiguity) with the constant now derived under the two tracks. Definedness is not extractability: whether the running agent can *produce* the diagnostic values needs instrumentation the theory names but does not supply (`#result-section-ii-survival`).

The structural conclusion is the chapter's load-bearing implication for practitioners, and it is worth stating at exactly the strength the sources earn. Two things are being recovered by a scaffolded loop, and they have different status. The *ordering* of the cascade can be imposed on a coupled component behaviorally — a structured-reasoning template or a multi-step prompt asks for "what did I learn" before "what does it mean for my plan" — and `#result-coupled-diagnostic-framework` classes this as a normative design pattern (discussion-grade), whose compliance is empirical. The *goal-blindness of step 1*, by contrast, cannot be produced inside a single goal-conditioned call at all: that is what Class 3 means (the goal is causally upstream of every computation), so the belief-update's bias is governed by the bias bound no matter how the output is formatted. Recovering a goal-blind step 1 therefore requires that the belief-update query be *separated* from the goal-conditioned one — the strict-wrapping regime W_1 of `#der-class-coercion-via-wrapping`, which yields Class 1 (Separated) status at the wrapper level by type signature (exact under its (C1)–(C3); approximate with a bounded leakage rate under (C1)–(C2)), at the cost of more component calls per macro-step and a residual selection-channel leakage whose *structural* bound additionally needs (C2′), no goal-correlated state across the call boundary — failing which W_1 too degrades to a behavioral bound. Partial wrapping W_2 (one goal-conditioned call, typed parsed response) gives the ordering but only *behavioral* separation, with no structural bound on leakage. So: **scaffolding is structurally necessary for a Class 3 (Coupled) substrate to run the orient cascade with a goal-blind belief-update — the only route to the causally-disciplined zero that Part II's sequential-resolution-dependent results consume — and W_1 is the regime in which that recovery is certifiable rather than merely complied with.** What scaffolding is *not* needed for is the large part of Part II that never invoked directed separation: 16 of 24 Part II results survive exactly at the statement level on a raw Class 3 agent (`#result-section-ii-survival`), including Prop B.5 of `#disc-credit-assignment-boundary` (persistence is credit-assignment-free). The necessity is about the sequential resolution and the results scoped to causal discipline, not about Part II wholesale. For a production LLM-agent practitioner, the claim is that structured-reasoning templates and multi-step prompts buy ordering, separate goal-blind calls buy an unbiased belief-update, and external monitors buy the instrumentation that makes the diagnostics computable at all — three different things, each doing real structural work, none of them aesthetic. *[Hypothesis]* The same mechanism is a natural reading of dual-process accounts of human cognition (Kahneman 2011) and of executive scaffolding in organizations, where an external structure recovers an ordering the coupled individuals do not enforce internally; the framework offers this as a structural analogy, not a derived transfer.

The engineering anchor worth carrying is the $\kappa \cdot \mathcal{A}$ decomposition (`#scope-observation-ambiguity-modulation`). The architectural coupling κ is *fixed* at the component level for a Class 3 substrate (you cannot un-couple an LLM by prompting; you can only wrap it); the ambiguity $\mathcal{A}$ is *variable* (you can engineer the prompt and environment). Prompt engineering, in the framework's vocabulary, is literally $\mathcal{A}$-reduction — and the bias bound says reducing $\mathcal{A}$ tightens the bound. The information term in both tracks is $I = I(G; \Omega_\tau \mid e_\tau, M_{\tau^-}) = \mathcal{A} \cdot H(\Omega_\tau \mid e_\tau, M_{\tau^-})$, and both tracks bound the *distance* as $\sqrt{I}$ — Track 1 bounds W_2^2 linearly in I, Track 2 bounds d_{FR} as $\sqrt{2I}$ — so halving $\mathcal{A}$ at fixed residual entropy shrinks the bias envelope by $\sqrt{2}$ under either track; $\mathcal{A}$ alone is the coarser envelope, I the quantity the theorems are stated in. The framework *predicts* that LLMs do best on low-$\mathcal{A}$ tasks (coding against tests, structured reasoning, well-defined transformations) and worst on high-$\mathcal{A}$ tasks (open interpretation, ambiguous goals, contested values); the prediction is consistent with the familiar practitioner pattern but has not been tested in the framework's own terms. The framework's prescription is concrete: where κ is structurally large, design $\mathcal{A}$ structurally small.

Deliberation is internal exploration that does not relax the bandwidth floor.

`#disc-exploit-explore-deliberate` extends the standard exploit / explore tradeoff into a three-way framing by surfacing deliberation as *internal exploration in model space rather than environment space* — the segment's own characterization, at discussion-grade, with the extended deliberation threshold as its one derived piece. What distinguishes deliberation from the other two activities in Pearl's terms is *where the interventional query is answered*: exploration obtains Level-2 data from the world through the loop (`#der-loop-interventional-access`, Part II Ch.2), whereas deliberation evaluates interventional and counterfactual queries *against the agent's own model* M_t — the answer is only as good as the model, which is why the source segment names model fidelity, not data quantity, as deliberation's ceiling. The contrast is what keeps "deliberation as internal exploration" from collapsing into "the agent thinks more": the two activities answer the same kind of question from different oracles, one of which can be wrong.

The composition with Part I Ch.4's bandwidth floor is the finding that has no other natural home, and it is this segment's composition, stated with the floor's scope: *deliberation does not relax the persistence-information-rate floor*. `#deriv-persistence-cost` shows (Model S, Gaussian-OU signal, high-resolution regime) that any process holding mismatch at the sector-persistence ultimate bound must acquire information *from observations* at rate $\dot{R} \geq n\alpha/2$ nats per unit time ($\alpha/2$ per dimension). That floor is a bound on observation-channel information about the environment, and internal computation contributes nothing to it: the model's information about the world is a function of what the observation channel delivered, and the data-processing inequality forbids processing from increasing it. So a deliberating agent is not supplying the floor from inside — it is *pausing* its supply, which is exactly the drift term $\rho_{\text{delib}} \cdot \Delta\tau$ of `#der-deliberation-cost`; the extended threshold from `#disc-exploit-explore-deliberate` says to deliberate until the marginal benefit (epistemic gain + strategic gain) equals that marginal drift cost. What deliberation *can* change is the target the floor is evaluated at: a better policy found by deliberation can lower ρ^{eff} and hence the correction rate α persistence demands (the action-value channel `#der-deliberation-cost` names but does not formalize), and structural adaptation can change the model class and its misspecification floor. The invariant is therefore precise — at a fixed correction rate and distortion target the floor is untouched by any amount of internal computation — and the posture that follows is that deliberation is *temporary reallocation of the cycle budget*, never a substitute for external bandwidth. The cross-domain transfer follows naturally to chess engines (search depth and observation freshness must both be funded), military command (war-gaming and reconnaissance must both be funded), and science (theory-building and experiment-running must both be funded).

The connection to Part II Ch.4's strategy-complexity-cost (9.9) is also worth surfacing here, at discussion-grade. Part of the cost of deliberation is the cognitive cost of holding Σ_t in the agent's representational capacity while operating on it — `#disc-exploit-explore-deliberate` notes that $\mathrm{DL}(\Sigma_t)$ lower-bounds the deliberation time a meaningful revision needs, and that for an LLM agent this surfaces as context-window competition. The reading this segment offers is that representational capacity and observation bandwidth are the same *kind* of constraint — both cap an information rate the agent must sustain — so the persistence-bandwidth vocabulary can be extended to cover environmental-observation bandwidth, strategy-representation capacity, and deliberation budget together. That unification is a framing, not a theorem: only the observation-channel floor is derived; the other two are resource constraints the framework names.

What Part II closes on, and the bridge into Part III.

The chapter is the structural close of Part II, and what Part II has delivered as a whole is now worth naming. The diagnostic core (satisfaction-gap and control-regret split, convention-indexed by the C1/C2/C3 hierarchy) gives the framework two orthogonal signals that distinguish "the world doesn't permit it" from "you're not doing it well enough." The strategy-DAG commitment with Markov factorization under causal sufficiency gives the formal representation the diagnostic operates on. The Pearl-Level-2 access from the loop, combined with the Correlation Hierarchy's scope statement and the chapter's directed-separation architectural classification, gives the agent the data substrate and the architectural class it sits in. The bias bound for Class 3 (Coupled) agents, with its conditional-theorem constant under named sub-scopes, gives the framework's quantitative reach into LLM-class substrates without overclaim. The orient cascade gives the corrective machinery; the survival-imperative drive gives the structural exploration mandate; the coupled-diagnostic framework and the wrapping construction give the scaffolding-as-structural-requirement claim in the form stated above.

The result of all this is what `#result-section-ii-survival` formalizes (*conditional* on its segment-by-segment analysis): Part II's definitional and structural architecture transfers exactly to fully-coupled (Class 3) agents in 16 of 24 cases *at the statement level* — the results' claims do not invoke directed separation — with the remaining results either approximating with bounded error (5/24) or naming the precise modification required (2/24 plus 1 boundary-defining failure). Statement-level survival is not operational extractability; computing the surviving quantities on an LLM agent needs the instrumentation that segment names. The bias-bound theorem is what lifts the approximate-survival category from order-of-magnitude guidance to conditional theorem *within* the Track 1 / Track 2 sub-scopes; outside them the order-of-magnitude reading stands. Part II's reach is wider than the most-restrictive scope condition suggests; this chapter is where the chapter-end can name that reach without overclaiming.

The bridge to Part III is direct. The directed-separation classification opens to *axis-decomposed composite-level class inheritance* (per the table in `#der-directed-separation` §"Composite-level class inheritance"): composite architectural class lifts through routing structure plus substrate sharing, *not* through goal alignment per se. Goal alignment vs strategic interaction operates on the parallel-but-independent *dynamic-regime* axis (`#disc-dynamic-regime-axis`), where strategic composition transitions composites from R0 contraction-regime to R1 equilibrium-regime or R2 cyclic-distributional-regime without architectural-class change (under goal-blind routing). The third instance of the identifiability-floor pattern (Instance 3, on the coupling-topology bit) lives in Part III: composite contraction *and regime* certification from component-marginal data is structurally impossible (Liberzon 2003 common-Lyapunov-nonexistence; cf. `#deriv-regime-marginal-indistinguishability` for the broader regime-axis witness), with five structural escapes including observer-on-sub-agent interventional access (Mode 2 of the loop-Level-2 pattern from Ch.2) and the passive escape (e) of composite-level convergence-rate-class observation. Part III is where AAT's machinery operates at the composite scope, with the chapter-end here as the last single-agent prerequisite to land.

What this chapter closes on, then, is the structural completion of Part II: the cascade that makes the diagnostics actionable, the dual-drive exploration that prevents dark-room collapse, the bias bound that quantifies Class 3 (Coupled) reach with conditional-theorem precision, the scaffolding requirement that makes a goal-blind cascade on a coupled substrate a matter of wrapper structure rather than prompt aesthetics, and the deliberation framing that grounds the three-way resource allocation in the same observation-channel floor as everything else. Part III's machinery operates on what Part II built; the implications work here is the last single-agent-scope synthesis before the composite chapters take over.

Part III

Agentic Composites

Scope: multiple agents interacting through a shared environment, or equivalently, the internal structure of composite agents. The composition-consistency commitment (11, recognized as a Part III Meta-Architecture II member in the chapter that opens this Part) requires that the theory apply at every level of description where the scope condition is met. This Part develops what "applies at every level" means formally (the composition closure criterion, which requires an additional contraction assumption beyond Part I's sector condition), what happens when composition is imperfect, and what the dynamics of inter-agent interaction look like.

Correlated observations as default; independence as the special case requiring justification. Adversarial dynamics are one end of a teleological unity spectrum, not a separate theory.

Three primary sources: the simulation-validated adversarial dynamics from TFT (TF-11/Appendix F, track-b simulations); the composition spike (`spikes/spike-agent-composition.md`*) which derives composition consistency from the scope condition's level-independence; and Miller's Coevolving Automata Model (Ex Machina, 2022), which provides constructive mechanisms for composition dynamics — how composites form, undergo phase transitions, and restructure through neutral drift and niche creation. Agent opacity (H_b) is adopted from Hafez et al. (2026).*

Chapter 11

Meta-Architecture II: Composition Consistency and the Dynamics-on-Architectural-Class Facet

*Part II opened with the certificate spine and its three Part-II-operative facets (M1 boundary / M2 scope-of-existence / M3 forced-identity). Part III opens with two cross-level meta-recognitions that organize the composite-agent material that follows: the composition-consistency commitment (how the framework *applies across levels*)* and the fourth certificate-spine facet (how architectural class *changes* under operations). Both are introduced before any Part III content chapter because both are framework-level recognitions — the composite-agent machinery in subsequent chapters is the instantiation, not the source.*

Composition consistency — cross-level compatibility as framework commitment *(11, methodological commitment). The framework's scope conditions (1.5, 1.6) do not restrict which level of description AAT applies to; the same observable can be analyzed at the individual / team / organizational level, and the theory must not return contradictory answers across those analyses. The commitment has three operational laws (tempo composition, persistence compatibility, mismatch consistency) and a three-layer decomposition (scope selection — 12.2; admissibility — 13.1; transfer — Tier-dependent via* `#result-contraction-template`*). Under Tier 1M sub-agents the composite contraction rate admits a closed form; outside Tier 1M the screening test $\tau_{eq} \ll \tau_{ext}$ is heuristic. Cross-level companion to the four certificate-spine facets — the spine names the geometry of a single certificate, composition-consistency names the relationship across certificates at different levels. Sister to 11.1: M4 names how architectural class *changes* under operations; composition-consistency names how the framework *applies across levels* regardless of architectural class.*

***M4 — Modularity-state dynamics** (11.1, dynamics-on-architectural-class facet): modularity is a *contested property* — composite agents do not sit fixed at one architectural class; they move under three structurally-distinct operations. **Truthification** (self-driven, modularity-increasing; restorative valence): the wrapping construction at the component level (13.4, in Ch.2 of this Part) and institutional scaffolding at the composite level (per 11.3 §"Defensive scaffolding"). **Strategic self-coupling** (self-driven, modularity-decreasing; offensive-enabling valence): operation leg 11.2. **Adversarial coupling pressure** (externally-driven, modularity-decreasing; vulnerability-creating valence): operation leg 11.3. Three pairwise dual relationships organize the cluster — truthification ↔ adversarial-pressure as direct dual (cultivation vs attack); truthification ↔ strategic-self-coupling as goal-belief-axis dual (both wrapping-construction-style commitments on opposite poles — goal-blind belief-update vs goal-conditioned action-selection); strategic-self-coupling ↔ adversarial-pressure as same-architecture-shape-opposite-driver. M2's Architecture ladder reads as the phase space across which M4 operations move populations.*

The two operation-leg segments. *11.2 develops the self-driven coupling-increasing operation with three structural extensions ((M1) coupling-dependent action space, (M2) strategy-DAG enabling-edge type, (M3) reversibility-cost asymmetry) and a four-mechanism prior-art adoption (Schelling 1960 external commitment devices, Ainslie 1992/2001 intertemporal personal rules, Akerlof-Kranton 2000/2010 identity coupling, Frank 1988 emotional commitment).* ***11.3*** *develops the externally-driven coupling-increasing operation — adversaries drive opponent coupling because coupling expands attack surface (identity-binding / affect / sunk-cost mechanisms); diagnostic capture is directional; orient-cascade inversion under pre-loaded O_t; defensive scaffolding as composite-agent restoration. The two operation legs are opposite-valence operations on the same architectural-class state $\kappa_{processing}$ per 6.3.*

This is a *meta-recognition* about how AAT applies, not a fact about how any individual agent works: AAT's predictions must be compatible across levels of description. If a system S and any decomposition $\{S_1, \dots, S_n\}$ both satisfy the scope condition, the theory's claims at the S-level cannot contradict its claims at the $\{S_i\}$-level. The scope condition does not restrict which level the theory applies to, so the theory itself must be level-invariant — otherwise the same observable could receive contradictory answers depending on where the analyst draws the modeling boundary. The commitment is methodological at the meta-requirement layer (a framework choice about applicability, not a physical fact about agents); its operational consequences are tiered.

This commitment is the cross-level companion to the four certificate-spine facets (5). Where the spine's facets describe the geometry of a *single* equilibrium-stability certificate (interior / scope-of-existence / forced-identity / boundary / projection), composition consistency describes the relationship *across* certificates at different levels — the same persistence argument must apply at every level where the scope condition is met, and the operational unpacking below is how that "applies at every level" claim cashes out. It sits alongside 11.1 as the second cross-level meta-recognition: M4 names how architectural class *changes* under operations; composition consistency names how the framework *applies across levels* regardless of architectural class. Together they organize the Part III meta-architecture.

The commitment has formal teeth: three composition laws must exist. *Tempo* at the composite level must be expressible as a function of sub-agent tempos plus their coordination structure; *persistence* of the composite must be derivable from the sub-agents' individual persistence plus that coordination structure; *mismatch* must compose consistently. The laws themselves — what tempo and persistence look like at the composite level, and when they transfer cleanly — are *derived* from this postulate in Part III under specific conditions, organized into three successively more specific layers:

- a *scope-selection* layer (12.2, teleological-alignment threshold) — *which* decompositions qualify as composites at all;
- an *admissibility* layer (13.1, conditions (A1)–(A4) and the bridge lemma) — *when* composite macro-dynamics are well-posed under those scope conditions;
- a *transfer* layer (13.1's tier-specific contraction assumption together with `#result-contraction-template`'s (CC-parallel)/(CC-cascade)/(CC-feedback) closed forms) — *which* Part I/II results lift faithfully, with sharpness depending on the tier of the sub-agents.

The transfer layer is where the deepest result lives. For **Tier 1** sub-agents (Bayesian updaters on exponential families, linear correctors with positive-definite gain, gradient descent on strongly convex losses), the composite contraction rate admits a closed-form lower bound — the parallel case yields $\lambda_c = \min_i \lambda_i$ (the composite is bounded below by its slowest sub-agent), cascade composition gives the same up to a coupling-gain adjustment, and heterogeneous negative-feedback composition gives a closed-form inequality involving feedback-loop gains and coordination costs. Macro-level persistence then reduces to the same persistence inequality used at the individual level, applied to the composite. For Tier 2 (locally convex, nonlinear) and Tier 3 (non-convex, discontinuous) sub-agents, this transfer holds only locally with degradation or per-domain, and the qualitative screening test $\tau_{\text{eq}} \ll \tau_{\text{ext}}$ — internal equilibration timescale short relative to external-dynamics timescale — remains useful but does not entail the formal persistence condition without further verification.

The boundary an analyst draws is itself a modeling choice. A development team is simultaneously an individual developer (an AAT agent), a team (a composite AAT agent), and part of an organization (a sub-agent in a larger composite); the scope condition is satisfied at every level. Composition consistency is what guarantees the theory does not return contradictory answers about observables regardless of which boundary is drawn, and the layered derivation above is what makes that guarantee precise rather than aspirational.

Commitment (composition-consistency)

For any system S satisfying the scope condition (1.6), and any decomposition of S into subsystems $\{S_1, \ldots, S_n\}$ where each S_i also satisfies the scope condition, AAT's predictions at the system level must be compatible with its predictions at the subsystem level. Specifically, composition laws must exist such that:

1. **Tempo composition**: $\mathcal{T}_S$ is expressible as a function of $\{\mathcal{T}_{S_i}\}$ and the coordination structure among them
2. **Persistence compatibility**: the system's persistence is derivable from the sub-agents' individual persistence conditions plus coordination structure
3. **Mismatch consistency**: δ_S is derivable from $\{\delta_{S_i}\}$ and their interaction structure

Structural consequence (derivation hierarchy)

Cross-level compatibility requires three successively more specific conditions, each with its own segment in Part III:

1. **Scope**: The decomposition is a *composite* (not merely a multi-agent system) when 12.2 is satisfied — teleological alignment via at least one of three disjunctive routes (shared objective, hierarchical derivation, or mutual benefit). Without scope-satisfaction via any route, level-compatibility is vacuous (composition quantities are not well-defined, so there is nothing to be consistent about).
2. **Admissibility**: Given a composite, its macro-dynamics are well-posed when the closure admissibility conditions (A1)-(A4) hold (13.1). These require AAT-shaped macro-state, macro-mismatch, macro-tempo, and sector-bounded correction at the composite level.
3. **Transfer of results**: Individual Part I/II results lift to the composite level through the bridge lemma in 13.1, which requires a *tier-specific contraction assumption* beyond (A4). For **Tier 1** agents (all Bayesian updaters on exponential families, linear correctors with positive-definite gain, gradient descent on strongly convex losses), the contraction holds and Part I results transfer exactly. For **Tier 2** agents (locally convex, nonlinear prediction models), transfer holds locally with factor degradation. For **Tier 3** agents (non-convex optimization, discontinuous corrections, non-mismatch-driven state), contraction must be verified per-domain and Part I results do not automatically lift.

[Derived (Conditional on Tier 1M + admissible composition topology, from `#result-contraction-template` *(CC-parallel) / (CC-cascade) / (CC-feedback))]*

Composite contraction rate — closed form under Tier 1M. When sub-agents satisfy the contraction-template preconditions (CT1)–(CT3) of `#result-contraction-template` (Tier 1M, equivalently Tier 1 of 13.1's bridge lemma under the DA2'-inc ≡ (CT2)-at-$M = I$ equivalence) and the composition topology falls under one of the closure cases of `#result-contraction-template`, the composite contraction rate λ_c admits a closed-form lower bound:

- **Parallel composition** (CC-parallel): $\lambda_c = \min_i \lambda_i$ — the composite contraction rate equals the slowest sub-agent's, with composite metric $M_c = \text{blockdiag}(M_1, M_2)$. Equivalently in timescale form, the composite relaxation time $\tau_c = 1/\lambda_c = \max_i \tau_i$ is bounded *below* by the slowest sub-agent's timescale.
- **Hierarchical / cascade** (CC-cascade): same bound, up to coupling-gain adjustment.
- **Negative-feedback heterogeneous** (CC-feedback) / (CM2-M): $(\lambda_1 - C_1)(\lambda_2 - C_2) > k_{12}k_{21}/4$ with feedback-loop gains k_{ij} and coordination costs C_i from 15.1.

Macro-level persistence then reduces to the standard persistence condition (4.5) applied to the composite: $\alpha_c > \rho_{\text{eff}}/R_c$, with α_c bounded by the topology-specific result above and effective disturbance $\rho_{\text{eff}} = \rho_{\text{ext}} + \varepsilon^* \nu_c$ from 13.2's bridge-lemma instantiation. Both Part I results (single-agent persistence) and the bridge-lemma's macro-tracking faithfulness flow from this single inequality. This is the formal counterpart to the screening test below: under Tier 1M the screening test is *derived*, not heuristic.

Heuristic (timescale separation — residual scope: Tier 2 / Tier 3)

Practical screening test for composites outside Tier 1M. Tier 2 sub-agents (extended-Kalman, locally-convex gradients, nonlinear prediction models) and Tier 3 sub-agents (non-convex optimization, discontinuous corrections, non-mismatch-driven state) do not in general carry the closed-form composite-rate bound above — the bridge-lemma contraction holds only locally with $\kappa(D\hat{o})^2$ degradation (Tier 2) or must be verified per-domain (Tier 3). For these composites, the qualitative condition

$$\tau_{\text{eq}} \ll \tau_{\text{ext}} \tag{11.1}$$

— internal equilibration timescale (the time for sub-agents to approximately synchronize models and coordinate actions) short relative to external-dynamics timescale — remains a useful *discussion-grade screening test*. When it holds, composite description is plausibly well-posed; when it fails, composite description is plausibly broken. The screening test is the operational analog of the persistence condition (4.5), but for Tier 2 / Tier 3 it does not entail the formal persistence condition without local-region or per-domain verification. Whether common organizational settings (software teams, military units) genuinely sit in Tier 1M or merely satisfy the heuristic without satisfying Tier 1M conditions is an empirical question, not a derived fact.

Axiomatic for the meta-requirement (cross-level compatibility). The postulate itself is a structural requirement for AAT's internal consistency: if the scope condition doesn't restrict which level the theory applies to, the predictions must not contradict across levels.

The operational consequences decompose into three layers with distinct epistemic statuses:

- **Scope selection** (12.2, robust-qualitative): which decompositions constitute composites. The scope condition is a disjunction of three qualitatively distinct routes (shared objective, hierarchical derivation, mutual benefit); whether they reduce to a common scalar threshold is open. Coverage of well-understood cases holds via the disjunction.
- **Admissibility of composite dynamics** (13.1, conditional): the (A1)-(A4) conditions are stated formally; the bridge lemma is conditional on the incremental sector bound (DA2'-inc), strictly stronger than (A4) alone.
- **Transfer of individual-agent results** (tier-dependent): Tier 1 composites transfer Part I/II results exactly; Tier 2 transfer with degraded factors; Tier 3 require per-domain verification. This is the sharpest scoping of "every result applies at every level" — it applies at every level *for Tier 1 composites*, degrades gracefully for Tier 2, and holds per-domain for Tier 3.

The composite contraction rate has two epistemic layers. **Tier 1M is derived (exact)**: under (CT1)–(CT3) on each sub-agent and an admissible composition topology, λ_c admits the closed-form lower bound from `#result-contraction-template`'s (CC-parallel) / (CC-cascade) / (CC-feedback) — the slowest-sub-agent / coupling-adjusted / heterogeneous (CM2-M) cases respectively. The screening condition $\tau_{\text{eq}} \ll \tau_{\text{ext}}$ is, in this regime, the formal persistence condition $\alpha_c > \rho_{\text{eff}}/R_c$ specialized to internal-vs-external rate balance — not heuristic. **Tier 2 and Tier 3 retain heuristic / discussion-grade status**: the closed-form composite rate does not transfer (Tier 2 with local degradation; Tier 3 per-domain), and $\tau_{\text{eq}} \ll \tau_{\text{ext}}$ remains a useful screening test without entailing macro-level persistence. The "small gap between screening and Tier 1M conditions in common settings" is an empirical claim about the population of real composites, not a derived fact.

The same pattern as individual persistence. The persistence condition says: there is a measurable threshold below which an individual agent's mismatch grows without bound and the agent degrades. Composition consistency says the same thing at the composite level: there is a measurable threshold below which the composite description breaks down. Under Tier 1M, the threshold is exact and closed-form — $\alpha_c > \rho_{\text{eff}}/R_c$ with α_c from (CC-parallel) / (CC-cascade) / (CC-feedback). Under Tier 2 / Tier 3, the threshold is only qualitatively captured by the screening ratio $\tau_{\text{eq}}/\tau_{\text{ext}}$ and exact transfer requires local-region or per-domain verification. The two-tier structure is the precise way "the theory applies broadly" — broadly under Tier 1M, qualitatively-with-conditions outside it.

What this buys the theory. With composition consistency stated early: - Part I/II results lift to the composite level *for Tier 1 composites passing the composition scope condition*, with degraded-factor lift for Tier 2 and per-domain verification for Tier 3 — the three-layer decomposition above makes this precise rather than leaving "applies at every level" as an unbounded claim - Part III becomes the study of *what happens near and beyond the threshold* — coordination overhead, unity dimensions, symbiogenic transitions, adversarial dynamics — rather than a separate multi-agent theory - The formal test for composition validity (13.1) develops the rigorous version of the timescale condition, with the incremental sector bound identifying which agent classes admit the contraction needed for cross-level transfer; `#result-contraction-template` lifts that bound into a metric-aware framework where (CC-parallel) / (CC-cascade) / (CC-feedback) yield the topology-indexed closed forms cited above

When composition fails. The persistence condition $\alpha_c > \rho_{\text{eff}}/R_c$ (Tier 1M) — or its qualitative shadow $\tau_{\text{eq}} \ll \tau_{\text{ext}}$ (Tier 2 / Tier 3) — fails when: - Internal coordination slows: C_{coord} from 13.2 rises through coupling overhead, conflict, or bureaucratic process, depressing α_c below the threshold; equivalently, τ_{eq} stretches. - External dynamics accelerate: ρ_{ext} rises (adversary acts faster, market shifts, crisis compresses decision timescales); equivalently, τ_{ext} shortens. - Both simultaneously (the classic organizational failure mode — internal friction increases while external demands intensify).

This is the formal analog of Brooks's Law: adding people to a late project increases $\varepsilon^* \nu_c$ in ρ_{eff} (13.2's coordination-overhead bound) and stretches τ_{eq} while ρ_{ext} and τ_{ext} stay fixed (the deadline doesn't move). Under Tier 1M the mechanism is formal — eventually the persistence inequality flips. Under Tier 2 / Tier 3 the mechanism is qualitative. Whether the specific mechanism (coordination-overhead saturation crossing the persistence threshold) is the dominant cause of Brooks's Law in practice is an empirical question.

The boundary is a modeling choice. A development team is simultaneously: individual developers (each an AAT agent), the team (a composite AAT agent), and part of an organization (a sub-agent within a larger composite). The scope condition is satisfied at every level. Composition consistency ensures the theory doesn't give contradictory answers about observable quantities (e.g., whether the team persists) regardless of which boundary is chosen.

What composition consistency does NOT say. It does not specify the form of the composition laws — those are derived in Part III (13.2). It does not say every decomposition is equally useful for analysis. And it does not require perfect internal coordination — only that internal equilibration is fast relative to external dynamics.

11.1 *Discussion:* Modularity-State Dynamics — Modularity as Contested Property Under Three Operations

`#der-directed-separation` characterizes modularity as architectural property — the static Class 1 (Separated) / Class 2 (Partial) / Class 3 (Coupled) partition under the $\kappa_{\text{processing}}$ measure. That partition is the *state space*. What this segment names is the *dynamics on the state space*: three structurally distinct operations that move an agent or composite agent across the architectural classes over time. **Truthification** is self-driven, coupling-decreasing — the agent (or its institutional environment) deliberately commits to architectural arrangements that restore directed separation. **Strategic self-coupling** is self-driven, coupling-increasing — the agent deliberately decreases its own modularity to acquire credibility-dependent actions. **Adversarial coupling pressure** is externally-driven, coupling-increasing — adversaries press the agent's coupling upward because coupling expands attack surface. The architectural class is not a fixed feature of an agent; it is a *contested property* that multiple structural operations act on, with different drivers and different valences. M4 names this dynamics pattern as the fourth meta-pattern of AAT alongside M1 (`#disc-identifiability-floor`), M2 (`#disc-separability-pattern`), and M3 (`#disc-additive-coordinate-forcing`) — all four sit as facets of the stability-certificate spine (`#disc-stability-certificate`); M4 specifically lives at the boundary of the architectural-class layer rather than the certificate-cone interior.

The recognition arose from convergent independent collaborator probes: multiple Opus instances arrived at the three-operation picture from different angles during the 2026-05-09 class-coercion-wrapping integration cycle. The convergence is itself evidence the pattern is *in the framework* rather than in any single agent's reading; M4's content is therefore *recognition and integration* — naming the pattern explicitly, mapping the three operations to their primary-segment grounds, and surfacing the dual relationships between them — not novel derivation.

The pattern.

Discussion

Each operation has the form: a structural mechanism acts on the agent's coupling state $\kappa_{\text{processing}}$ (`#der-directed-separation`) over some characteristic timescale, with a named driver (self vs external) and a named directional effect on the architectural class. The three operations together exhaust the framework's currently-recognized modularity-state operations. They are not symmetric — they differ in driver, in direction, and in the kind of strategic value (or vulnerability) they produce.

The three operations.

Table 11.1

Operation	*Driver*	*Direction on $\kappa_{processing}$*	*Strategic valence*	*Primary segment(s)*
Truthification	self-driven (agent's own deliberate commitment or institutional design)	coupling-*decreasing* — restores directed separation	restorative (recovers reality-tracking, reopens Part II results, closes adversarial attack surface)	`#der-class-coercion-via-wrapping` (component-level wrapping) + `#disc-adversarial-coupling-pressure` §"Defensive scaffolding" (composite-level institutional scaffolding)
Strategic self-coupling	self-driven (agent's own deliberate commitment)	coupling-*increasing*	offensive enabling (acquires credibility-dependent actions otherwise unavailable)	`#disc-strategic-self-coupling`
Adversarial coupling pressure	externally-driven (adversary's strategic action on the target)	coupling-*increasing*	vulnerability-creating (expands attack surface for the externally-acting adversary)	`#disc-adversarial-coupling-pressure`

The taxonomy along three axes — driver (self/external), direction (increasing/decreasing κ), and strategic valence (restorative/offensive/vulnerability) — already exhausts the three operations. A fourth cell in the (self/external) × (increasing/decreasing) grid — *externally-driven coupling-decreasing* — is not currently named as a distinct operation; the closest candidate is "external party imposing institutional architecture on the agent to enforce its modularity," which the framework treats as a special case of truthification under externally-supplied scaffolding rather than as a fourth operation. Whether that case warrants its own row is open framework-scope discussion; the current three-operation reading covers the currently-recognized cases cleanly.

Operational mechanisms across operations. Each operation is realized through specific operational mechanisms in the primary segments:

Truthification operates through two complementary mechanisms. *Component-level wrapping* (`#der-class-coercion-via-wrapping`) coerces a Class 2 (Partial) or Class 3 (Coupled) component into Class 1 (Separated) behavior at the wrapper level via external scaffolding — the W_0 / W_1 / W_2 regime hierarchy gives graded characterizations of how thoroughly the truthification has been applied (strict-W_1 structural commitment to goal-blind queries; W_2 output-structuring with typed parsed response; W_0 unwrapped baseline). *Composite-level institutional scaffolding* (`#disc-adversarial-coupling-pressure` §"Defensive scaffolding") constructs a composite agent whose modularity properties exceed those of its individual components — peer review, prediction registers, double-entry bookkeeping, adversarial procedure, prediction markets, source-authentication discipline, structured red-teaming. Both operate by externalizing channels of the orient cascade so resolution can run sequentially across the composite even when it cannot run sequentially within any single component. The two are not duplicate but *operationally distinct*: wrapping operates at the component level (one wrapper around one component); institutional scaffolding operates at the composite level (multiple sub-agents routed through scaffolded procedure).

Strategic self-coupling operates through four operational mechanisms (`#disc-strategic-self-coupling`'s mechanism table). *External commitment device* (Schelling 1960) couples to physical / institutional / public structure. *Intertemporal personal rule* (Ainslie 1992/2001) couples between current-self and future-self decisions. *Identity coupling* (Akerlof-Kranton 2000/2010) couples preferences to internalized identity-norms. *Emotional commitment* (Frank 1988) couples between rational calculation and affective response. Each mechanism differs in coupling target (external structure / intertemporal / identity / affect) but shares the structural shape: forgo flexibility now to acquire credibility-dependent actions later. The *amount of fungibility given up* is what the strategic interaction reads as credibility — Schelling's foundational structural observation.

Adversarial coupling pressure operates through three operational mechanisms (`#disc-adversarial-coupling-pressure`'s mechanism table). *Identity-binding* exploits the $O_t \to M_t$ pathway — make the target want the false belief to be true. *Affect / urgency* bypasses deliberate f_M — manufacture time pressure or affective load that prevents the orient cascade from running cleanly. *Sunk-cost engineering* exploits the $\Sigma_t \to M_t$ pathway — induce public or expensive commitment that subsequent M_t revisions cannot invalidate without coupled cost.

Dual relationships between operations. The three operations are not independent — they have structurally illuminating *dual relationships*.

Truthification and adversarial coupling pressure are direct duals at the cultivation-versus-attack axis. Adversarial pressure attacks what defensive scaffolding restores; the scaffolding restores what adversarial pressure attacks. The two operations are opposite-driver operations at opposite-direction effects on $\kappa_{\text{processing}}$. The asymmetric-advantage observation (`#disc-adversarial-coupling-pressure` §Discussion) is the in-repeated-interaction consequence of this dual: in adversarial settings, each side has structural incentive to (a) reduce its own coupling via truthification and (b) increase the opponent's coupling via adversarial pressure; the more-modular agent compounds informational advantage; the asymmetry produces strategic asymmetry over sustained engagement.

Truthification and strategic self-coupling are duals at the goal-belief separation axis (the structurally illuminating recognition surfaced in `#disc-strategic-self-coupling`'s §"Relationship to truthification"). Both are wrapping-construction-style structural commitments giving up flexibility along one axis for value along another, but operating on *opposite poles* of the goal-belief separation. *Truthification* structurally commits to goal-blind belief-update — the W_1 strict wrapper commits the composite to operate Class 1 at the belief-update layer by structurally foreclosing the goal-conditioned-update pathway. *Strategic self-coupling* structurally commits to goal-conditioned action-selection — the commitment-device move commits the agent to operate in a particular action subspace by structurally foreclosing the deliberation-can-revise-this pathway. Truthification gives up belief-pliability to gain reality-tracking; strategic self-coupling gives up action-pliability to gain commitment-credibility. The framework's epistemic-honesty discipline runs through both operations symmetrically — truthification commits to *not letting goals corrupt belief*; strategic self-coupling commits to *letting goals constrain action* in stable, credible ways.

Strategic self-coupling and adversarial coupling pressure share architectural shape but differ in driver and valence. Same operation on $\kappa_{\text{processing}}$ (both increasing), same architectural state acted upon, but opposite driver (self vs external) and opposite strategic valence (offensive enabling vs vulnerability-creating). The boundary between them blurs in practice (cult-formation literature: what begins as adversarial pressure can be internalized as identity-coupling, after which the agent perceives it as self-driven). The framework's posture: the *operations* are structurally distinct (driver is part of the operation's identity); their behavioral consequences can converge; the boundary is empirical when only behavioral signature is available, structural when driver-history is observable.

Position in the M-section meta-pattern family.

Discussion

The four M-section meta-patterns of AAT cover structurally distinct facets of the framework's analytical surface:

- **M1 — Identifiability floor** (`#disc-identifiability-floor`): the *boundary facet* of the stability-certificate spine. Structural impossibility results from external information-theoretic theorems (CHT, Cramér-Rao, common-Lyapunov nonexistence, Kalman-Ho); five-step constructive-impossibility shape (setting → external theorem → no-go → boundary characterization → strengthened consequence); rank-collapse subclass irreducibility via Sylvester's law of inertia.
- **M2 — Separability pattern** (`#disc-separability-pattern`): the *scope-of-existence facet*. Separable-core / structured-repair / general-open across eight ladders; positive-half complement to M1.
- **M3 — Additive coordinate forcing** (`#disc-additive-coordinate-forcing`): the *forced-identity facet*. AAT forces coordinates via uniqueness theorems on AAT-internally-motivated axioms at four layers (chain-log-additive identity; divergence-reverse-KL; update-log-odds via Cauchy-FE; metric-Fisher via (PI)/Čencov-invariance).
- **M4 — Modularity-state dynamics** (this segment): the *dynamics-on-architectural-class facet*. Three structurally distinct operations (truthification / strategic self-coupling / adversarial coupling pressure) acting on the static Class 1 / 2 / 3 partition of `#der-directed-separation`. The state space is M2's architecture ladder; M4 names what happens *to* an agent's location on that ladder over time under the three operations.

M4's relationship to M1 / M2 / M3 is sibling-meta-pattern rather than nested. M1 names what cannot be inferred; M2 names what can be inferred under what conditions; M3 names what AAT's representational choices commit to; M4 names what *changes* about an agent's architectural state under structurally-distinct operations. M4 specifically operates on the architecture state space M2 catalogues; the static M2 catalog and the dynamic M4 operations are complementary readings of the same architectural object.

The four meta-patterns together form AAT's analytical-surface vocabulary at the architectural layer. Adding M4 commits the framework to four meta-patterns as the canonical reading — a structurally weighty addition, justified by the cross-instance convergence on the three-operation pattern across the segments that ground M4's instances.

Why this pattern matters.

Recasts the architectural classification as dynamic, not static. `#der-directed-separation`'s Class 1 / 2 / 3 partition is often read as a *type* an agent belongs to. M4's recognition is that an agent's location on the partition is the *state*, and that multiple operations act on the state over time. An LLM is Class 3 (Coupled) by native architecture; under W_1 strict wrapping (`#der-class-coercion-via-wrapping`) the wrapped composite operates Class 1 at the wrapper level. A constitutional democracy is Class 1 by institutional design; under sustained adversarial pressure on truth-checking institutions, the composite-state functionally migrates toward Class 3 even though its constitutional architecture remains Class 1. The static-architecture reading captures the *type*; M4 captures the *dynamic location* and *the operations that move it*.

Identifies modularity as contested, cultivated, and constructed. Across the three operations, modularity is the strategic object — adversaries press it down; agents press it down themselves for offensive reach; defensive scaffolding presses it back up. None of these reads modularity as a fixed feature; all three treat it as a strategic resource being acted on. The recognition matches the empirical record across propaganda research, behavioral economics, institutional design literature, and the security and influence-operation literatures: modularity is what skilled actors fight over.

Names the boundary between strategic self-coupling and adversarial coupling pressure honestly. The behavioral signature at high $\kappa_{\text{processing}}$ may not distinguish a self-coupled committed agent from an adversarially-coupled captured one. The framework's machinery does not attempt to resolve the boundary at the level of agent experience; it characterizes the two operations as structurally distinct (driver: self vs external) and admits that behavioral convergence is honest scope. This is the same scope-honesty discipline AAT applies elsewhere — the framework names what it can structurally distinguish and acknowledges where empirical resolution is required.

Surfaces the dual-with-truthification recognition at the goal-belief separation axis. Of the three pairwise dual relationships across the operations, the truthification-vs-strategic-self-coupling dual is the most structurally illuminating: it reveals that the framework's two self-driven operations are wrapping-construction-style structural commitments operating on *opposite poles* of the goal-belief separation — truthification commits to goal-blind belief-update; strategic self-coupling commits to goal-conditioned action-selection. The framework's epistemic-honesty discipline runs through both symmetrically. This recognition could not have been seen at the single-operation level; it required all three operations to land before the dual became legible.

Connects naturally to the M2 separability ladder as a phase-space. The architectural classification ladder of `#disc-separability-pattern` (Class 1 separable-core / Class 2 structured-repair / Class 3 general-open) is, under M4's reading, a *phase-space across which the three operations move populations*. Truthification migrates the population toward Class 1; strategic self-coupling and adversarial pressure both migrate it toward Class 3 (under different drivers and valences). The static three-tier shape from M2 gains a dynamic reading from M4; the two meta-patterns compose naturally.

Provides a structural reading of historical and contemporary observations on institutional architecture. The framework's three-operation reading explains structural patterns observed across political science, organizational sociology, and consciousness-of-information-operations literatures: why authoritarian regimes that couple truth-checking to leadership goals experience long-run persistence-condition failures (strategic self-coupling at the composite-state level beyond the credibility-enabling range, into reality-foreclosing range); why constitutional democracies that commit institutionally to goal-blind truth-checking are structurally more durable than those that don't (truthification via institutional wrapping); why effective propaganda and information operations target the composite-state's O_t/Σ_t pathways rather than direct-observation channels (adversarial coupling pressure). The framework supplies the structural mechanism alongside the historical and contemporary records, not as replacement.

Epistemic Status.

Discussion-grade at the meta-pattern level. This segment names a structural pattern across three structurally-distinct operations, with the cross-operation pattern surfaced through convergent independent collaborator probes; the pattern itself is a presentational organizing principle, not a derived theorem.

Individual operations retain their own epistemic status. Truthification mechanism via wrapping is *exact* in the W_0 / W_1 / W_2 regime characterization of `#der-class-coercion-via-wrapping` (under stated assumptions); *robust qualitative* at the composite-scaffolding level of `#disc-adversarial-coupling-pressure` §"Defensive scaffolding". Strategic self-coupling is *discussion-grade* at the meta-pattern level (cross-mechanism recognition; the per-mechanism citations are at their respective tier in the prior-art literature) per `#disc-strategic-self-coupling`. Adversarial coupling pressure is *discussion-grade* with multiple sub-claims at *robust qualitative* per `#disc-adversarial-coupling-pressure`.

The cross-operation pattern claim — that the three operations *together* form the modularity-state-dynamics taxonomy — is *discussion-grade* recognition supported by: - Independent convergent collaborator probes arriving at the three-operation picture from different angles (the 2026-05-09 class-coercion-wrapping integration cycle's convergence finding). - The structural distinctness of the three operations along three axes (driver / direction / strategic valence). - The dual-relationship structure between them (truthification ↔ adversarial pressure direct dual; truthification ↔ strategic self-coupling dual at goal-belief axis; strategic self-coupling ↔ adversarial pressure same-architecture-shape opposite-driver). - The connection to M2's separability-pattern catalog as a phase-space across which the operations move populations.

The segment does *not* claim: - Completeness of the three-operation taxonomy (a fourth cell — externally-driven coupling-decreasing — is currently absorbed into truthification under externally-supplied scaffolding; whether it deserves its own row is open). - Formal derivation of the dual relationships (the dual-with-truthification recognition is structural, not derived). - Quantitative predictions about operation-rate or transition probabilities (M4 names *which operations exist*; their *quantitative parametrization* is downstream framework work).

Max attainable: *discussion-grade* for the meta-pattern recognition (it is a presentational organizing principle, not a derivation). The dual relationships could lift to *robust qualitative* under deeper analysis — the dual-with-truthification recognition at the goal-belief axis is the strongest candidate. Individual operation segments retain their own max-attainable as listed in their respective epistemic-status sections.

Discussion.

Relationship to `#disc-stability-certificate`. M4 sits alongside M1 / M2 / M3 as the fourth facet-recognition of the certificate spine. M1 is the *boundary facet*; M2 the *scope-of-existence facet*; M3 the *forced-identity facet*; M4 the *dynamics-on-architectural-class facet*. The four facets are sibling readings of the same underlying object (the stability-certificate spine + the architectural state space `#der-directed-separation` catalogs). M4's contribution is naming what *changes* about an agent's location in the certificate-cone's interior under the three structurally-distinct operations — the certificate exists (M1's boundary not breached), the scope is what it is (M2's tier), the coordinate is what it is (M3's forcing), and the agent's location *moves* under the three operations M4 names. Together the four meta-patterns characterize the architectural surface across its static structure (M1+M2+M3) and its dynamic operations (M4).

Connection to `#disc-separability-pattern` as phase-space. The Architecture ladder of M2 (Class 1 separable-core / Class 2 structured-repair / Class 3 general-open) acquires a *dynamic reading* under M4. Truthification migrates populations toward Class 1 (lower $\kappa_{\text{processing}}$); strategic self-coupling and adversarial pressure both migrate toward Class 3 (higher $\kappa_{\text{processing}}$), under different drivers. The structured-repair Class 2 tier is the natural home for *defensive scaffolding* under adversarial pressure — the explicit added machinery that restores identification under adversarial conditions but does not fully restore Class 1 by structural commitment. The three-tier shape from M2 is not just a static catalog of architectural types; M4 reads it as a phase-space across which the three operations move populations dynamically.

The asymmetric-advantage cross-operation pattern. Both `#disc-adversarial-coupling-pressure` (Discussion) and `#disc-strategic-self-coupling` (§"Asymmetric advantage in repeated strategic interaction") observe that in repeated strategic interaction with sustained modularity asymmetry, the more-modular agent compounds informational advantage (cleaner observations) and the more-self-coupled agent compounds strategic advantage (credible commitments the opponent cannot match). The two asymmetric-advantage observations are not redundant — they operate on different dimensions of strategic value (information vs commitment). Together they characterize the *full strategic-asymmetry consequence of modularity asymmetry*: the more-modular agent has cleaner reality-tracking; the more-self-coupled agent has stronger commitments; an agent with asymmetric advantage on *both* dimensions (defensive modularity against information attacks combined with offensive self-coupling on narrow strategic axes) compounds advantage on both axes. The framework's reading of this combined asymmetry is a structural prediction the M-section meta-pattern surfaces.

Findings.

Modularity as Contested Property Under Three Operations. Brief: Modularity — whether an agent's belief-update is *separated from* its goals or *coupled with* them — is often read as something an agent *is*. It is, instead, something an agent *is in*, at a moment, under operations that have moved it there. Three structurally distinct operations act on the modularity state: *truthification* (the agent commits to architectural arrangements — peer review, prediction registers, structured procedure, wrapping constructions — that restore goal-blind belief-update); *strategic self-coupling* (the agent deliberately decreases its own modularity to acquire credibility-dependent actions — Schelling-style commitment devices, Ainslie-style personal rules, identity coupling, emotional commitment); *adversarial coupling pressure* (an adversary presses the agent's coupling upward to expand attack surface — identity-binding, manufactured urgency, sunk-cost engineering). Each operation has a different driver (self vs external) and a different direction on $\kappa_{\text{processing}}$ (decreasing for truthification; increasing for the other two), and each has a distinct strategic valence (restorative vs offensive vs vulnerability-creating). The three operations are structurally related — truthification and adversarial pressure are direct duals at the cultivation-versus-attack axis; truthification and strategic self-coupling are duals at the goal-belief separation axis (both wrapping-construction-style commitments operating on opposite poles of the goal-belief separation, one giving up belief-pliability for reality-tracking, the other giving up action-pliability for commitment-credibility); strategic self-coupling and adversarial pressure share architectural shape but differ in driver and valence. A reader who carries this pattern away can predict where modularity is contested, where it is being attacked, where it is being deliberately surrendered, and where it is being structurally protected. The framework's static Class 1 / 2 / 3 partition under `#der-directed-separation` is the *state space*; M4 names the *dynamics on it*, and `#disc-separability-pattern`'s eight-ladder catalog is the *phase-space across which the operations move populations*.

Impact: Names the fourth meta-pattern of AAT alongside M1 (identifiability floor), M2 (separability pattern), M3 (additive coordinate forcing) — fourth facet-recognition of the stability-certificate spine, specifically the dynamics-on-architectural-class facet. Recasts the directed-separation property from static architectural type to dynamic state acted on by structurally-distinct operations. Connects the three previously-named operation legs (`#disc-adversarial-coupling-pressure`, `#disc-strategic-self-coupling`, `#der-class-coercion-via-wrapping` + `#disc-adversarial-coupling-pressure` §"Defensive scaffolding") into one cross-instance meta-pattern with explicit dual relationships and operational-mechanism table. Surfaces the dual-with-truthification recognition at the goal-belief separation axis — both self-driven operations are wrapping-construction-style structural commitments on opposite poles of the goal-belief separation, a structural recognition unavailable at the single-operation level. Identifies the M2 separability-ladder as a phase-space across which the M4 operations move populations dynamically — the static M2 catalog and the dynamic M4 operations are complementary readings of the same architectural object.

Novelty Claim: *Claim recognition* of a three-operation modularity-state-dynamics pattern across three structurally-distinct operations (truthification / strategic self-coupling / adversarial coupling pressure), each with its own driver, direction-on-κ, and strategic valence; *claim synthesis* of the three operations into a single meta-pattern alongside M1 / M2 / M3 with explicit dual-relationship structure (truthification ↔ adversarial direct dual; truthification ↔ strategic self-coupling dual at goal-belief axis; strategic self-coupling ↔ adversarial same-architecture-shape opposite-driver); *claim differentiation* on the boundary between strategic self-coupling and adversarial coupling pressure (structurally distinct at the driver axis; behaviorally convergent at high κ; honest scope on empirical resolution). The constituent recognitions live in the three operation-leg segments at their respective tiers; M4's contribution is the cross-instance unification and the dual-relationship structure.

Related Work:

Table 11.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
Truthification — component-level wrapping	`#der-class-coercion-via-wrapping` (this framework)	*internal antecedent* — operational mechanism for truthification at the component level; W_0 / W_1 / W_2 regime hierarchy
Truthification — composite-level institutional scaffolding	`#disc-adversarial-coupling-pressure` §"Defensive scaffolding as composition" (this framework)	*internal antecedent* — operational mechanism for truthification at the composite level; peer review, prediction registers, double-entry bookkeeping, adversarial procedure, structured red-teaming as Part III composition moves

ASF concern	*Prior-art language*	*Relationship / Positioning*
Strategic self-coupling — operation home	`#disc-strategic-self-coupling` (this framework)	*internal antecedent* — sister leg of M4; four-mechanism table adopting Schelling 1960, Ainslie 1992/2001, Akerlof-Kranton 2000/2010, Frank 1988 as first-class theory components
Adversarial coupling pressure — operation home	`#disc-adversarial-coupling-pressure` (this framework)	*internal antecedent* — sister leg of M4; mechanism table on identity-binding / affect-urgency / sunk-cost engineering; population-expansion claim for coupled formulation; cascade-inversion result
Static architectural classification (state space)	`#der-directed-separation` (this framework)	*internal antecedent* — Class 1 / 2 / 3 partition under $\kappa_{\text{processing}}$ measure; the state space on which M4's operations act
Architecture-ladder phase-space	`#disc-separability-pattern` Architecture ladder (this framework)	*internal antecedent* — three-tier shape (separable-core Class 1 / structured-repair Class 2 / general-open Class 3) reads as phase-space across which M4 operations move populations
Stability-certificate spine	`#disc-stability-certificate` (this framework)	*internal antecedent* — M4 is the dynamics-on-architectural-class facet of the spine; sibling to M1 boundary facet / M2 scope-of-existence facet / M3 forced-identity facet

Search Log:

- 2026-05-24 (*targeted at meta-pattern level*): The three-operation modularity-state-dynamics meta-pattern is named here at the cross-operation level for the first time; the constituent recognitions (each operation's primary segment) live in the operation-leg segments at their respective tiers. Targeted future search candidates: (a) whether any prior synthesis names truthification + strategic self-coupling + adversarial coupling pressure as instances of one structural meta-pattern under formal nomenclature; (b) whether the architecture-as-state-with-operations framing has been written down in active-inference, control-theoretic, or institutional-design literatures; (c) whether the dual-with-truthification recognition at the goal-belief separation axis has appeared in any framework engagement with the goal-belief question. Expected outcome: the constituent operations are well-precedented at the per-operation level (commitment devices, propaganda research, institutional architecture); the cross-operation unification as a meta-pattern of an integrated agent theory's analytical surface appears to be a fresh presentational recognition.

11.2 *Discussion:* Strategic Self-Coupling — Coupling as Action-Enabling

Some actions exist only under coupling. A threat is credible only when retreat has been foreclosed. The salesperson's sincere conviction enables persuasion that a modular-but-skeptical seller cannot match. The marathoner who runs to the finish is differently the same agent as the marathoner with backup plans. Religious practice, public commitment, vow-taking, oath-swearing, the binding of preferences to identity-norms — across these phenomena agents *deliberately decrease their own modularity* in exchange for access to actions otherwise unavailable. This is the *enabling polarity* of coupling-as-property — sister to `#disc-adversarial-coupling-pressure`'s *vulnerability polarity*, opposite-valence operation on the same architectural state. The framework's static Class 1 / 2 / 3 partition under `#der-directed-separation` is the state space; strategic self-coupling is the *self-driven operation that moves the agent toward higher $\kappa_{processing}$ for offensive strategic reach* (the third operation in the three-operation modularity-state-dynamics picture; cf. `#disc-adversarial-coupling-pressure` §"Scope" table). Where adversarial pressure drives an opponent's coupling upward to *exploit* the expanded attack surface, strategic self-coupling drives the agent's own coupling upward to *acquire* an expanded action surface — the same architectural transition, opposite-direction strategic value.

The phenomenon.

Discussion

The framework's policy specification — $\pi : (M_t, G_t) \to a_t$ selecting from a fixed action space $\mathcal{A}$ — implicitly treats actions as available independent of the agent's coupling state. In practice some actions are *credible only under coupling*: a commitment is a threat (or a promise) only when reversal is structurally costly; persuasion that depends on sincere conviction is unavailable to the deliberately-modular seller; coordination outcomes contingent on shared identity require the identity-coupling that makes deviation feel like self-betrayal rather than mere reconsideration. Treating $\mathcal{A}$ as fixed therefore mis-specifies the strategic landscape — it omits a class of actions rational agents may *choose to acquire by deliberately self-coupling*.

Four operational mechanisms recur across the literatures on commitment, intertemporal bargaining, identity economics, and emotion-as-commitment:

Table 11.3

Mechanism	*Coupling target*	*Strategic effect*
External commitment device (Schelling 1960 *The Strategy of Conflict*; 1966 *Arms and Influence*)	physical, institutional, or public structure	Foreclose retreat — burn bridges, delegate to mechanism that cannot be revoked, make public stake whose withdrawal is costly. The credibility paradox: a threat is credible only when reversal is structurally costly, so commitment value *equals* uncoupling cost.
Intertemporal personal rule (Ainslie 1992 *Picoeconomics*; 2001 *Breakdown of Will*)	between current-self and future-self decisions	Bind future-self behavior now — resolutions, personal rules, structured commitments — so the tempting-immediate-reward gradient cannot pull deviation when future-self encounters it. The agent treated as a population of temporally-displaced sub-agents who bargain; commitment via personal rule is the population-level coupling that stabilizes preference against hyperbolic-discount-induced reversal.
Identity coupling (Akerlof & Kranton 2000 *Economics and Identity*; 2010 *Identity Economics*)	between preferences and internalized identity-norms	Adopt "being a certain kind of person" — identity-norms foreclose action-space choices that would violate identity. Identity-coupled preferences are stable in a way incentive-based preferences are not; identity gives behavior a stickiness no per-decision optimization predicts, which is precisely what enables credible in-group coordination and costly signaling.
Emotional commitment (Frank 1988 *Passions Within Reason*)	between rational calculation and affective response	Cultivate emotional dispositions — anger pre-commits retaliation, love pre-commits loyalty, guilt pre-commits restitution — whose *ex-post irrationality is precisely the source of ex-ante commitment credibility*. The calculating rational actor who would defect under sufficient incentive cannot match the credibility of the angry actor who will retaliate regardless.

Each mechanism corresponds to a specific structural pathway through which the agent commits to a coupling that subsequently *enables* an action class. The mechanisms differ in coupling target (external structure vs intertemporal vs identity vs affect) but share the same structural shape: forgo flexibility now to gain credibility-dependent actions later.

The enabling polarity.

Discussion

`#disc-adversarial-coupling-pressure`'s core treatment names coupling as *vulnerability* — coupling expands attack surface; adversaries drive opponent coupling because the additional ports through O_t and Σ_t become exploitable. This segment names the structurally opposite valence on the same architectural state: coupling expands the agent's *own* action surface — the actions that only exist under coupling become available, and the agent acquires them by deliberately driving its own $\kappa_{\text{processing}}$ upward.

The two segments are *opposite-valence operations on the same architectural state*. Modularity is neither always-good nor always-bad; it is *contested*. Adversaries press it down for exploitation; agents drive it down themselves for offensive reach; defensive scaffolding presses it back up (truthification). The three-operation picture is the structural recognition that the directed-separation property is *not a fixed feature* of an agent but a *dynamic property* that multiple structural operations act on.

The credibility paradox is the operational core. Schelling's foundational observation: a commitment's value derives from its costliness to reverse. If reversal is free, the commitment is not a commitment; it is mere announcement of intention, which an interlocutor with adversarial or skeptical interest will discount. Costly-to-reverse means the agent has *given up something* — flexibility, optionality, the ability to renegotiate — and the *amount given up* is what the strategic interaction reads as credibility. The structural meaning: coupling enables actions because coupling is *evidence* of the agent's binding to a course of action; an uncoupled agent cannot produce that evidence even by trying, because the uncoupling itself is recoverable.

Strategic value — the action-space expansion.

Discussion

Three structural extensions to the framework's machinery characterize what strategic self-coupling commits to formally:

(M1) Coupling-dependent action space. The honest policy specification is

$$\pi : (M_t, G_t, \kappa_t) \to a_t, \quad a_t \in \mathcal{A}(\kappa_t) \tag{11.2}$$

with $\mathcal{A}(\kappa_t)$ non-decreasing in κ_t over a relevant range — additional actions become available as coupling increases. The set inclusion $\mathcal{A}(0) \subsetneq \mathcal{A}(\kappa_t)$ for $\kappa_t > 0$ is the structural recognition that the fixed-$\mathcal{A}$ framing in `#der-directed-separation` is a *scope-restriction*, not a complete specification. Strategic self-coupling is the operation that moves the agent from $\mathcal{A}(\kappa_0)$ to $\mathcal{A}(\kappa_1)$ with $\kappa_1 > \kappa_0$ in exchange for an enlarged feasible action set.

(M2) Coupling as strategic move in Σ_t. The strategy DAG of `#def-strategy-dag` carries causal-pathway claims about how actions affect objective-relevant outcomes. The enabling-polarity treatment requires a new edge type: nodes whose outcome variable is the agent's *own coupling state*, not an environmental quantity. An enabling edge represents "couple yourself to enable a^*" — a meta-strategic move that modifies the agent's own action space so that a downstream strategy node becomes feasible. Schelling's commitment-device move is exactly this: a strategy node whose direct effect is *not* objective-advancement but is *enabling-of-credible-strategy* at later DAG layers. Formally this extends the strategy DAG's node semantics to include self-coupling moves alongside causal-pathway moves; the existing confidence-weighted edge machinery (`#deriv-edge-credence-dynamics`) is unchanged.

(M3) Reversibility cost asymmetry. The strategic value of self-coupling depends on its irreversibility:

$$c_{\text{couple}}(\kappa_0 \to \kappa_1) \ll c_{\text{uncouple}}(\kappa_1 \to \kappa_0) \tag{11.3}$$

The asymmetry's magnitude is the parameter that determines coupling's credibility. Without the asymmetry, the commitment is not a commitment — it is a recoverable announcement. The reversibility-cost asymmetry imports adjacent to `#der-deliberation-cost` as a special case of the think-vs-act cost framework, but the *direction* of the asymmetry (uncoupling vastly more expensive than coupling) is the structural commitment that distinguishes strategic self-coupling from ordinary deliberation.

The three extensions are interdependent. (M1) without (M2) provides coupling-dependent actions but no mechanism by which the strategy can acquire them. (M1)+(M2) without (M3) makes self-coupling free and trivializes the strategic question — recoverable coupling is not credible, and credibility is the value the operation acquires. All three are needed for the formalization to track what the prior-art literature names commitment device.

The non-monotonicity question — coupling is not monotonically enabling.

Discussion

It is *not* the case that $\mathcal{A}(\varkappa_t)$ is monotone-increasing in $\varkappa_t$ across the full range. The structural intuition: initial coupling enables credibility-dependent actions (Schelling-style commitments, conviction-enabled persuasion, identity-enabled coordination); sustained over-coupling forecloses reality-dependent actions (the fully-captured cult member or motivated-reasoning subject loses access to actions that require accurate world-perception). At some interior point, $\mathcal{A}(\varkappa_t)$ achieves its maximum — coupling has gone far enough to enable commitment-dependent actions, but not so far that reality-tracking is foreclosed — and beyond that point further coupling subtracts from the feasible set.

The empirical shape of $\mathcal{A}(\varkappa_t)$ is plausibly **inverted-U**: increasing over $[0, \varkappa^*]$ (commitment-enabling range; the agent gains credibility-dependent actions); decreasing over $[\varkappa^*, 1]$ (over-coupling range; the agent loses reality-dependent actions). The inflection $\varkappa^*$ depends on the action class — for actions whose value depends primarily on credibility, $\varkappa^*$ is closer to 1; for actions whose value depends on accurate reality-tracking, $\varkappa^*$ is closer to 0. The shape is *open* — the framework does not currently derive its functional form, and this segment honestly flags the non-monotonicity as a structural commitment without committing to its quantitative shape.

The non-monotonicity has substantive implications: rational self-coupling is *bounded*. An agent that couples without bound forfeits the reality-tracking the commitment was meant to advance the agent's interests *in*. Schelling's commitment devices preserve flexibility along axes orthogonal to the commitment's target; effective intertemporal personal rules constrain a specific behavior while leaving general adaptive capacity intact; identity coupling that becomes total is the cult-formation pathology in which the agent loses access to non-identity-conforming action. The cluster's discipline — across all four mechanisms — is *partial* coupling along strategically-chosen axes, not maximal coupling everywhere.

Reversibility cost as the source of credibility.

Discussion

The (M3) asymmetry is not incidental to strategic self-coupling — it *is* the operational core. Schelling's foundational treatment is exactly this point: the strategic value of a commitment derives from the costliness of its reversal. The same structural observation runs through Ainslie's personal rules (rules whose violation is psychologically expensive because of the precedent-breaking cost across the population of future-selves), Akerlof-Kranton's identity coupling (defections from identity-norms incur the identity-cost of becoming "the kind of person who would do this"), and Frank's emotional commitment (the ex-post irrationality of emotional response is its ex-ante credibility — a calculating agent would defect under sufficient incentive; the emotional agent cannot match that calculating defection because the emotion overrides the calculation).

The recurring structural shape across the four mechanisms: the agent couples to something *less fungible* than the original preference state. Schelling's external mechanism is less fungible than verbal commitment (the mechanism cannot be talked out of executing); Ainslie's personal rule is less fungible than per-occasion choice (breaking the rule has cross-temporal precedent cost); Akerlof-Kranton's identity is less fungible than incentive-driven preferences (identity-norms carry self-recognition costs incentive-violation does not); Frank's emotion is less fungible than calculation (the emotional response is structurally compelled, not chosen per-instance). The strategic move is *exchange of fungibility for credibility* — give up the per-instance optimization choice in exchange for the credible-commitment value that per-instance optimization cannot produce.

This frames a useful diagnostic. A claimed commitment whose reversal would be *cheap* is not a commitment; an interlocutor with appropriate skepticism will discount it. The agent who wants to acquire commitment-credible actions must accept the structural cost of binding to something less fungible than their current preference state. The mechanism choice (external structure / intertemporal rule / identity / emotion) determines *what kind* of fungibility is given up; the amount of fungibility given up determines the credibility acquired.

Asymmetric advantage in repeated strategic interaction.

Discussion

Mirror to `#disc-adversarial-coupling-pressure`'s asymmetric-advantage section: in repeated interaction, when two agents face each other, each has structural incentive to (a) acquire credibility-dependent actions through strategic self-coupling and (b) deny the opponent access to those same actions by undermining their commitments. The self-coupled agent acquires actions that the uncoupled opponent cannot match; the uncoupled opponent faces commitments that incentive-based negotiation cannot dissolve. If the asymmetry is sustained, the self-coupled agent compounds strategic reach — its commitments are read as credible; its threats and promises bind; its negotiating position is bounded below by its commitment, not above by its current incentive-state.

The asymmetric-advantage shape is dual to the adversarial-pressure case: there, the more-modular agent compounds informational advantage (cleaner observations of the less-modular opponent); here, the more-self-coupled agent compounds *strategic* advantage (credible threats and promises the uncoupled opponent cannot match). The two asymmetries can coexist — an agent might be defensively modular against adversarial information attacks *and* offensively self-coupled in narrow strategic axes (the diplomat who is genuinely informed about reality but credibly committed to specific positions; the negotiator whose information-processing is clear but whose walk-away point is structurally fixed). The architectural states are not contradictory — different axes of $\kappa_{\text{processing}}$ can carry different operations.

This is *structural* asymmetry rather than tactical: the operations on modularity state are not single-instance moves but *durable architectural commitments* that shape long-run interaction. The agent who has self-coupled to specific commitments is *that agent* for the duration of the coupling; reversing requires paying the (M3) uncoupling cost; sustained interaction therefore reads the commitments as fixed and adapts strategy accordingly. The self-coupled agent's strategic position is fundamentally different from the uncoupled agent's, in a way that incentive-based negotiation cannot capture.

Boundary with adversarial coupling pressure.

Discussion

The boundary between strategic self-coupling (self-driven coupling-increasing) and adversarial coupling pressure (externally-driven coupling-increasing) blurs in practice. The cult-formation literature documents the canonical case: what begins as adversarial pressure (an external party manipulating the target's environment, social ties, information access) can be internalized as identity-coupling, after which the target perceives the coupling as self-driven and the line between operations is no longer cleanly drawn at the agent's experience. The behavioral signature of an agent at high $\kappa_{\text{processing}}$ may not distinguish the two; the *structural* distinction (driver: self vs external) is recoverable only with information beyond the agent's behavior.

The framework's posture on the boundary: the *operations* are structurally distinct (driver is part of the operation's identity); their behavioral consequences can converge; the boundary is empirical when only behavioral signature is available; the boundary is structural when driver-history is observable. Cult-formation, social engineering at the identity-modification level, and propaganda-induced identity-binding all sit at the boundary; sustained successful pressure produces an agent who would internally classify the resulting coupling as self-chosen. The framework's machinery does *not* attempt to resolve the boundary at the level of agent experience — it characterizes the two operations as structurally distinct and admits that behavioral convergence is honest scope.

The relationship to truthification.

Discussion

Strategic self-coupling is structurally the *dual* of truthification at a different layer of the agent. Truthification (instantiated via `#der-class-coercion-via-wrapping` and the institutional-scaffolding construction in `#disc-adversarial-coupling-pressure` §"Defensive scaffolding") is structural commitment to *goal-blind belief-update* — the W_1 strict wrapper commits the composite to operate Class 1 (Separated) at the belief-update layer by structurally foreclosing the goal-conditioned-update pathway. Strategic self-coupling is structural commitment to *goal-conditioned action-selection* — the commitment-device move commits the agent to operate in a particular action subspace by structurally foreclosing the deliberation-can-revise-this pathway.

Both are *wrapping constructions in the broader sense*. Both work by structural commitment that gives up flexibility along one axis in exchange for a different value along another. The two operations are duals: truthification gives up belief-pliability to gain reality-tracking; strategic self-coupling gives up action-pliability to gain commitment-credibility. The directed-separation property is foreclosed differently by each — truthification *restores* the separation at the belief-update layer; strategic self-coupling *deliberately weakens* it at the action-selection layer (specifically, the action-selection now depends on a coupled state κ_t that constrains $\mathcal{A}$, which is structurally a goal-conditioned narrowing of the action choice).

This dual relationship is structurally illuminating: both operations are deliberate departures from full architectural flexibility, in opposite directions on the goal-belief separation axis. The framework's epistemic-honesty discipline runs through both — truthification commits to *not letting goals corrupt belief*; strategic self-coupling commits to *letting goals constrain action* in stable, credible ways.

Epistemic Status.

Discussion-grade at the structural-recognition level. The segment articulates an extension of the framework's machinery (coupling-dependent action space; strategy-DAG enabling edges; reversibility-cost asymmetry) and adopts prior-art concepts (Schelling commitment devices, Ainslie personal rules, Akerlof-Kranton identity coupling, Frank emotions-as-commitment) as first-class theory components. The substantive claims are inherited from the prior-art literature at their original tiers:

- The credibility paradox (commitment value = uncoupling cost) is *exact* in Schelling's game-theoretic formalization for the stated mechanism-design setting; it is *robust qualitative* at AAT's level of abstraction (no AAT-internal derivation; the structural claim transfers across the four mechanisms).
- The inverted-U shape of $\mathcal{A}(\kappa_t)$ is *empirical / discussion-grade* — argued structurally, instantiated across the prior-art literatures, not derived from AAT primitives. Promotion would require a derived condition on the inflection point as a function of action-class composition.
- The reversibility-cost asymmetry is *robust qualitative* — observed across all four mechanisms, structurally derivable from the credibility-paradox observation.
- The dual relationship with truthification is *discussion-grade* — structural recognition that both operations are wrapping constructions in opposite directions on the goal-belief separation axis; the formal duality has not been derived.

Max attainable: *robust qualitative* for the structural pattern across the four mechanisms (the cross-mechanism convergence on the credibility-paradox shape is the kind of structural-recognition claim that promotes to robust-qualitative under careful prior-art search); *discussion-grade* for the inverted-U shape claim (promotion to *empirical* requires named domain-specific instantiation, e.g., the bounded-rationality literature's empirical work on self-control reversals); *conditional-derived* for the dual-with-truthification recognition only if the formal duality is worked out.

Discussion.

The mirror-image relationship with `#disc-adversarial-coupling-pressure`. The two segments cover opposite-valence operations on the same architectural state. Adversarial pressure: externally-driven, vulnerability-creating, coupling expanding attack surface. Strategic self-coupling: self-driven, action-enabling, coupling expanding the agent's own offensive reach. The two segments' structural parallelism is deliberate — same architectural state ($\kappa_{\text{processing}}$ per `#der-directed-separation`), same operation direction (increasing κ), opposite driver (external vs self) and opposite strategic valence (vulnerability vs enabling). The framework's third operation — truthification — is the opposite-direction operation (decreasing κ), instantiated via `#der-class-coercion-via-wrapping` (component-level wrapping) and `#disc-adversarial-coupling-pressure` §"Defensive scaffolding" (composite-level scaffolding). Together the three operations form the three-operation modularity-state-dynamics picture canonicalized at `#disc-modularity-state-dynamics`.

Connection to `#norm-explicit-strategy-condition`. The segment's central operation — choosing to self-couple — is a strategic decision the agent makes about its own architecture. This connects to the explicit-strategy normative condition: when does the agent gain from deliberate strategy formation vs reactive exploration? Strategic self-coupling adds a new dimension to that question:

when does the agent gain from deliberate *self-modification of architecture* vs leaving the architecture in its default state? The cost-benefit threshold for explicit strategy (`#norm-explicit-strategy-condition`) needs an analog for the self-coupling case — when does the credibility value of acquired actions exceed the structural cost of the coupling (the lost flexibility, the (M3) uncoupling cost amortized over the expected coupling duration)? An honest framing of the question is open; the segment names the structural dependency without deriving the threshold.

Composition consistency under self-coupling. Self-coupling within an agent may be analyzed as a composition move via `#disc-composition-consistency`. Ainslie's intertemporal-bargaining framing makes this explicit: the agent treated as a population of temporally-displaced sub-agents who bargain; the personal rule is a coupling between current-self and future-self that fixes the future-self's behavior in advance. The composite-agent reading: the agent at time t couples to its future-self at $t+1$ by committing now to constrain $t+1$ behavior. Formally this is a composite-agent move where the sub-agents are temporal slices of the same physical agent and the coupling is the personal-rule commitment. The connection is real but the formalization is open — the temporal-composite reading is suggestive but does not derive without explicit machinery for composite-agents-across-time. Worth following if the segment's downstream uses make this connection load-bearing.

The action-space-as-coupling-state-dependent formalization $\mathcal{A}(\kappa_t)$ as the load-bearing structural extension. Of the three structural moves (M1)–(M3), the (M1) extension is the most consequential for downstream framework reach. Once the framework admits coupling-dependent action spaces, a range of phenomena currently outside the framework's analytical surface become legible: commitment-device strategy; identity-driven coordination; emotion-as-strategic-resource; ritual and oath-taking as structural-commitment operations; institutional design that creates coupling-dependent action sets for collective actors. (M2) and (M3) are the supporting machinery. A future cycle that derives concrete results from (M1) — e.g., a sector-condition analog for coupling-dependent action sets, or a persistence-condition variant for agents with non-stationary $\mathcal{A}(\kappa_t)$ — would promote this segment from discussion-grade to derived-conditional. The structural extension is the recognition; downstream derivation is the natural follow-on.

Findings.

Strategic Self-Coupling as the Enabling Polarity of Coupling-as-Property. **Brief:** When you make a promise that everyone can see, when you burn the bridge behind your army, when you adopt an identity that makes certain choices feel like betrayal of who you are, when you cultivate the kind of anger that *will* retaliate even when retaliation costs you — you are deliberately giving up flexibility. The thing you give up has a strategic value the flexibility could not produce. A threat that you could costlessly withdraw is not a threat; an opponent will ignore it. A promise you could renegotiate at any moment is not a promise; a partner will not bet on it. Across very different domains — Schelling's commitment devices in international strategy, Ainslie's personal rules in self-control, Akerlof and Kranton's identity coupling in economics, Frank's emotions as commitment devices — the same structural shape recurs: agents acquire credibility-dependent actions by coupling to something less fungible than their current preference-state, and the strategic value of the action they acquire is precisely the cost of reversing the coupling. The framework's static picture of an agent's modularity as architectural property is incomplete; modularity is a *contested, cultivated, and constructed* property that agents act on for their own strategic reasons. Adversaries press it down for exploitation (`#disc-adversarial-coupling-pressure`); agents press it down themselves for offensive reach (this segment); defensive scaffolding presses it back up (the truthification operation). All three are operations on the same state-space; what they do to it differs in driver and direction.

Impact: Names the second of three operations in the modularity-state-dynamics three-operation picture (cf. `#disc-modularity-state-dynamics`). Surfaces the framework's previously-implicit fixed-$\mathcal{A}$ assumption (flagged in `#disc-adversarial-coupling-pressure` §"Scope") and characterizes the structural extension required to handle coupling-dependent action spaces — the (M1)–(M3) extensions (coupling-dependent action space; strategy-DAG enabling edges; reversibility-cost asymmetry). Recognizes the mirror-image relationship with `#disc-adversarial-coupling-pressure`: same architectural state ($\kappa_{\text{processing}}$), same operation direction (increasing κ), opposite driver and opposite strategic valence. Identifies the *dual relationship* with truthification — both are wrapping-construction-style structural commitments giving up flexibility along one axis for value along another, but operating on opposite poles of the goal-belief separation. The dual recognition is structurally illuminating: the framework's epistemic-honesty discipline runs through both — truthification commits to not letting goals corrupt belief; strategic self-coupling commits to letting goals constrain action in stable, credible ways. Opens the framework's reach to a class of phenomena (commitment-device strategy, identity-driven coordination, emotion-as-strategic-resource, ritual and oath-taking, institutional commitment design) that the fixed-action-space framing did not admit.

Novelty Claim: *Claim recognition* of strategic self-coupling as a structurally-distinct operation on modularity state, sister to and opposite-valence with adversarial coupling pressure (`#disc-adversarial-coupling-pressure`); *claim differentiation* on the (M1)–(M3) structural extensions required to formalize the operation in AAT's machinery (coupling-dependent action space, strategy-DAG enabling edges, reversibility-cost asymmetry); *claim transfer* of established commitment-device and identity-economics machinery into AAT's analytical surface as first-class theory components. The empirical phenomena (Schelling commitment, Ainslie personal rules, identity coupling, emotional commitment) are well-documented in the cited literatures; the contribution is naming their place in AAT's scope architecture, the structural extensions required, and the mirror-image relationship with adversarial coupling pressure that situates both operations within the three-operation modularity-state-dynamics taxonomy.

Related Work:

Table 11.4

ASF concern	*Prior-art language*	*Relationship / Positioning*
Commitment devices (external structure)	Schelling, T. C. (1960), *The Strategy of Conflict*, Harvard University Press; (1966), *Arms and Influence*, Yale University Press (found pre-2026)	*formal antecedent* — adopted with citation; the canonical commitment-device framework; the credibility paradox (commitment value = uncoupling cost) is Schelling's foundational structural observation. Primary-source verification scheduled for promotion past `draft`
Intertemporal bargaining and willpower	Ainslie, G. (1992), *Picoeconomics: The Strategic Interaction of Successive Motivational States Within the Person*, Cambridge University Press; (2001), *Breakdown of Will*, Cambridge University Press (found pre-2026)	*formal antecedent* — adopted with citation; the agent-as-population-of-temporally-displaced-sub-agents framing; personal rules as the population-level coupling that stabilizes preference. Closer to AAT's composition-consistency framing than Schelling's external-facing version
Identity economics	Akerlof, G. A. & Kranton, R. E. (2000), "Economics and Identity," *Quarterly Journal of Economics* 115(3):715–753; (2010), *Identity Economics: How Our Identities Shape Our Work, Wages, and Well-Being*, Princeton University Press (found pre-2026)	*formal antecedent* — adopted with citation; identity-utility-term as the formal hook for identity-coupled preferences; in-group coordination and costly signaling as the action class identity coupling enables
Emotions as commitment	Frank, R. H. (1988), *Passions Within Reason: The Strategic Role of the Emotions*, W. W. Norton (found pre-2026)	*formal antecedent* — adopted with citation; the ex-post-irrationality-equals-ex-ante-credibility structural observation; complement to Schelling for the *internal*-affective coupling case
Self-binding taxonomy	Elster, J. (1979), *Ulysses and the Sirens: Studies in Rationality and Irrationality*, Cambridge University Press; (2000), *Ulysses Unbound: Studies in Rationality, Precommitment, and Constraints*, Cambridge University Press (found pre-2026)	*adjacent literature* — survey treatment cataloguing forms of precommitment across psychology, politics, and law; useful for taxonomic comparison but not directly adopted

ASF concern	*Prior-art language*	*Relationship / Positioning*
Evolution of commitment in social contracts	Skyrms, B. (1996), *Evolution of the Social Contract*, Cambridge University Press; (2004), *The Stag Hunt and the Evolution of Social Structure*, Cambridge University Press (found pre-2026)	*adjacent literature* — evolutionary game-theoretic treatment of commitment supporting cooperation; useful for population-dynamics layer when self-coupling becomes a population-level phenomenon (norms, institutions); Part III population-scope work could engage
Sister segment — vulnerability polarity	`#disc-adversarial-coupling-pressure` (this framework, 2026-05)	*sibling internal segment* — opposite-valence operation on the same architectural state; the structural-parallelism is deliberate; both operations are part of M4's three-operation modularity-state-dynamics taxonomy at `#disc-modularity-state-dynamics`
Truthification operation — dual relationship	`#der-class-coercion-via-wrapping` (this framework); `#disc-adversarial-coupling-pressure` §"Defensive scaffolding" (this framework)	*internal antecedent* — the third M4 operation, structurally dual at the goal-belief separation axis (truthification commits to goal-blind belief-update; this segment's operation commits to goal-conditioned action-selection)

Search Log:

- 2026-05-24 (*targeted, segment-grade*): Prior-art adoption work per the spike (`spike-strategic-self-coupling.md` §2.1 direct adoption candidates + §2.2 adjacent literatures). Primary citations imported at segment-grade — one-line characterizations plus full bibliographic citations; primary-source verification scheduled when the segment advances past `draft`. The structural recognition (four mechanisms instantiating one cross-domain pattern) is named here for the first time at the AAT-internal level; targeted future search candidates: (a) whether any prior synthesis names commitment-device / identity coupling / willpower / emotions-as-commitment as instances of one structural operation under formal nomenclature; (b) whether the action-space-as-coupling-state-dependent formalization $\mathcal{A}(\kappa)$ has been written down in active-inference or bounded-rationality literatures; (c) whether the dual-with-truthification framing has appeared in any framework engagement with the goal-belief separation question. Expected outcome: the constituent moves are well-precedented at the textbook level; the cross-mechanism unification as an operation on modularity state alongside its adversarial-pressure dual appears to be a fresh presentational recognition.
- 2026-05-09 (*intuition-only on the cross-domain unification*): The structural recognition that the four mechanisms (Schelling commitment, Ainslie personal rules, Akerlof-Kranton identity coupling, Frank emotional commitment) instantiate one structural operation at different coupling targets is named here. The spike (`spike-strategic-self-coupling.md`) established the recognition; this segment lands it canonically with the cross-domain pattern documented.

11.3 *Discussion:* Adversarial Coupling Pressure

In adversarial settings, opponents do not merely deceive through the observation channel; they apply strategic pressure that drives a target *across architectural classes* — from Class 1 (Separated) toward Class 3 (Coupled). Coupling is itself an attack surface: a Class 1 (Separated) agent can be deceived only through the observation channel, while a GUC-Coupled agent exposes additional input ports through O_t and Σ_t. The result is a use-case expansion for `03-llm-core/`'s coupled formulation: the population that needs coupled-formulation analysis is broader than the architecturally-coupled. Any agent under sustained adversarial pressure that can drive its $\kappa_{\text{processing}}$ upward sits in the same scope, regardless of native class.

The phenomenon.

Discussion

A Class 1 (Separated) agent's f_M accepts inputs only through the event stream; its O_t and Σ_t have no legitimate input port from outside that stream. An adversary's only attack channel is engineered observation ambiguity, and the bias they can induce is bounded by `#deriv-observation-ambiguity-bias-bound`'s constant C relative to the ambiguity they can manufacture.

A GUC-Coupled agent — Class 3 (Coupled), or Class 2 (Partial) with active $O_t \to M_t$ pathways — exposes additional channels. Three operational mechanisms recur across the literatures on propaganda, social engineering, marketing, and influence operations:

Table 11.5

Mechanism	*Channel exploited*	*What the adversary does*
Identity-binding	$O_t \to M_t$	Make the target *want* the false belief to be true (bind belief to identity, group membership, self-image). Revising M_t now requires revising O_t, raising the cost of correction.
Affect / urgency	bypass deliberate f_M	Manufacture time pressure or affective load that prevents the orient cascade from running cleanly; the agent acts before M_t updates resolve.
Sunk-cost engineering	$\Sigma_t \to M_t$	Induce the target to commit publicly or expensively to a strategy; subsequent M_t revisions that would invalidate Σ_t now carry a coupled cost (admit prior commitment was wrong), and the coupling bends M_t revision instead of Σ_t revision.

Each mechanism corresponds to a specific structural pathway through which adversary-controlled signals act on M_t via O_t or Σ_t — not through the legitimate observation channel.

Diagnostic corruption is directional.

[Derived (Conditional on directional adversary objective; from 8.4, 8.5 + adversarial-pressure mechanisms above)]

The satisfaction-gap / control-regret split is silently corrupted under adversarial coupling pressure, and the corruption is *directional*: a competent adversary biases the diagnostic predictably rather than randomly.

If the adversary's interest is the target's continued effort along an unproductive axis, they bias the reading toward control regret — "the goal is right, try harder." The target compounds effort where it cannot pay, while the adversary extracts whatever value the effort produces (attention, emotional engagement, financial commitment, political mobilization).

If the adversary's interest is the target's *withdrawal* from an axis they could win, they bias the reading toward satisfaction gap — "the goal was wrong, give up." The target abandons achievable wins; the adversary captures the unclaimed ground.

Manufactured-outrage cycles instantiate the first; engineered-helplessness and demoralization campaigns instantiate the second. Both are *diagnostic capture*, not just deception about world-state — the target is reading a corrupted diagnostic and confidently acting on it.

Cascade inversion.

[Derived (Conditional on adversary controls O_t-targeted signal injection; from 10.1)]

The orient cascade resolves observation $\to M_t \to \Sigma_t \to O_t$ in that order, with O_t revising only when forced. Adversarial coupling pressure inverts the cascade: the adversary acts on O_t first (manufacture desire / identity-stake / urgency), which forces Σ_t to align, which then makes legitimate observations flow *against* the gradient and get rejected as inconsistent with the now-committed Σ_t. The cascade still runs, but its information dependency is reversed — and in that reversed form, observation contradicting the pre-loaded objective is the *least* legitimate input the agent will accept.

Population expansion for coupled formulation.

The Class 3 (Coupled) formulation handed off to `03-llm-core/` was originally scoped to agents whose native architecture is non-modular: LLMs whose attention processes goals and observations together; biological cognition with motivated-reasoning pathways. Adversarial coupling pressure expands this population: a Class 1 (Separated) organization without internal $O_t \to M_t$ pathways but operating in a hostile information environment that targets its leadership's identity and commitments is, for analytical purposes, in the coupled-agent population during that engagement. The native architecture is irrelevant once the coupling exists in operation.

This is a *use-case expansion*, not a derivation expansion: the coupled-formulation machinery does not need new content for this case. What's new is the recognition that the population is broader than architecture-determined.

Scope: implicit assumptions and the polarity of coupling.

Discussion

This segment covers one polarity of coupling-as-property — the *vulnerability* polarity, in which decreased modularity is externally driven and produces attack surface. Two implicit assumptions in the framework's machinery back the scope; both are worth surfacing rather than leaving hidden.

Bounded signaling. The framework formally restricts the $G_t \to$ world channel to action selection $a_t = \pi(M_t, G_t)$. A coarse $\mathcal{A}$ implicitly assumes π is sufficiently many-to-one — or the agent's behavioral signature sufficiently bandwidth-restricted — that observers cannot recover G_t from observable behavior alone. In practice the agent's full behavioral signature (micro-expressions, prosody, attention patterns, vocabulary choice, hormonal signals, response latencies) leaks G_t at much higher bandwidth than the discrete-action channel suggests. Against capable adversarial inferrers the bounded-signaling assumption fails, and this is part of *why* the bias bound's C becomes adversarially-tight in the §"Discussion" reframing: sophisticated adversaries effectively read G_t and target identity-bindings precisely. The assumption sits formally in `#der-directed-separation`'s agent-world interface; it is not yet surfaced there as a load-bearing scope statement, and surfacing it back-pressures that segment as a follow-on item.

Fixed action set independent of coupling. The action space $\mathcal{A}$ in the policy $\pi(M_t, G_t) \to a_t$ does not depend on the agent's coupling state. This implicitly assumes the actions available to the agent are independent of $\kappa_{\text{processing}}$. In practice, some actions exist *only* under coupling: Schelling-style commitment devices (where credibility requires foreclosed retreat); the salesperson whose sincere conviction enables persuasion that the modular-but-skeptical salesperson cannot match; the marathoner who can run the race only because retreat options have been burned; religious practice requiring coupled belief-and-action; identity-economics models where coupled identity unlocks coordination outcomes. Coupling is not just a *cost*; it is also *enabling*. Treating $\mathcal{A}$ as fixed therefore mis-specifies the strategic landscape — it omits a class of actions that rational agents may *choose* to acquire by deliberately self-coupling, and it forces the framework to read all coupling as failure mode.

Three operations on modularity state. Together with the segment's core treatment of adversarial coupling pressure, the two assumptions surface a clean three-operation picture of modularity-state dynamics:

Table 11.6

Operation	*Driver*	*Effect on modularity*	*Effect on agent*
Truthification	self-driven	increases	restores diagnostic; opens Part II results; instantiated by *defensive scaffolding* below
Strategic self-coupling	self-driven	decreases	*enables* actions otherwise unavailable (credibility, conviction, commitment)
Adversarial coupling pressure	externally-driven	decreases	*creates vulnerability* (this segment's core treatment)

This segment's primary subject is the third operation. The first operation is instantiated in §"Defensive scaffolding as composition" below — institutional scaffolding as the truthification leg of the three. The second operation is currently outside the framework's machinery; see Working Notes for the candidate sister segment and the candidate three-operation meta-segment.

Defensive scaffolding as composition.

Discussion

Restoring modularity under adversarial pressure is structurally a Part III move. Institutional scaffolding — peer review, prediction registers, double-entry bookkeeping, adversarial procedure, prediction markets, source-authentication discipline, structured red-teaming — constructs a composite agent *(individual + scaffolding)* with better Class 1 (Separated) properties than the bare individual, even when the individual's native architecture is Class 3 (Coupled) or driven there by adversarial pressure. The scaffolding externalizes channels of the orient cascade (belief-update checked by parties whose O_t doesn't share the individual's; prediction registers fixing M_t-claims before outcomes resolve; adversarial procedure forcing the feasibility check) so that resolution can run sequentially across the composite even when it cannot run sequentially within any single component.

The defensive composite recovers architectural modularity at the composite level whose properties exceed those of its sub-components — the routing-structure-driven side of the axis-decomposed composite-class-inheritance table in `#der-directed-separation` Discussion: when the routing (the scaffolding's institutional protocol) is structured to externalize orient-cascade channels, a composite of Class 3 (Coupled) sub-agents (or sub-agents under adversarial pressure toward Class 3) acquires Class 1 (Separated) or Class 2 (Partial) properties at the composite level. This is the operational dual of the wrapping construction (`#der-class-coercion-via-wrapping`): wrapping operates at the *component* level by externally scaffolding a single component; defensive scaffolding operates at the *composite* level by structuring the routing across components. Both move the composite class along Axis A (architectural class per `#der-directed-separation`), independently of the dynamic-regime axis (`#disc-dynamic-regime-axis`) where goal-alignment effects live.

Epistemic Status.

Discussion-grade at the structural-recognition level. The segment articulates a use-case expansion of existing machinery rather than a derivation of new machinery; the substantive claims are inherited from upstream segments at their tiers:

- The Class 1 / 2 / 3 (Separated / Partial / Coupled) partition and $\kappa_{\text{processing}}$ measure are *robust qualitative* per `#der-directed-separation`.
- The bias bound that adversarial pressure attempts to saturate is *conditional* per `#deriv-observation-ambiguity-bias-bound` (under named sub-scopes).
- The diagnostic-corruption directionality is *robust qualitative*: it follows from `#def-satisfaction-gap` and `#def-control-regret` being arithmetic identities once the value objects are fixed, plus the structural observation that adversary preferences

are typically directional rather than random; the directional bias is empirically observable across propaganda and influence-operation case studies.

- The orient-cascade inversion claim is *robust qualitative* — it follows from `#der-orient-cascade`'s information dependency under the explicit hypothesis that the adversary controls O_t-targeted signal injection.

Max attainable: *robust qualitative* for the structural pattern; the specific mechanism table is empirical (instantiated by propaganda, security, and influence-operation literatures) and would be promoted to *empirical* with named domain citations once a targeted prior-art search has been conducted.

Discussion.

This is the offensive complement to directed-separation-as-defense. `#der-directed-separation` characterizes modularity as architectural property. This segment names modularity as *contested* property under adversarial pressure: the architecture's modularity is what the adversary is trying to perforate, and what defensive scaffolding is trying to preserve or restore. The two segments together complete the picture — modularity is what the architecture enables (or fails to enable), and modularity is what the adversary attacks (or fails to attack).

Asymmetric advantage compounds in repeated interaction. When two agents face each other, each has structural incentive to (a) reduce its own coupling (defensive scaffolding) and (b) increase the opponent's coupling (offensive pressure). If the asymmetry is sustained, the more-modular agent gets clean observations of the less-modular agent's behavior — informative M_t updates that the bias bound's C does not impose — while the less-modular agent gets contaminated observations of the more-modular agent and bends M_t toward whatever pre-loaded objective the adversary engineered. The more-modular agent compounds informational advantage; the less-modular agent compounds bias. In repeated interaction with sustained asymmetry, this is *structural* rather than tactical advantage — modularity asymmetry produces strategic asymmetry.

Reframing the Class 3 (Coupled) bias bound's operational meaning. `#deriv-observation-ambiguity-bias-bound`'s constant C is presented in that segment as a worst-case bound. Adversarial coupling pressure reframes C as a *target* the adversary actively tries to saturate by engineering observation ambiguity to its maximum admissible value. Under sustained adversarial engineering, the bound is not slack — it is approached. This affects how the bound should be interpreted operationally: in non-adversarial settings, the bound is conservative; in adversarial settings, it is approximately tight under competent adversarial design. Whether the bound *can* be saturated depends on what the adversary controls (the observation channel's ambiguity-engineering bandwidth); under unrestricted adversarial control the saturation is essentially complete.

Connection to the separability pattern. The architectural classification ladder of `#disc-separability-pattern` reads adversarial coupling pressure as an *operational* driver of population movement along the ladder: Class 1 (Separated) agents under sustained adversarial pressure functionally migrate toward the Class 3 (Coupled) / general-open tier for the duration of the engagement, even though their native architecture remains Class 1 (Separated). The structured-repair tier (Class 2, Partial) is then the natural home for *defensive scaffolding* — the explicit added machinery that restores identification under adversarial conditions. The separability pattern's three-tier shape is therefore not just a static catalog of architectural types; it is a phase-space across which adversarial pressure and defensive scaffolding move populations.

Findings.

Adversarial Coupling Pressure as Use-Case Expansion of Coupled Formulation. **Brief:** Adversaries do not just lie; they reshape the *architecture* of the agent they are attacking. A Separated agent (Class 1) — one whose beliefs update from observation alone — can only be deceived through what it sees; the adversary's reach is bounded. But an agent whose beliefs are bent by what it wants to be true exposes new attack surface: every channel through which the adversary can act on what the target wants, or on what plan the target has committed to, becomes a way to reach the target's beliefs without going through legitimate observation. Practical mechanisms — identity-binding, manufactured urgency, sunk-cost engineering — are operational instantiations of this single structural move: drive the target across the architectural boundary so the additional ports open up. The framework's machinery for Coupled agents (Class 3) (originally scoped to architectural cases like LLMs and motivated cognition) therefore covers a broader population than its initial framing suggested: any agent under sustained adversarial pressure, regardless of native architecture. Defensive scaffolding — peer review, prediction registers, double-entry bookkeeping, adversarial procedure — is the structural counter-move: it constructs a composite agent whose modularity properties exceed those of its individual components, restoring at the system level what cannot be sustained at the individual level.

Impact: Names a phenomenon currently treated only obliquely in AAT: the strategic incentive in adversarial interaction to drive opponent coupling. Reframes the Class 3 (Coupled) bias bound (`#deriv-observation-ambiguity-bias-bound`) from passive worst-case to adversarially-saturated target — under competent adversarial design the bound is approximately tight rather than conservative. Connects the orient cascade's clean sequential resolution to its adversarially-inverted form, where observation flows against the information gradient and contradictory inputs are the least legitimate ones the agent will accept. Identifies institutional scaffolding (peer review, prediction markets, double-entry bookkeeping, adversarial procedure) as a Part III composition move that restores Class 1 (Separated) properties at the composite level even when individual sub-components are coupled or driven coupled. Expands the natural population for `03-llm-core/`'s coupled formulation beyond architecture-determined cases to include any agent under sustained adversarial pressure.

Novelty Claim: *Claim recognition* of adversarial coupling pressure as a structural phenomenon in AAT's existing scope architecture — adversaries strategically drive coupling because coupling expands attack surface — combined with *claim differentiation* on the population scope of coupled-formulation analysis: not just architecturally-coupled agents, but any agent under sustained adversarial coupling pressure. The empirical phenomena (manufactured outrage, identity-binding, social engineering, sunk-cost-driven commitment) are well-documented in propaganda, security, and behavioral-economics literatures; the contribution is naming their place in AAT's scope architecture, the corresponding population expansion for coupled formulation, and the asymmetric-advantage-compounding implication for repeated adversarial interaction.

Related Work:

- *Likely adjacent literatures, not yet searched* — propaganda research (Bernays 1928 *Propaganda*; Ellul 1965 *Propagandes*; Stanley 2015 *How Propaganda Works*), social engineering (Mitnick & Simon 2002 *The Art of Deception*; Hadnagy social-engineering corpus), behavioral economics on motivated reasoning (Kunda 1990 *The Case for Motivated Reasoning*; Kahan and the cultural-cognition project), influence operations (Singer & Brooking 2018 *LikeWar*), and Bayesian persuasion / cheap talk (Kamenica-Gentzkow 2011; Crawford-Sobel 1982). Relationship: *adjacent literature, intuition-only* — these literatures document the empirical phenomena instantiating the mechanism table; the contribution this segment claims is structural placement and population expansion, not phenomenon discovery.

Search Log:

- 2026-05-09 (*intuition-only*): Empirical phenomena (manufactured outrage, identity-binding, social engineering, sunk-cost commitment dynamics) are well-documented across propaganda research, social engineering, behavioral economics, and influence operations. The AAT-internal placement (adversarial pressure as driver of $\kappa_{\text{processing}}$ saturation) and the population-expansion claim for coupled formulation have not been searched. Targeted search recommended before promotion to draft, focusing on: (a) whether adversarial-cognitive-architecture framing exists in the active-inference, control-as-inference, or game-theoretic-deception literatures; (b) whether the "adversary drives opponent coupling" framing has been explicitly named in the cognitive-warfare or information-operations academic literatures; (c) whether the asymmetric-advantage-compounding result has been derived in any repeated-game treatment of cognitive influence.

Chapter 12

Scope and Composition Formation

12.1 *Scope:* Multi-Agent Scope

Part III's broadest scope: any system of multiple agents each satisfying 1.6 and interacting through a shared environment. Each agent has its own model and purposeful substate; each observes the environment and other agents' actions; each acts; their actions jointly determine the next environment state. The general case for organizations, teams, ecosystems, and adversarial encounters. *Coupling goes through the shared environment, not through direct state modification* — agents affect each other by affecting the environment.

A methodological default the framework adopts: **correlated observations are generic; independence is the special case requiring justification**. When agents share an environment, their observations are generically correlated — they see aspects of the same reality. Independence (uncorrelated observations) requires the agents to observe non-overlapping aspects, which is the special case.

Each agent's observation decomposes into environmental and *inter-agent* components — messages from other agents. The **multi-agent routing structure** is a key new object: the *topology* (who communicates with whom) plus the *protocol* (the rule governing what class of information flows between agents). The protocol is a *rule specifying the channel*, not the content of any message — individual messages reflect senders' states (including individual goals) through their policies. What the routing structure governs is the *infrastructure* — which channels exist and what kind of information they carry. **Routing is goal-blind** when neither topology nor protocol depends on the composite's goal state. *Goal-dependent routing* (crisis-specific channels, mission-specific intelligence-sharing protocols, reorganizing reporting chains based on objective) breaks goal-blindness and is the lever that distinguishes Case 1 from Case 2 in 13.3.

The framework draws a *load-bearing scope distinction* between two classes of machinery. **Agent-level machinery** (individual persistence, agent tempo, per-agent mismatch, sector-condition stability) applies to *every* agent in *every* multi-agent configuration — cooperative, adversarial, or indifferent — because each agent individually satisfies 1.6. **Composite-level machinery** applies only when 12.2 is satisfied. The adversarial regime spans the negative-teleological-unity end of the unity spectrum and is treated *both* ways: equilibrium-convergent adversarial pairs are captured by composite-level machinery (the strategic-composite route in 16.1); cyclic or non-convergent adversarial dynamics are captured by agent-level machinery applied across the pair.

Formal Expression.

Scope (multi-agent-scope)

A multi-agent system consists of N agents $\{A_1, \ldots, A_n\}$, each satisfying the scope condition (1.6), interacting through a shared environment with state $\Omega_t \in \mathcal{S}_{env}$:

- Each agent A_i has state $X_t^{(i)} = (M_t^{(i)}, G_t^{(i)})$

- Each agent observes: $o_t^{(i)} = h^{(i)}(\Omega_t, a_t^{(\neg i)}, \xi_t^{(i)})$ — observations may depend on other agents' actions
- Each agent acts: $a_t^{(i)} = \pi^{(i)}(X_t^{(i)})$
- The environment evolves: $\Omega_{t+1} = T(\Omega_t, a_t^{(1)}, \ldots, a_t^{(n)}, \omega_t)$

The coupling is through the environment: agent i's actions enter agent j's observation function and the shared environment transition. Agents may also communicate directly (a special case of action-observation coupling with a dedicated channel).

Definition (observation-decomposition)

Observation decomposition and routing.

Each agent's observation decomposes into environmental and inter-agent components:

$$o_t^{(i)} = \left(o_{\text{env},t}^{(i)},\ \{m_{ji,t}\}_{j \in \mathcal{N}_t(i)}\right) \tag{12.1}$$

where: - $o_{\text{env},t}^{(i)} = h_{\text{env}}^{(i)}(\Omega_t, \xi_t^{(i)})$: direct environmental observation (no inter-agent content) - $\mathcal{N}_t(i) \subseteq \{1, \ldots, N\} \setminus \{i\}$: the **communication neighborhood** — which agents send messages to i at time t - $m_{ji,t} = c_t^{(j \to i)}(X_t^{(j)})$: message from j to i, determined by the sender's full state and the communication protocol

Definition (multi-agent-routing-structure)

The **multi-agent routing structure** $R_t = (\mathcal{N}_t, \{c_t^{(j \to i)}\})$ specifies: - The **topology** $\mathcal{N}_t$: who communicates with whom - The **protocol** $c_t^{(j \to i)}$: the rule governing what class of information flows from j to i

Note: the protocol $c_t^{(j \to i)}$ is a *rule* specifying the channel, not the specific content of any message. Individual messages reflect the sender's state $X_t^{(j)}$ — including their individual goals — through the sender's policy. What the routing structure governs is the *infrastructure*: which channels exist and what kind of information they carry. *Bare-prose shorthand: the term "routing structure" is sanctioned within this segment after the first compound-form introduction; cross-segment citation should use the full "multi-agent routing structure" form.*

Definition (goal-blind-routing)

Routing is **goal-blind** when neither the topology nor the protocol depends on the composite's goal state:

$$\mathcal{N}_t \perp G_t^c \qquad \text{and} \qquad c_t^{(j \to i)} \perp G_t^c \quad \forall\, j, i \tag{12.2}$$

This means the communication infrastructure does not change based on what the composite is trying to achieve. Individual messages naturally reflect individual agents' goals through their policies — this is action, not routing. The routing condition is about the *structure* of information flow, not the *content* of individual messages.

Goal-dependent routing occurs when either the topology or the protocol varies with G_t^c. Examples: activating crisis-specific communication channels, changing intelligence-sharing protocols based on the current mission, reassigning reporting chains based on the operational objective.

Epistemic Status.

Axiomatic. This is a scope definition — it describes the class of systems Part III addresses. The only substantive choice is that coupling goes through the shared environment rather than through direct state modification. This follows from the agent boundary assumption (1.1): agents affect each other by affecting the environment, not by directly altering each other's internal states.

Discussion.

Correlated observations as default. When agents share an environment, their observations are generically correlated — they see aspects of the same reality. Independence (uncorrelated observations) requires the agents to observe non-overlapping aspects of the environment, which is the special case. Most multi-agent settings of interest involve substantial observation correlation.

The adversarial case sits along a spectrum — partly inside, partly outside the composite scope. Agents whose objectives conflict are multi-agent systems with negative teleological unity (14). Whether they form a composite depends on whether their strategic interaction admits an equilibrium structure: equilibrium-convergent adversarial pairs (potential or monotone games per Monderer-Shapley / Rosen) satisfy 12.2 via route (C-iv) as **strategic composites** with equilibrium-based macro-state — see 16.1 for the equilibrium-theoretic machinery and the (SC-1)–(SC-3) decomposition. Cyclic or non-convergent adversarial pairs satisfy none of (C-i)–(C-iv) and remain within 12.1 only. The asymmetric attacker / target case (one agent treated as an exogenous parameter rather than a sub-agent running its full AAT loop) is also a 12.1 phenomenon, handled by 15.2.

Two distinct classes of machinery apply across this spectrum:

- **Agent-level machinery** (individual persistence, agent tempo, per-agent mismatch, sector-condition stability) applies to *every* agent in *every* multi-agent configuration — cooperative, adversarial, or indifferent — because each agent individually satisfies 1.6 and Part I/II results apply directly. The adversarial tempo advantage (15.4) and adversarial destabilization (15.2) results are applications of this agent-level machinery to the case where one agent's actions are a disturbance source for another.
- **Composite-level machinery** applies *only* when 12.2 is satisfied via at least one of (C-i)–(C-iv). The (C-i)–(C-iii) alignment routes give composites with a shared objective O_c and admit closure-defect / team-persistence / composite-tempo / unity-closure machinery via 13.1. The (C-iv) strategic route gives composites with an equilibrium-based macro-state and admits the equilibrium-convergence / sector-template / closure-defect machinery in 16.1; the alignment-route and strategic-route forms are structurally distinct (an equilibrium-convergent adversarial composite has no shared O_c, only a shared $\mathcal{E}$).

The adversarial regime spans the negative-U_O end of the unity spectrum. Equilibrium-convergent adversarial pairs are captured by (C-iv) strategic-composite machinery; cyclic / non-convergent adversarial dynamics and asymmetric attacker / target configurations are captured by agent-level machinery applied across the pair. The existing adversarial segments (15.2, 15.4) operate at the agent-level for cases that fall outside (C-iv); 16.1 operates at the composite-level for cases that fall inside it.

Inter-agent communication as a special observation channel. Messages from j to i are actions by j and observations by i. The routing structure formalizes the infrastructure: who talks to whom ($\mathcal{N}_t$) and under what protocol ($c_t^{(j\to i)}$). The sender controls the content (unlike passive environmental observation), which introduces strategic manipulation (14.4). The gain from inter-agent communication enters the distributed tempo (14.4, Working Notes).

The routing/content distinction matters for directed separation. Individual messages reflect senders' goals — that's just action through policy. The directed-separation question at the composite level (13.3) is about the *routing structure*: does the infrastructure change based on the composite's goals? Goal-blind routing preserves directed separation; goal-dependent routing breaks it.

12.2 *Scope:* Composite Agent

The scope condition that distinguishes a *composite agent* from a *multi-agent system*. A set of purposeful sub-agents, each satisfying 1.6, constitutes a *composite agent* only when their objectives exhibit sufficient teleological alignment to define a coherent composite purpose. Without this condition, the sub-agents form a multi-agent system (12.1) that may still be analyzed — but the *composition machinery* (13.1, 15.1, 13.2) does not apply because there is no composite agent whose quantities to compute. This parallels the single-agent 1.6: before asking how an agent adapts, one must check that an agent is *present*.

The scope condition is satisfied via *any* of **four disjunctive routes**. **(C-i) Shared composite objective** — the strongest alignment: all sub-agents' policies are approximately optimal under a common objective. **(C-ii) Hierarchical derivation**: sub-objectives are derived by structure-preserving decomposition from a parent objective (military chain of command; corporate department structure) — sub-agents may not individually know the parent. **(C-iii) Mutual-benefit alignment** — the weakest route that still qualifies: a relevance variable exists such that joint actions raise each sub-agent's expected value above the non-cooperation baseline (symbiotic coexistence; commensal ecologies; trading partners). **(C-iv) Equilibrium-convergent strategic interaction**: an equilibrium concept (Nash, correlated, coarse correlated) exists such that coupled best-response dynamics converge to its support — qualitatively distinct from the first three because it requires *neither* shared objectives *nor* hierarchical derivation *nor* mutual benefit, only structural convergence of the strategic interaction. Composites satisfying (C-iv) are **strategic composites**; their macro-state is defined relative to the equilibrium structure rather than to a shared target. The routes are *not* shown to reduce to a common scalar threshold; each carries its own operationalization, and only one is required for scope-satisfaction.

A specific load-bearing role: the framework's third identifiability-floor instance (5.1, Instance 3 — composite contraction certification from component-marginal data is structurally impossible) has four structural escapes (observable coupling topology; matched Tier; passivity certificate; common contraction metric), but each escape needs a coherent target to certify. **Scope-satisfaction via at least one of the four routes is the enabling condition** for any of these escapes: without it, the "composite" is ill-defined (O_c does not exist under (C-i)-(C-iii); equilibrium structure $\mathcal{E}$ does not exist under (C-iv)) and the escapes have no coherent target.

Formal Expression.

Scope (scope-composite-agent)

Given sub-agents $\{A_1, \dots, A_N\}$ each satisfying 1.6, they constitute a *composite agent* iff there exists a common objective structure under at least one of the following routes:

(C-i) Shared composite objective. There exists O_c such that each sub-agent's effective policy is ϵ-compatible with O_c-optimal:

$$\exists\, O_c \;:\; \forall i,\; D\big(\pi_i,\; \pi_i^{O_c}\big) \le \epsilon \tag{12.3}$$

where $\pi_i^{O_c}$ is sub-agent i's optimal policy under the shared objective and D is an appropriate policy divergence. Strongest route; the sub-agents are all optimizing for the same thing, possibly with local decompositions.

(C-ii) Hierarchical derivation. There exists a parent objective O_c from which sub-objectives $\{O_i\}$ are derivable by decomposition consistent with 11:

$$\{O_i\} = \mathcal{D}(O_c) \tag{12.4}$$

where $\mathcal{D}$ is a structure-preserving decomposition. Each sub-agent optimizes its own O_i, but all O_i trace back to O_c. Military chain of command; corporate department structure. Sub-agents may not individually know O_c.

(C-iii) Mutual-benefit alignment. There exists a relevance variable Y such that the sub-agents' joint actions raise $\mathbb{E}[Y]$ above the non-cooperation baseline for each sub-agent:

$$\exists\, Y \;:\; \forall i,\; \mathbb{E}[Y \mid \text{joint}] > \mathbb{E}[Y \mid \text{non-coop}] \tag{12.5}$$

Weakest route. No explicit common objective, but interactions are positive-sum in some dimension. Symbiotic coexistence; commensal ecologies; trading partners who share no goals beyond mutual benefit.

(C-iv) Equilibrium-convergent strategic interaction. There exists an equilibrium concept $\mathcal{E}$ (Nash, correlated, or coarse correlated) such that coupled best-response dynamics of $\{A_i\}$ converge to (or cycle within the support of) $\mathcal{E}$:

$$\exists\, \mathcal{E}\ :\ \text{coupled best-response dynamics converge to the support of } \mathcal{E}. \tag{12.6}$$

Qualitatively distinct from (C-i)–(C-iii): requires neither shared objectives, nor hierarchical derivation, nor mutual benefit. Requires only **structural convergence** of the strategic interaction in the game-theoretic sense. The sub-agents' objectives may be partially opposing, but the interaction admits a stable joint behaviour. Composites satisfying (C-iv) are **strategic composites**, distinguished from alignment composites (C-i, C-ii) and mutual-benefit composites (C-iii). The composite's macro-state is defined relative to the equilibrium structure $\mathcal{E}$ rather than relative to a shared target state. See `#deriv-strategic-composition` for the A2'-analog transfer of `#result-sector-persistence-template` under potential-game (Monderer-Shapley 1996) or monotone-game (Rosen 1965) conditions, the (SC-1)–(SC-3) existence/stability/convergence decomposition, and the honest sub-scope β' scope exit for non-potential non-monotone games (VI existence + regret-minimization CCE set-convergence only).

Disjunctive form.

Scope (scope-composite-agent, disjunctive)

The scope condition is satisfied when **any** of (C-i), (C-ii), (C-iii), or (C-iv) applies. (C-i)–(C-iii) are progressively weaker qualitative requirements for what "teleological alignment sufficient to define a coherent composite purpose" means; (C-iv) covers strategic interaction with partially-opposing objectives via equilibrium convergence rather than alignment. The routes are *not* shown to reduce to a common scalar threshold. Each route carries its own operationalization and its own N-agent aggregation:

- (C-i) uses a value-function divergence $D(\pi_i, \pi_i^{O_c})$ aggregated across sub-agents.
- (C-ii) uses decomposition consistency of a parent objective O_c — a structural check, not a scalar.
- (C-iii) uses the existence of a relevance variable on which each sub-agent's marginal contribution is positive — a per-pair existential check, not a magnitude.
- (C-iv) uses existence of an equilibrium structure $\mathcal{E}$ under coupled best-response dynamics — a fixed-point check (Nash / VI / regret-minimization CCE), structurally distinct from the alignment checks in (C-i)–(C-iii).

The teleological unity measure U_O from 14 (pairwise value-correlation aggregated to the group) tracks one projection of alignment — primarily route (C-i) — but is not a reduction of all three routes to a single scalar. Downstream segments (14.1, 12.3, 15.1) describe quality *conditional on scope-satisfaction* without assuming a common threshold: they presume the scope condition holds via at least one route, then analyze composite quantities within that regime.

(C-i) gives the strongest alignment; (C-iii) gives the weakest that still qualifies. A composite may satisfy multiple routes simultaneously; only one is required for scope.

What fails the scope condition: sub-agents with orthogonal objectives that also fail to admit equilibrium convergence — no shared or derivable O_c, no relevance variable providing mutual benefit, and no equilibrium concept (pure Nash, mixed Nash, or CCE) whose support the coupled best-response or no-regret dynamics reach — or unclassifiable objective-structure coupling. Such systems remain within 12.1 but not 12.2. Adversarial pairs that admit equilibrium convergence via (C-iv) — whether pure-strategy Nash under α' (potential / monotone games), mixed Nash under β' (Nash 1950 existence for finite games), or CCE in distribution under β' (Hart-Mas-Colell 2000) — DO satisfy composition-scope-condition as strategic composites. Cyclic games (rock-paper-scissors, matching pennies) lack a pure-strategy Nash but admit mixed Nash and CCE convergence; they fall within β' and satisfy (C-iv). The narrow category that fails (C-iv) is games with no equilibrium concept whose support is reachable by any admissible dynamic — a genuinely small class within the standard game-theoretic landscape.

Epistemic Status.

Robust-qualitative. Max attainable: *robust-qualitative*. The composite-agent scope is a structural proposal paralleling the single-agent scopes (1.5 and 1.6, both axiomatic — the minimal conditions for adaptive and purposeful machinery to be non-vacuous). Unlike them, 12.2 is a theoretical choice: we could admit teleologically misaligned groups as composites (via the closure-defect framework alone) and separately track whether their composite objective is meaningful. The proposal here is that this separate tracking is better done as a scope restriction: composition quantities are defined for systems satisfying at least one of (C-i)–(C-iv), not for systems satisfying none.

The three alternative routes are not exhaustive but cover the well-understood cases. The relationship between the routes (whether they partition or overlap) and whether there is a single underlying scalar that reduces them all is open; the disjunctive form above is chosen because no such reduction has been demonstrated.

Discussion.

Why this is a scope condition, not merely a quality metric. The existing 13.1 framework treats any group admitting a low-ε^* projection as a composite. This is mechanically defensible but runs into a category problem: a composite's purposeful substate $G_c = (O_c, \Sigma_c)$ requires a well-defined composite objective O_c. If the sub-agents' objectives do not align under any common O_c, then G_c is ill-defined and the composite is a fiction. Closure defect can be computed (as a projection property), but calling the result a "composite agent" strains the term. Making this a scope condition resolves the category issue: composition applies where G_c is well-defined.

Relationship to the single-agent scopes. The parallel is tight. The single-agent scopes (1.5 + 1.6) together say: before asking whether an agent persists, check that it is an agent (it observes under residual uncertainty, and at least one action has causal effect). 12.2 says: before asking whether a composite persists, check that it is a composite (teleological alignment sufficient to define O_c). In both cases, the scope precedes the quantitative machinery.

Relationship to 14. U_O in 14 has a dual role: (i) scope tracking — primarily the (C-i) route, where value-correlation is the operationalization; (ii) quality parameter for action-closure ε_a, via 14.1. The two roles are not in tension, but they are also not identical across routes: scope-satisfaction via (C-ii) or (C-iii) does not directly correspond to a U_O value, so U_O is best read as a quality metric *conditional on scope*, not as the defining scope variable.

Four routes: three alignment routes plus one strategic-equilibrium route. Routes (C-i), (C-ii), (C-iii) — the **alignment routes** — differ in how directly the sub-agents share purpose. They are progressively weaker qualitative requirements — (C-i) strongest, (C-iii) weakest — but they are not shown to reduce to a single scalar. (C-i) uses value-correlation directly; (C-ii) uses hierarchical decomposition consistency; (C-iii) uses marginal benefit existence. A composite can satisfy multiple alignment routes simultaneously (a well-aligned team satisfies both (C-i) and (C-ii)). Route (C-iv) — the **strategic-equilibrium route** — is qualitatively distinct: it admits composites with partially-opposing objectives whose strategic interaction is equilibrium-convergent (potential or monotone games per Monderer-Shapley / Rosen; see `#deriv-strategic-composition`), and its composite macro-state is defined relative to the equilibrium structure $\mathcal{E}$ rather than a shared target. What matters for scope is that at least one of the four routes holds.

Miller's IAM definition. Miller (2022, *Ex Machina*, Ch 1) proposes that social behavior requires Interaction, Agency, and Mutual benefit (IAM). The mutual-benefit requirement in (C-iii) is the AAT analog of Miller's third element. Miller treats IAM as the definition of social behavior; the proposal here treats alignment as the scope condition for composite-agent analysis. The two align substantively: Miller's mutual benefit is a minimum teleological unity.

Multi-agent systems with and without composite status. Adversarial pairs partition along a finer line than alignment / non-alignment. Adversarial pairs that admit equilibrium convergence — via pure-strategy Nash under α' (potential or monotone strategic games per `#deriv-strategic-composition`), mixed Nash under β' (Nash 1950 existence for finite games), or CCE in distribution under β' (Hart-Mas-Colell 2000 under no-regret dynamics) — satisfy the scope condition via (C-iv) as strategic composites. Cyclic games (rock-paper-scissors, matching pennies) sit inside β': they lack pure-strategy Nash but admit mixed Nash and CCE convergence, so they satisfy (C-iv). The narrow category that genuinely fails (C-iv) is games with no equilibrium concept whose support is reachable by any admissible dynamic — a small class within the standard game-theoretic landscape. The asymmetric case — one agent as exogenous attacker, one as target — is a `#scope-multi-agent` phenomenon handled by `#der-adversarial-destabilization`; it does not form a composite in either (C-i)–(C-iii) or (C-iv) senses because the attacker is treated as a parameter rather than a sub-agent running its full AAT loop. Symmetric strategic composition under partially-opposing objectives is covered by (C-iv) and forms a strategic composite with equilibrium-based macro-state.

Load-bearing enabling role under `#disc-identifiability-floor` Instance 3. The composition-layer identifiability-floor instance establishes a no-go: composite contraction $\varkappa_c > 0$ is not in general certifiable from component-level data alone — the coupling-sign bit distinguishing cooperative from adversarial regimes is unidentifiable from component marginals. Four structural escapes are available (observable coupling topology; matched Tier at composite level; passivity certificate; common contraction metric). Scope-satisfaction via at least one of (C-i)–(C-iv) in this segment is the **enabling condition** for any of these escapes: without scope-satisfaction, the "composite" is ill-defined (O_c does not exist under (C-i)–(C-iii); equilibrium structure $\mathcal{E}$ does not exist under (C-iv); R_c is ill-typed), and the escapes have no coherent target to certify. The scope condition is therefore load-bearing apparatus that the composition-layer floor depends on — scope-satisfaction is what distinguishes a composite whose contraction or equilibrium convergence can be certified (under some escape) from a multi-agent system whose "composite contraction" is not a well-defined object.

How composites come into being: symbiogenesis. The formal mechanism by which a group passes from failing all three routes to satisfying at least one is described in 12.3. Briefly: in biological and social symbiogenesis, a host integrates an endosymbiont; the endosymbiont's objective is subsumed into the host's; the combined entity's objective O_c emerges where two independent objectives stood. After the transition, (C-ii) applies: the endosymbiont's objective is a sub-objective of O_c. This is the dynamical process of composite-agent identity creation; the scope condition here describes its result.

12.3 *Hypothesis:* Symbiogenic Composition

Symbiogenesis is an asymmetric composition mechanism in which one agent (the *host*) integrates another (the *endosymbiont*) as a specialized sub-component, with the endosymbiont's objective gradually subsumed into the host's. It is distinct from peer coupling (13.1) and from population-level restructuring (the extreme transition motif drawn from Miller 2022, discussed in 4.8): symbiogenesis is how composite agents *come into existence* by crossing the 12.2 from below. The mechanism is well-attested empirically (eukaryote formation, firm mergers, legal-precedent adoption, language families) but formally underspecified within AAT. This segment captures the phenomenon and flags the specific formalization gaps.

Formal Expression.

Hypothesis (symbiogenic-composition)

Given two purposeful agents A_h (host) and A_e (endosymbiont), each satisfying 1.6, symbiogenic composition is a process on the joint state space with three coupled dynamics:

(S-1) Objective absorption. The endosymbiont's objective O_e transforms toward alignment with or derivation from the host's objective O_h:

$$O_e(\tau) \xrightarrow{\tau \to \tau_{\text{consolidated}}} \mathcal{D}_e(O_h) \tag{12.7}$$

where $\mathcal{D}_e$ is a derivation functional (in the sense of route (C-ii) in 12.2): O_e becomes a sub-objective derived from O_h. Before: O_h and O_e are independent objectives, no route of 12.2 applies, and the pair is a multi-agent system (12.1) rather than a composite. After: O_e is a role within O_h; route (C-ii) applies; the composite (A_h, A_e) satisfies the composition scope condition.

(S-2) Function transfer. Structural content from the endosymbiont's state (elements of M_e or Σ_e) transfers to or becomes accessible by the host:

$$\{M_h, \Sigma_h\}(\tau) \xrightarrow{\tau \to \tau_{\text{consolidated}}} \{M_h, \Sigma_h\} \cup \mathcal{F}(M_e, \Sigma_e) \tag{12.8}$$

where $\mathcal{F}$ is a transfer mapping (structure-preserving integration of endosymbiont functions into host state). In biological symbiogenesis: gene transfer. In organizational symbiogenesis: acquired firm's processes, patents, know-how integrated into acquirer's operations. This is the grafting operation of 9.7 in its cross-agent form — the host grafts structure originating in the endosymbiont.

(S-3) Autonomy reduction. The endosymbiont's effective action space contracts; many of its choices become fixed by the host's coordination:

$$\mathcal{A}_e^{\text{effective}}(\tau) \xrightarrow{\tau \to \tau_{\text{consolidated}}} \mathcal{A}_e^{\text{restricted}} \subsetneq \mathcal{A}_e^{\text{initial}} \tag{12.9}$$

The endosymbiont retains enough autonomy to avoid catastrophic transfers (e.g., mitochondria retain some genome to handle local fast-timescale responses that would be hazardous to route through the host nucleus) but loses most independent decision-making.

Integrated transition. At consolidation, the joint system is a single composite agent A_c whose substate contains the integrated structure:

$$X_c = \big(M_c, G_c\big) = \big(M_h \cup \mathcal{F}(M_e, \Sigma_e),\ (O_c, \Sigma_c)\big) \quad \text{with } O_c \approx O_h \tag{12.10}$$

The endosymbiont persists as a specialized sub-component of the host, not as an independent agent. The 12.2 is now satisfied; the peer-coupling machinery of 13.1 applies to the resulting composite.

Epistemic Status.

Robust qualitative. Max attainable: *robust qualitative* — the phenomenon is well-attested empirically across biological and social domains, but a general mathematical formalization within AAT is open.

What is well-established (externally):

- The existence of symbiogenesis as a distinct evolutionary mechanism (Mereschkowsky 1905, 1910; Sagan 1967; Margulis & Sagan 1997). Mitochondria and chloroplasts are the paradigm cases.
- The social analog in firms, technology, language, legal systems, religions (Miller 2022, Appendix B).
- "Innovation by parts" as qualitatively different from "innovation by sparks" (gradual mutation) — Miller's framing.

What is *not* derived within AAT:

- A formal model of the objective-transfer dynamics (S-1). What evolutionary or optimization process drives $O_e \to \mathcal{D}_e(O_h)$?
- A formal specification of the transfer functional $\mathcal{F}$ in (S-2). What structure is preserved, what is lost, what is transformed?
- A precise characterization of autonomy reduction (S-3). Why does the endosymbiont retain some autonomy rather than becoming fully deterministic?
- Quantitative predictions — e.g., when symbiogenesis is favored over peer coupling, what governs the timescale of consolidation, under what conditions it reverses.

The three dynamics (S-1), (S-2), (S-3) are proposed schemas, not results. A follow-up development of an AAT-specific dynamical model is the natural next step.

Discussion.

The role of this mechanism in Part III. Three distinct composition mechanisms are now in scope:

1. **Peer coupling** (13.1, 15.1, 13.2) — sub-agents interact through shared environment; closure defect measures faithfulness of projection. Presumes scope-satisfaction via at least one route of 12.2 (not a scalar U_O threshold).
2. **Extreme transition motif** (Miller 2022; introduced in 4.8; pending dedicated segments for composition-transition dynamics, latent structural diversity, and endogenous coupling) — population-level restructuring via neutral drift / niche creation / cascading displacement. U_O shifts across a population as agent types replace one another.
3. **Symbiogenesis** (this segment) — hierarchical absorption. U_O crosses the composition scope condition from below, creating a composite that did not previously exist.

Before symbiogenesis, the sub-agents were a multi-agent system (12.1) but not a composite. After, the resulting composite is subject to all of AAT's composition machinery. The symbiogenic transition is the specific dynamical process of composite-agent identity creation.

Why this cannot be modeled as peer coupling. Peer coupling assumes pre-existing sub-agents being projected into a macro-description. The closure-defect framework presupposes the composite exists; it measures how faithfully the macro tracks the micro. Symbiogenesis is about a composite *coming into being* from two previously-independent agents. No projection Λ of the pre-symbiogenic system yields the post-symbiogenic composite, because the endosymbiont's objective *changes* during the process — its objective is different before and after. The transformation is intrinsic to the sub-agents' state, not an external projection choice.

Why this cannot be modeled as extreme transition. The extreme transition motif operates at the population level with many agents, neutral drift of types, and niche-construction dynamics. Symbiogenesis is typically between two specific agents (or a small number) and proceeds through a specific asymmetric integration rather than through statistical population dynamics. The mechanisms overlap — symbiogenesis often occurs as part of a larger transition — but the core mechanism of symbiogenesis (bilateral asymmetric integration) is distinct from the population-level dynamics of extreme transitions.

Examples across domains.

Table 12.1

Domain	*Host*	*Endosymbiont*	*Integrated composite*
Biology	Archaeal host cell	α-proteobacterium	Eukaryotic cell (mitochondrion persists as organelle)
Biology	Eukaryotic cell	Cyanobacterium	Plant cell (chloroplast persists as organelle)
Commerce	Acquiring firm	Acquired firm	Merged firm (acquired operates as division)
Technology	Base platform	Integrated component	Composite product (component operates within host system)
Linguistics	Host language	Adopted vocabulary/grammar	Creolized / evolved language
Law	Legal system	Adopted precedent	Evolved jurisprudence (precedent operates as doctrine)
Religion	Host tradition	Absorbed elements	Syncretic practice

In each case: asymmetric integration, autonomy reduction of the absorbed entity, gradual objective subsumption, functional specialization.

Connection to 9.7. The single-agent "grafting" operation in 9.7 is within-agent — an agent incorporates external structure into its own Σ_t. Symbiogenesis is cross-agent — the grafted structure originates in another agent, and the integration is accompanied by that other agent's objective being absorbed. These are related but distinct: grafting is the structural-change mechanism on the host side; symbiogenesis is the bilateral process that includes grafting plus objective-absorption plus autonomy reduction.

Rate-distortion interpretation (connecting to 14.1). Under the Information Bottleneck conjecture in 14.1, peer coupling is IB compression with the relevance variable defined by a shared composite objective. Symbiogenesis is the process by which the relevance variable itself shifts: from two separate IB problems (each sub-agent's own survival objective) to a single IB problem (the composite's survival objective). The symbiogenic transition creates the shared relevance variable, which in turn makes the IB frontier well-defined for the composite. This is a structural shift in the IB problem, not a compression along a fixed IB frontier.

Chapter 13

Composition Machinery

[Gap] Open question

13.1 *Formulation:* Composition Closure Criterion

The formal definition of when a group of interacting agents constitutes a *valid composite macro-agent*. The criterion: the closed-loop dynamics *approximately commute with coarse-graining* — projecting micro-states to macro-states and then running macro-dynamics yields approximately the same result as running micro-dynamics and then projecting. The **closure defect** ε^* is the minimum achievable mismatch between these two paths, taken over all admissible projections and macro-dynamics.

The construction accommodates a **timescale ratio** $K_c \geq 1$ — the number of micro-timesteps per macro-step — that lets the framework handle both lockstep composites ($K_c = 1$, synchronous-update) and composites that operate strictly slower than their sub-agents ($K_c \gg 1$, the regime 4.9 asserts as natural). When the ratio is large, the composite's description need not — and should not — track the micro-trajectory between macro-boundaries; timescale abstraction is recovered.

The closure-defect framework applies *only* to sets that satisfy 12.2. Given scope-satisfaction, a set forms a *meaningful* composite agent when $\varepsilon^* \leq \varepsilon_{\max}$. Without scope-satisfaction, the infimum is still well-defined as a projection property, but the resulting "composite" has ill-defined purposeful substate and composition-level machinery (team persistence, composite tempo) does not apply.

The formulation specifies **admissibility constraints** (A1)-(A4) on the macro-dynamics — conditions the macro must satisfy to be AAT-shaped: (A1) the macro-state decomposes into model and purposeful components with a recursive update; (A2) a macro-mismatch signal is well-defined; (A3) macro-tempo is well-defined; (A4) the macro-correction function satisfies the sector condition (4.4). Together these make the composite a proper AAT agent rather than just a low-dimensional projection. Parallel constraints (P1)-(P3) govern the projection (information preservation, Lipschitz continuity, dimensionality reduction). A **bridge lemma** is the technical heart of the result: Part I's sector-persistence template (`#result-sector-persistence-template`) applied at the composite level requires a *strictly stronger* contraction assumption on the macro-update map than (A4) alone supplies — the assumption is structurally motivated by (A4) but not formally derived from it. For different *Tiers* of sub-agent quality, Part I/II results transfer differently to the composite: Tier 1 transfers *exactly*; Tier 2 transfers locally with bounded degradation; Tier 3 requires per-domain verification.

The closure defect itself is the **effective disturbance** at the composite level — it enters the composite's persistence inequality as the rate at which macro-tracking diverges from micro-truth, combining with external environment perturbations to set the composite's total disturbance load (13.2). This makes precise the *cost of imperfect composition*: a composite with high closure defect carries that defect as a perpetual disturbance source against its own persistence.

Formal Expression.

Let a system consist of N sub-agents interacting in a shared environment with state space $\mathcal{S}_{\text{env}}$. The micro-state, micro-observations, and micro-actions are:

$$X_{\text{micro},t} = \{(M_{i,t}, G_{i,t})\}_{i=1}^{N} \in \mathcal{X}_{\text{micro}} \tag{13.1}$$

$$o_{\text{micro},t} = \{o_{i,t}\}_{i=1}^{N} \in \mathcal{O}_{\text{micro}} \tag{13.2}$$

$$a_{\text{micro},t} = \{a_{i,t}\}_{i=1}^{N} \in \mathcal{A}_{\text{micro}} \tag{13.3}$$

The coupled micro-dynamics form an action-observation loop:

$$X_{\text{micro},t} \xrightarrow{\pi_{\text{micro}}} a_{\text{micro},t} \xrightarrow{E} (\Omega_{t+1}, o_{\text{micro},t+1}) \xrightarrow{f_{\text{micro}}} X_{\text{micro},t+1} \tag{13.4}$$

We constrain our search to an admissible class of projections $\Lambda \in \mathcal{P}_{\text{adm}}$ mapping micro to macro, an admissible class of macro-dynamics $(\pi_c, E_c, f_c) \in \mathcal{M}_{\text{adm}}$, and a **timescale ratio** $K_c \geq 1$: the number of micro-timesteps per macro-step. Micro-time is indexed by $t \in \{0, \dots, H\}$; macro-time by $m \in \{0, \dots, \lfloor H/K_c \rfloor\}$ with macro-step m corresponding to micro-timestep $t = mK_c$. The projection components then have type signatures:

- $\Lambda_x : \mathcal{X}_{\text{micro}} \to \mathcal{X}_c = (M_c, G_c)$ — pointwise, evaluated at macro-boundaries.
- $\Lambda_\Omega : \mathcal{S}_{\text{env}} \to \mathcal{S}_{\text{env},c}$ — pointwise, evaluated at macro-boundaries.
- $\Lambda_o : \mathcal{O}_{\text{micro}}^{K_c} \to \mathcal{O}_c$ — aggregates the window of K_c micro-observations between successive macro-boundaries.
- $\Lambda_a : \mathcal{A}_{\text{micro}}^{K_c} \to \mathcal{A}_c$ — aggregates the window of K_c micro-actions similarly (used on the observation-comparison side; the macro-policy π_c emits one macro-action per macro-step).

Aggregation is part of the projection specification — common choices are mean/sum (linear systems), first/last (event-rate projections), or task-specific sufficient statistics. When $K_c = 1$ the windows collapse, Λ_o, Λ_a reduce to pointwise maps, and micro- and macro-tempos coincide (synchronous-update composites). When $K_c \gg 1$ the composite operates on the quasi-steady-state output of its sub-agents, the regime that 4.9 asserts as the natural one for composition. K_c is part of the problem specification; it does not appear in (A1)-(A4) or (P1)-(P3) below, but it determines at what granularity closure is measured.

Let $\mathcal{D}_{\text{micro}}$ be the distribution of reachable trajectories generated entirely by the true micro-system over horizon H, and let $o_{\text{micro},(m-1)K_c:mK_c}$ denote the window of K_c micro-observations $(o_{\text{micro},(m-1)K_c+1}, \dots, o_{\text{micro},mK_c})$ between macro-boundaries $m-1$ and m (similarly for actions).

[Definition (Composition Closure)] We define the minimal achievable closure defect ε^* over the admissible classes as:

$$\varepsilon^* = \inf_{\Lambda \in \mathcal{P}_{\text{adm}},\, (\pi_c, E_c, f_c) \in \mathcal{M}_{\text{adm}}} \left\| (\varepsilon_x, \varepsilon_a, \varepsilon_o) \right\| \tag{13.5}$$

where the expected component errors evaluated over true micro-trajectories $\tau \sim \mathcal{D}_{\text{micro}}$, measured **per macro-step**, are:

- $\varepsilon_x = \mathbb{E}_\tau\Big[\frac{1}{M_H}\sum_{m=1}^{M_H}\Big\|\Lambda_x\big(X_{\text{micro},\,mK_c}\big) - f_c\big(\Lambda_x(X_{\text{micro},\,(m-1)K_c}),\ \Lambda_o(o_{\text{micro},\,(m-1)K_c:mK_c})\big)\Big\|_{\mathcal{X}}\Big]$
- $\varepsilon_a = \mathbb{E}_\tau\Big[\frac{1}{M_H}\sum_{m=1}^{M_H}\Big\|\Lambda_a\big(a_{\text{micro},\,(m-1)K_c:mK_c}\big) - \pi_c\big(\Lambda_x(X_{\text{micro},\,(m-1)K_c})\big)\Big\|_{\mathcal{A}}\Big]$
- $\varepsilon_o = \mathbb{E}_\tau\Big[\frac{1}{M_H}\sum_{m=1}^{M_H}\Big\|\Lambda_o\big(E_{\text{obs}}(\Omega,\ a_{\text{micro}})|_{(m-1)K_c:mK_c}\big) - E_{c,\text{obs}}\big(\Lambda_\Omega(\Omega_{(m-1)K_c}),\ \pi_c(\Lambda_x(X_{\text{micro},\,(m-1)K_c}))\big)\Big\|_{\mathcal{O}}\Big]$

where $M_H = \lfloor H/K_c \rfloor$ is the number of macro-steps in horizon H, and $E_{\text{obs}}(\Omega, a_{\text{micro}})|_{(m-1)K_c:mK_c}$ denotes the window of micro-observations the environment produces across the macro-step.

Units. $\varepsilon_x, \varepsilon_a, \varepsilon_o$ carry units of distance — per-macro-step norm errors. Multiplied by the macro-update rate ν_c (units [time^{-1}]), they yield the disturbance-rate units [distance] · [time^{-1}] demanded by the sector-persistence template's ρ_ξ (`#result-sector-persistence-template`) and by the dimensional accounting in 13.2. The previous per-micro-step formulation conflated ε^*'s granularity with ν_c's — the macro-step formulation resolves the unit inconsistency.

$K_c = 1$ **special case.** When micro and macro update in lockstep, $M_H = H$, the observation/action windows collapse, and the formulas reduce to the synchronous form. This is the regime of tightly-coupled groups (distributed algorithms running in phase, ensemble filters where every member steps per tick) and is where the two-Kalman worked example below operates.

$K_c \gg 1$ **case.** When the composite lives at a strictly slower timescale than its sub-agents (the regime 4.9 asserts as natural), only micro-states at macro-boundaries enter ε_x, and macro-observations aggregate K_c micro-observations. The closure defect then measures how well the macro-dynamics predicts the *next* macro-state from the *previous* macro-state and the aggregated micro-observation window — not any intermediate micro-state. Timescale abstraction is recovered: the composite's description need not — and under $K_c \gg 1$ should not — track the micro-trajectory between macro-boundaries.

The closure-defect framework applies to sets that satisfy 12.2 — i.e., that form composites via at least one of the three alignment routes (shared objective, hierarchical derivation, mutual benefit). Given scope-satisfaction, a set forms a *meaningful* composite agent — distinguished from a multi-agent system whose low-ε^* projection is a happenstance of the environment rather than a reflection of composite coherence — when $\varepsilon^* \le \varepsilon_{\max}$. Without scope-satisfaction, the ε^* infimum is still well-defined as a projection property, but the resulting "composite" has ill-defined purposeful substate $G_c = (O_c, \Sigma_c)$ and composition-level machinery (team persistence, composite tempo) does not apply.

Admissibility constraints on macro-dynamics. *Formulation (macro-dynamics-admissibility)*

$\mathcal{M}_{\text{adm}}$ is the class of macro-dynamics (π_c, E_c, f_c) satisfying:

(A1) AAT agent structure. The macro-state decomposes as $X_c = (M_c, G_c)$. The macro-update is recursive at macro-time m:

$$X_{c,m+1} = f_c(X_{c,m}, o_{c,m+1}) \tag{13.6}$$

The macro-policy is state-dependent: $a_{c,m} = \pi_c(X_{c,m})$. This ensures the macro-agent has the same structural components as a single AAT agent (1.1, 3.2, 3.3).

(A2) Macro-mismatch. A mismatch signal is well-defined:

$$\delta_{c,m} = o_{c,m} - \hat{o}_{c,m}(M_c, a_{c,m-1}) \tag{13.7}$$

where $\hat{o}_{c,m}$ is the macro-model's prediction of the next macro-observation. This ensures the macro-agent can detect prediction errors — the foundation of adaptation (3.4).

(A3) Macro-tempo. The macro-update has well-defined adaptive tempo:

$$\mathcal{T}_c = \sum_k \nu_c^{(k)} \cdot \eta_c^{(k)*} \tag{13.8}$$

where k indexes the macro-agent's observation channels, $\nu_c^{(k)}$ is the event rate, and $\eta_c^{(k)*}$ is the optimal gain (3.8, 3.6). This ensures the persistence condition is formulable at the macro level. The additive form carries 3.8's noise-independence scope at the *macro* channel level (`#deriv-tempo-additivity`): when macro channels aggregate cross-correlated micro-streams, the exact macro tempo is the joint form, and the additive expression deviates with sign — the composite-level consequences are worked in 13.2.

(A4) Bounded macro-correction. The macro-correction function satisfies the sector condition (4.4):

$$\delta_c^T F_c(\mathcal{T}_c, \delta_c) \geq \alpha_c \|\delta_c\|^2 \quad \text{for } \|\delta_c\| \leq R_c \tag{13.9}$$

with $\alpha_c > 0$ (positive correction rate) and $R_c > 0$ (finite reserve). This ensures the macro-agent's corrections work — they reduce mismatch rather than amplifying it, within the macro-model's capacity.

What (A1)-(A4) prevent. Without these constraints, the infimum over $\mathcal{M}_{\text{adm}}$ is trivially zero (any dynamical system can curve-fit micro-trajectories over a finite horizon). The constraints force the macro-dynamics to be a genuine AAT agent — with decomposed state, prediction errors, adaptive tempo, and stable correction — not an arbitrary function approximator. ε^* then measures the irreducible cost of representing multiple agents as one AAT agent.

What (A1)-(A4) do NOT require. Directed separation (6.3) is not part of the admissibility constraints. It is an additional structural property that some composites have and others don't (13.3). Results that depend on directed separation declare it as an additional assumption. Similarly, strategy structure ($G_c = (O_c, \Sigma_c)$ with a DAG) is not required — simpler goal representations are admissible.

Definition (projection-admissibility, proposed from spike)

Admissibility constraints on projections.

The admissible class of projections $\mathcal{P}_{\text{adm}}(\epsilon_I, L)$ consists of projections $\Lambda = (\Lambda_x, \Lambda_o, \Lambda_a, \Lambda_\Omega)$ satisfying three conditions:

(P1) Information preservation. The projection retains at least a fraction $(1-\epsilon_I)$ of the predictive mutual information, evaluated across a macro-step:

$$I\big(\Lambda_x(X_{\text{micro},(m-1)K_c});\ \Lambda_o(o_{\text{micro},(m-1)K_c:mK_c}) \mid \Lambda_a(a_{\text{micro},(m-1)K_c:mK_c})\big) \geq (1-\epsilon_I)\cdot I\big(X_{\text{micro},(m-1)K_c};\ o_{\text{micro},(m-1)K_c:mK_c} \mid a_{\text{micro},(m} \tag{13.10}$$

The left side is the predictive information the macro-state has about the next macro-observation (aggregated over the K_c-window), conditioned on the macro-action. The right side is the micro-state's predictive information about the *same* macro-observation, conditioned on the same aggregated actions — both sides share a common target so the ratio is well-defined. The parameter $\epsilon_I \in [0, 1)$ controls how much predictive power may be sacrificed by projection. This is the information bottleneck (Tishby et al. 1999) applied to the projection map. When $K_c = 1$, the windows collapse to singletons and (P1) reduces to the per-micro-step form previously stated.

(P2) Lipschitz continuity. Each component of Λ is Lipschitz-continuous with constant L:

$$\|\Lambda_x(X) - \Lambda_x(X')\|_{\mathcal{X}_c} \leq L \cdot \|X - X'\|_{\mathcal{X}_{\text{micro}}} \quad \forall X, X' \in \mathcal{X}_{\text{micro}} \tag{13.11}$$

(and analogously for $\Lambda_o, \Lambda_a, \Lambda_\Omega$, each with its own Lipschitz constant). Without Lipschitz regularity, bounded closure defect does not imply bounded trajectory error — the bridge lemma requires this or something equivalent. When the projection is L-Lipschitz, the trajectory error bound becomes $L \cdot \varepsilon^*/\alpha_c$.

(P3) Dimensionality reduction. The macro-state space has strictly lower dimension than the micro-state space:

$$\dim(\mathcal{X}_c) < \dim(\mathcal{X}_{\text{micro}}) \tag{13.12}$$

This prevents the trivial identity projection, which achieves $\varepsilon^* = 0$ but is not a genuine abstraction.

Why three conditions. No single condition suffices alone. (P1) prevents degenerate projections (a constant map $\Lambda_x = c$ has zero mutual information). (P2) prevents pathological projections (discontinuous maps that amplify micro-errors into unbounded macro-errors). (P3) prevents trivial projections (the identity map satisfies (P1) and (P2) but achieves nothing). Together they constrain $\mathcal{P}_{\text{adm}}$ to projections that are informative, regular, and genuinely reductive.

The (ϵ_I, L) parameters are part of the problem specification, not derived quantities. The natural choice for many applications is $L = 1$ (non-expansive projection) and ϵ_I chosen as the minimum information loss compatible with genuine dimensionality reduction.

Independence from (A1)-(A4). The macro-dynamics admissibility (A1)-(A4) partially constrains the projection — (A1) requires Λ_x to preserve the M_c/G_c decomposition, and (A4) implicitly requires regularity through the sector condition. But (A1)-(A4) do not specify how much predictive information the projection must retain. A macro-agent with a very coarse projection can satisfy (A1)-(A4) while being a poor representation of the micro-system. The information-preservation condition (P1) fills this gap.

Bridge lemma: closure defect to trajectory error.

Derived (bridge-lemma, from sector-persistence-template + A4 + contraction assumption)

The bridge lemma is the sector-persistence template (`#result-sector-persistence-template`) applied with state variable $\xi = e_m = X_{c,m} - \Lambda_x(X_{\text{micro},\, mK_c})$ (trajectory error at macro-boundaries, between macro-evolved state and projected micro-state) and effective disturbance rate $\rho_\xi = \varepsilon^* \nu_c$ (closure defect per macro-step, multiplied by macro-update rate). The units are consistent by construction of the macro-step formulation above: ε^* is per-macro-step (distance) and ν_c is macro-step rate ([time^{-1}]), so ρ_ξ has the disturbance-rate units the template requires. The template yields directly:

$$\limsup_{m\to\infty} \|e_m\| \leq \frac{\varepsilon^* \nu_c}{\alpha_c} \tag{13.13}$$

This bound fits within the composite's sector-condition region iff $\varepsilon^* < \alpha_c R_c/\nu_c$, which is the persistence condition applied to the closure-error rate: if it fails, the macro-description diverges from micro-reality.

What the bridge lemma adds beyond the template. The template's precondition (T2) is the one-point sector bound. For the trajectory-error instantiation, the bound propagation requires a *strictly stronger* condition: the macro-update map $f_c(\cdot, o)$ must be contracting in its state argument (**incremental sector bound**, DA2'-inc — strong monotonicity of F_c), not merely one-point sector-bounded at each state. The per-step contraction factor is $\lambda = 1 - \alpha_c/\nu_c < 1$ (automatic when $\alpha_c < \nu_c$, true for any realistic agent that doesn't fully correct in a single step). This stronger condition is where the tier structure enters; see Epistemic Status for the Tier 1/2/3 taxonomy.

The sector condition (A4) does double duty — conditionally. (A4) ensures the macro-agent corrects external mismatch (single-agent persistence) AND — under the incremental sector bound — that the macro-description tracks micro-reality (this bridge). Both are instances of the same template applied to different state variables with different effective disturbance rates.

What Is Derived vs. What Is Chosen.

Table 13.1

Property	*Source*	*Strength*
Closure-defect quantity ε^* as infimum over admissible classes	Definition built from (A1)-(A4), (P1)-(P3), and K_c	Formulation choice
Three-component decomposition into $\varepsilon_x, \varepsilon_a, \varepsilon_o$	Match to the three arrows of the action-observation loop (state, action, observation)	Formulation choice
Per-macro-step formulation; ε^* has units of distance-per-macro-step	2026-04-22 temporal coarse-graining repair (Finding A); forced by dimensional consistency with $\rho_\xi = \varepsilon^* \nu_c$ in `#result-sector-persistence-template` and 13.2	Derived (dimensional consistency)
Timescale ratio $K_c \geq 1$ as independent problem-specification parameter	Application-specific; $K_c = 1$ (synchronous) and $K_c \gg 1$ (4.9 regime) both admissible	Formulation choice
(A1) Macro AAT structure $X_c = (M_c, G_c)$ with recursive update	Import from 1.1 + 3.2 + 3.3	Formulation choice (requirement)
(A2) Well-defined macro-mismatch	Import from 3.4	Formulation choice (requirement)
(A3) Well-defined macro-tempo	Import from 3.8 + 3.6	Formulation choice (requirement)
(A4) Sector-bounded macro-correction	Import from 4.4	Formulation choice (requirement)
(A1)-(A4) as a requirement *set* render the infimum non-trivial (exclude curve-fitting macro-dynamics)	Consequence of restricting $\mathcal{M}_{\text{adm}}$ to genuine AAT agents	Derived (under the chosen requirement set)
(P1) Information-preservation constraint	IB Lagrangian-dual with source X_{micro}, relevance $o_{\text{micro},t+1}$; formally connects to `#disc-compression-operations`	Formulation choice (requirement)
(P2) Lipschitz continuity of Λ	Required for bridge-lemma trajectory-error bound (without it, bounded ε^* does not imply bounded trajectory error)	Formulation choice (requirement)
(P3) Strict dimensionality reduction	Rules out identity projection (which trivially achieves $\varepsilon^* = 0$ but is not abstraction)	Formulation choice (requirement)
(P1)-(P3) independent of (A1)-(A4)	(A1) and (A4) constrain Λ_x partially but do not fix information-preservation level (§"Independence from (A1)-(A4)")	Derived
Bridge lemma: $\limsup \lVert e_m \rVert \leq \varepsilon^* \nu_c / \alpha_c$	Sector-persistence template (`#result-sector-persistence-template`) applied with state $\xi = e_m$, disturbance rate $\rho_\xi = \varepsilon^* \nu_c$	Derived (conditional on incremental sector bound DA2'a-inc)
Persistence-as-closure-boundedness condition $\varepsilon^* < \alpha_c R_c / \nu_c$	Sector-condition region fits the asymptotic error ball	Derived (under A4 + DA2'a-inc)
Incremental sector bound (DA2'a-inc) is strictly stronger than (A4)'s one-point sector bound	Witness $F_d(\delta) = 2\delta + 3 \sin \delta$ (oscillatory, globally inward, locally non-monotone), Discussion §"DA2'a-inc is strictly stronger than (A4)"	Proved

Property	*Source*	*Strength*
Three-tier agent classification (Tier 1 / 2 / 3) for bridge-lemma applicability	Taxonomy induced by whether DA2'a-inc holds globally (T1), locally (T2), or must be verified per-domain (T3)	Formulation choice (classification)
Tier 1: bridge lemma derived for Bayesian updaters on exponential families, gradient descent on strongly convex losses, linear correction with positive-definite gain-observation product	Standard monotone-operator / Kalman analysis; $H^\top H$-norm contraction derivation in Discussion §"Bridge-lemma contraction"	Derived (conditional on DA2'a-inc + linear prediction, i.e., C2)
Tier 2: local contraction with factor degraded by $\kappa(D\hat{o})^2$	Extended-Kalman / locally-convex analysis	Derived (local)
Tier 3: contraction must be verified per-domain	No general result available for non-convex / discontinuous / non-mismatch-driven components	Discussion-grade
Composite (A4) from sub-agent properties: $\alpha_c \geq \min_i(\alpha_i - \Delta\mathcal{T}_i^{\text{cost}})$, $R_c \leq \min_i R_i$	Weakest-link bound (Discussion §"Deriving composite (A4) from sub-agent properties")	Derived (conditional on bounded coordination cost per 15.1)
Composite critical-mass inequality $(\alpha - C)R > \rho + \gamma\mathcal{T}$ (symmetric-matched-Tier-1) — sign-sensitive refinement of the weakest-link bound; recovers 15.1 (cooperative) and 15.2 (adversarial) as signed special cases	`#deriv-critical-mass-composition`	Derived (conditional on Tier 1 + matched-symmetric + Model D + coupling model C1/C2)
Composite persistence as (contraction) $\wedge$ (scope-satisfaction): $\kappa_c(U_O) > 0 \wedge$ 12.2	`#deriv-critical-mass-composition` (CM4); U_O enters multiplicatively on γ plus scope-gate, not additively	Derived (with (UO-mult) discussion-grade)
Asymmetric-limit connection to 12.3 (S-3) autonomy reduction	`#deriv-critical-mass-composition` §asymmetric-limit; weighted Lyapunov V_μ with $\mu \to 0$	Sketch (S-2 function transfer not yet formalized; promotable when 12.3 closes (S-2))
Meta-machine exact composition ($\varepsilon^* = 0$) for finite-state Moore machines	Miller 2022 §3.3; product automaton is the micro-dynamics by construction	Derived (external theorem)
Two-Kalman instantiation: $\varepsilon^* = 0$ at all ρ_{corr} for the means-only projection at steady state	Analytic steady-state calculation (Discussion §"Two-Kalman instantiation (closed form)")	Proved (closed form)
ε_x depends on both sub-agent unity *and* update-rule heterogeneity (ΔK)	Heterogeneous-gain Kalman case; 14.1	Derived (specific closed form for the two-Kalman case)
Mahalanobis norm as natural choice for estimation-type agents	Two-Kalman verification; Kalman gain minimizes expected squared Mahalanobis distance	Formulation choice (canonical for estimation; general-domain norm specification open)
N-agent scaling of ε^*	A "polynomial vs. exponential in N" framing is mis-typed: ε^* is a per-step residue, while "scales in N" is an accumulation-typed predicate	Resolved: $\varepsilon^*(N)$ is dimension-free-zero in the benign linear-Gaussian-stationary regime (all N, all coupling), graph-Laplacian-bounded with **no exponential regime** under compression (the consensus operator `#deriv-critical-mass-composition` already names), and order-incompatibility-invariant ($\leq \lvert S\rvert \log 2$, N-free) for strategy-DAG composition. No exponential regime exists. Full statement: the next two rows + Discussion

Property	*Source*	*Strength*
Computability of (P1) for nonlinear / non-Gaussian systems	Requires Monte Carlo or variational bounds; closed-form only for linear-Gaussian	Open
Principled setting of the ϵ_I threshold	No formal criterion currently; candidate tying to team size and coupling structure	Open
Strategy-DAG projection under Λ_x	Causal-τ-abstraction frame; 8.3 §"Composing heterogeneous strategy DAGs" (topology / SCC-condensation), 14.1 (fixed-topology credence instance)	Resolved (*conditional*; one named sub-open Q1): the right object is a causal-τ-abstraction; exact Σ_c iff shared sub-orders jointly acyclic + AND/OR-compatible, else the minimal admissible Σ_c is the SCC-condensation with defect $\sum_{\lvert C\rvert>1} \mathrm{TC}(C) \le \lvert S\rvert \log 2$ — **bounded by shared-plan size, N-free.** Beckers–Eberhardt–Halpern (τ, ϵ)-abstraction ⇒ bounded ε_Σ (the *frame*, not term-for-term; the high-level distance remains AAT's to choose). Q1 (constant-tightness for non-deterministic intra-SCC coupling) is a named sub-open, not a reopening
Mori-Zwanzig zero-lag bound $\varepsilon^* \ge \lVert Q_\Lambda U P_\Lambda \rVert_{\mathrm{op}}$	Koopman/MZ projection (stationary coordinate-compatible case)	Derived (conditional on stationarity)
Full-kernel MZ bound on ε^*	Type mismatch (trajectory-accumulation vs. per-step)	Refuted at the ε^*-level; natural home is the bridge-lemma trajectory-error bound

The dividing line: (A1)-(A4) and (P1)-(P3) are *formulation choices of the requirement set* — alternative requirement sets are possible (e.g., entropy-preservation instead of MI-preservation; contraction bound in place of Lipschitz), and this one is chosen for parsimony, direct import from prior AAT segments, and downstream tractability. The *consequences* that follow under this requirement set — the non-triviality of the infimum, the bridge-lemma bound, the weakest-link composite-(A4) derivation, the two-Kalman $\varepsilon^* = 0$ result — are Derived, with bridge-lemma-level results carrying the additional DA2'a-inc condition (which is proved strictly stronger than (A4) alone). The Tier 1/2/3 classification is itself a formulation, but the per-tier results are derived under tier-specific conditions. The genuinely-open questions are now the computability / principled-setting questions for (P1) and ϵ_I. The N-agent-scaling and strategy-DAG-projection rows are **resolved** (re-typed; the "poly-vs-exp" / "domain-specific-unresolved" framings were mis-typed — see the two rows above): there is no exponential-in-N regime, and N-scaling does not decide whether large teams are representable as single AAT agents.

Epistemic Status.

Conditional. Max attainable: conditional (formulation choice).

The closure defect ε^* is well-defined given (A1)-(A4), (P1)-(P3), the timescale ratio K_c, and specified norms. The macro-dynamics admissibility (A1)-(A4) and the projection admissibility (P1)-(P3) are independent formulation choices — (A1)-(A4) specify what "AAT-shaped macro-dynamics" means; (P1)-(P3) specify what "informative, regular, genuinely reductive projections" means. Both are needed for ε^* to be an infimum over a non-trivial, non-degenerate class. The timescale ratio $K_c \ge 1$ is a third, independent piece of problem specification: it sets the granularity at which closure is measured and the type signatures of Λ_o, Λ_a. K_c does not constrain the macro-dynamics class and is not derived — it is chosen from the timescale separation of the application (per 4.9). Both $K_c = 1$ (tightly-coupled composites) and $K_c \gg 1$ (hierarchically composed composites on their own timescale) are admissible. The projection admissibility conditions have a proposed definition with one exact instantiation (two-Kalman case at $K_c = 1$, worked in the Discussion); general-case computability of (P1) and the dependence of ε^* on K_c for a fixed application remain open.

The bridge lemma is *conditional* — not fully derived from (A4) alone. The precise additional condition is the **incremental sector bound** (DA2'a-inc): the correction function F_d must be *strongly monotone* ($\langle \delta - \delta', F_d(\delta) - F_d(\delta') \rangle \geq c_{\min} \|\delta - \delta'\|^2$ for all pairs), not just satisfy the one-point sector bound. This is strictly stronger than (A4) — a witness exists: the oscillatory correction $F_d(\delta) = 2\delta + 3 \sin \delta$ is globally inward-pointing but locally non-monotone (Discussion §"DA2'a-inc is strictly stronger than (A4)"). Three tiers of agents result:

- **Tier 1 (contraction proved):** Kalman filters, exponential-family Bayesian updaters, gradient descent on strongly convex losses, all linear corrections with positive definite gain-observation product. For these agents the bridge lemma is *derived* from (A4) + DA2'a-inc + linear prediction (C2), with contraction factor λ_{eff} from `#deriv-discrete-sector-condition`.
- **Tier 2 (local contraction):** Extended Kalman, locally convex gradients, nonlinear prediction models. Contraction holds locally with factor degraded by $\kappa(D\hat{o})^2$.
- **Tier 3 (independent verification):** Non-convex optimization, discontinuous/rule-based corrections, agents with non-mismatch-driven state components. Contraction must be verified per-domain.

The discrete-time recurrence argument itself is standard (Elaydi 2005); the contribution is identifying the incremental sector bound as the precise condition that bridges sector-bounded correction to full-update-map contraction. The full $H^\top H$-norm contraction derivation is in the Discussion (§"Bridge-lemma contraction in the $H^\top H$-norm").

The norm choices ($\|\cdot\|_{\mathcal{X}}$, $\|\cdot\|_{\mathcal{A}}$, $\|\cdot\|_{\mathcal{O}}$, and the combination norm) are load-bearing. For estimation-type agents, the Mahalanobis norm (weighted by inverse prediction-error covariance) is the natural choice — verified exactly in the two-Kalman case. For general domains, norm specification remains open.

Discussion.

This criterion replaces intuitive questions about "where the boundary of an agent is" with a functional test: does a macroscopic AAT description preserve the underlying micro-dynamics well enough to remain predictive and capable? The core requirement is an **approximate dynamical homomorphism** — the macro-dynamics approximately commute with the projection.

Relationship to 11. The Part I postulate requires that AAT's machinery be scale-invariant — predictions at different levels of description must be compatible. This segment operationalizes "compatible" as "bounded closure defect under admissible coarse-graining." The admissibility constraints ensure the macro-description is genuinely AAT-shaped, so the same persistence condition, the same tempo framework, and the same mismatch dynamics apply at the macro level with macro-level parameters.

Projection-behaviour facet of the stability certificate (5). Coarse-graining is a non-invertible projection of the certificate (`#result-certificate-existence`): the certificate-as-metric survives (the Schur complement of a positive-definite form is positive-definite) but the *dynamic* guarantee does not — the closure defect ε^* is the certificate's projection-residue, equal to the Mori–Zwanzig zero-lag memory-commitator norm (derivation table above), zero exactly when the resolved subspace is invariant. Read through the spine, this is the question "does the agent's measuring-stick survive being viewed coarsely": its shape survives, its guarantee leaks by exactly the memory term. The composition-level identifiability floor (no common certificate across sub-agents; Liberzon, 5.1 Instance 3) is this same residue read as a boundary event — a *distinct* obstruction from the rank-collapse floors (a non-invertible projection, not a congruence), which is why the cross-sectional structure is several meta-patterns and not one.

The Jacobian-level observation. The incremental sector bound (DA2'-inc) used to prove the bridge lemma is mathematically equivalent to the contraction-metric condition (CT2) evaluated at the identity metric $M = I$, for continuously differentiable (C^1) correction functions F on convex domains. This equivalence (cf. Rockafellar & Wets 1998) means that AAT's bridge lemma is a specialized application of generalized contraction theory. See `#result-contraction-template` for the full lifting of this condition to arbitrary Riemannian metrics.

Topology-indexed composition closures (via `#result-contraction-template`). `#result-contraction-template` lifts the bridge-lemma's DA2'-inc condition into the contraction-metric framework of Lohmiller & Slotine 1998. Under its (CT1)–(CT3) preconditions, three topology-indexed closure results apply:

- **Parallel composition** preserves contraction under blockdiag metric at rate $\min(\lambda_1, \lambda_2)$ — recovers this segment's weakest-link bound.
- **Cascade composition** ($\dot{x}_1 = f_1(x_1), \dot{x}_2 = f_2(x_1, x_2)$) preserves contraction under lower-triangular weighted metric at rate bounded by $\min(\lambda_1, \lambda_2)$ up to coupling-gain adjustment (Slotine 2003 Thm 2). Handles hierarchical agent structures.
- **Negative feedback with bounded loop gain** (Slotine 2003 Thm 3) gives the heterogeneous critical-mass inequality $(\lambda_1 - C_1)(\lambda_2 - C_2) > k_{12}k_{21}/4$ via (CM2-M) in `#result-contraction-template` — extending `#deriv-critical-mass-composition`'s matched-symmetric (CM2) to heterogeneous sub-agents with different architectures and coupling strengths.

The Tier 1/2/3 bridge-lemma taxonomy of this segment maps cleanly to Tier 1M/2M/3M under `#result-contraction-template` (globally metric-contracting / locally metric-contracting / no globally valid metric). The observation that (CT2) at $M = I$ is *equivalent* to DA2'-inc means this segment has been carrying the Jacobian-level Euclidean contraction condition at the composite level all along; `#result-contraction-template` surfaces this at the single-agent level and adds the compositional theorems that generalize this segment's closure content beyond the matched-symmetric case.

DA2'-inc ≡ (CT2) at $M = I$ — structural-transparency equivalence. *[Derived, status: exact.]* For C^1 correction functions F on convex domains, the following three conditions are equivalent (standard monotone-operator theory; Rockafellar & Wets 1998 *Variational Analysis* Corollary 12.4; Nesterov 2004 *Introductory Lectures on Convex Optimization* §2.1.3 Theorem 2.1.9 — strong monotonicity ⇔ symmetric-part of Jacobian uniformly positive-definite):

(a) DA2'-inc (strong monotonicity): $(\delta - \delta')^\top(F(\delta) - F(\delta')) \geq c\|\delta - \delta'\|^2$ for all δ, δ'.

(b) Jacobian-symmetric-part PD: $(\partial F/\partial\delta)_{\text{sym}} \succeq cI$ pointwise on the domain.

(c) (CT2) at $M = I$ with $\lambda = c$: $-\partial F/\partial\delta - (\partial F/\partial\delta)^\top \preceq -2\lambda I$.

Derivation. (a) → (b): take $\delta' \to \delta$ along direction v, divide by $\|\delta - \delta'\|^2$, take the limit. (b) → (a): integrate $v^\top(\partial F/\partial\delta)_{\text{sym}}v \geq c\|v\|^2$ along the segment from δ' to δ. (b) ↔ (c): direct algebraic identity.

Implication for AAT-internal derivability. Under this equivalence, the Euclidean sub-scope metric-α_1 cases of `#result-contraction-template` (Kalman-scalar, Euclidean strongly-convex-gradient, L2-regularized, linear-PD-symmetric) are AAT-internally derived from the DA2'-inc commitment already carried at the composite level — no new axiom required. What the (CT2)-at-$M = I$ framing adds is *visibility*: the Jacobian-level condition was implicit under the DA2'-inc name; surfacing the equivalence converts the Euclidean sub-scope from "theorem-imported from Lohmiller-Slotine" to "AAT-internally derived via a standard monotone-operator identity." This is structural transparency, not new content. Non-Euclidean metric-α_2 cases (Fisher, Hessian, Lyapunov-metric) remain separately treated in `#result-contraction-template` — the equivalence is specific to $M = I$.

DA2'a-inc is strictly stronger than (A4): the witness. *[Derived, status: exact.]* The incremental sector bound (DA2'a-inc) does **not** follow from the one-point sector bound (A4) alone. A scalar witness exhibits a correction function satisfying the one-point bound while violating incremental monotonicity. Take

$$F_d(\delta) = 2\delta + 3\sin\delta \qquad \text{on } |\delta| \leq \pi. \tag{13.14}$$

One-point bound holds. Since $\delta \sin\delta \geq 0$ on $[-\pi, \pi]$ (the two factors share sign),

$$\delta F_d(\delta) = 2\delta^2 + 3\delta\sin\delta \geq 2\delta^2, \tag{13.15}$$

so the one-point sector bound (A4) holds with $\alpha = c_{\min} = 2$ on $\mathcal{B}_\pi$.

Incremental bound fails. In one dimension the incremental condition $(\delta - \delta')(F_d(\delta) - F_d(\delta')) \geq c_{\min}(\delta - \delta')^2$ is equivalent to $F_d'(\delta) \geq c_{\min}$ pointwise. Here $F_d'(\delta) = 2 + 3\cos\delta$, and at $\delta = \pi$ this is $2 + 3\cos\pi = 2 - 3 = -1 < 0$. So near $\delta = \pi$ there exist $\delta_1 < \delta_2$ with $F_d(\delta_1) > F_d(\delta_2)$, giving $(\delta_1 - \delta_2)(F_d(\delta_1) - F_d(\delta_2)) < 0$ — the incremental bound fails for *any* $c_{\min} > 0$.

The geometric reading: the one-point bound says "every correction points inward toward the equilibrium"; the incremental bound says "larger mismatches always receive larger corrections." An oscillatory correction can be globally inward-pointing (one-point bound holds with margin) while *locally* the correction magnitude decreases as the mismatch grows (it wobbles) — which breaks incremental monotonicity. This is why the bridge lemma's Tier-1 promotion carries the additional DA2'a-inc condition rather than resting on (A4): the witness shows the gap is real, not an artifact of proof technique.

Bridge-lemma contraction in the $H^\top H$-norm (Tier 1). *[Derived (bridge-lemma-contraction), conditional on (C1) incremental sector-Lipschitz, (C2) linear prediction, (C3) bounded gain.]* The Tier-1 promotion of the bridge lemma — from "f_c is assumed contracting" to "f_c is contracting, derived" — rests on the following. Let the macro-update be mismatch-driven, $f_c(X, o) = X + \eta\, g(o - \hat{o}(X))$, with observation-space correction $F_d(\delta) = Hg(\delta)$, and assume:

- **(C1)** F_d is incrementally sector-Lipschitz: $\langle \delta - \delta', F_d(\delta) - F_d(\delta') \rangle \geq c_{\min} \|\delta - \delta'\|^2$ (DA2'a-inc, strong monotonicity) and $\|F_d(\delta) - F_d(\delta')\| \leq c_{\max} \|\delta - \delta'\|$;
- **(C2)** linear prediction $\hat{o}(X) = HX$, so $\delta - \delta' = H(X' - X)$;
- **(C3)** bounded gain $\eta < 2c_{\min}/c_{\max}^2$.

Then the full update map contracts in the $H^\top H$-norm $\|v\|^2_{H^\top H} = v^\top H^\top H v$:

$$\|f_c(X, o) - f_c(X', o)\|_{H^\top H} \leq \lambda_{\text{eff}}\, \|X - X'\|_{H^\top H}, \qquad \lambda_{\text{eff}}^2 = 1 - 2\eta c_{\min} + \eta^2 c_{\max}^2 < 1. \tag{13.16}$$

Derivation. Applying H to the update difference $\Delta f = f_c(X, o) - f_c(X', o) = (X - X') + \eta[g(\delta) - g(\delta')]$ and writing $\Delta\delta = \delta - \delta' = H(X' - X)$,

$$H\Delta f = -\Delta\delta + \eta\big[F_d(\delta) - F_d(\delta')\big]. \tag{13.17}$$

Taking the squared Euclidean norm (which is the squared $H^\top H$-norm of Δf):

$$\|\Delta f\|^2_{H^\top H} = \|\Delta\delta\|^2 - 2\eta\big\langle \Delta\delta,\ F_d(\delta) - F_d(\delta') \big\rangle + \eta^2 \|F_d(\delta) - F_d(\delta')\|^2. \tag{13.18}$$

Applying (C1)'s incremental sector bound to the middle term and incremental Lipschitz bound to the last term:

$$\|\Delta f\|^2_{H^\top H} \leq \big(1 - 2\eta c_{\min} + \eta^2 c_{\max}^2\big) \|\Delta\delta\|^2 = \lambda_{\text{eff}}^2\, \|\Delta\delta\|^2, \tag{13.19}$$

and $\|\Delta\delta\|^2 = \|H(X' - X)\|^2 = \|X - X'\|^2_{H^\top H}$ closes the bound; (C3) makes $\lambda_{\text{eff}} < 1$. □

This λ_{eff} is exactly the mismatch-dynamics contraction factor of `#deriv-discrete-sector-condition` — the trajectory error e_m generates a virtual mismatch He_m contracted by the same mechanism that contracts actual mismatch, which is why the bridge bound $\limsup \|e_m\| \leq \varepsilon^* \nu_c / \alpha_c$ holds with no separate assumption once (C1)–(C3) do. When $H = I$ (direct observation) the norm is Euclidean and the factor is exactly λ_{eff}, recovering the scalar-Kalman case; for invertible H the Euclidean factor degrades by $\kappa(H)^2$. The classes for which (C1) holds — all linear corrections with positive-definite Hg, all exponential-family Bayesian updaters (strong monotonicity from log-partition convexity), all gradient descent on strongly convex losses — are the Tier-1 agents; (C2)'s linear-prediction requirement is what separates Tier 1 (exact) from Tier 2 (local, factor degraded by the prediction Jacobian's condition number squared).

The sector condition does double duty — conditionally. (A4) ensures the macro-agent's corrections work (structural persistence), AND — given the additional contraction assumption (see Epistemic Status) — it ensures the macro-description tracks micro-reality (bridge lemma). Both uses address *structural persistence* in the sense of `LEXICON.md` — the capacity of the correction machinery, not the current operating point or the composite's identity through time. The central insight is that structural persistence of the composite and faithfulness to its constituents are related through the same mechanism — contracting correction dynamics under bounded perturbation — but the bridge from sector-bounded correction to full-update-map contraction requires an assumption beyond (A4) that has been verified for estimation-type agents (Kalman) and remains open for general agents. When this assumption holds, the composite that persists in its own right also persists as a faithful representation of its constituents.

Deriving composite (A4) from sub-agent properties. If each sub-agent satisfies the sector condition with parameters (α_i, R_i), and coordination costs are bounded by $\Delta\mathcal{T}_i^{\text{cost}}$ per agent (15.1), then the composite satisfies (A4) with:

$$\alpha_c \geq \min_i(\alpha_i - \Delta\mathcal{J}_i^{\text{cost}}) \tag{13.20}$$

$$R_c \leq \min_i R_i \tag{13.21}$$

This is a weakest-link bound (conservative but clean): each sub-agent's effective correction rate is its own α_i degraded by its coordination overhead $\Delta\mathcal{J}_i^{\text{cost}}$, and the composite inherits the worst of these, while its reserve R_c cannot exceed the smallest sub-agent reserve. Cooperative coupling (15.1) can improve α_c beyond this bound by reducing effective disturbance. The key implication: (A4) is *verifiable* from micro-level properties — compute each sub-agent's sector-condition parameters, estimate coordination costs, and check whether the composite has positive correction rate. No need to compute f_c directly.

Sign-sensitive refinement (derived). The sign-blind weakest-link bound is subsumed by `#deriv-critical-mass-composition`'s closed-form inequality $(\alpha - C)R > \rho + \gamma\mathcal{T}$ (CM2) in the matched-symmetric-Tier-1 case, where the right-hand side makes the effective disturbance explicit and the sign of the coupling γ (cooperative $\gamma < 0$ vs adversarial $\gamma > 0$) enters directly. (CM2) can yield composite persistence even when the weakest-link bound alone fails — when cooperative coupling reduces $\rho + \gamma\mathcal{T}$ below what the raw $\alpha - C$ margin would permit. 15.1 (cooperative) and 15.2 (adversarial) are recovered as signed special cases. The honest composite-persistence statement is the conjunction of contraction (CM3) **and** scope-satisfaction (12.2) — (CM4) — making explicit that composite persistence can fail in two qualitatively different ways: the composite fails to contract, or the composite was never a composite. The asymmetric-parameter limit $\alpha_2 \to 0$ via weighted Lyapunov V_μ formalizes 12.3's (S-3) autonomy-reduction mechanism as a smooth deformation of (CM4), not a discontinuous regime change.

Connection to team-persistence. 15.1 derives persistence conditions for sub-agents in a cooperative-adversarial network. This segment provides the macro-level complement: the conditions under which the composite itself is a valid AAT agent. Together they close the loop: sub-agents persist individually (team-persistence) AND the composite is a meaningful macro-agent (composition closure with admissibility).

Constructive (A1)–(A4) via wrapping. The wrapping construction in `#der-class-coercion-via-wrapping` is a special-case instantiation where (A1)–(A4) hold *by construction* through the wrapper's type signatures rather than by post-hoc verification of an admissible projection Λ. The wrapper builds explicit $X_W = (M_W, G_W)$ with belief-update and strategy-update maps that have specified type signatures; (A1) is automatic; (A2)–(A4) follow from wrapper-design constraints (commit to a prediction map; choose f_M from the Tier-1 belief-update class). The closure-defect ε^* in this special case has a different reading than the standard fidelity reading — the wrapper *deliberately* changes the underlying component's behavior to enforce structural separation. Two distinct ε^* quantities appear in wrapper analyses: $\varepsilon^*_{\text{track}}$ (the standard fidelity quantity defined here, used in the bridge lemma) and $\varepsilon^*_{\text{coerce}}$ (the wrapper-vs-component behavioral divergence, distinct from $\varepsilon^*_{\text{track}}$ and not bounded by the bridge lemma). See `#der-class-coercion-via-wrapping` Discussion for the disambiguation.

Meta-machine: exact composition for finite automata (Miller 2022). When agents are finite-state automata (Moore machines), composition is *exact*: two machines with state sets S_1, S_2 interacting through their output/input channels form a **meta-machine** (Miller 2022, *Ex Machina*, §3.3) — itself a Moore machine with state set $S_1 \times S_2$, deterministic transitions (each machine's output determines the other's input), and a single starting state from the constituent starting states. The closure defect $\varepsilon^* = 0$ trivially, because the product automaton *is* the micro-dynamics — there is no approximation. The meta-machine always falls into a cycle (finite states with deterministic transitions must eventually revisit a state), so two interacting automata produce an eventually-periodic joint behavior. The composition-closure framework becomes interesting when we ask for a *compressed* macro-description: can a smaller automaton (fewer than $|S_1| \times |S_2|$ states) approximate the meta-machine? This is where $\varepsilon^* > 0$ and the admissibility constraints become non-trivial — (P3) requires genuine dimensionality reduction, and (P1) asks how much predictive information the compression preserves. Machine minimization (the theorem that every automaton has a unique minimized equivalent; Hopcroft et al. 2006) is a natural candidate for the optimal projection: the minimized meta-machine has the fewest states that reproduce the full joint behavior, achieving $\varepsilon^* = 0$ with (P3) satisfied whenever the minimized machine is smaller than the product. See `#worked-example-cam` (planned) for the full instantiation.

Two-Kalman instantiation (closed form). The simplest nontrivial worked case: two scalar Kalman filters tracking a bivariate random walk with cross-correlated process noise $\mathrm{Cov}(w_1, w_2) = \rho_{\text{corr}}\, q$ (common variance q, common observation noise r), each observing only its own component, no communication. This case sits at $K_c = 1$ — both sub-agents and the composite step together — so the macro-step formulation reduces to the pointwise form and the Λ_o aggregation is the identity. The natural projection keeps the state estimates and discards the covariance states (means-only projection, dimension 2 from micro-dimension 4). Each filter has the scalar steady state $P^* = \frac{1}{2}(-q + \sqrt{q^2 + 4qr})$, gain $K^* = P^*_{\text{pred}}/(P^*_{\text{pred}} + r)$, and marginal innovation variance $S^* = P^* + q + r$.

Although each filter is decoupled (agent *i*'s update uses only o_i), the composite innovation vector $\delta_c = (\delta_1, \delta_2)$ is *cross-correlated* whenever $\rho_{\text{corr}} \neq 0$, because the environment couples the processes. The steady-state cross-covariance of the estimation errors $e_i = \omega_i - \hat{\omega}_i$ solves the fixed point $C_e = (1 - K^*)^2 C_e + \rho_{\text{corr}} q$, giving

$$C_e = \frac{\rho_{\text{corr}}\, q}{1 - (1 - K^*)^2} = \frac{\rho_{\text{corr}}\, q}{K^*(2 - K^*)}, \tag{13.22}$$

and the innovation cross-covariance is $\Gamma = \text{Cov}(\delta_1, \delta_2) = C_e + \rho_{\text{corr}} q$, so the composite innovation covariance is

$$\Sigma_\delta = \begin{pmatrix} S^* & \Gamma \\ \Gamma & S^* \end{pmatrix}, \qquad \Gamma = \frac{\rho_{\text{corr}}\, q\, (1 + 2K^* - K^{*2})}{K^*(2 - K^*)}. \tag{13.23}$$

The per-step closure error of a macro-agent using gain K_c is $\varepsilon_X^2 = \text{tr}\big((K^* I - K_c)\, \Sigma_\delta\, (K^* I - K_c)^\top\big)$, minimized at $K_c = K^* I$ where it is exactly zero. So **at steady state the closure defect is exactly $\varepsilon^* = 0$ for all ρ_{corr}** — the means-only projection perfectly represents the micro-dynamics because the Kalman gains have converged to constants and the discarded covariance state carries no information. All three projection conditions are verified exactly: $\epsilon_I = 0$ (the discarded covariance is a deterministic constant, so it carries zero conditional mutual information; (P1) holds with equality), $L = 1$ (the projection is a coordinate extraction with operator norm 1), and $\dim(\mathcal{X}_c) = 2 < 4 = \dim(\mathcal{X}_{\text{micro}})$. The admissibility conditions (A1)–(A4) hold for the macro-Kalman with $\alpha_c = K^*$, $R_c = \infty$, $\mathcal{T}_c = 2K^*$. The transient defect (before the gains converge) decays geometrically at rate $(1 - K^*)^2$ and is exponentially small after $O(1/|\log(1 - K^*)|)$ steps.

The "cost of independence" — the estimation performance lost by not exploiting cross-correlations — is a separate quantity, the **performance gap** $\Delta_{\text{perf}} = 2(P^* - p)$ between the independent composite and the optimal joint filter (diagonal posterior covariance p). It is *not* a closure defect: it measures suboptimality relative to a centralized agent, not failure of the macro-description to track the micro-system. For small correlation it is second-order, $\Delta_{\text{perf}} \approx 2\rho_{\text{corr}}^2 q^2 r/(S^*)^2$ (the cross-gain is $O(\rho_{\text{corr}})$, and the information it extracts is $O(\rho_{\text{corr}}^2)$), growing monotonically to its maximum at $|\rho_{\text{corr}}| = 1$. The composition-closure framework diagnoses representability ("is this group a coherent AAT agent?" — yes), not optimality ("is this group as good as a centralized agent?" — no, when $\rho_{\text{corr}} \neq 0$). A group can be a perfectly valid composite agent while being suboptimal.

The first genuine $\varepsilon^* > 0$: purposeful sub-agents with Beta-Bernoulli strategy. The benign $\varepsilon^* = 0$ above turns on the discarded auxiliary state (the covariance) *converging* to a constant. Lift the sub-agents to purposeful agents, each carrying a single-edge strategy DAG with credence p_i updated Beta-Bernoulli on observed success/failure — $p_{i,t+1} = p_{i,t} + (\mathbb{1}[\cdot] - p_{i,t})/(n_{i,t} + 2)$ — and the discarded auxiliary state is the observation count n_i, which *diverges* rather than converging. The edge-update gain $1/(n_i + 2) \to 0$ as $n_i \to \infty$. A macro-agent that discards n_i must use a fixed macro-gain η_c, and the per-step strategy closure error $\varepsilon_\Sigma = \mathbb{E}\big[|(1/(n_i + 2) - \eta_c)(\mathbb{1}[\cdot] - p_i)|\big]$ then grows without bound (fixed η_c cannot match a decaying gain) unless $\eta_c = 0$, at which point the macro-agent stops learning. So this is the first genuine source of $\varepsilon^* > 0$: a decaying-gain micro-dynamics is non-representable by fixed-gain macro-dynamics. It is structurally distinct from the Kalman case precisely in the convergence-vs-divergence of the discarded state — the Kalman covariance converges (defect vanishes at steady state); the Beta-Bernoulli count diverges (defect persists). The repair is to keep an effective-sample-size n_c in the macro-state (option (b)), restoring $\varepsilon^* = 0$ at the cost of a less aggressive projection — which signals that the purposeful substate carries more load-bearing dimensions than the epistemic substate. When $\varepsilon^* > 0$, the bridge lemma becomes load-bearing: the macro-strategy eventually goes stale and must be re-projected, the composition-level analog of 4.8.

Norm specification for estimation-type agents. The two-Kalman case identifies the Mahalanobis norm as the natural choice for agents whose primary function is state estimation. The state norm weights by the inverse prediction-error covariance: $\|X_c - X_c'\|^2 = (\hat{\omega}_c - \hat{\omega}_c')^T (P^*_{\text{pred}})^{-1} (\hat{\omega}_c - \hat{\omega}_c')$. The observation norm weights by the inverse innovation covariance $S^{-1} = (P^*_{\text{pred}} + R)^{-1}$. The general principle: norms should weight by inverse uncertainty, so that differences in well-estimated components count more than differences in poorly-estimated ones. This is the norm the Kalman filter implicitly uses — the Kalman gain minimizes the expected squared Mahalanobis distance from truth. For domains without a natural covariance structure (discrete states, non-Gaussian models), the Euclidean norm remains the default.

Findings.

Composition-Closure Defect and Bridge Lemma. Brief: When does a group of interacting agents legitimately count as one larger agent? The closure-defect ε^* is the minimum residual prediction error from collapsing the micro-system into an AAT-shaped macro-description, taken over admissible coarse-grainings. The bridge lemma converts this prediction-level error into a trajectory-level guarantee: under sector-bounded macro-correction plus strong monotonicity of the macro-update, the macro-description tracks micro-reality with bounded asymptotic error $\varepsilon^* \nu_c / \alpha_c$. Composition is therefore not asserted by interface or by intuition — it is certified by an inequality, with explicit failure modes on both sides (the macro-dynamics fail to be AAT-shaped; the projection fails to be informative-and-regular; the macro-correction is not strongly monotone).

Impact: Closes the gap that Markov-blanket and active-inference accounts of nested agency leave open: those accounts give a statistical / thermodynamic boundary criterion but no control-theoretic loss bound on treating a collection as one agent. The closure-defect framework supplies the bound, attaches it to AAT's existing sector-condition machinery, and routes Part I and Part II results across composite boundaries on Tier-1-classified sub-agents. Downstream this lets `#deriv-critical-mass-composition` derive the composite sector constant from sub-agent parameters in closed form, lets `#der-tempo-composition` give Brooks's Law a formal expression in tempo units, and supplies the escape route (b) for `#disc-identifiability-floor` Instance 3. The criterion also draws an honest scope line: $\varepsilon^* = 0$ does not mean optimality (two non-communicating Kalman filters achieve $\varepsilon^* = 0$ while being suboptimal versus a joint filter); the framework diagnoses representability, not optimality.

Novelty Claim: *Claim differentiation* on bounded-loss composition as agent-boundary criterion. The bridge-lemma form is mathematically continuous with the approximate-information-state and influence-based-abstraction lines (Subramanian 2020; Congeduti 2020; Abel 2016; Taylor 2008), which prove approximation-loss results for compressed control under coarse-graining or aggregation. The differentiation is the *use* of such a bound as a criterion for when a multi-agent system can be re-described as a single agent — not as a control-policy compression result on a fixed agent — combined with explicit admissibility conditions (A1)-(A4) on the macro-dynamics that force the macro-description to itself be AAT-shaped, ruling out trivial curve-fits.

Related Work:

Table 13.2

ASF concern	*Prior-art language*	*Relationship / Positioning*
Statistical / thermodynamic boundary of a nested agent	Parr, da Costa & Friston 2019, "Markov blankets, information geometry and stochastic thermodynamics" *Phil. Trans. R. Soc. A* 378 (published 2019, found via Undermind 2026-04 Pillar 2)	*conceptual precursor* — supplies the boundary-and-nesting intuition; does not provide a coarse-graining error bound, a coordination-overhead theorem, or a strong-monotonicity condition for valid macro-agent treatment
Hierarchical-blanket nested agency with temporal depth	Kirchhoff, Parr, Palacios, Friston & Kiverstein 2018, "The Markov blankets of life" *J. R. Soc. Interface* 15 (published 2018, found 2026-04)	*conceptual precursor* — invokes synergetic ordering and self-organization at the compositional step but remains conceptual; no theorem bounding predictive loss of coarse-graining
Bound on control loss from compressed history	Subramanian, Sinha, Seraj & Mahajan 2020, "Approximate information state for approximate planning and reinforcement learning in partially observed systems" *arXiv:2010.08843* (published 2020, found 2026-04)	*closest mathematical neighbor* — proves a bound from predictive-compression error to control error in the same shape as the bridge lemma. ASF reuses the predictive-compression-to-control-loss mechanism and applies it at the coarser granularity of *agent boundary formation*, not single-agent state compression

ASF concern	*Prior-art language*	*Relationship / Positioning*
Loss bound from approximate influence-based abstraction	Congeduti, Mey & Oliehoek 2020, "Loss Bounds for Approximate Influence-Based Abstraction" *arXiv:2011.01788* (published 2020, found 2026-04)	*closest mathematical neighbor* — closer than the Markov-blanket line to coordination-overhead mathematics. ASF differs in framing: Congeduti bounds policy-value loss under approximate cross-agent influence; the bridge lemma bounds *macro-state trajectory error* under coarse-graining
Near-optimality under state aggregation	Abel, Hershkowitz & Littman 2016, "Near Optimal Behavior via Approximate State Abstraction" *ICML*; Taylor, Precup & Panangaden 2008, "Bounding Performance Loss in Approximate MDP Homomorphisms" *NeurIPS* (published 2016/2008, found 2026-04)	*adjacent literature* — single-agent MDP framing with aggregation bounds; same loss-from-coarse-graining mechanism, not framed as composite agency or organizational closure
Exact reduction of decentralized control via common information	Nayyar, Mahajan & Teneketzis 2012, "Decentralized Stochastic Control with Partial History Sharing" *IEEE TAC* 58 (published 2012, found 2026-04)	*adjacent literature* — names the formal space where a lossy composition theorem would live but proves an exact reduction, not a closure-defect bound
Approximate dynamical homomorphism / model reduction	Mori-Zwanzig projection (1965); balanced truncation (Moore 1981); Koopman-operator analysis (Koopman 1931; modern review Mezić 2005)	*formal antecedent* — supplies the approximate-dynamical-homomorphism framing of the closure defect. The MZ zero-lag bound surfaces in this segment's derivation table under stationarity. Adopted as standard machinery
Information bottleneck for the projection	Tishby, Pereira & Bialek 1999, "The information bottleneck method" *Allerton*	*formal antecedent* — admissibility condition (P1) is the IB Lagrangian-dual with source X_{micro} and relevance "next macro-observation given action"
Consistent abstractions of control systems	Pappas & Simic 2002, *Consistent abstractions of affine control systems* (IEEE TAC 47:745)[^cat-2026-05-22]	*formal antecedent* — control-theoretic foundation for defining macro-systems via surjective coarse-graining. The macro-dynamics admissibility condition (A1) is operationalizing exactly Pappas-Simic's surjectivity-of-coarse-graining requirement at the agent-state level
Compositional abstraction with explicit output-error bounds	Rungger & Zamani 2015, *Compositional Construction of Approximate Abstractions of Interconnected Control Systems* (IEEE TCNS 5:116)[^cat-2026-05-22]	*closest formal ancestor* — compositional abstraction with explicit output-error bounds is structurally the same shape as the closure-defect ε^* machinery this segment defines. The framework's bridge-lemma quantitative-error treatment is a refinement of the Rungger-Zamani approach applied at the agent-boundary-formation layer rather than the interconnected-control-system layer
Markov-chain lumpability	Buchholz 1994, *Exact and ordinary lumpability in finite Markov chains* (J. Appl. Probab. 31:59)[^cat-2026-05-22]	*adjacent literature* — Markov-aggregation ancestor of the commute-with-coarse-graining criterion underlying admissibility (P1)–(P3). The lumpability framework distinguishes *exact* (commutativity-of-projection-and-transition) from *ordinary* (statistical-equivalence-of-aggregated-states) — the former is the strong-monotonicity condition the bridge lemma uses

Search Log:

- 2026-04 (*nominally comprehensive*): Two-search Undermind pass over composite agency, approximate information states, Markov-blanket nested agency, decentralized control, state abstraction, and influence-based abstraction. Verdict: *Conceptual Precursor* (High confidence). Markov-blanket line (Parr 2019; Kirchhoff 2018) positioned as conceptual precursor on nested agency without bridge-lemma mathematics; approximate-information-states / state-abstraction / influence-based-abstraction line (Sub20; Con20; Abe16; Tay08; Nay12) positioned as closest mathematical neighbors with shared predictive-loss-to-control-error mechanism but applied to single-agent or already-decomposed control problems rather than to agent boundary formation.

- 2026-04 (*intuition-only*, prior to comprehensive search): adjacent literatures expected to host prior art were dynamical-systems model reduction (Mori-Zwanzig, balanced truncation), active-inference Markov-blanket work, and PAC-MDP state-abstraction. The comprehensive search confirmed all three families and added the influence-based-abstraction and decentralized-control-via-common-information lines that intuition did not surface.

13.2 *Derived (conditional):* Derived: Composite Tempo Inequality

The adaptive tempo of a composite agent is *bounded above* by the **joint tempo** of its sub-agents' pooled observation streams — not, in general, by the sum of their individual tempos. The gap between the joint potential and the realized composite tempo is the **coordination overhead** C_{coord} — tempo consumed by *internal reconciliation* rather than external mismatch correction. The relation between the joint baseline and the additive accounting $\sum_i \mathcal{T}_i$ is *signed*, inheriting `#deriv-tempo-additivity`'s two-sided deviation across agents: equal under cross-agent noise independence, smaller under common-source noise (the echo-chamber team), and **larger under anti-correlated noise — a composite can genuinely outrun the sum of its parts** (triangulating team), realizable at exactly zero closure defect by fusion as simple as averaging.

The intuition, corrected. Tempo is a rate of mismatch correction, and the composite's corrective capacity is set by the *information in its parts' pooled streams* — which is not the sum of the parts' individual capacities, because cross-agent noise structure is capacity no individual part has (anti-correlated errors cancel under fusion) or capacity the additive accounting double-counts (a shared upstream source). When closure defect is positive, sub-agents additionally spend tempo correcting *internal* mismatches generated by other sub-agents (Alice changes an API; Bob must update his model) or taking actions that counteract each other. That fraction is the coordination overhead and is *unavailable for external mismatch correction*.

The derivation connects closure defect to coordination overhead through *dimensional accounting*. The closure defect ε^* (units: distance) times the macro-update rate ν_c (units: inverse time) yields an internal disturbance rate matching the units of environmental disturbance. This internal mismatch must be corrected by the composite's correction capacity, and the correction consumes tempo that would otherwise address external mismatch. Conversion from disturbance rate to tempo-equivalent requires dividing by the task-adequacy distance scale $\|\delta_{\text{critical}}\|$ — mirroring the persistence-condition form used at the single-agent level. The composite therefore faces total *effective disturbance* combining external and closure-defect contributions, with coordination overhead bounded below by $C_{\text{coord}} \geq \varepsilon^* \nu_c / \|\delta_{\text{critical}}\|$.

This makes Brooks's Law precise. Adding sub-agents increases potential aggregate tempo $\sum \mathcal{T}_i$, but if it also increases closure defect (more coordination links to maintain), the realized composite tempo can *fall* — adding people to a late project makes it later when the closure-defect-driven tempo penalty from the additional links exceeds the new member's tempo contribution. The result is labeled *sketch* rather than exact because the closure-defect-to-coordination connection depends on the bridge lemma in 13.1 (which requires a contraction assumption on the macro-update map that is structurally motivated by (A4) but not formally derived from it). The remaining gap: whether the lower bound is tight or whether additional coordination costs (negotiation, synchronization, conflict resolution) contribute beyond the closure-defect mechanism.

Formal Expression.

This segment instantiates the sector-persistence template (`#result-sector-persistence-template`) at the composite level with state variable $\xi = \delta_c$ (composite mismatch) and a decomposed effective disturbance $\rho_{\text{eff}} = \rho_{\text{ext}} + \varepsilon^* \nu_c$ — external environment plus closure-defect contribution from 13.1's bridge lemma. The template supplies the Lyapunov machinery; this segment's distinctive content is the tempo-equivalent form of the coordination overhead lower bound $C_{\text{coord}} \geq \varepsilon^* \nu_c$.

Let $\mathcal{T}_i$ be the adaptive tempo of sub-agent i within a composite group of N agents. Let $\mathcal{T}_c$ be the adaptive tempo of the composite macro-agent A_c defined by an admissible coarse-graining Λ.

Let $\mathcal{T}_{\text{joint}}$ denote the joint tempo of the sub-agents' pooled observation channels — the exact object of `#deriv-tempo-additivity` ($U_c \cdot \mathbf{1}^\top \Sigma_n^{-1} \mathbf{1}$ in the Fisher-local regime, with Σ_n the pooled cross-agent noise covariance).

[Derived (Joint-Bounded Tempo, Conditional on the Fisher-local regime)] For any admissible composite agent A_c, the composite tempo is bounded by the joint tempo of its parts' pooled streams: $\mathcal{T}_c \leq \mathcal{T}_{\text{joint}}$ No admissible macro-dynamics extracts more corrective information per unit time than the pooled micro-observations carry (data processing at the Fisher level).

[Definition (Coordination Overhead)] The **coordination overhead penalty** C_{coord} is the difference between joint potential and realized macro-tempo: $C_{\text{coord}} := \mathcal{T}_{\text{joint}} - \mathcal{T}_c$

[Derived (Signed additive-baseline deviation, from `#deriv-tempo-additivity` *applied across agents)]* The additive accounting relates to the joint baseline with indefinite sign: $\mathcal{T}_{\text{joint}} \lessgtr \sum_{i=1}^{N} \mathcal{T}_i$, with equality iff the cross-agent noises are independent (plus the measure-zero cancellation set of `#deriv-tempo-additivity`), $<$ under common-source cross-agent noise, and $>$ under noise-cancelling (anti-correlated) structure. The historical form of this segment's claim, $\mathcal{T}_c \leq \sum_i \mathcal{T}_i$, is therefore **conditional on cross-agent noise independence** and false in general — with a closure-exact witness:

[Derived (Super-additive witness, exact)] Two sub-agents observing $o_i = \theta + n_i$ with $\text{Var}(n_i) = r$ and $\text{Corr}(n_1, n_2) = \rho < 0$, each running its own optimal filter; composite defined by the averaging projection $\Lambda_x = \frac{1}{2}(m_1 + m_2)$, macro-observation $\bar{o} = \frac{1}{2}(o_1 + o_2)$, and the recursive running-mean macro-update. The macro-update commutes with the micro-dynamics *identically* ($\varepsilon^* = 0$ exactly; (A1)–(A4) and (P1)–(P3) verified), and $\frac{\mathcal{T}_c}{\sum_i \mathcal{T}_i} = \frac{1}{1+\rho} > 1$, unboundedly as $\rho \to -1$ — tempos evaluated in the matched-uncertainty (Fisher-inflow) convention of `#deriv-tempo-additivity`; the convention is load-bearing (Epistemic Status). A faithful composite of a triangulating team outruns the sum of its parts; symmetrically, at $\rho > 0$ the same witness has $\mathcal{T}_c < \sum_i \mathcal{T}_i$ at zero closure defect and zero coordination activity — the additive baseline, not the composite, is what fails.

Epistemic Status.

Conditional. Max attainable: exact. Component strengths: the joint bound $\mathcal{T}_c \leq \mathcal{T}_{\text{joint}}$ is Fisher-level data processing, exact within the first-order tempo–Fisher regime of `#deriv-tempo-additivity` ((R1)–(R3) of `#deriv-fisher-local-update-gain` — the regime where tempo is a rate); the signed additive-baseline deviation is `#deriv-tempo-additivity`'s theorem applied across agents (same tier); the super-additive witness is *exact* — commutation is an identity, and the tempo ratio $1/(1+\rho)$ is closed-form, numerically confirmed (Working Notes). The ratio is stated in the matched-uncertainty (Fisher-inflow) convention — comparing information inflow at equal uncertainty, as throughout `#deriv-tempo-additivity`; the convention is load-bearing because static estimation makes literal own-trajectory fractional-contraction tempo estimator-blind. Independent verification probed convention-robustness: under dynamic θ (random-walk environment, each filter at its own steady state — the own-uncertainty reading) the witness's ratio becomes $\sqrt{1/(2(1+\rho))}$ with super-additivity threshold $\rho < -1/2$ rather than $\rho < 0$ — the refutation of the additive bound survives every tested reading; only the exact ratio and threshold are convention-tied. The historical sub-additive claim $\mathcal{T}_c \leq \sum_i \mathcal{T}_i$ survives only on the cross-agent noise-independence scope; its former grounding ("composition cannot create corrective capacity out of nothing") was refuted — cross-agent noise structure *is* corrective capacity that no individual part has, and the additive accounting cannot see it. The $\varepsilon^* \to C_{\text{coord}}$ connection remains *sketch-grade*: it depends on the bridge lemma in 13.1 (contraction assumption DA2'a-inc, structurally motivated by (A4) but not derived from it), and whether the lower bound is tight — or additional process costs (negotiation, synchronization, conflict resolution) contribute beyond the closure-defect mechanism — is open.

Intuition. Tempo is a rate of mismatch correction. When $\varepsilon^* = 0$ (exact closure) and the composite's fusion is information-lossless, $\mathcal{T}_c = \mathcal{T}_{\text{joint}}$ — and $\mathcal{T}_{\text{joint}}$ coincides with $\sum \mathcal{T}_i$ exactly when cross-agent noises are independent. When $\varepsilon^* > 0$, sub-agents spend part of their tempo correcting *internal* mismatches generated by other sub-agents (Alice changes an API; Bob must update his M_t) or taking actions that counteract each other. That fraction is C_{coord} and is unavailable for external mismatch correction.

Derived (coordination-overhead-bound, from composition-closure bridge lemma)

The closure-defect to coordination-overhead connection.

The closure defect ε^* introduces internal mismatch at rate $\varepsilon^* \nu_c$ — units of [distance] · [time^{-1}], a disturbance rate matching the units of environmental ρ. This internal mismatch must be corrected by the macro-agent's correction capacity, and the correction consumes tempo that would otherwise address external mismatch.

Dimensional accounting. Three quantities appear in the composite persistence analysis, each with its own units:

Table 13.3

Quantity	*Units*	*Role*
Tempo $\mathcal{T}, \mathcal{T}_c, \mathcal{T}_c^{\text{ext}}, C_{\text{coord}}$	[time^{-1}]	Rate of mismatch correction
Disturbance rate ρ, ρ_{eff}	[distance] · [time^{-1}]	Rate of mismatch injection
Closure defect ε^*	[distance]	Per-step norm error
Critical scale $\lVert\delta_{\text{critical}}\rVert$	[distance]	Task-adequacy boundary

The conversion between a disturbance rate and a tempo-equivalent requires division by a distance scale, mirroring the persistence condition form $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$.

Effective disturbance. The macro-agent faces total effective disturbance combining external and closure-defect contributions:

$$\rho_{\text{eff}} = \rho_{\text{ext}} + \varepsilon^* \nu_c \quad \text{(units: [distance]} \cdot \text{[time}^{-1}\text{])} \tag{13.24}$$

Coordination overhead (tempo-equivalent). The fraction of macro-tempo consumed by closure correction, expressed as a tempo:

Derived (coordination-overhead-tempo-form)

$$C_{\text{coord}} \;\geq\; \frac{\varepsilon^* \nu_c}{\|\delta_{\text{critical}}\|} \quad \text{(units: [time}^{-1}\text{])} \tag{13.25}$$

The closure error rate is converted to a tempo penalty by normalizing against the task-relevant distance scale. The inequality is a lower bound because additional coordination costs (negotiation, synchronization, conflict resolution — see Working Notes) contribute beyond the closure-defect mechanism. When the macro-agent's correction of internal mismatch is less efficient than its correction of external mismatch (because internal mismatches may be harder to diagnose — they look like environmental change from the macro perspective), the actual overhead exceeds this bound.

Consequences (dimensionally correct):

- The realized external tempo $\mathcal{T}_c^{\text{ext}}$ (the share of composite tempo directed at *external* mismatch) chains under the joint baseline — the coordination-overhead definition plus the bridge-lemma lower bound give the middle inequality, and externally-directed tempo cannot exceed total tempo: $\mathcal{T}_c^{\text{ext}} \;\leq\; \mathcal{T}_c \;\leq\; \mathcal{T}_{\text{joint}} - \frac{\varepsilon^* \nu_c}{\|\delta_{\text{critical}}\|} \qquad \left(= \sum_i \mathcal{T}_i - \varepsilon^* \nu_c / \|\delta_{\text{critical}}\| \;\text{ under cross-agent noise independence}\right)$
- **Brooks's Law.** Adding an agent increases $\mathcal{T}_{\text{joint}}$ (by its marginal joint contribution — which cross-agent correlation can make much smaller, or larger, than its solo tempo) but may also increase ε^*. The turning point — where adding a member lowers realized external tempo — occurs when: $\frac{\Delta \varepsilon^* \nu_c}{\|\delta_{\text{critical}}\|} > \Delta \mathcal{T}_i$ The closure-defect increase (expressed as tempo penalty) exceeds the new member's tempo contribution.
- **Composite persistence condition.** The composite persists iff $\mathcal{T}_c > \rho_{\text{eff}}/\|\delta_{\text{critical}}\|$; a *necessary* condition in joint-baseline terms is therefore $\mathcal{T}_{\text{joint}} > \frac{\rho_{\text{ext}} + \varepsilon^* \nu_c}{\|\delta_{\text{critical}}\|}$ which separates internal-coordination overhead from external challenge while keeping tempo units on the LHS and disturbance-rate units on the RHS (it reduces to the historical $\sum_i \mathcal{T}_i$ form exactly under cross-agent noise independence).

Discussion.

Equality conditions. $\mathcal{T}_c = \mathcal{T}_{\text{joint}}$ (i.e. $C_{\text{coord}} = 0$) requires: 1. **Orthogonal Routing:** The information dependency DAG exactly matches organizational boundaries (no costly cross-talk required). 2. **Perfect Shared Intent:** No tempo is lost to internal negotiation or conflicting pursuit of the macro-objective. 3. **Lossless Fusion:** No joint information in the pooled streams is discarded by the composite's aggregation.

These are stated as sufficient conditions; whether they are also necessary is open. The further equality $\mathcal{T}_{\text{joint}} = \sum_i \mathcal{T}_i$ is a *separate* condition — cross-agent noise independence — and its two-sided failure is what retired this segment's historical additive form: under common-source noise a zero-defect, zero-overhead composite still shows $\mathcal{T}_c < \sum_i \mathcal{T}_i$ (the additive baseline is inflated), and under anti-correlated noise a zero-defect composite shows $\mathcal{T}_c > \sum_i \mathcal{T}_i$ (triangulation). The old condition 3 ("no net macro-information loss") implicitly carried both the fusion condition and the independence condition; they are now separated.

Brooks's Law. Adding more agents (increasing $\mathcal{T}_{\text{joint}}$ by their marginal joint contribution — which cross-agent correlation can shrink toward zero or amplify) only increases the composite tempo $\mathcal{T}_c$ if the corresponding increase in closure defect ε^* doesn't let C_{coord} dominate. The model provides a formal analog of Brooks's Law: if coordination overhead (C_{coord}) grows faster than joint capability ($\mathcal{T}_{\text{joint}}$) when adding agents, then adding people to a late project makes it later — and a new member whose observation noise echoes the existing team's contributes almost nothing to $\mathcal{T}_{\text{joint}}$ while still adding coordination links, the sharpest Brooks case. Whether this specific mechanism (coordination overhead consuming tempo) is the dominant cause in practice is an empirical question.

Connection to 15.1. The persistence condition for the composite agent requires $\mathcal{T}_c > \rho_{\text{ext}}/\|\delta_{\text{critical}}\|$. Since $\mathcal{T}_c = \mathcal{T}_{\text{joint}} - C_{\text{coord}}$, high coordination overhead can push the composite below the persistence threshold, causing it to disintegrate as a coherent entity — even though each sub-agent individually persists.

Wrapping construction as a Brooks's-Law instance. Class coercion of a Class 2 (Partial) or Class 3 (Coupled) component via wrapping (`#der-class-coercion-via-wrapping`) is a concrete instance of the same coordination-overhead form. A wrapper makes $K \geq 2$ component calls per macro-step (one for each goal-blind belief-update query, one for the goal-conditioned strategy-update query, more in richer wrapper designs). The wrapper-level macro-update rate is $\nu_W = \nu_A/K$ where ν_A is the underlying component's nominal call rate, so the wrapper-level tempo is $\mathcal{T}_W \leq \mathcal{T}_A^{\text{nominal}} - C_{\text{coord}}^{\text{wrap}}$, with the coordination overhead $C_{\text{coord}}^{\text{wrap}}$ scaling with K. The cost of class coercion — the structural enforcement of directed separation at the wrapper level when the underlying component does not provide it — is paid in macro-tempo. Strict-wrapping (W_1) regimes have $K \geq 2$ minimum; partial-wrapping (W_2) regimes have $K = 1$ for the LLM call plus some parsing overhead, but the structural-separation guarantee is correspondingly weaker (see `#der-class-coercion-via-wrapping`).

13.3 *Hypothesis:* Directed Separation Under Composition

The question whether the composite-level directed-separation condition holds: if every sub-agent satisfies 6.3 individually, does the composite macro-agent also satisfy it? The answer is *conditional on the composite's routing structure* (12.1).

The framework draws a careful distinction first. Sub-agents' goal-driven actions shape the environment, which other sub-agents observe. This is **not** a violation of directed separation — it is the same mechanism the single-agent result explicitly allows: goals influence *events* through action, but processing of realized events is goal-blind. Agent B observing environmental changes caused by Agent A's goal-driven behavior is B processing a realized event goal-blindly. The observations carry information about A's goals, but that is a property of the event's content, not a failure of goal-blind processing. Similarly, individual messages reflecting individual goals is action through policy, not a routing-structure dependence on the composite's goals.

The substantive question is whether the **routing structure** depends on the composite's goal. Two cases. **Case 1 — goal-blind routing**: when neither communication topology nor protocol depends on the composite's goals (military command structures with doctrinal communication protocols; software teams with defined code-review processes; multi-agent AI systems with protocol-specified message passing), directed separation *survives* at the composite level. **Case 2 — goal-dependent routing**: when the topology changes based on objective (different reporting chains activated depending on mission) or the protocol changes (different intelligence products shared depending on objective), the composite's effective observation function acquires a goal argument. Even if each sub-agent processes its observations goal-blindly, *the set of observations reaching each sub-agent depends on the composite's goal through the routing function*. Directed separation *fails* at the composite level.

This refinement establishes that **composite-level class membership depends on routing structure, not just sub-agent class**. A composite of Class-1-(Separated) sub-agents with goal-blind routing remains Class 1 at the composite level. A composite of Class-1-(Separated) sub-agents with goal-*dependent* routing becomes effectively Class 3 (Coupled) at the composite level — and the coupled-formulation machinery applies at the macro level even though every sub-agent individually satisfies directed separation. The architectural classification is therefore *not preserved under composition with goal-dependent routing*. For the special case of a single-component composite (wrapper-around-component), 13.4 promotes the equivalent result from hypothesis to derived; the general N-agent case treated here remains a hypothesis.

Formal Expression.

Hypothesis (directed-separation-under-composition, extending directed-separation to composites)

Consider N agents $A_1, \ldots, A_N$, each satisfying directed separation individually: $M_{i,\tau^+} = f_M^{(i)}(M_{i,\tau^-}, e_{i,\tau})$ with no $G_t^{(i)}$ argument.

The question. Directed separation is about **processing**, not **selection** (6.3, scope condition). A single agent's goals affect which events it seeks (through π), but f_M processes whatever event arrives without reference to G_t. At the composite level, the analogous question is: does the composite's routing structure R_t (12.1) depend on the composite's goals G_t^c?

Note: sub-agents' goal-driven actions shape the environment, which other sub-agents observe. This is NOT a violation of directed separation — it is the same mechanism directed separation explicitly allows at the single-agent level: goals influence events through action, but processing of realized events is goal-blind. Agent B observing environmental changes caused by A's goal-driven behavior is B processing a realized event goal-blindly. The observations carry information about A's goals, but that is a property of the event's content, not a failure of goal-blind processing. Similarly, individual messages reflecting individual agents' goals is action through policy, not a routing-structure dependence on G_t^c.

Case 1: Goal-blind routing.

Hypothesis (Case 1)

If the routing structure satisfies $R_t \perp G_t^c$ (12.1, goal-blind routing) — neither the communication topology $\mathcal{N}_t$ nor the protocol $c_t^{(j \to i)}$ depends on the composite's goals — then:

- Each sub-agent processes observations goal-blindly (individual directed separation)
- The routing is goal-blind (by construction)
- Therefore f_M^c is G_t^c-independent

Directed separation **survives** at the composite level. Examples: military command structures with doctrinal communication protocols, software teams with defined code-review processes, multi-agent AI systems with protocol-specified message passing.

Case 2: Goal-dependent routing.

Hypothesis (Case 2)

If R_t depends on G_t^c — either the topology $\mathcal{N}_t$ changes (different reporting chains activated depending on the mission) or the protocol $c_t^{(j \to i)}$ changes (different intelligence products shared depending on the objective) — then the composite's effective observation function has a goal argument:

$$o_c = h^c(\Omega, a_{\text{micro}}, G_t^c, \xi) \tag{13.26}$$

Even if each sub-agent's $f_M^{(i)}$ processes o_i goal-blindly, the **set of observations reaching each sub-agent** depends on G_t^c through the routing function. Directed separation **fails** at the composite level. The composite is analogous to a Class 3 (Coupled) architecture: goal content shapes the information pathway, not through individual interpretation but through collective routing.

Epistemic Status.

Conditional on the routing structure of the composite. Max attainable: conditional (the two cases are genuine architectural alternatives). Previously discussion-grade; upgraded after routing formalization in 12.1 and architectural classification promotion in 6.3 (2026-04-01).

Foundation status.

1. The **architectural classification** (Class 1/2/3) is in 6.3's Formal Expression with formal operationalization ($\kappa_{\text{processing}}$). Status: robust qualitative.
2. The **routing structure** R_t is defined in 12.1 with a formal goal-independence condition ($R_t \perp G_t^c$). The definition decomposes into topology independence and protocol independence. Whether this captures all relevant ways that composite information flow can depend on goals is an open question.
3. The **admissible coarse-graining** Λ from 13.1 now has specified admissibility constraints: (A1)-(A4) for macro-dynamics and (P1)-(P3) for projections. The remaining gaps are validation (exercising the machinery on a purposeful multi-agent example) and computability (P1 requires conditional mutual information, tractable only for linear-Gaussian), not missing definitions.

The logic of each case is sound given the routing definition. Case 1: goal-blind routing + goal-blind processing = goal-blind composite. Case 2: goal-dependent routing means the composite's observation function depends on G_t^c, regardless of individual processing. Both follow from directed separation's scope condition.

The claim that the architectural classification lifts to composition is **structurally motivated** by 11 (the theory must give consistent answers at every level of description). This is an argument from theoretical coherence, not a derivation.

Discussion.

What this is and what it isn't. This segment provides a two-case taxonomy for directed separation under composition. It is not yet a derivation — the remaining caveat (admissibility placeholders) prevents that. It IS a useful classification that identifies which composites fall within Part III's scope (Case 1) and which require a coupled formulation (Case 2).

Constructive special case: wrapper-around-component. For the special case of a single-component composite — one underlying primitive component embedded inside an external scaffold — the question of when directed separation holds is *derived* by `#der-class-coercion-via-wrapping`. Under explicit conditions on the component (admissibility of goal-blind queries, stationary conditional, no implicit goal-inference) and on the wrapper's update maps (type signatures that exclude G_W from the belief-update path), directed separation holds at the wrapper level by structural commitment. The construction promotes the wrapper-around-component case from hypothesis to derived; the general N-agent composition case (Case 1 vs. Case 2 above) remains a hypothesis pending further work.

Most composites of interest are Case 1. Fixed communication structures are the norm in designed multi-agent systems: military doctrine specifies information flow regardless of the current mission; software development processes define code review, CI, and deployment pipelines independent of the feature being built; multi-agent AI protocols specify message formats and channels. Goal-dependent routing (Case 2) is rarer: crisis management (where communication structure changes with threat type), ad hoc teams assembled for specific objectives, and attention-based multi-agent architectures where query routing depends on the shared goal.

Connection to logogenic agents. LLM-based agents are individually Class 3 (Coupled; goal-conditioned attention). Whether composites of LLM agents are Case 1 or Case 2 depends on whether the inter-agent communication protocol is fixed or goal-dependent. A multi-agent LLM system with fixed API contracts between agents is Case 1 at the composite level even though each agent is individually Class 3 (Coupled). The internal architecture of each agent and the composition architecture are separate questions.

Adversarial implications. Goal-dependent routing (Case 2) makes the composite's objective partially observable through its communication patterns. An adversary who can observe which information is being routed where can infer the composite's current goal. This creates a meta-strategic incentive for fixed routing: it preserves not just epistemic hygiene but operational security.

13.4 *Derived:* Class Coercion via Wrapping

A *constructive* result of substantial practical importance. A Class 2 (Partial) or Class 3 (Coupled) component (one whose forward pass entangles belief-update and goal-conditioning) can be embedded inside an external scaffold whose state $X_W = (M_W, G_W)$ is updated by *structurally distinct query channels*: **goal-blind queries** to the component update M_W; **goal-conditioned queries** update G_W. Under stated conditions on the component, directed separation holds at the wrapper level *by construction*, and the composite system is Class 1 (Separated) per `#der-directed-separation` — even though the underlying component is not. This is the constructive direction of `#hyp-directed-separation-under-composition` for the wrapper-around-component special case: a procedure for *making* directed separation hold when the underlying component does not provide it.

The wrapper has four type-signed components: a *belief-side query selector* that chooses the model-update query from belief and observation only — *no goal argument*; a *strategy-side query selector* that may depend on the goal; a *belief-update map* that updates M_W from prior belief, observation, the query made, and the component's response — *no goal argument*; and a *strategy-update map* that may depend on the goal. The wrapper makes at least two component calls per macro-step: one goal-blind for the model update and one goal-conditioned for the purposeful-state update.

The result requires conditions on the component. **(C1) Goal-blind admissibility** — the component admits non-trivial goal-blind queries; the framework partitions components into *Class A* (goal-blind by design: POMDP belief-state filters, world models, sensory pipelines, retrieval systems), *Class B* (admit a goal-blind query mode alongside goal-conditioned ones: LLMs in summarization/fact-extraction modes, hybrid RL with separable value/policy), and *Class C* (fundamentally goal-conditioned: pure end-to-end goal-conditioned policy networks), with the construction applying to Classes A and B. **(C2) Stationary component conditional** — the component's output distribution conditional on input is fixed during operation (adaptation-during-deployment systems are out of scope). **(C2′) No goal-correlated cross-call state** — the component carries no information about the latent operator goal across the goal-blind / goal-conditioned call boundary; this is the condition on which the *structural* W_1 leakage bound depends. **(C3) No implicit goal-inference** — the component's response to a goal-blind query does not depend on G_W via inference from query patterns; for pretrained components like LLMs, (C3) holds *exactly* only when query content is statistically independent of G_W in the pretraining distribution. The framework supplies **two theorems**: an *exact* form under (C1)–(C3), and an *approximate* form under (C1)–(C2) plus a KL-leakage bound, where the wrapper-level leakage on M_W updates is bounded by the same bound (small leakage in, small leakage out via the data-processing inequality).

The construction supports three **wrapping regimes** of decreasing strictness, distinguished by where structural separation lives. $\mathbf{W_1}$ **(strict wrapping)** uses separate q_M and q_G calls per macro-step; separation lives at the *query boundary* and the residual leakage is the *selection-channel* quantity $\kappa_{W_1}^{\mathrm{sel}} = I(A(q_M); G^{\mathrm{op}}) \leq I(q_M; G^{\mathrm{op}})$ — bounded *structurally* by the goal-content of the wrapper's query-selection policy (under (C1)+(C2′)). $\mathbf{W_2}$ **(partial wrapping)** uses one goal-conditioned call per macro-step with a typed parsed response routing updates to M_W vs G_W slots; separation lives at the *write boundary* and leakage is bounded only *behaviorally* — by the component's compliance with the prompted instruction-to-separate. $\mathbf{W_0}$ **(no wrapping)** runs the raw Class 2 / Class 3 component with no separation commitment. What differs across the regimes is *what determines* the leakage rate — and, for W_1, whether the structural bound is available at all: it requires (C2′), failing which W_1 too degrades to a behavioral bound (`#disc-w1-structural-bound-boundary`). The hierarchy refines the Class 1 (Separated) cell of `#der-directed-separation` with a *Class-1-by-structure* (W_1 or natively goal-blind) vs *Class-1-by-behavior* (W_2) sub-distinction.

The result is *load-bearing* for the framework's treatment of LLM agents: an LLM is internally Class 3 (Coupled), but an LLM-agent *system* (LLM + tools + memory + monitoring) can be designed with modular topology that recovers Class 1 status at the system level. The construction's cost is paid in two places — *more component calls per macro-step* (the Brooks's-Law tempo overhead derived in the companion segment) and a *residual leakage rate* bounded structurally under W_1 or behaviorally under W_2. The companion segment `#der-class-coercion-in-composition` establishes that the wrapped system is also a valid AAT composite agent (satisfying (A1)–(A4) of `#form-composition-closure`) and inherits the sector-persistence template at the wrapper level.

Formal Expression.

Setup. Let $A : \mathcal{I}_A \to \mathcal{O}_A$ be a primitive component, treated by the wrapper as a black-box oracle: the wrapper issues queries (inputs) and consumes responses (outputs), without access to A's internal state. $\mathcal{Q}_A \subseteq \mathcal{I}_A$ is the set of admissible queries.

A **wrapper** W over A has state $X_W = (M_W, G_W) \in \mathcal{X}_M \times \mathcal{X}_G$ with $\mathcal{X}_G = \mathcal{X}_O \times \mathcal{X}_\Sigma$ per `#def-strategy-dimension`. The wrapper interacts with an environment via observations $o_W \in \mathcal{O}_W$ and actions $a_W \in \mathcal{A}_W$.

[Definition (wrapper-update-maps)] The wrapper's update at macro-step m uses four type-signed components:

- **Belief-side query selector:** $q_M : \mathcal{X}_M \times \mathcal{O}_W \to \mathcal{Q}_A$. The wrapper chooses the query for M_W updates from belief and observation only — *no G_W argument*.
- **Strategy-side query selector:** $q_G : \mathcal{X}_M \times \mathcal{X}_G \to \mathcal{Q}_A$. May depend on G_W.
- **Belief-update map:** $f_M : \mathcal{X}_M \times \mathcal{O}_W \times \mathcal{Q}_A \times \mathcal{O}_A \to \mathcal{X}_M$. Updates M_W from prior belief, observation, the query made, and the component's response. *No G_W argument.*
- **Strategy-update map:** $f_G : \mathcal{X}_G \times \mathcal{X}_M \times \mathcal{Q}_A \times \mathcal{O}_A \to \mathcal{X}_G$. May depend on G_W.

The external policy $\pi_W : \mathcal{X}_W \to \mathcal{A}_W$ selects the wrapper's external action.

A macro-step proceeds: construct $q_M(M_W, o_W)$ → query A → apply f_M; construct $q_G(M'_W, G_W)$ → query A → apply f_G; emit $\pi_W(X'_W)$. The wrapper makes $K \geq 2$ component calls per macro-step in this minimal form (more in richer wrapper designs).

Conditions. *[Conditions (component-admissibility)]* The theorem applies under three conditions on the component A:

(C1) Goal-blind admissibility. $\mathcal{Q}_A$ contains queries whose specification can be constructed from (M_W, o_W) alone — i.e., a non-trivial q_M exists. Components partition into three classes: - **Class A (goal-blind by design).** A's interface is goal-blind by construction — POMDP belief-state filters, world models, sensory pipelines, retrieval systems, calculators. (C1) holds trivially. - **Class B (admit a goal-blind query mode).** A supports goal-conditioned queries but also goal-blind ones. Large language models in summarization or fact-extraction modes; hybrid RL agents with separable value/policy; multi-modal models. (C1) holds operationally — the wrapper *chooses* to use the goal-blind mode. - **Class C (fundamentally goal-conditioned).** A's only operating mode requires goal-conditioning. Pure end-to-end goal-conditioned policy networks. (C1) fails; the construction does not apply.

(C2) Stationary component conditional. A's output distribution conditional on input is fixed during the wrapper's operation: $P(A(\cdot) \mid q)$ does not depend on prior queries or on side information beyond q. Adaptation-during-deployment systems are out of scope.

(C2′) No goal-correlated cross-call state. The component's hidden state does not carry information about the latent operator goal G^{op} across the boundary between the goal-blind (q_M) and goal-conditioned (q_G) calls of a macro-step. Equivalently, A's response to q_M is conditionally independent of G^{op} given q_M and the component's *pre-call* state, and that pre-call state is itself goal-uncorrelated — either reset across the call boundary, or stripped of Σ- and G-content. (C2′) is the condition on which the *structural* W_1 leakage bound below depends, and it strengthens (C2): the breaking condition for the structural bound is not online weight-adaptation but mere goal-correlated state persistence across the call boundary — which a frozen-weights LLM with a conversation cache exhibits with no weight adaptation at all. When (C2′) fails the structural bound is unavailable and only a behavioral bound remains; this boundary is derived in `#disc-w1-structural-bound-boundary`.

(C3) No implicit goal-inference. A's response to a goal-blind query does not depend on G_W via inference from query patterns:

$$P(A(q_M) \mid q_M, G_W) = P(A(q_M) \mid q_M) \quad \forall\, q_M, G_W \tag{13.27}$$

For pretrained components (notably LLMs), (C3) holds *exactly* only when query content is statistically independent of G_W in the pretraining distribution. The approximate form weakens (C3) to a leakage bound (Theorem 2 below).

Theorem 1: Directed separation at the wrapper level (exact form).

Derived (directed-separation-at-wrapper-exact, from C1+C2+C3)

Under (C1)–(C3), directed separation holds *exactly* at the wrapper level:

$$P(M_{W,m+1} \mid M_{W,m},\ o_{W,m+1},\ G_{W,m}) = P(M_{W,m+1} \mid M_{W,m},\ o_{W,m+1}) \tag{13.28}$$

Therefore W is a Class 1 (Separated) architecture per `#der-directed-separation`.

Proof. Identify all paths from $G_{W,m}$ to $M_{W,m+1}$ given $(M_{W,m}, o_{W,m+1})$. The update is

$$M_{W,m+1} = f_M\big(M_{W,m},\ o_{W,m+1},\ q_M(M_{W,m}, o_{W,m+1}),\ A(q_M(M_{W,m}, o_{W,m+1}))\big) \tag{13.29}$$

f_M has no G_W argument by type signature (D-pathway-1 closed). q_M has no G_W argument by type signature (D-pathway-2 closed). The remaining pathway is $A(q_M)$ depending on G_W given q_M. Under (C3), $P(A(q_M) \mid q_M, G_W) = P(A(q_M) \mid q_M)$ — the response is conditionally independent of G_W given q_M. Since q_M is itself a deterministic function of $(M_{W,m}, o_{W,m+1})$, conditioning on $(M_{W,m}, o_{W,m+1})$ determines q_M, and the integrand $P(M_{W,m+1} \mid M_{W,m}, o_{W,m+1}, q_M, A(q_M)) \cdot P(A(q_M) \mid q_M, G_W)$ no longer depends on G_W. The conditional distribution of $M_{W,m+1}$ given $(M_{W,m}, o_{W,m+1}, G_{W,m})$ equals that given $(M_{W,m}, o_{W,m+1})$. ∎

Theorem 2: Directed separation (approximate form, C3 weakened to leakage bound).

Derived (directed-separation-at-wrapper-approximate, from C1+C2+leakage-bound)

If (C3) is replaced by a KL-leakage bound

$$D_{\mathrm{KL}}\Big(P(A(q_M) \mid q_M, G_W)\ \Big\|\ P(A(q_M) \mid q_M)\Big) \le \kappa \quad \forall\, q_M, G_W \tag{13.30}$$

then the wrapper-level KL-divergence on M_W updates is bounded by the same κ:

$$D_{\mathrm{KL}}\Big(P(M_{W,m+1} \mid M_{W,m}, o_{W,m+1}, G_{W,m})\ \Big\|\ P(M_{W,m+1} \mid M_{W,m}, o_{W,m+1})\Big) \le \kappa \tag{13.31}$$

The wrapper is *almost-Class-1 (Separated)* with leakage rate $\le \kappa$. *Proof.* The wrapper-level M_W update is a deterministic function of the component response given the wrapper's other inputs; the data-processing inequality propagates the KL bound from response distribution to wrapper-state distribution. ∎

The W_1 structural leakage bound: a selection-channel quantity. The residual goal-leakage that survives in the W_1 regime is a *selection-channel* quantity, not a processing-channel one. The distinction is between two senses of the goal variable: the wrapper's internal purposeful-state register G_W — which q_M structurally does not take as an argument and which Theorem 1 closes by type signature — and the latent *operator goal* G^{op} that generated the situation the wrapper is in, which a competent component can infer from the *content* of the query the wrapper chose. The processing channel $(G_W \to A(q_M) \mid q_M)$ is shut structurally and exactly by Theorem 1; the entire residual leakage is the selection channel ($G^{\mathrm{op}} \to$ wrapper history $\to$ choice of $q_M \to A(q_M)$), and its magnitude is governed by the goal-content carried in the query the wrapper selects.

Derived (W1-selection-leakage-bound, from C1+C2′, data-processing inequality)

Under (C1) and (C2′), the goal-information reaching the wrapper's belief update through the goal-blind channel is bounded by the goal-content of the wrapper's query-selection policy:

$$\kappa^{\mathrm{sel}}_{W_1} \ :=\ I\big(A(q_M);\ G^{\mathrm{op}}\big)\ \le\ I\big(q_M;\ G^{\mathrm{op}}\big), \tag{13.32}$$

where G^{op} is the latent operator goal and q_M is the (random, history-dependent) goal-blind query drawn under the wrapper's operating policy.

Derivation. Under (C2′), $A(q_M)$ depends on G^{op} only through q_M — there is no goal-correlated cross-call state through which the goal could reach the response while bypassing the query — so $G^{\mathrm{op}} \to q_M \to A(q_M)$ is a Markov chain. The data-processing inequality along this chain gives $I(A(q_M); G^{\mathrm{op}}) \leq I(q_M; G^{\mathrm{op}})$. ∎

The bound is a property of the wrapper's q_M-selection policy, not of the component's input–output law, and is therefore design-controllable: maximally goal-blind queries (current observation only, no history, no system prompt) drive $I(q_M; G^{\mathrm{op}}) \to 0$ and recover exact directed separation on the merits; richer, history-laden queries raise $I(q_M; G^{\mathrm{op}})$ and with it the leakage ceiling. This is the formal content of the quality–separation tradeoff (§Discussion) — a choice of query-selection policy purchases a quantitative, structurally-derived leakage ceiling.

Propagating to the wrapper's belief state: the update $M_{W,m+1}$ is a deterministic function of $(M_{W,m}, o_{W,m+1}, q_M, A(q_M))$, so along $G^{\mathrm{op}} \to q_M \to (q_M, A(q_M)) \to M_{W,m+1}$ the data-processing inequality gives

$$I\big(M_{W,m+1};\, G^{\mathrm{op}} \,|\, M_{W,m},\, o_{W,m+1}\big) \;\leq\; I\big(A(q_M);\, G^{\mathrm{op}} \,|\, M_{W,m},\, o_{W,m+1}\big) \;\leq\; I\big(q_M;\, G^{\mathrm{op}}\big). \tag{13.33}$$

The conditioning set is $(M_{W,m}, o_{W,m+1})$ — deliberately *not* q_M. Conditioning on q_M would (correctly, for the closed processing path) zero the quantity out and miss the selection leak; the unconditional $\kappa^{\mathrm{sel}}_{W_1} = I(A(q_M); G^{\mathrm{op}}) \leq I(q_M; G^{\mathrm{op}})$ is the analyst-independent statement, with q_M drawn under the wrapper's actual operating policy.

Wrapping regime hierarchy. The construction supports three regimes, distinguished by where structural separation lives:

Table 13.4

Regime	*Construction*	*Leakage bound*	*Leakage source*
$\mathbf{W_0}$ (no wrapping)	Raw Class 2 (Partial) or Class 3 (Coupled) component	κ_{W_0} at the component's maximum goal-conditioning sensitivity	No constraint
$\mathbf{W_2}$ (partial wrapping)	One goal-conditioned call per macro-step; structurally typed parsed response routes updates to M_W vs. G_W slots	κ_{W_2} bounded *behaviorally* — by the component's compliance with the prompted instruction-to-separate; **no structural bound**	Component's instruction-following fidelity
$\mathbf{W_1}$ (strict wrapping)	Theorem 1 / 2 — separate q_M and q_G calls per macro-step	$\kappa^{\mathrm{sel}}_{W_1} = I(A(q_M); G^{\mathrm{op}}) \leq I(q_M; G^{\mathrm{op}})$ — bounded *structurally* by the goal-content of the wrapper's query-selection policy (under (C1)+(C2′))	Query-selection policy's goal-content (the goal G^{op} inferable from query content)

W_1 admits a structural bound — the selection-channel quantity $\kappa^{\mathrm{sel}}_{W_1} = I(A(q_M); G^{\mathrm{op}}) \leq I(q_M; G^{\mathrm{op}})$ under (C1)+(C2′); W_2 admits only a behavioral bound from the component's compliance fidelity. The two are different in kind — the structural bound is a property of the wrapper's query-selection policy (the goal-content of the queries it chooses) and is therefore design-controllable; the behavioral bound depends on the component's training and prompt-following. The W_1 structural bound is available exactly when the component carries no goal-correlated cross-call state ((C2′)); when (C2′) fails the leak routes through an unobservable channel and W_1 too degrades to a behavioral bound — the boundary derived in `#disc-w1-structural-bound-boundary`.

The W_0 / W_2 / W_1 distinction refines the Class 1 (Separated) cell of `#der-directed-separation`: within Class 1 (Separated), **Class-1-by-structure** (natively goal-blind components, or W_1 wrapping) has a structurally derivable directed-separation guarantee; **Class-1-by-behavior** (W_2 wrapping) has only an empirically estimable guarantee that depends on the component's instruction-following.

Epistemic Status.

Conditional on (C1), (C2), and (C3) (or its weakening to a leakage bound). The proofs are short conditional-independence reasoning (Theorem 1) and a single application of the data-processing inequality (Theorem 2); both are standard.

The W_1 structural leakage bound $\kappa^{\text{sel}}_{W_1} = I(A(q_M); G^{\text{op}}) \leq I(q_M; G^{\text{op}})$ is *conditional* on (C1) and (C2′). The data-processing step is standard; the load-bearing condition is (C2′) — the Markov chain $G^{\text{op}} \to q_M \to A(q_M)$ holds only when the component carries no goal-correlated state across the call boundary. The bound is an upper bound; its gap below $I(q_M; G^{\text{op}})$ is the "accuracy effect" of §Discussion "Two senses of component competence" and is not characterized here (Working Notes).

Max attainable: derived under stated conditions. (C3)'s exact form is a structural ideal that pretrained components (notably LLMs with goal-rich training data) generally satisfy only approximately; the realistic regime is Theorem 2 with κ characterized empirically. (C2′)'s exact form is likewise an ideal — a frozen-weights LLM with a conversation cache violates it without any weight adaptation — and where it fails only the behavioral bound of `#disc-w1-structural-bound-boundary` survives.

The wrapping regime hierarchy (W_0/W_2/W_1) is a *formulation* — the partition is made by the structural choice of where to place the separation commitment. The leakage bounds within each regime are derived once the regime is fixed.

Discussion.

Quality–separation tradeoff inside Class B. For Class-B components (admitting both goal-blind and goal-conditioned modes), the wrapper has a design choice: how aggressively to restrict q_M to goal-blind content, vs. how much context to allow that may carry goal-correlated information. Maximally goal-blind queries (only the current observation, no context, no history) drive the selection-channel mutual information $I(q_M; G^{\text{op}})$ — and with it the leakage ceiling $\kappa^{\text{sel}}_{W_1}$ — toward zero, but may produce information-poor responses that hurt f_M's update quality. Maximally informed queries (full history, retrieved context) produce richer responses but raise $I(q_M; G^{\text{op}})$ and therefore the upper bound on $\kappa^{\text{sel}}_{W_1}$. The tradeoff is real and resolved per application — and because $I(q_M; G^{\text{op}})$ is a property of the query-selection policy the wrapper controls, the tradeoff is a design dial, not a property of the component.

Component-admissibility partition. Class A components (goal-blind by design) satisfy (C1) trivially and don't need wrapping in the substantive sense — wrapping for Class A is organizational rather than structural. Class B components (LLMs, hybrid RL with separable value/policy, multi-modal models) are the substantive wrapping case — the wrapper *chooses* to use the goal-blind mode. Class C components (pure end-to-end goal-conditioned policy networks) fail (C1) and are scope-out for the basic theorem. Salvage paths for Class C — null-goal queries, goal-uniform averaging, auxiliary distilled goal-blind heads — exist but cost something (information loss, computation, training).

Resolution of the LLM scope question. The "Class 3 (Coupled) exit" framing — *directed separation violated by goal-conditioned agents (LLMs); handled as architectural scope, not approximation* — is refined by this segment from a scope exit to a constructive route through. Class 3 (Coupled) LLMs are scope-in *for the wrapper construction* (under Class-B admissibility). The cost is paid in residual leakage rate $\kappa^{\text{sel}}_{W_1}$ bounded by the goal-content of the wrapper's query-selection policy, $I(q_M; G^{\text{op}})$; the tempo cost is established separately in `#der-class-coercion-in-composition`. Whether this construction yields an operationally useful agent depends on how low the query-selection policy can drive $I(q_M; G^{\text{op}})$ while still licensing useful belief updates — and, where the component carries goal-correlated cross-call state, on whether (C2′) can be enforced at all (`#disc-w1-structural-bound-boundary`).

Relationship to `#hyp-directed-separation-under-composition`. The hypothesis is descriptive — when does directed separation hold under composition? This segment provides the constructive answer for the wrapper-around-component special case: directed separation holds whenever the wrapper's type signatures are respected and (C1)–(C3) hold (or their weakenings). The general N-agent composition question remains a hypothesis; the wrapper-around-component case is now derived.

Wrapping as a truthification mechanism. The wrapping construction is the *rigorous formal version* of what `#disc-adversarial-coupling-pressure` §"Defensive scaffolding as composition" gestures at informally — peer review, prediction registers, double-entry bookkeeping, adversarial procedure, structured red-teaming. Those external scaffolds are operational mechanisms for *increasing* the modularity of a composite agent in the face of forces that would couple it; the wrapping construction is the structural version of the same operation applied internally rather than externally. Both share the discipline: a goal-blind belief-update query path is structurally enforced (W_1 strict) or behaviorally bounded (W_2), at a definite cost (extra component calls per macro-step plus residual leakage rate). The W_0/W_2/W_1 regime hierarchy is the *graded* characterization of how thoroughly the truthification has been applied — W_0 is the un-truthified base state, W_2 behavioral truthification, W_1 structural truthification. Cross-reference: `#disc-modularity-state-dynamics` is the meta-segment in which the truthification operation sits as one of three operations on the modularity state — alongside strategic self-coupling (`#disc-strategic-self-coupling`, self-driven-decreasing) and adversarial coupling pressure (`#disc-adversarial-coupling-pressure`, externally-driven-decreasing). This segment is the canonical *formal* instance of the truthification operation at the component level (W_0 / W_1 / W_2 regime hierarchy), paired with the composite-level *defensive-scaffolding* instance from `#disc-adversarial-coupling-pressure` §"Defensive scaffolding as composition". The connection is also surfaced in `#impl-composition-machinery` §"Class-coercion as truthification mechanism."

Two senses of component competence: world-simulation vs goal-extraction. A subtle decomposition of *component competence* matters for how (C3) and the selection-channel bound $\kappa_{W_1}^{\text{sel}} = I(A(q_M); G^{\text{op}}) \le I(q_M; G^{\text{op}})$ scale with the component's capability. A high-capability component has at least two distinct competences that the classification machinery treats asymmetrically:

1. **World-simulation fidelity** — accuracy of A's response to the explicit content of a query, including counterfactual content. Sharpens f_M's update quality once a query is in hand.
2. **Input-structure extraction** — competence at inferring latent structure *from the query's surface form*, including inferring the operator's goal G^{op} from query patterns. This is the channel (C3) names.

The architectural classification of `#der-directed-separation` is *blind to (1) and defined by (2)*: class membership turns on whether the goal can causally reach f_M, not on how accurately f_M then computes given goal-blind input. For pretrained components the two competences correlate empirically — the same training that produces world-simulation fidelity also produces input-structure extraction — but conceptually they are distinct, and the distinction is load-bearing for (C3).

The two competences act on the two sides of the selection-channel bound. The bound's *ceiling* is $I(q_M; G^{\text{op}})$ — the goal-content present in the query, fixed by the wrapper's selection policy; the *realized* leak is $I(A(q_M); G^{\text{op}})$ — how much of that the response actually carries. The gap between them is governed by competence:

- *Accuracy effect (widens the gap, lowers the realized leak).* A more competent world-simulator returns posteriors closer to the true conditional $P(\cdot \mid q_M)$, which is goal-independent by definition; it leaves the goal-content sitting *in* q_M unexploited in its response, so $I(A(q_M); G^{\text{op}})$ can sit well below the ceiling $I(q_M; G^{\text{op}})$.
- *Inference effect (closes the gap, raises the realized leak).* A more competent input-structure extractor uses any query-content / goal-content correlation to infer G^{op} from the goal-blind query and bend its response accordingly, driving $I(A(q_M); G^{\text{op}})$ up toward the ceiling — even when no goal argument is structurally exposed.

For pretrained components where the two competences scale together (the typical case for LLMs trained on goal-rich corpora), where the realized leak sits between 0 and the ceiling in the fidelity limit is *empirical and component-specific*, not structural. The selection-channel ceiling is design-controllable (drive $I(q_M; G^{\text{op}})$ down by goal-blind query selection); the position within that ceiling is the accuracy-vs-inference balance, and separating the two appears to open more science than it closes.

Several lines worth pulling, kept open here rather than queued elsewhere so future cycles do not lose them:

- *Decorrelation by construction.* If world-simulation and input-structure extraction can be structurally decoupled at training time — by architectural separation of *what does the world look like given* q from *what is the latent context that produced* q — a Class 3 (Coupled) component could satisfy (C3) even at high world-simulation fidelity. The classification would then refine into "Class 3 but C3-respecting" vs "Class 3 and C3-violating" sub-cases that the current scalar κ_{W_1} does not surface, and the C3-respecting branch may be a more interesting design target than wrapping a C3-violating component.

- *Counterfactual capability is bi-side.* Counterfactual simulation (Pearl Level 2) requires both competences — accurate world-simulation under interventional queries *and* extraction of the intervention's intended scope from the query. Whether high counterfactual capability tends to raise or lower realized $\varkappa$ in practice is an open empirical question; the answer depends on how the two competences decompose in the specific component.
- *Decomposed empirical estimator.* The existing $\hat{\varkappa}_{\text{processing}}$ estimator from `#der-directed-separation` probes the *net* response divergence under varied goal-priming. An estimator that holds explicit query content fixed while varying only the implicit operator context might isolate the inference-effect contribution, giving a sharper diagnostic than the aggregate $\varkappa$ and surfacing whether a given high-fidelity component sits near $\varkappa \to 0$ (accuracy-dominated) or near $\varkappa$ ceiling (inference-dominated).
- *M4 amplification specificity.* The amplification of `#disc-modularity-state-dynamics` operations (strategic self-coupling, adversarial coupling pressure) under high component fidelity may track competence (2) specifically rather than competence (1). If so, M4 attack-surface measurement could be sharpened by measuring the inference effect rather than aggregate capability — a more competent world-simulator is not, *by that fact alone*, a more potent vector for the coupling-increasing operations.

The decomposition is named here at *discussion-grade* recognition — kept open as a place to pull on rather than as a closed result. Whether it lifts to *robust qualitative* or higher depends on whether the two competences can be operationalized as separately measurable quantities and whether the architectural-decorrelation conjecture holds for any realizable training regime.

Findings.

Constructive Directed Separation via Wrapping. **Brief:** When you have a component (like an LLM) whose belief-update and goal-conditioning are entangled in a single forward pass, you can build a scaffold around it that maintains explicit, separate stores for what the system believes and what it wants. The structural rule is that belief updates only see queries to the component that don't include the goal as input. Under reasonable conditions on the component, the wrapped system is goal-blind in its belief updates *by construction* — even though the underlying component isn't. The cost shows up as a residual leakage from the component's pretraining (the component might still infer the goal from query content, even when the goal isn't explicit in the input). Two practical regimes appear: strict wrapping with separate goal-blind and goal-conditioned calls (theoretically clean, with a structural leakage bound), and partial wrapping with one goal-conditioned call whose response is parsed into separate update fields (operationally common, with only a behavioral leakage bound — depending on the component's instruction-following fidelity rather than its query structure).

Impact: Promotes `#hyp-directed-separation-under-composition` to derived (in the wrapper-around-component special case). Refines the Class 1 (Separated) cell of `#der-directed-separation` with a structural-vs-behavioral sub-distinction (W_1 vs. W_2). Resolves the LLM scope question — Class 3 (Coupled) components are scope-in for the wrapper construction at a measurable cost, not scope-out. The composition-level consequences (wrapper as valid AAT composite agent, persistence-template inheritance, tempo cost) are derived in the companion segment `#der-class-coercion-in-composition`.

Novelty Claim: *Claim integration* of POMDP / cognitive-architecture prior art with the AAT Class 1/2/3 (Separated/Partial/Coupled) directed-separation taxonomy, plus the $W_0/W_2/W_1$ regime hierarchy that surfaces the structural-vs-behavioral leakage distinction and the LLM-specific (C1)–(C3) admissibility/leakage conditions. The wrapping move itself is rediscovery of patterns established in POMDP theory (Bayesian belief-update is goal-blind by construction) and cognitive architectures (modular agent design with separated belief/goal/action state, four decades). AAT's contribution is the structural-leakage analysis at the directed-separation level and the regime hierarchy that names where the separation guarantee lives.

Related Work:

Table 13.5

ASF concern	*Prior-art language*	*Relationship / Positioning*
Goal-blind belief-update by construction	Astrom 1965, "Optimal control of Markov processes with incomplete state information," *J. Math. Anal. Appl.* 10; Kaelbling, Littman, Cassandra 1998, "Planning and acting in partially observable stochastic domains," *Artificial Intelligence* 101	*formal antecedent* — POMDP belief-state filters are goal-blind by construction; the wrapping move recapitulates this in the AAT vocabulary. The closest formal prior art for the directed-separation guarantee.
Modular agent design with separated belief/goal/action	Newell 1990, *Unified Theories of Cognition*; Laird 2012, *The Soar Cognitive Architecture*; Anderson 2007, *How Can the Human Mind Occur in the Physical Universe?*; Sun 2016 *Anatomy of the Mind* (CLARION); Baars 1988 *A Cognitive Theory of Consciousness* / Dehaene 2014 *Consciousness and the Brain* (Global Workspace)	*formal antecedent* — cognitive architectures have done modular agent design with separated belief/goal/action state for 40+ years. The W_1 wrapping move is essentially the per-cycle commitment that cognitive architectures make at the system level.
Tool-using language-model agent frameworks	Yao et al. 2022, "ReAct: Synergizing reasoning and acting in language models"; Shinn et al. 2023, "Reflexion: language agents with verbal reinforcement learning"; Park et al. 2023, "Generative Agents: Interactive Simulacra of Human Behavior"; Packer et al. 2023, "MemGPT: Towards LLMs as operating systems"; Schick et al. 2023, "Toolformer: language models can teach themselves to use tools"	*empirical instantiation* — practical wrappers around language-model substrates. Most fall in W_2 (partial wrapping / output-structuring); Generative Agents' observation→memory step is the closest empirical instance of W_1. AAT's regime hierarchy gives these constructions a structural reading.
Hybrid deliberative/reactive architectures (prior art for class-coercion)	Gat 1992, *Integrating planning and reacting in a heterogeneous asynchronous architecture for controlling real-world mobile robots* (AAAI)[^cat-2026-05-22]; Simmons 1994, *Structured control for autonomous robots* (IEEE Trans. Robotics 10:34)[^cat-2026-05-22]; Au 2004, planner wrappers with external-query management[^cat-2026-05-22]	*formal antecedent* — hybrid deliberative/reactive architectures orchestrate reactive (entangled) layers beneath deliberative (separated) planners, structurally precedent for the wrapping construction's "scaffold an entangled component to recover separated behaviour at the wrapper level" move. AAT's contribution is *not* the wrapping move itself but the *theorem-shaped* wrapper-level directed-separation guarantee (Theorem 1 + Theorem 2) plus the explicit selection-channel leakage bound $\kappa^{\mathrm{sel}}_{W_1} = I(A(q_M); G^{\mathrm{op}}) \leq I(q_M; G^{\mathrm{op}})$ and the Brooks's-Law tempo cost in `#der-class-coercion-in-composition`

Search Log:

- 2026-05-09 (*targeted*): Web + training-data search across POMDP / cognitive-architecture / scaffolded-LLM threads. Verdict: **substantial overlap** with the POMDP and cognitive-architecture lines as the closest formal prior art. AAT's contribution is the structural-leakage analysis and regime hierarchy rather than novelty in the wrapping move itself.
- 2026-05-09 (*intuition-only*, prior to the targeted search): adjacent literatures expected to host prior art were active inference (Markov blankets), control theory (approximate simulation), and scaffolded-LLM frameworks. The targeted search confirmed all three and added the POMDP and cognitive-architecture lines as the formal precedents.

13.5 *Derived:* Class Coercion in Composition

Under the wrapper construction of `#der-class-coercion-via-wrapping`, the wrapped system W is a valid AAT composite agent: it satisfies (A1)–(A4) of `#form-composition-closure`, inherits the sector-persistence template from `#result-sector-persistence-template`, and incurs a tempo cost in the Brooks's-Law form of `#der-tempo-composition`. This segment establishes those composition-level consequences; the directed-separation guarantee that motivates the construction is established in the prerequisite segment.

Three additional design constraints on the wrapper itself are required for (A2)–(A4) to hold at the wrapper level: that the wrapper commits to a prediction map so macro-mismatch is well-defined; that its belief-update map supports a gain interpretation per `#def-adaptive-tempo`; and that the belief-update map satisfies the sector condition with positive correction rate. All three are automatic for Tier-1 belief-update maps via `#deriv-sector-condition` and `#der-gain-sector-bridge`; Tier-2/3 cases inherit the corresponding tier-restricted scope. (A1) holds by construction — the wrapper *builds in* the AAT-shaped state $X_W = (M_W, G_W)$.

Persistence at the wrapper level requires the wrapper's correction rate to exceed total effective disturbance, which now has *two* contributions: external environmental disturbance acting through the wrapper's observation channel, and internal disturbance from the component's response variance, bounded by the variance of the component's responses to goal-blind queries. The wrapping construction is honest about what it does *not* buy: the component's internal nondeterminism still acts as a disturbance on the wrapper level. What it eliminates is the *belief-goal coupling* that the directed-separation classification identifies as the load-bearing failure mode for Class 3 (Coupled) substrates.

The tempo cost is a **Brooks's-Law instance**: the wrapper makes $K \geq 2$ component calls per macro-step (one goal-blind for model update, one goal-conditioned for purposeful-state update; more in richer wrapper designs), so the wrapper's macro-event rate is at most $1/K$ of the component's underlying rate. The Brooks's-Law inequality of `#der-tempo-composition` predicts exactly this — more component calls per macro-step means lower macro-tempo, with the coordination overhead specific to maintaining the wrapper's (M_W, G_W) state separately from the component's internal state. The structural reading: a wrapped system gets directed separation (Class 1 status, the full Part II machinery applies) at the *structural price of admission* of (at minimum) halved macro-tempo plus an irreducible internal-disturbance term. This is the framework's quantitative content for the cost of converting a Class 3 (Coupled) substrate into a Class 1 (Separated) composite.

Formal Expression.

Setup (inherited). The wrapper structure, the four type-signed update components (q_M, q_G, f_M, f_G), the macro-step schedule ($K \geq 2$ component calls per macro-step), and the admissibility conditions (C1)–(C3) on the component A are defined in `#der-class-coercion-via-wrapping`. This segment uses that setup without redefinition.

Wrapper-design constraints. For (A2)–(A4) of `#form-composition-closure` to hold at the wrapper level, the wrapper's update structure must satisfy three additional constraints:

(D-A2) The wrapper commits to a prediction map $\hat{o}_W : \mathcal{X}_M \times \mathcal{A}_W \to \mathcal{O}_W$ so that macro-mismatch $\delta_W = o_W - \hat{o}_W$ is well-defined. *Conditions (wrapper-design)*

(D-A3) f_M supports a gain interpretation per `#def-adaptive-tempo`. Holds for Tier-1 belief-update maps — Bayesian on exponential families, gradient on strongly convex losses, linear-PD with bounded gain. Tier-2/3 cases inherit the corresponding tier-restricted scope from `#deriv-sector-condition`.

(D-A4) f_M satisfies the sector condition with positive correction rate. Automatic for Tier-1 belief-update maps via `#deriv-sector-condition` Prop A.1 and `#der-gain-sector-bridge`.

(A1) of `#form-composition-closure` holds by construction — $X_W = (M_W, G_W)$ has the AAT form because the wrapper *builds it in*.

Derived (wrapper-as-composite, from (D-A2)+(D-A3)+(D-A4))

Theorem: Wrapper as valid AAT composite agent.

Under (D-A2)–(D-A4) and the directed-separation conditions of `#der-class-coercion-via-wrapping`, the wrapper W satisfies (A1)–(A4) of `#form-composition-closure` and therefore qualifies as an AAT composite agent.

Proof. (A1) by construction of the type signatures; the wrapper's state is $X_W = (M_W, G_W)$ in the AAT shape. (A2) under (D-A2): the prediction map closes the mismatch definition. (A3) under (D-A3): Tier-1 belief-update maps inherit gain interpretation via `#deriv-sector-condition` Prop A.1, with the Tier-2/3 lifts deferring to the tier-restricted scope from `#form-composition-closure`'s bridge-lemma classification. (A4) under (D-A4): the sector condition transfers from the (Tier-1) belief-update map to the wrapper level via `#der-gain-sector-bridge`. ∎

Inheritance of the persistence template. Under (D-A4), the wrapper inherits `#result-sector-persistence-template` at the wrapper level: persistence holds when $\alpha_W R_W > \rho_W$. The wrapper-level effective disturbance has two contributions: external environmental disturbance ρ_{ext} acting through o_W, and internal disturbance from the component's response variance, ρ_{int}, bounded by the variance of A's responses to goal-blind queries. Total: $\rho_W = \rho_{\text{ext}} + \rho_{\text{int}}$. Persistence at the wrapper level requires $\alpha_W R_W > \rho_{\text{ext}} + \rho_{\text{int}}$.

Tempo cost — Brooks's-Law instance. The wrapper makes $K \geq 2$ component calls per macro-step (more in richer wrapper designs). If the component's nominal call rate is ν_A, the wrapper-level macro-update rate is $\nu_W = \nu_A/K$. By `#der-tempo-composition`,

$$\mathcal{T}_W \leq \mathcal{T}_A^{\text{nominal}} - C_{\text{coord}}^{\text{wrap}} \tag{13.34}$$

where $C_{\text{coord}}^{\text{wrap}}$ is the coordination overhead specific to the wrapping construction — the tempo consumed by maintaining the wrapper's (M_W, G_W) state separately from the component's internal state. This is the cost of class coercion paid in tempo: the same Brooks's-Law form whose general statement is in `#der-tempo-composition`. Adding state-management infrastructure reduces realized external tempo even when the underlying component's compute rate is unchanged.

Epistemic Status.

Conditional on the wrapper-design constraints (D-A2), (D-A3), (D-A4), and on the prerequisite directed-separation result of `#der-class-coercion-via-wrapping`. The proof is direct verification using inheritance from `#deriv-sector-condition` and `#def-adaptive-tempo`; both are standard. (D-A4) is automatic for Tier-1 belief-update maps; for Tier-2/3 belief-update maps the wrapper inherits the tier-restricted scope from `#deriv-sector-condition` and `#form-composition-closure`'s bridge-lemma classification.

Max attainable: derived under stated conditions. The composition-level claim is parametrized by the belief-update map's tier; Tier-1 cases inherit cleanly, Tier-2/3 cases inherit under additional restrictions.

Discussion.

Coercion-distance vs. tracking-distance. The closure-defect ε^* in `#form-composition-closure` is read as a *fidelity* measure: how well the macro-system tracks the projected micro-system. In the wrapping construction, the wrapper deliberately changes the underlying component's behavior to enforce separation — the wrapper *should* differ from the unwrapped component in the goal-conditioning direction. Two distinct quantities are involved:

- $\varepsilon^*_{\text{track}}$ — the standard fidelity quantity from `#form-composition-closure`. Used in the bridge lemma to bound trajectory error.
- $\varepsilon^*_{\text{coerce}}$ — the wrapping-specific quantity measuring behavioral divergence between the wrapped system and the unwrapped component. Higher means more aggressive coercion.

For wrapper analyses: the persistence-template inheritance (above) uses $\varepsilon^*_{\text{track}}$. Cost-of-class-coercion analyses use $\varepsilon^*_{\text{coerce}}$. The leakage analysis from `#der-class-coercion-via-wrapping` uses $\varkappa$ — a KL-divergence on response distributions, distinct from both trajectory-error quantities. Conflating them propagates downstream confusion.

Form-preservation reading. Read in form-preservation language, (A1)–(A4) of `#form-composition-closure` are the requirement that AAT's form survive coarse-graining: macro must itself be AAT. The wrapping construction is the *constructive* version — when the underlying component doesn't preserve form on its own (Class 2 (Partial) or Class 3 (Coupled)), the wrapper enforces form-preservation at the macro level by structural commitment of the type signatures (from `#der-class-coercion-via-wrapping`) plus the wrapper-design constraints here. The closure-defect $\varepsilon^*_{\text{track}}$ then plays the role of distance from the form-preserved fixed point in the constructive setup.

Findings.

Wrapper as Valid AAT Composite Agent with Brooks's-Law Tempo Cost. **Brief:** Once you've wrapped a coupled component to get directed separation (the structural rule from `#der-class-coercion-via-wrapping`), is the wrapped system actually a valid AAT agent that AAT's machinery applies to? The answer is yes, under reasonable design constraints on the wrapper's prediction map, gain interpretation, and sector behavior. This means the wrapped system inherits the persistence-condition machinery — the wrapper's own version of "can the agent keep up with disturbance" applies, with disturbance now decomposing into the external environment's drift plus the underlying component's response variance. There's a cost, and it's Brooks's Law in tempo coordinates: every additional component call per cycle (needed to maintain separate belief and goal stores) divides the macro-update rate. The wrapped system runs slower than the component would running alone, by a structurally determined factor.

Impact: Closes the loop on whether the wrapping construction yields a *usable* AAT agent: yes, the wrapper inherits `#result-sector-persistence-template` and `#der-tempo-composition` at the wrapper level. Makes the tempo cost of class coercion explicit and quantifiable — it's a Brooks's-Law instance, the same general form that governs all AAT compositions, not a special-case tax. Provides the basis for analyzing logogenic-substrate wrappers (e.g., PROPRIUM's auxilia hierarchy, agentic-tft's CONTEXTUALIZE-CHOOSE phase separation) as valid composite agents with computable persistence and tempo characteristics.

Novelty Claim: *Claim integration* of the AAT sector-Lyapunov persistence template, Brooks's-Law tempo accounting, and the form-composition-closure (A1)–(A4) discipline, applied to the wrapper-around-component construction. The composition-level inheritance is straightforward given the persistence-template and tempo-composition machinery; the contribution is establishing that the wrapped system is in the right shape for those results to apply, and naming the Brooks's-Law form of the coercion cost.

Related Work:

Table 13.6

ASF concern	*Prior-art language*	*Relationship / Positioning*
Predictive-loss bounds under coarse-graining of MDPs	Ravindran-Barto 2004, "An algebraic approach to abstraction in reinforcement learning"; Taylor, Precup, Panangaden 2008, "Bounding performance loss in approximate MDP homomorphisms," NeurIPS; Abel, Hershkowitz, Littman 2016, "Near optimal behavior via approximate state abstraction," ICML; Subramanian, Sinha, Seraj, Mahajan 2020, "Approximate information state for approximate planning," arXiv:2010.08843; Congeduti, Mey, Oliehoek 2020, "Loss bounds for approximate influence-based abstraction," arXiv:2011.01788	*adjacent literature* — control-theoretic predictive-loss bounds. AAT connects to this cluster via `#form-composition-closure`'s bridge lemma at the wrapper level.
Compositional structure for nested agents	Smithe 2024, "Structured Active Inference," arXiv:2406.07577; Capucci, Gavranović, Hedges, Rischel 2022, "Towards foundations of categorical cybernetics," ACT 2021	*adjacent literature* — categorical / lens framing of compositional agents. The wrapper construction is consistent with the lens framing; an alternative reading in categorical-systems-theory terms is available.
Form-preservation under coarse-graining	Friston 2019 *J. Theor. Biol.*, "On Markov blankets and hierarchical self-organisation"; Friston, Heins, Verbelen, Da Costa et al. 2025 *Front. Network Physiology*, "From pixels to planning: scale-free active inference"	*adjacent literature* — Friston's framing of the renormalization group as form-conservation under coarse-graining. The wrapping construction reads as a constructive form-preservation move.
IB-Lagrangian semigroup composition	Mehta, Schwab 2014, "An exact mapping between the Variational Renormalization Group and Deep Learning," arXiv:1410.3831; Kline, Palmer 2022, "Gaussian Information Bottleneck and the Non-Perturbative Renormalization Group," PMC8967309	*adjacent literature* — IB-as-RG. The wrapper's (P1) information-preservation condition (from `#form-composition-closure`) is IB-shaped; the semigroup composition rule is the closest analog of "AAT form preserved under iterated wrapping."
Singular-perturbation as unified RG	Chen, Goldenfeld, Oono 1996, "Renormalization group and singular perturbations," *Phys. Rev. E* 54:376; Chiba 2009, SIAM J. Appl. Dyn. Syst. 8:1066	*formal antecedent* — RG framework subsumes singular perturbation theory. The $K_c \gg 1$ timescale-separation regime in `#form-composition-closure` invokes this for free.

Search Log:

- 2026-05-09 (*targeted*): Search across MDP-homomorphism / hierarchical-control / categorical-systems-theory / FEP-RG / IB-RG / singular-perturbation-RG threads. Verdict: **substantial overlap** with the MDP-homomorphism and categorical-cybernetics lines as the closest formal prior art for composition under coarse-graining. AAT's contribution is the specific Brooks's-Law tempo form and the persistence-template inheritance pattern.

13.6 *Discussion:* Additional Implications & Discussion

The chapter is where AAT's per-agent machinery lifts to the composite level. Closure defect supplies the criterion for when a group of agents legitimately counts as one larger agent; the bridge lemma converts predictive defect into trajectory error; the wrapping construction supplies the constructive route from Class 3 (Coupled) components to Class 1 (Separated) composites; and the closed-form critical-mass derivation unifies four previously-separate persistence inequalities as signed special cases of a single composite-sector condition. The chapter's two segments with `## Findings` carry the load-bearing technical content well; the implications-side payoff is the cross-segment composition — how the four pieces produce composite-level results that no single segment carries — and what those results buy for `03-llm-core/`, AI-safety architecture, and the framework's identifiability-floor pattern at the composition layer.

Closure defect, bridge lemma, and the meaning of composition.

`#form-composition-closure`'s Findings cover the closure-defect / bridge-lemma machinery in full: when does a group of interacting agents legitimately count as one larger agent? — answered by an inequality, with explicit failure modes on both sides. The closure-defect ε^* is the minimum residual prediction error from collapsing the micro-system into an AAT-shaped macro-description; the bridge lemma converts that prediction-level error into a trajectory-level guarantee bounded by $\varepsilon^* \nu_c / \alpha_c$ under sector-bounded macro-correction plus strong monotonicity of the macro-update. The implication worth surfacing beyond the segment's own statement is *what* the framework is offering against adjacent positions, and *what* it is honest about not offering.

Against Markov-blanket and active-inference accounts of nested agency, AAT's bridge-lemma form supplies a *control-theoretic loss bound* on treating a collection as one agent — something the Markov-blanket apparatus does not produce, even when the boundary criterion is in hand. The closure-defect framework is consistent with the approximate-information-state and influence-based-abstraction lines from PAC-MDP work (Subramanian 2020, Congeduti 2020, Abel 2016, Taylor 2008) but applies the same predictive-loss-to-control-error machinery at the *agent boundary formation* scope rather than at the single-agent compression scope. The framework's distinctive content is the *use* of bounded-loss composition as an agent-boundary criterion, together with the admissibility conditions (A1)–(A4) on the macro-dynamics that force the macro-description to itself be AAT-shaped (ruling out trivial curve-fits where any composite could be called an "agent" by re-defining state).

The honest scope statement worth keeping in view is that $\varepsilon^* = 0$ does not mean *optimality* — two non-communicating Kalman filters achieve $\varepsilon^* = 0$ while being suboptimal versus a joint filter, because the joint filter could exploit cross-correlation the non-communicating pair cannot. The framework diagnoses *representability* of a composite as an AAT agent, not optimality of that composite against alternative architectures. Strong monotonicity is the hinge — sub-agents that are merely Lyapunov-stable do not compose into a coherent macro-agent, and this is one of the chapter's quieter surprises. Three agent tiers fall out of the strong-monotonicity requirement: Tier 1 where contraction is proved for the full class (Bayesian on exponential families, gradient descent on strongly convex losses, linear correctors with positive-definite gain); Tier 2 where it holds locally; Tier 3 where it must be verified per-domain.

The critical-mass closed form unifies four signed special cases.

The chapter's deepest cross-segment finding is what `#deriv-critical-mass-composition` produces by composing the bridge lemma with `#der-team-persistence`'s signed-coupling decomposition. For the symmetric-matched-Tier-1 dyad, the composite sector constant is now derived in closed form:

$$(\alpha - C)R > \rho + \gamma\mathcal{T} \tag{13.35}$$

with signed γ (negative cooperative, positive adversarial) and C the coordination-overhead cost. The single inequality recovers *four prior results as signed special cases*: single-agent persistence ($\gamma = 0, C = 0$, reducing to Part I Ch.4's central inequality); team-persistence with cooperative coupling ($\gamma < 0$, where the negative term enters as effective-disturbance reduction); adversarial destabilization ($\gamma > 0$, where the positive term amplifies disturbance); and coordination-dominated failure ($C > \alpha$, where the composite's effective correction rate goes negative regardless of individual quality — *Brooks's Law as a derived inequality* rather than a folkloric observation). The teleological unity U_O enters multiplicatively on γ plus scope-gate, not additively — a distinction with downstream implications for Ch.3's unity-closure-mapping result.

The implications-side payoff that the segment's Working Notes flag but is worth surfacing: this is the framework's *cleanest internal-economy demonstration*. One inequality, with signed parameters indexing the regime, recovers what previously read as four separate results. The result composes with the sector-persistence template economy (Part I Ch.4 implications, §4): the template absorbed the Lyapunov boilerplate; the critical-mass derivation specializes the template's persistence inequality with a signed-coupling effective-disturbance decomposition that handles single, cooperative, adversarial, and coordination-failure regimes simultaneously. For a practitioner, the engineering anchor is that Brooks's Law (adding people to a late project makes it later) has a *physics-level* derivation here: a development team's macro-correction rate is $\alpha - C$ where C is coordination overhead; once C exceeds α, adding people degrades persistence regardless of individual quality. The threshold is sharp, not gradual, and the inequality predicts exactly where it bites.

The closed form also produces `#disc-identifiability-floor`'s Instance 3 in its fullest form. Composite contraction certification from component-marginal data is structurally impossible — the coupling-sign bit ($\gamma < 0$ vs $\gamma > 0$) is unidentifiable from how each component looks in isolation (Liberzon 2003's common-Lyapunov-nonexistence is the external no-go). The five boundary routes named in `#disc-identifiability-floor` for Instance 3 escape this: under matched-Tier conditions the closed form supplies the coupling sign by interventional access at the composite scope, via Mode 2 of the loop-Level-2 pattern (observer-on-sub-agent, from Part II Ch.2 implications). The chapter is therefore where the *third* formal instance of the identifiability-floor pattern becomes statable; the four-instance pattern is completed with Instance 1 (Part II Ch.4), Instance 2 (Part II Ch.3), Instance 3 (here), and Instance 4 (Part II Ch.5 / Appendix A — `#der-architecture-noidentifiability`, architecture-noidentifiability from on-policy summary data).

Class coercion via wrapping — what the construction unlocks.

`#der-class-coercion-via-wrapping`'s Findings cover the constructive directed-separation result well: when a component (like an LLM) has belief-update and goal-conditioning entangled in a single forward pass, a scaffold around it can maintain explicit separate stores for belief and goal, with belief updates only seeing queries that don't include the goal as input. Under named admissibility conditions on the component, the wrapped system is goal-blind in its belief updates *by construction* — even though the underlying component is not. The $W_0/W_2/W_1$ regime hierarchy names the practical landscape: W_1 (strict wrapping, structural leakage bound, theoretically clean), W_2 (partial wrapping with goal-conditioned call and parsed response, behavioral leakage bound, operationally common), W_0 (no wrapping — the unconverted Class 3 case).

The implications-side payoff is what the construction unlocks beyond the segment's statement: it is the *constructive bridge* that resolves how AAT's Part II results — derived for Class 1 (Separated) agents — apply to Class 3 (Coupled) substrates. Without the wrapping construction, Class 3 components would sit outside Part II's persistence machinery as a scope exit; with the wrapping construction, the wrapped system at the *wrapper level* is Class 1 (Separated) by construction, inherits the sector-persistence template, and operates under the standard AAT results. The cost is paid in two places: a Brooks's-Law tempo overhead (more component calls per macro-step), and a residual leakage rate bounded structurally (W_1, by the wrapper's query design) or behaviorally (W_2, by the component's instruction-following fidelity).

The composition-level consequences are derived in `#der-class-coercion-in-composition`: the wrapped system is a valid AAT composite agent, satisfying admissibility conditions (A1)–(A4) of `#form-composition-closure` under wrapper-design constraints (D-A2)–(D-A4). It inherits the sector-persistence template at the wrapper level; its tempo cost is the Brooks's-Law composition cost from `#der-tempo-composition`. This is the chapter's quiet load-bearing claim for `03-llm-core/`: language-model substrates are *constructively* in AAT scope, at a measurable cost, rather than scope-out. The implications for AI-safety practice are concrete: scaffolded loops around LLMs are not just engineering convenience — they are the structural mechanism that converts a Class 3 (Coupled) substrate into a Class 1 (Separated) composite, and the leakage bounds tell the practitioner what residual coupling remains. The Generative Agents observation→memory step (Park et al. 2023) is the canonical empirical W_1 instance; most production tool-using LLM frameworks (ReAct, Reflexion, Toolformer, MemGPT) sit in W_2. The framework's regime hierarchy gives all of these constructions a *structural* reading rather than an engineering-pattern reading.

The cross-domain transfer follows naturally. Human bicameral / dual-process models (Kahneman 2011) are W_1-style internal scaffolding when the deliberative system constrains the intuitive system through structural commitment to goal-blind belief-update queries; organizations where executive scaffolding (boards, committees, decision rights) recovers cascade ordering across coupled individuals are W_1/W_2 at the organizational scale. The framework's structural claim is the same in all three cases: *separation at the wrapper level is what gives AAT-grade behavior; the wrapping construction is structurally non-optional for Class 3 (Coupled) substrates.*

Tempo composition, Brooks's Law, and the bandwidth connection.

`#der-tempo-composition` carries the sub-additive tempo inequality: $\mathcal{T}_c \le \sum_i \mathcal{T}_i$, with the gap bounded below by $C_{\text{coord}} \ge \varepsilon^* \nu_c$. The implication is that *coordination overhead acts as effective disturbance at the composite level* — closure defect $\varepsilon^* \nu_c$ is what the template's instantiation at the composite identifies as the internal contribution to effective disturbance, alongside the external environmental rate ρ_{ext}. The composite persists when $\alpha_c R_c > \rho_{\text{ext}} + \varepsilon^* \nu_c$, and Brooks's Law follows: adding agents increases $\sum_i \mathcal{T}_i$ but may increase $\varepsilon^* \nu_c$ faster, pushing the composite's effective disturbance past its persistence threshold even as nominal capacity grows.

The composition with Part I Ch.4's information-rate floor is worth surfacing as a cross-segment finding. Each agent in the composite pays its own $\dot{R} \ge n\alpha/2$ bandwidth cost; the composite as a whole pays the sum *minus what coordinated sub-agents save on shared observation*. But there is a complementary cost: the closure defect $\varepsilon^* \nu_c$ has its own information-rate signature, because maintaining the bridge-lemma's strong-monotonicity condition at the macro-update requires the composite to acquire information about how its sub-agents are interacting. The framework's prescription is that *composite persistence pays for both the sub-agents' individual bandwidth and the coordination-overhead bandwidth*, and the trade-off between adding agents (more $\sum_i \mathcal{T}_i$, more individual bandwidth) and increasing coordination overhead (more $\varepsilon^* \nu_c$, more coordination bandwidth) has a precise inflection point at $C_{\text{coord}} = \alpha_c$. Past the inflection point, additional sub-agents *cost more bandwidth than they buy*, and the composite's persistence-information-rate floor moves up faster than its persistence threshold. This is the implication of the chapter's machinery for distributed-systems design: the sweet spot for sub-agent count is bounded by the coordination-overhead growth rate, not by individual sub-agent capacity.

The chapter is also where Mode 2 of the loop-Level-2 deployment pattern (observer-on-sub-agent, from Part II Ch.2 implications) becomes structurally load-bearing. The Liberzon-anchored identifiability-floor Instance 3 says the coupling sign cannot be recovered from component marginals; the unique broadly-available escape is interventional access at the composite scope. Operationally, this means an external observer (or the composite agent itself acting on its sub-agents) can recover the sign by performing do-interventions on one sub-agent and observing the response of another. The closed-form critical-mass derivation gives the mathematical form of the recovered coupling; the loop-Level-2 access at the composite level gives the data substrate that recovery requires.

Closing notes and the bridge into Unity, Communication, and Shared Intent.

What this chapter closes on is the composite-agent foundation Part III needs: a representability criterion via closure defect, a trajectory-level loss bound via the bridge lemma, a closed-form persistence inequality that recovers single-agent, team, adversarial, and Brooks's-Law regimes as signed special cases, a constructive route from Class 3 (Coupled) components to Class 1 (Separated) composites with measured leakage bounds, and a sub-additive tempo inequality whose coordination-overhead term enters the persistence template as effective disturbance at the composite level. The chapter's two segment-level `## Findings` sections carry the load-bearing technical content; this implications segment surfaces the cross-segment composition and the engineering anchors that the segments individually cannot reach.

The bridge into Part III Ch.3 (Unity, Communication, and Shared Intent) is the closure-defect ε^* itself. The dimensions of unity — content axes (U_M, U_O, U_Σ, U_{obs}) and structural axis (U_f) — parametrize *what sub-agents can share*; Ch.3's `#result-unity-closure-mapping` shows that unity parametrizes a rate-distortion curve for closure defect. The chapter here gives the closure defect a precise form; Ch.3 gives unity its quantitative content; together they say *which dimensions of unity matter how much for which kinds of composite-agent persistence*. The cooperative-vs-adversarial machinery in

Ch.4 then operates on what Ch.2 and Ch.3 jointly build: cooperative coupling reduces γ (lowers effective disturbance); adversarial coupling raises γ (amplifies it); the signed-coupling structure is the same inequality viewed from opposite directions. Ch.5's strategic composition lifts the symmetric partially-opposing case from scalar coupling to equilibrium machinery (potential / monotone games), and the identifiability-floor pattern's Instance 3 instance becomes operational there in its strongest form.

Chapter 14

Unity, Communication, and Shared Intent

[**Gap**] Open question

The quality of a composite agent's composition — *conditional on 12.2 being satisfied via at least one of its four routes (three alignment routes + the strategic-equilibrium route (C-iv))* — is parametrized along **two architecturally distinct axes**:

- **Content axis (four unity dimensions).** What the sub-agents *share*: epistemic (U_M, shared model), teleological (U_O, shared objective), strategic (U_Σ, coordinated action), and perceptual (U_{obs}, shared observations).
- **Structural axis (update-rule homogeneity, U_f).** Whether sub-agents implement the *same* correction rule: how similar their f_M updates are across the population.

Together, the two axes parametrize the rate-distortion curves for the component closure defects (13.1, 14.1). Higher unity along either axis permits more aggressive compression at lower closure defect; neither axis alone is sufficient. In pure Part I composition (passive estimators, no G_t), agents with identical content can still produce non-zero ε_x if their update rules differ — the content axis cannot detect this, which is why the structural axis is required. Unity (in either sense) does not directly predict closure-defect magnitude; it controls the compressibility of the corresponding state, observation, or action component under projection.

Scope. The decomposition applies to composites that satisfy 12.2. U_O plays a role in the (C-i) route of the scope condition via value-correlation, but the scope condition is a disjunction of four routes — three alignment routes (shared objective, hierarchical derivation, mutual benefit) plus the strategic-equilibrium route (C-iv) — not a scalar threshold on U_O. Below scope-satisfaction (no route applies), the sub-agents are a multi-agent system per 12.1 and composition-level quantities are not well-defined.

For a composite agent A_c composed of sub-agents $\{A_1, \dots, A_n\}$, the unity profile consists of *four content dimensions* (this section, below) and *one structural dimension* (U_f, defined after the content dimensions). *Definition (definition-unity-dimensions)*

Epistemic unity U_M — how much of the reality model is shared:

$$U_M = \frac{I(M_t^{(1)}; \dots ; M_t^{(n)})}{(n-1)\, H(M_t^{(1)}, \dots , M_t^{(n)})} \tag{14.1}$$

The fraction of total model information that is shared (total-correlation ratio, normalized to $[0, 1]$). The numerator is the total correlation $I = \sum_i H(M_t^{(i)}) - H(M_t^{(1)}, \dots , M_t^{(n)})$; the $(n-1)$ factor in the denominator is its normalizer, since each marginal entropy is at most the joint entropy and hence $I \leq (n-1)\, H(M_t^{(1)}, \dots , M_t^{(n)})$, with equality iff each model determines all the others. So $U_M = 1$ for identical models and $U_M = 0$ for independent models, for every n. (Without the $(n-1)$ factor the ratio reaches $n-1$ at identity — the bare multi-information/joint-entropy form is normalized correctly only at $n = 2$.)

Teleological unity U_O — how aligned are the objectives:

$$U_O^{(i,j)} = \mathrm{corr}\Big(V_{O_t^{(i)}}(\tau),\ V_{O_t^{(j)}}(\tau)\Big) \tag{14.2}$$

over trajectories the composite encounters. $+1$ for identical objectives; -1 for perfectly opposed; 0 for orthogonal. The composite teleological unity is an aggregation over all pairs. The scalar ranges from fully cooperative to fully adversarial per objective dimension.

[Scope note] U_O plays a role in 12.2 — primarily along route (C-i), where value-correlation is the operationalization of teleological alignment. The scope condition itself is disjunctive: it is satisfied when *any* of the alignment routes (C-i), (C-ii), (C-iii) applies, or when the strategic-equilibrium route (C-iv) applies; not by U_O alone crossing a common scalar threshold. The quality-metric role of U_O captured in this segment presumes scope-satisfaction via *some* route; on the (C-iv) side it tracks at best the alignment projection of a strategic composite whose macro-state is defined by equilibrium structure rather than by a shared target. When the sub-agents fail all four routes, they form a multi-agent system (12.1) but not a composite, and composition-level quantities (closure defect, composite tempo, team persistence) are not well-defined.

Discussion

Strategic unity U_Σ — how coordinated is the joint policy:

$$U_\Sigma = 1 - \frac{D_{\mathrm{KL}}(\pi^c_{\mathrm{actual}} \| \pi^c_{\mathrm{optimal}})}{D_{\mathrm{KL}}(\pi^c_{\mathrm{independent}} \| \pi^c_{\mathrm{optimal}})} \tag{14.3}$$

where π^c_{optimal} is the jointly optimal policy. $U_\Sigma = 1$ when actual matches optimal; $U_\Sigma = 0$ when actual matches independent (no coordination). Requires knowing the jointly optimal policy, which is itself a strong assumption.

Perceptual unity U_{obs} — how much of the observation stream is shared:

The fraction of total observation information that reaches all sub-agents. Full perceptual unity means all agents observe the same signals; zero means private observations only. Enables epistemic convergence without explicit model-sharing.

Definition

Update-rule homogeneity U_f — how similar the sub-agent update rules are:

$$U_f = 1 - d\Big(f_M^{(1)}, \dots, f_M^{(n)}\Big) \tag{14.4}$$

where d is a distance over the space of update operators $f_M : (M, o, a) \mapsto M'$, normalized so that $U_f = 1$ when all sub-agents implement the same correction rule and $U_f = 0$ at maximal heterogeneity. The choice of d is case-specific — for parametric Kalman-like updates, $d \propto |\Delta K|/K_{\max}$ on the gain parameter; for Bayesian updates with shared structural form but different priors, d tracks divergence between the induced kernels; for arbitrary f_M in function space, candidates include operator-norm distance, Fisher-information-weighted distance, or IB-style comparison.

Discussion

Where the four content unities measure shared *information* across sub-agents (state, objective, policy, observation), U_f measures shared *structure* — whether the agents instantiate the same update law. The two axes are independent: agents can share a model ($U_M = 1$) while updating it differently ($U_f < 1$), and conversely. In purposeful settings (G_t present), U_Σ partially absorbs structural variation in the policy half of the cycle, but the model-update half remains uncovered without U_f. In pure Part I composition (passive estimators, no G_t), U_f is the only handle on structural homogeneity. The closed-form linear-Gaussian instance — heterogeneous Kalman gains $\Delta K = K_1^* - K_2^*$ producing $\varepsilon_x \propto |\Delta K|$ — is derived in 14.1 §Two-axis structure.

The achievable component closure defect $\varepsilon_d^{\min}(k_d)$ under a projection of macro-dimension k_d is a function of *both* axes — the relevant content unity U_d and the structural unity U_f — together with the projection-dimension parameter:

$$\varepsilon_d^{\min}(k_d) = f_d\big(k_d;\ U_d,\ U_f\big) \tag{14.5}$$

monotone decreasing in each unity argument and monotone increasing in compression aggressiveness (smaller k_d). The form is derived in 14.1; in linear-Gaussian scalar cases it admits closed-form expressions for $d \in \{x, o, a\}$. U_O and U_Σ enter ε_a jointly rather than separately — collapsing alignment to a single scalar (U_O alone) would mis-route the repair, since execution-path divergence (U_Σ) cannot be fixed by re-aligning targets. This is the **anti-collapse** discipline (5.7) at the composite layer: two distinct quantities a naive reading merges, routing to different corrections.

Discussion-grade. Max attainable: empirical. The four content dimensions are qualitatively motivated by correspondence with the four components of agent state (M_t, O_t, Σ_t, and the observation channel); the structural dimension U_f is forced by the linear-Gaussian two-Kalman case (14.1 §Two-axis structure), where heterogeneous gains produce non-zero ε_x that no content dimension can register. The two-axis architecture (content $\times$ structure) is therefore a definitional commitment of this segment, not a heuristic.

The specific metrics are sketches. The information-theoretic formulations (U_M, U_Σ) are well-defined in principle but require specifying distributions and distance measures for practical computation. U_f is even less prescriptive — the choice of operator distance d is case-specific, and a general theory of structural-variation measures across arbitrary f_M classes is open.

The claim that the dimensions are *substantially independent* is a hypothesis, not derived. Epistemic unity may enable strategic unity (shared models allow coordination without explicit planning); content unity along U_M does not constrain U_f (agents can share a posterior while updating it differently). Independence holds approximately, with documented joint dependencies: (U_O, U_Σ) jointly control ε_a and cannot be separated; U_f enters all three closure components $(\varepsilon_x, \varepsilon_o, \varepsilon_a)$ as a structural multiplier on what content unity alone would predict.

Clausewitz's three gaps. These dimensions map to the gaps identified by Clausewitz (systematized by Bungay in *The Art of Action*):

Table 14.1

Clausewitz Gap	*Unity Dimension*	*Formal Quantity*
Knowledge gap	Epistemic unity (U_M)	$1 - U_M$: fraction of model not shared
Alignment gap	Teleological unity (U_O)	$1 - U_O$: objective misalignment
Effects gap	Strategic + Perceptual unity	$1 - U_\Sigma$ + observation routing costs

The mapping is not perfect — Clausewitz's "effects gap" blends action coordination with observation feedback — but it provides 200+ years of organizational evidence for the qualitative decomposition.

Connection to closure defect. The unity dimensions parametrize a rate-distortion relation with the component closure errors in 13.1, not a direct correspondence. The formal statement is in 14.1: the achievable closure-defect component $\varepsilon_d(k_d)$ under a projection of macro-dimension k_d decreases monotonically in both the relevant content unity U_d and the structural unity U_f, with closed-form expressions in the linear-Gaussian case. Qualitative direction along the content axis: U_M governs the compressibility of state information (ε_x), (U_O, U_Σ) jointly govern action compressibility (ε_a), U_{obs} governs observation compressibility (ε_o). The naive reading "high U_d predicts low ε_d" fails: for non-compressing projections (e.g., the means-only projection in the two-Kalman case) $\varepsilon_x = 0$ regardless of U_M, while for heterogeneous-gain projections $\varepsilon_x > 0$ even at perfect content correlation. Both observations point at the same correction — closure defect lives on a rate-distortion surface with two unity arguments, not a single one.

What each dimension's absence costs.

- Low U_M: prediction conflicts → uncoordinated actions based on contradictory beliefs. Internal mismatch component from model disagreement.
- Low U_O: strategic friction → sub-agents pursue conflicting sub-goals. Effort wasted or counterproductive.
- Low U_Σ: redundancy and gaps → two agents fix the same bug while a critical one goes unnoticed.
- Low U_{obs}: information silos → critical signals observed by one agent but not actionable by the composite.
- Low U_f: structural drift → even agents sharing identical models, objectives, observations, and policy targets produce divergent macro-trajectories under aggressive projection, because their corrections respond to the same evidence with different gains. The composite cannot be summarized at low macro-dimension without residual error scaling with the gain mismatch.

14.1 *Result:* Unity-to-Closure Rate-Distortion Mapping

Unity dimensions parametrize rate-distortion curves for closure-defect components, not point-valued predictors. The achievable closure-defect component ε_d under projection of macro-dimension k_d is monotone decreasing in both the relevant content unity U_d and the structural unity U_f (update-rule homogeneity), with higher unity along either axis lowering achievable defect at a given compression. Closed forms hold in the linear-Gaussian case; structural monotonicity survives more broadly. The two-axis structure (content × structure) is forced by the heterogeneous-Kalman case below and is reflected definitionally in `#def-unity-dimensions`.

Several closed-form linear-Gaussian instances drive the structural conclusions. *Observation closure defect* scales as $1 - U_{\text{obs}}$ in the two-agent scalar case under 1D projection: higher perceptual unity means observations are more redundant, a 1D summary suffices, and ε_o shrinks. *Action closure defect* scales as $\kappa^2(1 - U_O)$ for independent policies under 1D state projection, with policy coordination (strategic unity) providing a further multiplicative reduction — U_O and U_Σ enter ε_a *jointly* rather than separately. A striking structural fact for the *state* closure defect: under linear dynamics with consistent linear projections, $\varepsilon_x \equiv 0$ regardless of U_M or compression dimension. State closure becomes non-trivial only when the projection's range is non-invariant under the micro-dynamics, the projection is inconsistent, the micro-dynamics are nonlinear — or *sub-agent update rules are heterogeneous*. This is the structural fact that forces the two-axis decomposition: the content axis cannot detect structural heterogeneity in updates; the structural axis is required.

The heterogeneous-Kalman closed form makes this concrete: two Kalman filters with different gains produce $\varepsilon_x^2 \propto (\Delta K/2)^2$ — a defect proportional to the gain mismatch even when priors are identical, even at perfect content correlation. The same structural-unity mechanism lifts to the *strategy layer* via the credence-composition case: N agents reasoning over a shared plan skeleton, working in the log-odds coordinate forced by `#deriv-edge-update-natural-parameter`, produce ε_Σ^{*2} driven by the *variance of edge-update gains across agents* — homogeneous gains yield zero defect, gain dispersion drives the defect, and the result is *dimension-free in N*. The structural-unity axis is forced — not optional — for any framework that wants to predict closure defect from agent-level properties.

The conceptual consequence of the rate-distortion framing is that **unity is a compressibility parameter**, not a magnitude that maps to closure-defect magnitude. Higher unity allows more aggressive compression at the same closure defect, or equivalently, lower closure defect at the same compression. The rate-distortion *surface* is what unity controls; closure-defect values sit on that surface depending on how aggressively the composite is compressed. This is the same shape as Shannon rate-distortion theory and as the Information Bottleneck (`#form-information-bottleneck`) — composition trades representational compactness against fidelity-to-the-micro-system, and unity dimensions name where the trade is cheap (high unity, small defect achievable under aggressive compression) versus expensive (low unity, defect grows fast under any non-trivial compression).

Formal Expression.

Rate-distortion framing (general).

Formulation (unity-rate-distortion)

Fix a composite agent satisfying the admissibility conditions (A1)-(A4) in 13.1. For each content unity dimension U_d (with $d \in \{M, \Sigma, \text{obs}\}$, and U_O contributing jointly with U_Σ — see below) and the structural unity U_f (update-rule homogeneity, defined in 14), the achievable component closure defect under a projection whose corresponding macro-dimension is k_d satisfies:

$$\varepsilon_d^{\min}(k_d) = f_d\big(k_d;\ U_d,\ U_f\big) \tag{14.6}$$

where f_d is monotone decreasing in both unity arguments, monotone increasing in aggressiveness of compression (smaller k_d). The mapping from unity to closure-defect *magnitude* is via the shape of this rate-distortion surface; unity does not directly predict closure-defect value. In the linear-Gaussian Kalman case the structural argument reduces to $1 - U_f \propto |\Delta K|/K_{\max}$ on the gain mismatch.

Linear-Gaussian closed forms (two-agent scalar case).

Derived (obs-closure-linear-Gaussian, from unity-dimensions, composition-closure)

For two agents with scalar observations correlated at $\rho_{o,\text{eff}}$ (combining ρ_{env} and ρ_{obs}), under 1D principal-component projection of observations, the minimum achievable observation closure defect is:

$$\varepsilon_o^2(k_o = 1) = \sigma_o^2 \cdot \frac{1 - \rho_{o,\text{eff}}}{2} \ \propto\ 1 - U_{\text{obs}} \tag{14.7}$$

Higher perceptual unity → observations are more redundant → 1D summary suffices → ε_o small. Exact in the linear-Gaussian scalar case.

Derived (action-closure-independent-policies, from unity-dimensions, composition-closure)

For two agents with scalar quadratic objectives, independent LQR policies ($\rho_\Sigma = 0$), scalar targets r_1, r_2 with correlation ρ_O, under 1D state projection, the minimum achievable action closure defect is:

$$\varepsilon_a^2 \propto \kappa^2 \cdot (1 - \rho_O) \ \propto\ \kappa^2 \cdot (1 - U_O) \tag{14.8}$$

where κ is the scalar LQR gain. Adding policy coordination ($\rho_\Sigma > 0$) further reduces ε_a through a multiplicative factor. The joint (U_O, U_Σ) dependence takes the form:

$$\varepsilon_a^2 \propto (1 - U_O) \cdot f_1(U_\Sigma) + g(U_\Sigma) \tag{14.9}$$

with f_1 decreasing in U_Σ and g capturing residual strategic-misalignment error even when targets coincide.

Derived (state-closure-linear-Gaussian-trivial, from composition-closure)

State closure in linear-Gaussian.

For linear-Gaussian micro-dynamics with consistent linear projections Λ_x and Λ_o (the macro observation projection is the same linear combination as the macro state projection), the state closure defect vanishes:

$$\varepsilon_x = 0 \tag{14.10}$$

regardless of U_M or compression dimension. Linear projections of linear dynamics are exact *when the range of* Λ_x *is invariant under the micro-dynamics matrix.* ε_x becomes non-trivial when:

- the projection's range is non-invariant under the dynamics matrix — even with linear dynamics, consistent projections, and homogeneous updates, cross-coordinate coupling or anisotropic noise scales that mix macro-subspace components with their orthogonal complement give $\varepsilon_x > 0$ (the Mori-Zwanzig zero-lag bound $\varepsilon^* \geq \|Q_\Lambda U P_\Lambda\|_{\text{op}}$ in 13.1 is the general expression of this obstruction),
- the projection is inconsistent (macro state and macro observation projections disagree),
- the micro-dynamics are nonlinear, or
- sub-agent update rules are heterogeneous (see Two-axis structure below).

Derived (two-axis-structure, from composition-closure, linear-Gaussian case)

Two-axis structure (update heterogeneity).

In the non-degenerate linear-Gaussian case with heterogeneous sub-agent update rules — e.g., two Kalman filters with different gains $K_1^* \neq K_2^*$ tracking correlated processes, projected to the 1D sum $\hat{\omega}_+ = (\hat{\omega}_1 + \hat{\omega}_2)/\sqrt{2}$ — the state closure defect has the closed form:

$$\varepsilon_x^2 = (\Delta K/2)^2\big[S_- - C_{+-}^2/S_+\big] \tag{14.11}$$

where $\Delta K = K_1^* - K_2^*$, $S_\pm$ are the innovation variances in the $\pm$ directions, and C_{+-} is their cross-covariance.

This exhibits two independent drivers of ε_x, one along each unity axis of 14:

1. **Content unity** (U_M, via process correlation ρ): higher correlation → lower ε_x.
2. **Structural unity** (U_f, via gain mismatch ΔK): when $\Delta K = 0$ (i.e., $U_f = 1$), $\varepsilon_x = 0$ at every ρ; when $\Delta K \neq 0$, $\varepsilon_x > 0$ even at perfect content correlation.

The four content unities measure shared information (goals, policies, observations, model state); U_f measures whether sub-agents implement the same correction rule. The two axes contribute to the closure-defect rate-distortion surface independently — content unity controls compressibility of what the agents agree on; structural unity controls whether projection induces memory by mixing the discarded subspace into the retained one.

Strategy-layer instance (credence composition). The same structural-unity axis has an exact closed form one level up, at the strategy layer. For N agents reasoning over a shared plan skeleton (V, E) and disagreeing only on edge-credences, work in the log-odds coordinate $\lambda_{ij} = \log \frac{p_{ij}}{1-p_{ij}}$ — the unique additive-evidence coordinate (8.3; forced by `#deriv-edge-update-natural-parameter`) in which edge updates are additive and the natural macro-projection is the log-odds centroid. The per-step closure defect is then driven by update-rule heterogeneity exactly as the Kalman case is, with the per-agent edge gains $\eta_{\Sigma,i}$ playing the role of the Kalman gains:

$$\varepsilon_\Sigma^{*2} = |E| \cdot \overline{\mathrm{Var}_i[\eta_{\Sigma,i}]} \cdot \mathrm{Var}[r], \tag{14.12}$$

where r is the per-edge evidence residual. This is the structural-unity (U_f) axis lifted from state to strategy: homogeneous gains ($\overline{\mathrm{Var}_i[\eta_{\Sigma,i}]} = 0$, $U_f = 1$) give $\varepsilon_\Sigma = 0$; gain dispersion across agents drives the defect, scaled by plan size $|E|$ and evidence-residual variance. It is *dimension-free in* N — a population variance is estimated more precisely, not enlarged, by adding agents from the same gain-distribution — the strategy-layer twin of the dimension-free state-composition regime. The complementary heterogeneous-*topology* case (incompatible shared sub-orders, an order-theoretic non-existence rather than a magnitude) is the SCC-condensation defect landed in 8.3's causal-abstraction composition subsection; the two together exhaust strategy-layer composition (credence axis here, topology axis there).

Epistemic Status.

Conditional. Max attainable: *exact* (linear-Gaussian scalar cases) to *robust qualitative* (general).

- The observation and action closed forms are *exact* in the linear-Gaussian scalar case with stated projection choices.
- The state closure form $\varepsilon_x^2 = (\Delta K/2)^2[S_- - C_{+-}^2/S_+]$ is *exact* in the two-Kalman heterogeneous case.
- The strategy-layer credence-composition form $\varepsilon_\Sigma^{*2} = |E| \cdot \overline{\mathrm{Var}_i[\eta_{\Sigma,i}]} \cdot \mathrm{Var}[r]$ is *exact* in the fixed-topology heterogeneous-credence case (log-odds additivity is forced by `#deriv-edge-update-natural-parameter`) and dimension-free in N.
- The rate-distortion framing (unity as compressibility parameter rather than direct predictor) is *robust qualitative* — it survives beyond linear-Gaussian, but concrete rate-distortion curves require case-by-case derivation.
- The joint $(U_O, U_\Sigma) \to \varepsilon_a$ formula is a *sketch* — the leading structure is derived; the precise forms of f_1 and g are mechanical extensions not fully computed here.

Ceiling-limiting factors: non-Gaussian cases require information-theoretic bounds (Gaussian IB is fully tractable; general IB is not), and the structural-unity axis U_f has worked closed forms in two cases — the linear-Gaussian Kalman gain-mismatch case (state layer) and the fixed-topology credence-composition case (strategy layer, via log-odds additivity) — while a general theory of f_M structural variation across arbitrary update operators is open.

Discussion.

Why a one-axis reading fails. A "high U_M predicts low ε_x" reading is wrong in the two-Kalman case with the standard means-only projection: $\varepsilon_x \equiv 0$ for every correlation value, irrespective of U_M. The closure-defect surface depends on the projection choice, on the content-unity axis, and on the structural-unity axis U_f — high content unity with mismatched update rules still produces $\varepsilon_x > 0$, while low content unity under a non-compressing projection still produces $\varepsilon_x = 0$. The rate-distortion framing is what makes the multi-parameter dependence explicit.

Connection to the Information Bottleneck (2.2). The rate-distortion shape is not coincidental. Projection admissibility condition (P1) in 13.1 is the Lagrangian-dual of the IB constraint: the projection sits on or above the IB frontier at rate $I(X;T) \le I_{\max}(\varepsilon_I)$ for the relevance variable "next observation given action" (`#disc-compression-operations` supplies the derivation). Unity dimensions — measured as mutual-information-like quantities between sub-agent state components — parametrize the frontier's shape. The four AAT compression operations (M_t, Σ_t, shared intent, Λ) share IB shape but are not shown to reduce to a single master problem (U-medium, per `#disc-compression-operations`); cross-instance theorems do not follow from shared shape alone. (P2) Lipschitz continuity is not naturally IB and remains a separate admissibility condition; (P3) dimensional reduction remains separate in the Gaussian case. The Gaussian-IB closed form applies to linear-Gaussian composition setups; beyond them, the IB frontier is definitional but requires variational or numerical approximation.

Two-axis structure. The unity profile in 14 decomposes into a content axis (four dimensions: U_M, U_O, U_Σ, U_{obs}) measuring shared information, and a structural axis (U_f) measuring shared correction rules. In purposeful-agent settings (G_t present), U_Σ already absorbs structural variation in the policy half of the cycle — agents with different action laws have different effective policies — but the model-update half remains uncovered without U_f. In pure Part I composition (passive estimators, no G_t), U_f is the only handle on structural homogeneity, and the heterogeneous-Kalman case in this segment is the canonical instance where it bites.

Interpretation of "low closure defect." Unity controls the rate-distortion curve; low closure defect is achievable with aggressive compression when unity is high. But closure defect alone does not measure composite *optimality* (see 13.1 §5.1): two independent Kalman filters can have $\varepsilon^* = 0$ (perfectly representable) while failing to exploit cross-correlations (suboptimal relative to a joint filter). The rate-distortion mapping is about representability, not optimality.

14.2 *Definition:* Shared Intent

When sub-agents within a composite must coordinate, they face a communication problem: transmitting the full purposeful state $G_t = (O_t, \Sigma_t)$ is expensive (high bandwidth, high latency), but acting without any shared purpose wastes coordination potential. The Information Bottleneck (`#form-information-bottleneck`) applied to inter-agent communication predicts an optimal compression: transmit enough of G_t to align behavior, not more. The **shared intent** is the IB-optimal compression of the sender's purposeful state — the minimal sufficient statistic for predicting the jointly optimal coordination behavior. The trade-off parameter controls the regime: at high weight, the agent communicates more detail (approaching full model sharing); at low weight, communication is minimal (approaching independent action).

The IB compression *preferentially preserves* three things, in order: **terminal objectives** (what the agent is trying to achieve — compact, slow-changing), then **high-level strategy** (which approach, not which specific steps — moderate size, moderate change rate), and last **strategic details** (specific edge credences in the strategy DAG — large, change fast, low coordination value). This gives the framework a structural reading of **commander's intent** in military doctrine — the commander communicates *what* to achieve and *why*, not *how*. The framework predicts this pattern *structurally*, from the information-bottleneck trade-off, not as a managerial convention: communicating objectives gives a long shelf life per bit transmitted when objectives change slowly and strategies change fast.

For agents with bounded communication capacity (bandwidth-limited channels, finite context windows), the strategy DAG must be summarized for transmission — a 500-node DAG cannot be shared in full; the IB compression identifies which structural features matter for coordination. The same compression is useful for an agent preserving its own state across context boundaries (the language-model 100% context turnover problem) — store the compressed version of the purposeful state, not the full state. The status is honestly *discussion-grade*: the formulation makes several strong assumptions (the sender must know the jointly optimal action, which requires knowing other agents' states; the compression is lossless in the IB sense; the trade-off parameter is fixed rather than dynamically adjusted), and the qualitative prediction (communicate purpose before plans before models) is more robust than the specific IB formulation.

Formal Expression.

Definition (shared-intent)

Let G_t^{full} be the source agent's complete purposeful state (O_t, Σ_t). Let G_t^{shared} be the compressed representation communicated to partners. The shared intent is the IB-optimal compression:

$$G_t^{\text{shared}} = \arg\min_{G_s} \left[I(G_t^{\text{full}}; G_s) - \beta \cdot I(G_s; a_t^{\text{coordinated}}) \right] \tag{14.13}$$

where $a_t^{\text{coordinated}}$ is the jointly optimal action and β controls the complexity-relevance tradeoff. At high β, the agent communicates more detail (approaching full model sharing). At low β, communication is minimal (approaching independent action).

The shared intent is the *minimal sufficient statistic* of the sender's purposeful state for predicting the jointly optimal coordination behavior.

Epistemic Status.

Discussion-grade. Max attainable: conditional (conditional on the IB framework being appropriate for inter-agent communication). The application of IB to inter-agent communication is structurally motivated — IB compresses optimally given a relevance criterion, and coordination relevance is the natural criterion — but the specific formulation assumes: (1) the sender knows the jointly optimal action (which requires knowing other agents' states), (2) the compression is lossless in the IB sense (real communication introduces noise, delay, and misinterpretation), (3) the β parameter is fixed rather than dynamically adjusted. These are strong assumptions. The qualitative prediction (communicate purpose before plans before models) is more robust than the specific IB formulation.

Discussion.

What gets compressed out. The IB compression preferentially preserves: 1. Terminal objectives (what the agent is trying to achieve) — these are compact and change slowly 2. High-level strategy (which approach, not which specific steps) — moderate size, moderate change rate 3. Strategic details (specific edge credences in Σ_t) — large, change fast, low coordination value

Connection to cognitive cost of Σ_t. For agents with bounded communication capacity (bandwidth-limited channels, finite context windows), the DAG must be summarized for transmission. A 500-node strategy DAG cannot be shared in full; the IB compression identifies which structural features of the DAG matter for coordination.

Organizational communication patterns. Commander's intent in military doctrine is an empirical instantiation: the commander communicates *what* to achieve and *why*, not *how*. This is IB-optimal if objectives change slowly (low ν_O) and strategies change fast (high ν_Σ) — communicating objectives gives a long shelf life per bit transmitted.

14.3 *Hypothesis:* Auftragstaktik Principle

A bandwidth-allocation hypothesis grounded in the shared-intent IB framework (`#def-shared-intent`). For a composite agent with limited communication bandwidth, the optimal allocation prioritizes sharing **objectives over strategies over models** — $B_O > B_\Sigma > B_M$. This captures the structural insight of *Auftragstaktik* (mission-type tactics): investing communication bandwidth in shared purpose (teleological unity) while accepting lower epistemic and strategic unity, granting sub-agents autonomy to adapt locally. The model predicts the same priority ordering that military doctrine discovered empirically over centuries — and the framework offers it as a *structural prediction*, not an organizational convention.

The priority ordering follows from the IB framework: the bits with the longest shelf life and highest coordination value per bit should be transmitted first. Objectives change slowly and enable autonomous coordination — sub-agents who share objectives can independently choose compatible strategies. Models change fast and provide diminishing coordination value — two agents with the same model but different objectives still conflict. The ordering holds under stated conditions: objectives change slowly relative to strategies, which change slowly relative to models; sub-agents have sufficient local adaptive capacity to maintain their own models.

The framework is also explicit about *when the ordering reverses*. When the environment is genuinely ambiguous and local observations are insufficient (fog of war, novel codebase, unprecedented market conditions), model synchronization may be worth more than objective sharing — sub-agents who share the same wrong model at least err consistently, which is sometimes better than each having a different wrong model. The framework predicts the reversal as a *regime-dependent prescription* rather than a universal ordering.

The status is *discussion-grade*. The priority ordering is a qualitative prediction supported by extensive military-organizational evidence (Bungay's analysis of Clausewitz, Wehrmacht doctrine, modern mission command), but it is not derived — the IB optimization would need to be solved explicitly with realistic cost functions to confirm the ordering. The empirical evidence is strong but comes primarily from one domain (military command); generalization to software teams, AI agent swarms, and other settings is plausible but unverified. Whether the IB-optimal-bandwidth-allocation *mechanism* is actually the reason Auftragstaktik works empirically — versus other mechanisms (psychological, organizational, traditional) — is an open question.

Formal Expression.

Hypothesis (auftragstaktik-principle)

Let a composite agent's total inter-agent communication bandwidth be $B = B_O + B_\Sigma + B_M$, allocated across objective sharing (B_O), strategy coordination (B_Σ), and model synchronization (B_M).

The hypothesis: the allocation that maximizes composite tempo $\mathcal{T}_c$ (or equivalently, minimizes coordination overhead C_{coord}) prioritizes:

$$B_O > B_\Sigma > B_M \tag{14.14}$$

when: - Objectives change slowly relative to strategies: $\nu_O \ll \nu_\Sigma$ - Strategies change slowly relative to models: $\nu_\Sigma \ll \nu_M$ - Sub-agents have sufficient local adaptive capacity: each $\mathcal{T}_i > \rho_i^{\text{local}}/\|\delta_{\text{critical}}^i\|$

The priority ordering follows from the IB framework (14.2): the bits with the longest shelf life and highest coordination value per bit should be transmitted first. Objectives change slowly and enable autonomous coordination (sub-agents who share objectives can independently choose compatible strategies). Models change fast and provide diminishing coordination value (two agents with the same model but different objectives still conflict).

Epistemic Status.

Discussion-grade. Max attainable: empirical. The priority ordering is a qualitative prediction grounded in the IB framework and supported by extensive military-organizational evidence (Bungay's analysis of Clausewitz, Wehrmacht doctrine, modern mission command). But it is not derived — the IB optimization would need to be solved explicitly with realistic cost functions to confirm the ordering, and the conditions under which the ordering reverses (e.g., when model synchronization is critical because the situation is genuinely ambiguous) are not characterized. The empirical evidence is strong but comes primarily from one domain (military command); generalization to software teams, AI agent swarms, and other settings is plausible but unverified.

Discussion.

When the ordering reverses. The prioritization $B_O > B_\Sigma > B_M$ assumes sub-agents can independently construct adequate local models. When the environment is genuinely ambiguous and local observations are insufficient (fog of war, novel codebase, unprecedented market conditions), model synchronization may be worth more than objective sharing — sub-agents who share the same wrong model at least err consistently, which is sometimes better than each having a different wrong model.

Bungay's evidence. In *The Art of Action*, Bungay documents that organizations consistently fail by inverting this priority: they over-invest in controlling *how* subordinates act (strategy sharing, B_Σ) rather than ensuring subordinates understand *why* (objective sharing, B_O). The result: subordinates who follow instructions precisely but cannot adapt when conditions change, because they lack the teleological context to improvise intelligently.

The software team instantiation. A well-functioning development team has: - High B_O: clear product goals, understood by all (sprint goals, feature objectives) - Moderate B_Σ: architectural decisions shared, implementation details autonomous - Low B_M: each developer builds their own mental model of the code they touch; full codebase understanding is neither expected nor efficient

When this inverts (micromanagement of implementation details, unclear product goals), team tempo drops — consistent with the Auftragstaktik prediction.

Connection to Conway's Law. Conway's Law (system structure mirrors communication structure) is a consequence: when B_Σ is low and B_O is high, sub-agents coordinate through shared objectives rather than explicit action coordination, producing systems whose boundaries reflect objective decomposition rather than communication channels.

14.4 *Hypothesis:* Communication Gain

The extension of the uncertainty-ratio gain principle (`#emp-update-gain`) to *inter-agent* communication channels. When an agent incorporates information from *another agent* (rather than from direct observation), the optimal update gain extends with additional terms in the denominator: **communication channel noise** (latency, ambiguity, compression loss, bandwidth limits), **source quality uncertainty** (the receiver's uncertainty about the sender's model calibration and domain competence), and **teleological-unity uncertainty** (the receiver's uncertainty about whether the sender's communications serve the receiver's interests or the sender's potentially conflicting objectives). When all additional terms are zero (perfect channel, calibrated and aligned source) the gain heads to 1 (full trust); when any term is large the gain heads to 0 (ignore the signal). When the "sender" is the environment (direct observation), the source-quality and alignment terms vanish and the standard single-agent gain formula is recovered.

The three additional denominator terms have *different natures*. Channel noise is a property of the *channel* — improvable by infrastructure. Source quality is a property of the *source* — improvable by the sender improving its model, or estimable by the receiver through calibration tracking. Alignment uncertainty is a property of the *relationship* — the game-theoretic variable. The three are independent levers; the framework's prescription is to track them separately.

The receiver's estimates of source quality and alignment constitute a **trust meta-model** — a model of models. This meta-model is itself subject to AAT's full apparatus: it has its own mismatch (trust prediction errors), should be updated with appropriate gain (not overreacting to single disagreements), and can be structurally inadequate (the agent's trust model class may not capture the actual reliability structure of its sources, per `#result-structural-adaptation-necessity`). The framework also names **risk-asymmetric trust**: the Bayesian posterior on source reliability gives the best *estimate*, but the *decision* about how much to trust should be risk-weighted. Trusting a deceptive agent in a high-stakes interaction can cause catastrophic model corruption (the adversarial-destabilization effects spiral, `#der-adversarial-destabilization`); mild miscalibration toward a reliable source causes only small ongoing inefficiency. For high-stakes interactions, the framework prescribes using a *conservative quantile* of the trust posterior rather than the mean. **Trust transitivity** is handled via a Bayesian mixture: when no direct experience exists with a candidate source but a trusted intermediary provides an assessment, the recommendation is discounted by the intermediary's own reliability — giving a principled three-phase trust formation (prior → transitive update → direct experience, which eventually swamps the prior).

The framework is explicit about a limitation: the additive denominator treats all uncertainty sources as independent zero-mean noise — appropriate for channel noise and source miscalibration, but *misses the adversary's actual strategy* of presenting as trustworthy to exploit high gain. The additive model captures the *defender's* response to detected misalignment; it does not model the *attacker's* optimization over the defender's trust dynamics. The status is honestly *hypothesis-discussion-grade* — the additive heuristic correctly drives the gain toward zero when alignment uncertainty is high, but a full treatment of the adversarial case requires game-theoretic equilibrium analysis external to AAT (AAT provides the state variables; the equilibrium analysis is external).

Formal Expression.

Hypothesis (communication-gain)

$$\eta_{ji}^* = \frac{U_{M_i}}{U_{M_i} + U_{o,ji} + U_{\text{src},j} + U_{\text{align},ji}} \tag{14.15}$$

where: - U_{M_i}: agent i's model uncertainty (same as 3.6) - $U_{o,ji}$: **communication channel noise** — latency, ambiguity, compression loss, bandwidth limitations of the channel between j and i - $U_{\text{src},j}$: **source quality uncertainty** — i's uncertainty about j's model calibration and domain competence - $U_{\text{align},ji}$: **teleological-unity uncertainty** — i's uncertainty about whether j's communications serve i's interests or j's potentially conflicting objectives

When all additional terms are zero (perfect channel, calibrated and aligned source): $\eta_{ji}^* \to 1$ (full trust). When any term is large: $\eta_{ji}^* \to 0$ (ignore the signal).

Connection to single-agent case. When j is the environment (direct observation): $U_{\text{src}} = U_{\text{align}} = 0$, recovering 3.6's standard form $\eta^* = U_M/(U_M + U_o)$.

Epistemic Status.

Hypothesis. The additive denominator treats all uncertainty sources as independent, zero-mean noise — a structural heuristic, not a strict variance derivation. This is appropriate for $U_{o,ji}$ (channel noise) and $U_{\text{src},j}$ (miscalibration), which are typically unstructured. For $U_{\text{align},ji}$ (deception), additivity is conservative: it correctly drives η^*_{ji} toward zero when teleological-unity uncertainty is high, but misses the adversary's *actual* strategy — presenting as trustworthy to exploit high η^*_{ji}. The additive model captures the *defender's* response to detected misalignment; it does not model the *attacker's* optimization over the defender's trust dynamics.

All four uncertainty terms must be expressed in a **common predictive-dispersion scale** before summation — the same units as U_{M_i} (variance of the predictive distribution over the observed quantity). When hard to estimate directly, a conservative approximation: set $U_{\text{src}} + U_{\text{align}}$ to the empirical variance of j's past prediction residuals as observed by i, minus the known channel noise $U_{o,ji}$.

Discussion.

The denominator terms have different natures. $U_{o,ji}$ is a property of the *channel* — improvable by infrastructure. $U_{\text{src},j}$ is a property of the *source* — improvable by j improving its model, or estimable by i through calibration tracking. $U_{\text{align},ji}$ is a property of the *relationship* — the game-theoretic variable.

Trust calibration as a meta-model. Agent i's estimates of $U_{\text{src},j}$ and $U_{\text{align},ji}$ constitute a **trust meta-model** — a model of models. This meta-model is itself subject to AAT's full apparatus: it has mismatch (trust prediction errors), should be updated with appropriate gain (not overreacting to single disagreements), and can be structurally inadequate (4.8 — the agent's trust model class may not capture the actual reliability structure of its sources).

Risk-asymmetric trust. The Bayesian posterior on source reliability gives the best *estimate*, but the *decision* about how much to trust should be risk-weighted. Trusting a deceptive agent (HILP — high impact, low probability) can cause catastrophic model corruption (15.2, effects spiral). Mild miscalibration toward a reliable source (LIHP) causes small ongoing inefficiency. For high-stakes interactions, use a conservative quantile of the trust posterior rather than the mean — require more evidence before granting high trust.

Trust transitivity. When agent i has no direct experience with agent k, but trusted intermediary j provides an assessment, the transitive trust question arises. A Bayesian mixture model discounts the recommendation by the intermediary's own reliability:

$$P_i(\theta_k \mid s_j) \propto \left[r_{ji} \cdot P(s_j \mid \theta_k) + (1 - r_{ji}) \cdot P_0(s_j)\right] \cdot P_i(\theta_k) \tag{14.16}$$

where r_{ji} is i's reliability estimate of j and θ_k is k's true alignment. When $r_{ji} \to 0$, the posterior collapses to the prior (no update); when $r_{ji} \to 1$, the full informative likelihood applies. This model gives a principled three-phase trust formation: prior → transitive update → direct experience (which eventually swamps the prior).

14.5 *Hypothesis:* Solicitable Escape from an Absorbing Region

The three escape routes from an absorbing observability region are not peers: two require no prior localization of the blind spot and one does, so where observability is estimated rather than known by construction, and under any resource bound, the only *solicitable* escape from a self-locking region is a differently-positioned observer.

Formal Expression.

9.3 establishes that unobservable regions of strategy are absorbing — frozen beliefs generate no mismatch signal, so no revision is triggered, and the agent "cannot learn and cannot recognize that it cannot learn." It names three escapes:

external shock, proactive observability investment (instrumenting previously unmonitored nodes), and another agent whose observations cover the blind spot (14.4).

Partition those routes by what the agent must already possess for the route to be *available to it as an action*.

[Definition (route-targeting)] A route is **targeted** if exercising it requires the agent to identify, in advance, the node or direction at which its observability is deficient. A route is **untargeted** otherwise.

[Hypothesis (solicitable-escape)] Of the three routes:

- **External shock** is untargeted and *unsolicitable*. It requires no prior localization, because the shock supplies the mismatch signal the frozen region cannot generate; but it also cannot be chosen — it is a thing that happens to the agent, not an action available to it.
- **Observability investment** is *targeted*, under the scope condition below. To instrument node v the agent must represent v as a node whose σ_v is worth raising. The absorbing region is case (c) of the zero-aporia trichotomy (3.4): near-zero mismatch because the channel is too noisy to detect model error, which is not distinguishable from within from case (a), a model that genuinely fits — *silence can mean peace or deafness*. What blocks the distinction is not absence of signal but the agent's inability to audit its own weighting: by 1.3 the agent does not know the noise distribution and must estimate U_o from innovations, and at a node with $\sigma_v \approx 0$ those innovations are exactly what is suppressed, so the estimate is unimprovable by observing v. The gain-collapse modes compound this — $\eta^* \to 0$ via $U_M \to 0$ (dogmatism) and via $U_o \to \infty$ (nihilism) are behaviorally identical (3.6), so even an agent that correctly registers "my gain here is zero" cannot read off whether the fault lies in the world or in itself. So for the blind spots that are actually self-locking, the precondition of the route is unavailable.
- **Another observer** is both untargeted *and* solicitable. A differently-positioned party's coverage does not require the blinded agent to know where its own deficiency lies — the coverage is a property of the other's position, not of the blinded agent's model — and soliciting it is an action the agent can take under its existing representational resources.

Scope (innovation-estimated observability)

The targeting claim above holds where the agent's knowledge of σ_v is **estimated from its own observations**. It fails where σ_v is known **architecturally** — an agent that knows by construction which nodes it has no sensor for can rank exactly those nodes without any signal from them, and for it the self-instrumentation route is both choosable and aimable. This is the same boundary flagged against 1.3's epistemic-opacity clause by several de-novo readers (a Kalman filter with a known noise covariance is handed its observation law rather than estimating it), so the scope condition tracks a distinction the framework already recognizes rather than introducing one. The interesting case is the estimated one, because that is where the region self-locks; the architecturally-known case is a design opportunity, not a trap.

[Derived (conditional on a resource bound)] The apparent counterexample is *blanket* instrumentation: raising σ_v everywhere requires no targeting. Blanket instrumentation is untargeted, and it does escape. But its cost scales with the node set while the value of instrumenting any particular node is the difference in sector parameters that node contributes (9.3's investment tradeoff), so under a bound on instrumentation capacity the agent must rank nodes — which reintroduces targeting. The claim is therefore conditional: *given* an instrumentation budget strictly below what blanket coverage requires, observability investment degrades to a targeted route and is unavailable for the self-locking case.

The consequence, stated as the hypothesis: **within an absorbing region, the only escape that is both untargeted and choosable is a differently-positioned observer.** Shock escapes but cannot be chosen; self-instrumentation can be chosen but — under estimated observability — not aimed.

Epistemic Status.

Discussion-grade. The argument rests on the absorbing condition as 9.3 states it (`robust-qualitative` there, and nothing here upgrades it) and on the targeting precondition, which is now argued from three upstream commitments rather than asserted: the (a)-vs-(c) ambiguity of 3.4, the epistemic opacity of 1.3, and the behavioral identity of the two gain-collapse modes in 3.6. That argument is assembled here in prose, not derived — which is what keeps the segment at `discussion-grade` rather than `conditional`.

Max attainable: conditional. Two steps remain, both now specific. The resource bound in the clause below is stated informally; formalizing it against the investment-tradeoff expression in 9.3 would make the blanket-instrumentation dismissal a derivation rather than an argument. And the targeting precondition wants the chain above written as a derivation: that an agent's estimate of σ_v is unimprovable by observation at v when $\sigma_v \approx 0$, so cases (a) and (c) are not separable from inside under estimated observability. The ingredients are all in canon and none of them is new; what is missing is the worked argument, which would also close the exception in the scope clause by stating exactly what architectural knowledge supplies that estimation cannot.

One thing this segment must not be read as claiming. The absorbing region is *not* one where the agent receives no signal. Observations continue; what collapses is the weight they carry ($\eta_{\text{edge}} \to 0$ via $U_{\text{obs}} \to \infty$). A zero-information channel is outside AAT's scope entirely (1.1, 1.5) — with no signal there is no mismatch, hence no correction and nothing to analyze — so reading the absorbing condition as blindness does not strengthen this segment, it removes its subject.

Scope, and what this is not. The result is about *strategy-edge observability*, which is the scope 9.3 carries; it says nothing directly about comprehension between agents, and any transfer to that setting is a further step this segment does not take. The other-observer route depends on 14.4, whose status is `discussion-grade`, so nothing this segment says about that route can be stronger than discussion-grade regardless of how the argument here is tightened — the two promotion steps above would raise the *partition*, not the route it favors. Nothing here asserts that a differently-positioned observer is *sufficient* for escape — only that it is the one route whose availability does not presuppose what the absorbing condition removes.

Falsifier. Exhibit a self-locking region escaped by self-instrumentation where the instrumented node was selected without prior signal distinguishing it from settled nodes — or show that some third mechanism supplies such a signal from inside the region, which would defeat the targeting premise directly.

Discussion.

The partition sharpens rather than softens the practical reading of 9.3. That segment's advice to prefer a weak-but-visible path over a strong-but-blind one is guidance available *before* a region becomes absorbing; it is a prophylactic. Once the region is absorbing, the same segment's own escape list is what remains, and this partition says which member of it an agent can actually reach for.

It also explains an asymmetry in how the three routes feel from inside, which is otherwise puzzling. An agent in an absorbing region has, by construction, no experience of being stuck; the region presents as settled knowledge rather than as a gap. That phenomenology is consistent with all three routes being nominally listed and only one being usable: the agent is not withholding effort, and more effort is genuinely not the missing ingredient — what is missing is a standpoint the agent does not occupy, and the only action that reaches one is to involve a party who does.

Two connections worth noting without leaning on them. The instrumentation route's failure has the same shape as the diagnostic-resolution loss the parent segment describes: an unobservable intermediate destroys per-edge identification and forces plan-level aggregation, so the agent knows a plan is failing without localizing the step — which is the targeting deficiency seen at a finer grain. And the partition gives a formal reason why a network of differently-positioned agents is not merely an efficiency over a solitary one: for the specific class of blind spots that are self-locking, it is the only member of the escape set that an agent can elect.

14.6 *Discussion:* Additional Implications & Discussion

The chapter sits between Ch.2's composition machinery (closure defect, bridge lemma, wrapping construction) and Ch.4's cooperative-vs-adversarial coupling, and its job is to give *unity* a quantitative content rather than leaving it as a verbal property of composites that work. Unity decomposes into named axes that parametrize a rate-distortion curve for closure defect; shared intent decomposes into an IB-optimal compression of the sender's purposeful state; communication gain decomposes the Bayesian update for inter-agent channels into source-noise, source-competence, and source-alignment terms. None of the five segments treated here carries a `## Findings` section, so the implications segment is their canonical catalog home — but the work is mostly synthesis: surfacing what the chapter delivers as a chapter, what's hypothesis-grade rather than derived, and what bridges into Ch.4 and the rest of Part III.

Proof provenance. Of the five chapter sources treated here (the chapter's sixth row, `#hyp-solicitable-escape`, is exploratory and out of this segment's scope), only `#result-unity-closure-mapping` carries derived content — *conditional*, with four *exact* closed forms (observation and action closure in the two-agent linear-Gaussian scalar case; state closure in the two-Kalman heterogeneous-gain case; strategy-layer credence composition in the fixed-topology case) and a rate-distortion *framing* that is robust-qualitative beyond them. `#def-unity-dimensions`, `#def-shared-intent`, and `#hyp-auftragstaktik-principle` are discussion-grade; `#hyp-communication-gain` is a hypothesis whose single-agent limit recovers `#emp-update-gain` (exact in the Fisher-local regime, robust-qualitative outside it). Each implication below is stated at the tier of the weakest source it uses, and the forward bridges into Ch.4, Ch.5, and Appendix A are marked as forward references — their homes come later in the OUTLINE.

Unity dimensions and the rate-distortion content of "well-coordinated".

`#def-unity-dimensions` decomposes unity into a *content axis* (four dimensions — U_M, U_O, U_Σ, U_{obs} — measuring sub-agent agreement on models, objectives, strategies, and observations respectively) and a *structural axis* (U_f, update-rule homogeneity). The implications-side payoff comes from `#result-unity-closure-mapping`: unity dimensions are not point-valued predictors of composite quality; they parametrize *rate-distortion curves* for closure-defect components. The achievable closure-defect component ε_d under projection of macro-dimension k_d is monotone decreasing in both the relevant content unity U_d and the structural unity U_f — higher unity along either axis lowers achievable closure defect at a given compression. The tiering is worth holding exactly: the closed forms are *exact* in their linear-Gaussian (and fixed-topology) instances; the rate-distortion framing — unity as a compressibility parameter rather than a defect predictor — is *robust-qualitative* beyond them; and monotonicity in U_f is established only in the two worked instances (Kalman gain mismatch at the state layer, edge-gain dispersion at the strategy layer), because U_f's operator distance d is case-specific and a general theory of structural variation across arbitrary update operators is open.

The two-axis structure is not aesthetic — it is *forced* by the heterogeneous-Kalman case, and the source carries the forcing as an exact closed form. Two agents can have identical content unity (same models, same objectives, same observations) and still produce composite closure defect if their update *machinery* differs: $\varepsilon_x^2 = (\Delta K/2)^2\,[S_- - C_{+-}^2/S_+]$ for Kalman gains $K_1 \neq K_2$, non-zero at perfect content correlation (non-degenerate case). The same axis has an exact twin one level up — N agents on a shared plan skeleton with dispersed edge-update gains give $\varepsilon_\Sigma^{*2} = |E| \cdot \overline{\text{Var}_i[\eta_{\Sigma,i}]} \cdot \text{Var}[r]$, dimension-free in N. The content axes alone miss both; the structural-unity term picks them up. The framework's prediction is concrete: composite agents with high content unity but low structural unity will show closure-defect signatures the content-only analysis cannot diagnose, and the remedy is *update-rule homogenization* (synchronizing how sub-agents update, not just what they believe). A candidate reading for multi-agent ML — offered as illustration, not as a derived prediction — is that an ensemble of identically-trained models and an ensemble whose members were trained under different optimizers or schedules differ in composite behavior through exactly this channel; making it a test requires exhibiting a content-unity measurement that comes out equal across the two ensembles while the update operators differ, which has not been done.

The composition with Ch.2's closure-defect framework is the implication that anchors Part III's coordination story. Unity's rate-distortion content is *what makes the bridge lemma actionable*: the lemma converts predictive closure defect into trajectory error, and unity tells the composite-design practitioner *which dimensions to invest in* to reduce ε^*. The routing is tiered like the source. Investing in U_O (objective alignment) reduces ε_a through the form $\varepsilon_a^2 \propto \kappa^2(1-U_O)$ — exact for two agents with scalar quadratic objectives, *independent* LQR policies ($\rho_\Sigma = 0$), under 1D state projection; investing in U_Σ (strategy coordination) reduces it further through a multiplicative factor $f_1(U_\Sigma)$ whose leading structure is derived but whose precise form is a sketch; investing in U_f (update homogeneity) is worked for the state and strategy layers, and `#def-unity-dimensions` commits definitionally to U_f entering all three components ($\varepsilon_x, \varepsilon_o, \varepsilon_a$) as a structural multiplier — a commitment, not yet a derivation, for ε_o and ε_a.

Shared intent and the Auftragstaktik bandwidth allocation.

`#def-shared-intent` formalizes commander's-intent / mission-command doctrine as an Information Bottleneck compression: the optimal inter-agent communication transmits enough of the sender's purposeful state $G_t = (O_t, \Sigma_t)$ to align coordination behavior, not more. The minimal sufficient statistic for predicting jointly optimal coordination is what gets transmitted; everything below that bar is discarded as IB-suboptimal at a given complexity-relevance tradeoff parameter β. The definition is discussion-grade on its own account — the sender is assumed to know the jointly optimal action, the compression is treated as IB-lossless, and β is fixed — and the qualitative ordering it motivates (purpose before plans before models) is more robust than the displayed objective.

The implications-side payoff is what `#hyp-auftragstaktik-principle` extracts from this: under named regime conditions (objectives change slowly relative to strategies which change slowly relative to models; sub-agents have sufficient local adaptive capacity), the hypothesis is that bandwidth-limited composites should prioritize objective sharing, then strategy coordination, then model synchronization — written there as $B_O > B_\Sigma > B_M$. The hypothesis's prediction is the priority ordering Auftragstaktik discovered empirically in 19th-century Prussian military doctrine (Moltke, Clausewitz) and that Bungay 2011 documents organizations consistently *fail* by inverting (micromanaging *how* subordinates act rather than ensuring they understand *why*). The mechanism the framework proposes — bits with longer shelf life and higher coordination value per bit transmitted first — is hypothesis-grade rather than derived, and the segment carries that honestly: the IB optimization has not been solved with realistic cost functions, and the ordering is a statement about which bits to send *first* under scarcity rather than a derived claim about total allocation. What the IB framework predicts and what military doctrine empirically converged on agree at the *qualitative* level; whether the underlying mechanism is in fact IB-optimal bandwidth allocation is open.

What the chapter *would* buy for AAT's composition machinery, if the link were derived, is a quantitative reading of the closure-defect cost in Ch.2's composite tempo inequality: maintaining ε^* small requires spending inter-agent bandwidth on the dimensions that close the closure defect, so $C_{\text{coord}} \geq \varepsilon^* \nu_c / \|\delta_{\text{critical}}\|$ would carry a bandwidth signature, front-loaded in the priority ordering. This is a *sketch*, and the missing link is identifiable: nothing in canon relates bandwidth spent on dimension d to the unity U_d it buys. The chain that would earn it is $B_d \mapsto U_d$ (a bandwidth-to-unity map, absent) composed with the rate-distortion surface $U_d \mapsto \varepsilon_d$ (present in `#result-unity-closure-mapping`); the front-loading claim additionally needs diminishing marginal unity per bit. The software-team picture the hypothesis offers is consistent with the ordering rather than evidence for the mechanism: well-functioning teams have high B_O (clear product goals understood by all), moderate B_Σ (architectural decisions shared, implementation autonomous), low B_M (each developer builds their own mental model of the code they touch), and the micromanagement inversion — implementation details controlled, product goals unclear — is where team tempo is reported to drop.

The cross-domain transfer the source offers is Conway's Law (system structure mirrors communication structure), presented there as a consequence: when B_Σ is low and B_O is high, sub-agents coordinate through shared objectives rather than explicit action coordination, producing systems whose boundaries reflect objective decomposition rather than communication channels. This is a structural *reading* at hypothesis-grade, not a derivation — moving from a bandwidth profile to the shape of system boundaries needs organizational and design assumptions (how objective decomposition is mirrored in artifact modularity) that neither home states. The empirical observation long predates AAT; what AAT adds is a candidate mechanism.

The scope statement that the hypothesis itself insists on: the ordering is *regime-dependent*, and when the regime conditions fail it reverses. The prioritization assumes sub-agents can independently construct adequate local models; when the environment is genuinely ambiguous and local observations are insufficient (fog of war, novel codebase, unprecedented market conditions), model synchronization may be worth more than objective sharing — sub-agents who share the same wrong model at least err consistently, which is sometimes better than each having a different wrong model. The hypothesis's prediction is that high-ambiguity regimes shift the IB-optimal allocation toward B_M, and the priority inversion is observable. A second reversal channel is the *cost structure* of the substrate: the ordering is dictated by the relative cost of transmitting each kind of bit, and for AI-agent teams — where model synchronization can be cheap (shared stores, synchronized weights) and objective alignment is the hard part — the human-derived template may invert outright. Neither reversal is characterized in closed form.

Trust as inverse effective noise — the communication-gain decomposition.

`#hyp-communication-gain` extends the single-agent uncertainty ratio from `#emp-update-gain` ($\eta^* = U_M/(U_M + U_o)$) to inter-agent channels by adding two new noise sources to the denominator: $U_{\text{src},j}$ (source-competence uncertainty, i's uncertainty about j's model calibration) and $U_{\text{align},ji}$ (teleological-unity uncertainty, i's uncertainty about whether j's communications serve i's interests or j's potentially conflicting objectives). The hypothesis's reading is that *trust formalizes as the inverse of effective noise on the inter-agent channel*: $\eta^*_{ji} \to 1$ when source-noise, source-competence, and source-alignment uncertainties are all small; $\eta^*_{ji} \to 0$ when any is large. The additive denominator is a structural heuristic — it treats the three sources as independent zero-mean dispersion on a common predictive scale, which is apt for channel noise and miscalibration and conservative-but-incomplete for alignment, where an adversary shapes the message *policy* rather than widening a noise distribution. The one part of the extension that is not a hypothesis is its limit: when j is the environment, $U_{\text{src}} = U_{\text{align}} = 0$ and the form reduces by construction to $\eta^* = U_M/(U_M + U_o)$, which is exact in the Fisher-local regime (Kalman, conjugate-Bayesian, smooth log-likelihood cases) and robust-qualitative outside it.

The structural payoff is that *trust calibration is itself an AAT process* — a discussion-grade recognition carried in the source. Agent i's estimates of $U_{\text{src},j}$ and $U_{\text{align},ji}$ constitute a *trust meta-model* — a model of models — and the source applies the framework's apparatus to it by analogy: mismatch (trust prediction errors), gain (not overreacting to single disagreements), and structural inadequacy (`#result-structural-adaptation-necessity` read at the meta-level — the agent's trust-model class may not capture the actual reliability structure of its sources; the result is not re-derived for the meta-model, it is invoked as the same shape one level up). This is the segment's quietest claim and one of the framework's more reach-y consequences: relationships between agents are AAT processes operating on agent-pair-indexed quantities, with structure mirroring the agent-environment loop one level up. For AI-agent practitioners the corresponding design suggestion is that trust assessments be Bayesian-update-style with explicit calibration tracking rather than threshold-based or static.

The decomposition also names a sharp distinction between the three uncertainty terms that downstream segments rely on. $U_{o,ji}$ is a property of the *channel* — improvable by infrastructure. $U_{\text{src},j}$ is a property of the *source* — improvable by j improving its model, or estimable by i through calibration tracking. $U_{\text{align},ji}$ is a property of the *relationship* — the game-theoretic variable, and the one that becomes load-bearing in Ch.4's cooperative-vs-adversarial coupling and Ch.5's strategic composition (forward references). The relation to the signed coupling γ of those chapters is a correspondence, not an identity, and three objects have to be kept apart: the coupling sign γ (a dynamical quantity — whether j's tempo contribution reduces or amplifies i's effective disturbance; `#deriv-critical-mass-composition`, Appendix A, forward reference), objective alignment U_O (content unity), and $U_{\text{align},ji}$ (i's *uncertainty* about j's alignment — an epistemic quantity in i's trust meta-model). Observed cooperative coupling is evidence that should move i's trust posterior toward high η^*_{ji}, and observed adversarial coupling evidence that should move it toward low; but γ is not determined by U_O (an auditor challenge recorded in `#deriv-critical-mass-composition`'s Working Notes — Gemini, AUDIT-WORKING-193847 — observes that identical targets under resource rivalry can *maximize* adversarial coupling, unadjudicated), and a confidently-known adversary has *low* alignment-uncertainty yet warrants low gain — a case the additive-variance form does not represent — distinct from the strategic-deception limitation the source names and defers to external game-theoretic analysis. What the decomposition makes operational at the trust-meta-model level is therefore the *routing*: which of the three terms an observed coupling sign should update.

The risk-asymmetric trust discussion in the source is worth surfacing as an implications-flavored point in its own right, with its status kept visible: it is a decision-policy recommendation, not a consequence of the reliability posterior. The Bayesian posterior on source reliability gives the best *estimate*; how much to *act* on it depends on a loss function the source names only qualitatively — trusting a deceptive agent (high impact, low probability) can cause catastrophic model corruption via #der-adversarial-destabilization's effects spiral (Ch.4, forward reference), while mild miscalibration toward a reliable source (low impact, high probability) causes small ongoing inefficiency. Under a loss that asymmetric, acting on a conservative quantile of the trust posterior rather than its mean is the standard Bayes-decision response (for asymmetric linear loss the optimal point decision *is* a quantile), so the source's prescription is decision-theoretically principled once the loss is specified; what is not yet in canon is the loss function and evidence model that would make the quantile explicit. The familiar pattern that high-trust relationships build slowly and break quickly is *consistent with* such a policy — slow accumulation under a conservative quantile, fast erosion on high-impact evidence — rather than derived from the framework.

Composition with the framework's other machinery.

The chapter's three findings compose with the rest of Part III in a way worth surfacing as a unifying point, with each link at its own tier. Ch.2's closure defect ε^* acquires its rate-distortion content from this chapter's #result-unity-closure-mapping (conditional; exact in the worked instances); Ch.2's coordination-overhead floor $C_{\text{coord}} \geq \varepsilon^* \nu_c / \|\delta_{\text{critical}}\|$ in #der-tempo-composition would acquire a bandwidth signature from the Auftragstaktik hypothesis if the bandwidth-to-unity link were derived (sketch, above). Ch.4's signed coupling γ acquires a trust-meta-model *reading* from this chapter's $U_{\text{align},ji}$ — a correspondence at hypothesis-grade, with the three-object distinction above; Ch.5's strategic composition under partially-opposing objectives is where the dynamics of $U_{\text{align},ji}$ under observed coupling sign would be worked, and that working is not yet done (Ch.4/Ch.5 and #deriv-strategic-composition: forward references). The chapter is the *bandwidth and trust layer* between Ch.2's composition machinery and Ch.4/Ch.5's coupling dynamics — and the framework's posture is that all four chapters are operating on the same composite-agent persistence machinery, with each chapter supplying a different parametric content (machinery, bandwidth/trust, coupling sign, equilibrium).

Two cross-domain transfers worth noting before leaving the chapter. *Multi-agent AI safety*: the communication-gain decomposition suggests that agent-to-agent communication be filtered through Bayesian-update-style assessment with explicit calibration tracking. What the displayed gain form actually licenses against flat trust models (one threshold applied uniformly to all interlocutors) is an *anti-collapse* point (#disc-anti-collapse) rather than a direction-of-error prediction: a flat model merges channel, source, and alignment uncertainty into one number, so when trust is miscalibrated it cannot tell whether the repair is better infrastructure, calibration tracking, or a relationship-level (game-theoretic) response — the three levers the source keeps separate. Whether flat models over- or under-trust in a given stakes regime is a question for the loss layer, which the gain formula alone does not carry. *Human-AI teaming*: the same apparatus names which dimensions of teaming to attend to, and in what order *under the human cost structure* — shared intent first (objective alignment), strategy coordination next (architectural decisions shared, implementation autonomous), model synchronization last (neither party needs the other's full mental model). The framework's contribution against the empirical-design literature on human-AI teaming is not that this ordering is fixed — it is regime-dependent by the hypothesis's own terms — but that it *names the variables the ordering turns on*: the change rates ν_O, ν_Σ, ν_M, local adaptive capacity, and the substrate's per-bit cost of each kind of sharing, the last of which is exactly where a human-AI team differs from a human team and where the ordering may not carry over.

Chapter 15

Cooperative and Adversarial Coupling

Once composites exist, the question becomes how their members interact. This chapter develops the dynamics — cooperative coupling that helps allies, adversarial coupling that hurts opponents, and the four-regime classification of how an arriving event lands on its recipient — under a single signed-coupling structure. Cooperation and adversarial dynamics are not two theories; they are the same machinery with opposite signs.

The framing comes from the disturbance decomposition. The disturbance rate an agent faces decomposes into environment, adversarial, and cooperative components,

$$\rho_i = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j \tag{15.1}$$

with $\gamma > 0$ in both cases, but opposite signs in the sum. Cooperative allies reduce agent i's effective disturbance by acting in the shared environment — stabilizing resources, preempting threats, absorbing perturbations. Adversaries amplify it by acting against. The machinery is identical; the sign is what differs. This is the unification that "adversarial dynamics as a separate theory" misses, and it is one of the clearer cases of AAT's framework character: not many results, one principle applied with a sign flip.

Two physically distinct cooperative mechanisms compose under this structure, and the chapter is careful about distinguishing them. *Communication tempo*: allies *tell* agent i things that improve its correction — the communication-tempo term that adds to $\mathcal{T}_i$ via 14.4. *Cooperative action*: allies *do* things that reduce the disturbance i faces — the negative term in ρ_i above. The two enter the persistence inequality at different points (one raising $\mathcal{T}$, the other lowering ρ) and warrant separate accounting. Counting a single ally-event in both would double-count its benefit. There is a useful concrete analogy: this is the mathematical difference between a consultant and an employee. A consultant boosts your tempo by telling you what's happening; if your α is too low to act on the information, the consultant doesn't save you — you'll just understand your failure more clearly. An employee lowers your effective disturbance by intercepting the volatility before it reaches your state. Communication and cooperative action are not interchangeable, and structurally failing agents need the second more than the first.

The adversarial side is where the famous results live. Under coupling-dominant deterministic disturbance — the canonical adversarial regime — the steady-state mismatch ratio between two agents scales as the *squared* tempo ratio:

$$\frac{\|\delta_B\|_{ss}}{\|\delta_A\|_{ss}} = \left(\frac{\mathcal{T}_A}{\mathcal{T}_B}\right)^2 \tag{15.2}$$

The faster agent both corrects its own mismatch faster *and* generates disturbance for the slower agent faster. Speed advantage is multiplicative, not additive: a 2:1 tempo ratio yields a 4:1 mismatch ratio; a 3:1 ratio yields 9:1; a 10:1

ratio yields 100:1. This is the formal analog of Boyd's "getting inside the opponent's OODA loop" — it turns military intuition into physics. The math itself is not mysterious; it is just the steady-state formula applied to both sides and ratioed. But the consequence is sharp. The squared scaling means that a sufficiently large tempo asymmetry in a coupling-dominant regime is not a quantitative disadvantage but a regime change: at 100:1 mismatch ratio, the slower agent is almost certainly past its operating reserve R, which means structural collapse rather than degraded performance. Speed advantage in a coupled regime *compounds*, and the consequence is qualitatively different from what additive-intuition predicts. Under stochastic adversarial coupling the exponent drops from 2 to 3/2; under non-coupling-dominant base disturbance it approaches 1. The exponent depends on the regime — the chapter names them explicitly — but the regime where it is sharpest is also where it matters most.

There is a caveat in the other direction: speed without coupling is useless. The destabilization threshold from 15.2 makes this explicit — if A's coupling to B is weak ($\gamma_A \to 0$), A can have infinite tempo and still fail to destabilize B. Being fast in a vacuum doesn't accomplish anything; you have to be fast *and* coupled. The two are jointly required and neither suffices.

The effects spiral — the corollary in 15.2 where B's degrading model causes B's actions to become erratic in ways that increase A's coupling effectiveness — is the mathematics of panic. When an agent is pushed past its adaptive reserve, its corrections become unreliable, which makes it more legible to the adversary, which increases the disturbance, which pushes the agent further past reserve. The spiral terminates only when the agent undergoes structural adaptation (a different model class, a different mode of operation) or ceases to function as an adaptive agent entirely. The structural implication for any agent under sustained adversarial pressure is that resilience is not just high α or high R — it is also having a mechanism to break the spiral before structural collapse, which usually means external intervention from outside the agent.

The other interesting move is the recipient-side decomposition. From the emitter's perspective, coupling is a single number — $\gamma_A \cdot \mathcal{T}_A$ — applied to the target's disturbance budget. From the recipient's perspective, that scalar collapses real structure. An arriving event from A lands on B as one of four qualitatively different things, determined by three independent boundary conditions in B's existing AAT quantities. It might be a *Regime I informative update* (small enough to be absorbed, large enough to register, within B's model class — the good case). It might be a *Regime II-a magnitude shock* (large enough to exit B's sector region — the destabilizing-fast-events case). It might be a *Regime II-b structural shock* (within sector but outside B's model class — the wrong-kind-of-event case, repairable only by structural adaptation rather than by faster correction). Or it might be a *Regime III ambient erosion* (below the observability floor — death by a thousand memos, where every event is small enough to be processed but the cognitive overhead of processing accumulates until the agent has no tempo left for real work; this is the formal version of why DDoS-of-low-priority-alerts is a real attack and not just inconvenience). Different regimes call for different repairs. Magnitude shocks call for more bandwidth; structural shocks call for a different model class; ambient erosion calls for better filtering, often at the infrastructure layer before signals reach the agent. Conflating them, which the scalar emitter view does, confuses diagnosis.

This is one of the chapter's load-bearing pedagogical moves. Most multi-agent frameworks treat "coupling" as a single coefficient. The recipient-side classification says: the coefficient is what the emitter cares about; the regime is what the recipient cares about; and they are not interchangeable for purposes of figuring out what to do.

The flow of the chapter: team persistence and the signed disturbance decomposition (15.1) → adversarial destabilization and the effects spiral (15.2) → the recipient-side four-regime classification (15.3) → the superlinear tempo advantage and its regime-dependent exponents (15.4). Chapter 5 extends to symmetric strategic composition (equilibrium-convergence framing for partially-opposing objectives), agent opacity as the formal dual of observation quality, and the qualifications that gate when the superlinear advantage actually obtains.

15.1 *Derived:* Team Persistence

Teams persist where individuals cannot through two physically distinct cooperative mechanisms: communication (allies share observations that improve correction) and action (allies act in the shared environment to reduce disturbance at its source). These mechanisms enter the persistence condition at different points — tempo and disturbance respectively — and a given cooperative interaction contributes through one mechanism or the other, not both.

The construction has two pieces. **Distributed tempo**: agent i's effective tempo sums direct-observation tempo *plus* communication tempo from allies, each ally contributing additively to correction capacity, subject to the noise-independence scope of `#deriv-tempo-additivity` (allies reporting from a shared upstream source overcount and saturate; allies with anti-correlated errors triangulate super-additively — the deviation is signed in both directions). **Cooperative-adversarial disturbance decomposition**: the disturbance rate decomposes into environment, adversarial, and cooperative components, with the cooperative term entering with a *negative sign* — allies acting in the shared environment *reduce* the disturbance i faces. Sub-agent persistence then requires correction rate exceed the *effective* disturbance rate (environment plus adversarial inputs minus cooperative inputs), divided by the critical-mismatch scale. This is the single-agent persistence inequality of 4.5 with a structured effective-disturbance decomposition substituted in.

The two cooperative mechanisms enter the persistence inequality at *different points* and warrant separate accounting: communication boosts $\mathcal{T}_i$ (the agent corrects faster); cooperative action lowers ρ_i (the agent has less to correct). A single ally-event contributes through *one* mechanism or the other; counting it in both would double-count the benefit. The structural consequence is the framework's answer to why agents form teams: not because teams are intrinsically valuable, but because *they shift the persistence inequality in favor of each member*. Cooperative coupling enters as a *negative* term in effective disturbance; adversarial coupling (Ch.4's 15.2) enters as a *positive* term. The signed structure is what unifies cooperative and adversarial dynamics under a single inequality.

Formal Expression.

This segment instantiates the sector-persistence template (`#result-sector-persistence-template`) at the multi-agent level with state variable $\xi = \delta_i$ (sub-agent i's mismatch) and a decomposed effective disturbance $\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$ that accounts for adversarial and cooperative coupling. The template supplies the Lyapunov machinery; this segment's distinctive content is the disturbance decomposition and the corresponding tempo extension.

Distributed Tempo. *Definition (distributed-tempo)*

Agent i's effective tempo includes contributions from both direct observation and communication from allies:

$$\mathcal{T}_i = \underbrace{\sum_k \nu_i^{(k)} \eta_i^{(k)*}}_{\text{direct observation tempo}} + \underbrace{\sum_{j \in \mathcal{N}(i)} \nu_{ji}^{\text{comm}} \eta_{ji}^*}_{\text{communication tempo}} \tag{15.3}$$

where ν_{ji}^{comm} is the rate of communication events from agent j to agent i, and η_{ji}^* is the communication gain (14.4). Faster team adaptation comes not only from faster individual sensing but from faster, more reliable knowledge transfer.

Noise-correlation scope. Both sums are additive, inheriting the cross-channel noise-independence condition of 3.8, whose exact scope is derived in `#deriv-tempo-additivity` — and the deviation under correlated sources is **signed**, in both directions. In the *common-source* direction (allies reporting from one upstream intelligence source, redundant status reports echoing the same dashboard), the additive communication tempo overcounts with a closed-form redundancy penalty, and under a *persistent* shared source it saturates outright: past the source's bias floor, additional correlated allies contribute zero marginal correction capacity — the multi-agent echo chamber. In the *synergistic* direction, allies whose report errors anti-correlate (or whose errors correlate with a reference ally's known failure modes) allow

triangulation: the joint information exceeds the additive accounting, so the additive formula *under*counts — a genuinely super-additive composition effect, and a structural reason to value *diverse* allies over *many* allies, including allies nearly useless in isolation whose errors track a good ally's blind spots. No scalar dependence summary (in particular the mutual information between communication sources) can carry the exact correction — it is sign-blind (`#deriv-tempo-additivity`, no-go N1); it survives only as a witness that the additive form is off. The exact joint object requires the cross-source noise covariance.

Formulation (disturbance-decomposition)

Cooperative-Adversarial Disturbance Decomposition.

The disturbance rate experienced by agent i decomposes into environment, adversarial, and cooperative components:

$$\rho_i = \rho_{i,\text{env}} + \sum_{j \in \mathcal{A}_i} \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_{j \in \mathcal{C}_i} \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j \tag{15.4}$$

where $\mathcal{A}_i$ is the set of agents adversarially coupled to i, $\mathcal{C}_i$ is the set cooperatively coupled, and the γ coefficients capture coupling effectiveness (as in 15.2).

The cooperative term is negative — but through action, not communication. Allies reduce agent i's effective disturbance by *acting in the shared environment* to prevent or mitigate disturbance at its source. Examples: an ally stabilizes a shared resource, neutralizes a threat before it reaches i, or absorbs environmental variation through their own actions. The mechanism is causal coupling through the shared environment, not information transfer.

Separation from communication tempo. The communication tempo term (above) captures allies *telling* agent i things that improve its correction. The cooperative disturbance term captures allies *doing* things that reduce the disturbance i faces. These are physically distinct: communication improves i's correction function (better α_i, higher $\mathcal{T}_i$); cooperative action reduces the disturbance ρ_i that i must correct against. A single cooperative event contributes through one channel or the other. An ally's message about a threat enters through communication tempo; an ally's action that eliminates the threat enters through disturbance reduction. Counting a single event in both terms would double-count the benefit and make the persistence threshold systematically optimistic.

Effective disturbance rate. The decomposition can yield $\rho_i < 0$ when cooperative coupling dominates both environment disturbance and adversarial coupling. The sector-condition analysis (4.4) assumes non-negative disturbance (GA-2). Define:

Definition (effective-disturbance)

$$\rho_i^{\text{eff}} = \max(\rho_i, 0) \tag{15.5}$$

When $\rho_i^{\text{eff}} = 0$, the agent's cooperative network fully absorbs all disturbance — the persistence condition is trivially satisfied and mismatch decays to zero. This is an idealized limit; in practice, $\rho_i^{\text{eff}} > 0$ because cooperative coupling is imperfect and environment disturbance is never fully preempted. All downstream uses of ρ_i in the persistence and reserve conditions should be read as ρ_i^{eff}.

Derived (team-persistence, from sector-condition-stability, persistence-condition)

Team Persistence Condition.

Applying the sector-condition framework (4.4) with ρ_i^{eff}, agent i persists iff:

$$\frac{\rho_i^{\text{eff}}}{\alpha_i} < R_i \tag{15.6}$$

Substituting the decomposition (the $\max(\cdot, 0)$ in ρ_i^{eff} is omitted to expose the three levers; the condition is trivially satisfied when the numerator is non-positive):

$$\frac{\rho_{i,\text{env}} + \sum_j \gamma^{\text{adv}}_{j\to i}\mathcal{T}_j - \sum_j \gamma^{\text{coop}}_{j\to i}\mathcal{T}_j}{\alpha_i} < R_i \tag{15.7}$$

This reveals three distinct levers for team persistence:

1. **Increase** α_i (individual correction efficiency) — better models, better gain calibration, including communication-improved tempo from allies (14.4)
2. **Increase cooperative disturbance reduction** ($\gamma^{\text{coop}}\mathcal{T}_j$) — more effective allied action in the shared environment: stabilizing shared resources, preempting threats, absorbing environmental variation. This is the action-based mechanism distinguished above, not the communication channel.
3. **Reduce adversarial coupling** ($\gamma^{\text{adv}}\mathcal{T}_j$) — better deception detection, reduced exposure to adversarial actions

Coordination Overhead Threshold.

Discussion — Coordination Threshold

Communication channels have costs: time to compose and parse messages, bandwidth limitations, synchronization requirements. These costs reduce the agent's effective tempo by diverting capacity from direct adaptation. Let $\Delta\mathcal{T}_i^{\text{cost}}(j)$ represent the tempo-equivalent coordination cost of maintaining the channel with j — the reduction in i's direct observation tempo caused by the overhead, in units of $[t^{-1}]$.

The net benefit of adding agent j to i's communication network is positive only when:

$$\nu^{\text{comm}}_{ji}\,\eta^*_{ji} > \Delta\mathcal{T}_i^{\text{cost}}(j) \tag{15.8}$$

Both sides have units $[t^{-1}]$: the LHS is communication tempo gained, the RHS is direct-adaptation tempo lost to coordination overhead. This implies a natural team-size limit: adding members increases communication tempo with diminishing returns (as U_{src} and U_o accumulate across diverse sources) while coordination costs grow, potentially superlinearly. The optimal team size occurs where the marginal communication tempo equals the marginal coordination cost.

Epistemic Status.

Conditional on the communication-gain hypothesis (14.4). The distributed tempo definition is a *formulation* — a representational choice extending 3.8 to the multi-agent case. The disturbance decomposition is a *formulation* — the additive structure and the sign convention are modeling choices, not derivations. The persistence condition is *derived* from the sector-condition framework (4.4) given the decomposition: the derivation is exact under the same assumptions (GA-2, GA-3 applied to ρ_i^{eff}). The coordination overhead threshold is *discussion-grade* — qualitatively clear but the claim about diminishing returns and superlinear costs is asserted, not derived.

Max attainable: *robust-qualitative* for the persistence condition (it inherits the sector-condition's robustness but the decomposition is a modeling choice). The coordination threshold could reach *conditional* with a concrete cost model.

Discussion.

Compositional analog of 4.5. Like the single-agent persistence condition, this segment addresses *structural persistence* (see Persistence in `LEXICON.md`) — whether the composite correction machinery can outpace the effective disturbance rate. It does not address operational persistence (whether any particular sub-agent is near its boundary) or continuity persistence (whether the team maintains coherent identity through personnel changes). A team can be structurally persistent as a composite while individual members are operationally fragile or while the team's continuity is interrupted by turnover. The single-agent persistence condition says an agent persists when $\mathcal{T} > \rho/\|\delta_{\text{critical}}\|$. This segment extends that condition to agents embedded in a cooperative-adversarial network. The formal structure is identical — the sector-condition machinery applies unchanged — but the *inputs* ($\mathcal{T}_i$ and ρ_i) now include inter-agent terms. This is consistent with 11: the same dynamical laws apply at every level of description; what changes between levels is which channels contribute to tempo and which sources contribute to disturbance.

Why teams can persist where individuals cannot. Two distinct mechanisms combine. First, communication tempo raises $\mathcal{T}_i$ — allies provide observations that improve correction. Second, cooperative action lowers ρ_i — allies act in the environment to reduce disturbance at its source. An individual agent facing $\rho_{i,\text{env}} > \alpha_i R_i$ fails the persistence condition. Adding cooperative allies can either raise the numerator's denominator (tempo) or lower the numerator directly (disturbance), or both — through physically distinct mechanisms.

Timescale separation and 11. The distributed tempo definition presumes that communication events and direct observation events are comparable — they enter additively into $\mathcal{T}_i$. This requires that the communication timescale is not so slow relative to the environment dynamics that communicated information is stale on arrival. When communication latency approaches $1/\rho_{i,\text{env}}$, the effective η^*_{ji} degrades (the observation uncertainty $U_{o,ji}$ increases with staleness), naturally suppressing the communication tempo contribution.

Complement to 15.2. That segment characterizes when an adversary can push an agent past its stability boundary. This segment characterizes the cooperative counterpart: when allies can pull an agent back from instability. The γ coefficients have the same structure — coupling effectiveness — but opposite sign in the disturbance decomposition. Both directions' effectiveness is opacity-modulated: 16.2 derives $\gamma^{\text{coop, effective}} \propto (1 - H_b/H_b^{\max})$ (an ally who cannot anticipate your state cannot preempt your disturbances — which is why low-H_b practices like auftragstaktik-style shared intent raise cooperative coupling without raising communication load).

Composite-level complement: `#deriv-critical-mass-composition`. This segment gives the *per-sub-agent* persistence condition within a team. `#deriv-critical-mass-composition` supplies the *composite-level* analog: a closed-form critical-mass inequality in the matched-symmetric-Tier-1 two-agent case, with the same signed-γ coupling structure used here but applied to the joint Lyapunov on the concatenated mismatch state. The two are complementary: the team persists at the sub-agent level when each i satisfies this segment's condition; the team persists at the composite level when `#deriv-critical-mass-composition`'s (CM4) holds. Cooperative coupling ($\gamma < 0$) reduces ρ_i^{eff} here and reduces $\rho + \gamma\mathcal{T}$ in (CM2) there — the same mechanism viewed at two scales.

15.2 *Derived:* Adversarial Destabilization

When two agents are coupled such that one's praxis contributes to the other's disturbance rate, the faster agent can generate aporia in the target faster than the target's epistrophe can resolve it — driving the target outside its invariant region and causing the correction mechanism to break down entirely.

The construction applies `#result-sector-persistence-template` with *coupling-amplified disturbance*: the target's effective disturbance becomes its base disturbance plus the attacker's coupling effectiveness times the attacker's tempo (Model D — deterministic drift coupling) or with a square-root variant (Model S — stochastic noise coupling). **Destabilization is the *negation* of the template's persistence condition** — the same inequality viewed in the opposite direction. Under Model D, the attacker destabilizes when its tempo times its coupling effectiveness exceeds the target's *adaptive reserve* (correction rate times reserve minus base disturbance). Under Model S the linear-in-correction-rate scaling is replaced by a square-root-in-correction-rate scaling — the same $1/\alpha$ versus $1/\sqrt{\alpha}$ split from `#result-sector-persistence-template` propagating through the destabilization negation. **This scaling difference is the structural source of the superlinear adversarial-advantage scaling** in 15.4's squared (Model D) versus three-halves-power (Model S) tempo-ratio result — not a separate derivation.

The segment also carries the **effects-spiral** corollary: an agent pushed past its adaptive reserve has unreliable corrections, which can make it more legible to the adversary, increasing coupling effectiveness, increasing disturbance, pushing the agent further past reserve. The spiral terminates only when the agent undergoes structural adaptation (a different model class) or ceases to function as an adaptive agent entirely. The functional form $\gamma_A(\|\delta_B\|)$ is not yet formalized, so the spiral is stated as *discussion-grade* rather than derived — the qualitative mechanism is clear but the closed-form instability proof is open work. A scope-honest counterpoint: **speed without coupling is useless**. If $\gamma_A \to 0$, the attacker can have infinite tempo and still fail to destabilize the target. Speed and coupling are *jointly required*; neither suffices.

Formal Expression.

This segment is the sector-persistence template (`#result-sector-persistence-template`) applied with coupling-amplified disturbance: $\rho_B = \rho_{B,\text{base}} + \gamma_A \mathcal{T}_A$ (Model D) or $\sigma_B = \sigma_{B,\text{base}} + \gamma_A \mathcal{T}_A$ (Model S). The destabilization threshold is the **negation** of the template's persistence condition for agent B: destabilization occurs precisely when the coupling-amplified disturbance violates $\alpha_B R_B > \rho_B$. Persistence and destabilization are the same inequality viewed in opposite directions. The superlinear adversarial scaling (15.4) follows from the template's $1/\alpha$ (Model D) versus $1/\sqrt{\alpha}$ (Model S) scaling, not from separate derivation.

Setup. Both agents satisfy the single-agent sector-persistence template (`#result-sector-persistence-template`) with parameters (α_A, R_A) and (α_B, R_B). Coupling amplifies B's effective disturbance rate by $\gamma_A \cdot \mathcal{T}_A$; destabilization is the negation of the template's persistence condition $\alpha_B R_B > \rho_B^{\text{eff}}$ for B. See 16.3 for regime taxonomy.

Derived (adversarial-destabilization, from sector-persistence-template)

Model D: deterministic drift coupling. *[Assumption (Coupling Model D)]* $\rho_B = \rho_{B,\text{base}} + \gamma_A \cdot \mathcal{T}_A$. The template's Model D conclusion $R_B^* = \rho_B/\alpha_B$ applied with the coupling model yields B's destabilization threshold $R_B^* > R_B$:

$$\boxed{\mathcal{T}_A > \frac{\alpha_B R_B - \rho_{B,\text{base}}}{\gamma_A}} \quad \text{(Model D)} \tag{15.9}$$

Denote $\Delta\rho_B^* = \alpha_B R_B - \rho_{B,\text{base}}$, B's adaptive reserve — the template's reserve quantity applied with the baseline disturbance. □

Model S: stochastic noise coupling. *[Assumption (Coupling Model S)]* $\sigma_B = \sigma_{B,\text{base}} + \gamma_A \cdot \mathcal{T}_A$ — the adversary's tempo increases unpredictability, not systematic direction. The template's Model S conclusion $R_B^* = \sigma_B\sqrt{n/(2\alpha_B)}$ (scalar $n = 1$) applied with the coupling yields the destabilization threshold:

$$\boxed{\mathcal{T}_A > \frac{R_B\sqrt{2\alpha_B} - \sigma_{B,\text{base}}}{\gamma_A}} \quad \text{(Model S)} \tag{15.10}$$

Scaling difference. The Model D threshold is linear in α_B; the Model S threshold is linear in $\sqrt{\alpha_B}$ — the same $1/\alpha$ versus $1/\sqrt{\alpha}$ split the template gives for the two disturbance models, propagated through the destabilization negation. This is the direct origin of the $b = 2$ versus $b = 3/2$ exponent distinction in 16.3, not a separate derivation. □

Unified view. Symmetrically, B destabilizes A when the analogous threshold on $\mathcal{T}_B$ is exceeded, using whichever model describes A's disturbance. The adversarial outcome depends on whether either agent can push the other past its stability limit.

Regime selection in practice. Model D fits situations where adversarial action produces persistent positional shifts (military maneuvering, API changes propagating through dependents, doctrinal initiative). Model S fits situations where adversarial action produces unpredictable perturbations around a stationary level (feints, randomized probing, market volatility). Mixed cases are handled by decomposing the disturbance into drift and noise components and applying both bounds additively.

Interpretation. "Getting inside the opponent's OODA loop" has a precise Lyapunov characterization: Agent A destabilizes Agent B when A's praxis, multiplied by coupling effectiveness, generates aporia in B faster than B's epistrophe can resolve it — specifically, when A's tempo times coupling exceeds B's adaptive reserve $\Delta\rho_B^*$. This captures:

- **Asymmetric coupling** ($\gamma_A \neq \gamma_B$): an agent with lower tempo but higher coupling effectiveness can still win.
- **Finite reserves**: an agent with very high $\mathcal{T}$ but operating near its model-class limit ($\Delta\rho^*$ small) is vulnerable despite high tempo.
- **Structural collapse**: when $R_B^* > R_B$, the failure mode is not merely "large mismatch" but "correction mechanism breakdown" — connecting to 4.8.

Corollary: The Effects Spiral. When Agent B is driven past its stability boundary ($R_B^* > R_B$), and B's degrading model causes B's actions to become erratic in a way that increases A's coupling effectiveness (γ_A increases with $\|\delta_B\|$), the result is a positive-feedback Lyapunov instability:

Discussion — Mechanism Schematic

$$\|\delta_B\| \uparrow \Rightarrow \text{ } B\text{'s actions become erratic } \Rightarrow \gamma_A \uparrow \Rightarrow \rho_B \uparrow \Rightarrow \|\delta_B\| \uparrow \tag{15.11}$$

With γ_A now an increasing function of $\|\delta_B\|$, the disturbance term in B's dynamics grows superlinearly. $\dot{V}_B > 0$ and increasing — mismatch accelerates away from the stability region. The spiral terminates only when B undergoes structural adaptation (4.8 — changing the model class) or ceases to function as an adaptive agent entirely.

The no-spiral converse (Cheung-Piliouras-Tao 2021, recognition-tier). Under *matrix-fixed* coupling — structurally different from the state-dependent $\gamma_A(\|\delta_B\|)$ this corollary requires — CPT 2021 (Theorem 19 + Proposition 17) *rules out* the Effects Spiral in a named sub-scope: for the FTRL + graphical-constant-sum + fully-mixed-Nash regime, the joint dynamics is Poincaré recurrent on bounded level sets of the storage function, so $\|\delta_B\|$ does not escape past R_B even with adversarial coupling. The Effects Spiral therefore requires the *state-dependent* coupling that CPT's matrix-fixed framework cannot model; under fixed-matrix coupling in that sub-scope, the spiral provably cannot occur. This is the *dual* of the spiral, not a derivation of it: CPT predicts bounded recurrence in strategy space; the Effects Spiral predicts unbounded escape in error space — the two are opposite dynamic phenomena, on different state spaces, with structurally different coupling assumptions. The state-dependent coupling formalization this corollary's open piece needs therefore remains AAT's open target — its closed-form derivation for concrete AAT agent classes remains open, not subsumed by the CPT result. Source: `ref/cheung-piliouras-tao-2021-passivity-regret.pdf` (Theorem 19 + Proposition 17).

Epistemic Status.

Both Model D and Model S destabilization thresholds are *exact* under their respective coupling assumptions (which treat $\mathcal{T}_A$ as exogenous). The Model D threshold follows from the deterministic sector-condition steady state $R^* = \rho/\alpha$ (Prop A.1); the Model S threshold follows from the stochastic sector-condition steady state $R_S^* = \sigma\sqrt{n/(2\alpha)}$ (Prop A.1S). Both coupling models (additive to ρ in Model D, additive to σ in Model S) are *assumptions* — they decouple the agents rather than modeling the fully coupled dynamical system where both agents' mismatch states co-evolve. The analysis therefore characterizes the *destabilization threshold* (the conditions under which A *can* push B past its stability boundary) rather than the full transient dynamics. This is a worst-case bound, treating A as operating at its steady-state tempo.

The effects spiral (corollary) is *discussion-grade* — the positive-feedback mechanism is qualitatively clear, but formalizing the $\gamma_A(\|\delta_B\|)$ functional form and proving instability under it requires specifying how an agent's degrading model affects its action quality, which the theory does not yet formalize.

A full coupled Lyapunov analysis with a joint function $V(\delta_A, \delta_B)$ would capture mutual feedback effects but requires specifying how each agent's mismatch state affects the other's disturbance in real time — an open extension.

Discussion.

Destabilization vs. steady-state ratio. The destabilization threshold is a failure of *structural persistence* (see Persistence in `LEXICON.md`) — the point where the correction machinery can no longer outpace the adversarially amplified disturbance. An adversary does not need to attack operational persistence (pushing the target near its boundary) or continuity persistence (disrupting identity) directly; destroying structural persistence is sufficient, because without it, operational persistence degrades to zero and the agent ceases to function regardless of its continuity stance. The linear analysis in 3.9 gives the steady-state mismatch ratio under coupling: a quantitative result about how much worse B does. This segment gives the qualitative result: under what conditions does B *fail entirely*, not merely fall behind. The linear analysis tells you the score; the Lyapunov analysis tells you when the game is over.

Connection to 15.4. The simulation results show the tempo advantage is superlinear (exponent ≈ 2 in pure adversarial regimes). This Lyapunov result explains WHY: the destabilization threshold creates a phase transition — below it, B persists (possibly with degraded performance); above it, B's correction mechanism collapses entirely, and the effects spiral accelerates the collapse.

Recipient-side refinement. This segment's $\gamma_A \mathcal{T}_A$ scalar increment compresses a richer structure. Per 15.3, events arriving at B fall into four regimes with three independent boundaries (sector-region / model-class / observability). This segment's destabilization story is the Regime II integration — specifically, the magnitude-shock sub-regime (II-a) where $\|e\|_B > R_B$. The structural-shock sub-regime (II-b), where the signal exceeds B's *model-class capacity* regardless of magnitude, produces destabilization via a different mechanism (per 4.8's trigger condition at truth) and admits a different repair (structural adaptation, not more tempo). Both sub-regimes manifest as "adaptive reserve exceeded" in the scalar view, but the distinction is load-bearing for diagnosis and repair. The 15.3 segment also surfaces an adversarial move this segment's formulation cannot express: *Regime-I-with-adversarial-content* — exploiting B's openness to informative updates by injecting misinformation with adversarially-chosen sign on the log-odds signal. See 15.3 Discussion for the full decomposition.

Agent opacity and coupling effectiveness. γ_A is stated as a parameter but its determinants are not decomposed. One key factor is how *legible* or *opaque* the target agent B is to the adversary A. We adopt from Hafez et al. (2026) the backward predictive uncertainty $H_b = H(S, A \mid S')$ as a measure of agent opacity — how many distinct (state, action) pairs produce indistinguishable environment transitions. High H_b means the agent is opaque; low H_b means it is legible. In adversarial settings, B's opacity directly affects A's ability to model B's correction function: low $H_b^{(B)}$ (transparent target) enables targeted disruption that maximizes γ_A, while high $H_b^{(B)}$ forces A to act against an uncertain model, reducing effective γ_A. Symmetrically, high $H_b^{(A)}$ (opaque adversary) degrades B's ability to anticipate A's disruptions — increasing the effective observation uncertainty U_o in B's model of A. The coupling effectiveness is thus modulated by opacity in both directions: $\gamma_A \propto 1/H_b^{(A)} \cdot 1/H_b^{(B)}$ is a qualitative relationship (precise functional form is open). H_b is the formal dual of observation quality U_o: where U_o characterizes how well the agent sees the world, H_b characterizes how well the world sees the agent. See `#der-agent-opacity` for the formal definition, the sign-flip derivation via signed coupling, the emitter-side four-regime classification, and the 16-cell emitter-recipient composition that operationalizes adversarial-edge-targeting.

Connection to extreme transition dynamics (Miller 2022). The effects spiral has a constructive counterpart: the same self-reinforcing coupling mechanism that drives destabilization here can drive *regime transitions* rather than collapse when the coupling is constructive rather than destructive. An environmentally neutral variant accumulates through drift, creating a niche that a mutant in the opposing population exploits — a positive-feedback cascade that rapidly transforms both populations. The Lyapunov coupling model applies to both signs: destructive coupling (this segment) increases ρ; constructive coupling (15.1) decreases it. The difference is sign, not structure. The endogenous emergence of coupling — where γ changes as population composition shifts — is the critical extension needed to formalize the full transition motif; it is flagged as a gap in Part III's dynamics (see 4.8 for the single-agent analog and the dynamics-level gaps enumerated in the OUTLINE).

15.3 *Derived:* Interaction-Channel Classification (Recipient-Side)

The same signal from agent A lands on recipient B as one of four qualitatively different things — informative update, magnitude-shock, structural-shock, or ambient noise — determined by three independent boundary conditions stated entirely in B's existing AAT quantities. The emitter-side collapse of this variation into a scalar $\gamma_A \mathcal{T}_A$ loses information that is load-bearing: the recipient's repair path depends on which regime the event falls into, and "more tempo" vs "different model class" address structurally different failure modes. This is the **anti-collapse** discipline (5.7) at the composition layer — the scalar collapse hides exactly the regime distinction that routes to opposite repairs.

Two event-level quantities enter the classification and *must not be conflated*: the **magnitude** of the event in B's observation space (how large a perturbation it produces in mismatch on arrival), and the **information content** of the event conditional on B's prior. A large-magnitude already-predicted event has large magnitude but small information content; a tiny-magnitude structurally novel event has small magnitude but large information content. Three independent boundaries — sector-region, model-class, observability-floor — partition events into four regimes. **Regime I (informative update)**: within sector, representable in the model class, above the observability floor — the good case. **Regime II-a (magnitude shock)**: event magnitude exceeds the sector region — too large for the correction mechanism; the repair is more bandwidth or tempo. **Regime II-b (structural shock)**: within sector but information content exceeds the model class — the class lacks representational capacity; the repair is *structural adaptation* (a different model class), and more tempo does not help. **Regime III (ambient erosion)**: below the observability floor — each event individually small enough to be processed but the cognitive overhead accumulates; the repair is *infrastructure-level filtering* before signals reach the agent.

The central methodological move: **the emitter sees a scalar; the recipient sees a regime**. Most multi-agent frameworks treat coupling as a single coefficient (the emitter-side view); the classification names what that coefficient hides — events of identical scalar magnitude can sit in any of the four regimes, with structurally different repairs. The Regime-III ambient-erosion case is the framework's structural reason that DDoS-of-low-priority-alerts is a real attack rather than just inconvenience: not because individual events are significant but because processing them consumes tempo that should go to actual work. For LLM-agent infrastructure this maps directly — push Regime-III filtering *below* the model level into wrapper or system infrastructure rather than processing every event at the model layer. The recipient-side decomposition also surfaces an adversarial move the scalar emitter formulation cannot express, treated in Discussion: *Regime-I-with-adversarial-content* — exploiting B's openness to informative updates by injecting sign-chosen misinformation on the log-odds signal.

Formal Expression.

Setup and Notation. Two purposeful agents A and B coupled through a shared environment. A's praxis produces an event e^A_τ that enters B's observation channel. On B's side the event is processed by the standard AAT machinery: h_B maps the A-induced environment state to observation o^B_τ (1.3); mismatch is $\delta^B_\tau = o^B_\tau - \hat{o}^B_\tau$; update absorbs δ^B_τ with gain $\eta^*_B = U_{M,B}/(U_{M,B} + U_{o,B})$ (3.6).

Two event-level quantities enter the classification and must not be conflated:

- $\|e^A_\tau\|_B$ — the **magnitude** of the event in B's observation space (how large a perturbation it produces in δ^B_τ on arrival).
- $\mathcal{I}(e^A_\tau)$ — the **information content** of the event conditional on B's prior, formally $I(e^A_\tau; \Omega \mid M_{B,\tau^-})$ per NOTATION.md's event-information quantity. A large-magnitude already-predicted event has large $\|e\|$ but small $\mathcal{I}$; a tiny-magnitude structurally novel event has small $\|e\|$ but large $\mathcal{I}$.

Let $\mathcal{F}(\mathcal{M}_B)$ denote B's model-class fitness (2.4), and $\mathcal{I}_{\max}(\mathcal{M}_B)$ the maximum per-event information content representable within the class (see Working Notes for the cleaner sufficient-statistics-span formulation).

Definition (regime-boundaries)

Classification Boundaries.

Event e^A_τ arriving at B falls into one of four regimes, determined by three independent boundary conditions:

Regime I (Informative update) when all three hold:

$$\text{(I-a)} \quad \|e^A_\tau\|_B \le R_B \qquad \text{(within sector-condition region)} \tag{15.12}$$

$$\text{(I-b)} \quad \mathcal{I}(e^A_\tau) \le \mathcal{F}(\mathcal{M}_B) \cdot \mathcal{I}_{\max}(\mathcal{M}_B) \qquad \text{(representable within model class)} \tag{15.13}$$

$$\text{(I-c)} \quad \mathcal{I}(e_\tau^A) \cdot \nu^{(k)} \geq U_{o,B}^{(k)} \cdot c_{\text{floor}} \qquad \text{(above observability floor)} \tag{15.14}$$

where k is the arrival channel, $\nu^{(k)}$ its event rate, and c_{floor} a detection-theory constant controlling the false-alarm tolerance.

Regime II-a (Magnitude-shock destabilization) when (I-a) fails:

$$\|e_\tau^A\|_B > R_B \tag{15.15}$$

The event exits B's sector-condition region on arrival. B's correction function does not point inward strongly enough to discharge the mismatch before the next event; under sustained rate $\nu > rsim\alpha_B$, destabilization proceeds per 15.2.

Regime II-b (Structural-shock destabilization) when (I-a) holds but (I-b) fails:

$$\|e_\tau^A\|_B \leq R_B, \qquad \mathcal{I}(e_\tau^A) > \mathcal{F}(\mathcal{M}_B) \cdot \mathcal{I}_{\max}(\mathcal{M}_B) \tag{15.16}$$

The event's information content exceeds what B's model class can represent. By 4.8, parametric update within $\mathcal{M}_B$ cannot close the mismatch; residuals retain systematic structure. Repair requires structural adaptation (a different model class), not more bandwidth.

Regime III (Ambient noise / slow erosion) when (I-a) and (I-b) hold but (I-c) fails:

$$\|e_\tau^A\|_B \leq R_B, \qquad \mathcal{I}(e_\tau^A) \leq \mathcal{F}(\mathcal{M}_B) \cdot \mathcal{I}_{\max}(\mathcal{M}_B), \qquad \mathcal{I}(e_\tau^A) \cdot \nu^{(k)} < U_{o,B}^{(k)} \cdot c_{\text{floor}} \tag{15.17}$$

The event is representable and within capacity but its information content sits below the observability floor. It contributes to δ_B's variance (enters Model S as part of $\sigma_{w,B}^2$) without triggering a usable update; B's adaptive reserve $\Delta\rho_B^*$ slowly drains.

Three Independent Boundaries. The three boundary conditions are structurally independent, each stated in quantities AAT already carries:

Table 15.1

Boundary	*AAT quantities*	*Failure mode*
(I-a) / (II-a): sector-region	$\|e\|_B$, R_B (from 2.4 / `#result-sector-persistence-template`)	*magnitude* — more capacity cures
(I-b) / (II-b): model-class	$\mathcal{I}(e), \mathcal{F}(\mathcal{M}_B), \mathcal{I}_{\max}(\mathcal{M}_B)$	*class* — structural adaptation cures
(I-c) / (III): observability	$\mathcal{I}(e), \nu_B^{(k)}, U_{o,B}^{(k)}$ (from 16.4)	*rate* — lower observation noise or higher event rate cures

No new ad-hoc thresholds are introduced. $\mathcal{I}_{\max}(\mathcal{M}_B)$ is the only new symbol; see Working Notes for its cleaner sufficient-statistics-span formulation.

Derived (regime-typed-rho-eff, from regime-boundaries + sector-persistence-template)

Regime-Typed Disturbance Decomposition.

Under a stream of events $\{e_\tau^A\}$, B's **regime-typed effective disturbance** rate decomposes into regime-typed contributions:

$$\rho_B^{\text{eff}} = \underbrace{\sum_{e \in \text{II-a}} \|e\|_B \cdot \nu_e}_{\text{magnitude disturbance}} + \underbrace{\text{floor}(\mathcal{M}_B) \cdot \sum_{e \in \text{II-b}} \nu_e}_{\text{structural mismatch floor}} + \underbrace{\sum_{e \in \text{III}} \sigma_e^2 \cdot \nu_e}_{\text{ambient variance}} - \underbrace{\sum_{e \in \text{I}} \iota_B(e)\, \mathcal{I}(e) \cdot \nu_e}_{\text{informative correction}} \tag{15.18}$$

The Regime-I term is **negative**: informative events reduce B's effective disturbance rate, not increase it. This generalizes 15.1's cooperative-action term $-\gamma^{\text{coop}}\mathcal{J}_j$: a cooperative event is precisely a Regime-I event from an aligned emitter, and the sign flip in the emitter-side decomposition falls out of the regime assignment on the recipient side. Adversarial events land in Regimes II-a/II-b; ambient-noise events in Regime III.

The emitter-side formulation $\gamma_A \mathcal{T}_A \to \rho_B^{\text{eff}}$ compresses the regime-typed sum into a single scalar, losing (i) the sign of cooperative coupling, (ii) the magnitude vs structural distinction in destabilization, and (iii) the observability-floor loss to Regime III.

Derivation (Kalman-over-Kalman, from regime-boundaries + update-gain)

Structured Derivation — Kalman-over-Kalman.

For a concrete check, take B as a Kalman filter on a scalar linear-Gaussian state with model class $\mathcal{M}_B = \{\theta \in [\theta_{\min}, \theta_{\max}]\}$, process noise q, observation noise r, sector parameter $\alpha_B = \eta_B^* = P_{\text{pred}}/(P_{\text{pred}} + r)$. B's sector-region radius is $R_B = \sqrt{q/(1-\theta_{\max}^2)}$ (stationary standard deviation at the class edge).

A's emitted perturbation ξ_A enters B's innovation as $\delta_\tau^B = \xi_A + \varepsilon_\tau + (\omega_\tau - \hat{\omega}_\tau)$. Four canonical distributions for ξ_A partition the classification:

Case 1 — Small-variance Gaussian $\xi_A \sim \mathcal{N}(0, s^2)$, $s^2 \ll r$ (expected Regime III). $\mathcal{I}(\xi_A) \approx s^2/(2r \ln 2)$ nats — small. (I-a) holds ($s \ll R_B$); (I-b) holds (Gaussian-within-Gaussian); (I-c) fails because $\mathcal{I}(\xi_A) < U_{o,B} \cdot c_{\text{floor}}$. Result: Regime III. Derived consequence — the contribution to ρ_B^{eff} is through $\sigma_{w,B}^2$; adaptive reserve drains by $\sum_e \eta_B^{*2} s_e^2 \cdot \nu_e$.

Case 2 — Moderate Gaussian $\xi_A \sim \mathcal{N}(\mu, s^2), \mu \ll R_B, s^2 \sim r$ (expected Regime I). $\mathcal{I}(\xi_A) = \frac{1}{2}\log(1+(s^2+\mu^2)/r)$ — substantial. All three (I-a)/(I-b)/(I-c) hold. Result: Regime I. Standard Kalman update; M_B refines; Regime-I term contributes negatively to ρ_B^{eff}.

Case 3 — Binary kick $\xi_A \in \{\pm\Delta\}$ with $\Delta > R_B$ (expected Regime II-a). (I-a) fails by construction. The Kalman update $\hat{x}^+ = \hat{x}^- + \eta^*\Delta$ undershoots by $(1-\eta^*)\Delta$ per event. If events arrive at rate $\nu > rsim\alpha_B$, lag accumulates and $\alpha_B R_B > \rho_B^{\text{eff}}$ is violated. Result: Regime II-a — destabilization per 15.2. Notice: the signal is *within* the model class (Gaussian handles $\pm\Delta$ mathematically), but correction cannot discharge it fast enough.

Case 4 — Heavy-tailed ξ_A with $\mathbb{E}[\xi_A^2] \sim r$ but kurtosis $\kappa \to \infty$ (expected Regime II-b). Mean contribution is fine; the problem is the distribution shape. The Kalman filter — Gaussian-optimal — mis-gains: too aggressive for small events, too conservative for genuine large ones. The per-event KL gap $D_{\text{KL}}(P_{\text{true}} \| P_{\mathcal{M}_B}) > 0$ for any heavy-tailed P_{true} against Gaussian. By 2.4, $\mathcal{F}(\mathcal{M}_B) < 1 - \varepsilon$ with ε lower-bounded by the KL gap; by 4.8, no parametric update within $\mathcal{M}_B$ closes the mismatch. Result: Regime II-b — residuals retain non-Gaussian structure (visible in kurtosis tests); repair requires expanding the model class (e.g., Student-t observation model), not more Kalman tuning.

Each case lands where the classification predicts. The derivation transfers to any recipient architecture in which the underlying AAT quantities are well-defined — this is the scope inherited from `#result-sector-persistence-template` + 2.4 + 3.8.

Recovery of Emitter-Side Results.

Derived (emitter-side-recovery)

Each existing emitter-side result is a restriction of the four-regime decomposition:

- **15.2** is Regime II-a integrated over a tempo-proportional event stream. The magnitude-shock sub-regime corresponds directly; the structural-shock II-b subcase is implicit in that segment, collapsed into "adaptive reserve exceeded" but here made explicit.
- **15.4** — superlinear tempo scaling follows from the sector-persistence template's $1/\alpha$ (Model D) vs $1/\sqrt{\alpha}$ (Model S) applied to Regime II events. The b-exponent drops toward zero in the high-$U_{o,B}$ limit because the fraction of A's events landing in Regime II drops (more fall into Regime III).
- **16.4** is the recipient-side expression of boundary (I-c): high $U_{o,B}$ pushes events into Regime III where they add to variance without contributing to destabilization.
- **12.3** corresponds to asymmetric classification: host's signals to endosymbiont contain high-$\mathcal{I}$ structure the endosymbiont's class cannot initially represent (Regime II-b for the endosymbiont, forcing structural adaptation toward the host's class); endosymbiont's signals to the host land in Regime I (host absorbs endosymbiont's accumulated structure). Consolidation is the fixed-point where both streams are Regime I.
- **Cooperative signaling** (in 15.1) — Regime-I events from aligned emitters contribute negatively to ρ_B^{eff} via the cooperative-action term. The communication-tempo $\nu_{ji}^{\text{comm}} \cdot \eta_{ji}^*$ is the rate of Regime-I events times the recipient's informative gain.

What Is Derived vs. What Is Chosen.

Table 15.2

Property	*Source*	*Strength*
Three independent boundaries (sector-region / model-class / observability)	Import from 2.4 + `#result-sector-persistence-template` + 16.4	Formulation choice (three-way partition; coarser / finer alternatives possible)
Four-regime partition (I / II-a / II-b / III)	The three boundaries yield four boundary-state combinations; only four are non-degenerate	Derived from the boundary structure
(I-a) / (II-a) boundary at R_B	`#result-sector-persistence-template`'s sector-region radius	Derived
(I-b) / (II-b) boundary at $\mathcal{F}(\mathcal{M}_B) \cdot \mathcal{I}_{\max}$	2.4 + class-capacity normalization	Formulation (the $\mathcal{I}_{\max}$ normalization is heuristic; sufficient-statistics-span form is cleaner — see Working Notes)
(I-c) / (III) boundary at $U_{o,B} \cdot c_{\text{floor}}$	16.4 + detection-theory threshold	Formulation (c_{floor} is a detection-power parameter)
Regime-typed ρ_B^{eff} decomposition with negative Regime-I term	Aggregation over event stream using regime classification	Derived (the sign of the Regime-I term is structural, not a choice)
Kalman-over-Kalman four-case derivation	Direct application of Kalman gain + sector + KL-gap + SNR analyses	Proved (for the stated case)
Recovery of 15.2, 12.3, 16.4, 15.1 as restrictions	Each emitter-side result is exhibited as a per-regime special case	Derived
Non-Gaussian Case 4 derivation (heavy-tailed → II-b via KL gap)	Informal argument grounded in robust-filtering literature (Huber, Masreliez)	Discussion-grade (rigorous version requires per-family KL computation)
Class 2 (Partial) approximation with $\kappa_{\text{processing}}$ degradation	Transfer from Class 1 (Separated) with goal-blind-update failure	Formulation (qualitative); exact form requires spelling out the goal-contamination coupling

Epistemic Status.

Conditional. Max attainable: *derived* for the classification structure in Class 1 (Separated) architectures; *exact* for the Kalman-over-Kalman worked case in sub-scope α; *robust qualitative* for general sub-scope β recipients.

The three boundaries are structurally independent and each is stated in an existing AAT quantity, so the four-regime partition is not ad-hoc; it is forced by the structure of the quantities already in the theory. Within Class 1 (Separated; goal-blind epistemic update per 6.3) and sub-scope α recipients (Kalman / conjugate-Bayesian / exponential-family / strongly-convex-gradient / linear-PD), the Kalman-over-Kalman derivation transfers with A2' derived per 4.3, and each case yields the predicted regime by direct substitution. In sub-scope β (PID / rule-based / human-judgment), the classification's form transfers but boundary (I-a) requires per-instantiation verification of the sector bound, inheriting from `#result-sector-persistence-template`'s sub-scope β caveat.

Scope limits. Class 3 (Coupled) recipients are out of formal scope: the coupled epistemic-purposeful update violates 6.3, so the regime assignment cannot be cleanly computed against M_B alone. This is `03-llm-core/` territory. Class 2 (Partial) recipients inherit with degradation proportional to $\kappa_{\text{processing}}$: the Regime-I update may be goal-contaminated, and Regime II-b may be misdiagnosed as II-a or III under goal-blind-update failure. Per-event classification is singular-trajectory-indexed by 4.10; aggregation over event streams is deferred to `#result-sector-persistence-template`.

What the classification does not claim. (i) It does not determine *semantic content*: knowing an event is Regime I tells you B will update, not *what* B will believe afterward — that lives in 9.6's default signal function. (ii) It does not make the four regimes sharp in practice: real recipients' sector regions and class capacities have estimation uncertainty, so borderline events oscillate across boundaries under parameter noise. (iii) It does not replace `#result-sector-persistence-template`'s temporal aggregation; it provides the per-event regime label that feeds into the aggregate disturbance bound.

Discussion.

Complement, not replacement, of the emitter-side segments. 15.2, 15.4, 16.4, 12.3, and 15.1 each describe what happens when A's praxis couples into B. They collapse the variation into a scalar increment on B's effective disturbance rate. The recipient-side classification decomposes that increment into regime-typed contributions with (importantly) a **signed** Regime-I component. The five emitter-side segments are recovered as restrictions — the classification provides the pattern-recognizer to match the emitter-side optimizer.

Why II-a vs II-b matters. Both destabilization regimes manifest as "adaptive reserve exceeded" in the emitter-side view. But the repairs are different: magnitude-shock (II-a) calls for more bandwidth — larger R_B, higher α_B, faster tempo; structural-shock (II-b) calls for a different model class — an expansion, grafting, or compression in the sense of 4.8. An organization running rapid-response incident drills against a bureaucratic-process adversary faces II-a and responds with more capacity; the same organization hit by a technology-regime change faces II-b and responds with restructuring. The two produce similar pain signals but admit opposite cures. Collapsing them confuses diagnosis.

Pairing with `#adversarial-edge-targeting` (Part III GAP). The classification provides the recipient-side pattern-recognizer; the emitter-side question — which of B's strategy edges are most valuable for A to attack — provides the adversary-optimization complement. The two compose: the emitter chooses which edge to target; the classification determines what happens at that edge. A particularly sharp move the classification makes visible but the emitter-side formulation hides: the **Regime-I-with-adversarial-content attack** — an emitter who can control the content y_G fed into B's default log-odds signal function (per 9.6) can choose the sign of $(y_G - \hat{P}_\Sigma)$ to push λ_k in the direction that degrades Σ_B most. This is not a destabilizing attack (which lands in Regime II); it is an *informational* attack that exploits B's openness to Regime-I updates to inject misinformation. The emitter-side formulation, which represents coupling as a scalar γ_A, cannot express the sign lever; the recipient-side decomposition does. The value is maximized at edges with large plan-sensitivity J_k and moderate credence p_k — exactly the edges load-bearing for Σ_B. The full arg-max — these two factors plus edge identifiability ι_k, observability σ_k^B, and the opacity targeting-fidelity factor $(1 - H_b^{B|A}/H_b^{\max})$ — is displayed and grounded in 16.2 §"The adversarial-edge-targeting arg-max" (the emitter-side optimizer this recipient-side classifier pairs with).

Agent opacity and the emitter-side regime distribution. Hafez et al. (2026)'s agent opacity $H_b^B = H(S, A \mid S')$ is primarily *emitter-side*: it gates A's ability to model which of B's regimes a given signal will land in. As $H_b^B \to 0$ the emitter can target specific regimes adversarially; as $H_b^B \to \infty$ the emitter's signals distribute across regimes and the targeting advantage collapses. Formally, $\gamma_A^{\text{effective}} = \gamma_A^{\max} \cdot f(H_b^B)$ with f monotonically decreasing. This is the dual of 16.4's story: where observation-gates-advantage says the emitter's advantage degrades when its observations of the *environment* are noisy, opacity says the advantage degrades when its observations of B are noisy. Cooperative signaling is the mirror: low H_b^B (legibility) enables aligned emitters to target signals into B's Regime I. This is 14.3's recipient-side translation: shared objectives give A a better model of B's prior and model class, which lets A produce signals that land in B's informative regime.

Organizational reception intuition. The four regimes correspond to a familiar organizational diagnostic: (Regime I) *heard* — the organization updated on the relevant edges of its Σ_{org}; (II-a) *shattered by shock* — signal was too large/fast, the organization's correction bandwidth was exceeded; (II-b) *couldn't hear* — signal required reorganization beyond the organization's structural capacity, matching Cohen & Levinthal's (1990) absorptive-capacity framing; (III) *absorbed as fatigue* — signal drained $\Delta\rho^*$ without producing structural change, the "death by a thousand memos" pattern. The cross-domain instantiation is cleanest where $\mathcal{F}(\mathcal{M}_B)$ maps to absorptive capacity and $U_{o,B}$ to communication-channel quality.

Future meta-segment candidate. The recipient-side classification is the *signal-reception* complement to `#result-sector-persistence-template`'s *persistence-under-disturbance* structure. If a future pattern emerges elsewhere (e.g., consolidation dynamics classifying inputs, or a more general theory of inter-agent coupling) a meta-segment `#signal-reception-pattern` could name the recipient-classification shape. Premature at this stage; log as speculation.

15.4 *Result:* Adversarial Tempo Advantage

Under adversarial coupling where one agent's actions contribute to the other's disturbance rate, the steady-state mismatch ratio scales superlinearly with the tempo ratio. The construction applies the sector-persistence template's steady-state formula to both agents and ratios them; in the *coupling-dominant limit* with symmetric coupling, the result reduces to a clean power law.

Under Model D (deterministic drift coupling), the exponent is $b = 2$ — the squared tempo advantage. A 2:1 tempo ratio yields a 4:1 mismatch ratio; 3:1 yields 9:1; 10:1 yields 100:1. The faster agent both corrects its own mismatch faster *and* generates disturbance for the opponent faster — the two effects compound multiplicatively rather than add. **Under Model S (stochastic noise coupling) the exponent drops to** $b = 3/2$, inheriting the $1/\sqrt{\alpha}$ steady-state scaling from `#deriv-sector-condition` Prop A.1S. **Under a non-coupling-dominant regime** (where base disturbance dominates coupling-induced disturbance) the exponent approaches 1 (Model D) or 1/2 (Model S). The regime where the squared scaling is sharpest is also where it matters most.

The central methodological framing is that this is **the formal analog of Boyd's "getting inside the opponent's OODA loop"** — military intuition becomes physics. The math itself is not mysterious; it is the steady-state formula applied to both sides and ratioed. The consequence is sharp: at 100:1 mismatch ratio the slower agent is almost certainly past its operating reserve R, which means **structural collapse rather than degraded performance**. Speed advantage in a coupled regime *compounds*, and the consequence is qualitatively different from what additive intuition predicts. The result is conditional on stated scope: the linear-correction case ($\alpha = \mathcal{T}$); the coupling-dominant limit; and the disturbance model. Paired with the scope-honest counterpoint from 15.2 — **speed without coupling is useless**: infinite tempo with zero coupling effectiveness destabilizes no one. Speed and coupling are *jointly required*; the squared scaling only kicks in when both are present in the coupling-dominant regime.

Formal Expression.

Setup. Two agents A, B with adaptive tempos $\mathcal{T}_A, \mathcal{T}_B$, each instantiating `#result-sector-persistence-template` with linear correction ($\alpha = \mathcal{T}$). The adversarial coupling of 15.2 enters each agent's effective disturbance:

Derived (adversarial-tempo-advantage, from sector-persistence-template + adversarial-destabilization coupling model)

$$\rho_A^{\text{eff}} = \rho_{\text{base}} + \gamma_B \cdot \mathcal{T}_B, \qquad \rho_B^{\text{eff}} = \rho_{\text{base}} + \gamma_A \cdot \mathcal{T}_A \tag{15.19}$$

with $\gamma_A, \gamma_B > 0$ coupling effectivenesses and ρ_{base} the shared background rate (asymmetric ρ_{base} generalizes straightforwardly).

Result (adversarial-tempo-advantage, Model D)

Model D: Deterministic coupling, $b = 2$.

The template's Model D conclusion $\|\delta\|_{ss} = \rho^{\text{eff}}/\mathcal{T}$ (linear case, $\alpha = \mathcal{T}$) applied to both agents and ratioed:

$$\frac{\|\delta_B\|_{ss}}{\|\delta_A\|_{ss}} = \frac{(\rho_{\text{base}} + \gamma_A\mathcal{T}_A)\,\mathcal{T}_A}{(\rho_{\text{base}} + \gamma_B\mathcal{T}_B)\,\mathcal{T}_B} \tag{15.20}$$

In the coupling-dominant limit ($\gamma\mathcal{T} \gg \rho_{\text{base}}$) with symmetric coupling ($\gamma_A = \gamma_B$):

$$\frac{\|\delta_B\|_{ss}}{\|\delta_A\|_{ss}} \to \left(\frac{\mathcal{T}_A}{\mathcal{T}_B}\right)^2 \tag{15.21}$$

The exponent is $b = 2$: a **squared** tempo advantage. A 2:1 tempo ratio yields a 4:1 mismatch ratio. The faster agent both (a) corrects its own mismatch faster and (b) generates disturbance for the opponent faster — the two effects compound rather than add. □

Derived (stochastic-tempo-advantage, from sector-persistence-template Model S + coupling)

Model S: Stochastic coupling, $b = 3/2$.

Under Model S the coupling enters the noise scale: $\sigma_B^{\text{eff}} = \sigma_{\text{base}} + \gamma_A\mathcal{T}_A$. Adversary tempo increases unpredictability, not systematic direction. The template's Model S steady state $\|\delta\|_{\text{rms}} = \sigma/\sqrt{2\mathcal{T}}$ (linear $\alpha = \mathcal{T}$, scalar $n = 1$) applied to both agents:

$$\frac{\|\delta_B\|_{\text{rms}}}{\|\delta_A\|_{\text{rms}}} = \frac{(\sigma_{\text{base}} + \gamma_A\mathcal{T}_A)\sqrt{\mathcal{T}_A}}{(\sigma_{\text{base}} + \gamma_B\mathcal{T}_B)\sqrt{\mathcal{T}_B}} \tag{15.22}$$

In the coupling-dominant, symmetric limit:

$$\frac{\|\delta_B\|_{\text{rms}}}{\|\delta_A\|_{\text{rms}}} \to \left(\frac{\mathcal{T}_A}{\mathcal{T}_B}\right)^{3/2} \tag{15.23}$$

The exponent is $b = 3/2$. □

Why 3/2, not 2. The half-power difference between the template's Model D ($1/\alpha$) and Model S ($1/\sqrt{\alpha}$) scalings propagates through the ratio. Numerator contributes $\mathcal{T}_A^1$ from the coupling; denominator contributes $\mathcal{T}_B^{1/2}$ from noise averaging; combined with the A-side $1/\mathcal{T}_A^{1/2}$ gives $\mathcal{T}_A^{3/2}/\mathcal{T}_B^{3/2}$.

Summary of Regime-Dependent Exponents.

Table 15.3

Regime	*Coupling type*	*Dominance*	*Exponent b*	*Source*
1	Deterministic drift (Model D)	Coupling-dominant	2	Derived above
2	Stochastic noise (Model S)	Coupling-dominant	3/2	Derived above
3	Either	Non-coupling-dominant	→ 1 (det.) or → 1/2 (stoch.)	Asymptotic limit

Regime 3 (non-coupling-dominant). When $\rho_{\text{base}} > rsim\gamma \cdot \mathcal{T}$ (or $\sigma_{\text{base}} > rsim\gamma \cdot \mathcal{T}$), the base disturbance dominates and the coupling terms become a perturbation. The mismatch ratio degrades toward $\mathcal{T}_A/\mathcal{T}_B$ (linear, $b = 1$) for Model D, or toward $(\mathcal{T}_A/\mathcal{T}_B)^{1/2}$ for Model S.

The simulation validation across all three regimes is in 16.3.

Epistemic Status.

Both coupling-dominant exponents are *exact* conditional on their respective disturbance models. The squared law ($b = 2$) is *exact* under Model D (deterministic bounded disturbance, GA-2) with coupling-dominant conditions. The 3/2 law ($b = 3/2$) is *exact* under Model S (stochastic disturbance, GA-2S) with coupling-dominant conditions, derived from the $1/\sqrt{\alpha}$ steady-state scaling (Prop A.1S in `#deriv-sector-condition`). Both derivations are straightforward algebra from the respective steady-state formulas and the coupling model. The coupling model itself is an *assumption* — the same one used in 15.2.

The non-coupling-dominant limits ($b \to 1$ for Model D, $b \to 1/2$ for Model S) are derived asymptotically. The smooth transition between regimes is confirmed by simulation (16.3) but the interpolation formula is empirical. The transition between regimes is smooth, not sharp.

Max attainable: exact conditional on the disturbance model and coupling model. The result is as strong as its assumptions; no additional work changes the epistemic status without changing the dynamical model.

Discussion.

Superlinearity is the key result. The naive expectation — twice as fast yields twice the advantage — is wrong under adversarial coupling. The mechanism is that the faster agent both (a) corrects its own mismatch faster and (b) generates disturbance for the opponent faster. These two effects multiply, producing the squared exponent. Speed advantage is not additive; it compounds.

Relationship to 15.2. The steady-state mismatch ratio quantifies how much worse the slower agent does *while both agents persist*. The destabilization threshold (15.2) marks where the slower agent fails entirely — its correction mechanism breaks down. Below the threshold, this segment's mismatch ratio applies. Above it, 15.2's Lyapunov divergence takes over. The two results are complementary: this one gives the score; that one gives the game-ending condition.

Regime dependence is operationally significant. Whether an adversary's tempo increase produces systematic drift (positional maneuvering, API changes, doctrinal initiative) or unpredictable noise (feints, randomized attacks, market volatility) determines the scaling law. The distinction is not academic — $b = 2$ vs. $b = 3/2$ means a 3:1 tempo ratio yields 9:1 vs. 5.2:1 mismatch ratio. The model predicts that consistent, directional pressure is more effective per unit of tempo than unpredictable disruption.

Formal analog of OODA-loop observations. The squared scaling is consistent with Boyd's observation that getting inside the opponent's decision cycle has disproportionate effects. The theory identifies a specific mechanism (multiplicative interaction of correction speed and disturbance generation) and a specific condition (coupling-dominant regime) under which this disproportionality holds. Whether this mechanism is the dominant one in actual adversarial interactions is an empirical question, not a mathematical one.

15.5 *Discussion:* Additional Implications & Discussion

The chapter is where AAT's signed-coupling structure first comes into formal view. Team persistence and adversarial destabilization turn out to be the same inequality viewed in opposite directions — both fall out of the sector-persistence template with effective disturbance decomposed into environmental and coupling-induced terms — and the chapter's headline result (superlinear adversarial tempo advantage) follows from the template's scaling without separate derivation. The recipient-side four-regime classification gives the chapter its distinctive contribution: what the emitter-side scalar $\gamma_A \mathcal{T}_A$ collapses, the recipient-side decomposition recovers, with three boundary conditions in *B*'s existing AAT quantities (applied under a precedence convention — four regimes, not eight) and operational consequence for what kind of repair the receiving agent needs. None of the chapter's four segments carries a `## Findings` section, so the implications segment is their canonical catalog home; the work is mostly synthesis: surfacing how the chapter's machinery composes with Ch.2's closure-defect framework and what the equilibrium hand-off to Ch.5 looks like.

Proof provenance for what follows — the status each implication inherits from its source. The persistence template itself is *exact* (`#result-sector-persistence-template`). The team-persistence condition is *derived given* a disturbance decomposition that is a *formulation* — the additive structure and sign convention are modeling choices (`#der-team-persistence`, conditional). The Model D / Model S destabilization thresholds are exact under coupling models that are *assumptions* treating the attacker's tempo as exogenous; the effects spiral is discussion-grade (`#der-adversarial-destabilization`, conditional). The four-regime classification's structure is derived for Class 1 recipients — exact in the Kalman sub-scope-α case, robust-qualitative for sub-scope β — with the model-class and observability boundaries themselves formulation choices and the four-way partition a precedence convention over three tests (`#der-interaction-channel-classification`, conditional). The superlinear exponents are exact only in the coupling-dominant, symmetric-coupling limit with linear correction (`#result-adversarial-tempo-advantage`, conditional). The contraction obstructions are theorem-imported scope statements (`#result-contraction-template`, conditional). No implication below is stronger than the weakest source it draws on; where a paragraph reaches into Ch.5 it says so.

Cooperative and adversarial are one signed coupling, not two theories.

The chapter intro `#cooperative-adversarial-intro` flags it as a chapter-level architectural commitment, and the chapter's segments deliver on it: cooperative coupling and adversarial coupling are not separate dynamics with their own machinery. They are the *same* signed-coupling structure with γ of opposite sign. `#der-team-persistence`'s composite-persistence condition decomposes effective disturbance for sub-agent i as $\rho_i^{\text{eff}} = \rho_{i,\text{env}} + \sum_j \gamma_{j\to i}^{\text{adv}} \mathcal{T}_j - \sum_j \gamma_{j\to i}^{\text{coop}} \mathcal{T}_j$ — adversarial contributions enter as positive terms (effective disturbance amplification), cooperative contributions as negative terms (effective disturbance reduction). The signed-coupling form is what makes cooperative coupling *quantitatively comparable* to adversarial coupling rather than addressing them in separate vocabularies, and it is what lets the framework's persistence template (Part I Ch.4 implications, §4) absorb both regimes as instantiations of a single Lyapunov argument.

The implications-side payoff is the duality between persistence and destabilization. `#der-adversarial-destabilization` makes this precise: B's persistence certificate is lost when $\rho_B^{\text{eff}} \geq \alpha_B R_B$, where ρ_B^{eff} includes the coupling-amplified term $\gamma_A \mathcal{T}_A$ — a certificate-level statement; escape is *forced* only in the strict case $\rho_B^{\text{eff}} > \alpha_B R_B$ with B's sector bound radially tight (`#deriv-sector-condition` Lemma A.1N — at equality the tight agent asymptotes to R_B without crossing). *This is the negation of B's persistence condition under the same coupling model.* The two are the same inequality, read in opposite directions: persistence asks whether the inequality holds, destabilization asks whether it fails, and the coupling parameter $\gamma_A \mathcal{T}_A$ is what determines which side of the threshold the composite sits on. The framework's posture is therefore that adversarial attacks are *not* a separate kind of phenomenon requiring separate machinery — they are the same machinery, with the attacker's contribution entering the target's effective disturbance via the signed-coupling structure.

The closed-form unification this enables, surfaced in Part III Ch.2 implications, deserves to be repeated as it bites here: the four prior persistence results — single-agent ($\gamma = 0$), team-persistence ($\gamma < 0$), adversarial-destabilization ($\gamma > 0$), coordination-failure ($C > \alpha$, Brooks's Law) — are signed special cases of one inequality. Cooperative and adversarial coupling sit at opposite ends of the same γ-axis; the framework's machinery does not pick a side, and the practitioner who understands one regime understands the other by sign change.

Universal taxonomy of inter-agent events — what the recipient sees.

`#der-interaction-channel-classification` is the chapter's distinctive AAT-internal contribution: the same signal from emitter A lands on recipient B as one of *four* qualitatively different things — *informative update*, *magnitude shock*, *structural shock*, or *ambient noise* — and which one is determined by three independent boundary conditions stated entirely in B's existing AAT quantities. The emitter-side scalar $\gamma_A \mathcal{T}_A$ that previous segments have used as a stand-in for "how strong the coupling is" loses this distinction; the recipient-side classification recovers it. The implication for the framework's repair machinery is concrete: the four regimes route to four different corrective paths, and "more tempo" works for some regimes but not for others.

The three boundary conditions deserve to be named even at the implications-segment grain, and the two event-level quantities they are stated in must not be conflated: the event's *magnitude* in *B*'s observation space (how large a mismatch perturbation it produces on arrival) and its *information content* conditional on *B*'s prior — a large already-predicted event has the first and not the second; a tiny structurally-novel event has the second and not the first. The first boundary, *sector-region*, separates informative updates from magnitude shocks: whether the event's magnitude stays within *B*'s sector-condition radius R_B. Inside, *B*'s correction discharges it; outside, the correction does not point inward strongly enough to discharge the mismatch before the next event, and under sustained arrival rate $\nu > rsim\alpha_B$ destabilization proceeds per `#der-adversarial-destabilization`. A magnitude shock can be perfectly representable in *B*'s model class — the classification's own worked case is a Gaussian filter hit by a $\pm\Delta$ kick it handles mathematically but cannot discharge fast enough — so the problem is discharge rate, not representability. The second, *model-class*, separates magnitude shocks from structural shocks: whether the event's information content exceeds what *B*'s model class can represent. When it does (`#result-structural-adaptation-necessity` applied locally), parametric update cannot close the mismatch and residuals retain systematic structure. The third, *observability-floor*, separates ambient noise from informative updates: whether the event's information content times its arrival rate clears *B*'s observation-noise-scaled detection threshold; below it the event produces no usable update — and it is not filtered for free, because it still enters as variance (next paragraph). The four named regimes follow from these three tests under a precedence convention (fail sector-region first, then model-class, then observability); they are the coarsest useful partition, not a forced one.

The framework's prescription is that *repair path depends on regime*, not on event magnitude or coupling strength alone — and the repair levers are the ones the classification segment names, on three distinct axes. Informative updates need no repair; they are the good case, absorbed by the standard gain-principle update (more $\mathcal{T}_B$ absorbs more of them per unit time, but that is ordinary adaptation, not a cure). Magnitude shocks respond on the *capacity* axis — larger R_B, higher α_B, faster tempo; in sub-scope-α instances where α_B is the update gain (the classification's Kalman case, $\alpha_B = \eta_B^*$), better observation quality raises α_B and so helps, but the axis is capacity, not observation quality as such. Structural shocks respond on the *class* axis — model-class expansion per `#result-structural-adaptation-necessity` — and no amount of capacity suffices; since both destabilization sub-regimes present identically as "adaptive reserve exceeded," misdiagnosing one as the other prescribes the wrong cure. Ambient noise responds on the *rate* axis — lower observation noise $U_{o,B}$ or a higher event rate lifts events over the floor — or, when the aggregate load is material, to *infrastructure-level filtering* below the agent proper: the events produce no usable update, but processing them consumes tempo and their variance drains adaptive reserve, so the repair is to keep them from reaching the core loop at all rather than to invest agent-level detection in them. The implication for adversarial robustness: an attacker who knows the target's regime boundaries can choose events that *miss* the target's repair vocabulary — structural shocks against a tempo-investment defender, magnitude shocks against a model-class-expansion defender. The framework's design prescription is that defenders should *cover all four repair paths*, not optimize for one.

The composite-level payoff is the regime-typed effective disturbance decomposition. The chapter's ρ_B^{eff} decomposes per regime, matching the displayed decomposition in `#der-interaction-channel-classification`: magnitude shocks contribute peak-load disturbance; structural shocks contribute the structural mismatch floor and set structural-adaptation cadence; ambient noise contributes *variance* — it enters Model S's $\sigma_{w,B}^2$ below the observability floor, yielding no usable update while slowly draining adaptive reserve (the drain-without-information is exactly why a Regime-III flood is an attack rather than an inconvenience); and informative updates enter with *negative* sign — they reduce effective disturbance, generalizing `#der-team-persistence`'s cooperative term. The aggregate effective-disturbance number from `#der-team-persistence` is the signed *sum* across regimes; the per-regime decomposition is what tells the practitioner *where the disturbance is coming from*. For composite-agent design under uncertainty about the disturbance regime, the framework's prescription is to instrument the four regime detectors independently — knowing ρ_B^{eff} is small is less useful than knowing which regime is contributing what.

Adversarial tempo advantage as superlinear scaling — and the disturbance-model split.

`#result-adversarial-tempo-advantage` carries the chapter's headline result, and the implication worth surfacing is *why* the scaling is superlinear. The template's Model D conclusion gives $\|\delta\|_{ss} = \rho^{\text{eff}}/\mathcal{T}$ in the linear case ($\alpha = \mathcal{T}$); substituting the coupling-amplified disturbance into both agents' steady states and ratioing gives — in the coupling-dominant limit $\gamma\mathcal{T} \gg \rho_{\text{base}}$ with symmetric coupling $\gamma_A = \gamma_B$ — the steady-state mismatch ratio scaling as $(\mathcal{T}_A/\mathcal{T}_B)^2$: superlinear by exponent $b = 2$ under deterministic drift coupling. Under Model S (stochastic) the template's $1/\sqrt{\alpha}$ scaling produces $b = 3/2$ in the same limit. The scope conditions travel with the exponents: outside the coupling-dominant limit the ratio is the full two-term expression and the exponent is a local elasticity, not a law. The two exponents are not arbitrary parameters: they fall out of Part I Ch.4's persistence-template structure, $1/\alpha$ versus $1/\sqrt{\alpha}$, applied to the coupling model. The framework's posture is that *the same template that explains why adaptive systems persist also predicts the precise rate at which they lose to a faster attacker* — the dual is structural.

The exponent splits further in Ch.5 — `#result-adversarial-exponent-regimes` gives three regime-typed values — $b = 2$ (Model D) and $b = 3/2$ (Model S) coupling-dominant, degrading smoothly toward $b \to 1$ (Model D) or $b \to 1/2$ (Model S) as base disturbance grows to dominate the coupling term. That non-coupling-dominant regime is a statement about the disturbance *budget*, not about observation noise. A second, distinct gating acts through the observation channel: `#obs-gated-tempo-advantage` records, as simulation evidence at observation tier, that noise on the agents' mismatch signals collapses the exponent (a faster attacker makes more corrections, each of them noisy), and the classification segment gives the recipient-side reading — higher $U_{o,B}$ pushes more of A's events into Regime III. Whether a defender can *exploit* this by adding noise to its own channel is not settled by either result: the same $U_{o,B}$ lowers the defender's gain $\eta_B^* = U_{M,B}/(U_{M,B} + U_{o,B})$, its tempo, and its ordinary persistence margin, and the Regime-III events it now receives still drain reserve through variance — the net is an unmodeled trade, and the defender-side prescription is a hypothesis, not a framework prediction. What the chapter establishes is narrower and firm: the superlinear exponent belongs to the coupling-dominant, low-observation-noise regime, and both gatings move the system out of it.

The operational anchor worth carrying is Boyd's OODA-loop aphorism, which the framework gives precise content — and the content is a *product* threshold, not a tempo ratio. *Inside the opponent's loop* means $\gamma_A\mathcal{T}_A > \Delta\rho_B^* = \alpha_B R_B - \rho_{B,\text{base}}$ (Model D): the attacker's tempo *times its coupling effectiveness* exceeds the target's adaptive reserve. Under linear correction ($\alpha_B = \mathcal{T}_B$) the Model D form is $\mathcal{T}_A > \mathcal{T}_B R_B/\gamma_A - \rho_{B,\text{base}}/\gamma_A$; the pure tempo ratio $\mathcal{T}_A > \mathcal{T}_B/k$ with $k = \gamma_A/R_B$ is its $\rho_{B,\text{base}} = 0$ case, earned only under linear Model D; under Model S the threshold is in $\sqrt{\alpha_B}$ and no ratio form survives, and in every case speed without coupling ($\gamma_A \to 0$) buys nothing. What the inside-the-loop position buys the attacker is not mere tempo advantage but the loss of B's persistence certificate. The chapter's intro framing — "superlinear tempo advantage as regime change rather than quantitative gap" — is the implications-flavored point: at the destabilization threshold the target does not gradually degrade; its certificate is lost, and strictly past it with a radially tight sector bound escape is forced — the way a balance held just barely beneath a tipping point is qualitatively different from one well above it. The threshold is the same one Part I Ch.4's central inequality names; the chapter's contribution is identifying the coupling-induced disturbance term that pushes the target past it.

Three adversarial obstructions to contraction — the equilibrium hand-off.

`#result-contraction-template` (Appendix A, depended on by Ch.4 via the persistence template) names the chapter's most consequential structural recognition: contraction analysis on a *shared trajectory-error state variable* — the machinery that runs the cooperative half of Part III — does not extend to strategic / adversarial regimes, and three obstructions from three different literatures say so, each scoped to a named assumption of the method. *Compositional contraction* (Slotine 2003) requires attracting fixed points of the composed dynamics; two-agent strategic dynamics with opposing objectives yield saddle-point equilibria of the joint best-response field, attracting in some directions and repelling in others, and the compositional theorems do not apply. *Passivity* (Khalil 2002 ch. 6) gives closed-loop stability only when the interconnection respects the port structure: the dissipation inequality $\dot{S} \le s(u, y)$ bounds storage growth by the supply rate, and a strategic opponent chooses its input to make that supply rate positive — passivity delivers cooperative and neutral robustness, not adversarial. *Last-iterate non-convergence* (Daskalakis et al. 2018): for generic non-zero-sum games, learning dynamics converge to equilibrium only in time-average, not last-iterate, sense — and there is no contraction metric for a system whose last iterates do not converge to an attractor.

The implications-side payoff is structural rather than parametric: three *unrelated* obstructions converge on the same boundary, which makes the limit a property of the regime rather than of AAT's framework — it is not that AAT's contraction machinery is too weak; it is that the attracting-fixed-point scope condition contraction analysis depends on fails in the strategic regime across multiple formal traditions. But the boundary is a boundary on the *state variable*, not on the mathematics. What Ch.5's `#deriv-strategic-composition` does is change the state variable to deviation-from-equilibrium, $\xi = \pi - \pi^*$, and under potential-game (Monderer-Shapley 1996) or monotone-game (Rosen 1965) conditions — its sub-scope α' — the *same* sector / strong-monotonicity machinery transfers, with the sector constant living at the joint potential's curvature or the joint Jacobian's symmetric part. So "contraction cannot handle strategic regimes" is true of contraction-to-shared-truth and false as a class claim: strictly monotone games are contracting in the right coordinates. The genuine no-go is sub-scope β' — cyclic, non-monotone games (bilinear saddles, rock-paper-scissors), where last iterates orbit and only set-convergence to coarse correlated equilibria survives (Hart–Mas-Colell 2000). The framework's posture is to recognize the limit precisely and hand off at the boundary.

The hand-off lands in Ch.5: `#deriv-strategic-composition` is where equilibrium machinery takes over for the symmetric partially-opposing case. The chapter's contribution is naming the limit; Ch.5's is supplying the replacement state variable and its sub-scope conditions. The posture across the boundary is that *scope-honesty is architecture*: knowing where each tool applies cleanly is more useful than extending one tool across all regimes, and a sector-condition argument run on a shared-truth state variable in a strategic regime would be certifying the stability of a point that is not an attractor.

The chapter also surfaces a quieter consequence, and it does not come from this chapter's signed-disturbance model — it comes from the contraction template's composition results: a *feedback interconnection* whose loop gain is too strong can break composite contraction even when every coupling is cooperative — a loop-gain topology statement, not one about the chapter's signed γ. In (CC-feedback) the closed loop contracts iff the loop gains satisfy $k_{12}k_{21} < 4\lambda_1\lambda_2$, a small-gain condition with no sign in it, and the heterogeneous critical-mass inequality (CM2-M), $(\lambda_1 - C_1)(\lambda_2 - C_2) > k_{12}k_{21}/4$, says the same with coordination cost included. In the signed-disturbance picture stronger cooperative coupling only lowers ρ^{eff}; in the composition picture it also raises loop gain, and past the small-gain bound the composite loses its certificate with no adversary present (conditional; Slotine 2003-imported). The case often conflated with this — tightly-coupled multi-agent learning where reward shaping creates implicit competition among nominal cooperators — is not strong cooperative coupling at all; it is partially-opposing objectives, and it belongs to Ch.5's strategic-composition machinery, where the effects spiral becomes a joint-Jacobian eigenvalue condition.

Closing the chapter and bridging to Strategic Composition.

The chapter delivers the signed-coupling structure that the rest of Part III operates on. Cooperative coupling reduces effective disturbance; adversarial coupling amplifies it; the same template handles both via the sign of γ. The four-regime taxonomy says what the recipient actually sees, and the framework's repair vocabulary diversifies accordingly. The super-linear adversarial tempo advantage falls out of the template's $1/\alpha$ versus $1/\sqrt{\alpha}$ scaling in the coupling-dominant limit; OODA's "inside the opponent's loop" acquires mathematical content as a coupling-times-tempo threshold against reserve. The three convergent obstructions to contraction analysis name precisely where the chapter's machinery hands off to equilibrium-theoretic tools.

What the chapter does *not* deliver is the equilibrium machinery itself, the agent-opacity formalism, the recipient-side classification's emitter-side dual, or the per-dimension persistence treatment that adversarial targeting depends on. Those all live in Ch.5, where the framework's machinery operates at full Part III scope: strategic composition with partially-opposing objectives, agent opacity as the formal dual of observation quality, the 16-cell emitter-recipient adversarial-targeting decomposition, the regime-typed exponent decomposition ($b = 2$ and $3/2$ coupling-dominant; $b \to 1$ or $1/2$ non-coupling-dominant), and the matrix-Loewner persistence formulation that adversarial concentration on weak dimensions requires. The chapter-end here is the last single-layer-of-coupling synthesis before Ch.5's strategic-equilibrium machinery takes over.

The bridge into Ch.5 runs through three threads. First, the contraction-template obstructions hand off to equilibrium machinery (potential / monotone games). Second, the recipient-side classification will compose with the emitter-side opacity classification in Ch.5 into a 16-cell joint composition with closed-form arg-max for adversarial-targeting decisions — the chapter here supplies one half, Ch.5 supplies the other. Third, the signed coupling γ structure feeds the *trust meta-model* of Ch.3's `#hyp-communication-gain`: the sign and persistence of the coupling i observes from j is *evidence* updating i's teleological-unity uncertainty $U_{\text{align},ji}$ — sustained cooperative coupling lowers it, observed adversarial coupling raises it — with the update carrying its own gain and its own mismatch, as any meta-model does. Coupling sign does not map directly onto alignment certainty, and the hypothesis segment's own caveat is the reason: an adversary's best move is to present cooperative coupling precisely so that U_{align} stays low. The implications-segment series continues to weave back across earlier chapters, with each chapter's machinery operating on what previous chapters have built.

Chapter 16

Strategic Composition and Channel Effects

[**Gap**] Open question

16.1 *Derivation:* Strategic Composition via Equilibrium Convergence

When two or more AAT agents interact through a shared environment with **partially-opposing objectives** $\{O_t^{(i)}\}$, the composition-level question is not "does the trajectory contract to zero closure-defect?" but "does the coupled best-response dynamics admit an equilibrium and converge to it?" Contraction to shared truth is a $U_O = 1$ special case; strategic composition is the $U_O < 1$ companion regime in which the correct primitive is **fixed-point existence and stability**, not Lyapunov contraction on a shared state. The framework names three composition-level questions distinctively here: **(SC-1)** does an equilibrium joint policy exist where no sub-agent has a unilateral mismatch reduction available; **(SC-2)** do coupled best-response dynamics initialized near it stay there; **(SC-3)** do they converge from arbitrary initialization to the equilibrium set? Fixed-point, local-stability, and reachability questions respectively — none a Lyapunov contraction question on a shared state variable.

Two sub-scopes carry progressively weaker conditions. Under potential-game (Monderer-Shapley 1996) or monotone-game (Rosen 1965) conditions — *sub-scope* α' — the sector-persistence template transfers to the gradient of the joint potential (resp. to a weighted-norm variational inequality on the joint Jacobian's symmetric part), and AAT's persistence machinery recovers at the equilibrium layer with state variable $\xi = \pi - \pi^*$ (deviation from Nash) and sector constant α_{joint} living at the *joint potential's curvature* rather than at any individual sub-agent's α. Outside α' — *sub-scope* β' — only set-convergence to coarse correlated equilibria is available (Hart-Mas-Colell 2000, rate $O(1/\sqrt{T})$), and the macro-state of a strategic composite is a *distribution* on joint strategy space rather than a state-space point. The right structural consequence is named on the **dynamic-regime axis** rather than the architectural-class axis (see `#disc-dynamic-regime-axis`): the strategic-composition move transitions the composite from contraction-regime (R0, under aligned objectives — scope routes C-i/ii/iii) to equilibrium-regime (R1, under partially-opposing objectives with potential/monotone structure — sub-scope α') or to cyclic-distributional-regime (R2, under partially-opposing objectives without potential/monotone structure — sub-scope β'). The composite's *macro-state type* changes correspondingly — state-variable in R0 / fixed-point object in R1 / distributional object in R2 — but the *architectural class* of the composite is preserved under goal-blind routing and distinct sub-agent substrates (it is Class 1 per the formal criterion when the sub-agents are Class 1; the Class 2 / Class 3 changes per `#hyp-directed-separation-under-composition` Case 2 require routing-structure goal-dependence or shared-substrate G^c-allocation, not strategic composition itself). This segment establishes the framing, derives the α'-transfer (R0 → R1 Lyapunov-machinery transfer per `#disc-dynamic-regime-axis`

§"Lyapunov-machinery transfer R0 → R1"), documents the β' scope limits honestly, and relates to `#der-adversarial-destabilization` (asymmetric adversarial) and `#deriv-critical-mass-composition` (aligned composition with shared target) as siblings on the composition axis.

Formal Expression.

Formulation (strategic-composition-framing)

The framing move.

For N purposeful sub-agents with partially-opposing $\{O_t^{(i)}\}$ running coupled AAT loops through a shared environment, the composition-level question is:

(SC-1) Existence of equilibrium. Does there exist a joint state $(X_{c,1}^*, \dots, X_{c,N}^*)$ — equivalently a joint policy profile $(\pi_1^*, \dots, \pi_N^*)$ — such that no sub-agent has a unilateral mismatch reduction available?

(SC-2) Stability of equilibrium. Do coupled best-response dynamics, initialised near $(X_{c,i}^*)$, remain there or return to it?

(SC-3) Convergence from interior. Do coupled dynamics, initialised away from any equilibrium, converge to the equilibrium set?

(SC-1)–(SC-3) are a fixed-point-existence question, a local-stability question, and a reachability question respectively. None is a Lyapunov contraction question on a shared state variable. The contraction framing in `#form-composition-closure` / `#deriv-critical-mass-composition` is recovered as the $U_O = 1$ special case (unique Nash equilibrium at the shared-objective optimum; best-response dynamics collapse to single-objective optimization).

Derived (A2'-analog-potential, from Monderer-Shapley 1996)

Sub-scope α': potential and monotone games (A2'-analog derived).

A strategic interaction $(\{A_i\}, \{O_t^{(i)}\})$ is a **potential game** (Monderer & Shapley 1996, *Games and Economic Behavior* 14) when there exists a scalar potential Φ such that each sub-agent's unilateral improvement matches the potential's unilateral improvement:

$$O_t^{(i)}(\pi_i', \pi_{-i}) - O_t^{(i)}(\pi_i, \pi_{-i}) = \Phi(\pi_i', \pi_{-i}) - \Phi(\pi_i, \pi_{-i}) \quad \forall i. \tag{16.1}$$

Under this condition plus each sub-agent's B1 directional fidelity (4.3) to $\nabla_{\pi_i} O_t^{(i)}$, the joint best-response dynamics satisfy

$$\frac{d\Phi(\pi)}{dt} = \sum_i \langle \nabla_{\pi_i} \Phi,\ \dot{\pi}_i \rangle \geq \alpha_{\text{joint}} \|\nabla \Phi\|^2 \quad \text{for some } \alpha_{\text{joint}} > 0 \tag{16.2}$$

whenever the joint configuration is not at a stationary point of Φ. This is (T2) transcribed to state variable ξ = gradient-of-potential, correction function F = joint best-response velocity field; the quadratic Lyapunov structure is the same. **The sector-persistence template transfers** with $\xi = \pi - \pi^*$ (deviation from Nash), $\alpha = \alpha_{\text{joint}}$, R = basin-of-attraction radius, and ρ_ξ the exogenous disturbance rate on strategy space. Equilibrium existence follows from potential-function compactness on compact strategy spaces; equilibrium stability follows from Φ's role as a joint Lyapunov function.

Derived (A2'-analog-monotone, from Rosen 1965)

Weaker than potential: a game is **diagonally strictly concave** (Rosen 1965, *Econometrica* 33) when the Jacobian of the joint gradient field is negative-definite on the joint strategy space. Under this condition, there exists a unique Nash equilibrium and the joint gradient dynamics converge to it exponentially. The convergence rate is bounded below by the smallest eigenvalue of the symmetric part of the joint Jacobian, playing the role of α_{joint}. No scalar potential need exist; a *weighted-norm* Lyapunov argument on the joint Jacobian's symmetric part substitutes.

Sub-scope α' comprises: potential games (Monderer-Shapley), monotone games (Rosen), strongly-monotone games, and exponential-family dual-averaging under concave objectives. For these classes, the sector-persistence template extends to equilibrium convergence with composite sector constant α_{joint} inheriting from the joint-gradient-field structure.

Sub-scope β': non-potential non-monotone games.

Derived (equilibrium-existence-via-VI, from Facchinei-Pang 2003)

Every strategic interaction with continuous strategy spaces and regular payoffs can be reformulated as a **variational inequality**: find $\pi^* \in \mathcal{K}$ such that $\langle F(\pi^*), \pi - \pi^* \rangle \geq 0$ for all $\pi \in \mathcal{K}$, where F is the joint pseudo-gradient field. When $\mathcal{K}$ is compact-convex and F is continuous, a solution exists (Hartman-Stampacchia theorem). **Pure-strategy Nash equilibrium existence is therefore guaranteed** for continuous-strategy games with compact convex strategy sets and continuous payoffs. But the VI framework gives *existence* only, not *convergence of any specific dynamic to the solution*; solutions may be non-unique.

Derived (regret-minimization-convergence-to-CCE, from Hart-Mas-Colell 2000)

Under no-regret learning (e.g., Hedge / multiplicative weights, Freund-Schapire 1997), the empirical joint distribution converges to the set of **coarse correlated equilibria** (CCE) at rate $O(1/\sqrt{T})$. This requires no structure on the game beyond each sub-agent computing its own regret.

Under β', the macro-state of a strategic composite is a *distribution* on the joint strategy space — the empirical joint play whose support is the CCE — rather than a state-space point. This is the structural shape of "convergence" in the β' regime: distributional convergence, not pointwise. Pure-strategy Nash may or may not exist (cyclic games — rock-paper-scissors, matching pennies — lack pure Nash but retain mixed Nash via Nash 1950 and CCE convergence via Hart-Mas-Colell 2000); the β' machinery's guarantees are at the distributional layer regardless.

Sub-scope β' scope-honesty: AAT can predict that long-run joint play lies in the CCE support; it cannot predict short-run trajectory, per-sub-agent mismatch convergence, or selection among multiple equilibria. The sector-persistence template does *not* apply in β'. This is a genuine scope limit shared with game theory as a whole, not a defect of AAT.

Cross-layer linearization fingerprint. The joint Jacobian of F at saddle-only Nash equilibria in β' has imaginary-axis (semisimple) spectrum — the linearization fingerprint of the R0-loss rung in the single-agent certificate-strength ladder of `#result-certificate-existence`. FTRL × graphical-constant-sum-with-fully-mixed Nash (Cheung-Piliouras-Tao 2021 Theorem 19: lossless DGS + Poincaré recurrence on bounded level sets) is the worked composite instance of R0-loss; the connection is between layers (single-agent linearized error space $\mathbb{R}^n$ vs composite joint-strategy space $\mathcal{X}^c$), not a collision of R-letter ladders.

Zero-sum scalar worked example.

Derived (zero-sum-scalar-instantiation)

Two agents A, B with scalar actions $a_i \in [-1, 1]$ and state $s_{t+1} = s_t + a_A - a_B + w_t$, $w_t \sim \mathcal{N}(0, \sigma^2)$. Objectives $O_t^{(A)}(s) = s$ (maximize s), $O_t^{(B)}(s) = -s$ (minimize s); zero-sum at the state-dependent payoff level.

Action-coefficient analysis. The agents' state-preferences are opposite, but the action-coefficients in s_{t+1} are also opposite: $\partial s/\partial a_A = +1$ and $\partial s/\partial a_B = -1$. Composing with the objectives' state-dependence:

$$\frac{\partial O^{(A)}}{\partial a_A} = \frac{\partial O^{(A)}}{\partial s} \cdot \frac{\partial s}{\partial a_A} = (+1)(+1) = +1 \tag{16.3}$$

$$\frac{\partial O^{(B)}}{\partial a_B} = \frac{\partial O^{(B)}}{\partial s} \cdot \frac{\partial s}{\partial a_B} = (-1)(-1) = +1 \tag{16.4}$$

Both agents marginal-prefer *increasing* their own action — the opposing state-preferences and opposing action-coefficients compose to aligned action-preferences.

Potential function. For Monderer-Shapley, $\partial\Phi/\partial a_i = \partial O^{(i)}/\partial a_i$ for each i. With both partial derivatives equal to $+1$, the potential is $\Phi(a_A, a_B) = a_A + a_B$ (modulo additive constant).

Nash equilibrium. Each agent's best response on $[-1, 1]$ is to push its own action to $+1$; the unique Nash equilibrium is $(a_A^*, a_B^*) = (1, 1)$. At equilibrium, $\Delta s = a_A^* - a_B^* = 0$: the substantive zero-sum property is that equal opposing action-coefficients cancel under the agents' aligned action-preferences, so joint action contributes no net displacement to s (state drift comes only from w_t). This is qualitatively different from the naive picture in which the agents push the state in opposite directions.

Sector-persistence template — interior-NE instantiation via Cournot-style payoffs. The unregularized example has a *linear* potential Φ and a *corner* equilibrium at $(1, 1)$; the unconstrained gradient field $\nabla\Phi = (1, 1)$ is constant in $\xi = \pi - \pi^*$, so there is no interior linear restoring force around the equilibrium and the template's (T2) sector lower bound $\xi^T F(\xi) \geq \alpha\|\xi\|^2$ does not hold without modification — the saturation, not a sector contraction, is what enforces equilibrium. The aligned-action-preferences finding is the conceptual payload of the unregularized form; the sector-template transfer requires a payoff structure whose curvature is genuinely quadratic, not merely projected onto a corner.

A Cournot-style duopoly (Cournot 1838; Monderer-Shapley 1996 §3 example) supplies that structure naturally. Two firms produce quantities $q_i \in [0, Q_{\max}]$; market price is $P(q_A, q_B) = a_0 - b(q_A + q_B)$; per-firm cost is linear at rate κ; per-firm profit is

$$O^{(i)}(q_A, q_B) = q_i\,(a_0 - b(q_A + q_B) - \kappa). \tag{16.5}$$

This is partially-opposing in the policy-conflict sense — each firm's marginal payoff is decreased by the other's production through the shared price — without being strictly zero-sum. It is a potential game with quadratic potential.

Derived (Cournot-as-potential, from Monderer-Shapley 1996 §3)

$$\Phi(q_A, q_B) = (a_0 - \kappa)(q_A + q_B) - b(q_A^2 + q_B^2) - b\,q_A q_B, \tag{16.6}$$

verified by $\partial\Phi/\partial q_i = (a_0 - \kappa) - 2bq_i - bq_{-i} = \partial O^{(i)}/\partial q_i$. The symmetric interior Nash equilibrium is $q^* = (a_0 - \kappa)/(3b)$, located strictly inside the action box whenever $0 < q^* < Q_{\max}$.

Under continuous-time best-response $\dot{q}_i = \partial O^{(i)}/\partial q_i$, writing $\xi = q - q^*\mathbf{1}$, the joint dynamics are $\dot{\xi} = -bM\xi$ with $M = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. Setting $F(\xi) = bM\xi$ (correction-strength sign convention; $\dot{\xi} = -F(\xi)$):

- (T1) $F(0) = 0$. ✓
- (T2) $\xi^T F(\xi) = b(2\xi_A^2 + 2\xi_B^2 + 2\xi_A\xi_B) \geq b\|\xi\|^2$, since M's eigenvalues are $\{1, 3\}$. So $\alpha_{\text{joint}} = b$ — the smallest eigenvalue of the symmetric part of the joint Jacobian, exactly the quantity Rosen 1965 identifies as the convergence rate. ✓
- (T2-upper) $\|F(\xi)\| \leq 3b\|\xi\|$ from M's largest eigenvalue.
- (T3) $R = \min(q^*, Q_{\max} - q^*)$ — the radius of the largest origin-centered ball in ξ-space contained in the feasible region; the closer of the two active constraints (zero-production and capacity) determines the basin.

The template's preconditions hold with α_{joint} inherited from the demand-side curvature parameter b rather than from any individual sub-agent's α. This is the substantive transfer: the composite-level sector constant lives at the *joint potential's curvature*, with economic interpretation (market-saturation slope) rather than ad-hoc parametrization. Other linear-quadratic strategic-substitutes games (LQR with coupled state, network-coordination with quadratic disagreement cost, public-goods with quadratic externalities) produce structurally equivalent template instantiations, with α_{joint} tracking the symmetric part of their respective joint Jacobians.

The unregularized scalar zero-sum and the Cournot instantiation are complementary: the former illustrates the framing's surprise — opposing state-preferences combined with opposing action-coefficients yield aligned action-preferences — while the latter exhibits the template transfer with genuine quadratic curvature and interior NE.

New scope route (C-iv) in `#scope-composite-agent`.

Proposed Scope (composition-scope-condition, route C-iv)

Strategic composition with partially-opposing $\{O_t^{(i)}\}$ admits a joint equilibrium structure $\mathcal{E}$ (Nash, correlated, or coarse correlated) such that coupled best-response dynamics converge to the support of $\mathcal{E}$. The composite exists as an AAT agent with macro-state defined relative to $\mathcal{E}$ rather than relative to a shared target state.

(C-iv) is qualitatively distinct from (C-i)–(C-iii): it does *not* require shared objectives, hierarchical derivation, or mutual benefit. It requires only structural convergence of the strategic interaction in the game-theoretic sense. Composites satisfying (C-iv) are **strategic composites**, distinguished from alignment composites (C-i, C-ii) and mutual-benefit composites (C-iii).

What Is Derived vs. What Is Chosen.

Table 16.1

Property	*Source*	*Strength*
Framing move (contraction → equilibrium convergence)	Standard game-theoretic positioning	Formulation choice
(SC-1) / (SC-2) / (SC-3) three-question decomposition	Parallel to fixed-point / stability / reachability in dynamical systems	Formulation choice
A2'-analog under potential-game condition	Monderer-Shapley 1996 transcribed into AAT notation with B1 directional fidelity	Derived (exact under potential-game + B1)
A2'-analog under monotone-game condition	Rosen 1965 transcribed with weighted-norm Lyapunov	Derived (exact under diagonal strict concavity)
Equilibrium existence via VI	Facchinei-Pang 2003 (Hartman-Stampacchia)	Derived (external theorem, applied to strategic composition setting)
Regret-minimization CCE convergence	Hart-Mas-Colell 2000	Derived (external theorem applied)
Sub-scope α' / β' partition	Parallel to `#deriv-sector-condition` α/β	Formulation choice
Zero-sum scalar instantiation (corner-NE conceptual lesson)	Direct substitution into potential-game framework	Exact (within stated setup)
Cournot-style sector-template instantiation (interior-NE quadratic)	Monderer-Shapley 1996 §3 + sector-persistence-template; demand-side curvature b supplies α_{joint}	Exact (within stated setup)
(C-iv) scope route	Extension to `#scope-composite-agent` disjunction	Formulation choice (scope extension)
Effects-spiral eigenvalue condition	Joint-Jacobian $\mathrm{Re}(\lambda_{\max}) > 0$ at candidate equilibria — formalizes `#der-adversarial-destabilization`'s discussion-grade effects spiral	Sketch (specific AAT instantiations open)
Strategic composition transitions composite from contraction-regime (R0) to equilibrium-regime (R1, under α') or cyclic-distributional-regime (R2, under β') per `#disc-dynamic-regime-axis`	Game-structure (potential / monotone / cyclic) under partially-opposing objectives, *not* architectural-class change	Derived (regime-axis, surfaced 2026-05-21; supersedes the earlier "Class 1 sub-agents → Class 2 composite" architectural-class framing)

Property	*Source*	*Strength*
Mechanism-design impossibility landed in `#disc-implementation-impossibility` (designer-side sister meta-pattern to identifiability-floor)	Gibbard-Satterthwaite 1973-75, Arrow 1951, Myerson-Satterthwaite 1983	Charter instances at `#deriv-strategy-proofness-impossibility` / `#deriv-bilateral-trade-impossibility` / `#deriv-social-welfare-aggregation-impossibility`; sub-scope α' named as adjacent-but-not-identical to the preference-domain-restriction escape
Potential function Φ as additive-coordinate-forcing instance	Φ's additivity is definitional consequence of being a potential game, not forced by AAT-internal axiom; adjacent family member	Discussion-grade
Bridge from strategic composition to `#form-composition-closure` ε^*	Macro-description is an equilibrium statistic, not an equilibrium state	Open
General equilibrium-selection under multiple Nash	Existence theorems do not pin down which equilibrium; selection (risk-dominance, payoff-dominance, Pareto) is partial	Open
Mean-field-game limit ($N \to \infty$)	Lasry-Lions 2007; Huang-Malhamé-Caines 2006; requires population scope condition	Open (pending Part III population gaps)

Epistemic Status.

Conditional. Max attainable: *exact* under potential-game + B1 (sub-scope α'); *derived* under monotone-game + diagonal strict concavity; *discussion-grade* for the framing and sub-scope β' set-convergence-only claim.

The A2'-analog results under Monderer-Shapley 1996 and Rosen 1965 are transcriptions of established game-theoretic theorems into AAT notation. AAT's contribution is not the mathematics but the recognition that the sector-persistence template transfers cleanly when these conditions hold, and that the composite-level sector constant α_{joint} lives at the *joint potential gradient* or *joint Jacobian's symmetric part*, not at any individual sub-agent's α. The framing move (contraction → equilibrium convergence) is a structural positioning move.

Sub-scope β' gives AAT substantially weaker predictive power than sub-scope α': *set-convergence to CCE only*, not trajectory convergence or per-agent mismatch convergence. This is an honest scope limit, not a defect — it mirrors game theory's own scope at the no-potential-no-monotonicity regime. AAT does not claim to predict equilibrium selection under multiple Nash, short-run dynamics in cyclic games (rock-paper-scissors), or convergence rates better than $O(1/\sqrt{T})$ in β'.

What this segment does not establish:

- General convergence proof for non-potential games under AAT-native update rules (imports Monderer-Shapley + Rosen as scope statements, doesn't re-prove them).
- Explicit joint-Jacobian analysis for specific AAT agent architectures (e.g., two Beta-Bernoulli agents on a shared DAG). Effects-spiral formalization is sketched, not derived per-instance.
- Bridge from strategic composition to `#form-composition-closure`'s closure-defect ε^* — the macro-description of a strategic composite is a distributional equilibrium statistic rather than a state, so the closure-defect formulation needs re-specification. Open.
- Full mechanism-design derivation (VCG, Bayesian-Nash) for specific AAT-relevant settings.

Honest Limits.

Scope honesty — strategic-composition-failures

- **No pure-strategy Nash equilibrium** in cyclic games (rock-paper-scissors; matching pennies). Mixed-Nash equilibrium exists universally for finite games (Nash 1950) but is a saddle point of best-response dynamics rather than a basin attractor; fictitious play orbits the mixed-Nash without converging pointwise to it. No-regret dynamics still drive the *empirical joint distribution* to the CCE set at rate $O(1/\sqrt{T})$ (Hart-Mas-Colell 2000), so β' machinery applies; the macro-state of a strategic composite in this regime is a distribution over the strategy space, not a state-space point.
- **Multiple equilibria with ambiguous selection** (coordination games; drive-left vs drive-right). Convergence to *some* equilibrium; AAT does not predict which.
- **Slow mixing / finite-sample failure** under regret-minimization: $O(1/\sqrt{T})$ cumulative regret implies finite-horizon play may be far from CCE support.
- **Non-compact strategy spaces** (unbounded prices, resource allocations). No equilibrium without compactification; compactification may be artificial.
- **Bayesian games with strategic + epistemic uncertainty compounding.** Existence and uniqueness become pathological under certain information structures.
- **Mean-field games** ($N \to \infty$) require population-scope machinery not covered here.

Discussion.

Sibling to `#der-adversarial-destabilization`. `#der-adversarial-destabilization` handles the *asymmetric* adversarial case: one agent is target; attacker's tempo is exogenous parameter; sector-persistence template applies to target's mismatch with coupling-amplified disturbance. This segment handles the *symmetric* strategic case: both agents running full AAT loops; state variable is joint deviation from equilibrium; correction function is joint best-response field. The two are siblings. `#der-adversarial-destabilization`'s Working-Notes-flagged "coupled Lyapunov analysis is the open problem" has its formal home here — but the analysis is *not* a Lyapunov problem; it is a fixed-point problem, and the correct primitive is equilibrium convergence rather than Lyapunov descent. `#der-team-persistence` is recovered as the cooperative limit (all γ^{coop} dominate; joint dynamics reduce to parallel single-agent sector-persistence with reduced effective disturbance); `#deriv-critical-mass-composition` sits in the middle with signed γ on a matched-symmetric-Tier-1 dyad under shared target.

The effects spiral's formal home. `#der-adversarial-destabilization`'s effects spiral — B's degrading model causes B's actions to destabilize A further — is a **joint-Jacobian eigenvalue condition** in the strategic-composition framing: the spiral exists iff $\max_{\text{candidate equilibria } \pi^*} \mathrm{Re}(\lambda_{\max}(\nabla F(\pi^*)$ 0, where F is the joint best-response field. This condition specializes monotone-game failure (the Jacobian's symmetric part fails to be negative-definite at equilibrium). The asymmetric formulation in `#der-adversarial-destabilization` cannot express this condition because it treats one agent's tempo as exogenous; the symmetric coupled formulation here does.

Dynamic-regime transition (not architectural-class change). `#der-directed-separation` classifies agents into Class 1 (Separated) / Class 2 (Partial) / Class 3 (Coupled) based on within-agent coupling between f_M and G_t. Strategic composition does *not* move a composite of Class 1 (Separated) sub-agents to Class 2 (Partial): the architectural class is set by the *processing pathway* (a $G^c \to f_M^c$ pathway bypassing e^c, which is the actual Class 1/3 boundary), not by *belief content* (each sub-agent's $M_t^{(i)}$ containing models of other agents' goal-state, which is a normal POMDP feature). Per the formal κ^c criterion of `#der-directed-separation`, under goal-blind routing and distinct sub-agent substrates the strategic composite is Class 1 (Separated) at the composite level — same architectural class as the sub-agents. Class change requires routing-structure goal-dependence (`#hyp-directed-separation-under-composition` Case 2) or shared-substrate G^c-allocation, neither of which is implied by strategic composition itself; the Cournot duopoly is the canonical witness for architectural-class invariance under strategic composition. What strategic composition *does* change is the **dynamic regime** (per `#disc-dynamic-regime-axis`): from R0 contraction-regime under aligned objectives to R1 equilibrium-regime under α' or R2 cyclic-distributional-regime under β'. The macro-state type changes correspondingly — state-variable (R0) / fixed-point object $\mathcal{E}$ (R1) / distributional object $\mu \in \Delta(\mathcal{X}^c)$ (R2). The dynamic-regime framing carries the structural load: the regime change is exactly what this segment derives in the α'/β' partition, while an architectural-class change is not derivable under the formal criterion.

Mechanism-design impossibility. If an outside designer can shape $\{O_t^{(i)}\}$, strategic composition becomes a **mechanism-design problem**: choose objectives so that the induced equilibrium *is* contraction to a designed state. Three classical impossibility results — Gibbard-Satterthwaite 1973-75 (no dominant-strategy non-dictatorial Pareto-efficient voting mechanism for ≥3 alternatives), Myerson-Satterthwaite 1983 (no efficient, individually-rational, incentive-compatible bilateral-trade mechanism without subsidies), Arrow 1951 (no social welfare function satisfying unrestricted-domain, Pareto-efficient, IIA, non-dictatorial) — land as the three charter instances of `#disc-implementation-impossibility`, the designer-side sister meta-pattern to `#disc-identifiability-floor`. Sub-scope α' of this segment is named there as *adjacent to but not identical with* the preference-domain-restriction escape (single-peaked / single-crossing): sub-scope α' secures best-response convergence under fixed reports, not strategy-proof revelation (the adjacency argument is developed in `#deriv-strategy-proofness-impossibility` Discussion). Supporting derivations carry the formal statements, AAT translations, and escape characterizations: `#deriv-strategy-proofness-impossibility` (GS), `#deriv-bilateral-trade-impossibility` (MS), `#deriv-social-welfare-aggregation-impossibility` (Arrow).

Meta-pattern positioning.

- `#disc-separability-pattern`*:* sub-scope α' (potential / monotone — template transfers) is separable-core; sub-scope β' (VI / regret-minimization — set-convergence only) is structured-repair; cyclic / non-convergent / multi-equilibrium-selection-ambiguous is general-open. Candidate additional ladder (strategic-interaction regime); decide whether to surface as 8th ladder or merge into an existing ladder.
- `#disc-identifiability-floor`*:* mechanism-design impossibility is *not* an identifiability-floor instance — there is an actor-positioning mismatch (designer-side vs agent-side), and the named AAT-side escape sub-scope α' addresses best-response-dynamics convergence, not strategy-proof revelation. The cluster lands in the designer-side sister meta-pattern `#disc-implementation-impossibility` instead. See Discussion above and the Related Work entry.
- `#disc-implementation-impossibility`*:* this segment's sub-scope α' is named there as adjacent-but-not-identical to the preference-domain-restriction escape in two charter instances (`#deriv-strategy-proofness-impossibility` for GS and `#deriv-social-welfare-aggregation-impossibility` for Arrow — both sharing the same adjacency).
- `#disc-additive-coordinate-forcing`*:* the potential function Φ plays an additive-coordinate role at the strategic layer, but its additivity is a *definitional consequence* of being a potential game (Monderer-Shapley require the additivity property by definition) rather than forced by a uniqueness theorem on an AAT-internal axiom. This positions Φ as an adjacent family member, parallel to Lyapunov quadratic and IB Lagrangian, not a primary instance.

Active-inference sharpening. Sun-Firestone 2020's dark-room argument observes that preferences-as-priors collapse under mutual prediction. Strategic composition gives this argument a formal substrate: two agents mutually predicting each other become a **fixed-point problem**, not a Lyapunov descent. The active-inference attempt to unify goal-seeking and prediction fails here in a derivable way — the fixed point is not at the prediction-optimum; it is at the strategic-equilibrium, which is structurally different. This strengthens `#def-satisfaction-gap`'s positioning against preferences-as-priors without requiring additional refutation machinery.

16.2 *Derived:* Agent Opacity (H_b)

Alongside AAT's heavily formalized *forward* observation quality (how well the agent sees the world — observation ambiguity, model-class fitness, identifiability floor on what the agent can infer), AAT carries a **dual quantity** measuring how well the world sees the agent: **backward predictive uncertainty** H_b, an observer-indexed, horizon-indexed, trajectory-indexed entropy of the agent's future actions given another agent's filtration. Adopted from Hafez et al. 2026 as a first-class multi-agent quantity. H_b is the dual of observation quality U_o: where U_o characterizes how well the agent sees the world, H_b characterizes how well the world sees the agent.

Each of the four indexing arguments — observer, horizon, trajectory, time — is load-bearing rather than ornamental. *Observer-indexed*: different observers (allies with shared infrastructure; adversaries with limited instrumentation; the environment itself) have different filtrations $\mathcal{F}_B^t$, and H_b varies accordingly — the same agent has different opacity to different observers. *Horizon-indexed*: immediate-next-action opacity ($\tau = 1$) and long-horizon-plan opacity ($\tau \gg 1$)

decouple — an agent may be predictable at immediate action but unpredictable at plan level, or vice versa. *Trajectory-indexed*: per `#scope-agent-identity`'s singular-trajectory scope, H_b is the opacity of *this* trajectory's continuation, not a type-level claim about a class of agents. *Time-indexed*: opacity drifts with learning as the observer's model of the agent improves.

H_b is **sign-flipped in value across regimes**: low H_b (legibility) enables cooperative coordination (15.1); high H_b (opacity) enables adversarial effectiveness (15.2, 15.4). The sign-flip is a direct consequence of AAT's existing signed-coupling structure rather than a separate posit — cooperative coupling wants low opacity (allies must predict to preempt); adversarial coupling wants high opacity (predicted attacks are neutralized). This segment's emitter-side four-regime classification is the dual of `#der-interaction-channel-classification`'s recipient-side theory; together they close `#adversarial-edge-targeting` as emitter-optimizer paired with recipient-classifier, with a 16-cell joint composition supplying closed-form arg-max for adversarial-targeting decisions.

Formal Expression.

Definition (agent-opacity-Hb)

For agent A on singular trajectory $\mathcal{C}_A$ and observer agent B with filtration $\mathcal{F}_B^t$ (per-trajectory observable history per `#scope-agent-identity`'s token-level commitment):

$$H_b^{A|B}(t,\tau) := H(a_{A,t+\tau} \mid \mathcal{F}_B^t) \tag{16.7}$$

the entropy of agent A's action at horizon τ conditional on observer B's filtration at time t. **Four indexing arguments:** observer B, time t, horizon τ, trajectory $\mathcal{C}_A$. Each is load-bearing:

- **Observer-indexed.** Different observers (allies with shared infrastructure; adversaries with limited instrumentation; environment itself) have different filtrations $\mathcal{F}_B^t$; H_b varies accordingly.
- **Horizon-indexed.** Immediate-next-action opacity ($\tau = 1$) and long-horizon-plan opacity ($\tau \gg 1$) decouple: an agent may be predictable at immediate action but unpredictable at plan level, or vice versa.
- **Trajectory-indexed.** Per `#scope-agent-identity`, AAT applies to agents on singular trajectories. $H_b^{A|B}$ is the opacity of *this* trajectory's continuation, not a type-level claim.
- **Time-indexed.** Opacity may drift with learning (as B's model of A improves, $H_b^{A|B}(t)$ decreases); steady-state values exist for ergodic regimes.

Under the IDT-observer specialization — B operates as Hafez's Information Digital Twin monitoring (S_A, a_A, S_A') from outside A's processing — and under ergodicity, $H_b^{A|B}(t,\tau) \to H(S, A \mid S')$ as defined in Hafez et al. 2026. AAT's added features (observer-indexing, horizon-indexing, trajectory-indexing) are the distinctive extensions.

Backward variance decomposition — the duality made structural.

Derived (Hb-variance-decomposition; exact in linear-Gaussian sub-scope α)

In the linear-Gaussian regime, write A's action as $a_{A,t+\tau} = \mu_A(X_{A,t+\tau}) + w_a$ with policy noise $w_a \sim \mathcal{N}(0, U_{a,A}^{(\tau)})$ independent of B's filtration. By the law of total variance, B's predictive variance splits into exactly two sources:

$$\mathrm{Var}\big(a_{A,t+\tau} \mid \mathcal{F}_B^t\big) = \underbrace{\mathrm{Var}\big(\mu_A(X_{A,t+\tau}) \mid \mathcal{F}_B^t\big)}_{U_{\pi,B\to A}^{(\tau)}} + \underbrace{U_{a,A}^{(\tau)}}_{\text{policy noise}}, \tag{16.8}$$

and since the predictive law is Gaussian here (Gaussian mean-uncertainty plus independent Gaussian noise — the same argument as the Kalman predictive),

$$H_b^{A|B}(t,\tau) = \tfrac{1}{2}\log(2\pi e) + \tfrac{1}{2}\log\Big[U^{(\tau)}_{\pi,B\to A} + U^{(\tau)}_{a,A}\Big]. \qquad (16.9)$$

$U_{\pi,B\to A}$ is B's uncertainty about A's policy conditional mean — **epistemic**, reducible as B learns A (observation, communication, shared architecture). $U_{a,A}$ is A's intrinsic action variability given its own state — **aleatoric from B's side**: no amount of observing A reduces it; only A making its policy more deterministic does. This is the Kalman forward decomposition $U_M + U_o$ (3.6) reflected across the agent-environment boundary: $U_{\pi,B\to A} \leftrightarrow U_M$ (the learnable model term) and $U_{a,A} \leftrightarrow U_o$ (the channel-noise floor) — with the channel noise living in the agent's *actuator* rather than its *sensor*. The headline duality claim is thereby structural rather than rhetorical: the forward and backward directions carry matching two-term epistemic-plus-aleatoric variance decompositions. (It is a structural correspondence, not an adjoint/optimization-theoretic duality theorem — see Working Notes.)

Two consequences. **(i) The transparency knobs are independent.** An agent can lower its opacity to allies by lowering U_π (publishing intent — 14.3 is precisely this knob) without touching U_a, or by lowering U_a (policy determinism) without publishing anything; conversely an adversary-facing agent can raise either. **(ii) Identifiability floors bound only the epistemic term.** When B's observations of A's action-consequences sit below an observability floor, $U_{\pi,B\to A}$ cannot be reduced even with unbounded data — a structural lower bound on $H_b^{A|B}$ from B's side, which is the observer-side floor reading in §Meta-pattern positioning.

Tier: exact in linear-Gaussian sub-scope α (law of total variance + Gaussian entropy); robust qualitative beyond (the two-source split survives; the closed log-form does not).

Derived (sign-flip-from-signed-coupling)

Sign-flip via signed coupling.

The value of H_b^A *to* A depends on the sign of A's coupling to other agents — the same signed-coupling structure that organizes `#der-team-persistence`, `#der-adversarial-destabilization`, and `#deriv-critical-mass-composition`'s (CM2) γ parameter.

- **Cooperative coupling ($\gamma^{\text{coop}} > 0$, reducing allies' disturbance).** For B to treat A's action as cooperation rather than disturbance, B must predict A's action well enough to preempt or complement it. Under `#der-interaction-channel-classification`'s recipient-side decomposition, unpredictable ally actions fall into Regime II (magnitude/structural shock) rather than Regime I (informative update). Therefore cooperative coupling effectiveness $\gamma^{\text{coop}}_{A\to B}$ is *increasing in legibility*, equivalently decreasing in $H_b^{A|B}$. Under sub-scope α Gaussian coupling: $\gamma^{\text{coop, effective}} \propto (1 - H_b^{A|B}/H_b^{\max})$.
- **Adversarial coupling ($\gamma^{\text{adv}} > 0$, amplifying target's disturbance).** Predicted attacks are neutralized; unpredicted attacks deliver effective disturbance. Adversarial coupling effectiveness is *increasing in opacity* — the mechanism of adversarial advantage (per `#result-adversarial-tempo-advantage`) operates *through* B's failure to predict A. Under the same sub-scope α setup: $\gamma^{\text{adv, effective}} \propto H_b^{A|B}/H_b^{\max}$.

The sign-flip on H_b's value-to-A lives in the sign of γ itself, not in a different sign on H_b. Cooperative regime ($\gamma^{\text{coop}} > 0$) rewards low H_b; adversarial regime ($\gamma^{\text{adv}} > 0$) rewards high H_b. The same H_b quantity, the same monotone dependence; opposite value-to-A because the signs of the coupling terms differ.

Emitter-side four-regime classification. *[Formulation (emitter-regimes, dual to 15.3)]*

Parallel to `#der-interaction-channel-classification`'s recipient-side four regimes, the emitter A sends events that fall into four emitter-side regimes based on A's opacity signal structure and self-model quality:

- **E-I Broadcast.** A emits actions transparently; $H_b^{A|B}$ is low for any observer B with standard instrumentation. Examples: public announcements, published decisions, legible industrial controllers.

- **E-II Selective-signal.** A is transparent to some observers and opaque to others (e.g., shared allied infrastructure gives allies lower H_b than adversaries without that infrastructure). Boundary: differential instrumentation in $\mathcal{F}_B^t$ across observers.
- **E-III Information-hide.** A is uniformly opaque to observers; actions are randomized, encrypted, or routed through dead-drops. $H_b^{A|B}$ near $H_b^{\max}$ for all observers lacking the key / pattern / channel.
- **E-IV Active-deceive.** A emits actions that mispredict — the observer's model of A converges to a *wrong* prediction that differs from the actual action by a larger margin than the same observer's model of the environment would accommodate. Boundary: A's self-model quality (for active-deceive, A must model the observer's model of A well enough to choose actions that exploit it).

The adversarial-edge-targeting arg-max. The 16-cell emitter-recipient composition (four emitter regimes × four recipient regimes) operationalizes the previously-reserved `#adversarial-edge-targeting` Part III gap: given emitter A choosing which edge k of target B's strategy DAG Σ_B to attack, the targeted-attack value is

Formulation (adversarial-edge-targeting arg-max; factor-wise grounded, multiplicative composition a modeling choice)

$$k^* = \arg\max_k \underbrace{p_k(1-p_k)}_{\text{credence leverage}} \cdot \underbrace{J_k^2}_{\text{plan sensitivity}} \cdot \underbrace{\iota_k}_{\text{edge identifiability}} \cdot \underbrace{\sigma_k^B}_{\text{observability}} \cdot \underbrace{\left(1 - H_b^{B|A}/H_b^{\max}\right)}_{\text{targeting fidelity}}. \tag{16.10}$$

The first four factors are B-**interior** — the value of edge k to a *perfectly-informed* adversary, each with a canonical home: credence leverage $p_k(1-p_k)$ is the Beta-Bernoulli update variance (`#deriv-edge-credence-dynamics`); plan sensitivity $J_k = \partial R_\Sigma / \partial p_k \geq 0$ is the plan-value Jacobian (`#deriv-edge-credence-dynamics`), entering squared because both the injected mismatch and its propagation to plan value scale with it; edge identifiability ι_k is the Regime-A interventional-access coefficient (9.5); and σ_k^B is B's observability of the edge (1.3). The **fifth factor is the opacity contribution**: A's targeting fidelity scales with A's legibility-*of*-B, i.e. with low $H_b^{B|A}$. Under full legibility ($H_b^{B|A} \to 0$) the arg-max is fully exploitable — A strikes the single highest-value edge. Under full opacity ($H_b^{B|A} \to H_b^{\max}$) the fifth factor vanishes and targeting collapses to the untargeted broadcast attack $\arg\max_k p_k(1-p_k)J_k^2$ — the edges most valuable *in expectation*, with no per-edge aim. This is the emitter-side optimizer paired with `#der-interaction-channel-classification`'s recipient-side classifier, closing the pairing that segment's §"Pairing with `#adversarial-edge-targeting`" advertised. The multiplicative composition is a first-order modeling choice (treating the factors as independent); the individual factors are the derived content, and a second-order interaction analysis (e.g., whether the targeting-fidelity factor should modulate J_k rather than multiply it) is the natural strengthening (Working Notes).

Tempo amplification by opacity.

Derived (tempo-amplification-by-opacity)

`#result-adversarial-tempo-advantage`'s tempo-multiplier $\gamma_A \mathcal{T}_A$ in `#der-adversarial-destabilization` decomposes into a tempo term and an opacity term:

$$\mathcal{T}_A^{\text{effective}} = \mathcal{T}_A \cdot \frac{H_b^{A|B}}{H_b^{\max}} \quad \text{(Model D adversarial coupling)} \tag{16.11}$$

The superlinear formula $(\mathcal{T}_A/\mathcal{T}_B)^2$ becomes $(\mathcal{T}_A/\mathcal{T}_B)^2 \cdot (H_b^{A|B}/H_b^{B|A})^2$ under bilateral opacity — a higher-order tensor product with the same exponent $b = 2$ (Model D) or $b = 3/2$ (Model S) from `#result-adversarial-exponent-regimes`. Whether b itself is reshaped under bilateral opacity is open; the leading-order scaling is the tempo-opacity product.

What Is Derived vs. What Is Chosen.

Table 16.2

Property	*Source*	*Strength*
$H_b^{A\|B}(t,\tau)$ definition	Adopted from Hafez et al. 2026; extended with observer / horizon / trajectory indexing per `#scope-agent-identity`	Formulation choice (adoption + AAT-extension)
Reduction to Hafez's $H(S, A \mid S')$ under IDT-observer + ergodic regime	Direct substitution	Derived (exact under IDT + ergodicity)
Sign-flip via signed coupling	Cooperative coupling requires predictability (allies preempt); adversarial coupling operates via disturbance-injection (predicted attack is neutralized)	Derived (from existing `#der-team-persistence` + `#der-adversarial-destabilization` signed-γ structure)
Emitter-side four-regime classification	Dual construction to `#der-interaction-channel-classification`'s recipient-side four regimes	Formulation choice
16-cell emitter-recipient composition closes `#adversarial-edge-targeting`	Five-factor arg-max: four B-interior factors (credence leverage, plan-Jacobian^2, edge identifiability, observability) each canonically grounded, times the opacity targeting-fidelity factor	Factors derived; multiplicative product-form a first-order modeling choice
Backward variance decomposition $H_b = \frac{1}{2}\log 2\pi e + \frac{1}{2}\log[U_{\pi,B\to A} + U_{a,A}]$	Law of total variance + Gaussian entropy; the structural dual of the Kalman $U_M + U_o$ split	Derived (exact in linear-Gaussian sub-scope α; robust qualitative beyond)
Tempo-amplification leading-order: $\mathcal{T}^{\text{eff}} = \mathcal{T} \cdot H_b / H_b^{\max}$	First-order substitution into `#result-adversarial-tempo-advantage`'s tempo-multiplier under Model D	Derived (conditional on Gaussian-coupling sub-scope α)
Parameterization-invariance of H_b	H_b is an action-marginal entropy; action space is coordinate-free per `#scope-agent-identity`	Derived
Candidate 4th `#disc-identifiability-floor` instance (generic observer-side form)	H_b's formal structure — "observer cannot predict agent's future action better than $H_b^{A\|B}$" — is a CHT-style no-go at the observer-side-inference task	Discussion-grade (framing; precise external theorem not yet identified)
Candidate opacity ladder for `#disc-separability-pattern`	Transparent-core / partial-transparency / full-opacity across observer filtrations	Formulation choice (ladder proposal)
Effects-spiral opacity amplification (higher $H_b \to$ higher $\gamma_A \to$ larger $\dot{V}_B \to B$'s actions become more erratic $\to$ observer's model of B degrades $\to$ higher $H_b^{B\|A}$)	Composition of sign-flip derivation with `#der-adversarial-destabilization`'s effects spiral	Sketch (discussion-grade; specific functional form open)
Dual-filtration apparatus (each agent's M_t carries an other-filtration as feature)	Would unify observer-indexing with `#scope-agent-identity`'s single-trajectory formalism more tightly	Open extension (mild architectural, orthogonal to derivations)
Sharp functional form for $\gamma_{\text{effective}}^{\text{adv}} = f(H_b)$	Leading-order: $\gamma \propto H_b$. Exact function depends on sub-scope — Gaussian-coupling linear; sigmoid-coupling saturating	Open per sub-scope

Epistemic Status.

Conditional. Max attainable: *exact* for the Hafez-reduction under IDT + ergodicity; *derived* for the sign-flip via signed-coupling; *formulation choice* for the emitter-regime classification structure; *conditional* for the tempo-amplification formula; *discussion-grade* for the meta-pattern candidate status.

Load-bearing: - The sign-flip derivation from existing signed-γ structure is the segment's core structural contribution — the adversarial/cooperative opacity duality is not a separate posit; it falls out of AAT's existing signed-coupling apparatus. - The emitter-side four-regime classification is a clean dual to `#der-interaction-channel-classification`; its derivation is parallel (boundaries in AAT-native quantities — emitter opacity signal structure, self-model quality, coupling regime). - The closure of `#adversarial-edge-targeting` via the 16-cell composition is a derived arg-max; the previously-stated GAP is filled.

Not established: - Sharp functional forms for $\gamma^{\text{adv}}_{\text{effective}}(H_b)$ outside Gaussian-coupling sub-scope α. - Whether b (the adversarial exponent from `#result-adversarial-exponent-regimes`) is reshaped under bilateral opacity. - Formal fourth `#disc-identifiability-floor` instance (requires external-theorem anchoring not yet identified); discussion-grade framing only. - The effects-spiral's opacity amplification (composition with `#der-adversarial-destabilization`'s spiral) is sketch-level.

Honest Limits.

- **Observer-indexing under complex information structures.** $\mathcal{F}^t_B$ may include shared memory, cryptographic keys, insider knowledge, or partial access to A's internal state. Enumerating all relevant observer-filtration structures for a specific application is task-specific; the segment provides the formal framework, not the per-instance enumeration.
- **Active-deceive (E-IV) requires A to model B's model of A.** Mutual-modeling regress under partially-opposing objectives connects to `#deriv-strategic-composition`'s joint-Jacobian analysis. Active-deceive is formally reachable only when A's GUC architecture admits modeling another agent's model — typically Class 3 (Coupled; LLM-style) or Class 2 (Partial) architectures.
- **Type-level vs token-level opacity.** H_b is trajectory-indexed per `#scope-agent-identity`. Statements about "the opacity of model X" (aggregated across deployments) are type-level claims outside AAT's formal scope; they require additional machinery (e.g., population-level dynamics per Part III gaps).

Discussion.

Dual to observation quality. U_o characterizes how well the agent sees the world: observation noise, ambiguity, model-class fitness. H_b characterizes how well the world sees the agent: predictability to observers. The duality is structural — both quantify information flow through the agent-environment boundary, in opposite directions. High U_o agent (observes the world well) and low H_b agent (is observed well) are independent properties; an agent can have one without the other.

Closing `#adversarial-edge-targeting`. The segment provides the emitter-side arg-max structure missing from the Part III adversarial machinery. Paired with `#der-interaction-channel-classification`'s recipient-side four-regime decomposition, the full adversarial-targeting problem has a closed-form: choose edges where emitter's H_b (to target) and target's recipient-side vulnerability (Regime II magnitude/structural shock) are jointly maximized. This operationalizes what "inside the opponent's loop" means at the targeting layer — Boyd's aphorism becomes an explicit optimization over the 16-cell emitter-recipient product.

Meta-pattern positioning. - `#disc-identifiability-floor`*:* H_b's structure suggests a generic observer-side floor — "the observer cannot predict the agent's action better than H_b" — that Instances 1/2/3 specialize on specific variables (causal structure / mixture parameters / coupling sign). The generic framing is candidate-status: it lacks a single external-theorem anchor clear enough to match F1's CHT or F13's Cramér-Rao, but H_b appears naturally in Instance 3's coupling-sign unidentifiability and in Instance 1's on-policy detection no-go (an observer watching the agent's on-policy play has non-zero H_b on the agent's interventional regime). - `#disc-separability-pattern`*:* candidate opacity ladder — transparent-core (E-I Broadcast; allies / public interfaces) / structured-repair (E-II Selective-signal; trust-weighted partial instrumentation) / general-open (E-III, E-IV; uniformly opaque or active-deceive). Adds to the ladder count if adopted. - `#disc-additive-coordinate-forcing`*:* H_b's logarithmic form is adopted from Shannon via Khinchin-Aczél axiomatics, imported as an applied external theorem rather than re-forced under an AAT-internal additivity axiom. Cross-agent additivity fails under the coupling regimes AAT cares about (correlated opacity structures break independence). Adjacent family member, parallel to the IB Lagrangian's position — not a primary instance.

Parameterization-invariance composes cleanly. H_b is an action-marginal entropy. Under `#scope-agent-identity`'s (PI) axiom, the action space is coordinate-free; H_b is invariant under change of the agent's internal-state parameterization. This composes with the (PI)/Čencov fourth primary instance of `#disc-additive-coordinate-forcing` without adding a new axiom.

Relation to `#der-directed-separation`. Class 3 (Coupled) agents have high structural opacity to any observer without internal access — their f_M and G_t are entangled, so predicting the next action requires joint state modelling. Class 1 (Separated) agents are more transparent at the interface level because the decomposed update admits separate modelling of epistemic vs. purposeful components. H_b therefore tends to be *architecturally higher* for Coupled (Class 3) agents — a structural consequence of architecture rather than choice, beyond what E-III Information-hide captures.

Hafez integration note. The IDT pattern (Hafez et al. 2026) uses bi-predictability P (how well a sidecar observer can predict the agent's next state-action) and entropy change ΔH as diagnostics. The IDT's reported 89% perturbation-detection accuracy (vs. 44% for reward-based monitoring) operates on the Level-2 structure per `#der-loop-interventional-access`. In AAT terms, the IDT is a low-H_b-preserving observation channel — its presence as a modular sidecar reduces H_b for the operator without increasing the agent's internal complexity. For `03-llm-core/`, this validates that modular monitoring of internally-merged agents is feasible and effective even when the agent itself is architecturally Class 3 (Coupled).

Findings.

Agent Opacity (H_b) as Dual to Observation Quality (U_o). **Brief:** How well the agent sees the world (observation quality U_o) and how well the world sees the agent (backward predictive uncertainty H_b) are formal duals — both quantify information flow through the agent-environment boundary, in opposite directions. H_b is observer-, horizon-, and trajectory-indexed, extending Hafez's $H(S, A \mid S')$ with these load-bearing indices. The same H_b quantity has opposite value-to-the-agent depending on coupling sign: cooperative coupling rewards low H_b (allies must predict to coordinate); adversarial coupling rewards high H_b (adversaries cannot neutralize what they cannot predict). The sign-flip is not a separate posit — it falls out of AAT's existing signed-coupling structure ($\gamma^{\text{coop}} > 0$ for ally-disturbance reduction; $\gamma^{\text{adv}} > 0$ for target-disturbance amplification). Adversarial tempo advantage decomposes into a tempo term and an opacity term: $\mathcal{T}^{\text{eff}} = \mathcal{T} \cdot H_b / H_b^{\max}$, so the superlinear $(\mathcal{T}_A / \mathcal{T}_B)^2$ advantage from `#result-adversarial-tempo-advantage` scales with the bilateral opacity ratio $(H_b^{A|B} / H_b^{B|A})^2$ under Model D.

Impact: Operationalizes Boyd's OODA-loop aphorism ("inside the opponent's loop") as a precise multiplicative relationship between tempo and opacity, replacing a metaphor with a coupled differential structure. Closes the previously-reserved `#adversarial-edge-targeting` gap as a 16-cell emitter-recipient composition (4 emitter regimes × 4 recipient regimes from the dual `#der-interaction-channel-classification`), giving a closed-form arg-max for "where to attack." The duality with U_o lets the same machinery cover opposite information regimes — high-U_o low-H_b agents (sees well, is seen well) versus low-U_o high-H_b agents (sees poorly, is hidden) — without parallel theorems for each. Direct relevance to multi-agent safety (legibility for cooperation), adversarial robustness (opacity as targeting-vulnerability shaping), and IDT-style monitoring of opaque AI systems (the IDT pattern is a low-H_b-preserving observation channel).

Novelty Claim: *Claim differentiation* on Hafez's H_b. The base quantity is *adopted* from Hafez et al. 2026 with citation and original name; the AAT-distinctive contributions are (i) the observer/horizon/trajectory indexing per `#scope-agent-identity`, (ii) the sign-flip derivation from existing signed-coupling structure rather than as a separate posit, (iii) the emitter-side four-regime classification dual to AAT's recipient-side classification, closing the adversarial-edge-targeting arg-max via the 16-cell composition, and (iv) the tempo-opacity decomposition of `#result-adversarial-tempo-advantage`. The $U_o \leftrightarrow H_b$ duality framing is the integrative move; predictability has been studied in games and information-theoretic security, but the formal-dual treatment with shared signed-coupling structure appears AAT-distinctive.

Related Work:

Table 16.3

ASF Concern	*Prior-art Language*	*Relationship / Positioning*
The base H_b quantity (entropy of agent's action conditional on observer filtration)	Hafez, Khan & Iqbal 2026 *A Mathematical Theory of Agency and Intelligence* — $H(S, A \mid S')$ under IDT-observer (published 2026, found 2026-03)	*formal antecedent* — adopted directly; AAT reduces to Hafez's form under IDT + ergodicity. The four-index extension is the AAT-distinctive layering
OODA loop and tempo-opacity advantage	Boyd 1986–1995 briefings: *Patterns of Conflict*, *The Essence of Winning and Losing* (delivered 1986–1995, found pre-2026)	*conceptual precursor* — supplies the "inside the opponent's loop" intuition for tempo and opacity advantage; the AAT result quantifies the coupling and decomposes the multiplier
Predictability as strategic resource in games	Camerer 2003 *Behavioral Game Theory*; Aumann & Maschler 1995 *Repeated Games with Incomplete Information* (published 1995/2003, found pre-2026)	*conceptual precursor* — predictability and information advantage are well-studied in game theory; the AAT treatment as a formal dual to sensor noise via signed coupling is distinctive
Information-theoretic security and channel opacity	Shannon 1949 *Communication Theory of Secrecy Systems*, Bell System Technical Journal 28(4); modern steganography literature (Cachin 1998) (published 1949/1998, found pre-2026)	*conceptual precursor* — opacity as a quantifiable information-channel property; AAT applies the same machinery to action-prediction rather than message-transmission
Causal Hierarchy Theorem and observer-side identifiability	Bareinboim, Correa, Ibeling & Icard 2022 (published 2022, found 2025)	*adjacent* — observer-side identifiability floor for causal structure; H_b plays an analogous role for action-prediction. Candidate fourth instance of `#disc-identifiability-floor` (discussion-grade)
Bi-predictability P as substrate-independent diagnostic	Hafez et al. 2026 (cited above)	*formal antecedent* — IDT-based bi-predictability complements H_b as the practitioner-facing diagnostic; AAT's observer-indexing makes the dependence on filtration explicit

Search Log: - 2026-04 (*intuition-only* on the U_o-H_b duality framing): no targeted Undermind-grade search has been conducted on whether information-theoretic agent frameworks elsewhere treat predictability-of-agent and observability-of-world as formal duals via shared signed-coupling structure. Hafez 2026's H_b is the explicit formal antecedent for the base quantity; the AAT-distinctive layering (observer/horizon/trajectory indexing + sign-flip derivation + emitter-regime dual) is the segment's contribution. Targeted future search candidates: epistemic game theory (Aumann's tradition); inverse reinforcement learning where the observer infers agent policy and predictability bounds the inference task (Ng & Russell 2000; Ziebart et al. 2008); legibility / readability literature in human-robot interaction (Dragan & Srinivasa 2013); multi-agent identifiability and observability literature. Pre-search expectation: the constituent moves are well-precedented; the integration as formal dual via shared coupling structure within an agent theory may be novel under nominally-comprehensive search. - 2026-03 (*targeted*): Hafez et al. 2026 confirmed as formal antecedent for the base H_b quantity and bi-predictability diagnostic; the IDT-as-low-H_b-channel framing follows directly.

16.3 *Observation:* Result: Adversarial Exponent Regimes

The adversarial tempo advantage exponent — the power b in $\|\delta_B\|/\|\delta_A\| \sim (\mathcal{T}_A/\mathcal{T}_B)^b$ — is not a single number. It depends on two structural features of the disturbance: whether the adversarial coupling enters as deterministic drift (Model D) or stochastic noise (Model S), and whether the coupling dominates the base disturbance rate. Three regimes, with the coupling-dominant exponents now derived analytically from the respective disturbance models: $b = 2$ under

Model D coupling-dominant (the framework's headline squared scaling); $b = 3/2$ under Model S coupling-dominant (the $1/\sqrt{\mathcal{T}}$ steady-state removes one half-power); $b \to 1$ (Model D) or $\to 1/2$ (Model S) when base disturbance dominates the coupling.

The exponent thus *degrades smoothly through intermediate values as base disturbance grows*, and the qualitative regime change that separates superlinear adversarial dynamics from cooperative-like dynamics is gated by the coupling-dominance condition rather than the presence of adversarial coupling per se. The defensive-design implication is concrete: noise injection into the defender's observation channel — which pushes the regime away from coupling-dominant deterministic drift toward the non-coupling-dominant exponent collapse — can *reduce* the attacker's superlinear advantage without changing tempo or gain. The framework's machinery for "where the squared law applies" is therefore the same machinery for "where the squared law does not apply"; the disturbance-model + coupling-dominance taxonomy carries both directions.

Formal Expression.

Derived (adversarial-exponent-regimes, from Model D/S steady states + coupling model; validated by simulation)

Regime 1: Model D (deterministic drift), coupling-dominant. When adversarial coupling enters as a persistent directional disturbance ($\rho_B = \rho_{\text{base}} + \gamma \cdot \mathcal{T}_A$, GA-2) and coupling dominates ($\gamma \cdot \mathcal{T}_B \gg \rho_{\text{base}}$):

$$b = 2 \qquad \text{(simulation: 1.999)} \tag{16.12}$$

Derived from the Model D steady state $\|\delta\|_{ss} = \rho/\mathcal{T}$ (Prop A.1). See 15.4.

Regime 2: Model S (stochastic noise), coupling-dominant. When adversarial coupling enters through the noise scale of zero-mean perturbations ($\sigma_B = \sigma_{\text{base}} + \gamma \cdot \mathcal{T}_A$, GA-2S) and coupling dominates:

$$b = \frac{3}{2} \qquad \text{(simulation: 1.481)} \tag{16.13}$$

Derived from the Model S steady state $\|\delta\|_{\text{rms}} = \sigma_w/\sqrt{2\mathcal{T}}$ (Prop A.1S). The $1/\sqrt{\mathcal{T}}$ scaling (vs. $1/\mathcal{T}$ for Model D) removes one half-power from the denominator, reducing the exponent from 2 to 3/2. See 15.4.

Regime 3: Non-coupling-dominant. When base disturbance is comparable to or exceeds the adversarial coupling ($\rho_{\text{base}} > rsim\gamma \cdot \mathcal{T}_B$):

$$b \to 1.0 \text{ (Model D)} \quad \text{or} \quad b \to 0.5 \text{ (Model S)} \tag{16.14}$$

The exponent degrades smoothly as the base-to-coupling ratio increases. The asymptotic limits are derived (they reflect the $1/\mathcal{T}$ or $1/\sqrt{\mathcal{T}}$ scaling without the coupling numerator); the smooth interpolation is empirical.

Table 16.4

$\rho_{base}/(\gamma \cdot \mathcal{T}_B)$	*Exponent (deterministic)*	*Exponent (stochastic)*
0.002	1.999	≈ 1.49
0.20	1.877	1.372
2.0	1.445	0.941
6.3	1.213	≈ 0.72

The two columns come from sweeps at different parameters — deterministic from the Variant-A coupling-dominance sweep ($\mathcal{T}_B = 0.1$), stochastic from the Variant-D sweep ($\eta = 0.01$) — aligned here on the shared base-to-coupling ratio axis. Stochastic cells at 0.20 and 2.0 are recorded values (rows $\gamma\mathcal{T}/q = 5.0$ and 0.5 of the Variant-D table); the 0.002 cell is near-asymptotic (recorded 1.481 at ratio 0.02; analytic asymptote 3/2) and the 6.3 cell is log-interpolated between the recorded values at ratios 4 and 10 (0.791, 0.645).

Epistemic Status.

Exact conditional on disturbance model. The coupling-dominant exponents are derived, not empirical: $b = 2$ follows from the Model D steady state (Prop A.1) and the coupling model; $b = 3/2$ follows from the Model S steady state (Prop A.1S) and the coupling model. The simulation results (6 variants, multiple parameter sweeps) now serve as validation of the derived exponents, not as their epistemic foundation. The non-coupling-dominant limits ($b \to 1$, $b \to 1/2$) are derived asymptotically; the smooth interpolation between coupling-dominant and non-coupling-dominant is empirical. What remains empirical is whether a given real adversarial interaction is better modeled as Model D or Model S — that is a domain question, not a theory question.

Discussion.

The disturbance model determines the exponent. The mismatch dynamics (3.9) now distinguish two disturbance models: Model D (bounded deterministic, GA-2) with steady-state $\rho/\mathcal{T}$, and Model S (stochastic zero-mean, GA-2S) with steady-state $\sigma_w/\sqrt{2\mathcal{T}}$. The different steady-state scaling is the root cause of the different exponents. This resolves the ambiguity that previously existed in the single-ρ formulation.

Why the squared law held for the coupling-dominance sweep. In Variant A, the coupling enters as deterministic drift: $\rho_B = \rho_{\text{base}} + \gamma \cdot \mathcal{T}_A$, and the steady state is $\|\delta_B\| = \rho_B/\mathcal{T}_B$. The ratio $\|\delta_B\|/\|\delta_A\|$ in the coupling-dominant limit gives $(\mathcal{T}_A/\mathcal{T}_B)^2$ directly.

Nonlinear correction creates thresholds, not lower exponents. For saturating, sigmoid, and breakdown correction functions under deterministic drift, the issue is not a reduced exponent but a catastrophic divergence when ρ exceeds the correction capacity ($\rho > \mathcal{T} \cdot R$). This is exactly the persistence threshold failure (4.5), observed directly in simulation.

Domain interpretation. Whether a given opponent's tempo increase causes deterministic drift or stochastic noise depends on the domain: - Military: an opponent who maneuvers faster creates systematic positional change (drift, $b \approx 2$) - Market: a competitor who acts unpredictably creates noise in signals ($b \approx 1.5$) - Software: a fast-changing API creates systematic drift in the codebase state (drift) - Adversarial ML: an opponent who varies attack vectors increases observation noise ($b \approx 1.5$)

16.4 *Observation:* Gated Tempo Advantage

Observation noise *gates* the adversarial tempo advantage from the prior result. When agents observe their mismatch through a noisy channel, the faster agent's additional corrections become noisy, partially offsetting its tempo advantage. The mechanism is direct: a faster attacker makes more corrections per unit time, each noisy because of observation noise; the *additional* corrections add proportional noise, partially canceling the additional speed. Simulation confirms this — at observation noise comparable to or larger than process noise, the steady-state mismatch-ratio exponent drops dramatically (fixed-gain agents lose nearly all of the squared scaling; optimal-gain agents matched to the noise level recover *some* but not all of it).

The framework's prescription: the optimal-gain principle (3.6) partially restores the advantage but cannot fully recover the noise-free level. The Riccati gain mitigates the loss by *reducing the gain to match the noise level* — correcting less aggressively but more accurately. *Gated* names the regime structure: the squared scaling that defines `#result-adversarial-tempo-advantage` applies cleanly only in the low-noise regime, while higher noise pushes the system into the non-coupling-dominant regime of `#result-adversarial-exponent-regimes` where the exponent collapses to linear or sub-linear. The defender-side payoff is concrete: noise on the defender's observation channel reduces the attacker's superlinear advantage without altering tempo or gain, which is the structural reason cheap noise injection into the defender's channel is a viable defense against a high-tempo attacker.

Formal Expression.

Observation (obs-gated-tempo-advantage, from track-b Variant E)

In a two-agent adversarial system with observation noise σ_{obs} added to each agent's mismatch signal:

Table 16.5

σ_{obs}	*Exponent (fixed η)*	*Exponent (optimal η^*)*
0.00	1.04	1.04
0.10	1.00	0.97
0.20	0.92	0.94
0.50	0.60	0.63
1.00	0.18	0.40

At $\sigma_{\text{obs}} = 1.0$ (10x the process noise), the fixed-gain adversarial exponent drops from ~ 1.0 to ~ 0.2 — tempo advantage nearly vanishes. The Riccati-optimal gain restores it to ~ 0.4, more than doubling the advantage but not recovering the noise-free level.

The mechanism. When observation noise is high, each correction step adds noise to the mismatch estimate. The faster agent makes more corrections per unit time, each noisy, partially offsetting the benefit of higher tempo. The optimal gain mitigates this by reducing η to match the noise level — correcting less aggressively but more accurately.

Epistemic Status.

Empirical. Max attainable: derived (the mechanism is analytically tractable via Riccati analysis of noisy AR(1) processes). The observation that noise degrades advantage is confirmed by simulation. The optimal gain's partial restoration is consistent with the uncertainty ratio principle (3.6: $\eta^* = U_M/(U_M + U_o)$). The quantitative degradation curve (b vs. σ_{obs}) is empirical at these parameters; a general analytical expression would require solving the coupled noisy-AR(1) system.

Discussion.

Observation quality gates tempo advantage. Boyd insisted that the quality of Orient (observation processing) matters more than raw OODA speed. The simulation results show a formal analog of this pattern: faster tempo with noisy observations (σ_{obs} high) gives nearly zero advantage over a slower agent with equally noisy observations. The tempo advantage is gated by observation quality — consistent with Boyd's emphasis, though the model captures a specific mechanism (noisy correction steps) rather than the full richness of Orient processing.

The optimal gain helps most in the moderate-noise regime. At $\sigma_{\text{obs}} = 0.05$ (observation noise half of process noise), the optimal gain cuts steady-state mismatch by 52% compared to fixed gain. At very high noise, the improvement is less dramatic in absolute terms but more important relatively (0.40 vs. 0.18 exponent).

Practical implication. An agent facing an adversary with superior tempo should invest in degrading the adversary's observation quality rather than trying to match their speed. Conversely, an agent with superior tempo should protect its observation channels — the tempo advantage is only as good as the observation quality that supports it.

Connection to code quality. In the software domain (`#der-code-quality-as-observation-infrastructure` — cross-component reference, see `02-tst-core/`), code quality IS observation infrastructure. A well-structured codebase provides low-noise observations (clear tests, readable code, explicit interfaces). A poorly structured codebase adds observation noise to every development cycle, degrading the developer's effective tempo regardless of how fast they work.

Recipient-side mechanism. High $U_{o,B}$ pushes adversarial events below the observability floor (boundary (I-c) in 15.3). Events that would otherwise land in Regime II (destabilizing) instead fall into Regime III (ambient noise): they contribute to $\sigma^2_{w,B}$ without producing destabilizing mismatch. The tempo-advantage exponent drops because the *fraction* of *A*'s events landing in Regime II shrinks — *A*'s tempo still matters, but more of it is dissipated into the noise floor. This is the recipient-side expression of the rate boundary.

16.5 *Result:* Per-Dimension Persistence

The scalar persistence condition overestimates adaptive capacity when the agent's correction gain varies across dimensions. *The weak dimension is the bottleneck*: it dominates the aggregate mismatch regardless of performance on strong dimensions. The correct condition is per-dimension, derived from the same Lyapunov argument applied dimension-by-dimension. Under Model D (bounded deterministic disturbance) persistence requires $\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$ for each dimension k; under Model S (Gaussian stochastic disturbance) the per-dimension AR(1) stationary distribution supplies three task-adequacy criteria (RMS / MAE / probability bound) related by exact constants. The exact thresholds differ by model and criterion, but the structural form — *weak dimension limits aggregate persistence* — is robust across all variants.

The result is *load-bearing for adversarial dynamics*. An attacker who identifies the target's weak dimension can concentrate disturbance there: the same total disturbance budget concentrated on the weak axis amplifies the mismatch ratio asymmetrically while aggregate metrics still look acceptable, and structural failure follows when that one dimension exceeds its $\delta_{\text{critical},k}$ threshold even with healthy aggregate $\|\delta\|_{L_2}$. Per-dimension monitoring is therefore not optimization polish but a structural requirement for any agent operating under directed pressure. The full adversarial-targeting machinery (the 16-cell emitter-recipient composition in `#der-agent-opacity` with closed-form arg-max) operates on this per-dimension structure, exploiting the asymmetry the scalar persistence condition hides. The matrix-Loewner persistence formulation `#deriv-matrix-persistence-condition` is the strictly sharper canonical form under cross-dimensional correction (off-diagonal $\mathcal{T}$, eigenbasis misalignment with coordinates); the per-dimension form here is the diagonal-$\mathcal{T}$ + axis-aligned-D_δ special case, sharp inside that regime and unsafe outside it.

Formal Expression.

Result (per-dimension-persistence)

For an agent with d-dimensional mismatch $\delta_t \in \mathbb{R}^d$, diagonal correction gain $\eta = \text{diag}(\eta_1, \dots, \eta_d)$, and per-dimension disturbance:

Model D: Deterministic Per-Dimension Threshold. Under bounded disturbance $|w_k(t)| \leq \rho_k$ (GA-2, per dimension), the per-dimension steady-state mismatch is:

$$|\delta_k|_{ss} = \frac{\rho_k}{\alpha_k} \tag{16.15}$$

Persistence requires $\alpha_k > \rho_k/R_k$ for each dimension, or in linear operational form:

$$\mathcal{T}_k > \frac{\rho_k}{\delta_{\text{critical},k}} \quad \text{for each dimension } k \tag{16.16}$$

This is the deterministic worst-case bound — exact under bounded disturbance by the same Lyapunov argument as Prop A.1, applied per dimension.

Model S: Stochastic Per-Dimension Steady State. Under stochastic disturbance $w_{k,t} \sim N(0, \rho_k^2)$ (GA-2S, per dimension), the discrete AR(1) process $\delta_{k,t+1} = (1-\eta_k)\delta_{k,t} + w_{k,t}$ has stationary distribution:

$$\delta_k \sim N\left(0,\ \frac{\rho_k^2}{2\eta_k - \eta_k^2}\right) \tag{16.17}$$

The stationary distribution supplies three task-adequacy criteria, each with its own threshold. The choice of criterion is an engineering decision; the three are related by exact constants for Gaussian δ_k.

(a) RMS bound (mean-square, matches the scalar form in 4.5):

$$\sqrt{E[\delta_k^2]} = \frac{\rho_k}{\sqrt{2\eta_k - \eta_k^2}} \tag{16.18}$$

Requiring $\sqrt{E[\delta_k^2]} < \delta_{\text{critical},k}$ and using the small-η_k approximation $2\eta_k - \eta_k^2 \approx 2\eta_k$:

$$\boxed{\eta_k > \frac{\rho_k^2}{2\,\delta_{\text{critical},k}^2}} \quad \text{(RMS criterion)} \tag{16.19}$$

This is the scalar Model S threshold in 4.5 ($\alpha > n\sigma_w^2/(2R^2)$) applied per dimension.

(b) MAE bound (mean absolute error; bounds the expected deviation rather than its square):

$$E[|\delta_k|] = \sqrt{E[\delta_k^2]} \cdot \sqrt{\frac{2}{\pi}} = \frac{\rho_k}{\sqrt{2\eta_k - \eta_k^2}} \cdot \sqrt{\frac{2}{\pi}} \tag{16.20}$$

Requiring $E[|\delta_k|] < \delta_{\text{critical},k}$:

$$\eta_k > \frac{\rho_k^2}{\pi\,\delta_{\text{critical},k}^2} \quad \text{(MAE criterion)} \tag{16.21}$$

MAE is smaller than RMS by the factor $\sqrt{2/\pi} \approx 0.798$, so the MAE threshold is $2/\pi \approx 0.637$ times the RMS threshold. The criteria differ by a constant but bound different quantities; applying the same numerical $\delta_{\text{critical},k}$ under both does not mean the same thing.

(c) Probability bound (tail-risk criterion, for applications where occasional excursions matter):

$$P\big(|\delta_k| > \delta_{\text{critical},k}\big) < \epsilon \iff \eta_k > \frac{\rho_k^2 \cdot z_{1-\epsilon/2}^2}{2\,\delta_{\text{critical},k}^2} \tag{16.22}$$

where $z_{1-\epsilon/2}$ is the two-sided Gaussian quantile. The probability bound at $\epsilon = 0.05$ (two-sided $z \approx 1.96$) is about $1.96^2 \approx 3.84$ times the RMS threshold — stricter because it bounds tail excursions rather than typical magnitudes.

Recommended primary form. The RMS criterion (a) is the canonical form for Model S persistence, matching the scalar treatment in 4.5 and the Lyapunov-based derivation in `#deriv-sector-condition` (Prop A.1S). The MAE and probability-bound variants are provided for applications where those are the natural task-adequacy measures. All three thresholds are quadratic in $\rho_k/\delta_{\text{critical},k}$ (not linear as in Model D), reflecting the $1/\sqrt{\alpha}$ scaling of the Model S stationary variance.

Common Structure. The aggregate L_2 mismatch $\|\delta\| = \sqrt{\sum_k \delta_k^2}$ is dominated by the dimension with the largest ρ_k/η_k ratio (Model S) or ρ_k/α_k ratio (Model D). The qualitative conclusion — the weak dimension is the bottleneck — holds for both models.

Epistemic Status.

Exact conditional on disturbance model. Both per-dimension forms are now derived from their respective disturbance models:

1. **Model D threshold** ($\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$) follows from Prop A.1 applied per dimension — the same Lyapunov argument with bounded disturbance, restricted to each coordinate. This is exact under GA-2 + GA-3.
2. **Model S steady state** follows from the AR(1) stationary distribution under Gaussian disturbance (GA-2S): $\delta_k \sim N(0, \rho_k^2/(2\eta_k - \eta_k^2))$. The RMS, MAE, and probability-bound thresholds are all exact under this distribution, differing by the constants 1, $\sqrt{2/\pi}$, and $z_{1-\epsilon/2}$ respectively. The RMS form $\eta_k > \rho_k^2/(2\delta_{\text{critical},k}^2)$ matches the scalar Model S treatment in 4.5 applied per dimension. The 4-significant-figure simulation match validates Model S, not Model D.

The previously noted "regime mixing" is resolved: the two threshold forms belong to different disturbance models. The Model D threshold is linear in ρ_k; the Model S threshold is quadratic. The 72% scalar overestimate and weak-dimension bottleneck are structural properties that hold under both models.

Discussion.

Scalar tempo overestimates. In a 3D system with gains $\eta = (0.15, 0.03, 0.03)$ and disturbances $\rho = (0.20, 0.20, 0.02)$:

Table 16.6

Dimension	η_k	ρ_k	ρ_k/η_k	$E[\|\delta_k\|]$
1 (well-tracked)	0.15	0.20	1.33	0.303
2 (weak)	0.03	0.20	6.67	0.656
3 (unimportant)	0.03	0.02	0.67	0.066

Scalar prediction: $\rho/\mathcal{T} = 0.284/0.21 = 1.35$. Actual $\|\delta\|_{L_2} = 0.785$. Overestimate: 72%. Dimension 2 alone accounts for 84% of the L_2 mismatch.

Isotropic allocation dominates. Equalizing the same total gain budget ($\eta = 0.07$ per dimension) reduces $\|\delta\|_{L_2}$ from 0.785 to 0.685 — a 13% improvement — because it reduces the bottleneck effect on the weak dimension.

Adversarial exploitation. An adversary who identifies the target's weak dimension can concentrate disturbance there. Targeted attack (80% on the weak dimension) amplifies the mismatch ratio by 17% (from 2.70 to 3.15). The real danger is structural: if the weak dimension's mismatch exceeds its critical threshold ($R_{\max}$), correction fails on that dimension while the aggregate $\|\delta\|_{L_2}$ may still look manageable. Per-dimension monitoring is essential.

Implications for the persistence condition. Like the scalar result, per-dimension persistence addresses *structural persistence* (see Persistence in `LEXICON.md`) — whether the correction machinery on each dimension can outpace that dimension's disturbance rate. An agent can be structurally persistent on every dimension while still being operationally fragile on one (near its per-dimension R_k boundary). The scalar persistence condition (4.5) remains correct as a *necessary* condition: if the aggregate tempo is insufficient, the agent fails. But it is not *sufficient* — an agent can satisfy the scalar condition while failing on a single dimension. The per-dimension condition has two forms: Model D ($\mathcal{T}_k > \rho_k/\delta_{\text{critical},k}$, exact under bounded disturbance) and Model S ($\eta_k > \rho_k^2/(\pi \cdot \delta^2_{\text{critical},k})$, exact under Gaussian disturbance). Both predict per-dimension failure correctly; the choice depends on the disturbance character in the domain.

Connection to multi-agent systems. The per-dimension result has a direct multi-agent analog: in a composite agent, each sub-agent's contribution to composite tempo may be strong in some dimensions and weak in others. The composite's persistence requires coverage across all relevant dimensions — a team of specialists who each handle one dimension well composes better than a team of generalists who are mediocre at everything, provided the dimension assignment matches.

Findings.

The Weakest-Link Dimensional Persistence Law. **Brief:** When mismatch is multi-dimensional, persistence is governed by the worst-served dimension, not by aggregate or average performance. The scalar persistence condition (`#result-persistence-condition`) overestimates adaptive capacity whenever per-dimension correction gains differ from per-dimension disturbance rates. The correct condition is *per-dimension*: $\alpha_k > \rho_k/R_k$ under bounded disturbance (Model D, linear in ρ_k), or $\eta_k > \rho_k^2/(2\,\delta^2_{\text{critical},k})$ under Gaussian disturbance (Model S, quadratic in ρ_k). The aggregate L^2 mismatch is dominated by the dimension with the largest ρ_k/η_k ratio. A simulated 3D system shows the scalar form overestimating adaptive capacity by 72% with a single dimension accounting for 84% of the L^2 mismatch; equalizing the gain budget across dimensions (isotropic allocation) reduces aggregate mismatch by 13% by raising the bottleneck dimension's gain.

Impact: Establishes that survival in any multi-attribute environment is a *min* operation, not a sum or average — a structural critique of scalar capability metrics in adversarial and high-stakes settings. Adversarial implication is sharp: an opponent who identifies the target's weak dimension can concentrate disturbance there, amplifying the mismatch ratio asymmetrically while the aggregate may still appear acceptable. Per-dimension monitoring is therefore not optimization polish but a structural requirement for any agent operating under directed pressure. Carries through composition: a team of specialists each handling one dimension well composes better than a team of generalists mediocre at everything — provided the dimension assignment matches the actual disturbance structure. Calibrates how much the scalar form misleads (Model D linear, Model S quadratic in ρ_k), so the cost of conflating the two is itself quantified rather than asserted.

Novelty Claim: *Claim differentiation* on per-dimension Lyapunov stability. The per-coordinate Lyapunov argument is standard (control theory routinely does diagonal stability analysis); weakest-link arguments appear throughout reliability theory; multi-attribute critiques of scalar evaluation are familiar in decision theory. The AAT-distinctive contributions are (i) the explicit Model D / Model S decomposition with the corresponding linear vs quadratic threshold scaling in ρ_k, (ii) the quantitative overestimate calibration (~72% in a simulated 3D system) showing the scalar form is not just incomplete but materially misleading, and (iii) the connection to adversarial *concentration* — an opponent's optimal targeting strategy is to maximize ρ_k at the weakest η_k, not to spread effort uniformly.

Related Work:

Table 16.7

ASF Concern	*Prior-art Language*	*Relationship / Positioning*
Per-coordinate Lyapunov stability and ultimate boundedness	Khalil 2002 *Nonlinear Systems* (3rd ed.), Prentice Hall, chapters 4 and 9 (published 2002, found pre-2026)	*formal antecedent* — supplies the per-coordinate Lyapunov machinery; the per-dimension result is its application to AAT's correction structure under both bounded and Gaussian disturbance
AR(1) stationary distribution under Gaussian forcing (Ornstein-Uhlenbeck stationary)	Uhlenbeck & Ornstein 1930 *Physical Review* 36:823; Karatzas & Shreve 1991 *Brownian Motion and Stochastic Calculus* (published 1930/1991, found pre-2026)	*formal antecedent* — supplies the AR(1) stationary form used in the Model S threshold derivation
Weakest-link reliability / serial-system reliability	Gnedenko, Belyayev & Solovyev 1969 *Mathematical Methods of Reliability Theory*; Barlow & Proschan 1975 *Statistical Theory of Reliability* (published 1969/1975, found pre-2026)	*conceptual precursor* — the min-operation intuition for serially-dependent failures; AAT instantiates this for adaptive correction rather than for component failure
Multi-attribute decision theory and aggregate-vs-attribute critique	Keeney & Raiffa 1976 *Decisions with Multiple Objectives*, Wiley (published 1976, found pre-2026)	*conceptual precursor* — recognizes that aggregate scoring obscures per-attribute deficits; AAT supplies the dynamics-side analog with explicit failure thresholds
Adversarial robustness and per-feature attack budgets	Szegedy et al. 2014 *Intriguing Properties of Neural Networks*, ICLR; Madry et al. 2018 *Towards Deep Learning Models Resistant to Adversarial Attacks*, ICLR (published 2014/2018, found pre-2026)	*adjacent* — adversarial-ML literature on per-feature perturbation budgets; AAT's concentration analysis (ρ_k at weakest η_k) gives an adaptive-system framing of the same vulnerability
Critique of scalar AI capability metrics	Various AI safety / evaluation literature (Hendrycks et al. 2021 benchmarks; capability evaluation methodology) (published 2021–, found 2026)	*adjacent* — the AAT result supplies a dynamics-grounded structural argument for why aggregate capability scores under-predict adversarial failure

Search Log: - 2026-04 (*intuition-only* on the integrated Model D / Model S framing): per-coordinate Lyapunov stability is standard control theory; weakest-link reliability and multi-attribute critiques of aggregate scoring are well-precedented in their own literatures. The AAT-distinctive contributions are (i) the side-by-side Model D / Model S threshold scaling difference (linear vs quadratic in ρ_k), (ii) the calibrated simulation overestimate as quantitative evidence the scalar form is misleading rather than merely incomplete, and (iii) the adversarial-concentration framing connecting weakest-dimension targeting to the per-dimension threshold. Targeted future search candidates: robustness in adversarial ML (per-feature adversarial attack literature, Goodfellow-Madry line); capability-evaluation methodology in AI safety (the critique-of-scalar-metrics framing has natural relevance there); portfolio-theory analogs of weakest-dimension exposure (Markowitz tradition with downside-risk concentration); reliability-theory weakest-link dynamics with stochastic forcing. - 2025 (*targeted*): Khalil 2002 confirmed as formal antecedent for per-coordinate Lyapunov machinery; the segment cites it inline.

16.6 *Discussion:* Additional Implications & Discussion

The chapter is where Part III closes, and several distinct strands converge here that have been building across the preceding chapters. The strategic-composition derivation lifts the symmetric partially-opposing case from Ch.4's scalar-coupling framing to equilibrium machinery, completing the contraction-to-equilibrium hand-off that Ch.4 surfaced as a boundary. Agent opacity formalizes the dual of observation quality — `#der-agent-opacity` is the chapter's strongest single segment, with rich `## Findings` content — and the 16-cell adversarial-targeting decomposition closes a previously-reserved gap. The adversarial-exponent regime taxonomy ($b = 2, 3/2, \sim 1$) gives the superlinear tempo advantage its disturbance-model decomposition; observation-noise gating gives the structural condition under which the superlinear advantage collapses to roughly linear. Per-dimension persistence — formally located here but content-wise belonging with Part I Ch.4 — gives adversarial concentration its mathematical form. The chapter delivers the framework's first operational instance of identifiability-floor pattern Instance 3 (composite-contraction certification from component-marginal data), and the closing notes hand off cleanly to TST and logogenic-agents work.

Strategic composition lifts contraction to equilibrium.

`#deriv-strategic-composition` is the chapter's central machinery move. When sub-agents are individually Class 1 (Separated — directed separation holds by construction) but pursue *partially-opposing* objectives, *the composite transitions on the dynamic-regime axis from R0 contraction-regime to R1 equilibrium-regime (under α') or R2 cyclic-distributional-regime (under β')*, per `#disc-dynamic-regime-axis`. The architectural classification from Part II Ch.1 is *preserved* under composition with goal divergence (under goal-blind routing and distinct sub-agent substrates, per `#der-directed-separation` Composite-level inheritance) — only the dynamic regime changes. The architectural class is set by the processing-pathway coupling, not by belief content (a sub-agent modeling another's goal-state is ordinary belief content, not a class-changing coupling). The framework's contribution is making the regime transition precise via equilibrium-convergence framing: under sub-scope α' (potential games — Monderer-Shapley 1996; monotone games — Rosen 1965 diagonally-strictly-concave), AAT's persistence machinery transfers cleanly with A2'-analog conditions derived; under sub-scope β', the framework reaches only CCE set-convergence rather than last-iterate convergence (the honest scope limit). The segment also provides the formal home for *effects spiral* via the joint-Jacobian eigenvalue condition — what previously read as a metaphor for cascading misalignment becomes a derivable structural property of the composite at R1.

The implications-side payoff has three threads worth surfacing. First, the result *closes the framework's contraction-to-equilibrium hand-off* that Ch.4's `#result-contraction-template` named as a boundary. Saddle-point equilibria break attracting-fixed-point under Slotine 2003; passivity universality fails under adversarial inputs; Daskalakis et al. 2018's last-iterate non-convergence completes the trio of obstructions. The strategic-composition derivation supplies the *replacement* machinery for those obstructions: equilibrium theory applies in the partially-opposing case where contraction analysis fails. The framework's posture is that *scope-honesty is load-bearing architecture* — knowing where each tool applies cleanly is more useful than trying to extend one tool across all regimes, and the chapter delivers the equilibrium-theoretic tool that handles where contraction stops.

Second, the regime-transition result composes with `#der-directed-separation`'s axis-decomposed inheritance table into a sharper structural commitment: *the framework cannot avoid equilibrium-regime / cyclic-distributional-regime treatment by limiting attention to architecturally-Class-1 sub-agents*. As soon as sub-agents have partially-opposing objectives — which is the empirical norm in multi-agent settings, including most human organizations and many ML systems — the composite operates in R1 or R2 on the dynamic-regime axis, requiring equilibrium-theoretic machinery rather than the contraction-machinery applicable in R0. The architectural-class axis and the dynamic-regime axis are independent: the wrapping construction (`#der-class-coercion-via-wrapping`) operates *only* on architectural class (Class 3 component → Class 1 composite via external scaffolding at structural-leakage cost); alignment work (`#hyp-symbiogenic-composition` symbiogenic absorption is the canonical mechanism) operates *only* on dynamic regime (R1 → R0 via objective re-alignment). The two axes interact but do not collapse — wrapping a strategic composite recovers architectural Class 1 at the wrapper level but the dynamics remain R1 (the equilibrium failure modes — multi-equilibria, saddle-Nash, equilibrium selection ambiguity — persist regardless of wrapping). The framework's posture is that the dynamic regime is a *property of the game structure* (objective alignment + potential/monotone structure + population scale), and goal divergence is one of two operations (alongside adversarial coupling pressure from `#disc-adversarial-coupling-pressure`) that move composites along their respective parallel-pattern axes — modularity-state on Axis A architectural class; dynamic regime on Axis B per `#disc-dynamic-regime-axis` §"Cross-axis interactions."

Third, the chapter formalizes mechanism-design impossibility (Gibbard-Satterthwaite 1973-75 / Myerson-Satterthwaite 1983 / Arrow 1951) as the *designer-side sister meta-pattern* `#disc-implementation-impossibility` to the agent-side `#disc-identifiability-floor`. The three classical theorems land as the cluster's charter instances (`#deriv-strategy-proofness-impossibility`, `#deriv-bilateral-trade-impossibility`, `#deriv-social-welfare-aggregation-impossibility`), with sub-scope α' of `#deriv-strategic-composition` named in two of them (GS and Arrow) as *adjacent to but not identical with* the preference-domain-restriction escape (single-peaked / single-crossing) — convergence-under-current-reports is not dominance-over-all-reports. The 2026-05-20 strengthen-first arm's three-reframing test (see `#deriv-strategy-proofness-impossibility` Discussion) is the documented argument for the non-identity. The agent-side / designer-side suffix discipline at meta-segment naming (-floor / -impossibility) tracks the actor-positioning distinction in the canon; the cluster's discipline filter is constructive-impossibility-shape only, deliberately excluding constructive mechanism design (VCG, revenue equivalence).

Symbiogenic composition and the asymmetric absorption mechanism.

`#hyp-symbiogenic-composition` (formally in Ch.1) gives the chapter its account of *how composite agents form* from sub-agents that were previously not composed: asymmetric absorption, where a host integrates an endosymbiont and the resulting composite's teleological unity U_O crosses the scope threshold from below. The implication is concrete: the framework recovers the empirical pattern of biological symbiogenesis (mitochondria, chloroplasts, gut microbiota-host integration) and organizational acquisition (objective absorption, function transfer, autonomy reduction) as the *asymmetric-parameter limit* of `#deriv-critical-mass-composition`'s closed form. Specifically, when sub-agent parameters $(\alpha_A, R_A, \gamma_A)$ and $(\alpha_B, R_B, \gamma_B)$ are *strongly asymmetric* (one sub-agent dominates), the composite sector constant α_c collapses toward the dominant sub-agent's α, and the absorbed sub-agent's objectives are progressively folded into the dominant sub-agent's via the U_O multiplier on γ. The framework's posture is that symbiogenesis is *not a separate kind of composition* — it is the asymmetric-parameter limit of the same composition machinery that handles the matched-Tier dyad and the heterogeneous-Tier cases.

The cross-domain transfer is concrete enough to ground in specific examples. The eukaryotic-cell formation (Margulis's serial endosymbiosis) is the canonical biological instance: an archaean host absorbing a bacterial endosymbiont produces a composite with the host's broader objective surface and the endosymbiont's specialized metabolic function. Corporate acquisition produces the same dynamics at a different scale: the acquirer's strategic surface absorbs the target's operational function, with the target's objectives folded in through the U_O alignment process. The framework's contribution is the *mathematical form* — the closed-form critical-mass derivation specialized to the asymmetric-parameter limit — that makes the symbiogenic process predictable rather than purely empirical.

Modular safety architectures fail under goal divergence.

The most consequential implication of strategic composition for AI-safety practice is what `#28` in the catalog names but does not yet have a segment home: modular safety architectures of the form "compose modular safety modules with a central planner" are *structurally guaranteed to fail under goal divergence*. The argument composes `#deriv-strategic-composition` (Class-1 sub-agents with partially-opposing objectives transition the composite from R0 contraction-regime to R1 equilibrium-regime or R2 cyclic-distributional-regime per `#disc-dynamic-regime-axis`) directly: when safety modules and the central planner have non-aligned objectives — exactly when the safety modules are *needed* (to constrain the planner against its own objectives that may produce undesired behavior) — the composite operates in R1 or R2 on the dynamic-regime axis, and the modular safety guarantees of the individual safety modules (designed for R0 contraction-regime dynamics with their unique-attracting-fixed-point machinery) do not transfer to the composite's equilibrium-regime or cyclic-distributional-regime dynamics. The failure mechanism is dynamic-regime change, *not* architectural-class change (the composite remains architecturally Class 1 under goal-blind routing per `#der-directed-separation` Composite-level inheritance; what fails is the *regime-dependent* machinery that the safety guarantees assumed).

The framework's reading of several empirical AI-safety patterns follows directly. Red-team penetrations of constitutional AI exploit the structural mismatch between the constitutional constraints (one set of objectives) and the model's training objectives (another set); under partial opposition the composite operates in R1 equilibrium-regime or R2 cyclic-distributional-regime — admitting saddle-Nash equilibria, multi-equilibria with distinct basins, and last-iterate non-convergence (Daskalakis et al. 2018) — regardless of each component's nominal modularity. Mesa-optimizer formation within nominally-modular systems (Hubinger et al. 2019) instantiates the same regime-axis failure: a sub-component developing its own objective surface that opposes the outer system's objective produces partial opposition that shifts the composite from R0 contraction-regime to R1 equilibrium-regime, and the composite's modular guarantees (designed for R0 contraction-machinery) collapse at the regime boundary. The recurring difficulty of "scalable oversight" without per-module identifiability is the chapter's identifiability-floor Instance 3 instance manifesting: composite contraction certification from component-marginal data is structurally impossible (Liberzon 2003), and oversight regimes that rely on component-only inspection cannot, in general, certify the composite's safety properties.

The framework's prescription is not that modular safety architectures cannot work — it is that they require either *goal-alignment commitment by design* (sub-agents constructed to share objectives, removing the partial-opposition condition; keeps the composite in R0 contraction-regime where modular guarantees compose cleanly) *or equilibrium-theoretic treatment from the start* (acknowledging the composite operates in R1 or R2 and applying `#deriv-strategic-composition`'s α'/β' machinery — joint-Lyapunov-on-deviation under potential/monotone structure for R1; CCE convergence at $O(1/\sqrt{T})$ for R2 — with explicit acknowledgment of equilibrium-selection ambiguity and saddle-Nash failure modes). The implications-side payoff for AI-safety methodology is concrete: any safety architecture that *assumes* modular guarantees compose at the composite level is making a structural commitment that holds only under R0 — i.e., only under goal alignment. Where goal alignment cannot be designed-in by construction (which is the empirical case for most current deployment settings), the framework prescribes the equilibrium-theoretic machinery at the composite scope (architectural wrapping per `#der-class-coercion-via-wrapping` remains available as an *independent* tool operating on the orthogonal architectural-class axis — it recovers Class 1 at the wrapper level but does not change the dynamic regime; the equilibrium-failure modes persist regardless of wrapping). See `#disc-dynamic-regime-axis` §"Cross-axis interactions" for the two-axis treatment.

Agent opacity, the $U_o \leftrightarrow H_b$ duality, and 16-cell adversarial targeting.

`#der-agent-opacity` carries the chapter's strongest single-segment finding, well-developed in its `## Findings` section. *How well the agent sees the world* (U_o) and *how well the world sees the agent* (backward predictive uncertainty H_b) are formal duals. Both quantify information flow through the agent-environment boundary, in opposite directions; both are observer/horizon/trajectory-indexed; and the same H_b quantity has *opposite value to the agent* depending on coupling sign — cooperative coupling rewards low H_b (allies must predict to coordinate), adversarial coupling rewards high H_b (adversaries cannot neutralize what they cannot predict). The sign-flip is not a separate posit; it falls out of the signed-coupling structure from Ch.4.

The implications-side payoff that the segment's Findings flag but is worth elaborating is the *16-cell joint composition* with the recipient-side classification from Ch.4. Four emitter regimes (high/low H_b, cooperative/adversarial) × four recipient regimes (informative / magnitude-shock / structural-shock / ambient-noise) produce 16 cells, each with a different optimization signature for "where to attack" or "where to coordinate." The closed-form arg-max for adversarial targeting routes to the cell where the attacker's opacity is high AND the recipient's regime detector treats the event as structural-shock (which the recipient cannot absorb without architectural change). The framework therefore replaces the OODA metaphor "inside the opponent's loop" with a precise bilinear optimization problem — the attacker chooses the cell that maximizes the recipient-side regime-induced repair cost, given the attacker's opacity budget.

The decomposition $\mathcal{T}^{\text{eff}} = \mathcal{T} \cdot H_b / H_b^{\max}$ gives the framework's quantitative reading of the OODA-loop tempo-opacity multiplier. The superlinear $(\mathcal{T}_A/\mathcal{T}_B)^2$ adversarial scaling from Ch.4 acquires a second multiplier: the *bilateral opacity ratio* $(H_b^{A|B}/H_b^{B|A})^2$ under Model D, with the corresponding 3/2 exponent under Model S. The framework predicts that high-opacity attackers (those who can act without being predicted by the target) get *quadratic* opacity-leverage on top of *quadratic* tempo-leverage — total quartic scaling in the opacity-times-tempo regime. This is the formal grounding for what Boyd's framing called "operating inside the OODA loop": tempo and opacity multiply, and the multiplication is quadratic-in-quadratic at the structural extreme.

The duality also unifies what would otherwise be parallel theorems. High-U_o low-H_b agents (sees well, is seen well) sit at one corner of the duality; low-U_o high-H_b agents (sees poorly, is hidden) sit at the opposite corner. The same machinery covers both via the signed-coupling structure rather than separate analyses, and the cross-domain transfer to multi-agent safety (legibility for cooperation), adversarial robustness (opacity as targeting-vulnerability shaping), and IDT-style monitoring of opaque AI systems all follow from the same dual.

Adversarial scaling regimes and the OODA self-bound.

`#result-adversarial-exponent-regimes` and `#obs-gated-tempo-advantage` together decompose the superlinear-tempo-advantage from Ch.4 into a regime-typed taxonomy: $b = 2$ under deterministic drift (coupling-dominant); $b = 3/2$ under stochastic noise (coupling-dominant); $b \to 1$ in the coupling-non-dominant regime where observation noise is high enough to gate the tempo advantage to near-linear scaling. Both segments are AAT-internal derivations; neither carries a `## Findings` section, so the catalog-grade treatment lives in this implications segment.

The implication worth surfacing is that the *non-superlinear regime* is operationally important and predictable from AAT's existing machinery. When observation noise $U_{o,B}$ is high enough that the attacker's tempo advantage doesn't penetrate the defender's observation channel cleanly, the gain-principle update $\eta_B^* = U_{M,B}/(U_{M,B} + U_{o,B})$ collapses toward zero, and the defender's effective correction rate α_B shrinks. But — counterintuitively — this *helps* the defender at the system level by gating the exponent. The framework predicts that *cheap noise injection* into the defender's observation channel can be a defensive strategy against a high-tempo attacker, by reducing the exponent from superlinear to linear, even at the cost of slowing the defender's individual updates. This is the structural reason the empirical pattern that highly-noisy environments produce more cooperative-game-like dynamics than expected — observation noise gates the adversarial advantage to near-linear scaling, where the attacker cannot exploit superlinear scaling to drive the defender past the destabilization threshold.

The composition with detection latency from `#deriv-update-detection-latency` produces a cross-segment finding (`#40` in the catalog — *OODA Tempo + Detection Latency Self-Bounds Adversarial Targeting*) that has no other natural home. The 16-cell adversarial-targeting matrix is *self-bounded*: the most damaging cell to attack is the one the defender's *detection latency* makes hardest to recognize. The composition is $\mathcal{T}^{\text{eff}} = \mathcal{T} \cdot H_b/H_b^{\max}$ (from `#der-agent-opacity`) and $\mathbb{E}[T_{\text{detect}}] = \Omega((n+1)/\varepsilon)$ (from `#deriv-update-detection-latency`): the cell with maximal damage is the one with maximal opacity and minimal ε given the defender's $n_{\min}$. Adversarial targeting therefore concentrates *where defender's response is slowest*, and the framework predicts the empirical pattern that cybersecurity incidents cluster on long-trusted dependencies (longest accumulated $n_{\min}$) and that adversarial-ML examples cluster on high-confidence model regions. Both observed empirically; the framework supplies the unifying mechanism.

The cross-segment finding's prescription is concrete: defenders should *short-circuit accumulated trust* on critical dependencies (forced forgetting to reset $n_{\min}$) and *probe high-confidence model regions* with explicit interventional checks. The detection-latency self-bound says these are not optional optimizations — they are the structural countermeasures the adversary's optimal-targeting argument forces the defender to deploy. The framework's contribution against the empirical adversarial-robustness literature is that the targeting concentration has a structural derivation, not merely a folkloric observation, and the countermeasures follow directly.

Closing Part III.

Per-dimension persistence (`#result-per-dimension-persistence`) is formally in this chapter though its content belongs with Part I Ch.4. The implications-segment treatment landed at Part I Ch.4; here it appears as the adversarial concentration mechanism: an opponent who identifies the target's weak dimension can concentrate disturbance there, amplifying the mismatch ratio asymmetrically while the aggregate L^2 metric still looks acceptable. The framework's prescription —

per-dimension monitoring is a structural requirement for any agent operating under directed pressure — bites hardest here, where strategic composition delivers attackers who can locate the weak direction and adversarial targeting decomposes the attack into recipient-side regimes that exploit the dimension-asymmetry.

The chapter is also where Part III's open population-dynamics questions surface as recognized gaps: latent structural diversity (variation invisible to current performance but consequential under regime change), endogenous coupling (γ as function of population composition rather than exogenous parameter), composition-transition dynamics (Miller 2022's extreme transition motif), and computational thresholds for social behavior (Miller's ICE framework). These four gaps are the bridge to *population-level* dynamics that the current pairwise/strategic-equilibrium scope does not cover. When fleshed out they would likely form their own chapter on population dynamics, possibly relocating the per-dimension result alongside. The implications-segment treatment flags this rather than over-claiming present coverage.

What Part III closes on is the composite-agent machinery in its operational form: closure-defect / bridge-lemma representability criterion (Ch.2), unity dimensions and Auftragstaktik bandwidth allocation (Ch.3), signed-coupling structure with cooperative-vs-adversarial as one inequality and the recipient-side four-regime taxonomy (Ch.4), strategic composition under equilibrium machinery and agent opacity with 16-cell adversarial targeting (Ch.5). The four chapters compose into AAT's full composite-agent treatment, with the identifiability-floor pattern's four formal instances (Instances 1–4) covering the framework's scope-honest commitments at the boundaries.

The bridge into TST (`02-tst-core/`) and the logogenic-agent stages (`03-llm-core/`, `04-eli-core/`) is direct from here. TST instantiates the AAT machinery in the software domain, with code quality as observation infrastructure, Lindy-Effect as the temporal-optimality grounding, and software development as the *privileged calibration laboratory* (high P1–P6 identification properties). Logogenic agents — Class 3 (Coupled) by construction — inherit the wrapping construction from Ch.2 and the bias bound from Part II Ch.5, with the framework's reach into LLM-class substrates running through those two segments. The orient-cascade-recovery-via-scaffolding argument from Part II Ch.5 implications is the structural claim that connects AAT's persistence machinery to production agentic systems.

The implications-segment series for AAT chapter-ends is complete with this segment. The TST and logogenic chapter-ends follow the same pattern; that work picks up under task 11 after Joseph's check-in.

> **[Gap]** Latent structural diversity: variation in correction architectures invisible to persistence analysis, consequential under regime change

> **[Gap]** Endogenous coupling: γ as function of population composition, not exogenous parameter; coupling emergence threshold

> **[Gap]** Composition transition dynamics: epochal stability → latent diversification → niche emergence → cascading restructuring → re-equilibration (adopts Miller 2022's extreme transition motif)

> **[Gap]** Computational thresholds for social behavior: minimum agent complexity and interaction depth for composition dynamics (adopts Miller 2022's ICE framework; grounds 9.9)

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